MATH 13–Skeletal Notes

MATH 13–Skeletal Notes

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Preface

These are **skeletal** notes. Every section carries the definitions, the statements of the theorems, and the outline of each example and each proof — but the details are left blank, and we fill them in together in class. What you are holding is the scaffolding of the course; the mathematics gets written into it as we do it.

So bring them with you. Before each class, print the section we are about to cover, or download it and annotate it on a tablet — the "PDF" button in the navigation bar gives you the whole book as one file to print from. That file is rebuilt every time new notes are posted, so download a fresh copy rather than printing from an old one. Then complete the blanks in class as we work through the material. A page you filled in yourself is worth more later than any set of notes handed to you finished.

Everything for this course lives here: the notes themselves, the assignments, the review problems for each exam, and the solutions. Each item is added as we reach it, so check back regularly rather than assuming a page is final. Solutions to an assignment are posted only after its due date has passed; the review sets are posted the same way.

The problems are all in Assignments and Review Problems. There are ten assignments, which are graded, and four sets of review problems, which are not — the review sets collect everything the assignments left over, and they are fair game on an exam. Assignments are due at 11:59 PM on the day shown below.

Assignments	**Due day, at 11:59 PM**
1, 2, 4, 5, 7, 8, 10	Friday
3, 6, 9	Wednesday (exam weeks 3, 6, and 9)

You hand your work in through Gradescope, not through this site. The steps — how to upload a scan or a photo of your written solutions, and how to tell Gradescope which page holds which problem — are under Submitting Your Assignments in Gradescope in the course syllabus. Read them once before the first assignment is due; a submission with the pages mismatched is the most common way points get lost for reasons that have nothing to do with the mathematics.

One more thing worth knowing before you start: each of the ten assignment pages carries a Socratic AI tutor that will coach you when you are stuck, without ever giving an answer away. How it works, what it will and will not do, and what to keep out of the chat are all set out in Using the AI Tutor.

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Chapter 1

Hyperbolic Functions

Objectives

After completing this section, you should be able to:

- Define the hyperbolic sine and cosine in terms of e^x and e^{-x}, and differentiate them.
- Sketch the graphs of $\sinh x$ and $\cosh x$, and state their domains and ranges.
- Use the identity $\cosh^2 x - \sinh^2 x = 1$ and relate it to the parametrization of the hyperbola $x^2 - y^2 = 1$.
- Work with the remaining hyperbolic functions $\tanh x$, $\coth x$, $\operatorname{sech} x$, and $\operatorname{csch} x$, and apply hyperbolic identities to integrals and equations.

This chapter is about certain combinations of e^x and e^{-x} which appear in multiple branches of science and engineering. In many ways they are analogous to the trigonometric functions, and they have a similar relationship to the hyperbola that the trigonometric functions have to the circle, which is why they are known as **hyperbolic functions**. We will study these relationships gradually during this chapter.

These are skeletal notes: the definitions, identities, and problem statements are given, and blank boxes and empty grids are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

1.1 Definitions and Derivatives

The **hyperbolic cosine** is defined as

$$\cosh x = \frac{e^x + e^{-x}}{2}, \tag{1.1.1}$$

and the **hyperbolic sine** is defined as

$$\sinh x = \frac{e^x - e^{-x}}{2}. \tag{1.1.2}$$

Find the derivatives $\dfrac{d}{dx}(\cosh x)$ and $\dfrac{d}{dx}(\sinh x)$ by differentiating the two definitions.

workspace

1.2 Graph, Domain, and Range of $\cosh x$ and $\sinh x$

Since $\sinh x$ and $\cosh x$ are built from the two exponentials $\frac{1}{2}e^x$ and $\frac{1}{2}e^{-x}$, their graphs are easiest to understand by sketching those exponentials first. See Figure 1.2.1 and Figure 1.2.2.

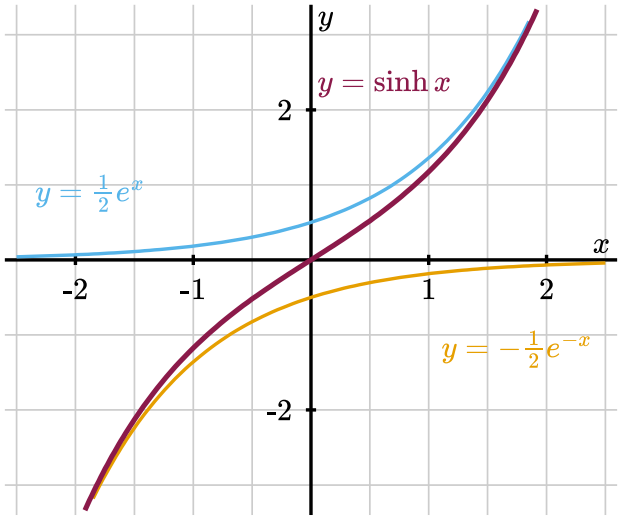

Figure 1.2.1 The graph of $y = \sinh x$. The curve is squeezed between the two exponentials $y = \frac{1}{2}e^x$ and $y = -\frac{1}{2}e^{-x}$, approaching the first as $x \to \infty$ and the second as $x \to -\infty$.

From the graph, read off the domain, the range, and the symmetry of $\sinh x$.

Domain:

Range:

Symmetry:

The graph suggests that $\sinh x$ hugs $\frac{1}{2}e^x$ when x is large and positive, and hugs $-\frac{1}{2}e^{-x}$ when x is large and negative. Confirm this observation.

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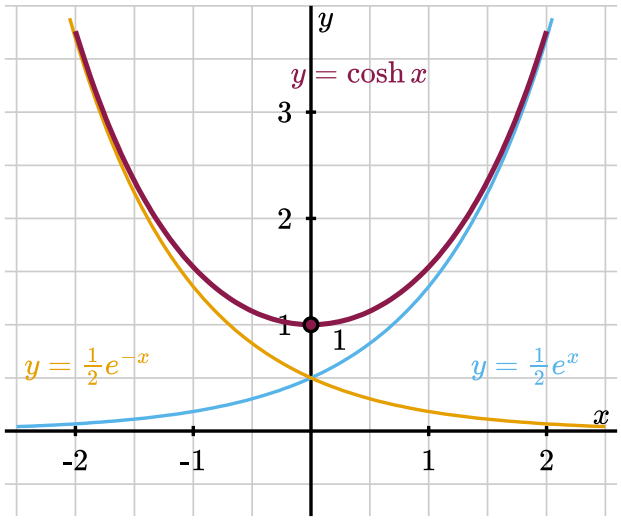

Figure 1.2.2 The graph of $y = \cosh x$. The curve is the sum of the two exponentials $y = \frac{1}{2}e^{x}$ and $y = \frac{1}{2}e^{-x}$, so it lies above both and has its minimum value 1 at $x = 0$.

From the graph, read off the domain, the range, and the symmetry of $\cosh x$.

Domain:

Range:

Symmetry:

The graph suggests that $\cosh x$ hugs $\frac{1}{2}e^{x}$ when x is large and positive, and hugs $\frac{1}{2}e^{-x}$ when x is large and negative, while never dropping below 1. Confirm these observations.

workspace

1.3 Identities and Other Hyperbolic Functions

A similar identity to the trigonometric identity $\cos^2 x + \sin^2 x = 1$ holds for the hyperbolic functions:

$$\cosh^2 x - \sinh^2 x = 1. \tag{1.3.1}$$

As you already know, any point $P(x, y)$ on the circumference of the unit circle ($x^2 + y^2 = 1$) can be described in terms of sine and cosine of an angle $t \in [0, 2\pi)$, i.e. $x = \cos t$ and $y = \sin t$, which results in the trigonometric identity $\cos^2 t + \sin^2 t = 1$. Mark this on the circle in Figure 1.3.1.

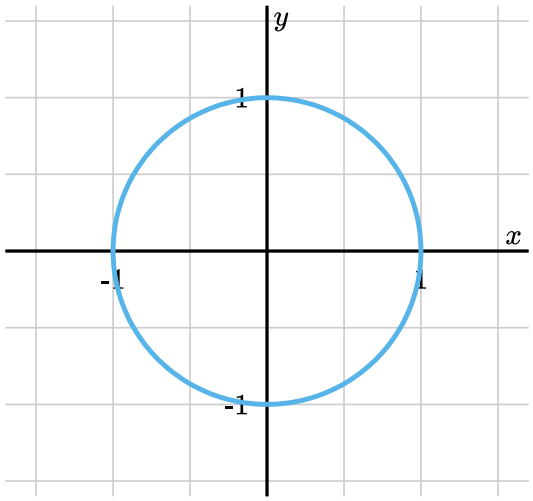

Figure 1.3.1 The unit circle $x^2 + y^2 = 1$. Mark a point $P(\cos t, \sin t)$ on it, and draw the right triangle with legs x and y and hypotenuse 1.

Similarly, any point $P(x, y)$ on the right branch of the hyperbola $x^2 - y^2 = 1$ can be represented as $x = \cosh t$ and $y = \sinh t$, where $t \in \mathbb{R}$. This follows directly from identity (1.3.1), i.e. $\cosh^2 t - \sinh^2 t = 1$. Mark this on the hyperbola in Figure 1.3.2.

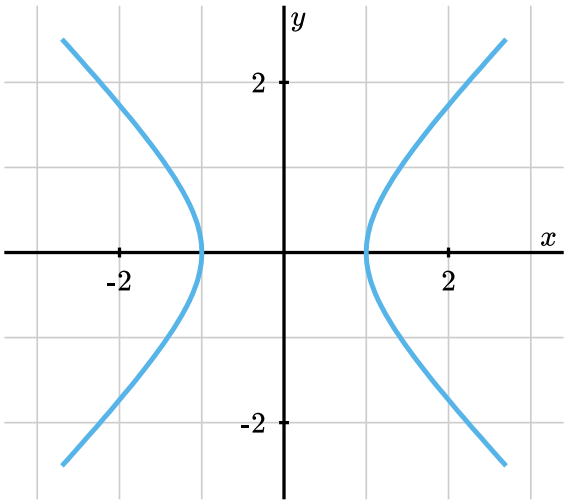

Figure 1.3.2 The hyperbola $x^2 - y^2 = 1$. Mark a point $P(\cosh t, \sinh t)$ on the right branch, and draw the segments from the origin to P and to the vertex.

Checkpoint 1.3.3 Parametrizing the Left Branch. Both branches of $x^2 - y^2 = 1$ appear in Figure 1.3.2, but the parametrization $P(\cosh t, \sinh t)$ traces only the right one, since $\cosh t \geq 1$ for every $t \in \mathbb{R}$. How would you parametrize the *left* branch, where $x \leq -1$?

Your answer and reasoning.

workspace

Hint. The identity $\cosh^2 t - \sinh^2 t = 1$ is unaffected if you change the sign of the first coordinate.

Checkpoint 1.3.4 True or False. The parametrization $x = -\cosh t$, $y = -\sinh t$, $t \in \mathbb{R}$, also traces the left branch of $x^2 - y^2 = 1$.

True or false? Your answer and reasoning.

workspace

As you might have guessed, the rest of the hyperbolic functions are defined

as follows.

$$\tanh x = \frac{\sinh x}{\cosh x} \qquad \operatorname{sech} x = \frac{1}{\cosh x}$$
$$\coth x = \frac{\cosh x}{\sinh x} \qquad \operatorname{csch} x = \frac{1}{\sinh x}$$

1.4 Hyperbolic Identities

Below are some identities that you may find useful in some problems, however, you are not expected to memorize them. We will prove some of them as an exercise later. Each hyperbolic identity is listed next to the trigonometric identity it resembles.

Table 1.4.1 Hyperbolic identities and their trigonometric counterparts.

Hyperbolic	Trigonometric
$\cosh^2 x - \sinh^2 x = 1$	$\cos^2 x + \sin^2 x = 1$
$1 - \tanh^2 x = \operatorname{sech}^2 x$	$1 + \tan^2 x = \sec^2 x$
$\sinh 2x = 2\sinh x \cosh x$	$\sin 2x = 2\sin x \cos x$
$\cosh 2x = \cosh^2 x + \sinh^2 x$	$\cos 2x = \cos^2 x - \sin^2 x$
$\cosh^2 x = \dfrac{\cosh 2x + 1}{2}$	$\cos^2 x = \dfrac{1 + \cos 2x}{2}$
$\sinh^2 x = \dfrac{\cosh 2x - 1}{2}$	$\sin^2 x = \dfrac{1 - \cos 2x}{2}$

Notice the pattern: each hyperbolic identity is its trigonometric counterpart with the sign changed wherever two sines (or two tangents) are multiplied together.

Example 1.4.2 Computing $\cosh^{-1}(x)$. Show that the inverse hyperbolic cosine can be written in terms of the natural logarithm as

$$\cosh^{-1}(x) = \ln\left(x + \sqrt{x^2 - 1}\right), \qquad x \geq 1.$$

Solution. (Restrict cosh to $x \geq 0$ so that it is one-to-one, set $y = \cosh(x)$, and solve for x.)

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Example 1.4.3 An integral via a hyperbolic substitution. Use hyperbolic functions to calculate the integral

$$\int \frac{\sqrt{x^2-1}}{x^2}\,dx, \qquad x \geq 1.$$

Hint: Similar to $\int \sec^2(x)\,dx = \tan(x) + C$, we have $\int \operatorname{sech}^2(x)\,dx = \tanh(x) + C$.

Solution.

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1.5 Hyperbolic Functions in Real Life

Hyperbolic functions are more than a formal analogue of the trigonometric functions: they describe a number of concrete physical situations. We look at three of them.

1.5.1 ∗Hanging Cables and the Catenary

Starred section. This one is for the interested reader. It will not be examined.

If a heavy flexible cable (such as a telephone line, a power line, or a chain) hangs freely from two supports, it settles into a curve called a **catenary**. Contrary to a common guess, this curve is not a parabola; it is the graph of a hyperbolic cosine,

$$y = a\cosh\left(\frac{x}{a}\right),$$

where x is measured horizontally from the lowest point of the cable, so that the y-axis is the axis of symmetry and the lowest point sits at height $y(0) = a$. Sliding the curve up or down, as in $y = c + a\cosh(x/a)$, only changes where we draw the x-axis.

The shape comes out of a balance of forces. Look at the piece of cable running from the lowest point to a point where the cable makes an angle θ with the horizontal, and let s be the arc length of that piece. Three forces act on it: the tension T_0 at the lowest point, which is horizontal; the tension T

along the cable at the other end; and the weight $\lambda g s$ of the piece, where λ is the mass per unit length; see Figure 1.5.1. Balancing the horizontal and the vertical components gives

$$T\cos\theta = T_0$$
$$T\sin\theta = \lambda g s.$$

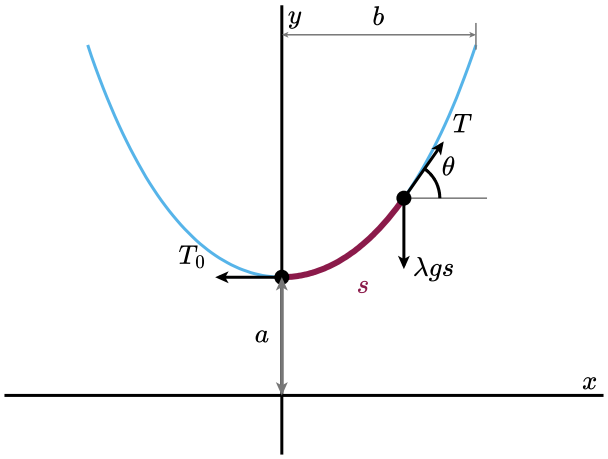

Figure 1.5.1 The three forces on the piece of cable of arc length s running from the lowest point of the cable to a point where the cable makes an angle θ with the horizontal. The lowest point sits at height $a = T_0/(\lambda g)$, and the supports are a distance $2b$ apart. (After Fig. 1 of Behroozi, cited below.)

The quantity $a = T_0/(\lambda g)$ has units of length, and dividing the second equation by the first eliminates T and leaves $s = a\tan\theta$. Since the cable makes the angle θ with the horizontal, $\tan\theta$ is its slope, so

$$\frac{dy}{dx} = \tan\theta = \frac{s}{a}.$$

Now use the arc length element $ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + (s/a)^2}\,dx$ and separate the variables:

$$dx = \frac{a\,ds}{\sqrt{a^2 + s^2}} \qquad \Longrightarrow \qquad x = a\sinh^{-1}\left(\frac{s}{a}\right),$$

where the constant of integration vanishes because $s = 0$ at $x = 0$.

Solving for s gives $s = a\sinh(x/a)$, and therefore

$$\frac{dy}{dx} = \frac{s}{a} = \sinh\left(\frac{x}{a}\right).$$

One last integration, together with $y(0) = a$, produces the catenary equation $y = a\cosh(x/a)$.

Two things are worth noticing. First, a is a pure scale factor: written as $y/a = \cosh(x/a)$, the equation shows that every catenary is a scaled copy of the single curve $y = \cosh x$, in exactly the same way that every circle is a scaled copy of the unit circle; Figure 1.5.2 shows four of them. Second, a is fixed by the cable itself. If the cable has half-length l and its two supports are a distance $2b$ apart, then putting $s = l$ at $x = b$ in $s = a\sinh(x/a)$ gives

$$l = a\sinh\left(\frac{b}{a}\right),$$

which determines a (numerically) from the two lengths. Since $a = T_0/(\lambda g)$, a cable pulled tight has a large a and hangs almost flat, while a slack one has a small a and sags sharply.

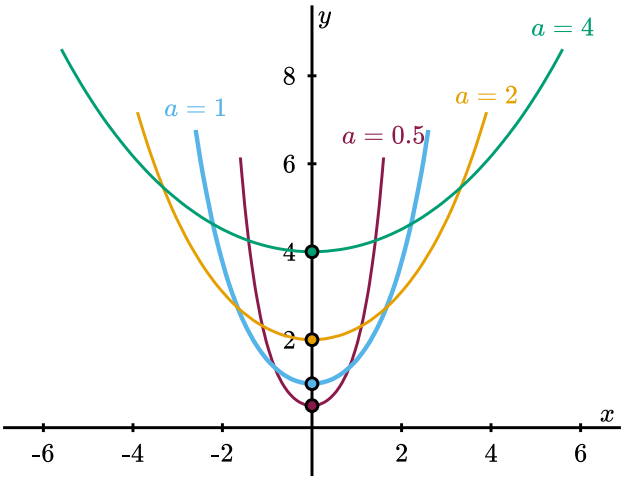

Figure 1.5.2 The catenaries $y = a\cosh(x/a)$ for $a = 0.5,\ 1,\ 2,\ 4$. Each curve meets the y-axis at its own value of a, and all four are scaled copies of the single curve $y = \cosh x$. (After Fig. 2 of Behroozi, cited below.)

A worked example of a hanging cable, in which we find the slope of the cable and the angle at which it meets its pole, appears in Example 1.6.1.

The derivation above follows F. Behroozi, In Praise of the Catenary, *The Physics Teacher* **56**, 214–217 (2018), which also discusses the sense in which all catenaries are similar to one another and suggests simple classroom demonstrations.

1.5.2 Celestial Mechanics

If a comet has enough speed, it can escape the gravitational pull of the sun, in which case one possible trajectory is a hyperbolic trajectory. The comet 2I/Borisov, discovered in 2019, is the first comet known to have come from outside our solar system. It was moving too fast for the sun to capture it, so its path is a hyperbola rather than an ellipse: it swung around the sun once and is now on its way back out. See Figure 1.5.3.

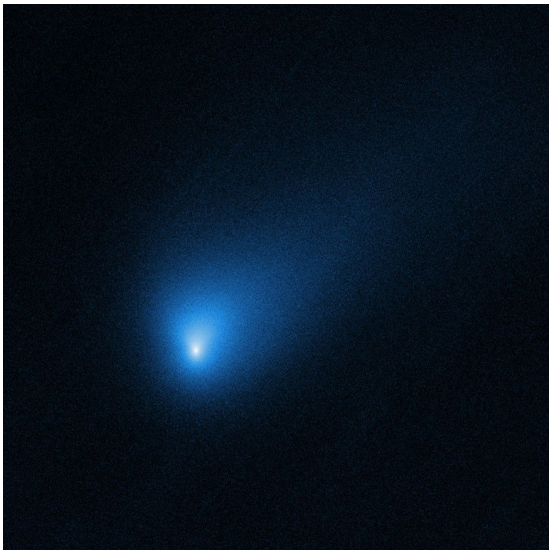

Figure 1.5.3 The interstellar comet 2I/Borisov, photographed by the Hubble Space Telescope in 2019. Because its speed exceeds the escape speed of the sun, its trajectory is a hyperbola and it passes through the solar system only once. (Image: NASA, ESA and D. Jewitt (UCLA).)

1.5.3 The Gateway Arch

The Gateway Arch in St. Louis, Missouri (designed in 1963 and completed in 1965) is a catenary turned upside down: flipping the curve converts the tension carried by a hanging chain into pure compression, which is what masonry and steel carry best. The geometric form of the gateway was set by Hannskari Bandel (structural engineer) and was expressed in the blueprints by the equation

$$y = A\left(\cosh\frac{Cx}{L} - 1\right),$$

where A, C, and L are constants. The arch is slightly flattened compared with a uniform hanging chain, because it is thicker at the base than at the top. It stands 630 feet tall and 630 feet wide at the base; the National Park Service describes its construction and its geometry at Gateway Arch National Park. See Figure 1.5.4.

Figure 1.5.4 The Gateway Arch in St. Louis, Missouri. Its centerline follows the curve $y = A\,(\cosh(Cx/L) - 1)$, an upside-down catenary. (Photograph by John Margolies, 1988; John Margolies Roadside America photograph archive, Library of Congress, Prints and Photographs Division.)

1.6 Further Examples

Example 1.6.1 A hanging telephone line. A telephone line hangs between two poles 14 m apart in the shape of the catenary $y = 20\cosh(x/20) - 15$, where x and y are measured in meters.

A. Find the slope of this curve where it meets the right pole.

B. Find the angle θ between the line and the pole.

Solution. (A sketch of the line between the poles at $x = -7$ and $x = 7$ may help.)

workspace

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Example 1.6.2 Rewriting $\sinh(\ln(x))$**.** Consider the function $f(x) = \sinh(\ln(x))$.

A. Express $f(x)$ as a fraction of two polynomials.

B. Calculate $\dfrac{df}{dx}$.

Solution.

workspace

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Example 1.6.3 A double-angle identity. Prove that $\sinh(2x) = 2\sinh(x)\cosh(x)$.

Proof (start from the definitions).

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Example 1.6.4 Solving a hyperbolic equation. Consider the equation $2\cosh(2x) + 10\sinh(2x) = 5$ and solve for x.

Solution.

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Chapter 2

Series and Sequences

2.1 Infinite Series

Objectives

After completing this section, you should be able to:

- Define an infinite sequence, an infinite series, and the sequence of partial sums, and decide convergence of a series from its partial sums.
- Recognize a geometric series, and determine whether it converges and what it converges to.
- Apply the nth Term Test for divergence, and understand what the test can and cannot conclude.
- Combine convergent series using the sum, difference, and constant multiple rules, and evaluate telescoping series from their partial sums.

What does it mean to add infinitely many numbers together? In this section we answer that question by defining an infinite series and its sequence of partial sums, so that convergence of a series becomes a statement about a limit we already know how to study. We then put the definition to work on the geometric series, whose behavior we can settle completely, and on the nth Term Test, which is our first quick check for divergence.

These are skeletal notes: the definitions, theorems, and problem statements are given, and blank boxes are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

2.1.1 Definitions

Definition 2.1.1 Sequence. A **sequence** is a list of numbers, $a_1, a_2, a_3, \cdots$. An **infinite sequence** of numbers is a function whose domain is the set of positive integers. ◆

Example 2.1.2 A Sequence of Halves. The numbers $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \cdots$ form an infinite sequence; its nth term is $a_n = \left(\frac{1}{2}\right)^n$. ▲

Definition 2.1.3 Infinite Series. The sum of the numbers in an infinite

sequence $\{a_n\}$, i.e. $a_1 + a_2 + a_3 + \cdots = \sum_{n=1}^{\infty} a_n$, is called an **infinite series**. Here a_n is the nth term of the series. ◆

Example 2.1.4 An Infinite Series with a Finite Sum. Infinite sequences can have finite sums. Consider the sum of the sequence from Example 2.1.2, i.e. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$.

It is most convenient to evaluate the result of this sum geometrically. Draw a square of side one in the space below. Shade half of it, then half of what is left, then half of what is left after that, and keep going. Label the pieces $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$. What is the total shaded area, and therefore what is the sum?

draw the unit square and its subdivisions here

▲

2.1.2 Partial Sums

Consider again the infinite sequence $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \cdots$. Let us denote the sum of the first n terms in this sequence by s_n, which are known as **partial sums**. Can we find a pattern in the sequence of partial sums?

Write out s_1, s_2, and s_3, each one both as a fraction and in the form $1 - \square$. Then guess a closed formula for $s_n = \sum_{k=1}^{n} \left(\frac{1}{2}\right)^k$.

$s_1 =$

$s_2 =$

$s_3 =$

$\vdots$

$s_n =$

Now use your formula for s_n to compute the infinite series, by taking the limit of the partial sum as $n \to \infty$. Compare the answer with the one you found geometrically from the square.

workspace

In the example above, the sequence of partial sums converged to a number. In general, if the sequence of the partial sums converges to a number, we say that the series **converges**, otherwise we say that the series **diverges**.

2.1.3 Geometric Series

An important example of infinite series is the geometric series. The **geometric series** is of the form

$$a + ar + ar^2 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1}, \tag{2.1.1}$$

where a, r are real numbers and $a \neq 0$. Note that we can re-write the series as $\sum_{n=0}^{\infty} ar^n$.

Let us compute the partial sum s_n for the geometric series, so that

$$s_n = a + ar + ar^2 + \cdots + ar^{n-1}. \tag{2.1.2}$$

Multiply (2.1.2) by r, subtract the result from (2.1.2), and solve for s_n. Almost every term should cancel.

$r\,s_n =$

$s_n - r\,s_n =$

$s_n =$

Next, take the limit of your partial sum as $n \to \infty$. Treat the cases $|r| < 1$, $|r| > 1$, and $r = 1$ separately, and in each case say what $\lim_{n\to\infty} r^n$ does.

$|r| < 1$:

$|r| > 1$:

$r = 1$:

Checkpoint 2.1.5 What Happens When $r = -1$? We have now handled $|r| < 1$, $|r| > 1$, and $r = 1$, but one case is still missing: $r = -1$. Write out the partial sums s_1, s_2, s_3, s_4 of the series $a - a + a - a + \cdots = \sum_{n=1}^{\infty} a(-1)^{n-1}$ for $a \neq 0$. Does the sequence of partial sums approach a single number as $n \to \infty$? What does that tell you about the series?

Your answer and reasoning.

workspace

Hint. The partial sums do not grow without bound here, the way they do

when $r = 1$. Look instead at whether they settle down to one value.

Putting all of the cases together, we can say that if $|r| \geq 1$, the geometric series is divergent.

Summary.

The geometric series (2.1.1) converges to $\dfrac{a}{1-r}$ if $|r| < 1$, i.e.

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \qquad |r| < 1, \tag{2.1.3}$$

and diverges if $|r| \geq 1$.

Next, as an application of the geometric series, we will go through the following example, which is from our textbook.

Example 2.1.6 A bouncing ball. You drop a ball from a meters above a flat surface. Each time the ball hits the surface after falling a distance h, it rebounds a distance rh, where r is positive but less than 1. Find the total distance the ball travels up and down.

Solution. (Use Figure 2.1.7 to write the total distance as an infinite series, then match it against (2.1.3). Watch the first drop: it is travelled only once.)

workspace

▲

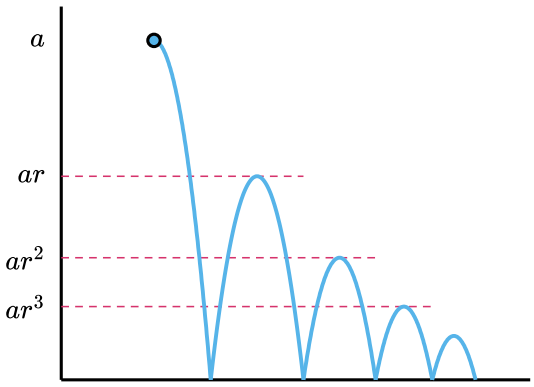


Figure 2.1.7 The ball falls a distance a, then rises and falls a distance ar, then ar^2, and so on, so the total distance travelled is $a + 2ar + 2ar^2 + \cdots$.

2.1.4 The nth Term Test

Theorem 2.1.8 *If* $\sum_{n=1}^{\infty} a_n$ *converges, then* $\lim_{n\to\infty} a_n = 0$.

Important note: If $\lim_{n\to\infty} a_n = 0$, we *cannot* conclude that $\sum_{n=1}^{\infty} a_n$ converges. See parts D and E in the example below. Also see Example 2.1.12.

The following test is a consequence of the above theorem.

The nth Term Test.

If $\lim_{n\to\infty} a_n$ does not exist or $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Example 2.1.9 Testing series for convergence. Determine whether the series is convergent or divergent. If it is convergent, find its sum.

A. $\sum_{n=1}^{\infty} n$

B. $\sum_{n=1}^{\infty} (-1)^n$

C. $\sum_{n=1}^{\infty} \frac{n^3+1}{n^2+n}$

D. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

E. $\frac{4}{9} - \frac{8}{27} + \frac{16}{81} - \frac{32}{243} + \cdots$

Solution.

A.

B.

C.

D.

E.

▲

2.1.5 Combining Series

Theorem 2.1.10 *If* $\sum a_n = A$ *and* $\sum b_n = B$ *are convergent series, then*

1. Sum Rule: $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. Difference Rule: $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. Constant Multiple Rule: $\sum ka_n = k\sum a_n = kA$ *(Any number* k*).*

Example 2.1.11 Using the difference rule. Evaluate $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)} - \frac{1}{2^n}$.

Solution. (You have already summed both of these series in this section: one in Subsection 2.1.2 and the other in part D of Example 2.1.9.)

workspace

▲

Example 2.1.12 A telescoping series with logarithms. Evaluate $\sum_{k=0}^{\infty} \ln\left(\frac{k+2}{k+1}\right)$, if it converges and if it diverges, show that it does.

Solution.

workspace

▲

2.2 Taylor and Maclaurin Series

Objectives

After completing this section, you should be able to:

- Approximate a function near a point by its tangent line and by higher order polynomials built from its derivatives.
- Write down the Taylor series generated by f at $x = a$ and the Maclaurin series generated by f.
- Compute the Taylor polynomial of order n generated by f at $x = a$.
- Find Maclaurin series of standard functions such as $\sin(x)$ directly from the definition.

2.2.1 Definitions of Taylor Series, Maclaurin Series, and Taylor Polynomials

In this section we will answer the following question:

> *If all we know about a function is information at $x = a$, i.e. $f(a), f'(a), f''(a), \cdots$, how can we approximate $f(x)$ with a polynomial $p(x)$?*

These are skeletal notes: the definitions and problem statements are given, and blank boxes are left wherever a derivation or a solution belongs, so that you can fill them in by hand.

Throughout, let f be a function with derivatives of all orders throughout some interval containing a as an interior point.

Before stating the definitions, let us see where the coefficients of such a polynomial have to come from. Suppose the only things we know about f are its readings at $x = a$: the value $f(a)$, the slope $f'(a)$, the second derivative $f''(a)$, and so on. We look for a polynomial written in terms of $x - a$,

$$p_n(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n, \tag{2.2.1}$$

and we pin down the unknown coefficients $c_0, c_1, \ldots, c_n$ by requiring p_n to agree with f at $x = a$ in as many derivatives as it has coefficients. Writing it in terms of $x - a$ rather than x is what makes this manageable: every term after the first vanishes at $x = a$.

Work out the first few coefficients. In each row, differentiate the polynomial as many times as it has coefficients, evaluate at $x = a$, and match against f.

a line, $p_1(x) = c_0 + c_1(x-a)$: $c_0 =$ $c_1 =$

a parabola, $p_2(x) = c_0 + c_1(x-a) + c_2(x-a)^2$: $c_2 =$

a cubic, $p_3(x) = \cdots + c_3(x-a)^3$: $c_3 =$

Notice what should *not* have happened: c_0 and c_1 should come out the same every time. Each new condition is the first one in which the next coefficient appears, so it determines that coefficient and leaves the earlier ones untouched. That is why p_2 is the tangent line with a single new term added to it, rather than a fresh approximation built from scratch. The pattern in the denominators should now be visible: $0!, 1!, 2!, 3!, \ldots$

One computation settles every coefficient at once. Differentiate (2.2.1) exactly k times and evaluate at $x = a$. Say what happens to the terms of degree below k, to those of degree above k, and to the term $c_k(x-a)^k$ itself; then impose $p_n^{(k)}(a) = f^{(k)}(a)$ and solve for c_k.

$p_n^{(k)}(a) =$

$c_k =$

So there is nothing to choose. Once we ask a polynomial to match f and its first n derivatives at $x = a$, its coefficients are determined, and they are the numbers $f^{(k)}(a)/k!$. Reading $f^{(0)}$ as f and $0!$ as 1, the first coefficient $c_0 = f(a)$ fits the same formula. Letting n grow without bound leads to the following definitions.

Definition 2.2.1 Taylor Series. The **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots$$

◆

Definition 2.2.2 Taylor Polynomial of Order n. The **Taylor polynomial of order n** generated by f at $x = a$ is the polynomial

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

◆

Remark 2.2.3 Maclaurin Series. The centre $a = 0$ is common enough to have its own name. The Taylor series generated by f at $x = 0$ is known as the **Maclaurin series generated by f**, which is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots .$$

Example 2.2.4 Approximating e^x near $x = 0$. Write down the Taylor polynomials of orders 1, 2, and 3 generated by $f(x) = e^x$ at $x = 0$.

Solution. Every derivative of e^x is e^x again, so the coefficients of Definition 2.2.2 are easy. Check your answers against Figure 2.2.5.

$p_1(x) =$

$p_2(x) =$

$p_3(x) =$

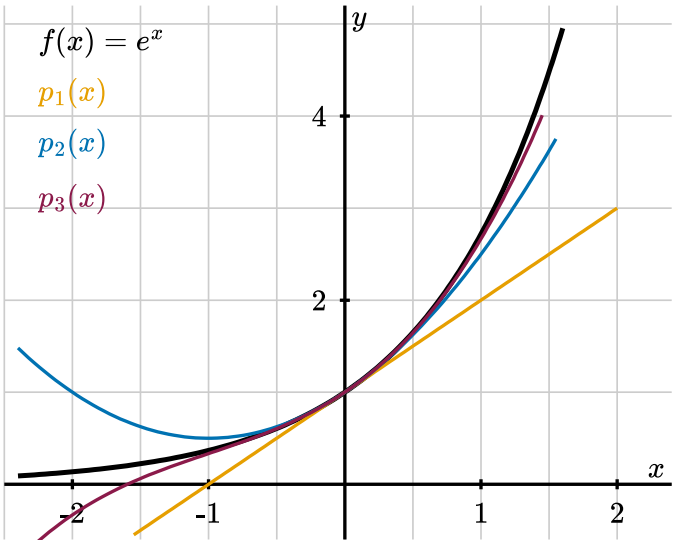


Figure 2.2.5 The function $f(x) = e^x$ together with the Taylor polynomials p_1, p_2, and p_3 at $x = 0$. Near $x = 0$ each polynomial hugs the curve more closely than the one before it.

An animation of this construction accompanies the online version of these notes, with sliders for the centre a and the order n. Two things are worth noting from it. Moving a rebuilds the same construction at a new point of the curve, where p_n again touches f. Raising n adds exactly one term, $\frac{f^{(n)}(a)}{n!}(x-a)^n$, which vanishes at $x = a$ — so it changes the shape of the polynomial everywhere else but never moves it off the point $(a, f(a))$.

The interval on which p_n stays within 0.1 of f widens with every term: about $[-0.5, 0.4]$ for p_1, about $[-1.3, 1.2]$ for p_3, and wider still by p_8. So the Taylor polynomials do not merely improve at the centre; the region where they are usable grows. Whether that region eventually covers everything, and how large the error is at a given x, are the questions of the next section.

2.2.2 More Examples

Example 2.2.6 The Maclaurin series of $\sin(x)$**.** Consider the function $f(x) = \sin(x)$.

A. Find the Taylor series generated by f at $x = 0$. Note that this is the same as the Maclaurin series generated by f.

B. Calculate the first four Taylor polynomials $p_n(x)$.

C. Plot the original function $f(x) = \sin(x)$ and the Taylor polynomials obtained in part B to confirm that the higher order polynomials provide a better approximation.

A. Differentiate f repeatedly, evaluate at $x = 0$, find the pattern, then use Remark 2.2.3.

derivatives:

at $x = 0$:

pattern:

series:

B. Keep the first four non-zero polynomials.

$p_1(x) =$

$p_3(x) =$

$p_5(x) =$

$p_7(x) =$

C. Check your polynomials against Figure 2.2.7. ▲

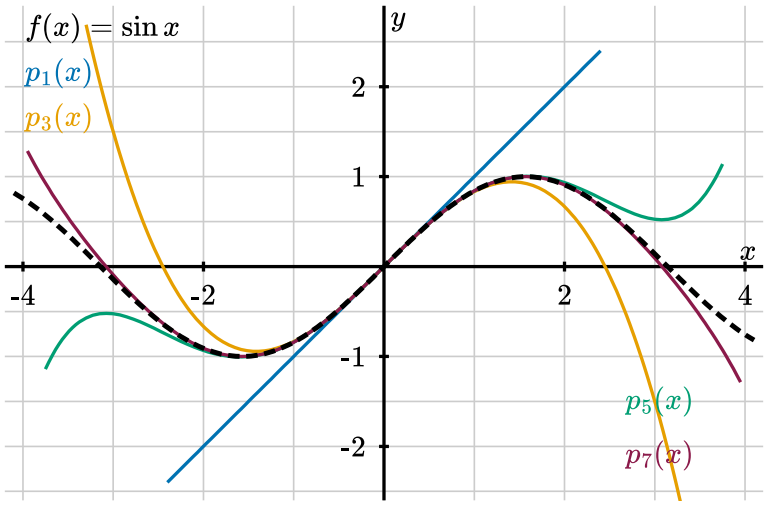


Figure 2.2.7 The function $f(x) = \sin x$ together with the Taylor polynomials at $x = 0$, where $p_1(x) = x$, $p_3(x) = x - \frac{x^3}{3!}$, $p_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$, and $p_7(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$.

Checkpoint 2.2.8 Consider the function $f(x) = \dfrac{x}{1-9x}$.

A. Compute the Maclaurin series generated by f. Express the result in sigma notation.

B. Find the interval of convergence for this series.

A.

B.

Hint. Write $\dfrac{1}{1-9x}$ as a geometric series first.

Checkpoint 2.2.9 Compute the Taylor series of $f(x) = \dfrac{1}{3x+7}$ at $x = 1$.

workspace

Hint. Either differentiate repeatedly at $x = 1$, or write $3x + 7$ in terms of $x - 1$ and reduce to a geometric series.

A list of Taylor series. The Maclaurin series below are derived or used throughout this book, and the first four are the ones on the formula sheet of the exams. Each of them is a Taylor series centered at $a = 0$, so each can be obtained from Definition 2.2.1 by computing derivatives, as we did for e^x and $\sin(x)$. Most of the time, though, it is faster to start from one of these and substitute, differentiate, integrate, or multiply: that is how the series for $\cos(2x)$, e^{-x^2}, $\dfrac{\sin x}{x}$, and $\arctan(x)$ are found in this book.

Table 2.2.10 A list of Taylor series: the Maclaurin series derived or used in this book.

Series	Interval of convergence	Where in the book
$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ $= 1 + x + x^2 + x^3 + \cdots$	$(-1, 1)$	The geometric series, Subsection 2.1.3. Replacing x by $-x$, x^2, or $-x^2$ gives the series
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ $= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	$(-\infty, \infty)$	Derived in Example 2.2.4. Convergence proved in Subsection 2.3.2.
$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$ $= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$	$(-\infty, \infty)$	Derived in Example 2.2.6. Divided by x in Example 2.3.13.
$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$ $= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$	$(-\infty, \infty)$	The derivative of the series of $\sin(x)$. Used, with $2x$ and $5x$ in place of x, in Assignn
$\sinh(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$ $= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$	$(-\infty, \infty)$	Derived in Sample Past Exam 1; it is the odd
$\cosh(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$ $= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$	$(-\infty, \infty)$	The derivative of the series of $\sinh(x)$; it is the
$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ $= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$(-1, 1]$	Its Taylor polynomials, for $\ln(1+2x)$ and $\ln(1$
$\arctan(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$ $= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$	$[-1, 1]$	Obtained by integrating the series of $\frac{1}{1+t^2}$ in S Evaluated at $x = 1$ to approximate π in Exam

Two more series in this book come from the table by substitution, and both converge on $(-\infty, \infty)$: $e^{-x^2} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{n!}$, used in Example 2.3.12 to approximate an integral, and $\frac{\sin x}{x} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n+1)!}$, used in Example 2.3.13 to evaluate one exactly.

One more standard series, the binomial series $(1+x)^m = \sum_{k=0}^{\infty} \binom{m}{k} x^k$, with interval of convergence $(-1, 1)$, is the subject of Section 2.4.

2.3 Taylor Series and its convergence

Objectives

The main objectives of this section are to answer the following questions:

1. Under what conditions is the function $f(x)$ equal to the Taylor series generated by f at $x = a$? That is, when can we say $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$?

2. If the Taylor polynomial of degree n, $p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$, is used as an approximation to $f(x)$, how can we estimate the error in this approximation? That is, how can we estimate the magnitude of $R_n(x) = f(x) - p_n(x)$?

Theorem 2.3.1 and Theorem 2.3.4 will enable us to answer these questions. We begin by stating Taylor's formula and the remainder estimation theorem, and then use them to bound the error in a Taylor approximation, discuss some applications of these theorems, and close with the proof of the remainder theorem.

These are skeletal notes: the theorems and problem statements are given, and blank boxes are left wherever a derivation or a solution belongs, so that you can fill them in by hand.

2.3.1 Taylor's Formula and The Remainder Estimation Theorem

Theorem 2.3.1 Taylor's Formula. *Let f be a function that has $n+1$ continuous derivatives on an open interval I containing a. Then for each $x \in I$ and for each positive integer n, there exists a number c between a and x such that*

$$f(x) = p_n(x) + R_n(x), \tag{2.3.1}$$

where

$$p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k, \tag{2.3.2}$$

and

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}. \tag{2.3.3}$$

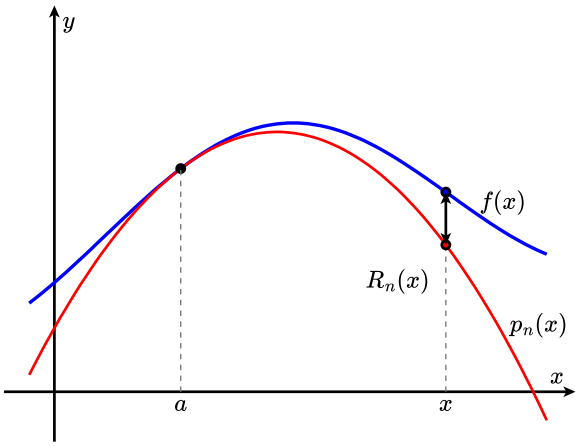


Figure 2.3.2 Taylor's Formula: $f(x) = p_n(x) + R_n(x)$. The Taylor polynomial p_n agrees with f at $x = a$; away from a, the vertical gap between them is the remainder $R_n(x)$.

Remark 2.3.3 Finding an upper bound for the error term without knowing the value of c. Usually the value of c is not explicitly known. However, we may manage to find ***an upper bound*** for the error term $R_n(x)$ without knowing the exact value of c. This is achieved by finding an upper bound M for $|f^{(n+1)}(c)|$, where c is between a and x, and then using this upper bound to estimate the error term.

This is the idea behind every error estimate in this section. It is used in Subsection 2.3.2 to prove that the Taylor series of e^x converges to e^x for every x, and in each of the three examples of Subsection 2.3.3.

Theorem 2.3.4 The Remainder Estimation Theorem. *Let f be a function that has $n+1$ continuous derivatives on an open interval I containing a. Then for each $x \in I$ and for each positive integer n, there exists a number c between a and x such that*

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}, \tag{2.3.4}$$

where M is an upper bound for $|f^{(n+1)}(c)|$ on the interval between a and x.

(Derive (2.3.4) from (2.3.3).)

workspace

You are not expected to know the proof of Taylor's formula itself, only to understand the statement and its implications. The proof is worked through in Subsection 2.3.5.

2.3.2 An example of a Taylor series that converges

Consider the function $f(x) = e^x$. The Taylor series generated by f at $x = 0$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = _____, \tag{2.3.5}$$

and its remainder is

$$R_n(x) = _____. \tag{2.3.6}$$

(Compute (2.3.5) and (2.3.6).)

Taylor series of e^x at $x = 0$:

$R_n(x) =$

(Show that (2.3.5) converges to e^x for every $x \in \mathbb{R}$.)

workspace

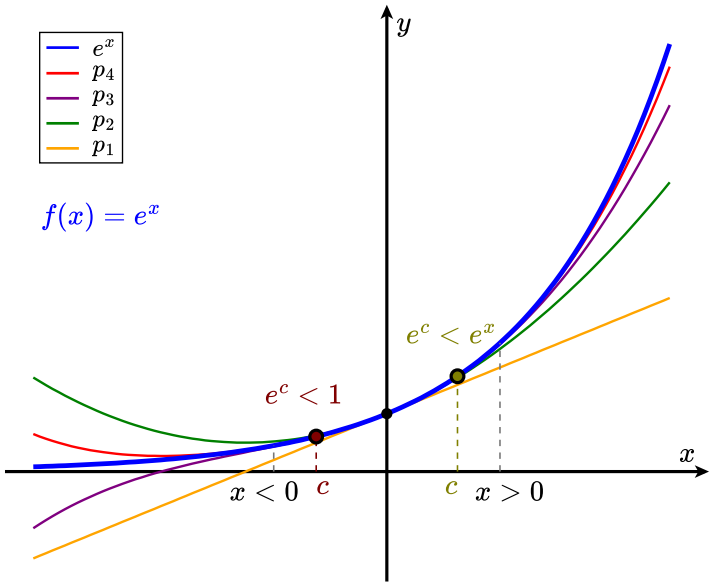


Figure 2.3.5 Schematic graph of $f(x) = e^x$ and its Taylor polynomials $p_1, \ldots, p_4$ about 0.

An animation of this convergence — the Taylor polynomials p_n hugging the graph of $f(x) = e^x$ over a wider and wider interval as n grows — accompanies the online version of these notes.

2.3.3 The error in using a Taylor polynomial

Suppose we approximate $f(x)$ by the Taylor polynomial (2.3.2) of degree n at $x = a$. The error is $R_n(x) = f(x) - p_n(x)$, which is (2.3.1) rearranged, and is given by (2.3.3). To bound it we need an upper bound M for $|f^{(n+1)}(c)|$ on the interval between a and x, as in Remark 2.3.3. The three examples below carry out that step in three different settings.

Example 2.3.6 Approximating e^x using a Taylor polynomial. Approximate $e^{0.5}$ using the Taylor polynomial of degree 2 at $x = 0$. Find an upper bound for the error in this approximation.

Solution.

$p_2(x) =$

bound on $|R_2(0.5)|$*:*

actual error:

▲

Example 2.3.7 Given a desired error bound, find values of x for which the approximation is guaranteed to be valid. Assume that we use the Taylor polynomial of degree 3 at $x = 0$ to approximate $\sin(x)$. For approximately what values of x can you replace $\sin(x)$ by such a Taylor polynomial with an error of magnitude no greater than 4×10^{-3}?

Solution.

$p_3(x) =$ *and* $R_4(x) =$

bound on $|R_4(x)|$*:*

solve for $|x|$*:*

▲

Example 2.3.8 Finding the value of n such that the error is less than a given tolerance. Find the smallest value of n for which the polynomial approximation for $\sin(x)$ is accurate to ± 0.005 for values of x in the interval $[-\pi/2, \pi/2]$.

Solution.

bound on $|R_n(x)|$:

inequality to solve:

test values of n:

▲

2.3.4 Applications of Taylor Series and Remainder Theorem

We begin with two uses of Taylor series that belong to single-variable calculus: computing limits of the indeterminate form $\frac{0}{0}$, which along the way gives a proof of L'Hôpital's rule, and approximating definite integrals whose integrands have no elementary antiderivative.

The remaining three parts of this subsection are starred. Physicists often use Taylor series to approximate functions in order to simplify calculations. The first example is the kinetic energy of an object in relativistic mechanics. The second example is the approximation of the period of a pendulum. We will discuss the error in using these approximations and when it is valid to use them. Additionally, we will discuss how to use the Taylor series to estimate the value of π in the last example.

2.3.4.1 Computing Limits and Integrals with Taylor Series

Taylor's formula Theorem 2.3.1 replaces a function near a point by a polynomial plus a remainder whose size the Remainder Estimation Theorem controls. That is exactly what two problems from single-variable calculus call for: a limit of the indeterminate form $\frac{0}{0}$, where numerator and denominator both vanish and the question is *how fast*, and a definite integral whose integrand has no elementary antiderivative.

We will use two Maclaurin series. The series of $\sin(x)$ was found in Example 2.2.6:

$$\begin{aligned}\sin(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \\ &= x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots, \qquad x \in \mathbb{R}.\end{aligned} \tag{2.3.7}$$

The series of $\arctan(x)$ is

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots, \qquad |x| \le 1; \tag{2.3.8}$$

it comes from the geometric series (2.1.1): since $\frac{d}{dt}\arctan(t) = \frac{1}{1+t^2}$, and $\frac{1}{1+t^2} = 1 - t^2 + t^4 - t^6 + \cdots$ for $|t| < 1$, integrating this from 0 to x term by term gives (2.3.8).

Example 2.3.9 A $\frac{0}{0}$ limit by Taylor series. Compute

$$\lim_{x\to 0} \frac{\arctan(x) - x}{\sin(x) - x}.$$

Solution.

numerator and denominator as series:

factor and cancel:

limit:

why the remaining terms vanish:

▲

Example 2.3.10 Proving L'Hôpital's rule with Taylor's formula. Let f and g have continuous second derivatives on an open interval containing a, with $f(a) = g(a) = 0$ and $g'(a) \neq 0$. Use Taylor's formula to prove L'Hôpital's rule:

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}. \tag{2.3.9}$$

Solution.

Taylor's formula for f and for g:

after cancelling the common factor:

the limit as x tends to a:

the second equality:

▲

Remark 2.3.11 Why the rule sometimes has to be applied several times. If $f'(a) = g'(a) = 0$ as well, the quotient $f'(x)/g'(x)$ is again of the form $\frac{0}{0}$ and L'Hôpital's rule has to be applied again. Taylor's formula explains what is going on. Suppose f and g have $k+1$ continuous derivatives near a, their first $k-1$ derivatives vanish at a, and $g^{(k)}(a) \neq 0$. Then Taylor's formula with $n = k$ gives

$$\frac{f(x)}{g(x)} = \frac{\frac{f^{(k)}(a)}{k!}(x-a)^k + R_k(x)}{\frac{g^{(k)}(a)}{k!}(x-a)^k + \widetilde{R}_k(x)} \longrightarrow \frac{f^{(k)}(a)}{g^{(k)}(a)} \qquad \text{as } x \to a,$$

because each remainder is at most a constant times $|x-a|^{k+1}$. The limit is the ratio of the first nonzero Taylor coefficients, which is exactly what k applications of L'Hôpital's rule compute. In Example 2.3.9 the first nonzero coefficients were those of x^3. Check that $f'''(0)/g'''(0)$, with $f(x) = \arctan(x) - x$ and $g(x) = \sin(x) - x$, agrees with the limit you found.

Example 2.3.12 A definite integral with no elementary antiderivative. The function e^{-x^2} has no elementary antiderivative, so the integral

$$I = \int_0^1 e^{-x^2}\,dx$$

cannot be evaluated with the fundamental theorem of calculus. Use the Taylor polynomial of order n of e^u to approximate I, bound the error with the Remainder Estimation Theorem, and find a value of n that guarantees an error of at most 10^{-4}.

Solution.

series with remainder, and the bound on the remainder:

integrate from 0 to 1:

error bound:

choice of n:

▲

Example 2.3.13 A non-elementary integral computed exactly. In Example 2.3.12, term-by-term integration produced a series that we could only *estimate*. Sometimes, however, the series produced by term-by-term integration is one we already recognize, and then the integral can be evaluated *exactly*, even though the integrand has no elementary antiderivative. Show that

$$\int_0^\infty e^{-x} \frac{\sin x}{x}\, dx = \frac{\pi}{4}.$$

Hint. Each power of x has to be integrated against e^{-x} on $[0, \infty)$. Use the following fact from Calculus II, proved by integration by parts[1]: for every integer $k \geq 0$,

$$I_k = \int_0^\infty x^k e^{-x}\, dx = k!. \tag{2.3.10}$$

Solution.[2]

a series for the integrand:

integrate term by term:

recognize the series:

▲

2.3.4.2 ∗Approximating Relativistic Kinetic Energy

Starred section. This one is for the interested reader. It will not be examined.

In relativistic mechanics, the mass of an object moving with velocity v is given by:

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \tag{2.3.11}$$

where m_0 is the rest mass of the object and c is the speed of light. Then the kinetic energy of an object of mass m moving with velocity v is:

$$K = mc^2 - m_0c^2 \tag{2.3.12}$$

In the case when $v \ll c$, we can use the Taylor series to approximate the kinetic energy. In Example 2.3.15 we will show that the kinetic energy can be

[1] Let $I_k = \int_0^\infty x^k e^{-x}\,dx$. Then $I_0 = \int_0^\infty e^{-x}\,dx = 1$. For $k \geq 1$, integration by parts with $u = x^k$ and $dv = e^{-x}\,dx$ gives $I_k = \Big[-x^k e^{-x} \Big]_0^\infty + k\,I_{k-1} = k\,I_{k-1}$, since $x^k e^{-x} \to 0$ as $x \to \infty$. By induction, $I_k = k(k-1)\cdots 1 \cdot I_0 = k!$.

[2] Integrating an infinite series term by term over the unbounded interval $[0, \infty)$ needs a justification, and the Remainder Estimation Theorem supplies one. Let $S_N(x) = \sum_{n=0}^{N} \frac{(-1)^n x^{2n}}{(2n+1)!}$ be the partial sum of the series for $\frac{\sin x}{x}$, a polynomial of degree $2N$. For $f(x) = \sin x$ every derivative satisfies $|f^{(k)}(t)| \leq 1$. The Taylor polynomial of $\sin x$ of degree $2N+2$ has no x^{2N+2} term, so it equals $x\,S_N(x)$, and the Remainder Estimation Theorem gives $|\sin x - x\,S_N(x)| \leq \frac{|x|^{2N+3}}{(2N+3)!}$. Dividing by $x > 0$, $\left|\frac{\sin x}{x} - S_N(x)\right| \leq \frac{x^{2N+2}}{(2N+3)!}$. Multiplying by e^{-x}, integrating, and using $I_k = k!$ with $k = 2N+2$, the integral differs from the partial sum $\sum_{n=0}^{N} \frac{(-1)^n}{2n+1}$ by at most $\frac{(2N+2)!}{(2N+3)!} = \frac{1}{2N+3}$. This bound tends to 0 as $N \to \infty$, so the partial sums of the Leibniz series converge to the integral. Since they also converge to $\pi/4$, the integral equals $\pi/4$. A numerical evaluation of the integral gives $0.785398\ldots$, which agrees with $\pi/4 = 0.785398\ldots$. The bound $\frac{1}{2N+3}$ also shows why the Leibniz series converges slowly: to guarantee an error below 10^{-3} we need roughly 500 terms.

approximated by the formula $K \approx \frac{1}{2}m_0v^2$ when $v \ll c$. See Figure 2.3.14 for a comparison of the relativistic kinetic energy and its Newtonian approximation.

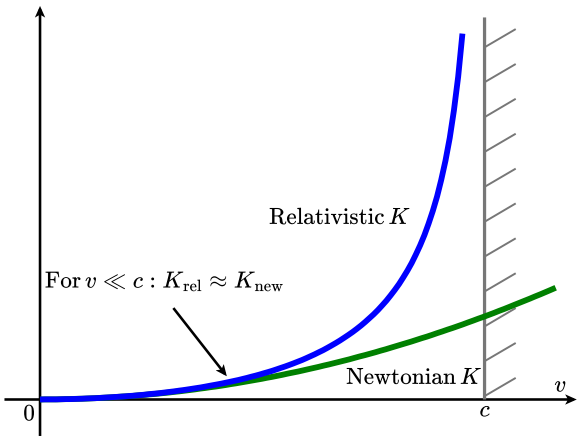


Figure 2.3.14 Relativistic versus Newtonian kinetic energy. The relativistic energy races toward a wall at the speed of light c, while the Newtonian energy follows a gentle parabola; the two are approximately in agreement only when $v \ll c$.

Example 2.3.15 Newtonian kinetic energy as an approximation to relativistic kinetic energy when $v \ll c$. Show that the kinetic energy of an object moving with velocity v can be approximated by the formula $K \approx \frac{1}{2}m_0v^2$ when $v \ll c$.

Solution.

workspace

When $v \ll c$, we can therefore approximate the kinetic energy as:

$$K \approx \frac{1}{2}m_0v^2. \tag{2.3.13}$$

▲

Example 2.3.16 Estimating the error in using the Newtonian kinetic energy formula to approximate the relativistic kinetic energy. Assume

that a car is moving with a velocity of $v = 30m/s$ (≈ 67 miles per hour). Use the remainder's theorem to estimate the error in using the Newtonian kinetic energy formula (2.3.13) to approximate the relativistic kinetic energy of the car. The speed of light is $c = 3 \times 10^8 m/s$.

Solution.

workspace

▲

2.3.4.3 ∗Small-Angle Approximation for a Pendulum

Starred section. This one is for the interested reader. It will not be examined.

We begin this section by briefly reviewing the forces acting on a simple pendulum and how the small-angle approximation allows us to treat its motion as simple harmonic motion.

The bob moves along the arc, so only the component of gravity *tangent* to that arc drives the motion. Resolving the weight $m\vec{\mathbf{g}}$ into a component along the string ($mg\cos\theta$, balanced by the string tension $\vec{\mathbf{F}}_T$) and a component tangent to the arc, as shown in Figure 2.3.17, gives the restoring force

$$F = -mg\sin\theta. \tag{2.3.14}$$

The minus sign indicates that the force always points back toward the equilibrium (straight-down) position. This is *not* Hooke's law: the force is proportional to $\sin\theta$, not to the displacement itself, so the motion is not exactly simple harmonic. Writing the arc displacement as $x = \ell\theta$, we would need F to be proportional to x—that is, to θ—for the motion to be simple harmonic.

The small-angle approximation bridges this gap. From the Taylor series

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots, \tag{2.3.15}$$

when θ is small (in radians) the higher-order terms are negligible and $\sin\theta \approx \theta$. The restoring force then becomes

$$F \approx -mg\theta = -\frac{mg}{\ell}\,x,$$

which *is* Hooke's law $F = -kx$ with effective spring constant $k = \dfrac{mg}{\ell}$.

This is exactly what fixes the period. Newton's second law $F = ma = m\dfrac{d^2x}{dt^2}$ turns Hooke's law into the equation of motion

$$m\frac{d^2x}{dt^2} = -\frac{mg}{\ell}\,x, \qquad \text{that is,} \qquad \frac{d^2x}{dt^2} = -\frac{g}{\ell}\,x.$$

Notice that the mass cancels. The equation says that x is a function whose second derivative is a negative multiple of itself, and the functions with that property are the sines and cosines: writing $\omega^2 = \dfrac{g}{\ell}$, every solution has the form

$$x(t) = A\cos(\omega t) + B\sin(\omega t), \qquad \omega = \sqrt{\frac{g}{\ell}},$$

which you can verify by differentiating twice. The number ω is the angular frequency, and $\cos(\omega t)$ and $\sin(\omega t)$ repeat when ωt increases by 2π. So the motion repeats after a time T with $\omega T = 2\pi$, giving

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{\ell}{g}}. \tag{2.3.16}$$

Thus, for small swings the pendulum behaves as a simple harmonic oscillator, with the period (2.3.16), which is independent of both the amplitude and the mass. Taylor's Remainder Theorem (see Example 2.3.18) enables us to quantify how small θ must be for this approximation.

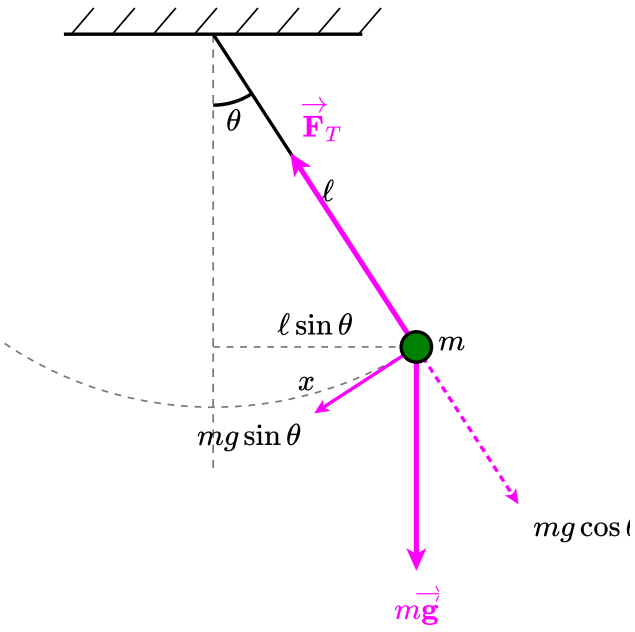


Figure 2.3.17 Forces on a simple pendulum. The weight $m\vec{\mathbf{g}}$ resolves into a component $mg\cos\theta$ along the string (balanced by the tension $\vec{\mathbf{F}}_T$) and a component $mg\sin\theta$ tangent to the arc, which acts as the restoring force.

An animation of these forces — the weight resolving into its radial and tangential components as the pendulum swings — accompanies the online version of these notes.

Example 2.3.18 Simple Harmonic Motion of a Pendulum as an Approximation. Use the remainder theorem to analyze the claim made in Giancoli's textbook that for small angles, $\sin\theta \approx \theta$. Specifically, show that the error is less than 1% for angles below 15°.

Here is the exact quote from Giancoli's textbook:

> "For angles less than 15°, the difference between θ (in radians) and $\sin\theta$ is less than 1%."

Solution.

workspace

▲

2.3.4.4 ∗Approximating the value of π using Taylor series

Starred section. This one is for the interested reader. It will not be examined.

In this subsection, we will discuss how to approximate the value of π using Taylor series. We will use the Taylor series for $\arctan(x)$, which centered at $a = 0$ is

$$\begin{aligned} \arctan(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1} \\ &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots, \qquad |x| \leq 1. \end{aligned} \tag{2.3.17}$$

Example 2.3.19 Approximating π using the Taylor series for $\arctan(x)$. Use the Taylor series for $\arctan(x)$ to approximate the value of π. Use Taylor's Remainder Theorem to find an upper bound for the error in this approximation, and show that the error decreases as the order n of the Taylor polynomial increases.

Solution.

workspace

For the record: the derivatives of $f(x) = \arctan(x)$ satisfy

$$\left|f^{(n+1)}(t)\right| \leq n! \quad \text{for all } t \in \mathbb{R}, \tag{2.3.18}$$

and the resulting error bound is

$$|\pi - 4\,T_n(1)| \leq \frac{4}{n+1}. \tag{2.3.19}$$

The bound goes to zero as $n \to \infty$, but very slowly: to guarantee an error of at most 0.01 one needs $n \geq 399$. ▲

As we saw in the previous example, Taylor series approached π very slowly. In the project below, we will see how to use the so-called Euler's formula to approximate π much faster. As you may know, there are many other methods to approximate π, which we will not cover here.

Project 2.3.4.1 Computing π with Euler's identity. In Example 2.3.19 we approximated π by evaluating the Taylor series (2.3.17) at $x = 1$, and Taylor's Remainder Theorem gave the error bound (2.3.19), which decreases very slowly. In this guided problem we compute π far more efficiently using **Euler's identity**

$$\arctan\left(\tfrac{1}{2}\right) + \arctan\left(\tfrac{1}{3}\right) = \frac{\pi}{4}, \tag{2.3.20}$$

which lets us evaluate the Taylor series at the small arguments $x = \frac{1}{2}$ and $x = \frac{1}{3}$, where it converges much faster.

(a) *Proving Euler's identity.*

Let $\alpha = \arctan\left(\frac{1}{2}\right)$ and $\beta = \arctan\left(\frac{1}{3}\right)$. Use the addition formula

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\,\tan\beta}$$

to prove that $\alpha + \beta = \dfrac{\pi}{4}$.

$\tan(\alpha+\beta) =$

why $\alpha+\beta$ must be $\pi/4$ and not another angle:

(b) *The approximation.*

Let $T_n(x)$ be the Taylor polynomial of order n for $\arctan(x)$ centered at 0. Use Euler's identity (2.3.20) to explain why

$$\pi \approx 4\left[T_n\left(\tfrac{1}{2}\right) + T_n\left(\tfrac{1}{3}\right)\right].$$

Write out this approximation explicitly for $n = 3$.

why the approximation holds:

the case $n = 3$, as a number:

(c) *Bounding the error with the Remainder Theorem.*

In Example 2.3.19 we showed that the derivatives of $f(x) = \arctan(x)$ satisfy (2.3.18). Use (2.3.4) to show that for $|x| \le 1$

$$|\arctan(x) - T_n(x)| \le \frac{|x|^{n+1}}{n+1},$$

and conclude that

$$\left|\pi - 4\left[T_n\left(\tfrac{1}{2}\right) + T_n\left(\tfrac{1}{3}\right)\right]\right| \le \frac{4}{n+1}\left[\left(\tfrac{1}{2}\right)^{n+1} + \left(\tfrac{1}{3}\right)^{n+1}\right].$$

the bound on $|\arctan(x) - T_n(x)|$*:*

combining the two remainders:

(d) *How much better is it?*

Evaluate the error bound for $n = 9$ and compare it with the bound $\frac{4}{10} = 0.4$ obtained in Example 2.3.19 for the same order. Then find the smallest n for which the bound guarantees an error of at most 10^{-6}.

the bound at $n = 9$*, versus* 0.4*:*

the smallest n *with bound at most* 10^{-6}*:*

An animation of the approximation $4\left[T_n\left(\frac{1}{2}\right) + T_n\left(\frac{1}{3}\right)\right]$ converging to π, with its error compared against the much slower method at $x = 1$, accompanies the online version of these notes.

2.3.5 Proof of the remainder theorem

In this subsection, we will provide a proof of the remainder theorem.

Let f be a function that has $n+1$ continuous derivatives on an open interval I containing a. We want to show that for each $x \in I$ and for each positive integer n, there exists a number c between a and x such that

$$f(x) = p_n(x) + R_n(x),$$

where

$$p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k,$$

and

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

To prove this, we will first prove the following lemma:

Lemma 2.3.20 *Let $f(x)$ be a function that is $(n+1)$-times differentiable. Also, suppose that $f(a) = f'(a) = f''(a) = \ldots = f^{(n)}(a) = 0$ and $f(x) = 0$, where*

$x \neq a$. Then there exists a number c between a and x such that $f^{(n+1)}(c) = 0$.

Proof.

workspace

Remark 2.3.21 The error function and its derivatives are zero at the point of expansion. Note that for the error function $R_n(x) = f(x) - p_n(x)$, we have $R_n(a) = R_n'(a) = R_n''(a) = ... = R_n^{(n)}(a) = 0$.

Completing the proof.

workspace

2.4 The Binomial Series and Applications of Taylor Series

Objectives

After completing this section, you should be able to:

- Write down the binomial series for $(1+x)^m$ and compute the binomial coefficients $\binom{m}{k}$ for any real power m.
- Explain why the series terminates when m is a non-negative integer and is infinite otherwise.
- Expand functions such as $(x+1)^5$, $(x+1)^{-2}$, and $(x+1)^{1/2}$ as binomial series.

We have already seen one of the applications of the Taylor series, namely approximating a function by a polynomial of degree n. In this section we study two more applications of the Taylor series: the binomial series and estimating/ evaluating non-elementary integrals.

These are skeletal notes: the definitions, theorems, and problem statements are given, and blank boxes are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.

2.4.1 Binomial Series

Start with a positive integer exponent, where nothing beyond multiplication is needed. Multiply out by hand.

$(1+x)^1 =$

$(1+x)^2 =$

$(1+x)^3 =$

$(1+x)^4 =$

Two things to notice: how many terms does $(1+x)^n$ have, and what pattern do the coefficients follow? Strip the coefficients out and stack one row per exponent, each row centered below the one before it.

$n = 0$:

$n = 1$:

$n = 2$:

$n = 3$:

$n = 4$:

write each row centered below the previous one

Figure 2.4.1 Pascal's triangle. Row n holds the coefficients of $(1 + x)^n$.

This array is called **Pascal's triangle**. Every row starts and ends with 1. State the rule that produces each remaining entry from the row above it, use it to write row $n = 5$, and read off $(1 + x)^5$.

the rule:

row $n = 5$:

$(1 + x)^5 =$

The triangle tells us *what* the coefficients are, but not yet *why*. Written out in full, the fourth power is

$$(1 + x)^4 = (1 + x)(1 + x)(1 + x)(1 + x), \tag{2.4.1}$$

and every term of the expansion comes from picking either the 1 or the x out of each factor. So x^3 needs the x from three of the four factors and the 1 from the one left over. List the ways, and read off the coefficient of x^3.

the ways to pick three x's:

coefficient of x^3:

In general the coefficient of x^k in $(1+x)^n$ counts the ways of choosing k of the n factors to supply an x. That count is written $\binom{n}{k}$, read "n choose k", and equals

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}. \tag{2.4.2}$$

Checkpoint 2.4.2 Why Does the Addition Rule Hold? The rule you wrote down for Pascal's triangle is a consequence of the same count. Explain why $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

workspace

Hint. A choice of k factors out of n either uses the last factor or it does not. Count the two kinds of choice separately.

Putting the count together with the expansion gives the **Binomial Theorem**:

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k = 1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots + nx^{n-1} + x^n. \tag{2.4.3}$$

What if the exponent is *not* a positive integer — say -2, or $\frac{1}{2}$? The factorial formula (2.4.2) makes no sense there, but the equivalent "falling product" form

$$\binom{m}{k} = \frac{m(m-1)(m-2)\cdots(m-k+1)}{k!} \tag{2.4.4}$$

asks only that we start at m and multiply k numbers, each one less than the previous, and that recipe works for *any* real m. Use it to explain why the expansion stops in one case and not in the other.

m a non-negative integer:

m negative or fractional:

So a negative or fractional m gives an infinite power series rather than a polynomial. Newton took that step; Taylor series justify it. Check that the coefficients (2.4.4) predicts really are the Taylor coefficients of $f(x) = (1+x)^m$ at $x = 0$.

$f(x) = (1+x)^m$ $\quad f(0) =$

$f'(x) =$ $\quad f'(0) =$

$f''(x) =$ $\quad f''(0) =$

$f'''(x) =$ $\quad f'''(0) =$

$f^{(k)}(x) =$ $\quad f^{(k)}(0) =$

$\dfrac{f^{(k)}(0)}{k!} =$

The resulting series

$$1 + mx + \frac{m(m-1)}{2!}x^2 + \cdots + \frac{m(m-1)(m-2)\cdots(m-k+1)}{k!}x^k + \cdots \tag{2.4.5}$$

is known as the **binomial series**.

Binomial Series.

For $-1 < x < 1$,

$$(1+x)^m = 1 + \sum_{k=1}^{\infty} \binom{m}{k} x^k \tag{2.4.6}$$

where

$$\binom{m}{1} = m, \qquad \binom{m}{2} = \frac{m(m-1)}{2},$$

and

$$\begin{aligned}\binom{m}{k} &= \frac{m!}{k!(m-k)!} \\ &= \frac{m(m-1)(m-2)\cdots(m-k+2)(m-k+1)}{k!} \quad \text{for } k \geq 3.\end{aligned}$$

Remark: The binomial series (2.4.6) converges for $|x| < 1$. We will not prove this in these notes.

Example 2.4.3 The power is a positive integer. Use binomial series to write the function $f(x) = (x+1)^5$ as a series.

Solution. (Apply (2.4.6) with $m = 5$, writing out the terms through $k = 7$. Then compute $\binom{5}{6}$, and compare your answer with the expansion you got from Pascal's triangle.)

the series with $m = 5$:

$\binom{5}{6} =$ $\binom{5}{k}$ *for $k > 5$:*

$(1+x)^5 =$

▲

Example 2.4.4 The power is a negative integer. Use binomial series to write the function $f(x) = (x+1)^{-2}$ as a series.

Solution. (Apply (2.4.6) with $m = -2$. In the general coefficient every

factor is negative, so pull the signs out. State where the series converges.)

$\binom{-2}{1} =$ $\binom{-2}{2} =$

$\binom{-2}{k} =$

$(1+x)^{-2} =$

▲

Example 2.4.5 The power is not an integer. Use binomial series to write the function $f(x) = (x+1)^{\frac{1}{2}}$ as a series. Compute the five non-zero terms.

Solution. (Apply (2.4.6) with $m = \frac{1}{2}$. Only the first factor of each coefficient is positive.)

$\binom{\frac{1}{2}}{1} =$

$\binom{\frac{1}{2}}{2} =$

$\binom{\frac{1}{2}}{3} =$

$\binom{\frac{1}{2}}{4} =$

$(1+x)^{\frac{1}{2}} =$

Check your answer against Figure 2.4.6, which plots $y = \sqrt{1+x}$ with the partial sums of this series on the interval of convergence $-1 < x < 1$.

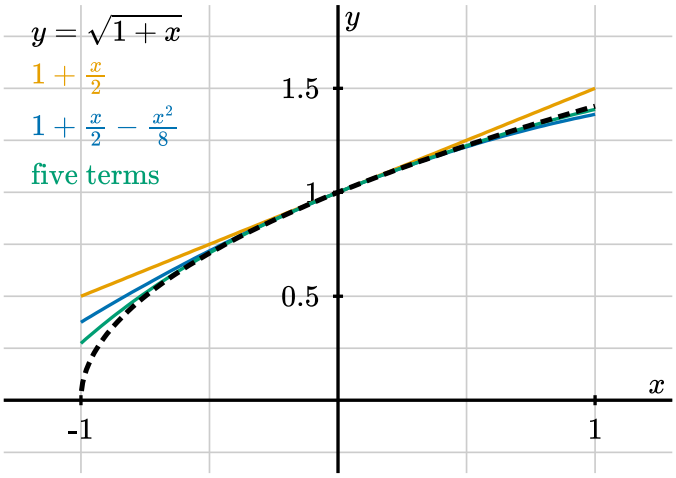


Figure 2.4.6 The function $y = \sqrt{1+x}$ and the partial sums of its binomial series with two, three, and five terms. On the interval $-1 < x < 1$ the partial sums approach the function.

▲

2.4.2 Applications

Many integrals that arise in practice cannot be expressed in terms of elementary functions. Replacing the integrand by its Taylor series and integrating term by term produces a series for the integral, and the partial sums of that series give numerical estimates whose accuracy we can control. Taylor series also give an efficient way to evaluate limits of indeterminate form, often faster than repeated applications of l'Hôpital's rule.

Example 2.4.7 Estimating a non-elementary integral. Use the binomial series to estimate

$$\int_0^{1/3} \frac{1}{\sqrt{1-x^3}}\,dx,$$

and compute an upper bound for the error of your estimate.

Solution. (The antiderivative of $\dfrac{1}{\sqrt{1-x^3}}$ is not an elementary function, so replace the integrand by its series: apply (2.4.6) with $m = -\frac{1}{2}$, substituting $-x^3$ for x. Integrate the result term by term, evaluate at the endpoints, and keep the first two terms as your estimate. For the error, note that all the terms are positive, so the error is exactly the sum of the omitted terms; show that each omitted term is less than $\frac{1}{27}$ times the one before it and compare with a geometric series, using (2.1.3).)

series for $(1-x^3)^{-1/2}$*:*

integrate term by term:

two-term estimate:

upper bound for the error:

▲

Example 2.4.8 Evaluating a limit. Use series to evaluate

$$\lim_{x\to 0} \frac{\sqrt{1+\sin x} - 1 - \frac{x}{2}}{x^2}.$$

Solution. (Start from the binomial series for $\sqrt{1+u}$ that you found in Example 2.4.5. Near $x=0$ the quantity $\sin x$ is small, so substitute $u=\sin x$, then replace $\sin x$ by its Maclaurin series $\sin x = x - \frac{x^3}{3!} + \cdots$ and collect powers of x up to x^3. Subtract $1+\frac{x}{2}$, divide by x^2, and take the limit. Which coefficient decides the answer?)

$\sqrt{1+u} =$

$\sqrt{1+\sin x} =$ *(collect powers of x)*

numerator:

the limit:

▲

In the second example the constant terms and the $\frac{x}{2}$ terms cancel exactly, so the limit is decided by the coefficient of x^2 in the series for $\sqrt{1+\sin x}$. Evaluating the same limit by l'Hôpital's rule would require differentiating the numerator twice, a much longer computation.

Chapter 3

3D Space and Vector Algebra

3.1 Three-Dimensional Coordinate Systems

Objectives

After completing this section, you should be able to:

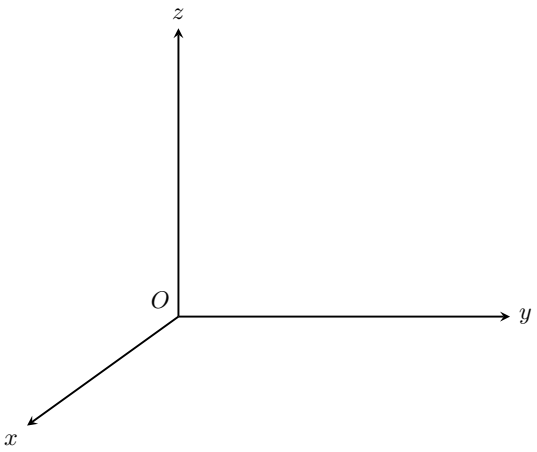
- Locate a point $P(a, b, c)$ in space using the rectangular (Cartesian) coordinate system, and check that the axes are arranged according to the right-hand rule.
- Identify the three coordinate planes and the eight octants, and recognize the surfaces described by equations such as $x = a$, $y = b$, and $z = c$.
- Compute the distance between two points in space, and write the equation of a sphere with a given center and radius.
- Describe the region of space determined by a collection of equations and inequalities, and use completing the square to recognize the equation of a sphere.

3.1.1 The Rectangular Coordinate System

To locate a point in the plane we need two numbers; to locate a point in space we need three. In this section we set up the rectangular coordinate system in space, which is the setting for everything that follows. We then measure distance, describe spheres, and practice translating between an equation and the surface or region it represents.

We will study the Cartesian coordinate system, which is also known as the **rectangular coordinate system**. We choose a point O in space, called the **origin**, and three mutually perpendicular lines through it, called the **coordinate axes** and labelled the x-, y-, and z-axis. A point P in space is then described by an ordered triple (a, b, c), where a, b, and c are the signed distances from the origin along the three axes.

On the axes below, mark off a units along the x-axis, then b units parallel to the y-axis, then c units parallel to the z-axis, and label the point $P(a, b, c)$ you reach.



The three axes must follow the **right-hand rule**: if the index finger of the right hand points along the positive x-axis and the middle finger points along the positive y-axis, then the thumb points along the positive z-axis. Figure 3.1.1 illustrates this convention; the square corner marks in that figure are a reminder that the three axes are mutually orthogonal, so each pair of axes meets at a right angle. Every coordinate system in this course is arranged this way.

The right-hand rule will return in the section on the cross product, where we use it in a second form to determine the direction of the cross product $\mathbf{u} \times \mathbf{v}$ of two vectors.

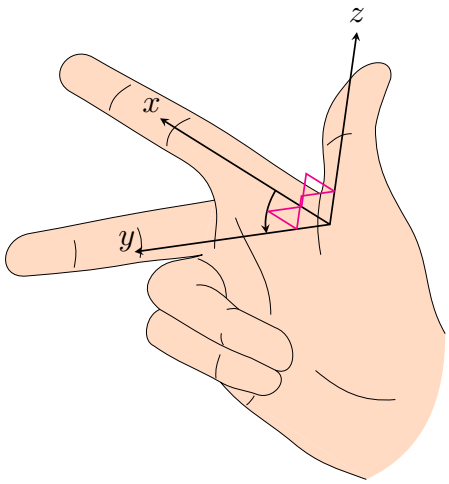
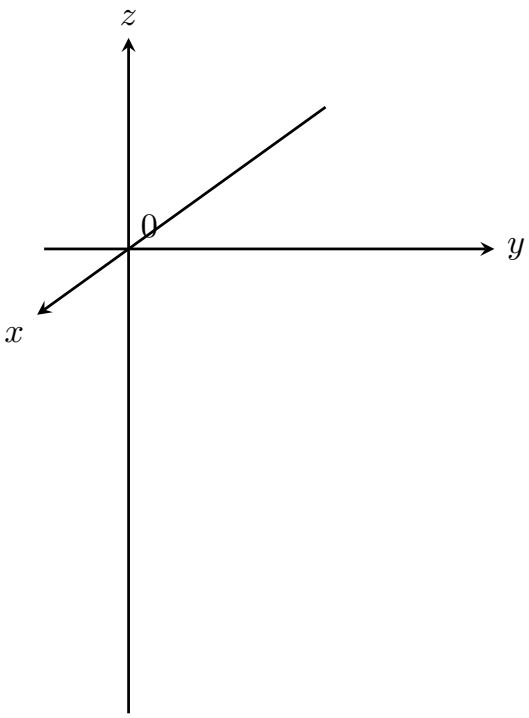


Figure 3.1.1 The right-hand rule. With the index finger of the right hand along the positive x-axis and the middle finger along the positive y-axis, the thumb points along the positive z-axis. The square corner marks record that the three axes are mutually orthogonal: each pair of axes meets at a right angle, even though the drawing on the page must distort those angles.

Now plot the point $(-4, 3, -5)$ on the axes below.



3.1.2 Coordinate Planes and Octants

The three coordinate axes determine three **coordinate planes**. The xy-plane is the plane containing the x- and y-axes, and it is described by the single equation $z = 0$; similarly the xz-plane is $y = 0$ and the yz-plane is $x = 0$. These three planes are shown in Figure 3.1.2.

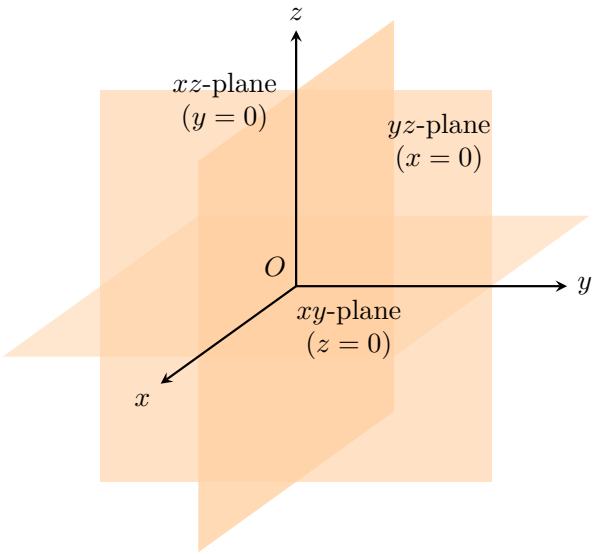


Figure 3.1.2 The three coordinate planes: the xy-plane ($z = 0$), the xz-plane ($y = 0$), and the yz-plane ($x = 0$). They divide space into eight octants.

In the plane the two axes create four quadrants; in space the three coordinate planes create eight **octants**. The **first octant** is the one in which x, y, and z are all positive.

From Figure 3.1.2, record the sign pattern of the octant a point lies in.

First octant, signs of (x, y, z):

The octant directly below it:

Octants with a negative coordinate:

More generally, equations such as $x = a$, $y = b$, and $z = c$ represent planes parallel to the yz-, xz-, and xy-planes respectively. Each of these equations places one restriction on a point of space and leaves the other two coordinates free, so each describes a plane. Consequently a point can be thought of as the intersection of three such planes: the point $P(1, 1, 1)$ of Figure 3.1.3 is the intersection of the planes $x = 1$, $y = 1$, and $z = 1$.

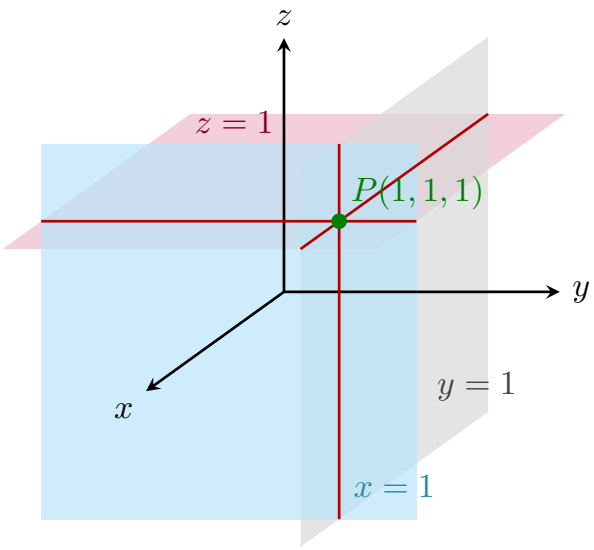


Figure 3.1.3 A point as the intersection of three planes. The planes $x = 1$, $y = 1$, and $z = 1$ meet pairwise in three lines, and all three lines meet in the single point $P(1, 1, 1)$.

3.1.3 Distance Between Two Points

The distance between two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is denoted by $\|P_1P_2\|$ and can be computed with the Pythagorean theorem. As Figure 3.1.4 shows, the segment P_1P_2 is the diagonal of a rectangular box whose edges are parallel to the coordinate axes.

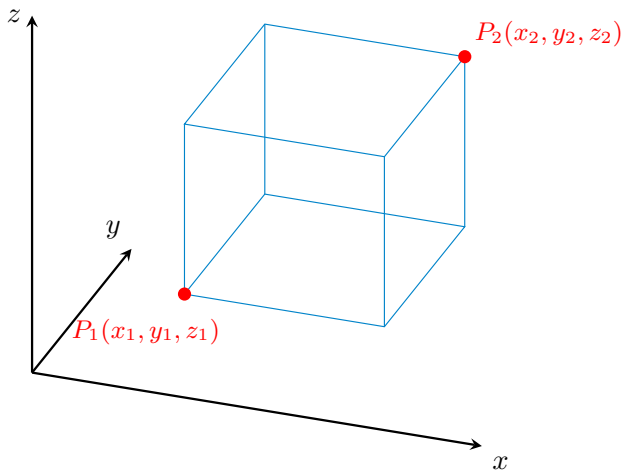


Figure 3.1.4 The segment P_1P_2 is the diagonal of a box with edges parallel to the coordinate axes.

Derive the distance formula from that box.

Step 1, in the base: $\|P_1B\|^2 =$

Step 2, up the vertical edge: $\|P_1P_2\|^2 =$

Substituting, $\|P_1P_2\| =$

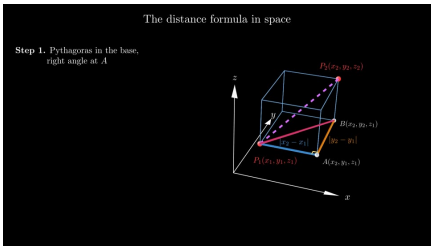
Record the formula you obtained here, so it is easy to find later.

Distance in Space.

For $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$,

$$\|P_1P_2\| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}. \tag{3.1.1}$$

The video in Figure 3.1.5 builds the box of Figure 3.1.4 one edge at a time and applies the Pythagorean theorem twice, in the same two steps. Use it to check the derivation you just wrote.



Standalone

Figure 3.1.5 Deriving the distance formula in space by applying the Pythagorean theorem twice: first in the base triangle P_1AB, then in the vertical triangle P_1BP_2.

3.1.4 Spheres

A **sphere** of radius a is the set of all points $P(x, y, z)$ whose distance from a fixed center $P_0(x_0, y_0, z_0)$ equals a, as in Figure 3.1.6.

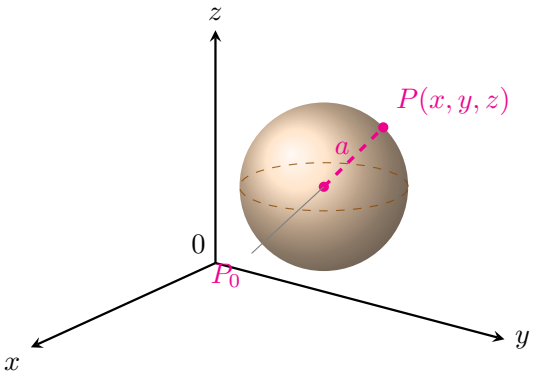


Figure 3.1.6 A sphere of radius a centered at $P_0 = P_0(x_0, y_0, z_0)$: the set of all points $P(x, y, z)$ at distance a from the center.

Turn that description into an equation.

The condition, before squaring:

Equation of the sphere:

Equation of a Sphere.

The sphere of radius a centered at $P_0(x_0, y_0, z_0)$ is

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2. \tag{3.1.2}$$

Example 3.1.7 Recognizing a Sphere by Completing the Square.

Describe the geometric surface represented by the following equation.

$$x^2 + y^2 + z^2 + 2x + 4y = 0$$

Solution.

Completing the square:

Center: *Radius:*

▲

3.1.5 Describing Regions in Space

A single equation in x, y, and z normally describes a surface, two simultaneous equations describe the curve where two surfaces meet, and inequalities describe solid regions. The next example collects several of these situations.

Example 3.1.8 Describing Equations and Inequalities Geometrically. Describe the geometrical meaning of the following equalities and inequalities.

A. $x = 1$

B. $x = 1,\ y = 2$

C. $x^2 + y^2 + z^2 = 1$

D. $1 \leq x^2 + y^2 + z^2 \leq 4,\ z \leq 0$

E. $x^2 + y^2 = 1$

F. $x^2 + y^2 = 1,\ -1 \leq z \leq 1$

G. $x^2 + y^2 = 1,\ z = 3$

Solution.

A.

B.

C.

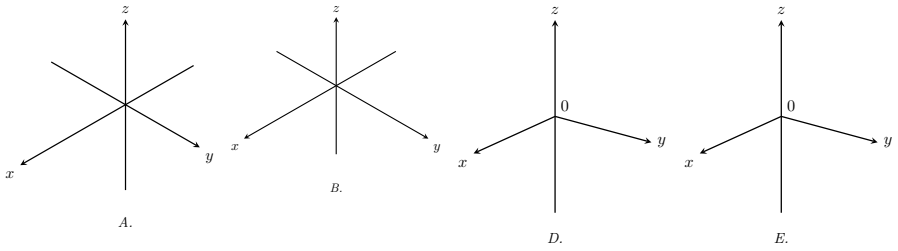
D.

E.

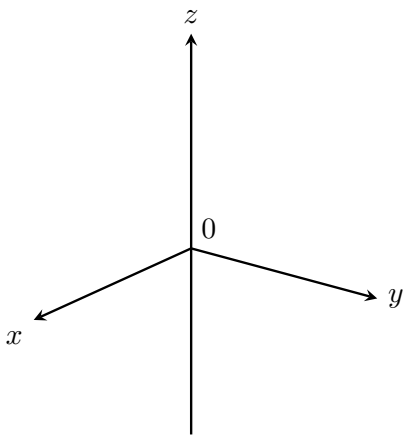
F.

G.

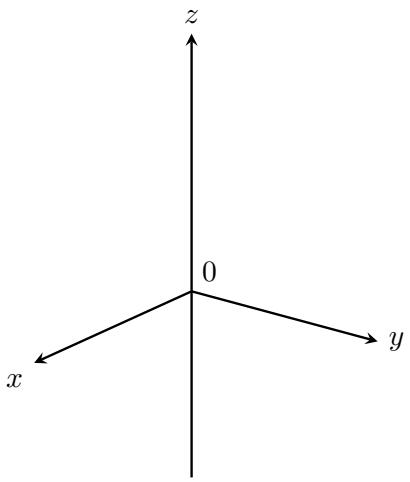
Now sketch parts ***A***, ***B***, ***D***, and ***E*** on the four sets of axes below.



Parts ***F*** and ***G*** are both built from the cylinder of part ***E***. Sketch them on the axes below, and say in one line what changed.



F. $x^2 + y^2 = 1,\ -1 \le z \le 1$



G. $x^2 + y^2 = 1,\ z = 3$

▲

3.2 Vectors

Objectives

After completing this section, you should be able to:

- Explain the difference between a scalar quantity and a vector quantity, and describe a vector by its initial and terminal points.
- Write a vector in component form, find the standard position vector of $\overrightarrow{PQ}$, and compute the length $\|\mathbf{v}\|$ of a vector.
- Add vectors and multiply them by scalars, both algebraically and geometrically using the parallelogram law.
- Write a vector as a linear combination of the standard unit vectors $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$, and find the unit vector in the direction of a given vector.

3.2.1 Definitions, Terminology, and Notation

Some quantities can be described using real numbers alone, such as time or temperature. To describe other quantities we need a direction as well as a magnitude. Consider the velocity of a car: we need to know both the speed of the car and its direction of motion to specify its velocity completely. For instance, we might say that the car is moving with speed 65 miles per hour in the direction of north. Such quantities are referred to as **vector quantities**, and they are the subject of this section.

A **vector** is a directed line segment from an **initial point** A to a **terminal point** B, and is denoted by $\overrightarrow{AB}$. We denote the length of a vector by $\|\overrightarrow{AB}\|$. See Figure 3.2.1.

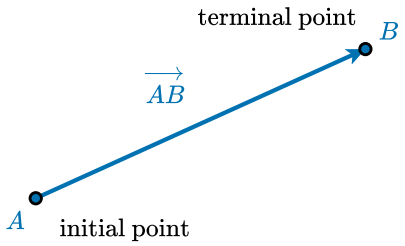


Figure 3.2.1 A vector is a directed line segment. It carries two pieces of information: a direction, shown by the arrowhead, and a magnitude, which is its length $\|\overrightarrow{AB}\|$.

Two vectors are **equal** if they have the same direction and the same magnitude. Notice that this definition says nothing about where a vector is located: a vector may be moved around the plane freely, and as long as its direction and length are unchanged it is the same vector. The vectors shown in Figure 3.2.2 are all equal to each other, $\overrightarrow{AB} = \overrightarrow{CD} = \overrightarrow{OP} = \overrightarrow{EF}$.

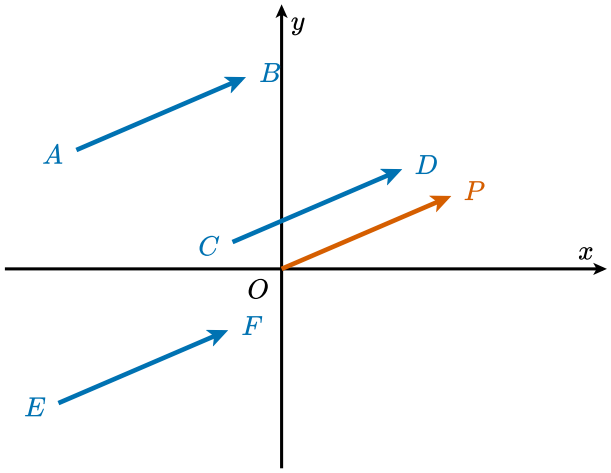


Figure 3.2.2 Four equal vectors. They have different initial points, but the same direction and the same length, so $\overrightarrow{AB} = \overrightarrow{CD} = \overrightarrow{OP} = \overrightarrow{EF}$. Only $\overrightarrow{OP}$ begins at the origin.

Only one of the vectors in Figure 3.2.2 starts from the origin. We say that $\overrightarrow{OP}$ is in **standard position**, and we denote it by a bold letter such as $\mathbf{v} = \overrightarrow{OP}$, which is also referred to as the **standard position vector**.

In 2D, if the initial point of a vector is the origin, i.e. $(0, 0)$, and its final point is (v_1, v_2), then the **component form** of the vector is $\mathbf{v} = \langle v_1, v_2 \rangle$. In 3D, if the initial point of a vector is the origin, i.e. $(0, 0, 0)$, and its final point is (v_1, v_2, v_3), then the component form of the vector is $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$.

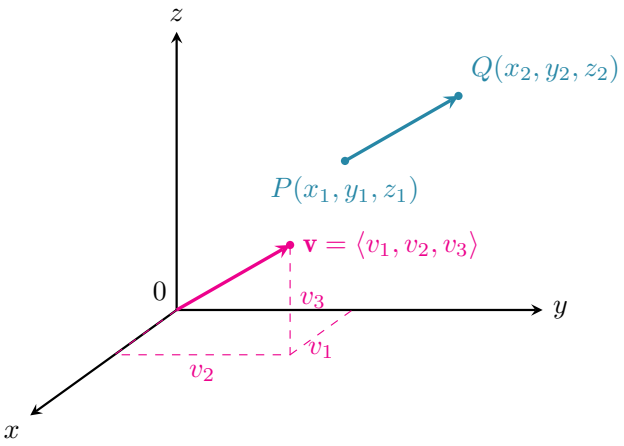


Figure 3.2.3 The vector $\overrightarrow{PQ}$ and its standard position vector $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$. The two arrows are equal as vectors; only the second one begins at the origin.

Read the components of $\mathbf{v} = \overrightarrow{PQ}$ off Figure 3.2.3.

Components: $\mathbf{v} = \langle v_1, v_2, v_3 \rangle =$

Length: $\|\mathbf{v}\| =$

Components and Length.

For $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, the standard position vector of $\overrightarrow{PQ}$ is

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle = \langle x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1 \rangle, \tag{3.2.1}$$

and its magnitude is

$$\begin{aligned} \|\mathbf{v}\| = \|\overrightarrow{PQ}\| &= \sqrt{v_1^2 + v_2^2 + v_3^2} \\ &= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}. \end{aligned} \tag{3.2.2}$$

Example 3.2.4 Length and Standard Position Vector.

A. Compute the length of the vector with the initial point $P(-1, 0, 1)$ and the final point $Q(1, 1, -1)$.

B. Find the standard position vector corresponding to this vector.

Solution.

A. $\|\overrightarrow{PQ}\| =$

B. $\mathbf{v} =$

Check: $\|\mathbf{v}\| =$

▲

3.2.2 Vector Algebra Operations

Definition 3.2.5 Vector Addition. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$. The **sum** of $\mathbf{u}$ and $\mathbf{v}$ is the vector

$$\mathbf{u} + \mathbf{v} = \langle u_1 + v_1,\ u_2 + v_2,\ u_3 + v_3 \rangle. \tag{3.2.3}$$

◆

Parallelogram law. The parallelogram law tells us how to add two vectors geometrically, and is described in Figure 3.2.6. Placing the tail of $\mathbf{v}$ at the head of $\mathbf{u}$ produces the sum as the third side of a triangle; equivalently, drawing both vectors from a common initial point makes the sum the diagonal of the parallelogram they span.

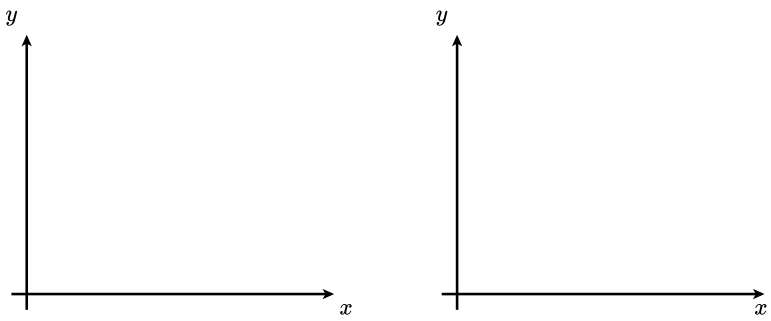


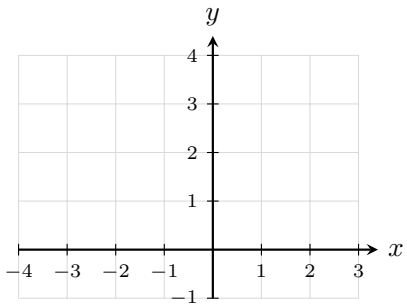
Figure 3.2.6 The parallelogram law: head to tail on the left, the diagonal of the parallelogram on the right.

Example 3.2.7 Adding Two Vectors. Add the following two vectors geometrically and algebraically.

$$\mathbf{u} = \langle 1, 1 \rangle \quad \text{and} \quad \mathbf{v} = \langle -2, 1 \rangle$$

Solution.

$\mathbf{u} + \mathbf{v} =$



▲

Definition 3.2.8 Scalar Multiplication. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and let $k \in \mathbb{R}$ be a scalar. The **scalar multiple** $k\,\mathbf{u}$ is the vector

$$k\,\mathbf{u} = \langle ku_1,\ ku_2,\ ku_3 \rangle. \tag{3.2.4}$$

◆

Note that both definitions can be reduced to the 2D case simply by removing the last components of the vectors.

Scaling a vector by k scales its length by $|k|$. Show this.

$\|k\mathbf{u}\| =$

Length of a Scalar Multiple.

$$\|k\mathbf{u}\| = |k|\,\|\mathbf{u}\|. \tag{3.2.5}$$

The vector $-\mathbf{u}$ has the same length as $\mathbf{u}$; however, the two vectors point in opposite directions. In general the length of $k\mathbf{u}$ is $|k|$ times the length of $\mathbf{u}$, and if $k > 0$ the two vectors point in the same direction, whereas if $k < 0$ they point in opposite directions. Figure 3.2.9 shows several examples.

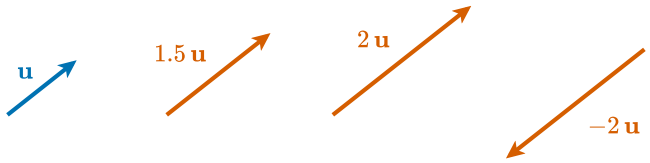


Figure 3.2.9 Scalar multiples of $\mathbf{u}$. Multiplying by 1.5 or 2 stretches the vector without turning it; multiplying by -2 stretches it and reverses its direction.

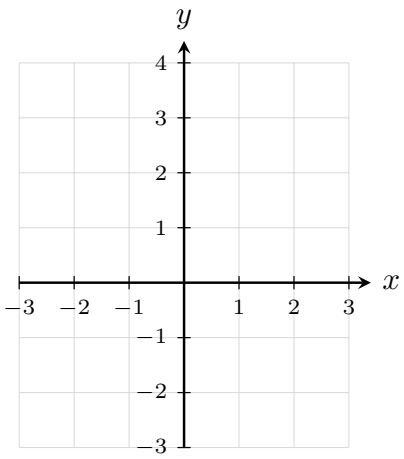
Definition 3.2.10 Vector Subtraction. Subtraction is handled with the same idea: the **difference** $\mathbf{u} - \mathbf{v}$ means

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}), \tag{3.2.6}$$

so we reverse $\mathbf{v}$ and then add. Geometrically, $\mathbf{u} - \mathbf{v}$ is the vector that points from the head of $\mathbf{v}$ to the head of $\mathbf{u}$ when both are drawn from a common initial point. ◆

Example 3.2.11 Adding and Subtracting Geometrically. Given the two vectors $\mathbf{u} = \langle 1, 1\rangle$ and $\mathbf{v} = \langle -1, 2\rangle$, find $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ geometrically.

Solution.



$\mathbf{u} + \mathbf{v} =$

$\mathbf{u} - \mathbf{v} =$

▲

The operations satisfy the familiar algebraic rules collected in Theorem 3.2.12.

Theorem 3.2.12 Properties of Vector Operations. *Let* $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ *be vectors and* a, b *be scalars. Then*

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
3. $\mathbf{u} + \mathbf{0} = \mathbf{u}$
4. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
5. $0\,\mathbf{u} = \mathbf{0}$
6. $1\,\mathbf{u} = \mathbf{u}$
7. $a(b\mathbf{u}) = (ab)\mathbf{u}$
8. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
9. $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$

3.2.3 Unit Vectors

Vectors of length one are referred to as **unit vectors**. The **standard unit vectors** are the ones pointing in the positive direction of x, y, and z, and they are denoted by $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$ respectively, as in Figure 3.2.13.

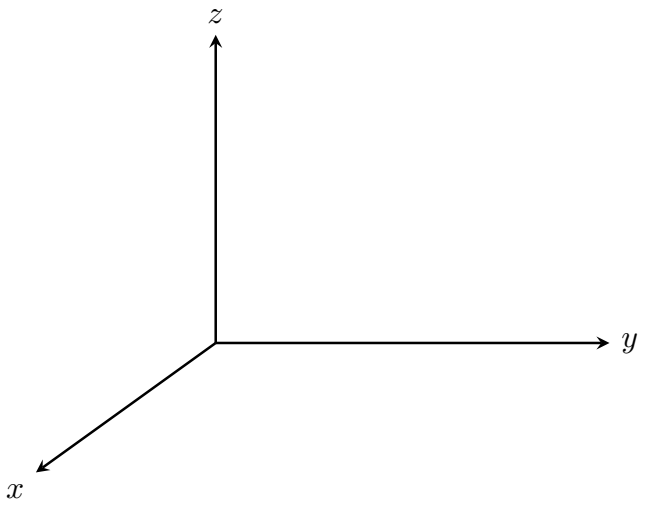


Figure 3.2.13 Empty axes for the standard unit vectors $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$, and for the decomposition $\mathbf{a} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$ of a vector into components along the axes.

Combining (3.2.3) and (3.2.4), any vector $\mathbf{a} = \langle a_1, a_2, a_3\rangle$ can be written as a linear combination of $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$ as

$$\mathbf{a} = \langle a_1, a_2, a_3\rangle = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}. \tag{3.2.7}$$

To find a unit vector pointing in the same direction as a nonzero vector $\mathbf{a}$, we divide $\mathbf{a}$ by its own length. We write $\hat{a}$, read "a hat", for the resulting unit vector:

$$\hat{a} = \frac{\mathbf{a}}{\|\mathbf{a}\|}. \tag{3.2.8}$$

From now on a hat always marks a vector of length one, as it does for the standard unit vectors $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$.

Check that (3.2.8) really does produce a unit vector.

$\|\hat{a}\| =$

Example 3.2.14 Standard Unit Vectors and a Unit Direction. Consider the two vectors $\mathbf{v} = \langle 1, 0, 1\rangle = \hat{\imath} + \hat{k}$ and $\mathbf{u} = \langle -1, 1, 0\rangle = -\hat{\imath} + \hat{\jmath}$.

A. Express $\mathbf{u} + \mathbf{v}$ in terms of the standard unit vectors.

B. Find the unit vector in the direction of $\mathbf{u} + \mathbf{v}$.

Solution.

A. $\mathbf{u} + \mathbf{v} =$

$\|\mathbf{u} + \mathbf{v}\| =$

B. $\widehat{\mathbf{u} + \mathbf{v}} =$

▲

3.3 The Dot Product

Objectives

After completing this section, you should be able to:

- Understand the dot product as a way of combining two vectors into a single number that measures the extent to which they point in the same direction.
- Interpret the sign of the dot product geometrically—positive for an acute angle, negative for an obtuse angle, and zero for perpendicular vectors—and recognize orthogonality as the vanishing of the dot product.
- Understand how the dot product encodes the angle between two vectors, connecting the algebra of vectors to the Law of Cosines.
- Interpret the projection of one vector onto another as the component of one vector in the direction of the other, and understand the decomposition of a vector into parts parallel and perpendicular to a given direction.
- Understand work as a physical measure of the effect of a force along a direction of motion, and appreciate why the dot product is the natural tool for expressing it.
- Explain how the dot product is used in data science to measure similarity: represent preferences as vectors, compute the cosine similarity and the angular distance between them, interpret the result, and extend the computation from two dimensions to $\mathbb{R}^n$.

3.3.1 The Dot Product and the Angle between Two Vectors

Consider two vectors $\mathbf{u}$ and $\mathbf{v}$, and let $\mathbf{w} = \mathbf{u} - \mathbf{v}$, as shown in Figure 3.3.1. By the law of cosines,

$$\|\mathbf{w}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\| \cos\theta. \tag{3.3.1}$$

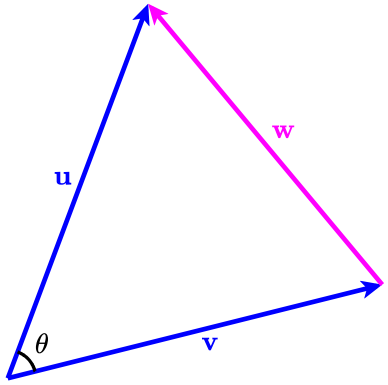
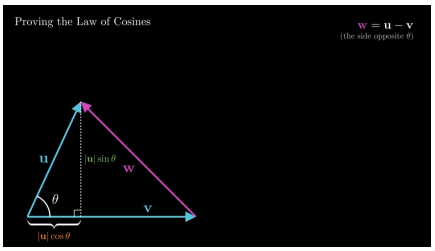


Figure 3.3.1 The vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w} = \mathbf{u} - \mathbf{v}$, together with the angle θ between $\mathbf{u}$ and $\mathbf{v}$.



Standalone

Figure 3.3.2 Proving the law of cosines: drop a perpendicular from the tip of $\mathbf{u}$ and apply the Pythagorean theorem to the right triangle whose hypotenuse is $\mathbf{w} = \mathbf{u} - \mathbf{v}$.

Now compute $\|\mathbf{w}\|^2$ a second way, in components, and compare the two answers.

In components: $\|\mathbf{w}\|^2 =$

Setting the two expressions equal:

Solving for $\cos\theta$*:*

Theorem 3.3.3 Angle between Two Vectors. *Let* $\mathbf{u} = \langle u_1, u_2, u_3\rangle$ *and* $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ *be two nonzero vectors, and let* θ*, with* $0 \le \theta \le \pi$*, be the angle between them. Then*

$$\begin{aligned} \cos\theta &= \frac{u_1v_1 + u_2v_2 + u_3v_3}{\|\mathbf{u}\|\|\mathbf{v}\|} \\ \iff \quad \theta &= \cos^{-1}\left(\frac{u_1v_1 + u_2v_2 + u_3v_3}{\|\mathbf{u}\|\|\mathbf{v}\|}\right). \end{aligned} \tag{3.3.2}$$

Definition 3.3.4 Dot Product. The term $u_1v_1 + u_2v_2 + u_3v_3$ in the numerator of (3.3.2) is known as the **dot product** of the two vectors $\mathbf{u} = \langle u_1, u_2, u_3\rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3\rangle$, and is denoted $\mathbf{u}\cdot\mathbf{v}$.

$$\mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3. \tag{3.3.3}$$

◆

Example 3.3.5 Computing a dot product. Let $\mathbf{u} = \langle 1, 2, 3\rangle$ and $\mathbf{v} = \langle -2, 0, 4\rangle$. Compute $\mathbf{u}\cdot\mathbf{v}$, and use it to find the angle between $\mathbf{u}$ and $\mathbf{v}$.

Solution.

$\mathbf{u} \cdot \mathbf{v} =$ *so the angle is:*

$\|\mathbf{u}\| =$ $\|\mathbf{v}\| =$

$\cos\theta =$

$\theta =$

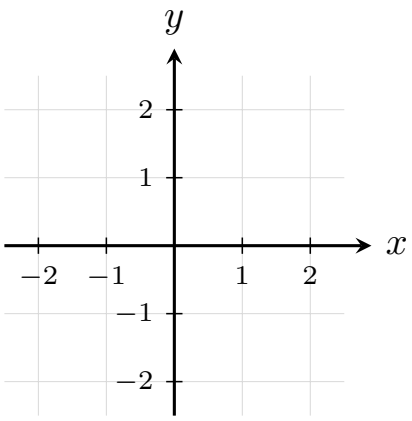
▲

Remark 3.3.6 Orthogonal vectors. If the two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, then $\theta = \pi/2$, which means $\mathbf{u} \cdot \mathbf{v} = 0$. Conversely, if $\mathbf{u}$ and $\mathbf{v}$ are two vectors such that $\mathbf{u} \cdot \mathbf{v} = 0$, then $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

$$\mathbf{u} \cdot \mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v}. \tag{3.3.4}$$

Example 3.3.7 A right triangle detected with the dot product. Consider the triangle with vertices $A(1, -1)$, $B(1, 1)$, and $C(2, 0)$. Show that this is a right triangle using the dot product and the orthogonality criterion (3.3.4).

Solution.



The two vectors and their dot product:

Conclusion:

▲

Example 3.3.8 The remaining angles of the triangle. For the triangle of Example 3.3.7, calculate the remaining angles.

Solution.

The angle at A:

The angle at B:

▲

Properties of the Dot Product.

Fact 3.3.9 *If* $\mathbf{u}$, $\mathbf{v}$, *and* $\mathbf{w}$ *are any vectors and* c *is a scalar, then*

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$
3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
4. $\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2$
5. $\mathbf{0} \cdot \mathbf{u} = 0$

Prove properties 1, 3, and 4.

1.

3.

4.

3.3.2 Projection of u in the Direction of v

The dot product measures how much two vectors point in the same direction. This section turns that measurement into a vector: given $\mathbf{u}$ and $\mathbf{v}$, we ask how much of $\mathbf{u}$ lies along $\mathbf{v}$. The answer is the **projection** of $\mathbf{u}$ onto $\mathbf{v}$, and it lets us split any vector into a part that points along $\mathbf{v}$ and a part perpendicular to it. That decomposition is what makes the dot product useful in practice—it is how we find the component of a force along a direction of motion, as in Subsection 3.3.3, or the component of gravity down a slope.

The projection of $\mathbf{u}$ in the direction of $\mathbf{v}$ is denoted $\text{proj}_{\mathbf{v}}\mathbf{u}$, and is shown in Figure 3.3.10.

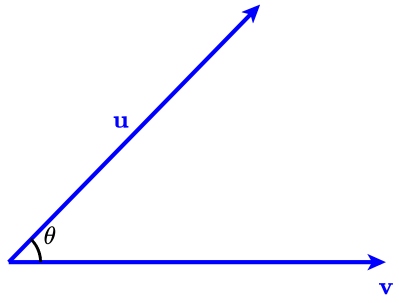


Figure 3.3.10 The vectors $\mathbf{u}$ and $\mathbf{v}$, for the projection $\text{proj}_{\mathbf{v}}\mathbf{u}$ of $\mathbf{u}$ in the direction of $\mathbf{v}$ and the perpendicular component $\mathbf{u} - \text{proj}_{\mathbf{v}}\mathbf{u}$.

Build the projection formula from that picture.

Length × direction:

Replacing $\cos\theta$*:*

$proj_{\mathbf{v}}\mathbf{u} =$

Fact 3.3.11 Projection Formula.

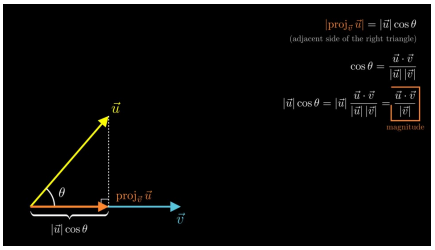
$$proj_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\right)\mathbf{v}. \tag{3.3.5}$$

Writing u as Two Vectors, One Parallel and One Perpendicular to v. As can be seen in Figure 3.3.10, $\text{proj}_{\mathbf{v}}\mathbf{u}$ is parallel to $\mathbf{v}$, and the vector $\mathbf{u} - \text{proj}_{\mathbf{v}}\mathbf{u}$ is perpendicular to it. Also, the sum of these two vectors equals $\mathbf{u}$. This means we can write $\mathbf{u}$ as

$$\mathbf{u} = \underbrace{\qquad\qquad}_{\text{parallel to } \mathbf{v}} + \underbrace{\qquad\qquad}_{\text{orthogonal to } \mathbf{v}}. \tag{3.3.6}$$

$\mathbf{u} = proj_{\mathbf{v}}\mathbf{u} + (\mathbf{u} - proj_{\mathbf{v}}\mathbf{u}) =$

with the projection formula:



Standalone

Figure 3.3.12 The projection of $\vec{u}$ onto $\vec{v}$ as a shadow cast by light from above.

Example 3.3.13 Decomposing a vector into parallel and perpendicular parts. Consider the two vectors $\mathbf{v} = \hat{\imath} + \hat{\jmath}$ and $\mathbf{u} = 2\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$. Write $\mathbf{u}$ as a sum of two vectors, one of which is parallel to $\mathbf{v}$ and the other perpendicular to it.

Solution.

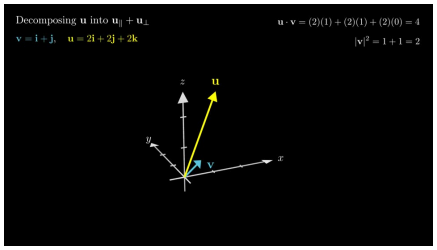
$\mathbf{u} \cdot \mathbf{v} =$ $\|\mathbf{v}\| =$

$\mathbf{u}_{\parallel} =$

$\mathbf{u}_{\perp} =$

Check: $\mathbf{u}_{\perp} \cdot \mathbf{v} =$

To achieve a visual understanding of this decomposition, watch the video below.



Standalone

Figure 3.3.14 Decomposing $\mathbf{u} = 2\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$ into $\mathbf{u}_{\parallel} = 2\hat{\imath} + 2\hat{\jmath}$, lying in the xy-plane along $\mathbf{v}$, and $\mathbf{u}_{\perp} = 2\hat{k}$, pointing straight up.

▲

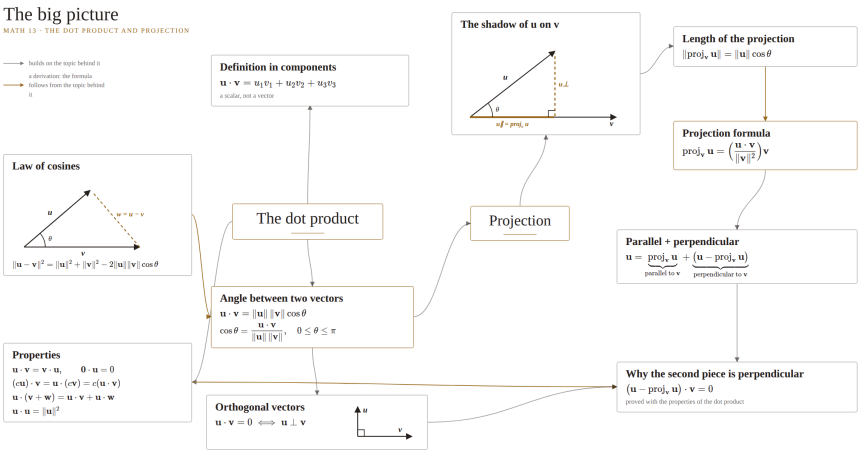
Checkpoint 3.3.15 Concept Check. Everything in (3.3.6) rests on the claim read off from Figure 3.3.10: that $\text{proj}_{\mathbf{v}}\mathbf{u}$ is parallel to $\mathbf{v}$, and that $\mathbf{u}-\text{proj}_{\mathbf{v}}\mathbf{u}$ is perpendicular to $\mathbf{v}$. A picture is not a proof. Explain why each half of the claim is true.

parallel half:

perpendicular half:

The Big Picture. The map below gathers the ideas of this section so far on one page: the dot product and the angle between two vectors, and the projection of one vector onto another. An arrow between two topics means the second builds on the first, and the gold arrows mark a derivation, where a formula follows from the topic behind it.

The online version of the map is interactive: every topic on it is a link to the part of the section where it is covered.



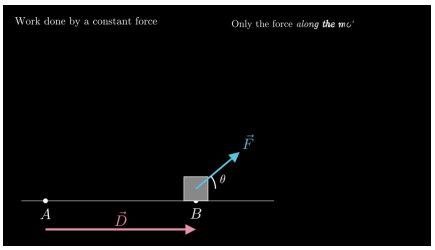
3.3.3 Application in Physics: Work

Consider the scenario in which a constant force $\mathbf{F}$ causes an object to move from point P to point Q, as shown in Figure 3.3.18. The vector $\overrightarrow{PQ}$ is often denoted $\mathbf{D}$ and is referred to as the **displacement vector**. The work done by the force is then

Definition 3.3.16 Work.

$$\text{Work} = \mathbf{F}\cdot\mathbf{D} = \|\mathbf{D}\|\|\mathbf{F}\|\cos\theta, \tag{3.3.7}$$

where $\|\text{proj}_{\mathbf{D}}\mathbf{F}\| = \|\mathbf{F}\|\cos\theta$ is the magnitude of the force in the direction of motion, as computed in Subsection 3.3.2. ◆



Standalone

Figure 3.3.17 A constant force $\mathbf{F}$ acting on an object causing a displacement from A to B.

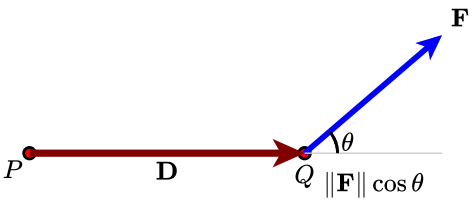


Figure 3.3.18 A force $\mathbf{F}$ displaces an object from P to Q along the displacement vector $\mathbf{D}$. The reference segment shows the magnitude of the force in the direction of motion, $\|\mathbf{F}\|\cos\theta$.

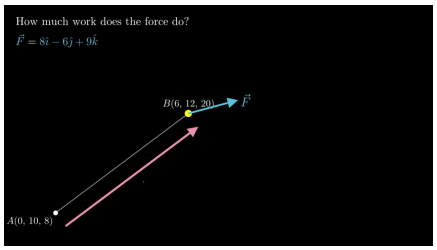
Example 3.3.19 Computing work done by a force. A force is given by the vector $\mathbf{F} = 8\hat{\imath} - 6\hat{\jmath} + 9\hat{k}$ and moves a particle from the point $A(0, 10, 8)$ to the point $B(6, 12, 20)$. Find the work done.

Solution.

D =

Work =

For more help, watch the video below.



Standalone

Figure 3.3.20 Set up the work computation: build the displacement vector first, then take the dot product. What do you get?

▲

Example 3.3.21 Gravity on an incline. Suppose a mass of 10 kg is resting on an inclined plane. Gravity exerts a force equivalent to mg on the object, where $g = 9.8\ \text{m/s}^2$ is the gravitational acceleration. Suppose the incline is tilted at a 30° angle. Compute the components of the force that are parallel and perpendicular to the inclined plane.

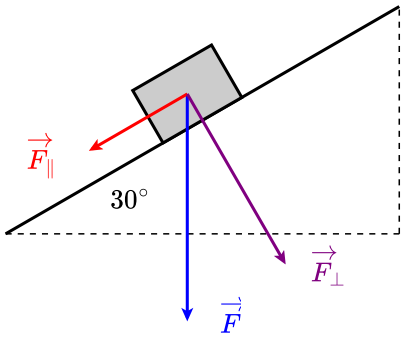


Figure 3.3.22 Forces acting on a mass on a 30° incline: the weight $\overrightarrow{F}$ resolves into a component $\overrightarrow{F}_{\parallel}$ down the slope and a component $\overrightarrow{F}_{\perp}$ into the surface, with $\overrightarrow{F} = \overrightarrow{F}_{\parallel} + \overrightarrow{F}_{\perp}$.

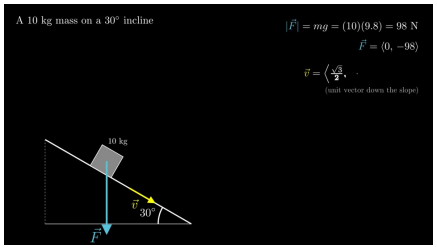
Solution.

$\|\overrightarrow{F}\| =$ $\overrightarrow{F} =$

$\hat{v} =$

$\overrightarrow{F}_{\parallel} =$

$\overrightarrow{F}_{\perp} =$



Standalone

Figure 3.3.23 Decomposing gravity on a 30° incline: the parallel component $\vec{F}_{\parallel} = \text{proj}_{\vec{v}} \vec{F}$ pulls the mass down the slope, and the perpendicular component $\vec{F}_{\perp} = \vec{F} - \vec{F}_{\parallel}$ presses it into the surface.

▲

3.3.4 Application in Data Science: Measuring Similarity in Tastes

Every time a streaming service tells you "because you watched *Sinners*…", some piece of software has decided that your taste resembles the taste of other people in its database. Deciding whose taste resembles whose can be framed as a geometry problem, and one tool that can help us decide is the dot product.

The first step is to turn taste into a vector. Fix a list of movies and ask each person to rate every one of them on a scale from -5 to 5, where 5 means "loved it", -5 means "I want those two hours back", and 0 means indifference. Reading the ratings off in the same order for everybody turns each person into a vector, and the question "do these two people have similar taste?" becomes a question about the angle between two vectors, which by Theorem 3.3.3 is exactly what the dot product measures.

Definition 3.3.24 Angular Distance and Cosine Similarity. Let $\mathbf{u}$ and $\mathbf{v}$ be two nonzero vectors. Their **angular distance** is the angle θ between them, obtained from (3.3.2),

$$d_A(\mathbf{u}, \mathbf{v}) = \theta = \cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right). \tag{3.3.8}$$

The quantity $\dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \cos\theta$ appearing inside is called the **cosine similarity** of $\mathbf{u}$ and $\mathbf{v}$. The smaller the angular distance, the more similar the two vectors.

◆

Angular distance asks "how nearly do these two vectors point the same way?" Notice that it is completely blind to the lengths of $\mathbf{u}$ and $\mathbf{v}$: multiplying either vector by a positive number leaves θ untouched. We will come back to what that means for moviegoers.

Two Movies at a Time. Two movies give vectors in $\mathbb{R}^2$, which we can draw. Take two of the ten films nominated for Best Picture at the 2026 Academy Awards, *Sinners* and *Hamnet*, and record each person's rating of *Sinners* as the first component and of *Hamnet* as the second. Mehdi, Norm, and Popcorn rate them as follows:

$$\begin{aligned} \text{Mehdi:} \quad & \mathbf{u} = \langle 4, 5 \rangle \\ \text{Norm:} \quad & \mathbf{v} = \langle 5, 3 \rangle \\ \text{Popcorn:} \quad & \mathbf{w} = \langle -4, -2 \rangle. \end{aligned}$$

Mehdi and Norm both enjoyed the two films; Popcorn sat through both of them and regretted it. Figure 3.3.25 shows the three taste vectors.

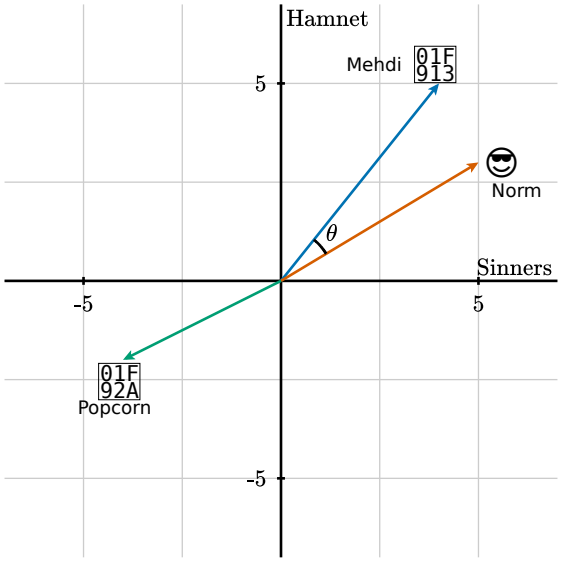


Figure 3.3.25 The taste vectors **u** (Mehdi), **v** (Norm), and **w** (Popcorn). The angle θ is the angular distance between **u** and **v**.

Example 3.3.26 Who should watch a movie together? Using the taste vectors above, rank the three pairs by angular distance.

Solution.

Table 3.3.27 Angular distance between the three taste vectors.

Pair	$\mathbf{u} \cdot \mathbf{v}$	$\cos\theta$	d_A
Mehdi, Norm			
Mehdi, Popcorn			
Norm, Popcorn			

Lengths and dot products:

Who are the movie buddies, and what did the signs already tell you?

▲

Angular Distance Ignores Enthusiasm. Suppose Ray, Vera, and Gus rate the same two films

$$\begin{aligned} \text{Ray:} \quad & \mathbf{a} = \langle 2, 2 \rangle \\ \text{Vera:} \quad & \mathbf{b} = \langle 5, 4 \rangle \\ \text{Gus:} \quad & \mathbf{c} = \langle 5, 5 \rangle, \end{aligned}$$

as in Figure 3.3.28.

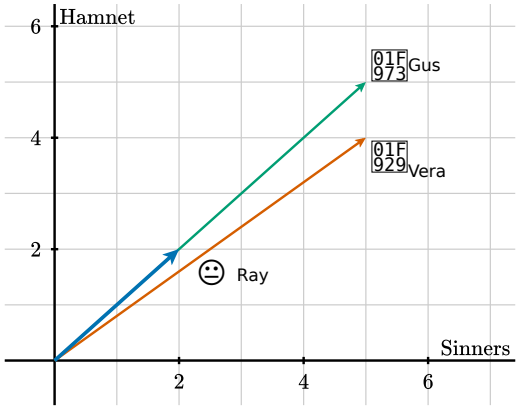


Figure 3.3.28 Ray, Vera, and Gus. Ray's vector **a** lies along Gus's vector **c**, so their angular distance is 0, even though Gus handed out far more generous ratings.

Find $d_A(\mathbf{a}, \mathbf{c})$ and $d_A(\mathbf{b}, \mathbf{c})$.

$d_A(\mathbf{a}, \mathbf{c}) =$ *because*

$d_A(\mathbf{b}, \mathbf{c}) =$

What this says about enthusiasm:

More Movies, More Dimensions. Nothing above depended on there being exactly two movies. With n movies, a person's taste vector is $\mathbf{u} = \langle u_1, u_2, \ldots, u_n \rangle$ in $\mathbb{R}^n$, the dot product of (3.3.3) becomes a sum of n products, $\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i$, and the length of a vector becomes $\|\mathbf{u}\| = \sqrt{\sum_{i=1}^{n} u_i^2}$. The angular distance of

Definition 3.3.24 carries over verbatim:

$$d_A(\mathbf{u},\mathbf{v}) = \cos^{-1}\left(\frac{\sum_{i=1}^{n} u_i v_i}{\sqrt{\sum_{i=1}^{n} u_i^2}\,\sqrt{\sum_{i=1}^{n} v_i^2}}\right). \tag{3.3.9}$$

We can no longer draw the picture, but every formula still applies, and the language of Remark 3.3.6 still means something: two taste vectors with $\mathbf{u}\cdot\mathbf{v} = 0$ are orthogonal, an angular distance of $90°$, which here says that knowing one person's ratings tells you nothing about the other's. This is what makes the dot product useful in practice. A streaming service with 50,000 titles works in $\mathbb{R}^{50{,}000}$, and finding the users nearest to you is still nothing more than the arithmetic in (3.3.9).

Example 3.3.29 All ten nominees. List the ten 2026 Best Picture nominees alphabetically—*Bugonia*, *F1*, *Frankenstein*, *Hamnet*, *Marty Supreme*, *One Battle After Another*, *The Secret Agent*, *Sentimental Value*, *Sinners*, *Train Dreams*—and suppose Mehdi, Norm, and Popcorn rate all ten:

$$\begin{aligned}\mathbf{u} &= \langle 3,\ -1,\ 2,\ 5,\ 0,\ 5,\ 4,\ 3,\ 4,\ 2\rangle\\ \mathbf{v} &= \langle 1,\ 4,\ 3,\ 3,\ 5,\ 4,\ 0,\ 1,\ 5,\ -2\rangle\\ \mathbf{w} &= \langle -2,\ 3,\ -3,\ -2,\ 4,\ -1,\ -4,\ -5,\ -4,\ -3\rangle.\end{aligned}$$

(The fourth and ninth entries are the *Hamnet* and *Sinners* ratings from before.) Which pair has the most similar taste?

Solution.

Table 3.3.30 Angular distance across all ten nominees.

Pair	$\mathbf{u}\cdot\mathbf{v}$	$\cos\theta$	d_A
Mehdi, Norm			
Mehdi, Popcorn			
Norm, Popcorn			

Lengths and dot products:

Compared with the two-movie table:

▲

Ten components are a lot to handle by hand, and a real recommender system deals with thousands. The cell below carries out the computation of Example 3.3.29. Press *Evaluate* to run it and check the table you just filled in,

then edit the ratings and run it again to see whose taste your own vector is closest to.

```
# ratings of the ten nominees, in alphabetical order
u = vector([ 3, -1,  2,  5,  0,  5,  4,  3,  4,  2])
v = vector([ 1,  4,  3,  3,  5,  4,  0,  1,  5, -2])
w = vector([-2,  3, -3, -2,  4, -1, -4, -5, -4, -3])

def cosine_similarity(a, b):
    return a.dot_product(b) / (a.norm() * b.norm())

# the angular distance, converted from radians to degrees
def angular_distance(a, b):
    return arccos(cosine_similarity(a, b)) * 180 / pi

for name, a, b in [("Mehdi, Norm", u, v),
                   ("Mehdi, Popcorn", u, w),
                   ("Norm, Popcorn", v, w)]:
    print(name, N(cosine_similarity(a, b), digits=3),
        N(angular_distance(a, b), digits=4))
```

```
Mehdi, Norm 0.549 56.71
Mehdi, Popcorn -0.762 139.6
Norm, Popcorn -0.0744 94.27
```

3.4 The Cross Product

Objectives

After completing this section, you should be able to:

- Understand the cross product as an operation that produces a vector perpendicular to two given vectors, with its direction fixed by the right-hand rule.
- Interpret the magnitude of the cross product as the area of the parallelogram spanned by the two vectors, and recognize that the cross product is zero exactly when the vectors are parallel.
- Understand why the geometric and component definitions of the cross product describe one and the same object.
- Contrast the cross product with the dot product: one returns a vector and detects parallelism, while the other returns a scalar and detects perpendicularity.
- Understand torque as a measure of the turning effect of a force, and appreciate the role of cross product in determining torque.
- Describe circular motion with the moving unit vectors $\hat{r}$ and $\hat{\theta}$, and represent the rate of rotation by the angular velocity vector $\vec{\omega}$ along the axis of rotation.
- Use the cross product to obtain the velocity of a rotating object from its angular velocity and position, $\vec{v} = \vec{\omega} \times \vec{r}$.

3.4.1 Definition and Properties of the Cross Product

Definition 3.4.1 The Cross Product (First Definition). Given two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$, the **cross product** $\mathbf{u} \times \mathbf{v}$ is the *vector* defined by

$$\mathbf{u} \times \mathbf{v} = (\|\mathbf{u}\|\|\mathbf{v}\| \sin\theta)\, \hat{n}, \tag{3.4.1}$$

where θ, with $0 \leq \theta \leq \pi$, is the angle between $\mathbf{u}$ and $\mathbf{v}$, and $\hat{n}$ is the unit vector orthogonal to both $\mathbf{u}$ and $\mathbf{v}$ whose direction is given by the right-hand rule. ◆

The magnitude of $\mathbf{u} \times \mathbf{v}$ is equal to the area of the parallelogram built on the two vectors $\mathbf{u}$ and $\mathbf{v}$, as shown in Figure 3.4.2.

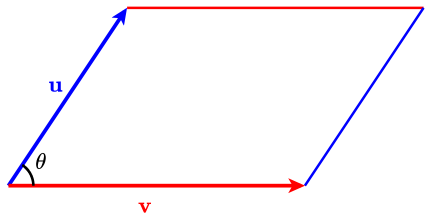


Figure 3.4.2 The parallelogram built on the vectors $\mathbf{u}$ and $\mathbf{v}$, with base $\|\mathbf{v}\|$ and height $\|\mathbf{u}\||\sin(\theta)|$.

Read the area of that parallelogram off the figure, and compare it with (3.4.1).

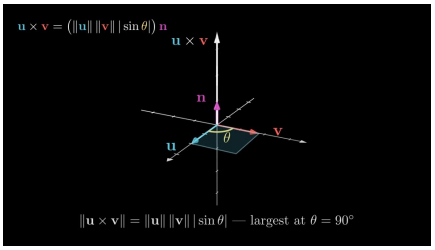
Area = base × height =

Why the absolute value:

Fact 3.4.3 Area of the Parallelogram.

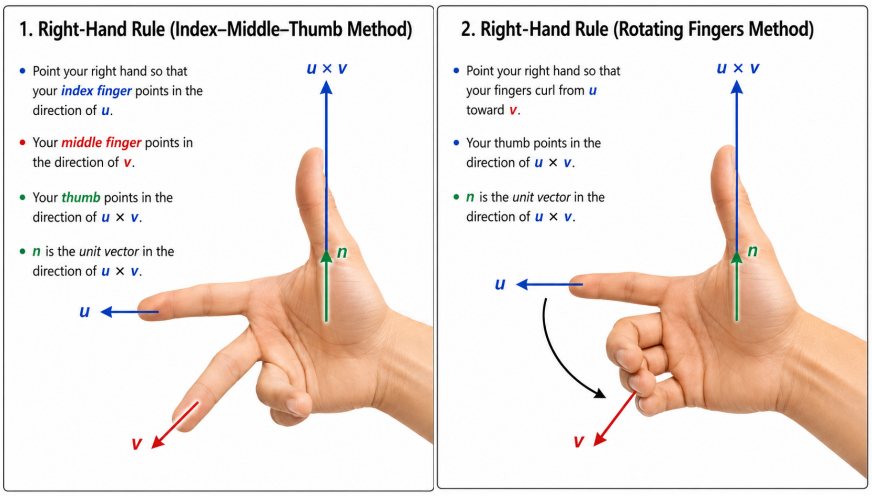
$$\textit{Area of the parallelogram} = \|\mathbf{u}\|\|\mathbf{v}\||\sin\theta| = \|\mathbf{u} \times \mathbf{v}\|. \tag{3.4.2}$$

The vector $\hat{n}$ is the unit vector that determines the direction of $\mathbf{u} \times \mathbf{v}$ using the right-hand rule. Note that $\mathbf{u} \times \mathbf{v}$ is orthogonal (perpendicular) to both vectors, as shown in Figure 3.4.4.



Standalone

Figure 3.4.4 The cross product $\mathbf{u} \times \mathbf{v}$ points in the direction of the unit vector $\hat{n}$, given by the right-hand rule, and is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$; its length is $\|\mathbf{u}\|\|\mathbf{v}\||\sin\theta|$.



(a) Index–Middle–Thumb method. **(b)** Rotating-fingers method.

Figure 3.4.5 Two equivalent right-hand rules for $\mathbf{u} \times \mathbf{v}$.

Fact 3.4.6 Parallel Vectors. *From (3.4.1), it follows that the cross product of two parallel vectors is zero.*

$$\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} \times \mathbf{v} = \mathbf{0}, \qquad \mathbf{u}, \mathbf{v} \neq \mathbf{0}. \tag{3.4.3}$$

Say why (3.4.3) follows, in one line.

workspace

Theorem 3.4.7 Properties of the Cross Product. *If* $\mathbf{u}$, $\mathbf{v}$, *and* $\mathbf{w}$ *are any vectors and* r, s *are scalars, then*

1. $(r\mathbf{u}) \times (s\mathbf{v}) = (rs)(\mathbf{u} \times \mathbf{v})$
2. $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$
3. $(\mathbf{v} + \mathbf{w}) \times \mathbf{u} = \mathbf{v} \times \mathbf{u} + \mathbf{w} \times \mathbf{u}$
4. $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$

5. $\mathbf{0} \times \mathbf{u} = \mathbf{0}$

Property 4 is the one that has no analogue for the dot product, so prove it.

First components of $\mathbf{v} \times \mathbf{u}$ *and* $\mathbf{u} \times \mathbf{v}$*:*

In words:

Checkpoint 3.4.8 Show that $(\mathbf{u} - \mathbf{v}) \times (\mathbf{u} + \mathbf{v}) = 2(\mathbf{u} \times \mathbf{v})$.

Solution.

Expanding:

Simplifying:

Example 3.4.9 Computing a cross product using the first definition. Use Definition 3.4.1 to compute

1. $\hat{\imath} \times \hat{\jmath}$, and also $\hat{\jmath} \times \hat{\imath}$;
2. $\mathbf{u} \times \mathbf{v}$, where $\mathbf{u} = 4\hat{\imath}$ and $\mathbf{v}$ is a vector of length 3 lying in the xy-plane, making an angle of $30°$ with $\mathbf{u}$ measured counterclockwise.

Solution.

1. $\hat{\imath} \times \hat{\jmath} =$

$\hat{\jmath} \times \hat{\imath} =$

2. $\mathbf{u} \times \mathbf{v} =$

Area check:

▲

3.4.2 Second Definition of the Cross Product

Writing the vectors in components gives another definition for the cross product.

Definition 3.4.10 The Cross Product (Second Definition). If $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$, then

$$\mathbf{u} \times \mathbf{v} = \langle u_2 v_3 - u_3 v_2,\ u_3 v_1 - u_1 v_3,\ u_1 v_2 - u_2 v_1 \rangle. \tag{3.4.4}$$

◆

One way to remember the second definition of the cross product is to write it as a determinant:

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} \\ &= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \hat{\imath} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \hat{\jmath} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \hat{k}, \end{aligned} \tag{3.4.5}$$

where

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc. \tag{3.4.6}$$

Example 3.4.11 Computing a cross product. Calculate the cross product $\mathbf{u} \times \mathbf{v}$ if $\mathbf{u} = \langle 1, -1, 2 \rangle$ and $\mathbf{v} = \langle 3, -2, 1 \rangle$.

Solution.

$\mathbf{u} \times \mathbf{v} =$

Check:

▲

Example 3.4.12 Computing a vector perpendicular to two other vectors. Consider the points $A(1, -1, 2)$, $B(2, 0, -1)$, and $C(0, -2, 3)$. Find a unit vector which is orthogonal to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

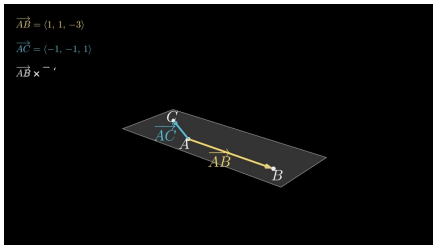
Solution.

$\overrightarrow{AB} =$ $\overrightarrow{AC} =$

$\overrightarrow{AB} \times \overrightarrow{AC} =$

$\hat{n} =$

The other one:



Standalone

Figure 3.4.13 The unit vector $\hat{n}$ is orthogonal to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$, and hence to the plane through the points A, B, and C.

▲

Example 3.4.14 Computing the area of a triangle. Find the area of the triangle with the vertices $A(1,-1,2)$, $B(2,0,-1)$, and $C(0,-2,3)$. Note that these are the same points that we had in the previous example.

Solution.

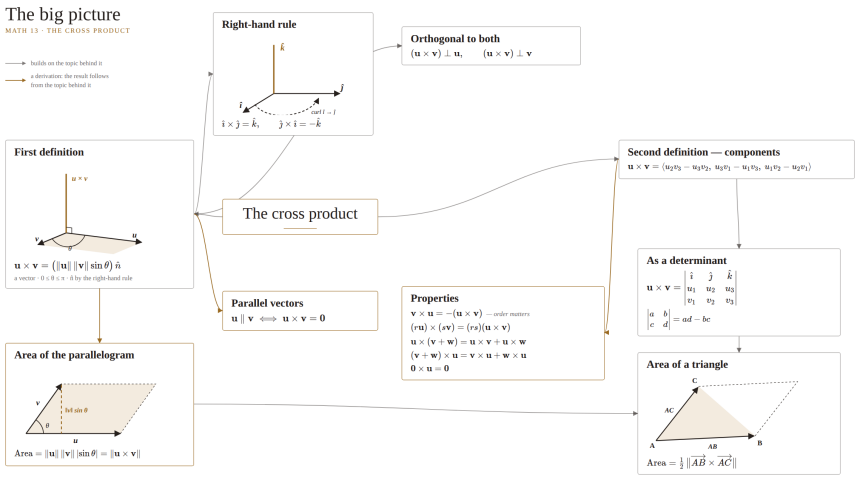
Area of the parallelogram =

Area of the triangle =

▲

The Big Picture. The map below gathers the ideas of this section so far on one page: the two definitions of the cross product and what each one tells us. An arrow between two topics means the second builds on the first, and the gold arrows mark a derivation, where a formula follows from the topic behind it.

The online version of the map is interactive: every topic on it is a link to the part of the section where it is covered.



3.4.3 True or False?

Checkpoint 3.4.15 True or False. Let $\overrightarrow{u}$ and $\overrightarrow{v}$ be two vectors in $\mathbb{R}^3$. Then, $\overrightarrow{u}\times\text{proj}_{\overrightarrow{v}}\overrightarrow{u}$ is perpendicular to $\overrightarrow{u}-\text{proj}_{\overrightarrow{v}}\overrightarrow{u}$.

Answer. True.

Solution. True.

Justify your answer.

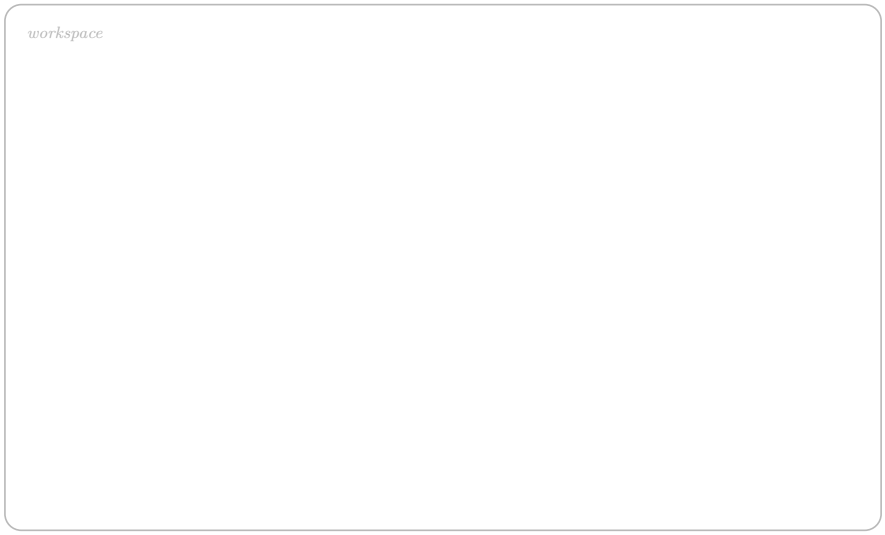
workspace

Checkpoint 3.4.16 True or False. Let $\hat{u}$ and $\hat{v}$ be two standard unit vectors in $\mathbb{R}^3$. Then, $(\hat{u}\cdot\hat{v})^2+\|\hat{v}\times\hat{u}\|^2=1$.

Answer. True.

Solution. True.

Justify your answer.



3.4.4 ∗Application: Torque

Starred section. This one is for the interested reader. It will not be examined.

One of the well known physical applications of the cross product is **torque**, the rotational analog of force. A force makes an object accelerate; a torque makes it *rotate*. How effective a force is at producing rotation depends not only on its magnitude, but also on *where* it is applied and in *what direction*. Think of pushing a door: pushing hard near the hinge barely moves it, while a light push at the doorknob swings it open easily. Pushing at the doorknob but *toward* the hinge does nothing at all. The cross product captures all three scenarios and more.

Definition 3.4.17 Torque. Suppose a force $\vec{F}$ is applied at a point whose position vector relative to the axis of rotation is $\vec{r}$. The **torque** produced by $\vec{F}$ is the vector

$$\vec{\tau} = \vec{r} \times \vec{F}. \tag{3.4.7}$$

◆

An interactive view of the torque $\vec{\tau} = \vec{r} \times \vec{F}$ — the force swung around the head of $\vec{r}$, with the torque vector responding — accompanies the online version of these notes.

Everything we know about the cross product now translates into physics. By (3.4.1), the magnitude of the torque is

$$\|\vec{\tau}\| = \|\vec{r}\|\|\vec{F}\| |\sin\theta|, \tag{3.4.8}$$

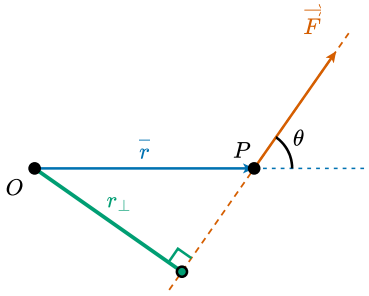
where θ is the angle between $\vec{r}$ and $\vec{F}$, and the direction of $\vec{\tau}$ is given by the right-hand rule: $\vec{\tau}$ points along the axis of rotation, and curling the fingers of the right hand from $\vec{r}$ toward $\vec{F}$ gives the sense of the rotation. Torque has units of meter-newtons ($\mathrm{m} \cdot \mathrm{N}$) in SI.

The quantity in (3.4.8) can be grouped in two useful ways, shown in Figure 3.4.18. Write down the two groupings.

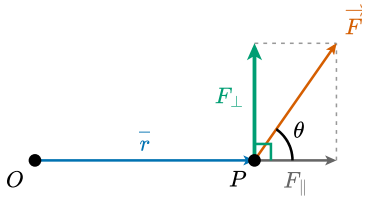
Grouped as a lever arm: $\|\vec{\tau}\| =$

Grouped as a perpendicular force: $\|\vec{\tau}\| =$

Why they agree:



$r_\perp = \|\vec{r}\| \sin\theta$
$\|\vec{\tau}\| = r_\perp \|\vec{F}\|$



$F_\parallel$ produces no rotation

$F_\perp = \|\vec{F}\| \sin\theta$
$\|\vec{\tau}\| = \|\vec{r}\| F_\perp$

(a) Force times lever arm: $\|\vec{\tau}\| = r_\perp \|\vec{F}\|$.

(b) Distance times perpendicular force: $\|\vec{\tau}\| = \|\vec{r}\| F_\perp$.

Figure 3.4.18 The two groupings of (3.4.8). In both pictures the axis of rotation passes through O, perpendicular to the page, the force $\vec{F}$ is applied at P, and θ is the angle between $\vec{r}$ and $\vec{F}$; the two products are equal because each is $\|\vec{r}\|\|\vec{F}\| \sin\theta$.

The first grouping, shown in Figure 3.4.18(a), says the torque is the force times the **lever arm** $r_\perp$, the perpendicular distance from the axis to the line along which the force acts. The second, shown in Figure 3.4.18(b), says it is the distance $\|\vec{r}\|$ times $F_\perp$, the component of the force perpendicular to $\vec{r}$; the parallel component $F_\parallel$ points straight at the axis and produces no rotation.

Fact 3.4.19 Zero Torque. *By (3.4.3), if $\vec{F}$ is parallel to $\vec{r}$ then $\vec{\tau} = \vec{r} \times \vec{F} = \vec{0}$: a force directed straight at (or away from) the axis produces no rotation. This is why you cannot open a door by pushing toward its hinge.*

Example 3.4.20 Biceps torque. The biceps muscle exerts a vertical force of $\|\vec{F}\| = 700$ N on the lower arm, attached at a point 5.0 cm from the elbow joint (the axis of rotation). Calculate the torque about the elbow when

1. the forearm is horizontal, and
2. the forearm makes a 30° angle below the horizontal.

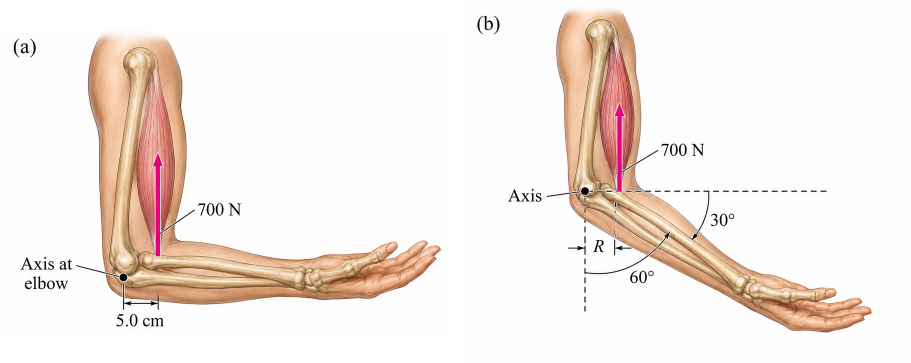


Figure 3.4.21 The biceps exerts a 700 N force on the forearm: (a) forearm horizontal, with the muscle attached 5.0 cm from the elbow axis; (b) forearm at 30° below horizontal, with lever arm R.

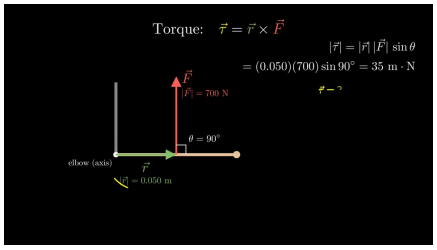
Solution.

Setup: $\vec{F} =$

1. *Forearm horizontal:* $\vec{r} =, \quad \vec{\tau} =$

2. *Forearm* 30° *below:* $\vec{r} =, \quad \vec{\tau} =$

Which is larger, and why it matters:



Standalone

Figure 3.4.22 The torque $\vec{\tau} = \vec{r} \times \vec{F}$ of the biceps force about the elbow. When the forearm drops 30° below the horizontal, the angle between $\vec{r}$ and $\vec{F}$ becomes 120° and the lever arm $r_\perp = \|\vec{r}\| \sin\theta$ shortens.

▲

3.4.5 ∗Application: Circular Motion: Velocity and Angular Velocity

Starred section. This one is for the interested reader. It will not be examined.

A second physical application of the cross product describes the motion of an object travelling in a circle: a rider on a circular track, a ball whirled on a string, a point on a spinning wheel, or the moon on its (nearly circular) orbit. In each case the object turns about a fixed centre, and the natural way to keep track of where it is at time t is not by its coordinates x and y but by the single angle $\theta(t)$ it has swung through. Our goal is to study how the velocity and the rate of change of that angle are related.

3.4.5.1 A Frame that Turns with the Particle

Let an object move on a circle of radius r centred at the origin of the xy-plane, and let $\theta(t)$ be the angle its position makes with the positive x-axis at time t, as in Figure 3.4.23. At the point P where the particle sits we attach two unit vectors:

- $\hat{r}(t)$, pointing radially outward, away from the centre, and
- $\hat{\theta}(t)$, perpendicular to $\hat{r}(t)$ and tangent to the circle, pointing in the direction of increasing θ.

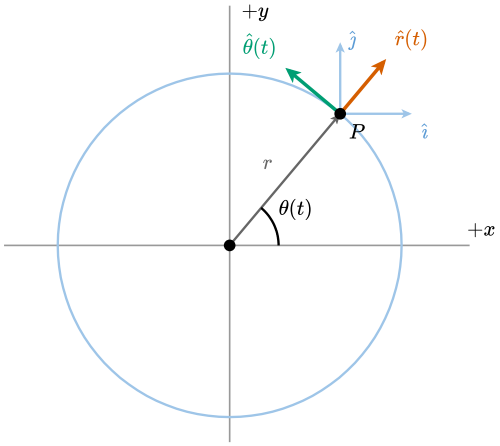


Figure 3.4.23 The moving frame at the point P: $\hat{r}(t)$ points radially outward and $\hat{\theta}(t)$ is tangent to the circle. The fixed vectors $\hat{\imath}$ and $\hat{\jmath}$ are drawn at P for comparison; unlike $\hat{r}$ and $\hat{\theta}$, they do not turn as the particle moves.

Read the components of the moving frame off Figure 3.4.23, and write down the position vector. ($\hat{r}$ makes the angle θ with the x-axis, and $\hat{\theta}$ is a quarter turn counterclockwise from it. Then check that $\hat{r}$, $\hat{\theta}$, $\hat{k}$ is right-handed by computing $\hat{r} \times \hat{\theta}$ with (3.4.5); a Pythagorean identity finishes it.)

$\hat{r}(t) =$

$\hat{\theta}(t) =$

$\vec{r}(t) =$ $\hat{r} \times \hat{\theta} =$

The Moving Frame.

$$\hat{r}(t) = \cos\theta(t)\,\hat{\imath} + \sin\theta(t)\,\hat{\jmath}, \tag{3.4.9}$$

$$\hat{\theta}(t) = -\sin\theta(t)\,\hat{\imath} + \cos\theta(t)\,\hat{\jmath}, \tag{3.4.10}$$

$$\overrightarrow{r}(t) = r\,\hat{r}(t). \tag{3.4.11}$$

The pair $\hat{\imath}$, $\hat{\jmath}$ is fixed once and for all, while the pair $\hat{r}(t)$, $\hat{\theta}(t)$ rides along with the particle and turns as it goes; this is exactly what makes the second pair convenient.

3.4.5.2 The Velocity in Circular Motion

Because the particle moves, the angle θ depends on t, and so do the two moving unit vectors. Differentiate the frame, and then the position.

$\frac{d\hat{r}}{dt} =$

$\frac{d\hat{\theta}}{dt} =$

$\overrightarrow{v}(t) = \frac{d\overrightarrow{r}}{dt} =$

Why the velocity has no radial component:

Velocity in Circular Motion.

$$\overrightarrow{v}(t) = r\,\frac{d\theta}{dt}\,\hat{\theta}(t), \tag{3.4.12}$$

whose $\hat{\theta}$-component

$$v_\theta = r\,\frac{d\theta}{dt} \tag{3.4.13}$$

is called the **tangential component of the velocity**. It is positive when the particle turns counterclockwise and negative when it turns clockwise.

Definition 3.4.24 Angular Speed. The **angular speed** of the particle is the magnitude of the rate of change of its angle,

$$\omega = \left| \frac{d\theta}{dt} \right|, \tag{3.4.14}$$

measured in radians per second. By (3.4.12), the speed of the particle and its angular speed are related by

$$v = \|\vec{v}\| = r\,\omega. \tag{3.4.15}$$

◆

It is worth seeing where (3.4.15) comes from geometrically, without any differentiation of unit vectors. Figure 3.4.25 shows the particle moving from $\vec{r}(t)$ to $\vec{r}(t + \Delta t)$ through a small angle $\Delta\theta$. Recover (3.4.15) from that picture.

Chord length: $\|\Delta\vec{r}\| =$

Taking the limit:

The direction of the chord as $\Delta t \to 0$*:*

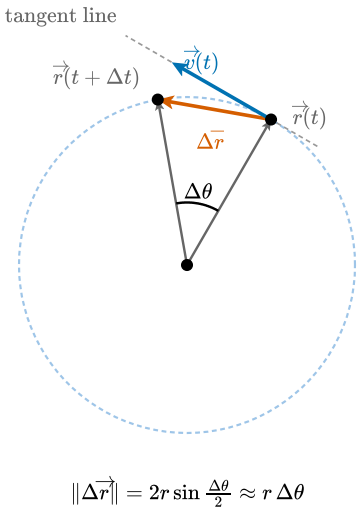


Figure 3.4.25 The displacement $\Delta\vec{r}$ is the chord subtending the angle $\Delta\theta$. For small $\Delta\theta$ the chord is nearly the arc, $\|\Delta\vec{r}\| \approx r\,\Delta\theta$, and its direction approaches that of the tangent line, along which $\vec{v}$ points.

3.4.5.3 Angular Velocity as a Vector

So far the rate of turning has been a number, $d\theta/dt$. But a rotation has a direction as well as a rate: the plane of the circle can be tilted any which way, and the particle can go around it either way. Both pieces of information can be carried by a single vector, provided we point it along the axis of rotation rather than along the motion.

Definition 3.4.26 Angular Velocity Vector. For an object moving in a circle in the xy-plane about the origin, the **angular velocity vector** is

$$\vec{\omega} = \frac{d\theta}{dt}\,\hat{k} = \omega_z\,\hat{k}, \tag{3.4.16}$$

directed along the axis of rotation. Its SI units are $\text{rad}\cdot\text{s}^{-1}$, and the angular speed of (3.4.14) is its magnitude, $\omega = \|\vec{\omega}\| = |\omega_z|$. ◆

The sign of ω_z records which way the object goes around, and the right-hand rule turns that sign into a direction in space: curl the fingers of your right hand in the direction of the motion and your thumb points along $\vec{\omega}$. The two cases are shown in Figure 3.4.27.

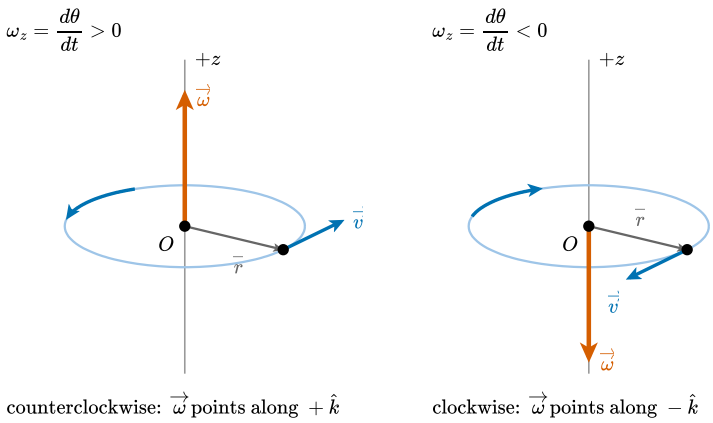


Figure 3.4.27 The angular velocity vector points along the axis of rotation, in the direction given by the right-hand rule: up the z-axis for counterclockwise motion (left), down it for clockwise motion (right). Note that $\vec{\omega}$ is *not* tangent to the orbit — the velocity $\vec{v}$ is.

3.4.5.4 Velocity as a Cross Product

We now have two vectors attached to the motion: the angular velocity $\vec{\omega}$, along the axis, and the position $\vec{r}$, in the plane of the circle. The velocity is perpendicular to both, which is precisely the job of the cross product.

Fact 3.4.28 Velocity and Angular Velocity. *For an object in circular motion about the origin,*

$$\vec{v} = \vec{\omega} \times \vec{r}. \tag{3.4.17}$$

Verify (3.4.17).

$\hat{k} \times \hat{r} =$

$\vec{\omega} \times \vec{r} =$

The geometry of the cross product delivers all three features of the velocity at once. Fill them in from Figure 3.4.29, with no computation.

Magnitude:

Direction:

Sense:

There is a bonus. A piece of $\vec{r}$ running along the axis is parallel to $\vec{\omega}$ and so contributes nothing to the cross product, since $\hat{k} \times \left(z\,\hat{k}\right) = \vec{0}$. Consequently (3.4.17) remains true when $\vec{r}$ is measured from *any* point on the axis of rotation, not just from the centre of the circle.

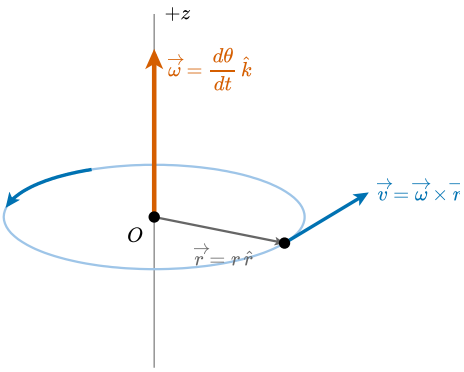


Figure 3.4.29 The velocity of a particle in circular motion is $\vec{v} = \vec{\omega} \times \vec{r}$. Because $\vec{\omega}$ and $\vec{r}$ are perpendicular, the speed is ωr, and the right-hand rule places $\vec{v}$ tangent to the orbit in the direction of travel.

3.4.5.5 Examples

Example 3.4.30 Angular velocity from an angle function. A particle moves in a circle of radius R centred at the origin of the xy-plane; at $t = 0$ it is on the positive x-axis. The angle it makes with the positive x-axis is $\theta(t) = At - Bt^3$, where A and B are positive constants.

1. Find the angular velocity vector.
2. Find the velocity vector, in polar coordinates.
3. At what time t_1 is the angular velocity zero?
4. In which direction does the angular velocity point for $t < t_1$, and for $t > t_1$?

Solution.

$\frac{d\theta}{dt} =$

1. $\vec{\omega}(t) =$

2. $\vec{v}(t) =$

3. $t_1 =$

4. *Before* t_1*:* *After* t_1*:*

▲

Example 3.4.31 A turntable. A record turntable spins at $33\frac{1}{3}$ revolutions per minute, counterclockwise when viewed from above. Take the axis of rotation to be the z-axis, with the turntable lying in the xy-plane, distances in meters.

1. Find the angular velocity vector $\vec{\omega}$.
2. Find the velocity of the point of the record at $\vec{r} = \langle 0.12,\, 0,\, 0\rangle$, and its speed.

Solution.

1. $\omega =$ $\vec{\omega} =$

2. $\vec{v} =$

Speed, and the check:

▲

Checkpoint 3.4.32 A carousel. A carousel turns *clockwise* when viewed from above, making one complete turn every 12 seconds. Taking the axis of rotation as the z-axis, find the angular velocity vector, and the velocity of a horse located at $\vec{r} = \langle 0, 4, 0 \rangle$ meters.

Solution.

$\omega =$ $\vec{\omega} =$

$\vec{v} =$

Direction check:

3.5 Parametrization of Plane Curves

Objectives

After completing this section, you should be able to:

- Describe the motion of a point in the plane with a pair of parametric equations $x = f(t)$ and $y = g(t)$, and understand the parameter t as recording *when* the point visits each place on the curve.
- Sketch a parametric curve from a table of values, and indicate the direction of increasing t in which the curve is traced.
- Find parametric equations for common curves, such as lines, circles, and ellipses, and eliminate the parameter to recover a Cartesian equation when possible.
- Model the motion of a projectile with parametric equations, and derive the equation of its parabolic trajectory.
- Distinguish the **intersection points** of two parametric curves from their **collision points**, where two moving objects reach the same place at the same time.

3.5.1 Parametric Equations

Imagine that you would like to describe the motion of an object whose motion is confined to a plane. If you know how its Cartesian coordinates, x and y, change as time passes, you will be able to draw a curve that represents the trajectory of its motion, as shown in Figure 3.5.2. Mathematically, we will need two equations to demonstrate how the x- and y-coordinates evolve in time t, which are known as the **parametric equations**.

Definition 3.5.1 Parametric Equations. If x and y are given as functions of a third variable t, called a **parameter**, by

$$x = f(t), \qquad y = g(t), \tag{3.5.1}$$

then the points $(x, y) = (f(t), g(t))$ trace out a curve in the plane, called a **parametric curve**. The equations $x = f(t)$ and $y = g(t)$ are the parametric equations of the curve. ◆

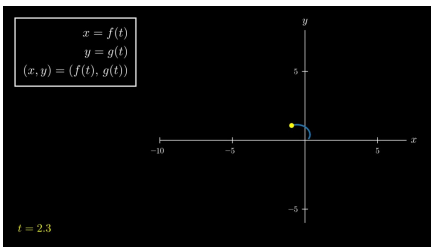


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Figure 3.5.2 A point moves in the plane. As time t passes, its coordinates $x = f(t)$ and $y = g(t)$ change, and the point traces out the trajectory of the motion.

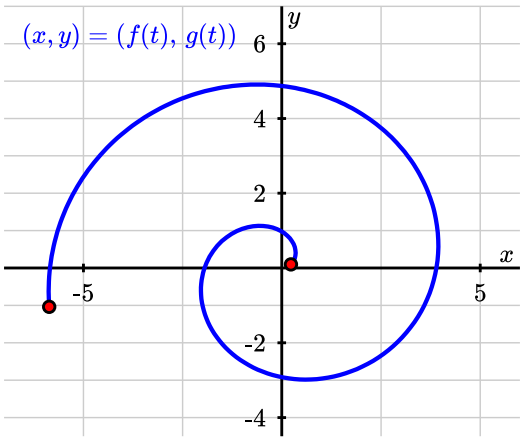
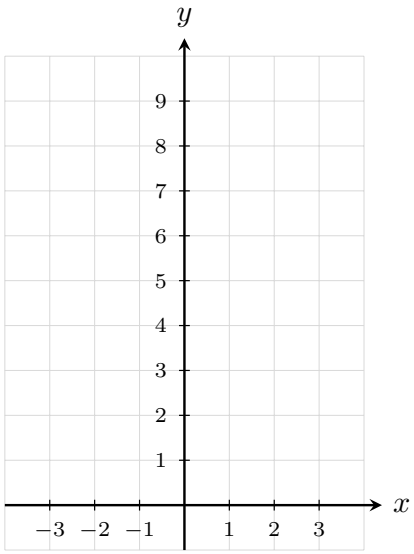


Figure 3.5.3 The trajectory of a moving point whose coordinates are given by the parametric equations $x = f(t)$, $y = g(t)$.

As an example, consider $x = t$ and $y = t^2$. Fill in Table 3.5.4, plot the seven points on the grid below, and join them.

Table 3.5.4 Points on the curve $x = t$, $y = t^2$

t	x	y
-3		
-2		
-1		
0		
1		
2		
3		



Eliminating t:

Name of the curve, and its direction:

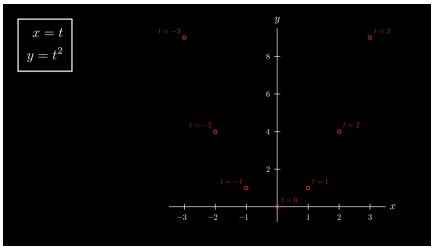


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Figure 3.5.5 The parametric curve $x = t$, $y = t^2$: the points from the table appear first, and then the moving point traces out the parabola $y = x^2$ in the direction of increasing t. Use it to check your sketch.

As our second example, consider the equation of the circle centered at $(0, 0)$ with unit radius,

$$x^2 + y^2 = 1. \tag{3.5.2}$$

Here, we would like to parametrize x and y in terms of a single parameter t, as in (3.5.1), in such a way that by varying the value of t we can reconstruct the circle (3.5.2). Find such a pair.

The identity, and the parametrization:

Interval of t, and the check:

The Unit Circle.

$$x = \cos(t), \qquad y = \sin(t), \qquad t \in [0, 2\pi). \tag{3.5.3}$$

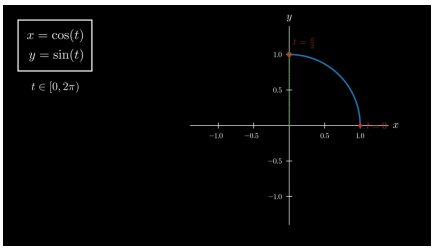

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Figure 3.5.6 Parametrizing the unit circle: as t increases from 0 to 2π, the point $(\cos(t), \sin(t))$ travels counterclockwise around the circle, passing through the marked points at $t = 0$, $\pi/2$, π, and $3\pi/2$.

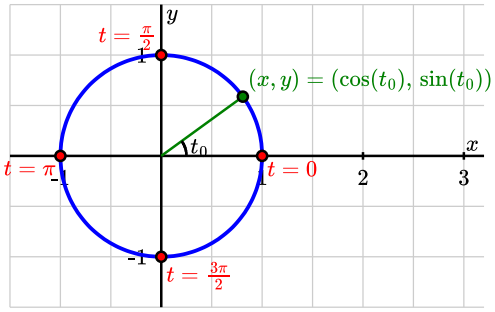


Figure 3.5.7 Parametrizing the unit circle $x^2+y^2 = 1$ by $x = \cos(t)$, $y = \sin(t)$, $t \in [0, 2\pi)$. A generic point of the circle corresponds to the parameter value t_0.

Checkpoint 3.5.8 Question. Can you think of another way to write a parametric form of the circle $x^2 + y^2 = 1$?

workspace

3.5.2 Some Examples of Parametrizing Curves

Example 3.5.9 Parametrizing Curves. Find the parametric equations corresponding to the following curves.

A. The line segment connecting the two points $(1, 0)$ and $(0, 1)$.

B. The ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{9} = 1$.

C. The circle of radius 2 centered at $(2, 0)$.

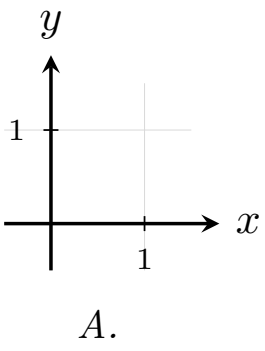
Solution.

A.

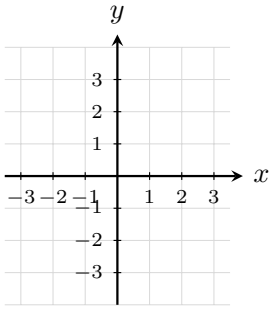
B.

C.

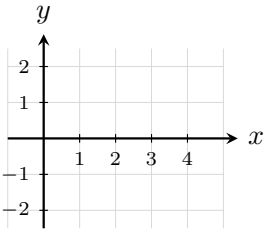
Now sketch the three curves on the grids below, marking the point reached at $t = 0$ and the direction of increasing t on each.



A.



B.



C.

▲

The three videos below trace the curves out. Use them to check your sketches and, in particular, the directions you marked.

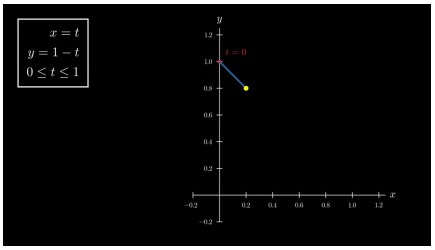


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Figure 3.5.10 Part A: the line segment from $(0, 1)$ to $(1, 0)$.

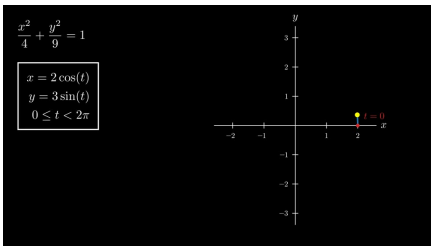

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Figure 3.5.11 Part B: the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$, traced counterclockwise, with the points at $t = 0$, $\pi/2$, π, and $3\pi/2$ marked.

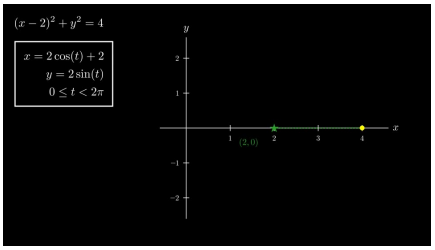

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Figure 3.5.12 Part C: the circle of radius 2 centered at $(2, 0)$.

3.5.3 ∗Projectile Motion

Starred section. This one is for the interested reader. It will not be examined.

A type of parametric equation you may have already seen is the equation of motion for a projectile. Here the parameter is time, and $(x(t), y(t))$ gives the position of the projectile at time t.

Recall two basic facts about motion in one dimension from your introductory physics course. For motion along a straight line at constant speed v, the position is

$$x(t) = v\,t + x_0. \tag{3.5.4}$$

For motion with uniform acceleration a, the position is

$$x(t) = \frac{1}{2}at^2 + v_0t + x_0. \tag{3.5.5}$$

Now consider an object moving under the influence of gravity alone, ignoring air resistance. Write down its equations of motion.

Components of the initial velocity:

$x(t) =$

$y(t) =$

Equations of Motion for a Projectile.

$$x(t) = v_0 \cos(\theta)\, t + x_0, \tag{3.5.6}$$

$$y(t) = -\frac{1}{2} g t^2 + v_0 \sin(\theta)\, t + y_0, \tag{3.5.7}$$

where v_0 is the initial speed, θ is the launch angle, and g is the acceleration due to gravity.

Now eliminate the parameter to find the trajectory.

$t =$

Substituting, $y =$

The trajectory is a:

Differentiating the equations of motion gives the velocity of the projectile. Write down its two components, and use them to say what happens at the

apex.

$v_x =$ $v_y =$

Why v_x is constant:

At the apex:

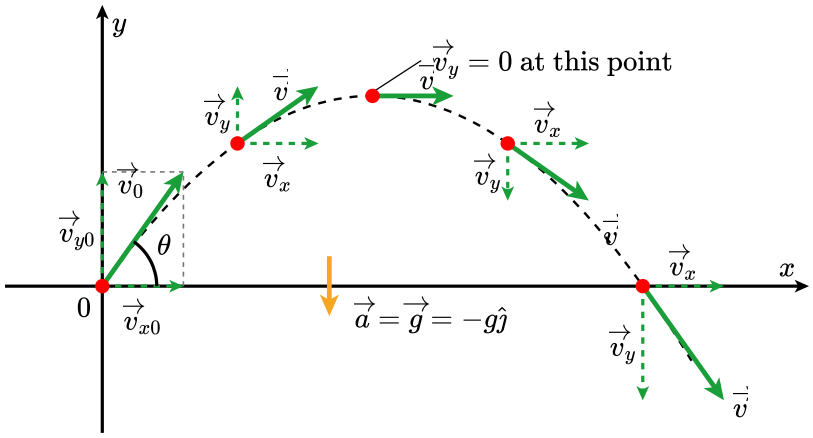

Figure 3.5.13 The velocity of a projectile at five instants along its parabolic path. The horizontal component $\vec{v}_x$ stays constant while the vertical component $\vec{v}_y$ changes under the constant downward acceleration $\vec{g}$; at the apex the vertical component is zero.

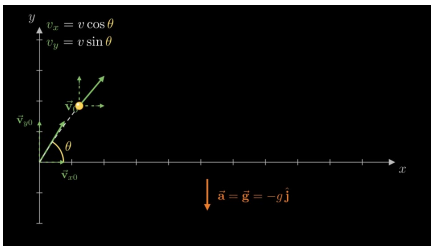

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Figure 3.5.14 A projectile launched from the origin traces out its parabolic trajectory. The horizontal velocity component v_x is constant, while the vertical component v_y shrinks to zero at the apex and then grows in the downward direction as the projectile falls.

Activity 3.5.3.1 Projectiles on an Inclined Plane. A particle is launched from a point O on an inclined plane and travels in the vertical plane that contains the line of greatest slope through O. The plane makes an angle β with the horizontal, and the particle leaves O with initial speed v_0 at an elevation θ to the horizontal. Taking the origin at O with horizontal and vertical axes

and neglecting air resistance, the trajectory is the projectile parabola

$$y = x\tan\theta - \frac{g\,x^2}{2\,v_0^2\cos^2\theta}.$$

Let P be the point where the particle meets the plane again, and let $R = OP$ be the range measured *along the plane*, so that $P = \big(R\cos\beta,\ R\sin\beta\big)$. Work through the tasks below to find the range on the plane, the maximum range for a given v_0, and the time of flight, and then to read off the corresponding results when the particle is fired *down* the plane.

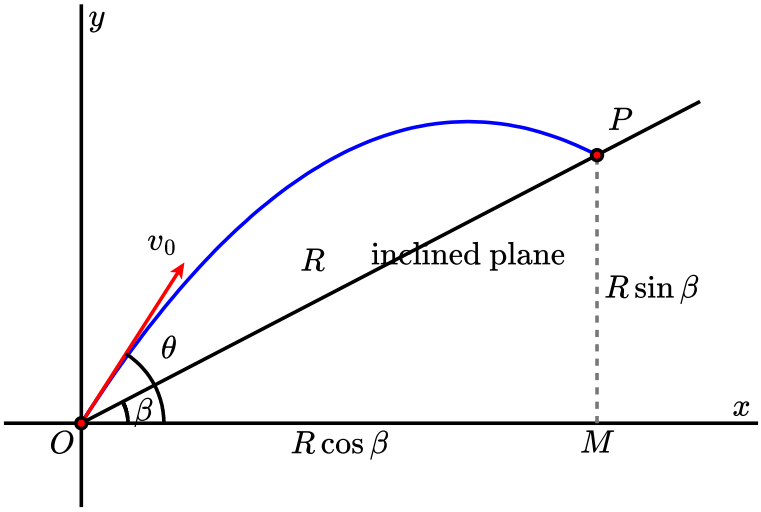

Figure 3.5.15 A particle projected up the plane with speed v_0 at elevation θ, striking the incline (angle β) at P after travelling a range $R = OP$ along the slope.

(a) Since $P = \big(R\cos\beta,\ R\sin\beta\big)$ lies on the trajectory, substitute these coordinates into the parabola and cancel a factor of R to show that

$$\sin\beta = \cos\beta\,\tan\theta - \frac{g\,R\,\cos^2\beta}{2\,v_0^2\cos^2\theta}.$$

workspace

(b) Rearrange the relation from the previous task to isolate R, and use $\sin\theta\cos\beta - \cos\theta\sin\beta = \sin(\theta-\beta)$ to obtain the **range up the plane**

$$R = \frac{2\,v_0^2\,\sin(\theta-\beta)\cos\theta}{g\cos^2\beta}.$$

Collecting the R-term:

Solving for R:

(c) Using the identity $2\sin(\theta-\beta)\cos\theta = \sin(2\theta-\beta) - \sin\beta$, rewrite the range as

$$R = \frac{v_0^2}{g\cos^2\beta}\big[\sin(2\theta-\beta) - \sin\beta\big].$$

For a fixed launch speed v_0, deduce the elevation θ that maximizes R, and show that the maximum range up the plane is

$$R_{\max} = \frac{v_0^2}{g\,(1+\sin\beta)}.$$

The rewritten range:

The optimal elevation, $\theta =$

$R_{\max} =$

Hint. With v_0, g, and β held constant, only $\sin(2\theta-\beta)$ varies, and it is largest when it equals 1. Afterwards use $\cos^2\beta = (1-\sin\beta)(1+\sin\beta)$.

(d) The horizontal distance OM to the foot of P is covered at the constant horizontal speed $v_0 \cos\theta$. Using $OM = R\cos\beta$, show that the **time of flight** is

$$T = \frac{2\,v_0\,\sin(\theta - \beta)}{g\cos\beta}.$$

$T =$

(e) When the particle is projected *down* the plane, the geometry is the same with β replaced by $-\beta$. Write down the range, maximum range, and time of flight down the inclined plane.

$R_{down} =$

$R_{\max,down} =$

$T_{down} =$

Larger or smaller than up the plane, and why:

The maximum range up the plane occurs when $2\theta - \beta = \dfrac{\pi}{2}$. Rearrange that condition into the form $\theta - \beta = \dfrac{\pi}{2} - \theta$, and interpret each side geometrically: one is the angle the launch direction makes *above the incline*, the other the angle it makes *below the vertical*. What does their being equal say about the direction of projection for maximum range?

workspace

3.5.4 ∗Intersection Points versus Collision Points

Starred section. This one is for the interested reader. It will not be examined.

A parametrization carries more information than the curve it traces: it also records *when* the moving point visits each place on the curve. Because of this, there are two different questions we can ask about two parametric curves

$$\mathbf{r}_1(t) = \big\langle f_1(t),\, g_1(t)\big\rangle, \qquad \mathbf{r}_2(t) = \big\langle f_2(t),\, g_2(t)\big\rangle,$$

thought of as the trajectories of two particles.

Definition 3.5.16 Intersection Points and Collision Points. A point P is an **intersection point** of the two curves if P lies on both curves; that is, if there are parameter values t and s, *not necessarily equal*, with

$$\mathbf{r}_1(t) = \mathbf{OP} = \mathbf{r}_2(s). \tag{3.5.8}$$

A point P is a **collision point** of the two particles if both particles are at P at the *same* time; that is, if there is a single value of t with

$$\mathbf{r}_1(t) = \mathbf{r}_2(t) = \mathbf{OP}. \tag{3.5.9}$$

◆

An intersection point is a statement about the two *paths*: they cross, like two roads on a map. A collision point is a statement about the two *motions*: the cars are at the crossroads at the same instant.

Say how the two definitions differ when you set up the equations, and how they are related.

To find intersection points, solve:

To find collision points, solve:

Does one imply the other?

In the next two examples the two *paths* are always the same: the parabola $y = x^2$ and the line $y = x + 2$, which meet at the two points $(-1, 1)$ and $(2, 4)$. Only the *schedule* of the second particle will change.

Example 3.5.17 The paths cross, but the particles never meet. Two particles move in the plane with position vectors

$$\mathbf{r}_1(t) = \langle t,\ t^2 \rangle, \qquad \mathbf{r}_2(t) = \langle -t,\ 2 - t \rangle, \qquad t \in \mathbb{R}.$$

Find all intersection points of their paths, and all collision points of the particles.

Solution.

Paths:

Collisions:

Times at which each particle visits $(-1, 1)$ *and* $(2, 4)$*:*

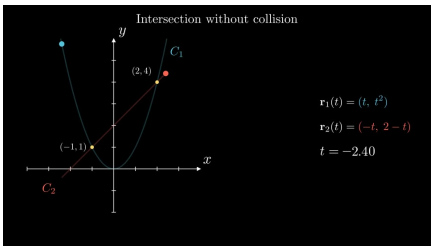

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Figure 3.5.18 The particles $\mathbf{r}_1(t) = \langle t, t^2 \rangle$ and $\mathbf{r}_2(t) = \langle -t, 2 - t \rangle$ move simultaneously. Their paths cross at $(-1, 1)$ and $(2, 4)$, but the clock shows that the two particles are never at either crossing at the same time.

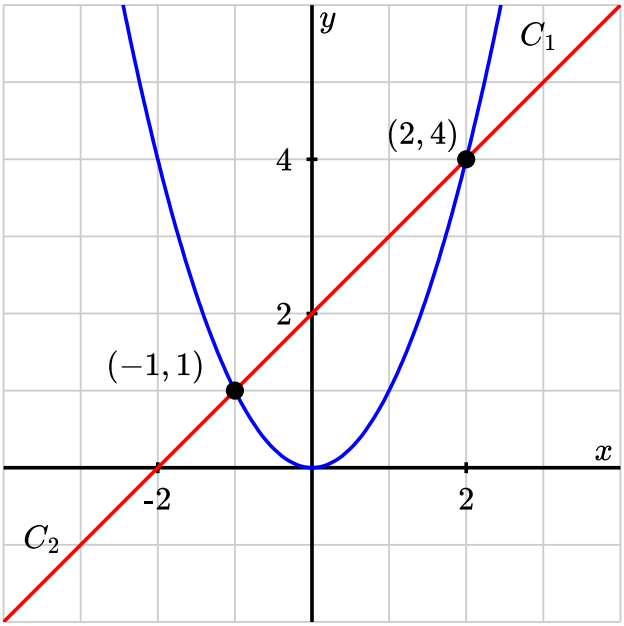


Figure 3.5.19 The parabola C_1 and the line C_2 intersect at $(-1, 1)$ and $(2, 4)$. At each of these points the two particles arrive at different times, so neither point is a collision point.

Example 3.5.20 The same paths, a different schedule—now they collide. Keep the first particle, but re-schedule the second one:

$$\mathbf{r}_1(t) = \langle t,\ t^2 \rangle, \qquad \mathbf{r}_2(t) = \langle 2t - 2,\ 2t \rangle, \qquad t \in \mathbb{R}.$$

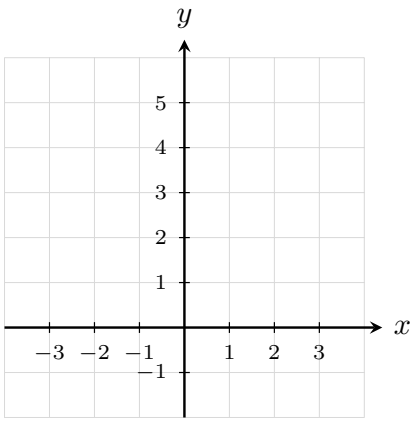
Find the intersection points of the paths and the collision points of the particles.

Solution.

Paths:

Collisions:

The other crossing:



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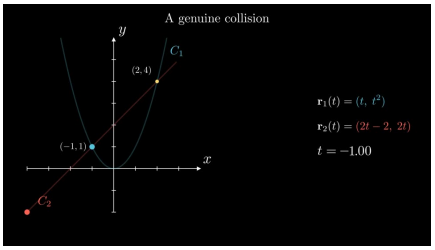

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Figure 3.5.21 With $\mathbf{r}_2(t) = \langle 2t - 2, 2t \rangle$ the second particle follows the same line as before, but on a different schedule. Use it to check which crossing is the collision, and when it happens.

Insight 3.5.22 Every Collision Is an Intersection, but Not Conversely. If two particles collide at P, then certainly P lies on both paths, so every collision point is an intersection point. The converse fails: the paths may cross at a point that the two particles visit at different times. Changing the parametrization of a curve does not move the curve, but it can create or destroy collisions.

Checkpoint 3.5.23 Keep $\mathbf{r}_1(t) = \langle t, t^2 \rangle$. Find a parametrization $\mathbf{r}_2$ of the line $y = x + 2$ for which the two particles collide at *both* intersection points.

$\mathbf{r}_2(t) =$

The collision equations, and their solutions:

Hint. Try to arrange that the second particle has the same x-coordinate as the first one at every time t.

3.6 Lines and Planes in Space

Objectives

After completing this section, you should be able to:

- Write the vector equation and the parametric equations of a line in space from a point on the line and a vector parallel to it.
- Compute the distance from a point to a line using the cross product.
- Find the equation of a plane from a point on the plane and a normal vector, and find the line of intersection of two planes.
- Compute the distance from a point to a plane using the projection onto the normal vector.

3.6.1 Vector Equation for a Line in Space

Consider a line that passes through the point $P_0(x_0, y_0, z_0)$ and is parallel to the vector $\mathbf{v}$, as in Figure 3.6.1. Write down the vector equation of that line.

$\overrightarrow{P_0P} =$

$\mathbf{r}(t) =$

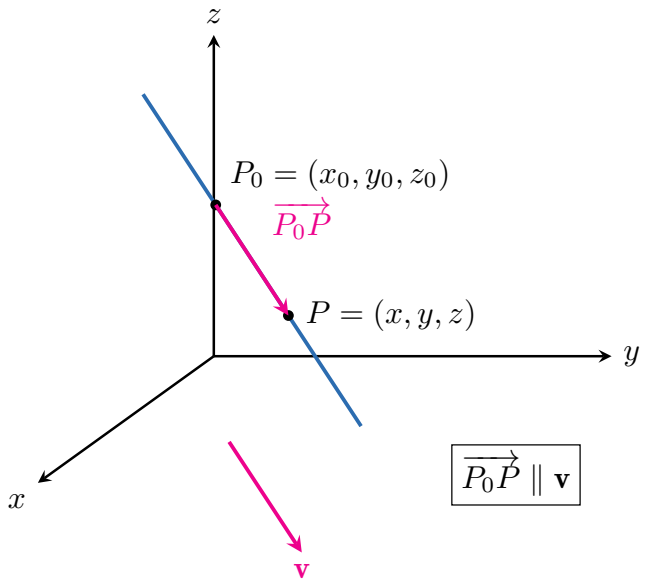


Figure 3.6.1 The line through P_0 parallel to $\mathbf{v}$. For any point P on the line, the vector $\overrightarrow{P_0P}$ is parallel to $\mathbf{v}$.

Vector Equation of a Line.

$$\mathbf{r}(t) = \mathbf{r}_0 + t\,\mathbf{v}, \quad t \in \mathbb{R}. \tag{3.6.1}$$

lets you move P_0 and $\mathbf{v}$ and slide the parameter t to see the point $\mathbf{r}(t) = \mathbf{r}_0 + t\,\mathbf{v}$ travel along the line.

An interactive figure of this line — drag P_0 or $\mathbf{v}$, and slide t to move the point along it — accompanies the online version of these notes.

3.6.2 Parametric Equations for a Line

The vector equation (3.6.1) can be interpreted as the following parametric equations, where $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$:

$$\begin{aligned} x &= x_0 + t\,v_1 \\ y &= y_0 + t\,v_2 \qquad t \in \mathbb{R}. \\ z &= z_0 + t\,v_3 \end{aligned} \tag{3.6.2}$$

When none of v_1, v_2, v_3 is zero, each parametric equation can be solved for t; equating the three expressions eliminates t and gives the **standard equations** (also called the **symmetric equations**) of the line:

$$\frac{x - x_0}{v_1} = \frac{y - y_0}{v_2} = \frac{z - z_0}{v_3}. \tag{3.6.3}$$

If one of the components of $\mathbf{v}$ is zero, say $v_3 = 0$, that coordinate stays constant along the line, and the standard equations are written as $\dfrac{x - x_0}{v_1} = \dfrac{y - y_0}{v_2}$, $z = z_0$.

Example 3.6.2 The line through two points. Find the vector equation, the parametric equations and the standard equations of the line that passes through the points $P_0(2, 0, 0)$ and $P(1, -1, 1)$.

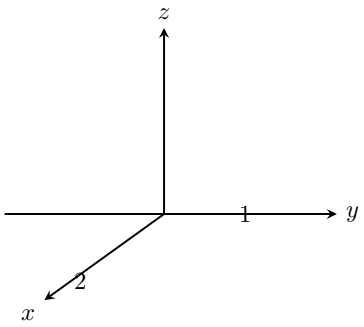
Solution.

$\mathbf{v} =$

$\mathbf{r}(t) =$

Parametric equations:

Standard equations:



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3.6.3 The Distance from a Point to a Line

Our goal is to find the distance d of the point S from the line L, as shown in Figure 3.6.3. Derive a formula for it.

From the triangle, $d =$

As a cross product, $d =$

Distance from a Point to a Line.

$$d = \frac{\|\overrightarrow{PS} \times \mathbf{v}\|}{\|\mathbf{v}\|}. \tag{3.6.4}$$

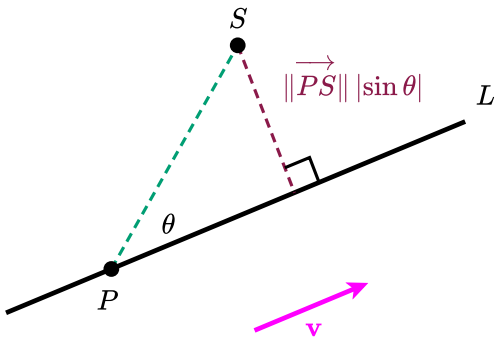


Figure 3.6.3 The distance of the point S from the line L is $d = \|\overrightarrow{PS}\|\,|\sin\theta|$, where P is any point on the line.

Example 3.6.4 Distance from a point to a line. Calculate the distance between the point $S(1,1,0)$ and the line with the vector equation $\mathbf{r}(t) = (2-t)\,\hat{\imath} - t\,\hat{\jmath} + t\,\hat{k}$, $t \in \mathbb{R}$.

Solution.

$\mathbf{v} =$ $P =$ $\overrightarrow{PS} =$

$\overrightarrow{PS} \times \mathbf{v} =$

$d =$

▲

3.6.4 Equation of a Plane in Space

Let us assume that we have computed the normal vector $\overrightarrow{n}$ to a plane and that we have a point P_0 on the plane. A normal vector is perpendicular to the plane and hence perpendicular to any vector lying in it. Denoting an arbitrary

point of the plane by P, as in Figure 3.6.5, this gives

$$\vec{n} \cdot \overrightarrow{P_0P} = 0. \tag{3.6.5}$$

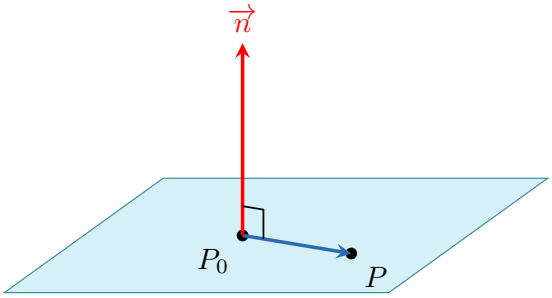


Figure 3.6.5 The normal vector to a plane is perpendicular to all the vectors in the plane.

Turn (3.6.5) into an equation in x, y, and z.

Point-normal form:

Standard form:

$D =$

Equation of a Plane.

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0, \tag{3.6.6}$$

equivalently

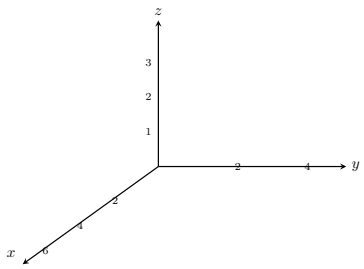
$$Ax + By + Cz = D, \qquad D = Ax_0 + By_0 + Cz_0. \tag{3.6.7}$$

Notice that the coefficients of x, y, and z in the standard form are the components of the normal vector.

Example 3.6.6 Finding the equation of a plane. Find an equation of the plane through the point $P_0(2, 4, -1)$ with normal vector $\vec{n} = \langle 2, 3, 4 \rangle$. Find the intercepts and sketch the plane.

Solution.

Equation of the plane:



x-intercept: *y-intercept:* *z-intercept:*

▲

3.6.5 Lines of Intersection

Suppose we would like to find the line of intersection L of two planes with normal vectors $\vec{n}_1$ and $\vec{n}_2$, as in Figure 3.6.7. To write down the equation of L we need a direction vector $\mathbf{v}$ for it, and the two normal vectors are enough to produce one.

Explain why, and write down $\mathbf{v}$.

Each normal vector is … to L, because:

$\mathbf{v} =$

When is this nonzero?

Direction of the Line of Intersection.

$$\mathbf{v} = _____. \tag{3.6.8}$$

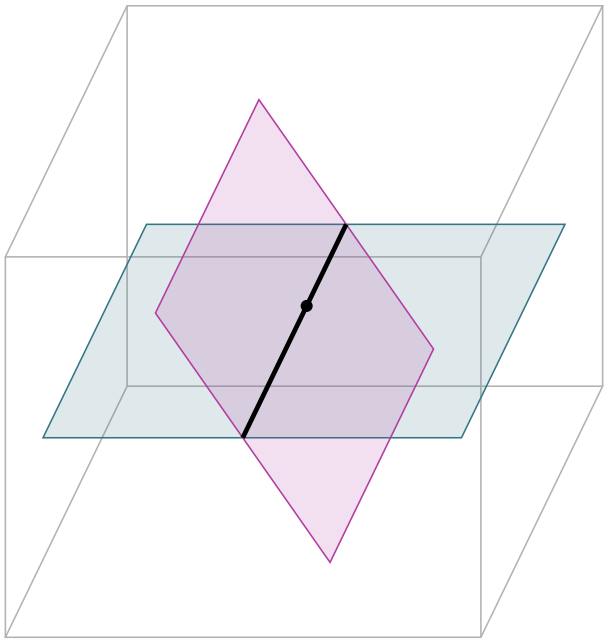


Figure 3.6.7 The direction vector of the line.

Example 3.6.8 The line of intersection of two planes. Find the parametric equations of the line of intersection of the two planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Solution.

$\vec{n}_1 =$ $\vec{n}_2 =$

$\mathbf{v} =$

A common point $P_0 =$

Parametric equations:

▲

3.6.6 Distance of a Point from a Plane

Our goal here is to find the distance d of the point S from a plane, given a point P on the plane and the normal vector $\vec{n}$, as shown in Figure 3.6.9. Derive a formula for it.

$d = \left\|\text{proj}_{\vec{n}}\, \overrightarrow{PS}\right\| =$

Simplified, $d =$

Distance from a point to a plane.

$$d = \frac{\left|\overrightarrow{PS} \cdot \vec{n}\right|}{\|\vec{n}\|} \tag{3.6.9}$$

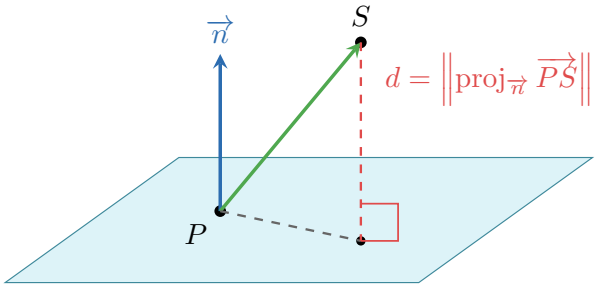


Figure 3.6.9 The distance of the point S from the plane is the magnitude of the projection of $\overrightarrow{PS}$ onto the normal vector $\vec{n}$. The corner mark records that the dashed segment meets the plane at a right angle, so it is parallel to $\vec{n}$.

Example 3.6.10 Distance from a point to a plane. Find the distance from the point $S(2, 3, 0)$ to the plane with equation $5x + y + z = -1$.

Solution.

$\vec{n} =$	$P =$
$\overrightarrow{PS} =$	$\overrightarrow{PS} \cdot \vec{n} =$
$d =$	

▲

3.7 Conic Sections

Objectives

After completing this section, you should be able to:

- Understand the parabola, ellipse, and hyperbola as curves defined by distance conditions relative to foci and a directrix, rather than merely as graphs of particular equations.
- Recognize how a single geometric idea—a locus of points satisfying a distance condition—unifies the three conic sections.
- Interpret the roles of the focus, directrix, vertices, and axes in determining the shape and orientation of each conic.
- Understand the reflection properties of the conics and why they explain applications such as parabolic reflectors and whispering galleries.
- Appreciate how the defining property of the hyperbola underlies position-finding methods such as hyperbolic navigation.

Conic sections are the curves obtained by intersecting a double cone with a plane. There are three types: **parabolas**, **ellipses**, and **hyperbolas**. In this section we give the focus–directrix and focal definitions of each, read the key features off the standard equations, and work examples that change the orientation.

An interactive figure of the slicing itself — the plane swinging from horizontal to steep, the section stretching from circle to ellipse, opening into a parabola and breaking into a hyperbola — accompanies the online version of these notes.

Record what decides which of the three curves you get.

Ellipse when the tilt is:

Parabola when the tilt is:

Hyperbola when the tilt is:

3.7.1 Parabola

Definition 3.7.1 A **parabola** is the set of points in a plane that are equidistant from a fixed point F (the **focus**) and a fixed line (the **directrix**). ◆

The standard equation follows from that definition and nothing else. Choose coordinates that make the picture symmetric: put the focus at $F(0, p)$ and the directrix at $y = -p$, as in Figure 3.7.2. Derive the standard form.

The equidistance condition:

After squaring:

Standard form:

The Standard Parabola.

$$x^2 = 4py. \tag{3.7.1}$$

The vertex is at the origin, the focus at $(0, p)$, and the directrix is the line $y = -p$; the vertex sits halfway between them. If $p < 0$ the curve opens downward, and interchanging the roles of x and y gives the sideways parabola

$$y^2 = 4px. \tag{3.7.2}$$

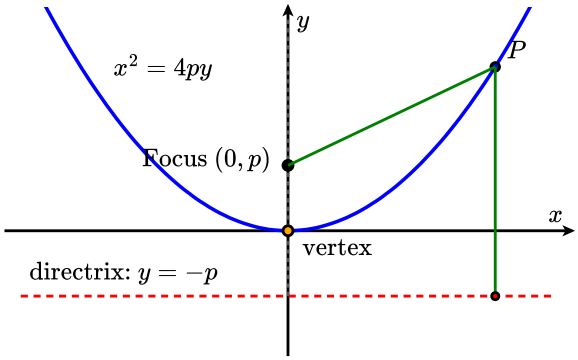
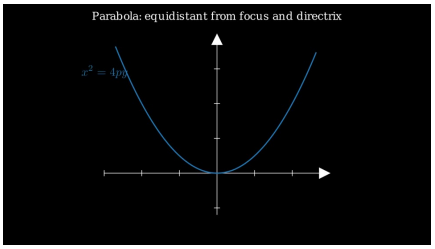


Figure 3.7.2 The parabola $x^2 = 4py$. The point P is equidistant from the focus $(0, p)$ and the directrix $y = -p$.



Standalone

Figure 3.7.3 Animation: as P moves along the parabola, its distance to the focus always equals its distance to the directrix.

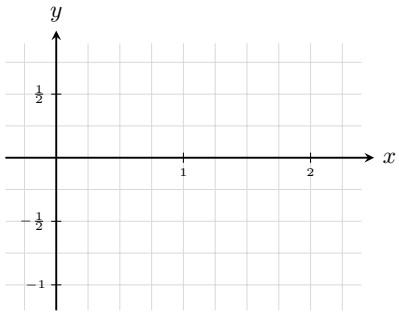
Example 3.7.4 From equation to focus and directrix. Find the focus and the directrix of $y = -\frac{1}{2}x^2 + x - \frac{1}{2}$.

Solution.

Completing the square:

$4p =$ *so* $p =$ *vertex:*

Focus: *Directrix:*

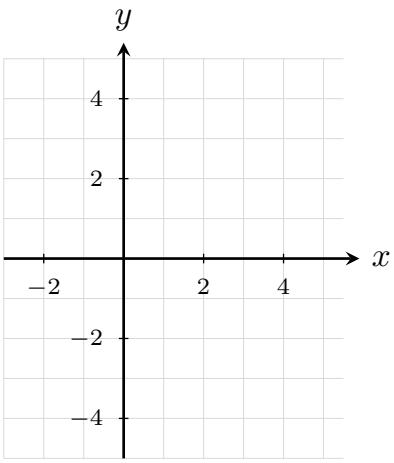


▲

Example 3.7.5 From focus and directrix to equation (a sideways parabola). Find the equation of the parabola whose focus is $(2, 0)$ and whose directrix is the vertical line $x = -2$. In which direction does it open?

Solution.

The equidistance condition, squared and expanded:



Equation: *Opens:* $p =$

▲

3.7.2 Ellipse

Definition 3.7.6 An **ellipse** is the set of points in a plane the sum of whose distances from two fixed points F_1 and F_2 (the **foci**) is constant. ◆

The Standard Ellipse.

An ellipse centred at the origin with its foci on the x-axis has the standard form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > b. \tag{3.7.3}$$

The segment joining $(a, 0)$ and $(-a, 0)$ is the **semi-major axis**, the segment joining $(0, b)$ and $(0, -b)$ is the **semi-minor axis**, and the foci satisfy $c^2 = a^2 - b^2 > 0$.

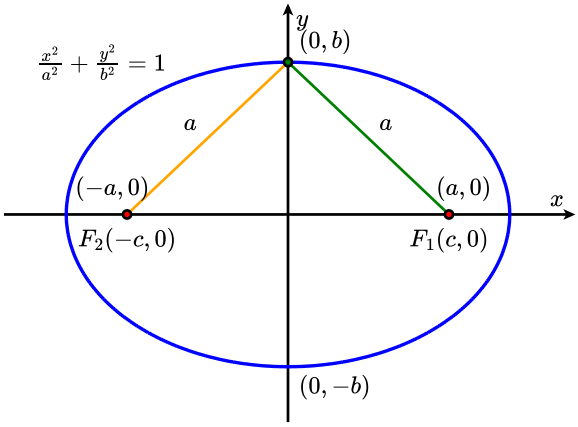
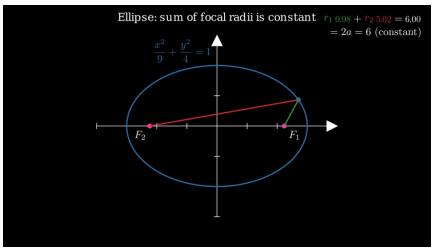


Figure 3.7.7 An ellipse with foci $F_1(c, 0)$ and $F_2(-c, 0)$. Each slanted segment from $(0, b)$ to a focus has length a.



Standalone

Figure 3.7.8 Animation: as P travels around the ellipse, the sum $r_1 + r_2$ of its distances to the two foci stays fixed at $2a = 6$.

Where do the formulas for a come from? Two evaluations of Definition 3.7.6 do all the work. Carry them out.

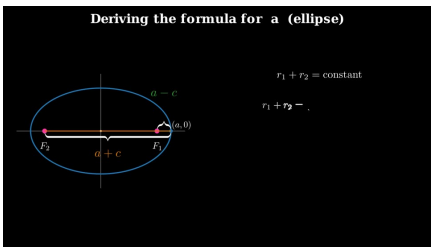
At the vertex: $r_1 + r_2 =$

At the co-vertex:

Which is the hypotenuse, and why the minus sign:

Focal Relation for the Ellipse.

$$a^2 = b^2 + c^2 \quad \Longleftrightarrow \quad c^2 = a^2 - b^2. \tag{3.7.4}$$



Standalone

Figure 3.7.9 Animation: deriving the formula for a. Sliding P to the vertex shows the constant equals $2a$; sliding it to the co-vertex builds the right triangle giving $a^2 = b^2 + c^2$.

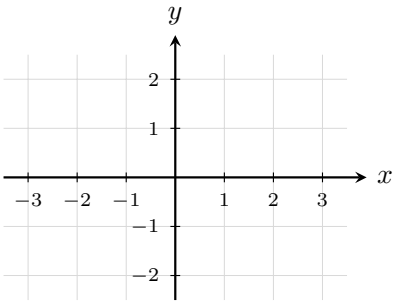
Example 3.7.10 Horizontal ellipse. Sketch the graph of $4x^2 + 9y^2 = 36$ and locate the foci.

Solution.

Standard form:

$a =$ $b =$ *major axis:*

$c =$ *foci:*



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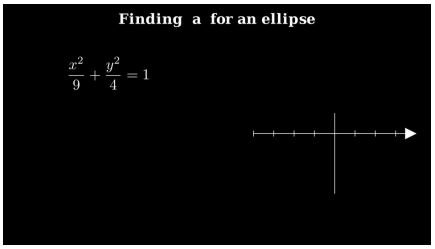


Standalone

Figure 3.7.11 Animation: in practice, put the equation in standard form and read a^2 off the *larger* denominator; then a is the semi-major axis, measured from the center to a vertex.

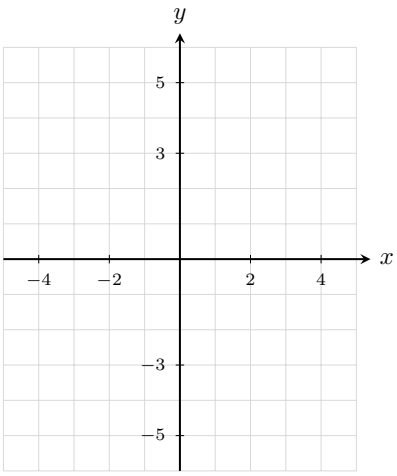
Example 3.7.12 A vertical ellipse (foci on the y-axis). Sketch the graph of $25x^2 + 16y^2 = 400$ and locate the foci.

Solution.

Standard form:

$a =$ $b =$ *major axis:*

$c =$ *foci:*



▲

3.7.3 Hyperbola

Definition 3.7.13 A **hyperbola** is the set of points in a plane the *difference* of whose distances from two fixed points F_1 and F_2 (the **foci**) is constant. ◆

The Standard Hyperbola.

A hyperbola centred at the origin with its foci on the x-axis has the standard form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1. \tag{3.7.5}$$

Its vertices are $(\pm a, 0)$, the foci satisfy $c^2 = a^2 + b^2$, and the asymptotes are $y = \pm \dfrac{b}{a}\, x$.

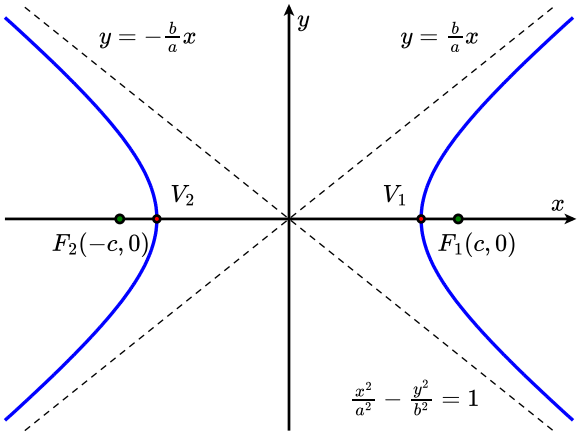
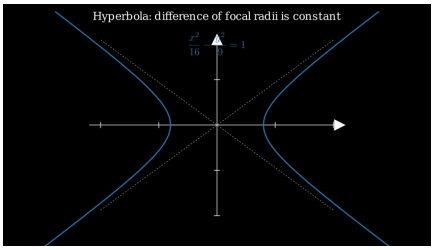


Figure 3.7.14 A hyperbola opening left and right, with vertices $(\pm a, 0)$, foci $(\pm c, 0)$, and asymptotes $y = \pm \frac{b}{a} x$.



Standalone

Figure 3.7.15 Animation: as P moves along a branch, the absolute difference $|r_1 - r_2|$ of its distances to the foci stays fixed at $2a = 8$, while the branch hugs its asymptotes.

The formulas for a arise exactly as they did for the ellipse, with sums replaced by differences. Work them out.

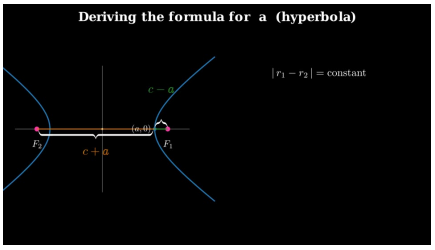
At the vertex: $|r_1 - r_2| =$

From the central box:

Contrast with the ellipse:

Focal Relation for the Hyperbola.

$$c^2 = a^2 + b^2 \qquad \Longleftrightarrow \qquad a^2 = c^2 - b^2. \tag{3.7.6}$$



Standalone

Figure 3.7.16 Animation: deriving the formula for a. Sliding P to the vertex shows the constant equals $2a$; the central box then builds the right triangle giving $c^2 = a^2 + b^2$.

The same box explains the asymptotes. Show where they come from.

Solving for y:

The limit as $|x| \to \infty$:

Why the curve never touches its asymptotes:

Slopes for a vertical hyperbola:

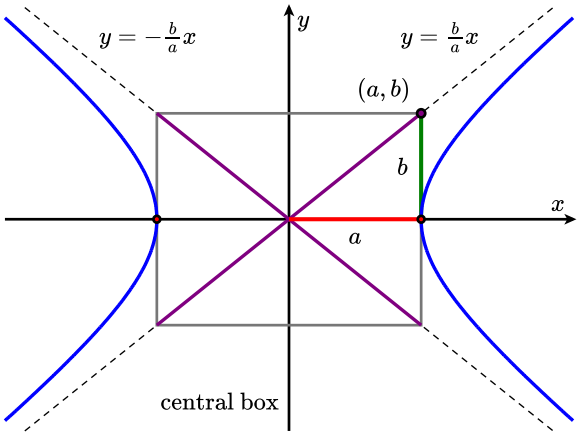


Figure 3.7.17 The central box has half-width a and half-height b. Its diagonals, extended, are exactly the asymptotes: the diagonal to the corner (a, b) has run a and rise b, hence slope $\frac{b}{a}$.

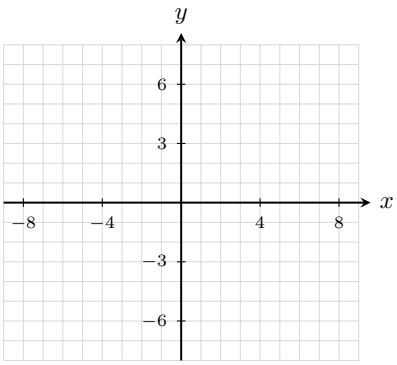
Example 3.7.18 Horizontal hyperbola. Find the foci and asymptotes of the hyperbola $9x^2 - 16y^2 = 144$ and sketch its graph.

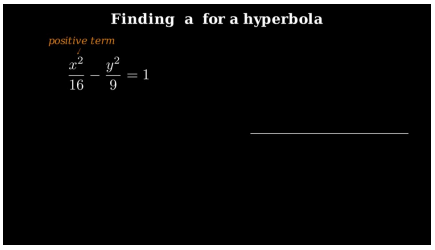
Solution.

Standard form:

$a =$ $b =$ *opens:*

$c =$ *foci:* *asymptotes:*





Standalone

Figure 3.7.19 Animation: in practice, put the equation in standard form and read a^2 off the denominator under the *positive* term; then a is measured along the transverse axis, from the center to a vertex.

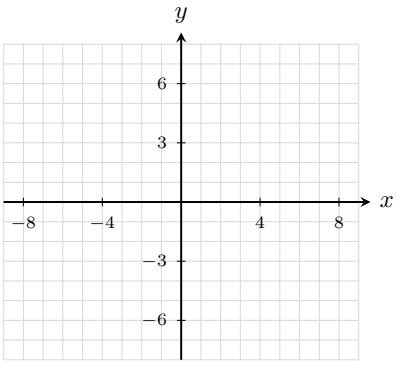
Example 3.7.20 A vertical hyperbola (opens up and down). Find the vertices, foci, and asymptotes of $16y^2 - 9x^2 = 144$ and sketch its graph.

Solution.

Standard form:

$a =$ $b =$ *vertices:*

$c =$ *foci:* *asymptotes:*



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3.7.4 ∗Application: Parabolic Reflectors

Starred section. This one is for the interested reader. It will not be examined.

Satellite dishes, radio telescopes, headlights, and solar cookers are all parabolic in cross-section, and for one reason: a parabola has a **reflection property**. Every ray travelling parallel to the axis of the parabola reflects off the curve and passes through the *focus*. Run the argument backwards and a source placed at the focus emits a perfectly parallel beam—which is why a headlight is shaped this way too.

This is not a coincidence of the shape; it follows from the equidistance definition (Definition 3.7.1). For $x^2 = 4py$ the tangent at the point $(x_0,\, x_0^2/4p)$ has slope $x_0/2p$, and a short computation shows that the incoming vertical ray and the segment from that point to $(0, p)$ make equal angles with the normal. The law of reflection then sends every such ray straight to the focus, so that is where the receiver goes.

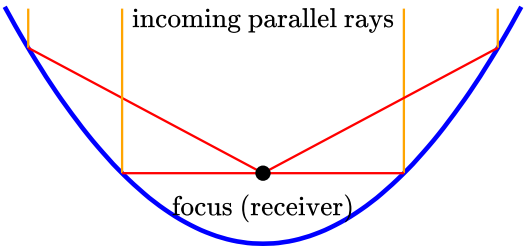
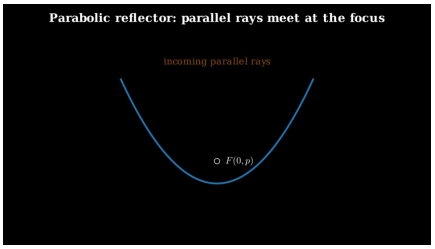


Figure 3.7.21 A parabolic dish. Rays arriving parallel to the axis (orange) reflect off the dish (red) and all converge on the focus, where the receiver sits.



Standalone

Figure 3.7.22 Animation: parallel rays strike the dish at different points and all reflect through the single focus.

Example 3.7.23 Placing the receiver. A satellite dish has a parabolic cross-section that is 4 feet wide and 1 foot deep at its center. How far from the vertex should the receiver be mounted?

Solution.

A point on the rim:

$p =$

The receiver goes:

▲

3.7.5 ∗Application: Whispering Galleries

Starred section. This one is for the interested reader. It will not be examined.

An ellipse has a reflection property of its own: a ray leaving one focus reflects off the ellipse and passes through the *other* focus. In a room whose ceiling is a half-ellipse—the dome of St. Paul's Cathedral, or Statuary Hall in the U.S. Capitol—a whisper released at one focus is gathered up by the whole ceiling and delivered to a listener standing at the other focus, while people in between hear nothing.

The constant-sum definition (Definition 3.7.6) explains why it works so well. Every path from one focus to the wall and on to the other focus has the *same* total length, no matter where it strikes. So the reflected sound does not merely arrive at the far focus: it arrives from every direction *at the same instant*, in phase, and the pieces reinforce one another instead of smearing out. That is what makes a whisper audible across the room.

Write down that common length, and say why the phrase "in phase" is the key to the effect.

Every path has length:

Why that matters:

Figure 3.7.24 The interior of St. Paul's Cathedral beneath Wren's dome. Hand-coloured aquatint by Thomas Rowlandson and Augustus Pugin, from Ackermann's *Microcosm of London* (1809). Public domain, via The Metropolitan Museum of Art.

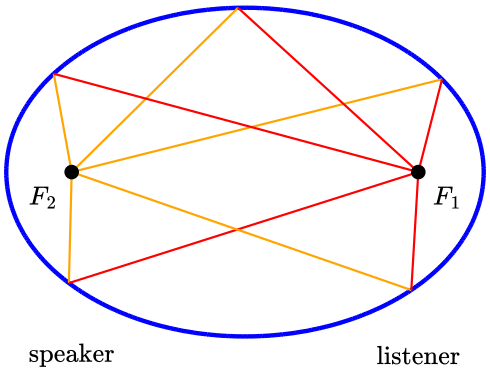
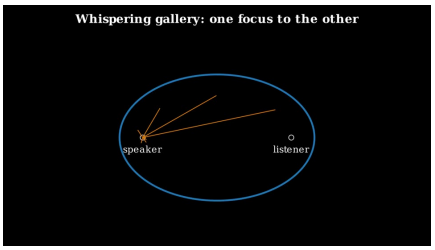


Figure 3.7.25 A whispering gallery seen from above. Five sound paths leave the speaker at one focus, bounce off the elliptical wall, and all arrive at the listener at the other focus. Each path has the same total length $2a$.



Standalone

Figure 3.7.26 Animation: sound leaves one focus in many directions, reflects off the wall, and reconverges at the other focus—every path the same length $2a$.

Example 3.7.27 Where should the listener stand? A whispering gallery is 100 feet long and 60 feet wide. Where should the speaker and the listener stand, and how far does the whisper travel on its way across?

Solution.

$a =$ $b =$ $c =$

They stand at:

The whisper travels:

▲

3.7.6 ∗Application: Hyperbolic Navigation (LORAN)

Starred section. This one is for the interested reader. It will not be examined.

Progress in electrical engineering enabled navigation systems based on the transmission of electromagnetic waves. One example is LORAN-C marine navigation, developed in the United States during World War II. A vessel receives a synchronized signal from a pair of transmitters; the signal from the more distant transmitter arrives later, so the measured delay Δt determines the *difference* between the vessel's distances to the two transmitters,

$$|r_2 - r_1| = v\,\Delta t,$$

where v is the signal speed.

Say what that single measurement tells the vessel, and what a second one adds.

One delay places the vessel on:

The sign of the delay tells it:

A second pair of stations gives:

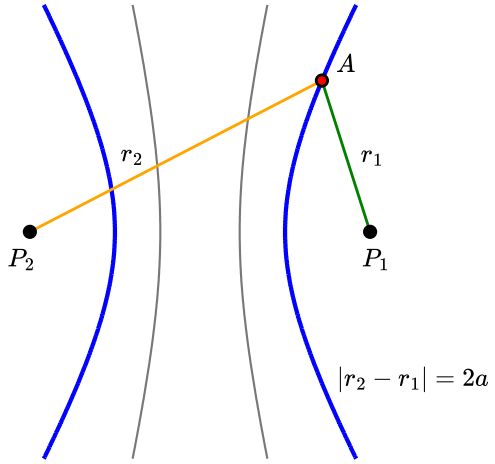
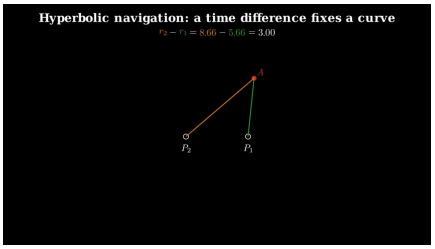


Figure 3.7.28 Hyperbolic navigation. The measured delay places the unknown position A on the solid hyperbola with foci at the receivers P_1 and P_2; a different delay would place it on the dashed one instead.



Standalone

Figure 3.7.29 Animation: the point A moves, and both signal distances r_1 and r_2 change—but their difference does not, tracing out the hyperbola.

Example 3.7.30 Locating a hiker from three receivers. Three receivers P_1, P_2, and P_3 are deployed in the landscape. The figure captures the distances we know. Emma's tourist navigation device sends a signal to all three receivers. The signal arrives at receivers P_1 and P_3 at the same time, and at receiver P_2 it arrives 80 microseconds later. Where is Emma located? Assume the signal travels 300,000 km per second, and determine the position in a suitably chosen coordinate system.

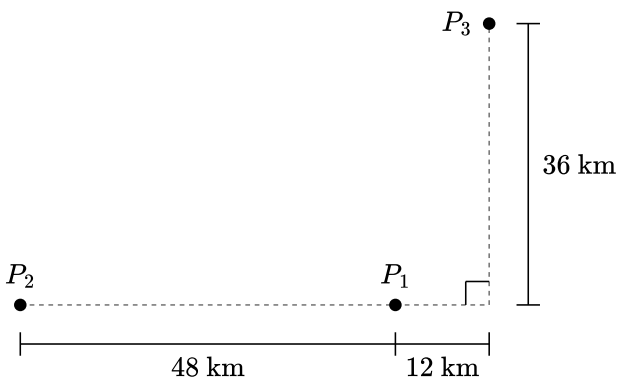


Figure 3.7.31 The exercise data: P_1 and P_2 are 48 km apart, and P_3 is 36 km directly above the point 12 km beyond P_1.

Solution.

The two timing facts, as geometry:

Coordinates, and the three receivers in them:

The line o:

The hyperbola h:

Solving:

Emma is at: *Check:*

▲

Remark 3.7.32 What if Emma were not equidistant from P_1 and P_3? The exercise was arranged so that the signal reached P_1 and P_3 simultaneously, and that is what kept the algebra easy: a distance difference of *zero* is the degenerate case of the hyperbola, and the locus collapses to a straight line, the perpendicular bisector of P_1P_3. We then only had to substitute a linear parametrization into one quadratic equation.

If the two arrival times differed, the delay would instead give a *nonzero* difference $\big|\,\|AP_3\| - \|AP_1\|\,\big| = v\,\Delta t$, and Emma would lie on a genuine second hyperbola, with foci P_1 and P_3—tilted, since its focal axis P_1P_3 is not parallel to a coordinate axis. Locating her would then mean intersecting the correct branches of *two* hyperbolas, as in the figure below. Conceptually nothing changes—each measurement contributes one curve, and the position is their common point—but algebraically we would be solving two quadratic equations together rather than a linear one and a quadratic one.

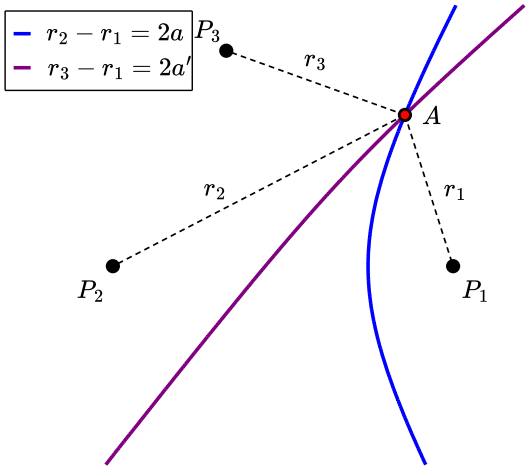


Figure 3.7.33 Two measurements fix the position. The first delay (receivers P_1, P_2) puts A on the blue branch; the second (receivers P_1, P_3) puts it on the purple branch. The position A is at the crossing, and the dashed segments show the three distances r_1, r_2, r_3.

3.8 Quadric Surfaces

Objectives

1. We know that Conic Sections are defined by second-degree equations in two variables. What occurs when we extend this idea to three variables?
2. How are conic sections and quadric surfaces related? How are they different?
3. What conditions on the coefficients of a second-degree equation in three variables will produce a quadric surface? What are the different types of quadric surfaces?

The conic sections are the curves cut out by second-degree equations in two variables. Asking the same question one dimension higher leads to the **quadric surfaces**: the surfaces in space described by a second-degree equation in three variables,

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0, \quad (3.8.1)$$

where A through J are constants.

The general equation (3.8.1) carries more freedom than we need. The cross terms xy, xz, and yz only tilt the surface away from the coordinate axes, and the linear terms Gx and Hy only slide it off the origin; neither affects the *shape* of the surface. Setting those coefficients to zero therefore costs us nothing essential, and it leaves a surface in standard position. Every example in this section comes from the resulting reduced equation

$$Ax^2 + By^2 + Cz^2 + Dz = E. \quad (3.8.2)$$

Each surface below is obtained by one choice of the five remaining coefficients. What tells the surfaces apart is which of them vanish and, above all, the signs of those that do not.

The method is the same every time: fix the value of one variable, which cuts the surface with a plane, and identify the **trace** that results. Before starting, write down what you expect a trace to be, and why.

A trace is:

It must be a conic section because:

3.8.1 Elliptical Paraboloid

We begin with the case in which x and y appear squared with coefficients of the same sign, while z appears only to the first power.[1] Equation (3.8.2) then reduces to

$$z = \frac{x^2}{4} + \frac{y^2}{9},$$

the surface known as an **elliptical paraboloid**.

Find its traces in the planes $x = 2$, $y = -3$, and $z = 1$.

Trace in the first plane:

Trace in the second plane:

Trace in the third plane:

So the surface is called:

[1] In equation (3.8.2), take $A = 1/4$, $B = 1/9$, $C = 0$, $D = -1$, and $E = 0$.

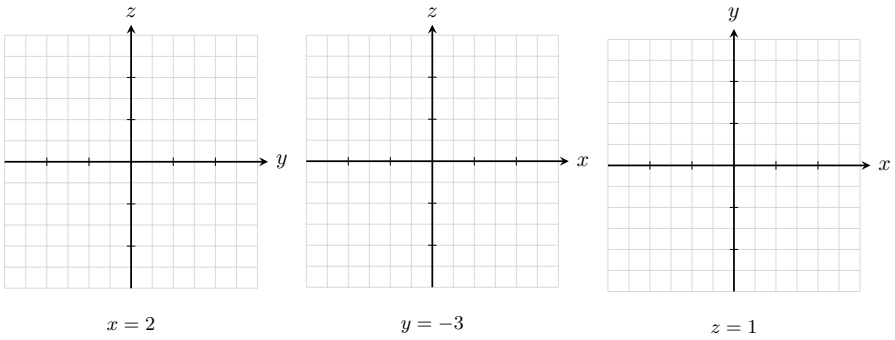
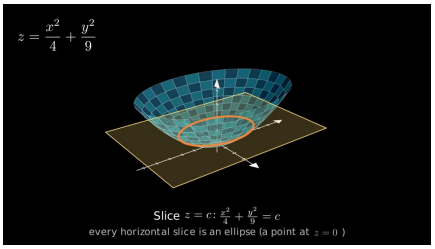


Figure 3.8.1 Grids for the traces of the elliptical paraboloid $z = \frac{x^2}{4} + \frac{y^2}{9}$ in the planes $x = 2$, $y = -3$, and $z = 1$.

An interactive figure slicing the elliptical paraboloid $z = \frac{x^2}{4} + \frac{y^2}{9}$ with the planes $x = c$, $y = c$ and $z = c$ accompanies the online version of these notes.



Standalone

Figure 3.8.2 Slicing $z = \frac{x^2}{4} + \frac{y^2}{9}$ with planes $z = c$, $x = c$, and $y = c$. Use it to check the traces you found.

3.8.2 Hyperbolic Paraboloid

Now change a single sign: keep z to the first power, but let the squared terms in x and y carry *opposite* signs.[2] Equation (3.8.2) becomes

$$z = \frac{x^2}{4} - \frac{y^2}{9},$$

a **hyperbolic paraboloid**. That one sign is enough to change the surface completely.

Take the same three planes $x = 2$, $y = -3$, and $z = 1$ as in Subsection 3.8.1, and find the traces.

[2] In equation (3.8.2), take $A = 1/4$, $B = -1/9$, $C = 0$, $D = -1$, and $E = 0$.

Trace in the first plane:

Trace in the second plane:

Trace in the third plane:

So the surface is called:

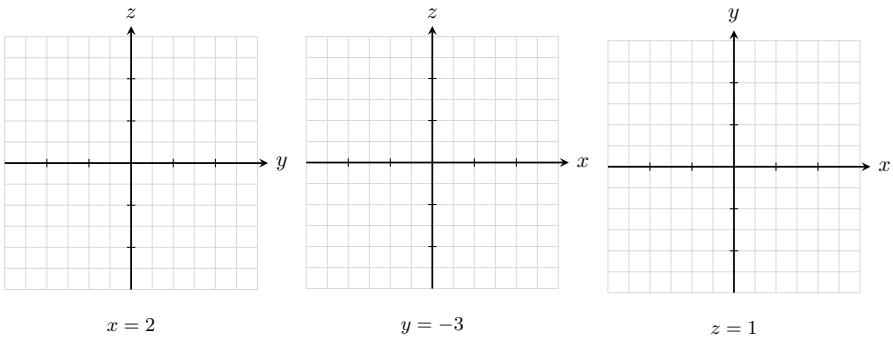
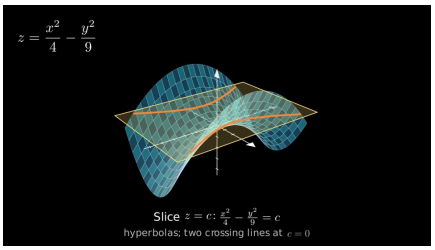


Figure 3.8.3 Grids for the traces of the hyperbolic paraboloid $z = \frac{x^2}{4} - \frac{y^2}{9}$ in the planes $x = 2$, $y = -3$, and $z = 1$.

An interactive figure slicing the hyperbolic paraboloid $z = \frac{x^2}{4} - \frac{y^2}{9}$ with the planes $x = c$, $y = c$ and $z = c$ accompanies the online version of these notes.



Standalone

Figure 3.8.4 Slicing $z = \frac{x^2}{4} - \frac{y^2}{9}$ with planes $z = c$, $x = c$, and $y = c$. Use it to check the traces you found.

3.8.3 Ellipsoid

Suppose next that all three variables appear squared with coefficients of the same sign, and that no linear term is present at all.[3] Equation (3.8.2) reduces to

$$\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{16} = 1,$$

an **ellipsoid**.

Find the traces in the three coordinate planes $x = 0$, $y = 0$, and $z = 0$.

Trace in the first plane:

Trace in the second plane:

Trace in the third plane:

So the surface is called:

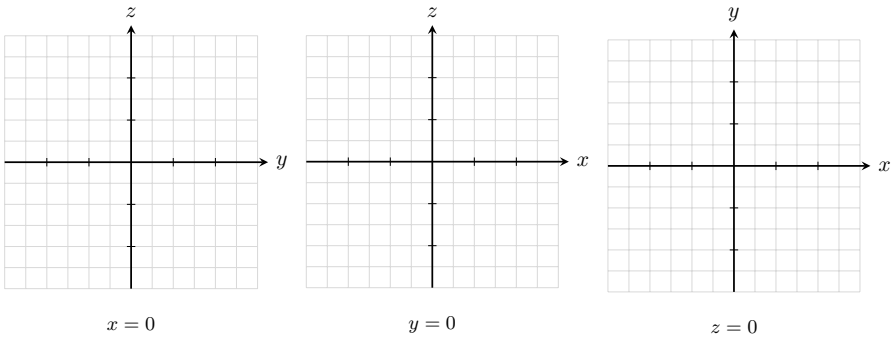
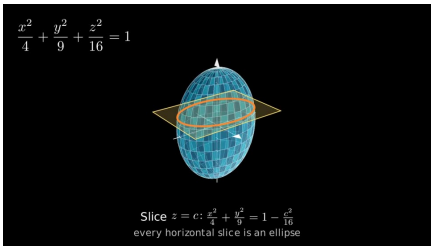


Figure 3.8.5 Grids for the traces of the ellipsoid $\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{16} = 1$ in the coordinate planes $x = 0$, $y = 0$, and $z = 0$.

An interactive figure slicing the ellipsoid with the planes $x = c$, $y = c$ and $z = c$ accompanies the online version of these notes.

[3] In equation (3.8.2), take $A = 1/4$, $B = 1/9$, $C = 1/16$, $D = 0$, and $E = 1$.



Standalone

Figure 3.8.6 Slicing the ellipsoid with planes $z = c$, $x = c$, and $y = c$. Use it to check the traces you found.

3.8.4 Elliptic Cone

Finally, keep all three squared terms, but let the coefficient of z^2 be negative and set the constant on the right-hand side to zero.[4] Equation (3.8.2) becomes

$$z^2 = \frac{x^2}{4} + \frac{y^2}{9},$$

an **elliptic cone**.

Find the traces in the planes $x = 2$, $y = -3$, and $z = 1$.

Trace in the first plane:

Trace in the second plane:

Trace in the third plane:

So the surface is called:

[4] In equation (3.8.2), take $A = 1/4$, $B = 1/9$, $C = -1$, $D = 0$, and $E = 0$.

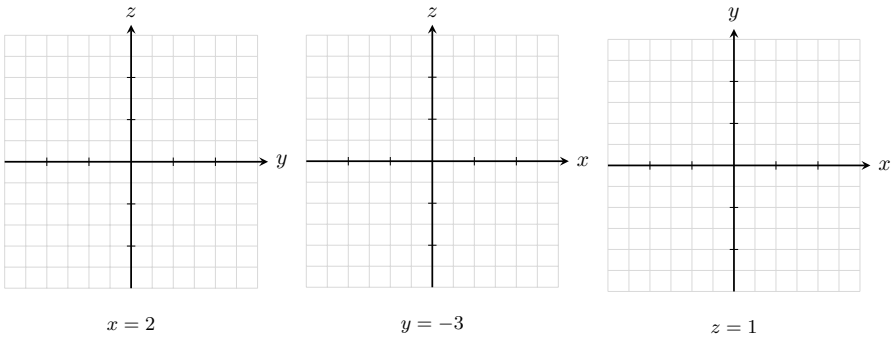
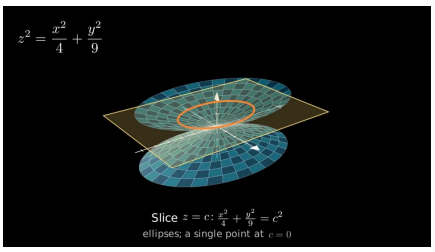


Figure 3.8.7 Grids for the traces of the elliptic cone $z^2 = \frac{x^2}{4} + \frac{y^2}{9}$ in the planes $x = 2$, $y = -3$, and $z = 1$.

An interactive figure slicing the elliptic cone with the planes $x = c$, $y = c$ and $z = c$ accompanies the online version of these notes.



Standalone

Figure 3.8.8 Slicing the elliptic cone with planes $z = c$, $x = c$, and $y = c$. Use it to check the traces you found.

3.8.5 Hyperboloids

The two remaining quadric surfaces are left for you to explore. The aim of this subsection is to practice the slicing method of the previous subsections on a new pair of surfaces and, in doing so, to see what separates a hyperboloid of one sheet from a hyperboloid of two sheets.

Activity 3.8.5.1 Slicing the Hyperboloids. Use the same slicing method as in the previous subsections to study the two surfaces

$$\frac{x^2}{4} + \frac{y^2}{9} - \frac{z^2}{16} = 1 \qquad \text{and} \qquad \frac{z^2}{16} - \frac{x^2}{4} - \frac{y^2}{9} = 1,$$

both of which come from equation (3.8.2).[5] Discover why the first is called a **hyperboloid of one sheet** while the second is called a **hyperboloid of two sheets**. Pay particular attention to the traces in the horizontal planes $z = c$: they are what tells the two surfaces apart.

(a) For the surface $\frac{x^2}{4} + \frac{y^2}{9} - \frac{z^2}{16} = 1$, find the trace in the plane $z = c$ for an arbitrary constant c, and the traces in the coordinate planes $x = 0$ and $y = 0$. For which values of c does the plane $z = c$ actually intersect the surface? Identify each trace as an ellipse, parabola, or hyperbola.

Trace in $z = c$:

Which c give a trace?

Traces in $x = 0$ and $y = 0$:

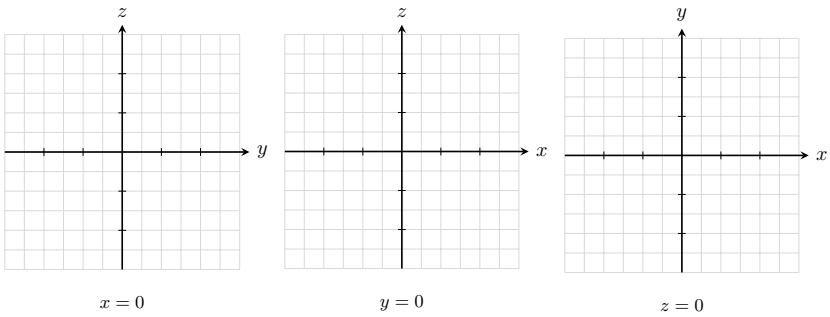
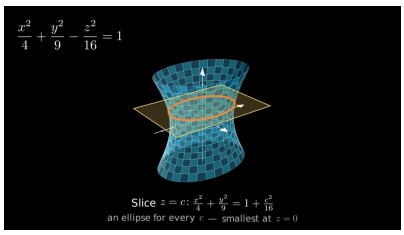


Figure 3.8.9 Grids for the traces of the hyperboloid of one sheet $\frac{x^2}{4} + \frac{y^2}{9} - \frac{z^2}{16} = 1$ in the coordinate planes $x = 0$, $y = 0$, and $z = 0$.

An interactive figure slicing the hyperboloid of one sheet with the planes $x = c$, $y = c$ and $z = c$ accompanies the online version of these notes.



Standalone

Figure 3.8.10 Slicing the hyperboloid of one sheet with planes $z = c$, $x = c$, and $y = c$. Use it to check the traces you found.

(b) Repeat the analysis for the surface $\frac{z^2}{16} - \frac{x^2}{4} - \frac{y^2}{9} = 1$. For which values of c does the plane $z = c$ intersect the surface this time? What happens for c between those values?

Trace in $z = c$:

Cases:

Traces in $x = 0$ and $y = 0$:

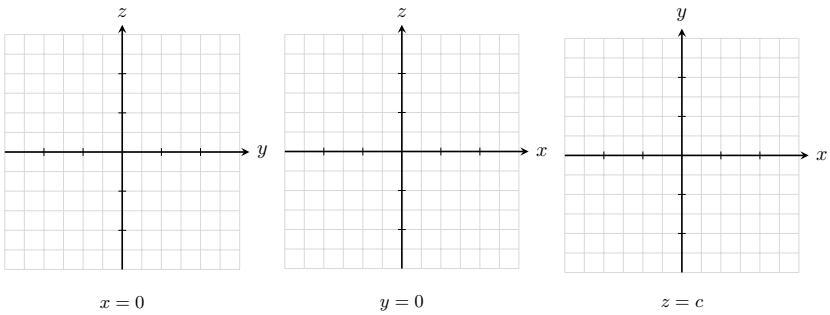
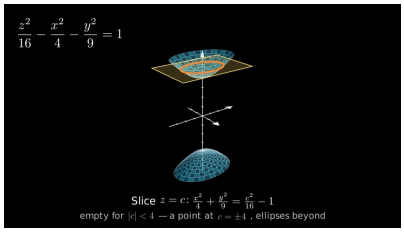


Figure 3.8.11 Grids for the traces of the hyperboloid of two sheets $\frac{z^2}{16} - \frac{x^2}{4} - \frac{y^2}{9} = 1$ in the coordinate planes $x = 0$, $y = 0$, and a plane $z = c$ that meets it.

An interactive figure slicing the hyperboloid of two sheets with the planes $x = c$, $y = c$ and $z = c$ accompanies the online version of these notes.



Standalone

Figure 3.8.12 Slicing the hyperboloid of two sheets with planes $z = c$, $x = c$, and $y = c$. Use it to check the traces you found.

(c) Using only the traces in the planes $z = c$, explain why the first surface is called a hyperboloid of *one sheet* and the second a hyperboloid of *two sheets*. Can you predict the number of sheets directly from the signs in the equation?

Why one sheet, and why two:

The rule from the signs:

The axis of the hyperboloid:

3.8.6 Summary of the Quadric Surfaces

We collect the six surfaces of this section in Figure 3.8.13. For each one the table gives its equation, a picture of the surface, and the equation and type of the cross-section cut by the planes $x = c$, $y = c$, and $z = c$; a cell also notes any value of c for which that slice is just a point or a pair of lines. Every entry is one you worked out above, so use the table to check your completed notes, and keep it afterwards as a reference. A printable copy is available as a PDF.

[5] For the first surface $A = 1/4$, $B = 1/9$, $C = -1/16$, $D = 0$, and $E = 1$; for the second surface $A = -1/4$, $B = -1/9$, $C = 1/16$, $D = 0$, and $E = 1$. For the ellipsoid all three squared terms were positive; here exactly *one* squared term is negative for the first surface and exactly *two* are negative for the second. Keep this count in mind as you work through the traces.

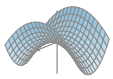
Equation	Shape	$x = c$ slices	$y = c$ slices	$z = c$ slices
elliptical paraboloid $z = \frac{x^2}{4} + \frac{y^2}{9}$		$z = \frac{c^2}{4} + \frac{y^2}{9}$ *parabola* opens up	$z = \frac{x^2}{4} + \frac{c^2}{9}$ *parabola* opens up	$\frac{x^2}{4} + \frac{y^2}{9} = c$ *ellipse* point $(0,0,0)$ if $c = 0$
hyperbolic paraboloid $z = \frac{x^2}{4} - \frac{y^2}{9}$		$z = \frac{c^2}{4} - \frac{y^2}{9}$ *parabola* opens down	$z = \frac{x^2}{4} - \frac{c^2}{9}$ *parabola* opens up	$\frac{x^2}{4} - \frac{y^2}{9} = c$ *hyperbola* opens left–right; two lines if $c = 0$
ellipsoid $\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{16} = 1$		$\frac{y^2}{9} + \frac{z^2}{16} = 1 - \frac{c^2}{4}$ *ellipse* point if $c = \pm 2$	$\frac{x^2}{4} + \frac{z^2}{16} = 1 - \frac{c^2}{9}$ *ellipse* point if $c = \pm 3$	$\frac{x^2}{4} + \frac{y^2}{9} = 1 - \frac{c^2}{16}$ *ellipse* point if $c = \pm 4$
elliptic cone $z^2 = \frac{x^2}{4} + \frac{y^2}{9}$		$z^2 - \frac{y^2}{9} = \frac{c^2}{4}$ *hyperbola* opens up–down; two lines if $c = 0$	$z^2 - \frac{x^2}{4} = \frac{c^2}{9}$ *hyperbola* opens up–down; two lines if $c = 0$	$\frac{x^2}{4} + \frac{y^2}{9} = c^2$ *ellipse* point $(0,0,0)$ if $c = 0$
hyperboloid of one sheet $\frac{x^2}{4} + \frac{y^2}{9} - \frac{z^2}{16} = 1$		$\frac{y^2}{9} - \frac{z^2}{16} = 1 - \frac{c^2}{4}$ *hyperbola* opens left–right; two lines if $c = \pm 2$	$\frac{x^2}{4} - \frac{z^2}{16} = 1 - \frac{c^2}{9}$ *hyperbola* opens left–right; two lines if $c = \pm 3$	$\frac{x^2}{4} + \frac{y^2}{9} = 1 + \frac{c^2}{16}$ *ellipse*
hyperboloid of two sheets $\frac{z^2}{16} - \frac{x^2}{4} - \frac{y^2}{9} = 1$		$\frac{z^2}{16} - \frac{y^2}{9} = 1 + \frac{c^2}{4}$ *hyperbola* opens up–down	$\frac{z^2}{16} - \frac{x^2}{4} = 1 + \frac{c^2}{9}$ *hyperbola* opens up–down	$\frac{x^2}{4} + \frac{y^2}{9} = \frac{c^2}{16} - 1$ *ellipse* point if $c = \pm 4$; empty if $\lvert c \rvert < 4$

Figure 3.8.13 The six parent quadric surfaces, with the equation and type of the slice cut by each family of coordinate planes $x = c$, $y = c$, and $z = c$.

3.8.7 Quadric Surfaces in the World

The surfaces of this section are more than abstract shapes. Each one appears in the physical world, and often it appears *because* of the very geometric properties we have been discovering by slicing.

Elliptical paraboloid. A paraboloid has a striking reflective property: every ray travelling parallel to its axis bounces off the surface and passes through a single point, the focus. This is why the dishes of radio telescopes and satellite receivers, and the reflectors behind car headlights and solar concentrators, are shaped like paraboloids — they gather parallel incoming rays to one point, or, run in reverse, send rays out in a single parallel beam. The benefit is sensitivity: because parallel rays converge on the focus no matter where they strike the surface, enlarging the dish simply gathers more of a faint signal onto the same receiver, letting a radio telescope detect fainter and more distant objects. The

National Radio Astronomy Observatory explains how a parabolic dish bounces incoming radio waves to a focus.

Figure 3.8.14 The dishes of the Very Large Array are elliptical paraboloids, focusing incoming radio waves onto a single receiver. (Photo: Wikimedia Commons, CC BY 3.0.)

Hyperbolic paraboloid. The saddle is **doubly ruled**: through every point pass two straight lines that lie entirely on the surface, just like the flat traces — the two crossing lines in the plane $z = 0$ — that appear when the surface is sliced with the plane $z = 0$. A doubly curved roof can therefore be built out of straight beams, or poured over straight formwork, which makes the hyperbolic paraboloid both strong and inexpensive to build. It is the shape of many "saddle" roofs and of Félix Candela's thin concrete shells — and, on a smaller scale, of a Pringles potato chip. The benefit is efficiency: a thin shell of this shape carries its load by compression and tension acting within the surface rather than by bending, so it can roof a very wide span with only a few inches of material. Encyclopaedia Britannica's article on shell structures describes how such curved roofs are engineered from straight lines.

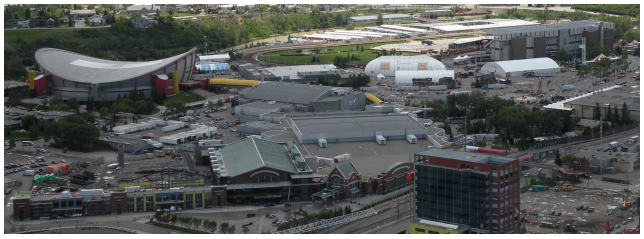


Figure 3.8.15 The roof of the Scotiabank Saddledome in Calgary is a hyperbolic paraboloid. (Photo: Wikimedia Commons, CC BY-SA 3.0.)

Ellipsoid. An ellipsoid has two focal points, and a wave leaving one focus reflects off the surface straight toward the other. In a "whispering gallery" a whisper at one focus is heard clearly across the room at the other; in medical **lithotripsy** a shock wave generated at one focus is focused onto a kidney stone placed at the other, breaking it apart without surgery. On the largest scale, the Earth itself is modeled as a slightly flattened ellipsoid — the reference ellipsoid against which GPS coordinates are measured. The benefit of this two-focus focusing is precision without contact: a lithotripter can concentrate its energy on a stone deep inside the body while sparing the surrounding tissue. Modeling the Earth as an ellipsoid brings a different benefit — a single smooth equation

captures its slight flattening, giving GPS a far more accurate reference than a sphere would. NOAA's National Geodetic Survey describes the reference ellipsoids that anchor those coordinates.

Figure 3.8.16 The Earth is modeled as an oblate ellipsoid, the reference surface behind GPS coordinates. (Image: NASA.)

Elliptic cone. When an aircraft flies faster than sound, the pressure waves it creates pile up into a cone that trails behind it, the Mach cone. The **sonic boom** you hear is the instant this cone of compressed air sweeps past you. This cone is not a design choice but an unavoidable consequence of the motion; even so, its geometry is informative: the sine of its half-angle equals $1/M$, where M is the Mach number, so a narrower cone means a faster aircraft. NASA's Glenn Research Center describes how supersonic disturbances stay confined within this cone.

Figure 3.8.17 A fighter jet at transonic speed, wrapped in a cone-shaped condensation cloud that traces out the Mach cone. (Photo: U.S. Navy.)

Hyperboloid of one sheet. Like the saddle, this surface is **ruled** — recall the two straight-line traces we found in the planes $x = \pm 2$ in Activity 3.8.5.1. Because straight members can be arranged to sweep out its double curvature, it can be built cheaply from straight beams or straight formwork while remaining very stiff. This is exactly why the cooling towers of power plants, and open lattice towers such as Vladimir Shukhov's, are hyperboloids

of one sheet. The benefit is that stiffness and wind resistance come almost for free: because the surface is ruled, the whole tower is assembled from straight, easily fabricated members, so it reaches great height with remarkably little material. The University of Houston's Engines of Our Ingenuity tells the story of Shukhov and these ruled towers.

Figure 3.8.18 A power-plant cooling tower is a hyperboloid of one sheet, assembled from straight structural members. (Photo: Wikimedia Commons, CC BY 3.0.)

Hyperboloid of two sheets. Like the ellipse, a hyperbola has two foci, and a ray aimed at one focus reflects toward the other. Reflecting telescopes of the Cassegrain and Ritchey–Chrétien type — including the Hubble Space Telescope — use mirrors ground to hyperboloidal shapes to fold a long optical path into a short tube. The same two-focus idea drives **hyperbolic navigation**: comparing the arrival times of two signals places a receiver on a hyperboloid, and intersecting several such surfaces is how systems from LORAN to GPS pin down a location. The benefit is sharpness in a compact instrument: the paired hyperboloidal mirrors fold a long focal length into a short tube and cancel the aberrations that would blur a simpler design, keeping Hubble's images crisp across its whole field of view. NASA explains how these curved mirrors gather starlight in its overview of the Hubble Space Telescope's optics.

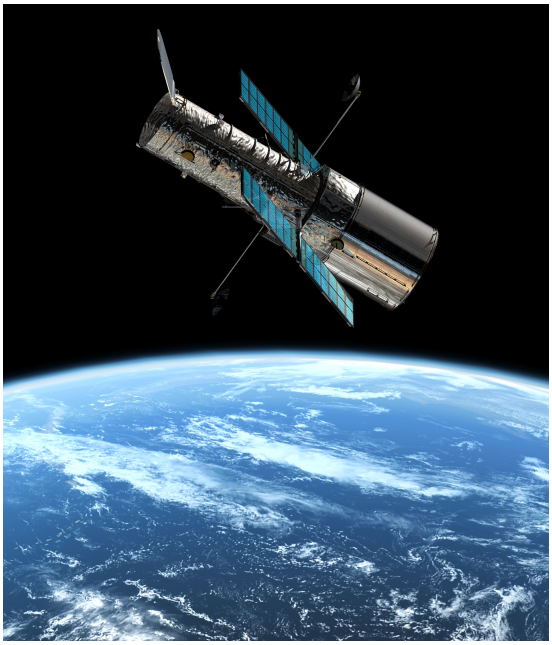


Figure 3.8.19 The Hubble Space Telescope focuses light with hyperboloidal mirrors. (Image: NASA.)

Chapter 4

Functions of Multiple Variables

4.1 Functions of Several Variables

Objectives

After completing this section, you should be able to:

- Evaluate a function $z = f(x, y)$ of two variables, describe its domain as a region of the plane, and recognize its graph as a surface in space.
- Find and sketch the level curves $f(x, y) = c$ of a function of two variables, and read a contour map.
- Explain the relationship between the level curves drawn on the surface and the contour map drawn in the xy-plane.
- Describe the level surfaces $f(x, y, z) = c$ of a function of three variables.

So far we have studied functions that take one real number as input and return one real number as output. Many quantities, however, depend on more than one input: the temperature at a point of a metal plate depends on two coordinates, and the temperature in a room depends on three. In this section we study functions of two variables, whose graphs are surfaces in space, and we develop two ways of visualizing them, namely level curves and contour maps. At the end we look at functions of three variables, which we visualize by their level surfaces.

4.1.1 Functions of Two Variables

As an example consider $z = f(x, y) = 1 - x^2 - y^2$. This function takes a pair of real numbers $x, y \in \mathbb{R}$ as its input and as an output gives a real number $z \in \mathbb{R}$. Evaluate it at the point $P_0(0.5, 0.5)$, and describe its domain. (Substituting is one line. For the domain, ask which pairs (x, y) the formula accepts, and then which of those keep z from going negative — the disk drawn in Figure 4.1.1 is the second of these.)

$f(0.5, 0.5) =$

Domain:

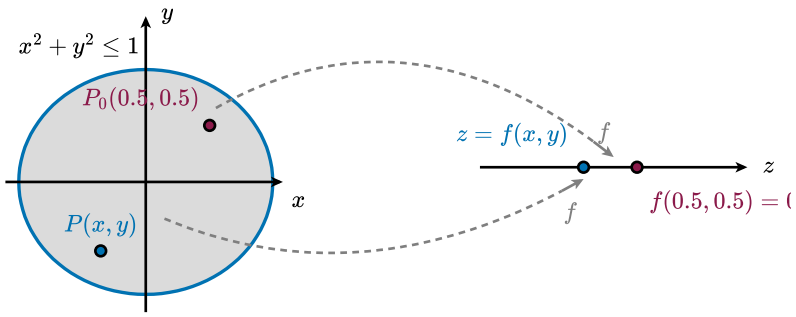


Figure 4.1.1 The function f takes a point (x, y) of the disk $x^2 + y^2 \leq 1$ and returns the single number $z = f(x, y)$. The point $P_0(0.5, 0.5)$ is sent to $f(0.5, 0.5) = 0.5$.

If we repeat this process for all the points inside the disk, we have a set of points in 3D of the form $P(x, y, f(x, y))$, which represent a surface known as a **paraboloid**. The surface is the **graph** of the function, and it is drawn in Figure 4.1.2. Notice that the point $P_0(0.5, 0.5)$ of the domain sits in the xy-plane, while the corresponding point of the graph lies directly above it.

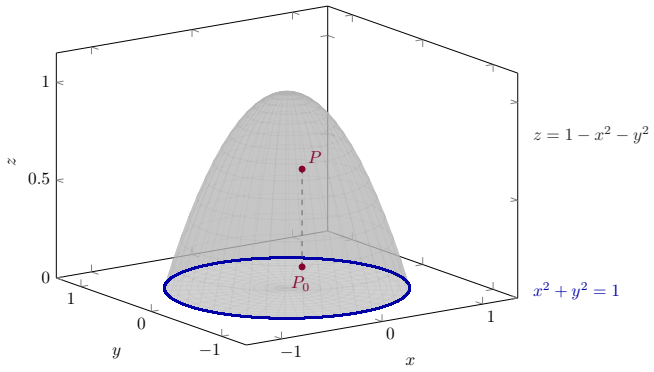


Figure 4.1.2 The graph of $z = f(x, y) = 1 - x^2 - y^2$ over the disk $x^2 + y^2 \leq 1$. The point $P_0(0.5, 0.5, 0)$ of the domain lies in the xy-plane and the corresponding point $P(0.5, 0.5, 0.5)$ lies on the surface directly above it.

Read the coordinates of that point of the graph off Figure 4.1.2, and say in one line how the graph of a function of two variables differs from the graph of a function of one variable. (The third coordinate is the number you computed

above. For the comparison, count dimensions: what does the input live in, and what does the graph live in?)

P =

A graph of one variable is a …, of two variables a …

4.1.2 Level Curves and Contour Maps

Functions of two variables can be visualized by means of **level curves**. These are curves that connect points where the function $z = f(x, y)$ takes on a given value c. Take again $z = f(x, y) = 1 - x^2 - y^2$ and find its level curves. (Set $f(x, y) = c$ and rearrange until only $x^2 + y^2$ is on the left; the shape should then be recognizable at a glance. Say what curve you get and what its radius is, and note for which values of c there is no curve at all.)

Setting $f(x, y) = c$:

The curve is a … of radius …

No curve when:

Level Curves of $f(x, y) = 1 - x^2 - y^2$.

$$1 - x^2 - y^2 = c \quad \Longleftrightarrow \quad x^2 + y^2 = 1 - c. \tag{4.1.1}$$

Fill in the radius for each of the equally spaced values $c = \frac{1}{2}$, 0, $-\frac{1}{2}$, -1, $-\frac{3}{2}$ below, then look at Figure 4.1.3, where these curves are drawn on the surface itself as intersections with the horizontal planes $z = c$. (Notice which way round it goes: larger c gives the *smaller* circle. Say why, from the shape of the surface.)

$c = \frac{1}{2}$: $c = 0$: $c = -\frac{1}{2}$:

$c = -1$: $c = -\frac{3}{2}$:

Why larger c gives a smaller circle:

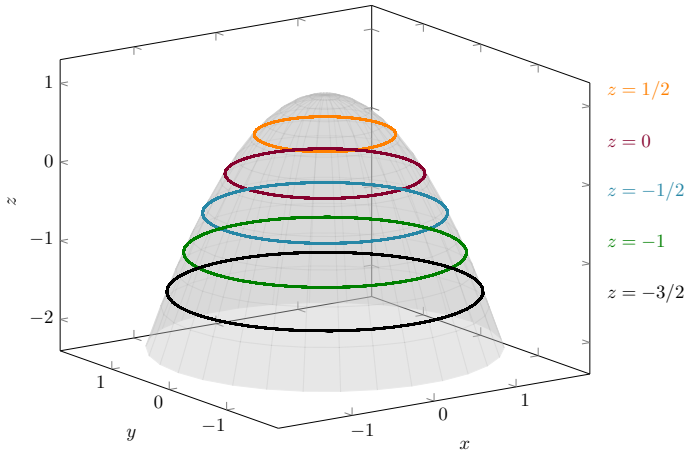
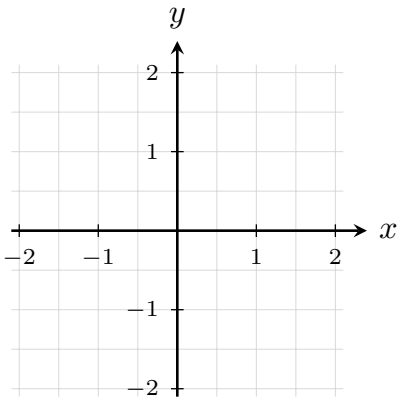


Figure 4.1.3 Level curves of $f(x, y) = 1 - x^2 - y^2$ drawn on the surface at the equally spaced heights $c = \frac{1}{2}, 0, -\frac{1}{2}, -1, -\frac{3}{2}$. The curve at height c is the intersection of the graph with the horizontal plane $z = c$; higher values of c give smaller circles.

Projecting those level curves down into the xy-plane gives the **contour map** (or **contour plot**) of the function. Draw it on the grid below, labelling each circle with its value of c. (Use the same five values, so the contour interval is constant at $\Delta c = \frac{1}{2}$ — that is what makes the spacing of the circles meaningful. After drawing it, say what close-together contours mean and what far-apart ones mean.)



Contours close together:

Contours far apart:

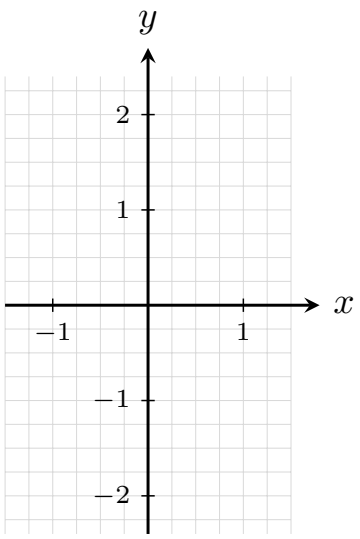
Example 4.1.4 Sketching Level Curves. Sketch some level curves of the function $g(x, y) = x^2 + \dfrac{y^2}{4}$.

Solution. (Set $g(x, y) = c$. For $c > 0$ the result is a conic you met in Section 3.7; name it, and give its two semi-axes in terms of c. Then sketch the curves for several values of c on the grid, and say which direction they are stretched in and by how much. What happens when $c = 0$, and when $c < 0$?)

Setting $g(x, y) = c$:

Shape, and its semi-axes:

$c = 0$: $c < 0$:



The surface $z = g(x, y)$ itself is shown in Figure 4.1.5; it is the **elliptic paraboloid** of Subsection 3.8.1. Use it to check the curves you drew.

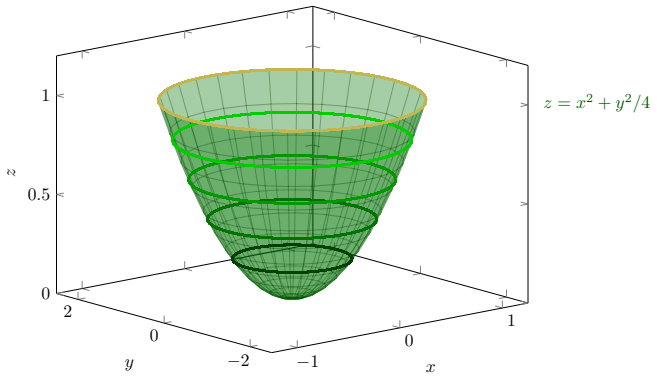


Figure 4.1.5 The elliptic paraboloid $z = g(x, y) = x^2 + \frac{y^2}{4}$. Slicing it with a horizontal plane $z = c$ produces the ellipses you drew above; the rings drawn here are at the equally spaced heights $c = 0.2, 0.4, \ldots, 1$.

▲

4.1.3 More Than Two Variables: Level Surfaces

It is not possible to draw the graph of a function of more than two variables. The graph of a function $f(x, y, z)$ would consist of the set of points $(x, y, z, f(x, y, z))$ in four-dimensional space $\mathbb{R}^4$. However, it is possible to draw the **level surfaces** of a function of three variables $f(x, y, z)$. These are the

surfaces with equation

$$f(x, y, z) = c. \tag{4.1.2}$$

Identify the level surfaces of $f(x, y, z) = x^2 + y^2 + z^2$, then check your answer against Figure 4.1.6. (Write out (4.1.2) for this f and compare it with (3.1.2). Give the radius in terms of c, and say what happens for $c < 0$ and for $c = 0$.)

The level surfaces are … of radius …

$c = 0$: $c < 0$:

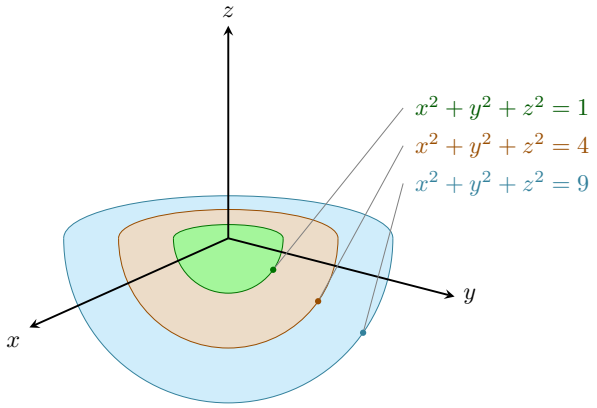


Figure 4.1.6 The level surfaces of $f(x, y, z) = x^2 + y^2 + z^2$ are spheres. Shown here are $x^2 + y^2 + z^2 = 1$, $x^2 + y^2 + z^2 = 4$, and $x^2 + y^2 + z^2 = 9$, cut away so the inner ones are visible.

For functions of four or more variables we can no longer visualize the graph or the level surfaces, and we must rely on intuition developed through the study of functions of two and three variables.

Example 4.1.7 Level Surfaces of a Function of Three Variables. Describe the level surfaces of $g(x, y, z) = x^2 + y^2 - z^2$.

Solution. (Three cases, decided by the sign of c. Take $c = 0$ first and solve for z. For $c \neq 0$ the surfaces are the hyperboloids of Subsection 3.8.5; decide the number of sheets not by drawing but by asking, for each c, which values of z are possible — rearrange so that one side is a square and see whether it can vanish.)

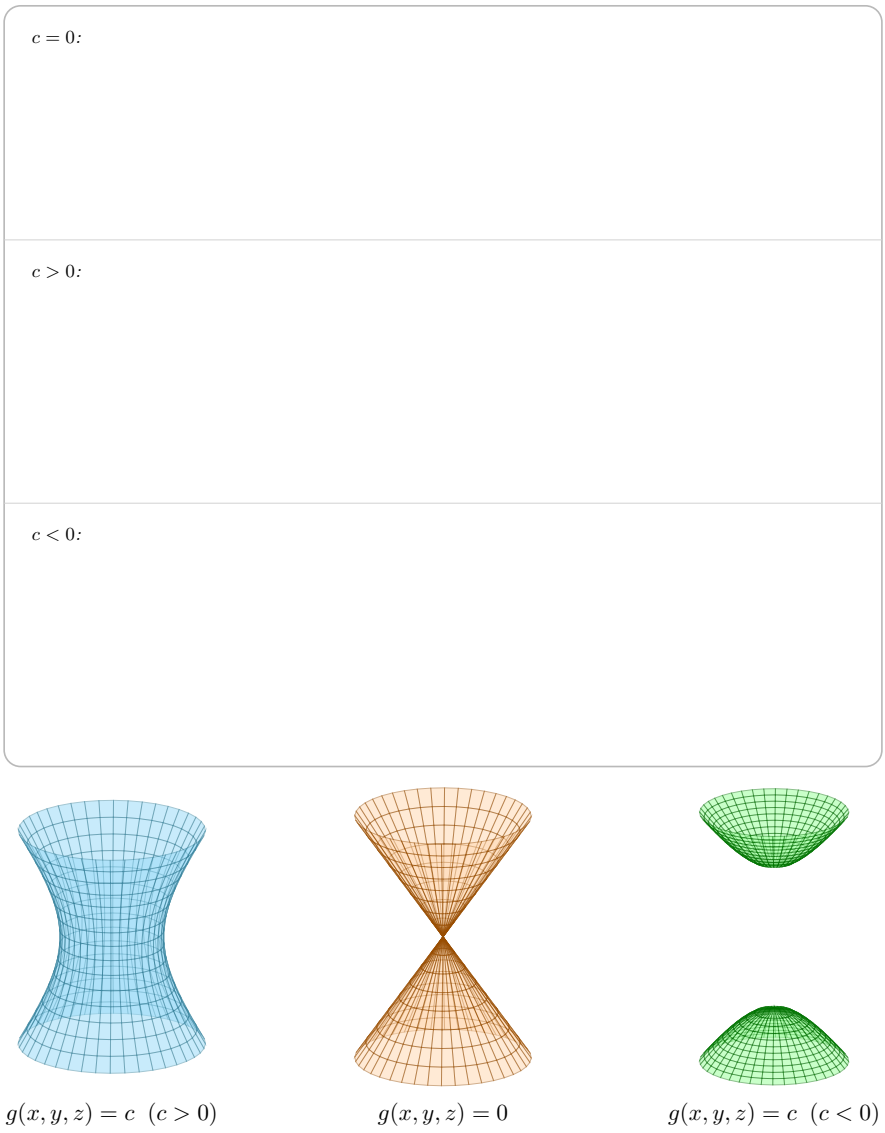


Figure 4.1.8 Level surfaces of $g(x, y, z) = x^2 + y^2 - z^2$. For $c > 0$ the surface is a hyperboloid of one sheet, for $c = 0$ it is a cone, and for $c < 0$ it is a hyperboloid of two sheets.

An animation of the transition — the hyperboloid of two sheets closing up into the cone and opening out into the hyperboloid of one sheet as c passes through zero — accompanies the online version of these notes. ▲

4.2 Limits and Continuity in Higher Dimensions

Objectives

After completing this section, you should be able to:

- Explain what it means for $\lim_{(x,y)\to(a,b)} f(x, y)$ to exist, and why the value must be the same along every path of approach.
- Evaluate limits of functions of two variables using direct substitution, algebraic simplification, and the limit laws.

- Show that a limit does not exist by finding two paths along which the function approaches different values.
- Decide whether a function of two variables is continuous at a point, and identify the points at which a piecewise-defined function fails to be continuous.

4.2.1 Limits of Two-Variable Functions

For $y = f(x)$, we could ask what value the function approaches when x approaches $a \in \mathbb{R}$, and there were two possible directions (from the left and from the right) to approach $x = a$, which we denoted by $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$. If $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L \in \mathbb{R}$, then $\lim_{x\to a} f(x) = L$.

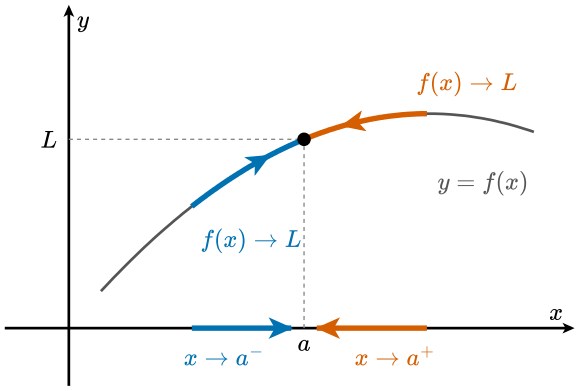


Figure 4.2.1 For $y = f(x)$ there are only two ways to approach $x = a$: from the left and from the right. When both one-sided limits agree, their common value L is $\lim_{x\to a} f(x)$.

In two variables the situation is richer, because there are infinitely many directions, and indeed infinitely many curved paths, along which a point can approach (a, b). This section makes that precise and develops the tools for computing such limits, and for showing that a limit fails to exist.

Figure 4.2.2 shows a few of the infinitely many paths along which a point can approach (a, b). With that picture in front of you, write down what it takes for the limit to exist, and what it takes to show that it does not.

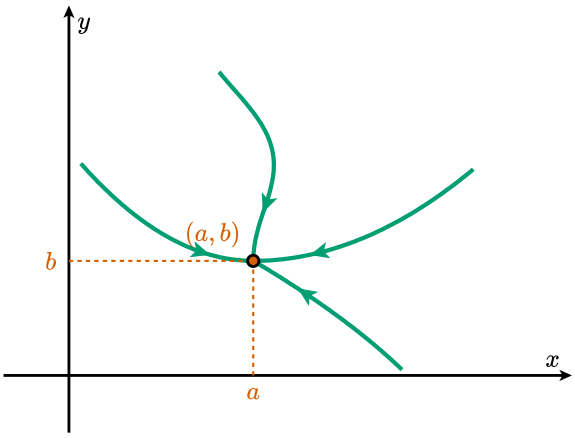


Figure 4.2.2 A few of the infinitely many paths along which a point can approach (a, b). The paths may be straight lines or curves, and they may arrive from any direction.

(One of the two is a statement about *every* path and the other about just *two* of them; that asymmetry is the whole working method of this section, so make it explicit. Then say why the one-variable test, checking the limit from the left and from the right, is not enough here.)

The limit exists if:

To show it does not exist, it is enough to:

Why left and right are not enough:

Theorem 4.2.3 Properties of Limits. *The following rules hold if L, M, and k are real numbers and*

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L \qquad \text{and} \qquad \lim_{(x,y)\to(x_0,y_0)} g(x,y) = M.$$

1. ***Sum Rule:*** $\lim\limits_{(x,y)\to(x_0,y_0)} \big(f(x,y) + g(x,y)\big) = L + M$
2. ***Difference Rule:*** $\lim\limits_{(x,y)\to(x_0,y_0)} \big(f(x,y) - g(x,y)\big) = L - M$
3. ***Product Rule:*** $\lim\limits_{(x,y)\to(x_0,y_0)} \big(f(x,y) \cdot g(x,y)\big) = L \cdot M$
4. ***Constant Multiple Rule:*** $\lim\limits_{(x,y)\to(x_0,y_0)} kf(x,y) = kL$ *for any number* k

5. Quotient Rule: $\lim_{(x,y)\to(x_0,y_0)} \dfrac{f(x,y)}{g(x,y)} = \dfrac{L}{M}$, *provided* $M \neq 0$

6. Power Rule: *If* r *and* s *are integers with no common factors and* $s \neq 0$, *then* $\lim_{(x,y)\to(x_0,y_0)} \big(f(x,y)\big)^{r/s} = L^{r/s}$, *provided* $L^{r/s}$ *is a real number. (If* s *is even, we assume that* $L > 0$.*)*

Example 4.2.4 Evaluating a Limit by Direct Substitution. Evaluate

$$\lim_{(x,y)\to(2,2\pi)} \frac{x}{y} + \cos(xy).$$

Solution. (Nothing here is indeterminate, so take the two pieces separately by direct substitution and then reassemble them. Name the rule from Theorem 4.2.3 that lets you split the limit in the first place.)

The two pieces:

Recombining (rule used:):

▲

Example 4.2.5 A Limit That Does Not Exist. Consider the function $f(x,y) = \dfrac{xy}{x^2+y^2}$. Does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist?

Solution. (Substituting gives $0/0$, so approach along the three paths drawn in Figure 4.2.6 instead. Restrict f to each path first — along $y = 0$ it becomes a function of x alone — and notice that each restriction is *constant*, so no limit needs taking. Then apply the test you wrote down above.)

Along $y = 0$: $f(x, 0) =$

Along $x = 0$: $f(0, y) =$

Along $y = x$: $f(x, x) =$

Conclusion:

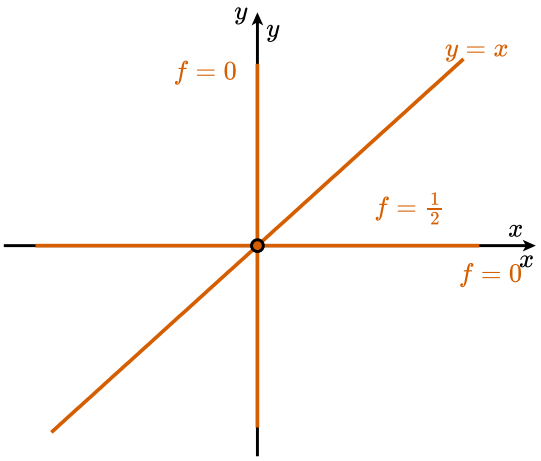


Figure 4.2.6 Three paths of approach to the origin for $f(x, y) = \frac{xy}{x^2+y^2}$. The function is constantly 0 along the two axes and constantly $\frac{1}{2}$ along the line $y = x$, so no single value is approached and the limit does not exist.

An interactive figure lifting those same three paths onto the surface itself — each path drawn in at its own height, so the two different values are visible at once — accompanies the online version of these notes.

What the lifted paths show:

▲

Example 4.2.7 Evaluating a Limit by Simplifying. Evaluate

$$\lim_{(x,y)\to(1,1)} \frac{\frac{x^2}{2} - xy + \frac{y^2}{2}}{x^2 - y^2}.$$

Solution. (Direct substitution gives $0/0$ again, but this time the expression factors: pull $\frac{1}{2}$ out of the numerator to reveal a perfect square, and factor the denominator as a difference of squares. One factor then cancels — say why that is legitimate as $(x, y) \to (1, 1)$ — and what remains submits to direct substitution.)

Factoring:

Why the cancellation is legitimate:

The limit:

▲

4.2.2 Continuity

Definition 4.2.8 Continuity at a Point and on the Domain. A function $f(x, y)$ is **continuous at the point** (x_0, y_0) if

1. f is defined at (x_0, y_0),
2. $\lim_{(x,y)\to(x_0,y_0)} f(x, y)$ exists,
3. $\lim_{(x,y)\to(x_0,y_0)} f(x, y) = f(x_0, y_0)$.

A function is **continuous** if it is continuous at every point of its domain.

◆

Example 4.2.9 Continuity Everywhere Except at the Origin. Show that the function

$$f(x,y) = \begin{cases} \dfrac{2xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

is continuous everywhere except at $(x,y) = (0,0)$.

Solution. (Two separate arguments. Away from the origin, say what kind of function this is and which rule of Theorem 4.2.3 then settles continuity in one step. At the origin, the work is Example 4.2.5 again with a factor of 2; give the two paths and the value along each, and name the condition of Definition 4.2.8 that fails.)

Away from the origin:

At the origin, two paths:

Condition that fails:

▲

Example 4.2.10 A Discontinuity Detected by a Curved Path. Show that the function

$$f(x,y) = \begin{cases} \dfrac{xy^2}{x^2+y^4} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

is discontinuous at $(x,y) = (0,0)$.

Solution. (Start with the whole family of straight lines $y = mx$ at once, keeping m as a letter; simplify $f(x, mx)$ and take $x \to 0$. You should get the same answer for every m — which is exactly why this example is here, so say what it does *not* prove. Then look at the degrees in the denominator to see which curved path would make the two terms balance, and try it.)

Along $y = mx$: $f(x, mx) =$

What this does not prove:

A curved path that balances the denominator:

Conclusion:

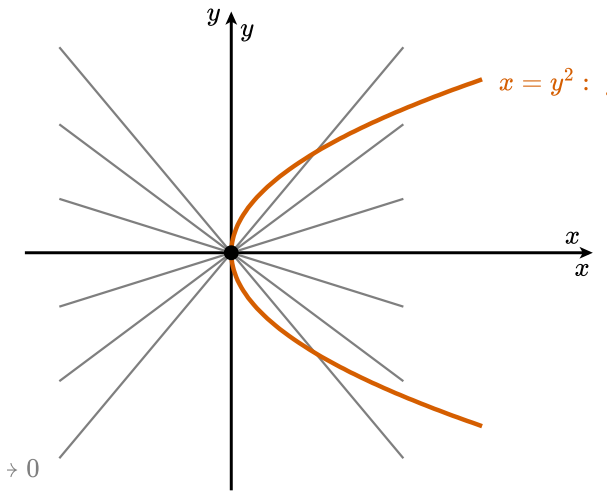


Figure 4.2.11 Every straight line through the origin gives the limit 0, but along the parabola $x = y^2$ the function is constantly $\frac{1}{2}$. A limit must agree along *all* paths, not merely all straight ones.

There is a second route to the same conclusion. Give it. (Once you know the value along the curved path, you do not need a second path at all: compare that value with $f(0,0)$ and use condition ***3*** of Definition 4.2.8 directly.)

workspace

▲

4.3 Partial Derivatives

Objectives

After completing this section, you should be able to:

- State the definition of the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ as limits, and interpret each as the slope of a tangent line to a curve cut from the surface by a vertical plane.
- Compute partial derivatives by treating all other variables as constants, and use the standard notations f_x, $\frac{\partial f}{\partial x}$, and $\frac{\partial f}{\partial x}(x_0, y_0)$.
- Compute second partial derivatives, including the mixed partials f_{xy} and f_{yx}.
- Apply the mixed derivative theorem to choose the more convenient order of differentiation.

A function of one variable has one derivative, which measures the rate at which the output changes as the single input changes. A function of two variables has two inputs, and so it has two rates of change: one obtained by moving in the x-direction and one by moving in the y-direction. These are the partial derivatives, and each is an ordinary one-variable derivative in disguise, taken while the other variable is held fixed.

4.3.1 Definition of Partial Derivatives

Consider the point $P(x_0, y_0, f(x_0, y_0))$ on the surface $z = f(x, y)$, and the two curves through it cut by the vertical planes $y = y_0$ and $x = x_0$. Each is drawn on its own below, with its tangent line at P: Figure 4.3.1 for the first and Figure 4.3.2 for the second.

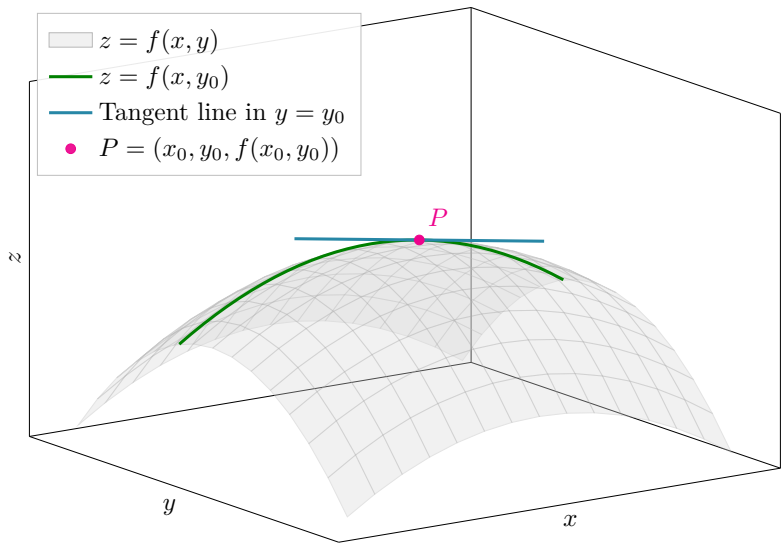


Figure 4.3.1 The curve $z = f(x, y_0)$ cut from the surface by the vertical plane $y = y_0$, and its tangent line at P. The slope of that tangent line is $\left.\frac{\partial f}{\partial x}\right|_{(x_0, y_0)}$.

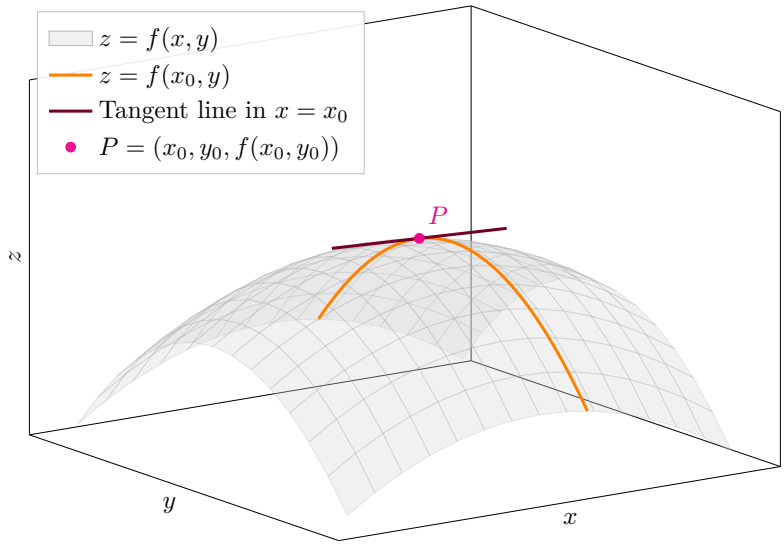


Figure 4.3.2 The curve $z = f(x_0, y)$ cut from the surface by the vertical plane $x = x_0$, and its tangent line at P. The slope of that tangent line is $\left.\frac{\partial f}{\partial y}\right|_{(x_0, y_0)}$.

Write down the limit definition of each of the two slopes. (Both are the ordinary difference quotient you already know, with one input held still: for $\partial f/\partial x$, freeze y at y_0 and give x_0 an increment h. Say which of the two figures each one belongs to, and which variable is doing the moving.)

$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} =$

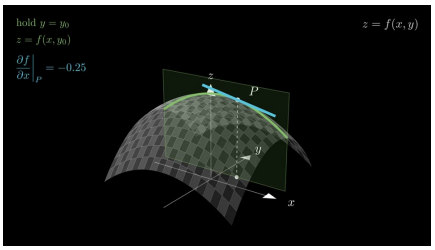
$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} =$

The Partial Derivatives.

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \lim_{h\to 0} \frac{f(x_0+h,\, y_0) - f(x_0,y_0)}{h}, \tag{4.3.1}$$

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \lim_{h\to 0} \frac{f(x_0,\, y_0+h) - f(x_0,y_0)}{h}. \tag{4.3.2}$$

Figure 4.3.3 builds both pictures up one piece at a time — the surface, the cutting plane, the parabola it cuts out, and the tangent line to that parabola — first in the plane $y = y_0$ and then in the plane $x = x_0$, closing with both in place. Use it to check the two limits you just wrote down.



Standalone

Figure 4.3.3 The two partial derivatives at P, built one at a time: the cut by $y = y_0$ with its green parabola $z = f(x, y_0)$ and blue tangent line of slope $\frac{\partial f}{\partial x}$, then the cut by $x = x_0$ with its amber parabola $z = f(x_0, y)$ and red tangent line of slope $\frac{\partial f}{\partial y}$, and finally the two together.

In practice we never evaluate those limits. Say what we do instead. (Compare (4.3.1) with the one-variable definition: if y never changes, then $f(x, y_0)$ is a function of x alone. State the resulting rule for each variable in a single sentence.)

To find f_x:

To find f_y:

Partial Derivatives as Ordinary Derivatives.

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \left.\frac{d}{dx} f(x, y_0)\right|_{x=x_0}, \tag{4.3.3}$$

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \left.\frac{d}{dy} f(x_0, y)\right|_{y=y_0}. \tag{4.3.4}$$

Here are some common notations for the partial derivative with respect to x:

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} \qquad \frac{\partial f}{\partial x}(x_0, y_0) \qquad f_x(x_0, y_0).$$

Note that similar notation can be used for the partial derivative with respect to any variable such as y and z.

4.3.2 Examples

Example 4.3.4 Partial Derivatives of $1 - x^2 - y^2$. Consider the function $f(x, y) = 1 - x^2 - y^2$ and evaluate the following partial derivatives at the point $(x_0, y_0) = (1/2, 1/2)$.

A. f_x

B. f_y

Solution. (Do it twice. First straight from the definitions (4.3.1) and (4.3.2): freeze one variable, give the other an increment h, and simplify the numerator before dividing — every term left carries a factor of h. Then again by (4.3.3) and (4.3.4), holding the other variable constant, and check the two routes agree. This is the surface of Subsection 4.1.1; finish by saying what the two numbers say about it at that point, and why symmetry made them equal.)

A. From the definition, $f_x(1/2, 1/2) =$

B. From the definition, $f_y(1/2, 1/2) =$

The shortcut: $f_x =$ $f_y =$ *and what they say:*

▲

Example 4.3.5 A Quotient with a Trigonometric Term. Consider the function $f(x, y) = \dfrac{2y}{y + \cos x}$ and evaluate the following partial derivatives at the point $(x_0, y_0) = (0, 0)$.

A. f_x

B. f_y

Solution. (For f_x the numerator $2y$ is a constant, so only the reciprocal needs differentiating; for f_y the whole thing is a quotient in y. Watch that $\cos x$ is a constant in the second case but not the first.)

A. $f_x =$

$f_x(0, 0) =$

B. $f_y =$

$f_y(0, 0) =$

▲

Example 4.3.6 Partial Derivatives as Slopes on a Surface. Consider the elliptic paraboloid $z = f(x, y) = 4 - x^2 - 2y^2$. Evaluate $f_x(1, 1)$ and $f_y(1, 1)$ and explain what these numbers represent.

Solution. (Compute both partials as before. Then, for each, substitute the fixed value to get the equation of the trace curve in that vertical plane, and say which curve of Figure 4.3.7 and Figure 4.3.8 it is. Compare the two slopes: one tangent line is steeper, and the coefficients in f say why.)

$f_x(1, 1) =$ *trace in* $y = 1$*:*

$f_y(1, 1) =$ *trace in* $x = 1$*:*

What each number represents:

Which is steeper, and why:

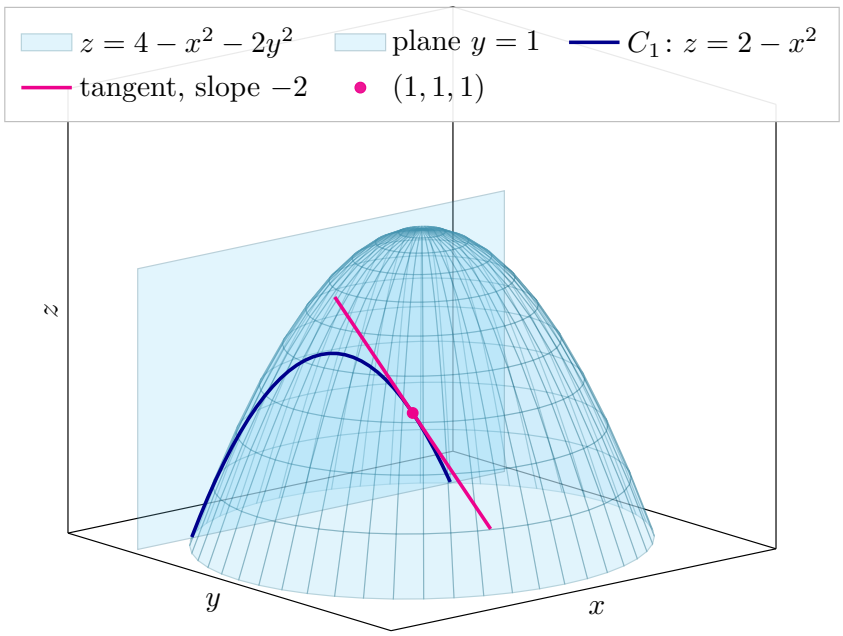


Figure 4.3.7 The trace C_1 cut from $z = 4 - x^2 - 2y^2$ by the plane $y = 1$ is the parabola $z = 2 - x^2$. Its tangent line at $(1, 1, 1)$ has slope $f_x(1, 1) = -2$.

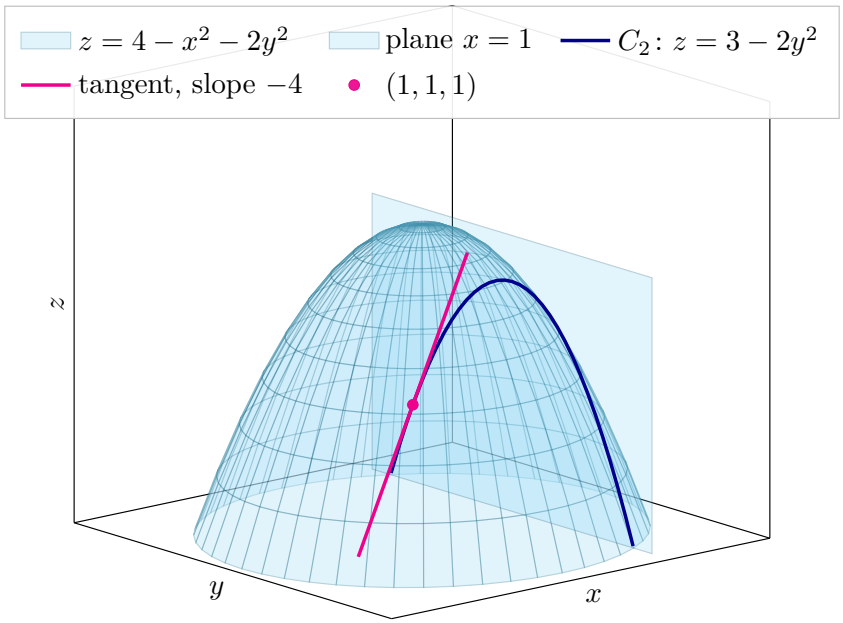


Figure 4.3.8 The trace C_2 cut from $z = 4 - x^2 - 2y^2$ by the plane $x = 1$ is the parabola $z = 3 - 2y^2$. Its tangent line at $(1, 1, 1)$ has slope $f_y(1, 1) = -4$.

An interactive figure putting the two slices in one picture — the dome cut open by either plane, with the parabola it leaves and the tangent line at P — accompanies the online version of these notes. ▲

4.3.3 Second Derivatives and the Mixed Derivative Theorem

We also define second derivatives such as

$$\frac{\partial^2 f}{\partial x \partial y},$$

by which we mean that we first differentiate with respect to y and then with respect to x. We can alternatively denote such a second derivative by f_{yx}. Likewise

$$\frac{\partial^2 f}{\partial x^2}$$

means that we differentiate with respect to x twice, and we can alternatively denote such a second derivative by f_{xx}.

Example 4.3.9 Computing Second Partial Derivatives. Consider the function $f(x, y) = \cos(xy^2)$ and evaluate the following partial derivatives.

A. f_{yx}

B. f_{xy}

C. f_{xx}

Solution. (Mind the order: in the subscript notation the *left* letter is differentiated first. Parts ***A*** and ***B*** start from different first derivatives, so do them independently and then compare the answers — the comparison is the point of the exercise. Both need the product rule at the second step.)

First derivatives: $f_x =$ $f_y =$

A. $f_{yx} =$

B. $f_{xy} =$

C. $f_{xx} =$

▲

Theorem 4.3.10 The Mixed Derivative Theorem (Clairaut's Theorem). *If $f(x, y)$ and its partial derivatives f_x, f_y, f_{xy}, and f_{yx} are defined throughout an open region containing a point (a, b) and are all continuous at (a, b), then*

$$f_{xy}(a, b) = f_{yx}(a, b). \tag{4.3.5}$$

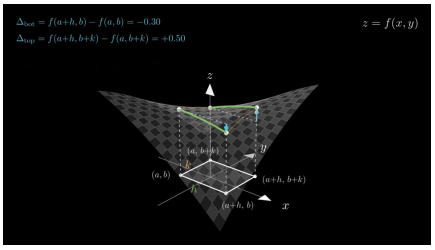
Why should the order of differentiation not matter? There is a geometric reason, with no derivative rules involved at all. Work it out. (Start at (a, b), step h in the x-direction and k in the y-direction, and write down the four heights of the surface above the corners of the resulting rectangle. Group them by the two edges running in the x-direction: the difference of those two rises measures how f_x changes as y increases. Then group the *same* four heights by the two edges running in the y-direction instead. Both groupings collapse to one four-corner sum Δ; write it out and notice how it treats x and y. Then let $h, k \to 0$.)

The four corner heights:

Grouped along the x-edges:

Grouped along the y-edges:

$\Delta =$ *and letting* $h, k \to 0$*:*



Standalone

Figure 4.3.11 Why $f_{xy} = f_{yx}$: the four heights of the surface over the corners of a small rectangle with sides h and k. Differencing along the two x-edges (green, with blue rises) gives approximately $hk\,\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$; differencing along the two y-edges (amber, with red rises) gives approximately $hk\,\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$. Both are the same four-corner sum Δ, so shrinking the rectangle to the point gives $f_{xy}(a,b) = f_{yx}(a,b)$.

Check Theorem 4.3.10 against your answers to parts ***A*** and ***B*** of Example 4.3.9, and say what practical freedom the theorem gives you.

The two answers agree because:

So in practice we may:

Example 4.3.12 Choosing the Convenient Order. Consider $f(x,y) = \cos(x^2y) + \dfrac{e^{(y^2+y+1)}}{y^2+5}$ and calculate f_{xy}.

Solution. (Do not differentiate in the order the subscript suggests. Look at the second term: it depends on y alone, so one of the two orders kills it immediately, and Theorem 4.3.10 says you are entitled to use that order instead. State which order you chose and why before computing.)

Order chosen, and why:

First derivative:

$f_{xy} =$

▲

4.4 The Chain Rule

Objectives

After completing this section, you should be able to:

- State and apply the chain rule for a function $w = f(x,y)$ whose intermediate variables x and y are functions of a single independent variable t.

- Extend the chain rule to three intermediate variables, and to two independent variables r and s.
- Draw a tree diagram for a composition and read the corresponding chain rule formula off the diagram.
- Use the chain rule in applied settings, for instance to find the rate of change of temperature experienced by an object moving along a path.

The chain rule for functions of one variable tells us how to differentiate a composition. When a function of several variables has inputs that themselves depend on other variables, there is a corresponding rule, and it has one term for each intermediate variable. Tree diagrams give a reliable way of writing these formulas down without memorizing them.

4.4.1 One Independent Variable

Theorem 4.4.1 Chain Rule for Functions of Two Independent Variables. *Consider the two-variable function $w = f(x, y)$. If the following conditions are satisfied:*

1. *$w = f(x, y)$ has continuous partial derivatives f_x and f_y,*

2. *$x = x(t)$ and $y = y(t)$ are differentiable functions of t,*

then the function $w = f(x(t), y(t))$ is a differentiable function of t and

$$\frac{dw}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}. \tag{4.4.1}$$

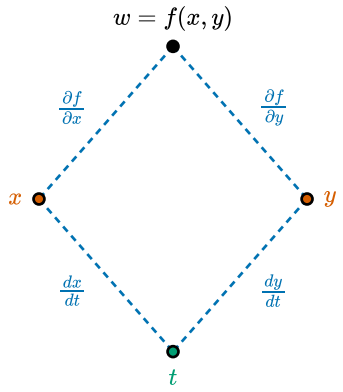


Figure 4.4.2 Tree diagram for $w = f(x(t), y(t))$. There are two paths from w down to t, one through x and one through y, and the chain rule adds one term for each.

Figure 4.4.2 is how to remember that formula without memorizing it. Work out the reading rule, and label the parts. (Which variable is **dependent**, which are **intermediate**, and which is **independent**? Then describe how to get from the picture to (4.4.1): something happens along each path, and something else happens between paths. Count the paths and compare with the number of terms.)

Dependent: *Intermediate:* *Independent:*

Along each path we …, and between paths we …

Number of paths = number of terms =

Theorem 4.4.3 Chain Rule for Functions of Three Independent Variables. *Consider the three-variable function $w = f(x, y, z)$. If the following conditions are satisfied:*

1. *$w = f(x, y, z)$ has continuous partial derivatives f_x, f_y, and f_z,*

2. *$x = x(t)$, $y = y(t)$, and $z = z(t)$ are differentiable functions of t,*

then the function $w = f(x(t), y(t), z(t))$ is a differentiable function of t and

$$\frac{dw}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}. \tag{4.4.2}$$

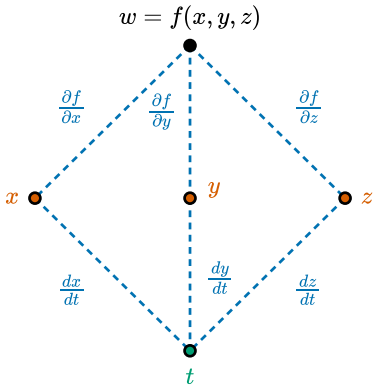


Figure 4.4.4 Tree diagram for $w = f(x(t), y(t), z(t))$. Each of the three intermediate variables contributes one term to (4.4.2).

Example 4.4.5 Differentiating Along a Circle. Consider the function $w = f(x, y) = xy$, where $x = \cos t$ and $y = \sin t$. Evaluate $\left.\dfrac{dw}{dt}\right|_{t=\pi/2}$.

Solution. (Do it twice. First with (4.4.1): the two partials of xy are easy, and a double-angle identity tidies the result. Then again by substituting $x(t)$ and $y(t)$ into f first and differentiating the one-variable function that results. The two answers must agree; say what the second method shows about when the chain rule is worth using at all.)

Method 1, with the chain rule:

Method 2, substituting first:

When is the chain rule worth it?

Geometrically, the constraint $x = \cos t$, $y = \sin t$ confines the point (x, y) to the unit circle $x^2 + y^2 = 1$, so the composition traces a curve on the saddle $z = xy$, as shown in Figure 4.4.6. The derivative $\frac{dw}{dt}$ is the rate at which the height of that curve changes; check the sign of your answer against the picture at $t = \pi/2$.

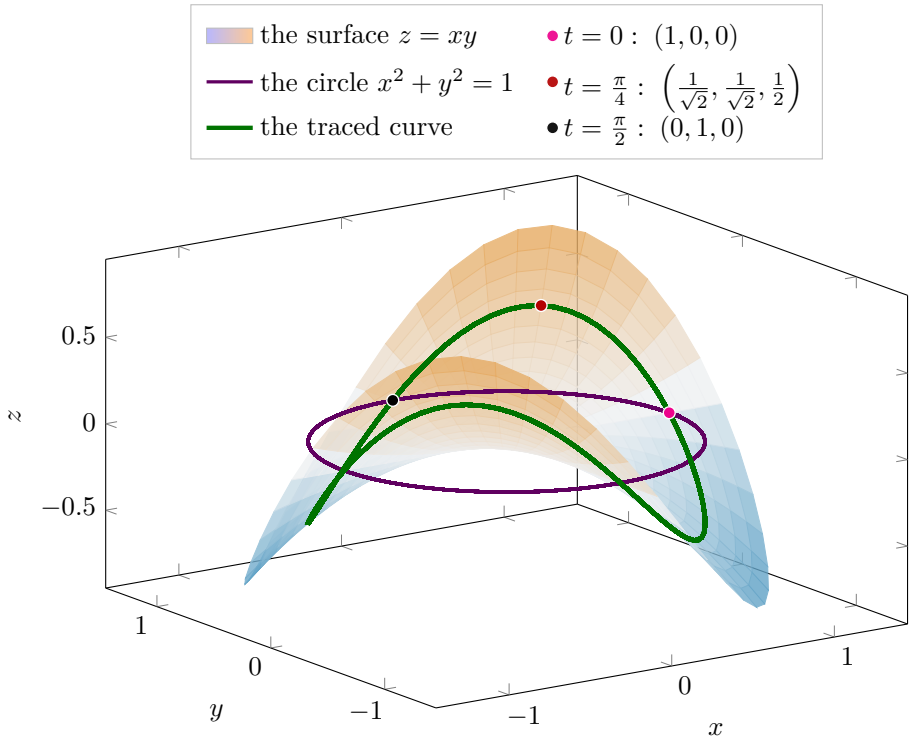


Figure 4.4.6 The curve traced on the surface $z = xy$ as (x, y) runs around the unit circle $x^2 + y^2 = 1$. Marked are the points at $t = 0$, $t = \pi/4$, and $t = \pi/2$.

An animation of that picture — the point going round the circle with the height w and the rate dw/dt read out, so the value you computed can be checked at $t = \pi/2$ — accompanies the online version of these notes. ▲

Example 4.4.7 A Second Application of the Chain Rule. Consider the function $w = f(x, y) = xy + y^2$, where $x = \cos t$ and $y = 1 + 2\sin t$. Evaluate $\left.\dfrac{dw}{dt}\right|_{t=\pi}$.

Solution. (Same routine as before, but here substituting first would be messy, so use (4.4.1). Compute the two partials and the two ordinary derivatives separately before combining, and substitute $t = \pi$ only at the very end.)

$f_x =$ $f_y =$ $dx/dt =$ $dy/dt =$

$\frac{dw}{dt} =$

At $t = \pi$:

▲

4.4.2 Two Independent Variables

Theorem 4.4.8 Chain Rule for Two Independent and Two Intermediate Variables. *Consider the two-variable function $w = f(x, y)$. If $w = f(x, y)$, $x = x(r, s)$, and $y = y(r, s)$ are differentiable functions, then the partial derivatives of the function $w(r, s) = f(x(r, s), y(r, s))$ are given by*

$$\frac{\partial w}{\partial r} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r} \qquad \textit{and} \qquad \frac{\partial w}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s}. \tag{4.4.3}$$

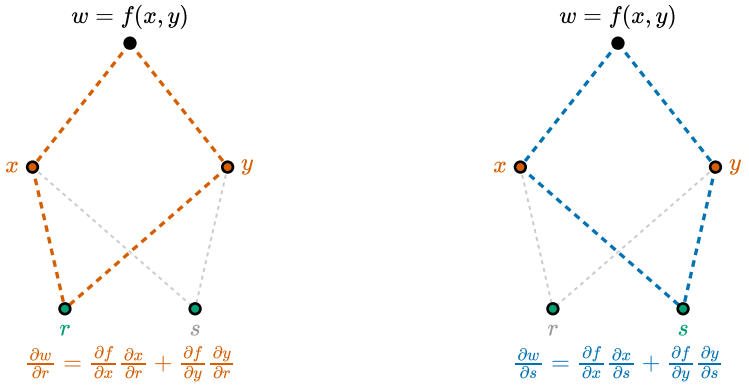


Figure 4.4.9 Two readings of the same tree for two independent variables. On the left, the branches leading to r give $\frac{\partial w}{\partial r}$; on the right, the branches leading to s give $\frac{\partial w}{\partial s}$.

Say what changed. (The tree now has two variables at the bottom, so it

is read twice. State which branches you follow for each reading, and why the derivatives at the top of the tree became partial derivatives when they were ordinary ones in (4.4.1).)

Reading the tree for r, then for s:

Why the lower derivatives are now partial:

Example 4.4.10 Two Independent Variables. Consider the function $z = f(x, y) = e^x \sin y$, where $x = st^2$ and $y = s^2t$, and evaluate $\dfrac{\partial z}{\partial s}$ and $\dfrac{\partial z}{\partial t}$.

Solution. (Here s and t are the independent variables. Lay out all six ingredient derivatives first — two for f, and four for x and y — then assemble each answer with (4.4.3). Finish by writing everything back in terms of s and t, so no x or y is left.)

The six ingredient derivatives:

$\dfrac{\partial z}{\partial s} =$

$\dfrac{\partial z}{\partial t} =$

▲

4.4.3 More Examples

Example 4.4.11 The Temperature Experienced by a Moving Ant. Suppose that the temperature on a metal plate is given by the function T with

$$T(x, y) = 100 - (x^2 + 4y^2),$$

where the temperature is measured in degrees Fahrenheit and x and y are each measured in feet. Now suppose that an ant is walking on the metal plate in such a way that it walks in a straight line from the point $(1, 4)$ to the point $(5, 6)$.

a. Find parametric equations $(x(t), y(t))$ for the ant's coordinates as it walks the line from $(1, 4)$ to $(5, 6)$.

b. What can you say about $\frac{dx}{dt}$ and $\frac{dy}{dt}$ for every value of t?

c. Determine the instantaneous rate of change in temperature with respect to t that the ant is experiencing at the moment it is halfway from $(1, 4)$ to $(5, 6)$, using your parametric equations for x and y. Include units on your answer.

Solution. (For ***a***, parametrize the segment as in Section 3.5; state the range of t and check both endpoints. For ***b***, the answer is a property of straight-line motion — say what it is and why. For ***c***, the tree is Figure 4.4.2 with T in place of w; find the halfway value of t first, and do not forget the units, which come from °F and the unit of t.)

a. $x(t) =$ $y(t) =$ *range of* t*:*

b.

c. $\frac{dT}{dt} =$

Halfway, $t =$ *so the rate is* *units:*

▲

Example 4.4.12 Proving an Identity with the Chain Rule. Suppose that $w = f(xz, yz)$, where f is a differentiable function. Show that

$$x\frac{\partial w}{\partial x} + y\frac{\partial w}{\partial y} = z\frac{\partial w}{\partial z}.$$

Solution. (Name the intermediate variables $u = xz$ and $v = yz$ first, so that $w = f(u, v)$, and draw the tree in the box below. Notice that z appears under *both* intermediate variables while x and y appear under only one each — that asymmetry is what makes the identity work. Read the three partial derivatives off your tree, then form each side of the identity and compare.)

Tree diagram:

$w_x =$ $w_y =$ $w_z =$

Left side:

Right side:

▲

4.5 Directional Derivatives and Gradient Vectors

Objectives

After completing this section, you should be able to:

- Understand the directional derivative as the rate of change of $f(x, y)$ in an arbitrary direction, generalizing the partial derivatives f_x and f_y, which are the special cases in the $\hat{\imath}$ and $\hat{\jmath}$ directions.
- Interpret the directional derivative geometrically as the slope of the tangent line to the curve in which a vertical plane through P_0, in the direction of $\hat{u}$, cuts the surface $z = f(x, y)$.
- Understand the gradient vector ∇f and its relationship to the directional derivative through $(D_{\hat{u}} f)_{P_0} = (\nabla f)_{P_0} \cdot \hat{u}$.
- Interpret the directional derivative through the angle between ∇f and $\hat{u}$, and understand why f changes most rapidly in the direction of the gradient.
- Understand why the gradient vector is perpendicular to the level curves of f.

Recall from Section 4.3 that the partial derivatives f_x and f_y correspond to the rate of change of $z = f(x, y)$ in the $\hat{\imath}$ and $\hat{\jmath}$ directions. In this section, we generalize this notion to the rate of change of $z = f(x, y)$ in an arbitrary direction $\hat{u}$, which is a unit vector of the form $\hat{u} = u_1\hat{\imath} + u_2\hat{\jmath}$.

4.5.1 The Definition of the Directional Derivative

Definition 4.5.1 Directional Derivative. The **derivative of f at $P_0(x_0, y_0)$ in the direction of the unit vector** $\hat{u} = u_1\hat{\imath} + u_2\hat{\jmath}$ is the number

$$\left(\frac{df}{ds}\right)_{\hat{u},\, P_0} = \lim_{s \to 0} \frac{f(x_0 + su_1,\ y_0 + su_2) - f(x_0, y_0)}{s}, \tag{4.5.1}$$

provided the limit exists. It can alternatively be denoted by $(D_{\hat{u}} f)_{P_0}$. ◆

The geometry behind this definition is best seen in a picture. Starting at the point $P_0(x_0, y_0)$, we move a distance s in the direction of the unit vector $\hat{u}$ to reach a second point, and we compare the values of the surface $z = f(x, y)$ above the two points. Before exploring the figures, write down where that second point is and what the quotient in (4.5.1) measures. (The displacement from P_0 is $s\hat{u}$, so read its two components off. Then say what the quotient is the slope of, and what it becomes in the limit. Why must $\hat{u}$ be a *unit* vector for s to be the distance actually travelled?)

The second point $P =$

The quotient is the slope of …, and in the limit …

Why $\hat{u}$ must be a unit vector:

The four figures that follow show the construction from complementary angles: the first lets you explore it in space, the second records the displacement in the xy-plane, the third animates the secant line rotating onto the tangent line, and the fourth reports the difference quotient itself as s shrinks, so that the limit can be watched rather than only read.

An interactive view of this definition — the point P moving the distance s from P_0 along $\hat{u}$, with the secant slope approaching the directional derivative as $s \to 0$ — accompanies the online version of these notes.

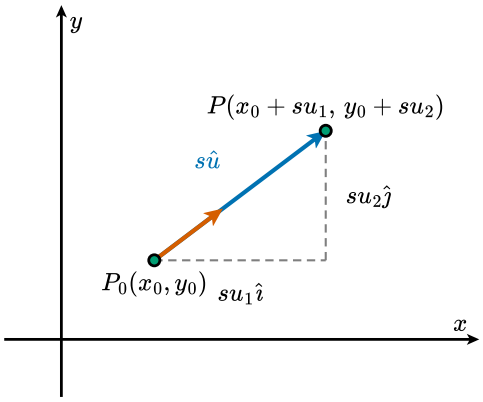


Figure 4.5.2 The two points P_0 and P used in the definition of the directional derivative. As $s \to 0$, the point P slides along the direction $\hat{u}$ back toward P_0.

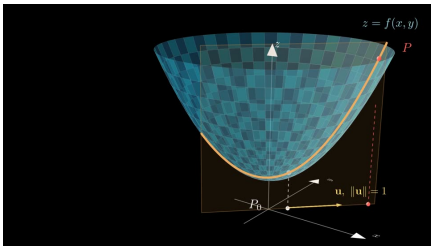


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Figure 4.5.3 Animation of the definition of the directional derivative. The vertical plane through P_0 in the direction $\hat{u}$ cuts the surface $z = f(x, y)$ in a curve, and the point $P = P_0 + s\hat{u}$ moves on this plane. The secant line through $(P_0, f(P_0))$ and $(P, f(P))$ has slope $\big(f(P) - f(P_0)\big)/s$, and as $s \to 0$ it rotates onto the tangent line, whose slope is the directional derivative $(D_{\hat{u}} f)_{P_0}$.

An interactive figure of the difference quotient as a function of the direction angle and the step s accompanies the online version of these notes.

Example 4.5.4 **Example I.** Consider the function $f(x, y) = 1 + x^2 + y^2$, the point $P_0(1, 1)$, and the unit vector $\hat{u} = \dfrac{1}{\sqrt{2}}\hat{\imath} + \dfrac{1}{\sqrt{2}}\hat{\jmath}$. Use the definition of directional derivative to compute $(D_{\hat{u}} f)_{P_0}$.

Solution. (Substitute the given x_0, y_0, u_1, u_2 into (4.5.1) and expand. Both squares are the same, which halves the algebra; after expanding, every surviving term carries a factor of s, so the quotient simplifies before the limit is taken. Keep this number — you will recover it in two lines in Example 4.5.9 once the gradient is available.)

Substituting and expanding:

$(D_{\hat{u}} f)_{P_0} =$

The computation has a useful geometric picture. Along the ray through $P_0(1,1)$ in the direction $\hat{u}$, the values of f are a single-variable function $g(s)$. Write it down, and connect it to what you just computed. (Substitute the same two expressions into f, but keep s as a variable rather than letting it go to zero. Then say which derivative of g your answer above is, and check it by differentiating g directly.)

$g(s) =$

$g'(0) =$ *which is:*

An animation of the secant rotating onto the tangent line at P_0 as $s \to 0$ accompanies the online version of these notes. ▲

Activity 4.5.1.1 The Partial Derivatives Are Directional Derivatives. Definition 4.5.1 measures the rate of change of f in an arbitrary direction $\hat{u}$. Two directions are already familiar from f_x and f_y. In this activity you will check that the definition reproduces exactly those two derivatives when $\hat{u}$ points along a coordinate axis. Use only (4.5.1); nothing beyond the definition is needed.

(a) Take $\hat{u} = \hat{\imath}$, so that $u_1 = 1$ and $u_2 = 0$. Substitute these into (4.5.1) and simplify. Which single-variable limit is left, and what is its value?

workspace

Hint. Ask yourself which of the two inputs of f actually changes as s varies.

(b) Repeat the computation with $\hat{u} = \hat{\jmath}$, and state the conclusion.

workspace

(c) Test this on the function of Example 4.5.4. With $f(x, y) = 1 + x^2 + y^2$ and $P_0(1, 1)$, compute $(D_{\hat{\imath}} f)_{P_0}$ straight from the definition, then compute $f_x(1, 1)$ in the usual way and compare. Compare both with the value found in Example 4.5.4: which direction climbs fastest, and does that match your intuition about the surface?

From the definition, and by differentiating:

Comparison with the diagonal direction:

(d) Finally, take $\hat{u} = -\hat{\imath}$. Show that $(D_{-\hat{\imath}} f)_{P_0} = -f_x(x_0, y_0)$, and say in a sentence why this is what you should expect.

The computation (try the substitution $h = -s$):

Why this is expected:

Every partial derivative is a directional derivative, so nothing already known about f_x and f_y is lost. What the definition adds is the freedom to point $\hat{u}$ anywhere.

4.5.2 The Gradient Vector

Evaluating that limit every time would be painful, and the chain rule removes the need to. Note first that the line through $P_0(x_0, y_0)$ in the direction $\hat{u}$ can be written as

$$x(s) = x_0 + su_1, \qquad y(s) = y_0 + su_2. \tag{4.5.2}$$

Use that, together with the chain rule, to find a formula for the directional derivative. (This is (4.4.1) with s in place of t, and the dependency diagram is Figure 4.5.6. Differentiate (4.5.2) to get the two easy factors, and then *recognize the result as a dot product* — that last step is the whole point, so write the two vectors out separately.)

$dx/ds =$ *$dy/ds =$*

By the chain rule, $\frac{df}{ds} =$

As a dot product:

Definition 4.5.5 Gradient Vector. The **gradient vector** of the function $f(x, y)$ at $P_0(x_0, y_0)$ is the vector

$$(\nabla f)_{P_0} = \left(\frac{\partial f}{\partial x}\right)_{P_0} \hat{\imath} + \left(\frac{\partial f}{\partial y}\right)_{P_0} \hat{\jmath}. \tag{4.5.3}$$

The gradient vector and the directional derivative of $f(x, y)$ at $P_0(x_0, y_0)$ in the direction $\hat{u} = u_1\hat{\imath} + u_2\hat{\jmath}$ are related via the following dot product:

$$\left(\frac{df}{ds}\right)_{\hat{u},\, P_0} = (\nabla f)_{P_0} \cdot \hat{u}. \tag{4.5.4}$$

◆

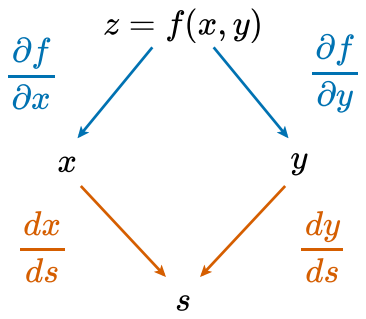


Figure 4.5.6 The chain rule dependency diagram: multiplying down each branch and adding the two products gives $\frac{df}{ds} = \frac{\partial f}{\partial x}\frac{dx}{ds} + \frac{\partial f}{\partial y}\frac{dy}{ds}$.

Theorem 4.5.7 The Gradient is Perpendicular to the Level Curves. *Let $f(x, y)$ be a differentiable function, and let C be the level curve $f(x, y) = c$ for a constant c. At every point of C where $\nabla f \neq \mathbf{0}$, the gradient vector ∇f is perpendicular to the level curve C.*

Prove it. (Parametrize C as $\mathbf{r}(t) = \langle x(t), y(t)\rangle$. The one fact you have is that f is *constant* along C; differentiate that statement with respect to t and see what the right-hand side becomes. Then recognize the left-hand side as a dot product of two vectors you can name, and say what a vanishing dot product means. Finish by explaining why the velocity vector is tangent to C, and where the hypothesis $\nabla f \neq \mathbf{0}$ is needed.)

Along C, $f(x(t), y(t)) =$

Differentiating:

As a dot product:

Conclusion, and where $\nabla f \neq \mathbf{0}$ is used:

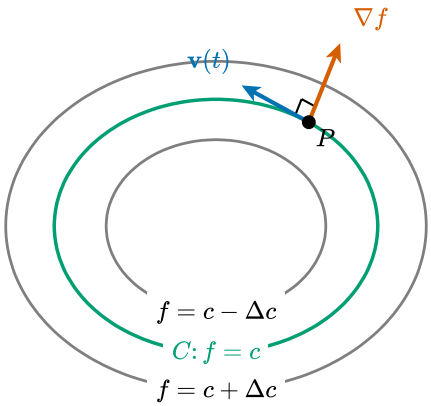


Figure 4.5.8 The gradient ∇f meets the level curve C: $f = c$ at a right angle: it is perpendicular to the velocity vector $\mathbf{v}(t)$, which is tangent to C.

Activity 4.5.2.1 Checking that the Gradient Meets Level Curves at Right Angles. Theorem 4.5.7 claims that ∇f is perpendicular to the level curve through a point. This activity asks you to verify that claim by hand at one point, and then to test it at many points using the interactive figure that follows.

(a) Let $f(x, y) = 1 + x^2 + y^2$. Find the level curve of f through $P_0(1, 1)$, and

give a vector tangent to that curve at P_0.

The level curve:

A tangent vector at P_0:

Hint. Level curves of this f are circles centred at the origin. A tangent to a circle is perpendicular to its radius.

(b) Compute $(\nabla f)_{(1,1)}$ and then the dot product of that gradient with your tangent vector. What does the result say?

$(\nabla f)_{(1,1)} =$ *dot product =*

Which means:

(c) Repeat the argument at an arbitrary point $(a, b) \neq (0, 0)$ to show the right angle is not a coincidence of the point $(1, 1)$.

workspace

(d) An interactive figure — the gradient arrow drawn at a draggable point, together with the level curve through it — accompanies the online version of these notes. Here, use the formula $\nabla f = 2x\,\hat{\imath} + 2y\,\hat{\jmath}$ to describe how the length of the arrow changes as the point moves away from the origin,

and to say where the gradient vanishes.

How the arrow length changes, and why:

Where the gradient vanishes, and what happens there:

Example 4.5.9 Example II. Consider the function $f(x, y) = 1+x^2+y^2$, the point $P_0(1, 1)$, and the unit vector $\hat{u} = \frac{1}{\sqrt{2}}\hat{\imath} + \frac{1}{\sqrt{2}}\hat{\jmath}$. Evaluate the directional derivative $(D_{\hat{u}}f)_{P_0}$ and interpret your result.

Solution. (Same function, point and direction as Example 4.5.4, but now use (4.5.4): find ∇f, evaluate it at P_0, and dot it with $\hat{u}$. It should take two lines and agree with the limit you computed earlier. Then read the answer geometrically off Figure 4.5.10: $\hat{u}$ points along a line in the xy-plane, that line lies in a vertical plane, and that plane cuts the surface in a curve — name the curve, and say which line's slope your number is.)

$\nabla f =$ $(\nabla f)_{(1,1)} =$

$(D_{\hat{u}}f)_{P_0} =$

Geometric interpretation:

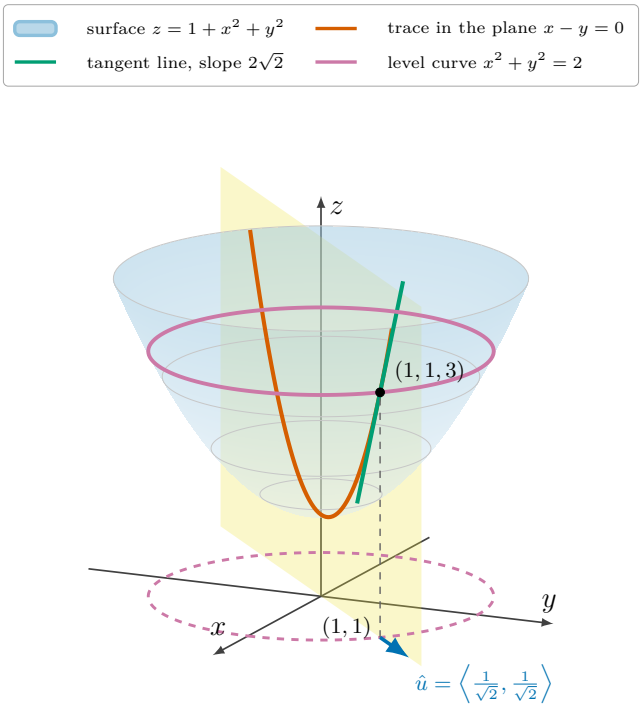


Figure 4.5.10 The surface $z = 1 + x^2 + y^2$ cut by the vertical plane $x - y = 0$, which contains the direction $\hat{u}$. The intersection is a parabola, and the directional derivative $(D_{\hat{u}} f)_{(1,1)} = 2\sqrt{2}$ is the slope of its tangent line at the point $(1, 1, 3)$. The level curve $f = 3$ through $(1, 1)$, the circle $x^2 + y^2 = 2$, is drawn on the surface at height $z = 3$, with its dashed projection in the xy-plane.

An animation of the tangent to the sliced surface tipping from steepest ascent through level to steepest descent as $\hat{u}$ turns accompanies the online version of these notes. ▲

4.5.3 Directions of Fastest Increase, Fastest Decrease, and No Change

Rewrite the directional derivative so that the direction enters through a single angle. (Start from (4.5.4) and expand the dot product geometrically, as in (3.3.2), with θ the angle between ∇f and $\hat{u}$. One factor then drops out; say which and why.)

$D_{\hat{u}} f =$

The Directional Derivative and the Angle.

$$D_{\hat{u}} f = \|\nabla f\| \cos\theta. \tag{4.5.5}$$

Only the factor $\cos\theta$ depends on the direction, and it runs over $[-1, 1]$ as $\hat{u}$ turns through the circle of directions at P_0. Its three extreme values single out three directions, drawn in Figure 4.5.11. Fill them in.

$\cos\theta = 1$: *direction:* $D_{\hat{u}} f =$

$\cos\theta = -1$: *direction:* $D_{\hat{u}} f =$

$\cos\theta = 0$: *direction:* $D_{\hat{u}} f =$

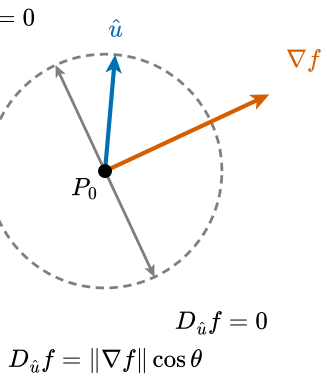


Figure 4.5.11 Directions around P_0: f increases fastest along ∇f, decreases fastest along $-\nabla f$, and does not change in the two directions orthogonal to ∇f.

Activity 4.5.3.1 Verifying the Three Special Directions. The three cases above are all consequences of the single formula (4.5.5). In this activity you will derive them yourself. An interactive figure for checking your answers — $\hat{u}$ dragged around the circle of directions, with $D_{\hat{u}}f$ read out — accompanies the online version of these notes.

(a) Beginning with $D_{\hat{u}}f = \nabla f \cdot \hat{u}$, explain why the dot product can be rewritten as $\|\nabla f\| \cos\theta$. What would go wrong if $\hat{u}$ were not a unit vector?

The rewriting:

If $\hat{u}$ were not a unit vector:

(b) Treat $\|\nabla f\|$ as fixed and let θ run over $[0, 2\pi)$. What are the largest and smallest values of $D_{\hat{u}}f$, and at which angles do they occur? For which angles is $D_{\hat{u}}f = 0$?

Largest and smallest, and where:

Zero when $\theta =$

(c) Now take $f(x, y) = 1 + x^2 + y^2$ at $P_0(1, 1)$, the function of Example 4.5.4. Compute ∇f and $\|\nabla f\|$ at P_0, write $D_{\hat{u}} f$ as a function of θ, and give the three special unit vectors explicitly.

$(\nabla f)_{(1,1)} =$ $\|\nabla f\| =$

$D_{\hat{u}} f(\theta) =$

The three special unit vectors:

(d) Check your work against the figure below. Drag $\hat{u}$ around the circle and confirm that the meter agrees with your formula at several angles. Then use the three buttons and compare the readings with your answers to the previous task. As you drag, at what angle does the reading change most quickly, and at what angle is it momentarily stationary?

Changes fastest at $\theta =$ *because:*

Stationary at $\theta =$ *which means, for a walker:*

Example 4.5.12 Example III. Consider the function $f(x, y) = \dfrac{x^2}{2} + \dfrac{y^2}{2}$. Find the directions in which, at the point $P_0(1, 1)$,

A. f increases most rapidly,

B. f decreases most rapidly,

C. there is no change in f.

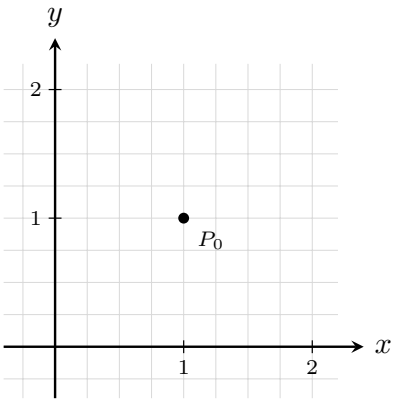
Solution. (Compute $(\nabla f)_{(1,1)}$ first; parts ***A*** and ***B*** then follow from the list above, but the question asks for *directions*, so normalize. For part ***C*** set $(\nabla f)_{(1,1)} \cdot \hat{u} = 0$ and solve together with $\|\hat{u}\| = 1$; there are two answers, and they are related in a way worth stating. Finally draw all four directions on the grid, together with the level curve through P_0, and say how each is placed relative to that curve.)

$(\nabla f)_{(1,1)} =$

A.

B.

C.

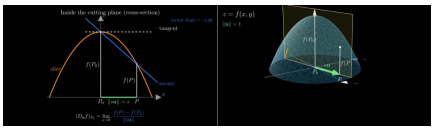


An animation of $D_{\hat{u}}f$ sweeping between $\|\nabla f\|$ and $-\|\nabla f\|$ as $\hat{u}$ turns accompanies the online version of these notes.

A matching animated 3D view of the tangent line tipping as $\hat{u}$ turns accompanies the online version of these notes. ▲

4.5.4 Summary

We collect here the main ideas of this section.



Standalone

Figure 4.5.13 A directional derivative viewed together in the plane and in space: the unit vector $\hat{u}$ at P_0 in the xy-plane selects a slice of the surface $z = f(x, y)$, and $(D_{\hat{u}}f)_{P_0}$ is the slope of the resulting tangent line to that slice.

- *The directional derivative.* Definition 4.5.1 defines $(D_{\hat{u}}f)_{P_0}$ as the limit in (4.5.1); it measures the rate of change of f at P_0 in the direction of a unit vector $\hat{u}$. It is computed directly from this limit in Example 4.5.4.
- *The gradient vector.* Using the chain rule we obtain Definition 4.5.5, which gives the shortcut $(D_{\hat{u}}f)_{P_0} = (\nabla f)_{P_0} \cdot \hat{u}$ in (4.5.4). This is applied in Example 4.5.9.

- *Fastest increase, fastest decrease, and no change.* Writing $D_{\hat{u}} f = \|\nabla f\| \cos\theta$ in (4.5.5) shows that f increases most rapidly in the direction of ∇f, decreases most rapidly in the direction of $-\nabla f$, and does not change in the directions orthogonal to ∇f. These directions are found in Example 4.5.12.
- *The gradient is perpendicular to the level curves.* Theorem 4.5.7 proves that ∇f is perpendicular to the level curve through each point. Equivalently, the directions of no change are tangent to the level curve, while the directions of most rapid increase and decrease are perpendicular to it.

4.5.5 Scalar Fields, Potentials, and the Direction Nature Chooses

A **scalar field** is a rule that assigns a single number to every point of a region: the temperature $T(x, y)$ in a city, the pressure $P(x, y)$ over an ocean, the elevation $h(x, y)$ on a hillside, the concentration $c(x, y)$ of a pollutant in a corridor. A scalar field carries no direction of its own. It is only a number at each location.

Physics, by contrast, is full of quantities that point somewhere: heat flows, forces push, wind blows. These are **vector fields**. The gradient is what ties them back to scalar fields — at least for the **conservative** fields taken up in MATH 14 at SCU, where the whole vector field is recovered from a single scalar potential by differentiation. Because such fields are so common in nature, the pattern in the table below appears again and again across physics.

When a scalar field governs a physical force or flux, we call the field a **potential**, and the associated vector quantity is, up to a constant, the *negative* gradient of that potential.

Setting	Scalar field	Vector quantity	Level curves are
Heat conduction	temperature T	heat flux $\mathbf{q} = -k\nabla T$ (Fourier's law)	isotherms
Electrostatics	potential V	electric field $\mathbf{E} = -\nabla V$	equipotentials
Mechanics	potential energy U	conservative force $\mathbf{F} = -\nabla U$	contours of constar
Meteorology	pressure P	pressure-gradient force $-\nabla P$ per unit volume	isobars

The minus sign in each of these laws encodes one physical fact: *nature moves downhill.* Heat drains from hot to cold, charges move toward low electric potential, a ball rolls toward low gravitational potential energy, air accelerates toward low pressure. So the direction the physics actually selects is the direction of most rapid *decrease*, which is exactly the case $\theta = \pi$ of $D_{\mathbf{u}} f = \|\nabla f\| \cos\theta$.

Three consequences carry over unchanged from Subsection 4.5.3, and they are worth restating in physical language.

- $\theta = 0$: fastest increase, at the rate $\|\nabla f\|$. This is the direction heat comes *from*.
- $\theta = \pi$: fastest decrease, at the rate $-\|\nabla f\|$. This is the direction the flux, the force, or the wind actually points.
- $\theta = \pi/2$: no change at all. These directions trace the level curves of the field — isotherms, equipotentials, isobars, the contour lines on a topographic map. Since ∇f is orthogonal to the level curve through a point, *the flux, the force, or the wind always crosses the contours at right angles.*

Example 4.5.14 Two Runners in a City Heat Island. On a still summer afternoon the pavement downtown is far hotter than the outlying neighborhoods. Model the surface air temperature, in degrees Fahrenheit, by

$$T(x, y) = 74 + 14\, e^{-(x^2+3y^2)/40},$$

where x and y are measured in miles east and north of downtown. The core reaches 88°F, and the outskirts settle near 74°F. Two runners meet at the point $P_0(1, 1)$.

1. The first runner is overheating and wants cooler air as quickly as possible. Which direction should she take, and how fast does the temperature fall along it?
2. The second runner has settled into the temperature at P_0 and wants a route on which it never changes at all. What path should she run?

Solution. (Differentiate the exponential with the chain rule; the same factor $e^{-(x^2+3y^2)/40}$ comes out of both components, so keep it factored. For part 1 the heading is $-\nabla T$ normalized, and the rate is $-\|\nabla T\|$ — notice how modest that number is, and say what that tells the runner. For part 2, requiring $D_{\mathbf{u}}T = 0$ at a single instant gives two directions, but following that condition *continuously* gives a curve; find its equation by asking when T is constant, and name it.)

$\nabla T =$ $(\nabla T)_{(1,1)} =$

1. Heading: *rate:*

2. The two directions:

Her route, as a curve:

In the animation below, switch between the two runners and check your answers: on the loop the thermometer should hold steady while the gradient

arrow stays perpendicular to her heading, and on the escape route the same reading should fall steadily, the path cutting across every isotherm at a right angle.

An interactive figure of the two routes from P_0 in the temperature field — the runner on the level curve holding a steady reading while the other route heats up — accompanies the online version of these notes. ▲

4.6 Tangent Planes and Differentials

Objectives

After completing this section, you should be able to:

- Derive the equation of the tangent plane to the surface $z = f(x, y)$ at a point $P_0(x_0, y_0, f(x_0, y_0))$ from the slopes $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ of the two tangent lines in the planes $y = y_0$ and $x = x_0$.
- Find the normal vector $\mathbf{n} = \nabla w\Big|_{P_0}$ of the tangent plane, where $w(x, y, z) = f(x, y) - z$, and write the parametric equations of the normal line to the surface at P_0.
- Estimate the change Δf in the value of f produced by moving a small distance ds away from P_0 in the direction of a unit vector $\hat{u}$, using the differential $df = \left(\nabla f\Big|_{P_0} \cdot \hat{u}\right) ds$.
- Find the linearization $L(x, y)$ of a differentiable function at a point and use the tangent plane as a linear approximation of the surface.

Recall from Section 4.3 that the partial derivative $f_x(x_0, y_0)$ is the slope of the tangent line to the curve cut from the surface $z = f(x, y)$ by the plane $y = y_0$, and that $f_y(x_0, y_0)$ is the slope of the tangent line to the curve cut by the plane $x = x_0$. In this section we put these two tangent lines together: the plane containing both of them is the tangent plane to the surface at $P_0(x_0, y_0, f(x_0, y_0))$. We find its equation, use its normal vector to write the equation of the normal line to the surface, estimate the change in f caused by moving a small distance away from P_0 in a given direction, and finally use the tangent plane to linearize f near a point.

4.6.1 Equation of the Tangent Plane

We use what we learned about the partial derivatives and slopes of tangent lines to calculate the equation of the tangent plane to the surface $z = f(x, y)$ at $P_0(x_0, y_0, f(x_0, y_0))$. Start with a general plane through P_0,

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0, \qquad z_0 = f(x_0, y_0). \tag{4.6.1}$$

Identify A, B and C by looking at the plane one slice at a time. (Solve (4.6.1) for $z - z_0$. Setting $y = y_0$ leaves the tangent line lying in the plane $y = y_0$, so its slope must be one of the two partial derivatives; do the same with $x = x_0$. Then substitute the two slopes back and clear the denominator.)

solved for $z - z_0$:

the slice $y = y_0$, and the slope it names:

the slice $x = x_0$, and the slope it names:

the tangent plane:

The Tangent Plane.

The equation of the **tangent plane to the surface** $z = f(x, y)$ at $P_0(x_0, y_0, f(x_0, y_0))$ is

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0. \qquad (4.6.2)$$

Figure 4.6.1 shows the two tangent lines inside the planes $y = y_0$ and $x = x_0$, together with the plane that contains both of them.

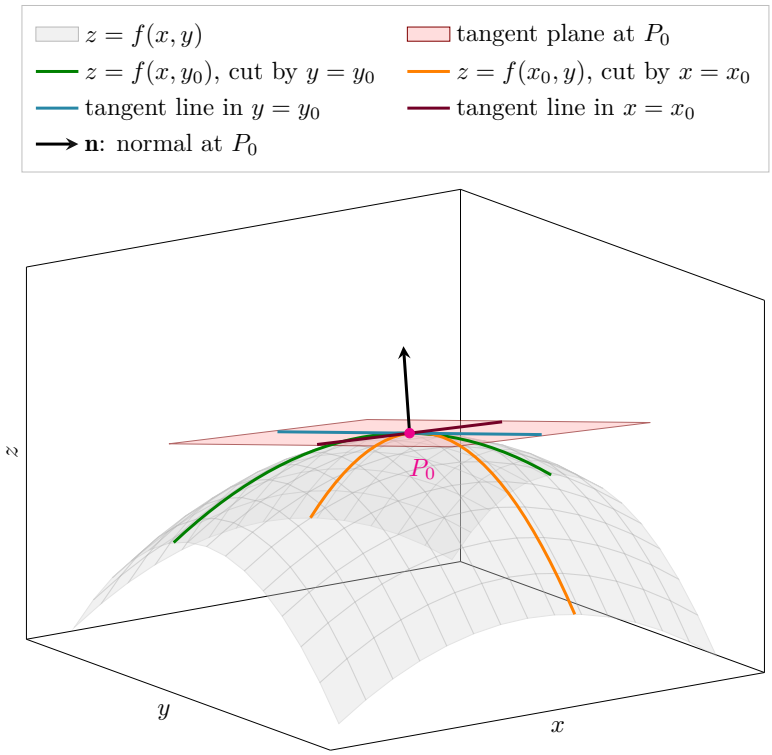


Figure 4.6.1 The two vertical planes $y = y_0$ and $x = x_0$ cut the surface $z = f(x, y)$ along the green and amber curves of Figure 4.3.1 and Figure 4.3.2, whose tangent lines at P_0 have slopes $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$. The tangent plane at P_0 contains both tangent lines, and its normal vector is $\mathbf{n} = \nabla w\Big|_{P_0}$. Equivalently, $\mathbf{n}$ is given (up to sign) by the cross product of the direction vectors $\hat{\imath} + f_x(x_0, y_0)\,\hat{k}$ and $\hat{\jmath} + f_y(x_0, y_0)\,\hat{k}$ of the two tangent lines.

The same plane can be reached a second way, through a gradient. Write the equation of the surface as $w(x, y, z) = f(x, y) - z = 0$, so that the surface is a level surface of w.

Take the gradient of that level function. (Compute the three partial derivatives of w and assemble ∇w; the third one is where the $-\hat{k}$ comes from. Evaluate at P_0 and compare the result with the coefficients of (4.6.2) — they should be the same three numbers. Say why that makes $\nabla w\Big|_{P_0}$ normal to the tangent plane.)

$\nabla w =$

at P_0, and why it is normal to the plane:

The Normal Vector.

The **normal vector to the tangent plane** of $z = f(x, y)$ at P_0 is

$$\mathbf{n} = \nabla w\Big|_{P_0} = f_x(x_0, y_0)\,\hat{\imath} + f_y(x_0, y_0)\,\hat{\jmath} - \hat{k}, \qquad w(x, y, z) = f(x, y) - z. \tag{4.6.3}$$

Equivalently, $\mathbf{n}$ is given (up to sign) by the cross product of the direction vectors $\hat{\imath} + f_x(x_0, y_0)\,\hat{k}$ and $\hat{\jmath} + f_y(x_0, y_0)\,\hat{k}$ of the two tangent lines.

An interactive figure that rebuilds the picture above one layer at a time — the two slices with their tangent lines, then the tangent plane that contains both lines, then the normal vector — accompanies the online version of these notes.

Finally, $\mathbf{n}$ is the direction vector of the line normal to the surface at P_0. (A line is determined by a point on it and a direction vector; you have both. Write its parametric equations, using (4.6.3) for the direction.)

the normal line at P_0:

The Normal Line.

The parametric equations of the **normal line to the surface** $z =$

$f(x, y)$ at $P_0(x_0, y_0, f(x_0, y_0))$ are

$$\begin{aligned} x(t) &= x_0 + f_x(x_0, y_0)\, t \\ y(t) &= y_0 + f_y(x_0, y_0)\, t \qquad t \in \mathbb{R}. \\ z(t) &= f(x_0, y_0) - t \end{aligned} \tag{4.6.4}$$

4.6.2 Tangent Plane and Normal Line to a Surface

In the first example, we find the normal line and the tangent plane to a paraboloid at a given point.

Example 4.6.2 Tangent Plane and Normal Line to a Paraboloid. Consider the surface $z = f(x, y) = -x^2 - y^2$ and the point $P_0(1, 1, -2)$.

A. Find the equation of the normal line to the surface at P_0.

B. Find the equation of the tangent plane to the surface at P_0.

▲

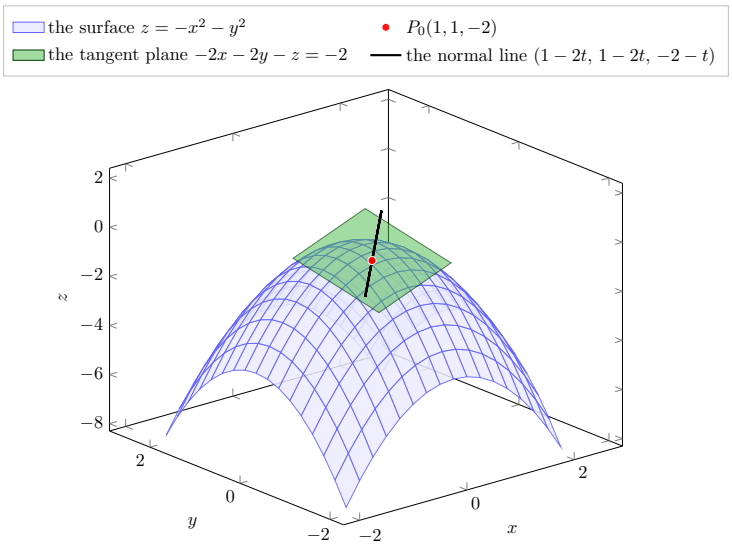


Figure 4.6.3 The graph of $z = -x^2 - y^2$, the tangent plane $-2x - 2y - z = -2$ (green), and the normal line $x = 1 - 2t$, $y = 1 - 2t$, $z = -2 - t$ (black) at the point $P_0(1, 1, -2)$.

Figure 4.6.4 works through this computation from the start: it builds the gradient ∇w, evaluates it at P_0 to get the normal vector, and then draws the normal line and the tangent plane on the surface. Note that the video takes the normal vector in the outward direction, $\mathbf{n} = -\nabla w\big|_{P_0} = \langle 2, 2, 1 \rangle$, which is the opposite of the $\nabla w\big|_{P_0}$ used above; either choice gives the same normal line and the same tangent plane, since the two vectors differ only by the factor -1.

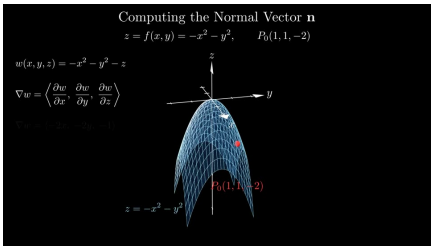


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Figure 4.6.4 Computing the normal vector for Example 4.6.2. The surface, the point $P_0(1, 1, -2)$, the gradient computation, the normal vector, the normal line, and the tangent plane are built up in turn, followed by a slow rotation of the finished figure.

4.6.3 Estimating the Change in a Particular Direction

Before turning to functions of two variables, it is worth recalling the analogous idea from single-variable calculus. For a differentiable function $y = f(x)$, moving a small distance dx away from a point x changes the height of the curve by an amount that can be estimated from the slope $f'(x)$: the resulting **differential** is

$$df = f'(x)\,dx. \tag{4.6.5}$$

As Figure 4.6.5 shows, this differential closely approximates the actual change in the function, $\Delta f = f(x + dx) - f(x)$, as long as dx is small.

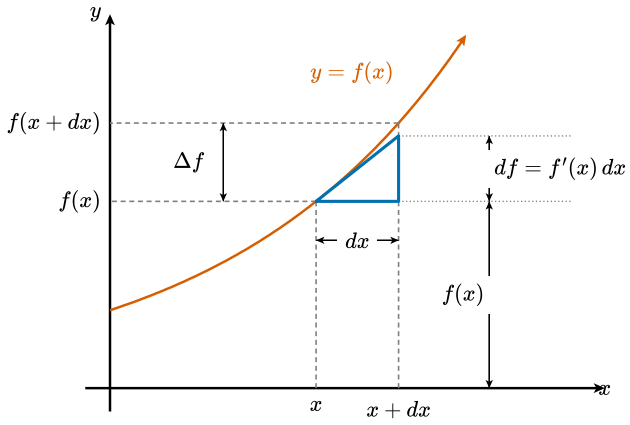


Figure 4.6.5 The tangent line at x approximates $y = f(x)$ near $x + dx$. The differential $df = f'(x)\,dx$, found by walking along the tangent line, closely approximates the actual change $\Delta f = f(x + dx) - f(x)$, found by walking along the curve.

The same question arises for a function of two variables. The difference is that near a point in the plane there is no longer a single direction to move in — moving away from a point requires both a distance and a direction. Consider the function $z = f(x, y)$, a point $P_0(x_0, y_0)$ in its domain, and a unit vector $\hat{u} = u_1\,\hat{\imath} + u_2\,\hat{\jmath}$. The question is:

> *How can we estimate the change in f (Δf) as a result of moving a small distance (ds) away from P_0 in the direction $\hat{u}$?*

Answer the question by following the displacement. (Write the displacement of length ds in the direction $\hat{u}$ in components, add it to P_0 to name the point it reaches, and write down the change it causes in the value of f. Check the

length of your displacement vector: it has to come to ds. Then follow it in Figure 4.6.6.)

the displacement $\hat{u}\,ds$, and its length:

the point P it reaches:

$\Delta f =$

The Exact Change.

Moving away from $P_0(x_0, y_0)$ by the distance ds in the direction $\hat{u}$ takes us to $P(x_0 + u_1\,ds,\ y_0 + u_2\,ds)$ and causes the change

$$\Delta f = f(x_0 + u_1\,ds,\ y_0 + u_2\,ds) - f(x_0, y_0) \tag{4.6.6}$$

in the value of the function.

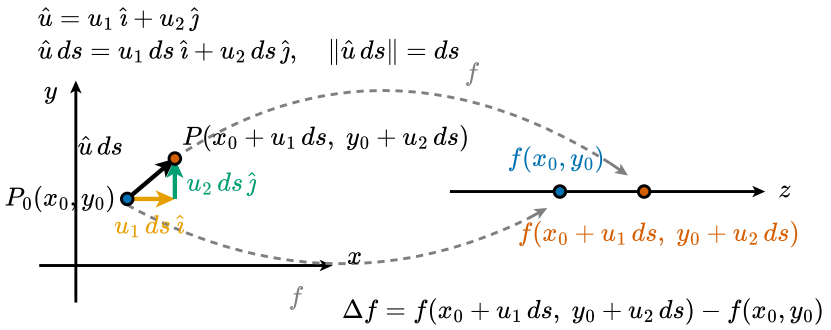


Figure 4.6.6 Moving away from $P_0(x_0, y_0)$ by the distance ds in the direction $\hat{u} = \langle u_1, u_2 \rangle$ causes the change $\Delta f = f(x_0 + u_1\,ds,\ y_0 + u_2\,ds) - f(x_0, y_0)$ in the value of the function.

Computing (4.6.6) exactly means evaluating f twice. The directional derivative gives an estimate that costs only one gradient. ((4.5.4) gives the rate at which f changes per unit distance travelled in the direction $\hat{u}$. Multiply by the distance actually travelled, and say why the result is only an estimate of (4.6.6) rather than equal to it.)

the rate, and the estimate it gives:

why it is only an estimate:

The Differential.

The rate of change of f at P_0 in the direction $\hat{u}$ is

$$\left(\frac{df}{ds}\right)_{\hat{u},P_0} = \nabla f\Big|_{P_0} \cdot \hat{u}, \tag{4.6.7}$$

so the change in f over a distance ds is estimated by

$$\Delta f \approx df = \left(\nabla f\Big|_{P_0} \cdot \hat{u}\right) ds. \tag{4.6.8}$$

The quantity df is called the **differential** of f.

Example 4.6.7 Estimating the Change in a Given Direction. Consider the function $f(x, y) = 1 + x^2 + y^2$, the point $P_0(1, 1)$, and the unit vector $\hat{u} = \frac{1}{\sqrt{2}}\,\hat{\imath} + \frac{1}{\sqrt{2}}\,\hat{\jmath}$. Estimate the change in the value of f as a result of moving away from $P_0(1, 1)$ in the direction of $\hat{u}$ by 0.1 units. ▲

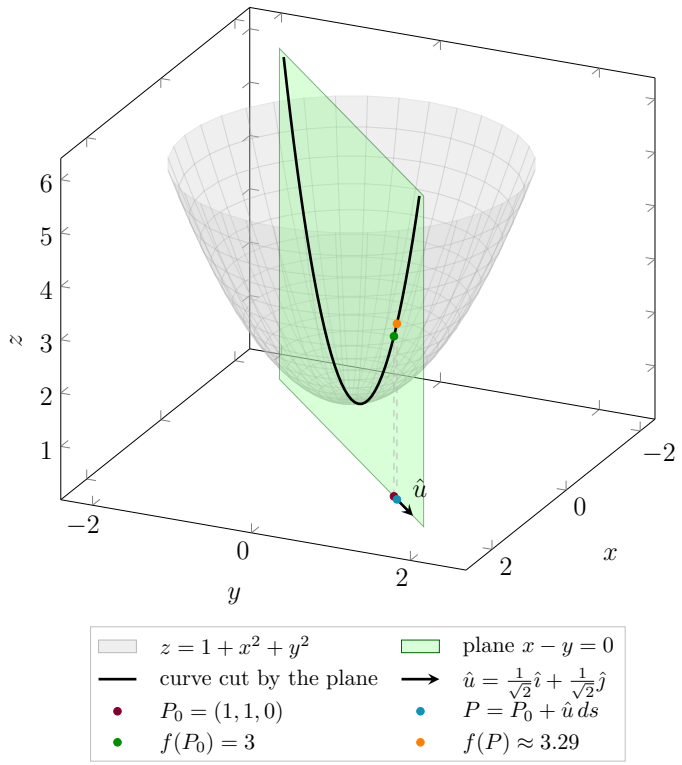


Figure 4.6.8 The graph of $z = 1 + x^2 + y^2$ and the vertical plane $x - y = 0$ that contains the direction $\hat{u}$. Moving from $P_0(1, 1, 0)$ by $ds = 0.1$ along $\hat{u}$ moves the input point (light blue), and the value of f climbs along the black intersection curve from $f(1, 1) = 3$ (green) to the orange point.

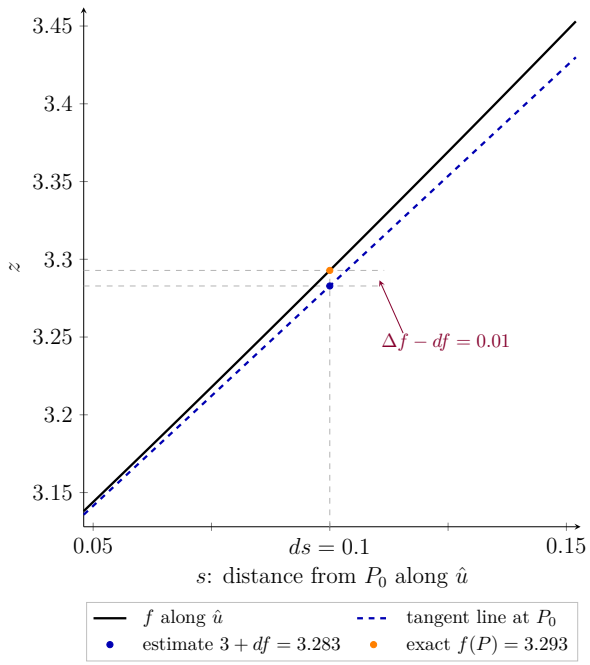


Figure 4.6.9 The curve of Figure 4.6.8 plotted against the distance s travelled along $\hat{u}$, together with its tangent line at P_0. The window starts at $s = 0.05$, past the stretch where the two are indistinguishable, so that the gap between them can be seen: at $s = ds = 0.1$ the tangent line has risen by $df \approx 0.283$ to 3.283, while the curve itself has risen by $\Delta f \approx 0.293$ to 3.293.

4.6.4 Linearizing $z = f(x, y)$

Before generalizing to two variables, recall how a differentiable function $y = f(x)$ is approximated near a point x_0 by its tangent line. This line passes through $\big(x_0, f(x_0)\big)$ with slope $f'(x_0)$, so it is the graph of the linear function

$$L(x) = f(x_0) + f'(x_0)(x - x_0). \tag{4.6.9}$$

As Figure 4.6.10 shows, $L(x)$ matches both the value and the slope of f at x_0, so its graph hugs the curve nearby.

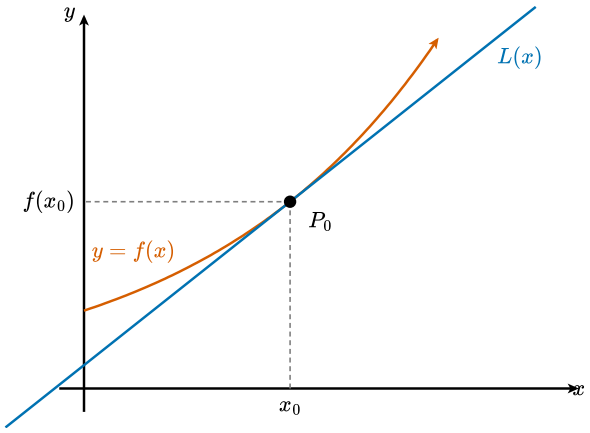


Figure 4.6.10 The tangent line $L(x) = f(x_0) + f'(x_0)(x - x_0)$ approximates $y = f(x)$ near x_0.

The same idea extends to a function of two variables $z = f(x, y)$. Near a point $P_0(x_0, y_0, f(x_0, y_0))$ on the surface, the **tangent plane** plays the role that the tangent line played above, and the single slope $f'(x_0)$ is replaced by the two partial derivatives $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ — one for the rate of change in each coordinate direction. The resulting linear approximation is called the **linearization** of f, defined as follows.

Definition 4.6.11 The **linearization** of a differentiable function $f(x, y)$ at a point (x_0, y_0) is the function

$$L(x, y) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + f(x_0, y_0). \tag{4.6.10}$$

The linearized function $L(x, y)$ can be used as an approximation for $f(x, y)$ near (x_0, y_0). ◆

This is not a new object. (Solve (4.6.2) for z and compare what you get with (4.6.10). Then say, in one sentence, what the graph of L is.)

the tangent plane solved for z, and what the graph of L is:

Example 4.6.12 Linear Approximation of a Function. Consider the function $f(x, y) = -2x^2 - 2xy^3 - 2x$. Approximate $f(x, y)$ near the point $P_0\,(-0.5,\, -0.5,\, f(-0.5, -0.5))$ with a linear function $L(x, y)$. ▲

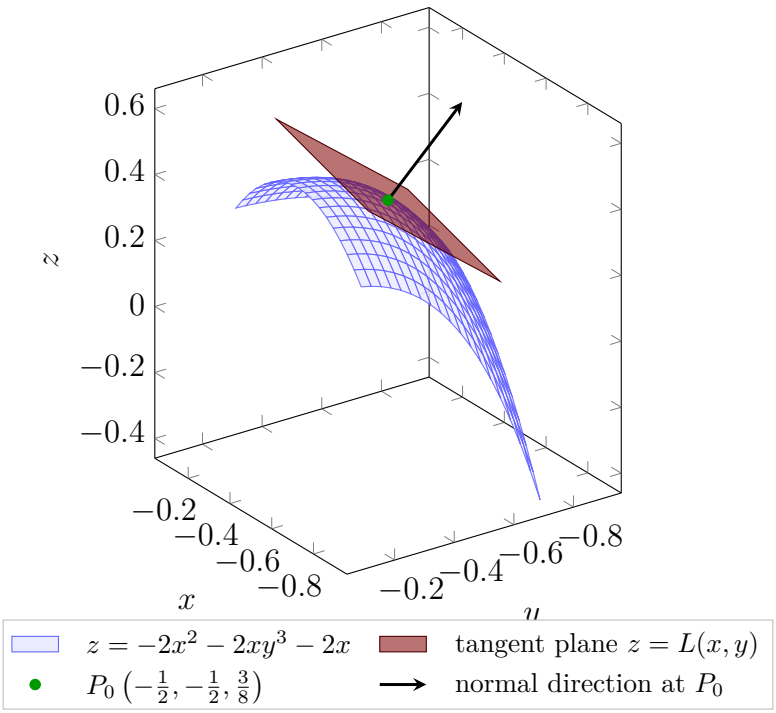


Figure 4.6.13 The graph of $z = -2x^2 - 2xy^3 - 2x$ and the tangent plane at $P_0\left(-\frac{1}{2}, -\frac{1}{2}, \frac{3}{8}\right)$, whose equation is $z = L(x, y) = \frac{1}{4}\left(x + \frac{1}{2}\right) + \frac{3}{4}\left(y + \frac{1}{2}\right) + \frac{3}{8}$. The view is close in on P_0, where the plane hugs the surface, so that $L(x, y)$ approximates $f(x, y)$; farther out the surface curves away from the plane. The arrow at P_0 is the normal direction to the surface.

4.7 Extreme Values and Saddle Points

Objectives

After completing this section, you should be able to:

- Understand what it means for a function of two variables to have a local maximum, a local minimum, or a saddle point, and visualize each on the surface $z = f(x, y)$.
- Recognize critical points as the candidates for local extrema, and understand why a vanishing gradient is necessary but not sufficient for an extremum.
- Understand the idea behind the Second Derivative Test and what it reveals about the local shape of a surface near a critical point.
- Distinguish local extrema from absolute extrema, and understand why the absolute extrema of a function on a closed, bounded region may occur on the boundary.
- Interpret extreme values in a physical setting, such as the equilibrium and stability of a system.

4.7.1 Local Maximum and Local Minimum

Definition 4.7.1 Local Maximum and Local Minimum. Let the domain of the function $f(x, y)$ contain the point (a, b). Then

- $f(a, b)$ is a **local minimum** if $f(a, b) \leq f(x, y)$ for all the points (x, y) in an open disk centered at (a, b).
- $f(a, b)$ is a **local maximum** if $f(a, b) \geq f(x, y)$ for all the points (x, y) in an open disk centered at (a, b).

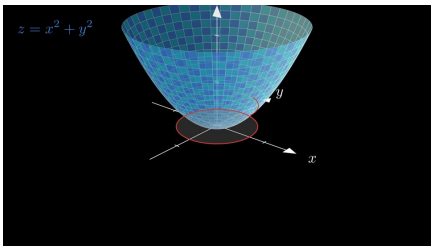
◆

Test the definition on two surfaces you already know, $f(x, y) = x^2 + y^2$ and $f(x, y) = -x^2 - y^2$, at the origin. (For each, decide which of the two inequalities of Definition 4.7.1 holds on a disk around $(0, 0)$, and say why it holds for *every* point of the disk rather than just some. The word "local" is doing work here: say what would change if the disk had to be the whole plane.)

$x^2 + y^2$ at the origin:

$-x^2 - y^2$ at the origin:

What <q>local</q> is doing:



Standalone

Figure 4.7.2 The paraboloid $z = x^2 + y^2$. Over an open disk centered at $(0, 0)$, the value $f(0, 0) = 0$ is smaller than every other value of the function, so $f(0, 0)$ is a local minimum.

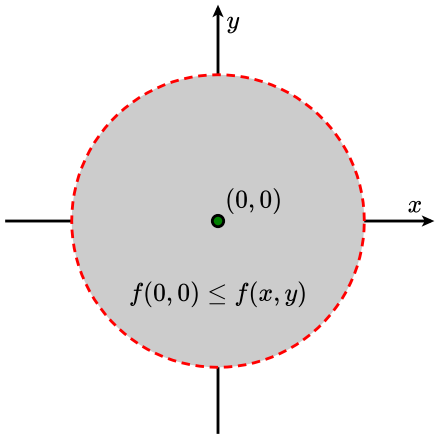


Figure 4.7.3 An open disk in the domain of $f(x, y) = x^2 + y^2$ centered at $(0, 0)$. For every point (x, y) inside the disk we have $f(0, 0) \leq f(x, y)$, so $f(0, 0)$ is a local minimum.

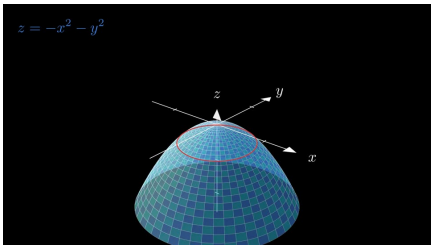


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Figure 4.7.4 The paraboloid $z = -x^2 - y^2$ opens downward. Over an open disk centered at $(0, 0)$, the value $f(0, 0) = 0$ is larger than every other value of the function, so $f(0, 0)$ is a local maximum.

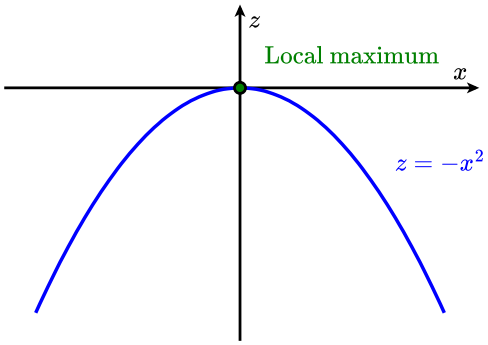

Figure 4.7.5 The trace of the surface $z = -x^2 - y^2$ in the plane $y = 0$ is the parabola $z = -x^2$, which has a maximum at the origin. By symmetry, every vertical cross-section through the origin has the same shape.

Theorem 4.7.6 Theorem I. *If $f(x, y)$ has a local maximum or minimum at an interior point (a, b) of its domain and if the first partial derivatives exist there, then*

$$f_x(a, b) = f_y(a, b) = 0. \tag{4.7.1}$$

Check Theorem 4.7.6 on $f(x, y) = x^2 + y^2$, and then read the theorem carefully in the other direction. (The check is two derivatives. Then ask: does the theorem say that a point where both partials vanish *must* be an extremum? It does not — state which way the implication runs, and why that matters for

how you will use it.)

$f_x(0,0) =$ $f_y(0,0) =$

The implication runs:

Definition 4.7.7 Critical Point. An interior point of the domain of a function $f(x,y)$ where both f_x and f_y are zero or where one or both of f_x and f_y do not exist is a **critical point** of f. Note that not every critical point is a local extremum. ◆

Definition 4.7.8 Saddle Point. A differentiable function $f(x,y)$ has a **saddle point** at a critical point (a,b) if in every open disk centered at (a,b) there are domain points (x,y) where $f(a,b) < f(x,y)$ and domain points (x,y) where $f(a,b) > f(x,y)$. The corresponding point $(a,b,f(a,b))$ on the surface $z = f(x,y)$ is called a saddle point of the surface. ◆

4.7.2 A Saddle Point Example

Example 4.7.9 Saddle Point. Consider the function

$$f(x,y) = \frac{y^2}{3} - \frac{x^2}{3}. \tag{4.7.2}$$

Show that f has a saddle point at $(0,0)$.

Solution. (This is the hyperbolic paraboloid of Subsection 3.8.2. Find the partials and confirm that $(0,0)$ is the only critical point. Then, to match Definition 4.7.8, you need points of *both* kinds in every disk around the origin: restrict f to each coordinate axis in turn and look at the sign. Note at the end that both partials vanish here and yet there is no extremum.)

$f_x =$ $f_y =$ *critical point:*

Along the x-axis: $f(x, 0) =$

Along the y-axis: $f(0, y) =$

Conclusion:

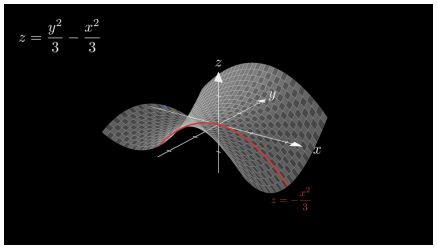

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Figure 4.7.10 The hyperbolic paraboloid $z = y^2/3 - x^2/3$. Along the x-axis the surface falls below the origin, and along the y-axis it rises above the origin, so $(0, 0)$ is a saddle point.

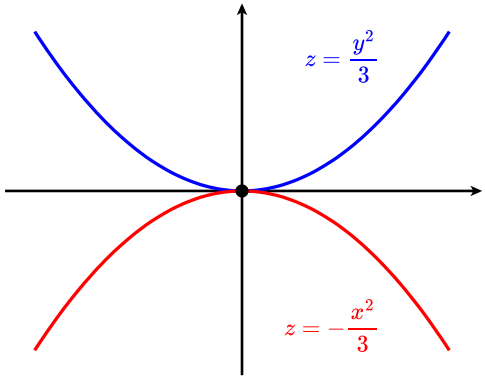


Figure 4.7.11 The traces of $z = y^2/3 - x^2/3$ along the two coordinate planes. In the plane $y = 0$ the trace $z = -x^2/3$ opens downward, while in the plane $x = 0$ the trace $z = y^2/3$ opens upward, so $(0, 0)$ is a saddle point.

▲

4.7.3 The Second Derivative Test

Theorem 4.7.12 Theorem II (The Second Derivative Test). *Suppose that $f(x, y)$ and its first and second partial derivatives are continuous throughout a disk centered at (a, b) and that $f_x(a, b) = f_y(a, b) = 0$. Then*

*i. f has a **local maximum** at (a, b) if $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b).*

*ii. f has a **local minimum** at (a, b) if $f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b).*

*iii. f has a **saddle point** at (a, b) if $f_{xx}f_{yy} - f_{xy}^2 < 0$ at (a, b).*

iv. The test is inconclusive *at (a, b) if $f_{xx}f_{yy} - f_{xy}^2 = 0$ at (a, b). In this case, we must find some other way to determine the behavior of f at (a, b).*

The expression $f_{xx}f_{yy} - f_{xy}^2$ is called the **discriminant** or the **Hessian** of $f(x, y)$, and it is easiest to remember as a determinant:

$$H(x, y) = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - f_{xy}^2. \tag{4.7.3}$$

Before using it, get the shape of the test clear. (Which quantity is consulted *first*, and what does its sign decide? Only in one of those branches does a second quantity get consulted — which, and what does it decide there? Note also which entry of (4.7.3) is used twice, and why Theorem 4.3.10 permits that.)

First look at: *which decides:*

Then, if needed, look at: *which decides:*

Why f_{xy} may be used for both off-diagonal entries:

Example 4.7.13 Finding Local Extrema and Saddle Points. Find the local maximum and minimum values and saddle points of the function

$$z = f(x, y) = x^4 + y^4 - 4xy. \tag{4.7.4}$$

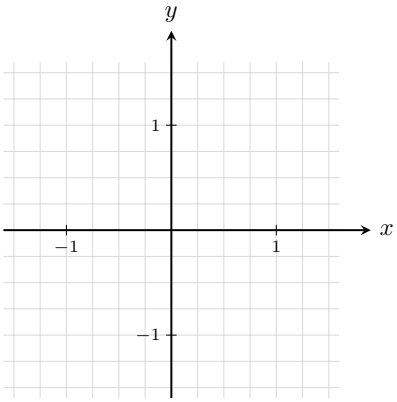
Solution. (Set both first partials to zero. The resulting pair is symmetric in x and y; substitute one into the other to get a single equation in one variable, and factor it completely — only the real roots count. Then form (4.7.3) once, as a formula in x and y, and evaluate it at each critical point rather than recomputing. Mark the critical points on the grid, which is the intersection of the two curves you solved.)

$f_x =$ $f_y =$

Solving:

$H(x, y) =$

Classification of each point:



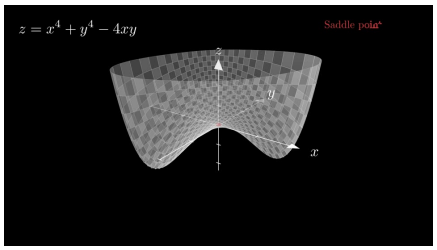


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Figure 4.7.14 The surface $z = x^4 + y^4 - 4xy$ has a saddle point at $(0, 0, 0)$ and two local minima, at $(1, 1, -2)$ and $(-1, -1, -2)$.

▲

Example 4.7.15 Shortest Distance to a Plane. Find the shortest distance from the point $(1, 0, -2)$ to the plane $x + 2y + z = 4$.

Solution. (Write the distance from $(1, 0, -2)$ to a general point of the plane using (3.1.1), then use the plane's equation to eliminate one variable. Minimizing d^2 instead of d avoids the square root — say why the two have their minimum at the same place. Find the single critical point and confirm with the Hessian, then argue that this local minimum is the absolute one. Finally substitute back to get the distance itself.)

$d^2 = f(x, y) =$

Why minimizing d^2 is enough:

Critical point:

$H =$ *so the point is a:*

$d =$

▲

4.7.4 Absolute Maximum and Absolute Minimum

In the next example, we will see how to find absolute extrema of a function $f(x, y)$ over a closed bounded region R.

Example 4.7.16 Absolute Extrema over a Rectangle. Find the absolute maximum and minimum values of the function $f(x, y) = x^2 - 2xy + 2y$ on the rectangle

$$R = \{(x, y) : 0 \le x \le 3,\ 0 \le y \le 2\}. \tag{4.7.5}$$

Solution. (Three steps. First the critical points of f lying *inside* R. Then the boundary, which is not one curve but four edges: on each, one variable is fixed, so f becomes a one-variable function on a closed interval — handle each with single-variable calculus, and do not forget the endpoints. Finally compare every value collected. Note that no second derivative test is needed anywhere here; say why. Label the four edges on Figure 4.7.18 as you go.)

I. Interior critical points:

II. On L_1: $f =$ min: max:

On L_2: $f =$ min: max:

On L_3: $f =$ min: max:

On L_4: $f =$ min: max:

III. Absolute max: absolute min:

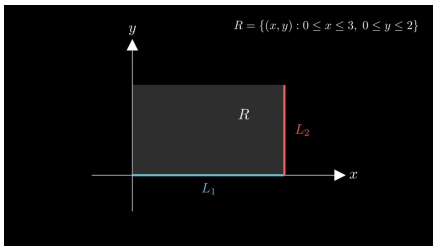

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Figure 4.7.17 The closed bounded rectangle $R = \{(x, y) : 0 \le x \le 3,\ 0 \le y \le 2\}$ with its four edges L_1, L_2, L_3, and L_4, and the critical point $(1, 1)$ in its interior.

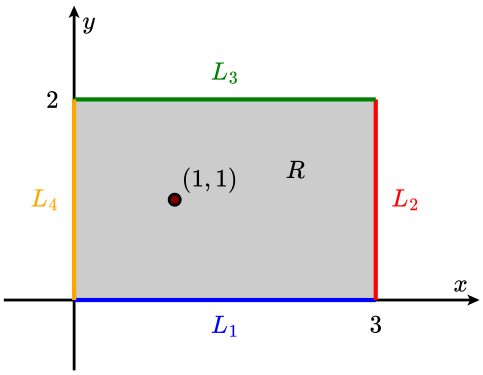

Figure 4.7.18 The rectangle R and its four edges. To find the absolute extrema, we evaluate f at the critical points inside R and compare with the extreme values of f along each edge.

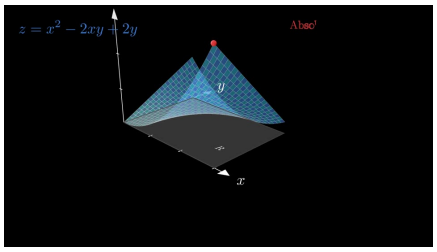

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Figure 4.7.19 The surface $z = x^2 - 2xy + 2y$ over the rectangle R. The absolute maximum $f(3,0) = 9$ and the absolute minimum $f(0,0) = f(2,2) = 0$ occur on the boundary of R.

▲

4.7.5 A Physical Application: Equilibrium and Stability

Consider a marble that rolls on the surface $z = f(x,y)$ under the influence of gravity. If we measure heights from the xy-plane, the potential energy of the marble at the point (x,y) of the surface is

$$U(x,y) = mgz = mg\,f(x,y), \tag{4.7.6}$$

where m is the mass of the marble and g is the acceleration due to gravity. The marble is in **equilibrium** at a point where the force along the surface vanishes, which happens exactly where both partial derivatives of the potential energy are zero:

$$U_x = mg\,f_x = 0 \quad \text{and} \quad U_y = mg\,f_y = 0. \tag{4.7.7}$$

Say what (4.7.7) means mathematically, and what the second derivative test then tells you physically. (Since mg is a nonzero constant, the equilibrium points of the marble coincide with something you have already named. Then match each of the three conclusions of Theorem 4.7.12 to the behaviour of a marble nudged from rest.)

The equilibrium points are exactly:

Local min → *local max →* *saddle →*

Example 4.7.20 Equilibrium and Stability of a Marble. A marble rests at the origin on each of the following three surfaces:

$$\text{(i) } z = x^2 + y^2, \qquad \text{(ii) } z = -x^2 - y^2, \qquad \text{(iii) } z = \frac{y^2}{3} - \frac{x^2}{3}. \tag{4.7.8}$$

Show that $(0, 0)$ is an equilibrium point in each case, and determine whether the equilibrium is stable or unstable.

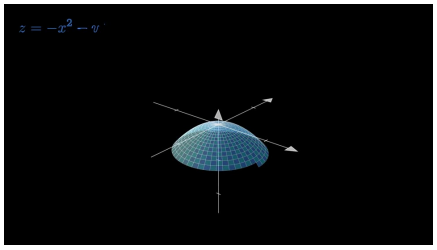
Solution. (All three surfaces have appeared already in this section. For each, check the two first partials vanish at the origin, then compute $H(0,0)$ and $f_{xx}(0,0)$ and read off Theorem 4.7.12. Translate each verdict into what the marble does. Case (iii) is the interesting one: the marble rolls back along one axis and away along the other — say which, and why that still counts as unstable.)

(i) $H(0,0) =$ $f_{xx} =$ *verdict:*

(ii) $H(0,0) =$ $f_{xx} =$ *verdict:*

(iii) $H(0,0) =$ $f_{xx} =$ *verdict:*

Why a saddle is unstable even though one direction restores:



Standalone

Figure 4.7.21 A marble at an equilibrium point on each of the three surfaces. On the bowl $z = x^2 + y^2$ a displaced marble rolls back (stable); on the dome $z = -x^2 - y^2$ it rolls away (unstable); on the saddle $z = y^2/3 - x^2/3$ it rolls back along the y-direction but away along the x-direction (unstable).

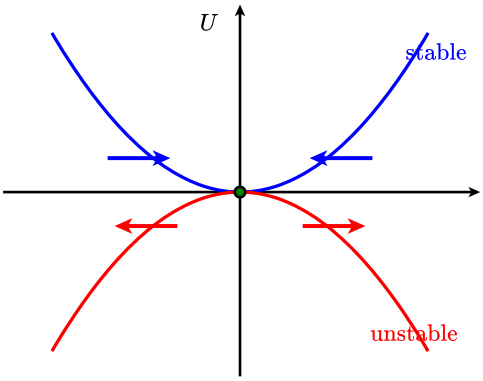


Figure 4.7.22 Cross-sections of the potential energy near an equilibrium point. Along a direction where U has a minimum (blue), the force is restoring and pushes the marble back: stable. Along a direction where U has a maximum (red), the force pushes the marble away: unstable. For the bowl, every cross-section is the blue type; for the dome, every cross-section is the red type; a saddle has one of each.

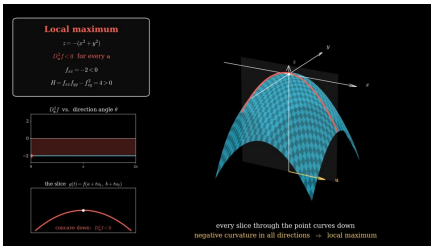
State the general principle these three cases illustrate.

An equilibrium is stable exactly when:

▲

4.7.6 Why the Second Derivative Test Works

Where do the conditions of Theorem 4.7.12 come from? The key idea is to take the directional derivative *twice*. At a critical point every direction looks flat to the first derivative, so we ask the second derivative how the surface *curves* as we walk away from the point in each direction. The video animates the argument; the equations that appear in it are the ones you will derive below.



Standalone

Figure 4.7.23 Slicing a surface through a critical point in a rotating unit direction $\mathbf{u}$. The second directional derivative $D^2_{\mathbf{u}}f$ is the curvature of the slice: one sign for all directions gives a local extremum, both signs give a saddle point, and completing the square shows that the discriminant $H = f_{xx}f_{yy} - f_{xy}^2$ decides between the two.

Slicing the surface in a direction u. Suppose $f(x, y)$ has continuous first and second partial derivatives near a critical point (a, b), so that $f_x(a, b) = f_y(a, b) = 0$. Fix a unit vector $\mathbf{u} = (u_1, u_2)$ and walk away from the critical point along the line through (a, b) in the direction of $\mathbf{u}$. Write down the height of the surface above that line as a single-variable function $g(t)$. (Parametrize the line as in (4.5.2), then feed it into f. This g is the **slice** cut by the vertical plane through (a, b) containing $\mathbf{u}$.)

$g(t) =$

The first directional derivative. Differentiate g once, and evaluate at the critical point. (Use (4.4.1); the two inner derivatives are constants, read straight off the parametrization. Recognize the result as something already named in Subsection 4.5.2. Then set $t = 0$ and use the hypothesis — the answer is the same for *every* $\mathbf{u}$, which is exactly why one derivative cannot tell the three cases apart. Say that in a sentence.)

$g'(t) =$

$g'(0) =$ *for every* $\mathbf{u}$*, because:*

So the first derivative cannot distinguish:

The second directional derivative. Differentiate once more. (Apply the chain rule to each of the two terms of $g'(t)$, remembering that f_x and f_y are themselves functions of x and y. Four second partials appear, but Theorem 4.3.10 lets two of them be combined; the result is a quadratic in u_1 and u_2.)

Differentiating again:

$D^2_{\mathbf{u}} f =$

The Second Directional Derivative.

$$D^2_{\mathbf{u}} f = f_{xx}\, u_1^2 + 2 f_{xy}\, u_1 u_2 + f_{yy}\, u_2^2. \tag{4.7.9}$$

The number $g''(0) = D^2_{\mathbf{u}} f(a, b)$ is the **curvature of the slice** at the critical point. Say what its sign means, and then what the single-variable second derivative test, applied to g, gives in each of three cases. (A positive value means the slice is concave one way, a negative value the other. Then ask what

must be true *across all directions* for a local minimum, for a local maximum, and for a saddle.)

Positive means … *negative means …*

Local minimum if:

Local maximum if:

Saddle point if:

Writing $\mathbf{u} = (\cos\theta, \sin\theta)$ turns (4.7.9) into a function of the direction angle alone. Write it down, and evaluate it for the three surfaces of Example 4.7.20. (Two of the three come out *constant* in θ, which is the whole story for those surfaces; the third does not, and a double-angle identity puts it in a form where you can count how often it changes sign in a full turn.)

$D_{\mathbf{u}}^2 f(\theta) =$

For (i): *for (ii):* *for the saddle:*

Completing the square: the discriminant appears. Checking the sign of (4.7.9) one direction at a time is impossible; there are infinitely many directions. Instead, assume $f_{xx} \neq 0$, multiply (4.7.9) by f_{xx}, and complete the square in u_1. (Multiplying first is what makes the square come out with integer

coefficients. Group the u_1^2 and u_1u_2 terms into a perfect square, and see what has to be added back. Check your work by expanding the square again.)

$f_{xx} \cdot D_{\mathbf{u}}^2 f =$

Completing the square:

Check by expanding:

The Completed Square.

$$f_{xx} \cdot D_{\mathbf{u}}^2 f = (f_{xx}u_1 + f_{xy}u_2)^2 + \left(f_{xx}f_{yy} - f_{xy}^2\right) u_2^2. \qquad (4.7.10)$$

Now say what has been gained. (Every term on the right of (4.7.10) is a square except one coefficient — name it, and match it against (4.7.3). How many numbers now control the sign of $D_{\mathbf{u}}^2 f$ in *all* directions at once, and which are they?)

The one coefficient that is not a square:

The sign in all directions is controlled by:

Reading off the four cases. Work through the three cases from (4.7.10).

(For $H > 0$: both terms on the right are non-negative, and they cannot vanish together — check the two possibilities $u_2 = 0$ and $u_2 \neq 0$ separately. Conclude that $D_{\mathbf{u}}^2 f$ has a fixed sign, and say which. Note also that $H > 0$ rules out $f_{xx} = 0$; show why.)

(For $H < 0$: the two terms now compete, so find *two* directions giving

opposite signs. One is $\mathbf{u} = (1, 0)$; for the other, choose $\mathbf{u}$ so that the perfect square vanishes — the bracket in (4.7.10) tells you which direction that is.)

(For $H = 0$: what does (4.7.10) collapse to, and in which direction does it vanish? Say what the second derivative can and cannot tell you there.)

$H > 0$:

$H < 0$:

$H = 0$:

These are precisely conditions (i)–(iv) of Theorem 4.7.12. One technical remark: the argument above examines only straight-line slices through (a, b), which by itself is not quite enough (a function can increase along every line yet fail to have a local minimum). The full proof replaces f near (a, b) by its second-order Taylor expansion; continuity of the second partials guarantees that when $H \neq 0$ the quadratic form $D^2_{\mathbf{u}} f$ keeps its sign uniformly on a small disk around (a, b), which upgrades the line-by-line conclusion to a genuine local one.

4.8 Lagrange Multipliers

Objectives

After completing this section, you should be able to:

- Recognize a constrained optimization problem: finding the maximum and minimum values of a function f when its inputs are restricted by a constraint of the form $g = 0$, such as a curve in the plane or a surface in space.
- Explain geometrically why, at a constrained extremum, the gradient vectors ∇f and ∇g must be collinear, and why the level curve of f through the extremum is tangent to the constraint curve there.
- State the method of Lagrange multipliers and use it to find the extrema of a function subject to a constraint.
- Solve shortest-distance problems with Lagrange multipliers, and compare the method with the direct approach that uses the normal vector.

Constrained optimizations are everywhere. A robot tracing a surface must keep its hand on that surface, so its coordinates must satisfy the equation of the surface at all times; planning the robot's motion is then a problem of optimizing time, distance, or energy expense subject to that constraint. Fitting a line to data points in an image can also be posed as a constrained optimization problem. In Section 4.7 we found the extreme values of a function $f(x, y)$ over regions of the plane. In this section we learn how to find the extrema of a function whose domain is restricted to a subset of the plane such as a curve, namely the problem

$$\begin{aligned} &\text{maximize or minimize } f(x, y) \\ &\text{subject to the constraint } g(x, y) = 0. \end{aligned} \tag{4.8.1}$$

More generally, one can impose several constraints on a function of many variables; here we work with a single constraint on a function of two or three variables.

4.8.1 The Method of Lagrange Multipliers

4.8.1.1 Why the Gradients Must Be Collinear

The set of points satisfying the constraint $g(x, y) = 0$ is a curve C in the plane: it is precisely the level curve of g of value 0. We showed in Theorem 4.5.7 that the gradient of a function is perpendicular to its level curves, so at every point of C where $\nabla g \neq \mathbf{0}$,

$$\nabla g \cdot \hat{T} = 0, \tag{4.8.2}$$

where $\hat{T}$ is a unit vector tangent to C at that point. To stay on the curve, any small motion away from the point must be along the tangent direction $\hat{T}$.

How does f change as we move along C? By (4.5.4), the rate of change of f in the direction of the unit tangent vector $\hat{T}$ is the directional derivative

$$D_{\hat{T}} f = \nabla f \cdot \hat{T}. \tag{4.8.3}$$

Figure 4.8.1 shows a point of C where this rate is not zero.

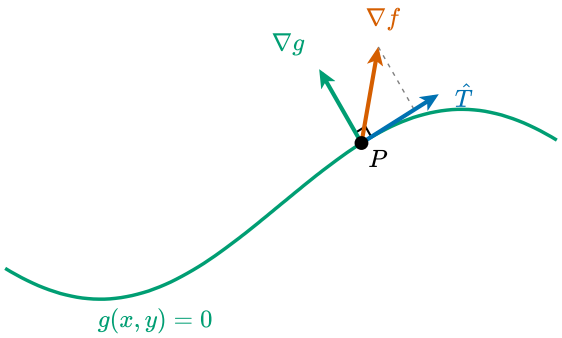


Figure 4.8.1 A point of the constraint curve $g(x, y) = 0$ that is not a constrained extremum. The gradient ∇g is perpendicular to the curve by Theorem 4.5.7, but ∇f is not: it has a nonzero component (dashed) along the tangent direction $\hat{T}$, so f is still changing as we move along the curve.

Turn the picture into an argument. (Say first why the point drawn in Figure 4.8.1 cannot be a constrained extremum — what happens to f along $\hat{T}$, and what happens along $-\hat{T}$. Then write what (4.8.3) must equal at a point that *is* a constrained extremum, put it beside (4.8.2), and finish the argument: in the plane, the vectors perpendicular to one nonzero vector all lie on a single line.)

why the point in the figure is not an extremum:

at a constrained extremum:

what the two conditions together force:

Collinear Gradients at a Constrained Extremum.

At a constrained local maximum or minimum, a small motion along the curve produces no change in f, so

$$\nabla f \cdot \hat{T} = 0. \tag{4.8.4}$$

Both gradients are then perpendicular to $\hat{T}$, hence collinear: there exists some number $\lambda \in \mathbb{R}$ such that

$$\nabla f = \lambda \nabla g. \tag{4.8.5}$$

Figure 4.8.2 puts (4.8.5) in a picture, by superposing the constraint curve $g(x, y) = 0$ onto the family of level curves of f, that is, the curves $f(x, y) = c$. In the figure, $c_1 < c^* < c_3 < c_4 < c_5$.

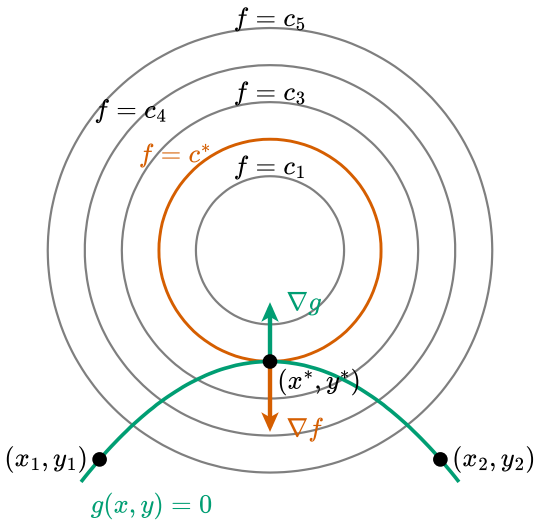


Figure 4.8.2 The constraint curve $g(x, y) = 0$ superposed on the family of level curves of f, with $c_1 < c^* < c_3 < c_4 < c_5$. At the constrained minimum (x^*, y^*), the level curve $f = c^*$ is tangent to the constraint curve and the two gradients are collinear: $\nabla f = \lambda \nabla g$. Here they point in opposite directions, so $\lambda < 0$; it is only the collinearity that matters.

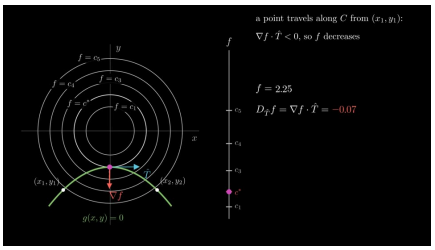
Read the figure by imagining a point moving along the constraint curve from (x_1, y_1) to (x_2, y_2). (At each stage compare the direction of travel with ∇f: the sign of that comparison is the sign of the change in f. Say which value of f is attained at (x^*, y^*), why the motion there is perpendicular to both gradients, and what that makes the level curve $f = c^*$ and the constraint curve do to each other at that point.)

on the way in, from (x_1, y_1)*:*

at (x^*, y^*)*:*

on the way out, to (x_2, y_2)*:*

the two curves at (x^*, y^*)*:*



Standalone

Figure 4.8.3 Figure 4.8.2, set moving. A point travels along the constraint curve from (x_1, y_1) to (x_2, y_2) carrying the unit tangent $\hat{T}$ (blue) and ∇f (red); the shaded band on $\hat{T}$ is the component of ∇f along the direction of travel, and the gauge beside the picture reads off the value of f. The component is negative and shrinking on the way in, zero at (x^*, y^*), and positive on the way out, so f falls to the constrained minimum c^* and rises again. Since $\hat{T}$ is perpendicular to ∇g as well, the two gradients are collinear there, (4.8.5), and the level curve $f = c^*$ is tangent to the constraint curve.

Suppose we find the set S of points (x, y) satisfying the two equations

$$\begin{aligned} g(x, y) &= 0, \\ \nabla f &= \lambda \nabla g \quad \text{for some } \lambda. \end{aligned}$$

Then S contains the local extrema of f subject to the constraint $g(x, y) = 0$. The same reasoning applies to functions of three variables: there the constraint $g(x, y, z) = 0$ defines a level surface of g, the gradient ∇g is perpendicular to that surface, and at a constrained extremum ∇f can have no component tangent to the surface, so once again ∇f and ∇g must be collinear.

4.8.1.2 Stating the Method

Theorem 4.8.4 The Method of Lagrange Multipliers. *Let $f(x, y, z)$ and $g(x, y, z)$ be differentiable functions. To find the local maximum and minimum values of f subject to the constraint $g(x, y, z) = 0$, find the values of x, y, z, and λ that simultaneously satisfy the equations*

$$\nabla f = \lambda \nabla g \qquad \text{and} \qquad g(x, y, z) = 0. \tag{4.8.6}$$

For functions of two variables $f(x, y)$ and $g(x, y)$, the same equations apply, with the variable z omitted.

The number λ is called a **Lagrange multiplier**. The equation $\nabla f = \lambda \nabla g$ is a vector equation, so it holds component by component; together with the constraint it gives four equations in the four unknowns x, y, z, λ (three equations in x, y, λ for functions of two variables). Notice that we are usually not interested in the value of λ itself: it is an auxiliary unknown that we eliminate along the way while solving for the coordinates of the candidate points.

4.8.2 Examples: Functions of Two Variables

4.8.2.1 Extrema on an Ellipse

In the first example, we find the greatest and smallest values of a function whose domain is restricted to a curve in the plane, namely an ellipse.

Example 4.8.5 Extrema of $f(x, y) = xy$ on an Ellipse. Find the greatest and smallest values that the function $f(x, y) = xy$ takes over the ellipse

$$\frac{x^2}{8} + \frac{y^2}{2} = 1. \tag{4.8.7}$$

▲

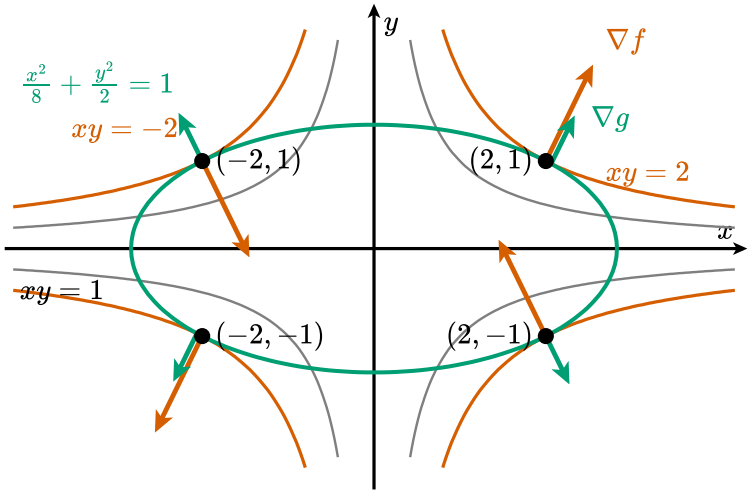


Figure 4.8.6 The ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$ and the level curves of $f(x, y) = xy$. The hyperbolas $xy = \pm 2$ are tangent to the ellipse at the four extremal points, where $\nabla f = \lambda \nabla g$. The two gradients are drawn at all four points: at the maxima $(\pm 2, \pm 1)$ they point the same way ($\lambda = 2$), and at the minima $(\mp 2, \pm 1)$ they point in opposite directions ($\lambda = -2$). Where they overlap, the two arrows are drawn slightly apart so both are visible.

The figure is worth checking against the answer. (Pick the point $(2,1)$, compute ∇f and ∇g there, and confirm that one is a multiple of the other with exactly the λ you found. Then say what the level curves $xy = \pm 2$ do to the ellipse at the four extremal points, and what the gray level curves $xy = \pm 1$ do instead.)

the two gradients at $(2,1)$, and λ:

the extremal level curves, and the others:

4.8.2.2 Absolute Extrema on a Circle

Example 4.8.7 Absolute Extrema on a Circle. Find the absolute maximum and minimum of $f(x,y) = x^2 + y^2 - 2x + 2y + 5$, given the constraint $x^2 + y^2 = 4$. ▲

There is a second, purely geometric route to the same two numbers. (By (**??**), on any point of the plane f is 3 plus the square of the distance to $(1,-1)$. Find the points of the constraint circle farthest from and nearest to $(1,-1)$ — the center of that circle is the origin, and $(1,-1)$ sits inside it — and recompute the two extreme values from their distances.)

farthest point and its distance, and the value of f:

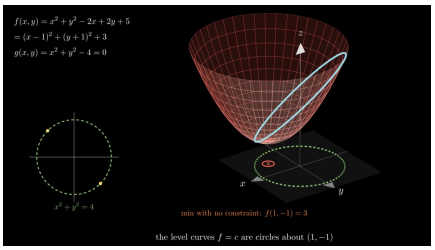
nearest point and its distance, and the value of f:

Note also that in Figure 4.8.9 the gradients ∇f and ∇g point in the same direction at both extrema, but with different multipliers. (Get each one from the component equation $x - 1 = \lambda x$.)

λ *at the minimum:*

λ *at the maximum:*

The video below animates this picture. The completed-square form (**??**) shows that, with no constraint, f would take its least possible value 3 at the vertex $(1, -1)$—the bottom of the paraboloid. But that vertex lies *inside* the constraint circle (a distance $\sqrt{2} < 2$ from the origin), so it is not available to us here: the method of Lagrange multipliers instead locates the highest and lowest points of f *along* the circle. As the level circles of f expand outward from $(1, -1)$, the first one to reach the constraint circle gives the constrained minimum and the last gives the maximum; at each of these contacts the two circles are tangent and ∇f is parallel to ∇g.



Standalone

Figure 4.8.8 The extrema of f on the constraint circle $x^2 + y^2 = 4$ (green). Over that circle the paraboloid carries the light blue curve, and we are after its highest and lowest points. The level circles of f (red), centered at the vertex $(1, -1)$ where $f = 3$, expand until they touch the constraint circle; the first contact is the constrained minimum B and the last is the maximum A, and at both the gradients ∇f and ∇g line up, $\nabla f = \lambda \nabla g$.

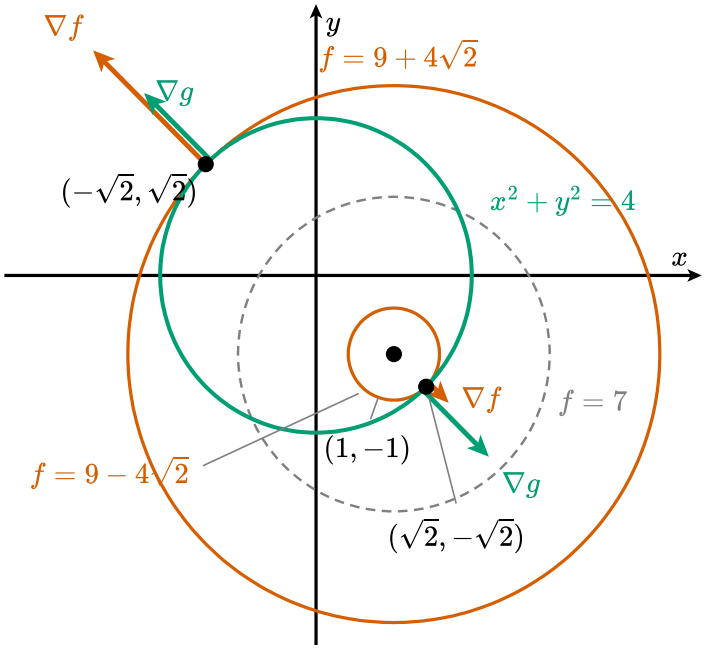


Figure 4.8.9 The constraint circle $x^2 + y^2 = 4$ and three level curves of $f(x, y) = x^2 + y^2 - 2x + 2y + 5$, which are circles centered at $(1, -1)$. The extreme level curves $f = 9 \pm 4\sqrt{2}$ are tangent to the constraint circle, while an intermediate level curve (dashed) crosses it. At each point of tangency $\nabla f = \lambda \nabla g$.

4.8.3 Examples: Functions of Three Variables

4.8.3.1 Shortest Distance to a Plane

In Example 4.7.15 we found the shortest distance from a point to a plane by substituting the equation of the plane into the distance function and minimizing the resulting function of two variables with the second derivative test. In the next example we solve the same problem with Lagrange multipliers, and then once more using the normal vector to the plane.

Example 4.8.10 Shortest Distance to a Plane, Revisited. Find the shortest distance from the point $(1, 0, -2)$ to the plane $x + 2y + z = 4$. ▲

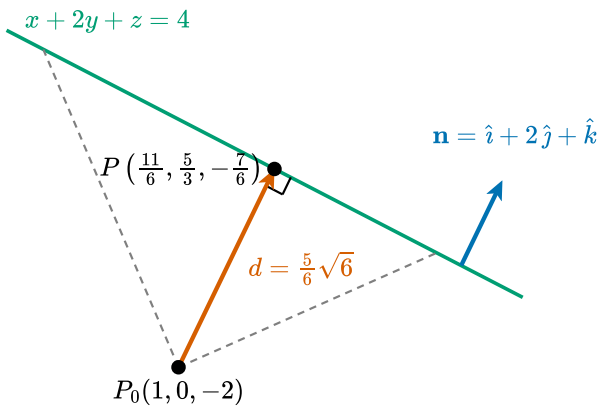


Figure 4.8.11 A schematic, edge-on view of the plane $x + 2y + z = 4$ and the point $P_0(1, 0, -2)$. The shortest segment from P_0 to the plane is along the normal vector $\mathbf{n}$; the dashed segments, which reach the plane in any other direction, are longer.

Another solution, using the normal to the plane. (Figure 4.8.11 says that the shortest segment meets the plane at a right angle, so $\overrightarrow{P_0P}$ is parallel to the normal vector $\mathbf{n}$ of the plane — read $\mathbf{n}$ off the coefficients. Write that parallelism with a scalar, substitute into the equation of the plane, and check that you land on the same point as before.)

$\mathbf{n}$, *and* $\overrightarrow{P_0P}$ *parallel to it:*

substituting into the plane:

the point, and the distance:

The two solutions are more closely related than they look. (Write ∇f at the closest point in terms of $\overrightarrow{P_0P}$, and ∇g in terms of $\mathbf{n}$. Then say in one sentence what the Lagrange condition (4.8.5) is asserting geometrically.)

∇f, ∇g, and what $\nabla f = \lambda \nabla g$ says here:

4.8.3.2 Minimizing the Cost of a Box

Checkpoint 4.8.12 Minimizing the cost of a box. A jewelry box is to be constructed of material that costs \$1 per square inch for the bottom, \$2 per square inch for the sides, and \$5 per square inch for the top. If the total volume is to be 96 in^3, what dimensions will minimize the total cost of construction?

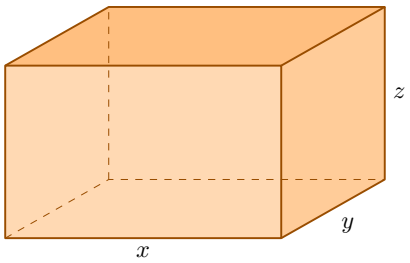


Figure 4.8.13 The box with dimensions x, y, z: bottom and top of area xy, and sides of areas xz and yz.

Solution. (Write the total cost of the six faces as $C(x, y, z)$ — the bottom and the top both have area xy but carry different prices — and the volume requirement as $g = 0$. Set up the Lagrange system; multiplying its three equations by x, y, z respectively makes every right-hand side λxyz, so the left-hand sides can be compared in pairs. Finish with the constraint and report the three dimensions.)

the cost $C(x, y, z)$, and the constraint $g = 0$:

the Lagrange system:

comparing the equations in pairs:

the constraint, and the dimensions:

4.8.4 Functions of n Variables

4.8.4.1 The Method in n Variables

Nothing in the reasoning behind Theorem 4.8.4 depended on the inputs being two or three numbers. The constraint $g = 0$ still describes a set of points on which we are allowed to move; ∇g is still perpendicular to that set; and at a constrained extremum f still cannot be changing along any direction we are allowed to move in, so ∇f can have no component tangent to the constraint set. As before, the only way for a vector to be perpendicular to every allowed direction is for it to point along ∇g. So the method reads the same way for any number of variables.

Theorem 4.8.14 The Method of Lagrange Multipliers in n Variables. *Let f and g be differentiable functions of the n variables $x_1, x_2, \ldots, x_n$. To find the local maximum and minimum values of f subject to the constraint $g(x_1, x_2, \ldots, x_n) = 0$, find the values of $x_1, x_2, \ldots, x_n$ and λ that simultaneously satisfy the equations*

$$\nabla f = \lambda \nabla g \qquad \text{and} \qquad g(x_1, x_2, \ldots, x_n) = 0. \tag{4.8.8}$$

Here the gradient has n components,

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n} \right),$$

and likewise for ∇g. Reading (4.8.8) one component at a time turns it into

$$\frac{\partial f}{\partial x_j} = \lambda \frac{\partial g}{\partial x_j}, \qquad j = 1, 2, \ldots, n,$$

which together with $g = 0$ is a system of $n+1$ equations in the $n+1$ unknowns $x_1, x_2, \ldots, x_n, \lambda$. For $n = 2$ and $n = 3$ this is exactly (4.8.6). The bookkeeping grows, but the strategy does not: eliminate λ, solve for the candidate points, and then compare the values of f there.

One habit does need adjusting. In Example 4.8.5 and Example 4.8.7 the constraint curve was closed and bounded, so the Extreme Value Theorem guaranteed in advance that the absolute maximum and minimum existed and were among the candidate points. In n variables a constraint set is often unbounded, and then no such guarantee comes for free: after solving the system we must argue separately that the point we found really is the extremum we are after. The example below does exactly that.

4.8.4.2 An Inequality Between Two Means

Our last application of the method is not an optimization problem at first sight: it is an inequality, valid for any number of positive numbers, which turns into a constrained optimization problem as soon as the right quantity is held fixed. For positive numbers $x_1, x_2, \ldots, x_n$, the **harmonic mean** and the **geometric mean**[1] are

$$H = \frac{n}{1/x_1 + 1/x_2 + \cdots + 1/x_n}, \qquad G = \sqrt[n]{x_1 x_2 \cdots x_n}. \tag{4.8.9}$$

The example below proves that $H \le G$ always, with equality exactly when all the numbers are the same. If you are curious about where these two averages come from, and about what the inequality amounts to for two numbers, where it can be both checked by hand and seen in a picture, read Remark 4.8.16 after the example; it is optional.

Example 4.8.15 The Harmonic Mean Never Exceeds the Geometric Mean. Let $n \ge 2$ and let $x_1, x_2, \ldots, x_n$ be positive real numbers. Use the method of Lagrange multipliers to prove that

$$\frac{n}{1/x_1 + 1/x_2 + \cdots + 1/x_n} \le \sqrt[n]{x_1 x_2 \cdots x_n}, \tag{4.8.10}$$

that is, $H \le G$, with equality if and only if $x_1 = x_2 = \cdots = x_n$. ▲

One last observation costs nothing and doubles the yield. (Apply (4.8.10) not to the numbers x_j themselves but to their reciprocals $y_j = 1/x_j$. Work out what the harmonic mean and the geometric mean of the y_j are in terms of the arithmetic mean A and the geometric mean G of the x_j, then invert the resulting inequality — both sides are positive, so the direction flips.)

[1] You are not expected to have met these averages before, and nothing later in the course depends on them. The **arithmetic mean** $A = \frac{x_1 + \cdots + x_n}{n}$ is the ordinary average. The **geometric mean** $G = \sqrt[n]{x_1 \cdots x_n}$ is the number that, repeated n times, has the same *product*: for $n = 2$ it is the side of a square with the same area as an a-by-b rectangle, and it is the right average for growth factors, since an investment that grows by factors $x_1, \ldots, x_n$ in successive years grows in total as though it had grown by the constant factor G each year. The **harmonic mean** $H = \frac{n}{1/x_1 + \cdots + 1/x_n}$ is the reciprocal of the average of the reciprocals, and it is the right average for rates: drive one mile at 30 mph and the next at 60 mph and your average speed over the two miles is the harmonic mean 40 mph, not the arithmetic mean 45 mph.

H and G of the reciprocals:

the inequality that results:

The Three Means.

Applying (4.8.10) to the reciprocals gives the celebrated inequality between the geometric and arithmetic means,

$$\sqrt[n]{x_1 x_2 \cdots x_n} \;\le\; \frac{x_1 + x_2 + \cdots + x_n}{n}, \tag{4.8.11}$$

again with equality exactly when all the x_j are equal. Together with the example this establishes the full chain

$$H \le G \le A \tag{4.8.12}$$

for any number of positive numbers—the general form of what Figure 4.8.18 shows for two.

Remark 4.8.16 For the Curious: the Two-Variable Case and a Picture. This remark is optional, and nothing later in the course depends on it. It is here for the reader curious about what the inequality just proved actually says. For two numbers it can be checked with nothing but algebra and, better, read off a single picture, along with the arithmetic mean for good measure.

For $n = 2$ no calculus is needed at all. Write the two positive numbers as a and b. Then

$$H = \frac{2}{\frac{1}{a} + \frac{1}{b}} = \frac{2ab}{a+b}, \qquad G = \sqrt{ab}, \tag{4.8.13}$$

and since $a + b > 0$ and $\sqrt{ab} > 0$ we may multiply and divide freely without reversing the inequality:

$$\begin{aligned} \frac{2ab}{a+b} \le \sqrt{ab} &\iff 2ab \le \sqrt{ab}\,(a+b) \\ &\iff 2\sqrt{ab} \le a + b \\ &\iff 0 \le a - 2\sqrt{ab} + b = \left(\sqrt{a} - \sqrt{b}\right)^2, \end{aligned}$$

where the second line divides through by $\sqrt{ab} > 0$, using $ab = \sqrt{ab} \cdot \sqrt{ab}$. The last line is true for every choice of a and b, and it is an equality exactly when $\sqrt{a} = \sqrt{b}$, that is, when $a = b$.

The middle line of that chain is worth a second look. It says $\sqrt{ab} \le \frac{a+b}{2}$, so for two numbers the statement $H \le G$ is the very same statement as $G \le A$, where A is the arithmetic mean. The two-variable case therefore carries the whole chain (4.8.12) on its own, without any appeal to (4.8.11).

All three means can be read off a single picture, and the geometric mean is the interesting one. It comes from a fact about right triangles: the altitude drawn from the right angle to the hypotenuse is the geometric mean of the two pieces into which it cuts the hypotenuse. In Figure 4.8.17 the right angle is at C, the altitude CD has length h, and it splits the hypotenuse into pieces of lengths a and b. The two small triangles ADC and CDB are both similar to the whole triangle ACB, hence to each other, so matching the short leg of one against the long leg of the other gives

$$\frac{a}{h} = \frac{h}{b} \quad \Longrightarrow \quad h^2 = ab \quad \Longrightarrow \quad h = \sqrt{ab} = G. \tag{4.8.14}$$

Now lay the two lengths a and b end to end and draw the semicircle having the resulting segment as its diameter, as in Figure 4.8.18. The radius is $\frac{a+b}{2} = A$, drawn vertically at the center O. Any triangle inscribed in a semicircle has a right angle at the point on the arc, so the vertical segment DC erected where a meets b is precisely the altitude of (4.8.14): its length is $G = \sqrt{ab}$. Finally drop a perpendicular from D to the radius OC, meeting it at M. Applying (4.8.14) a second time, now to the right triangle ODC with its right angle at D, gives $DC^2 = CM \cdot CO$, so

$$CM = \frac{DC^2}{CO} = \frac{G^2}{A} = \frac{ab}{\frac{a+b}{2}} = \frac{2ab}{a+b} = H.$$

The chain (4.8.12) is now visible rather than computed: $CM \le CD$ because a leg of the right triangle CMD cannot exceed its hypotenuse, and $CD \le CO$ for the same reason in the right triangle ODC. Both become equalities exactly when D coincides with O, that is, when $a = b$.

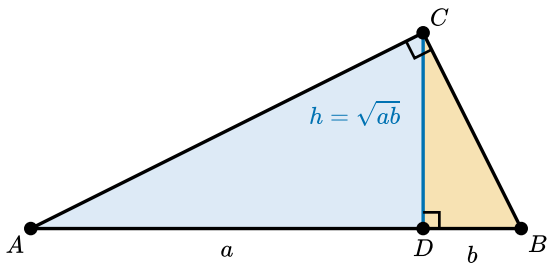


Figure 4.8.17 The altitude from the right angle of a right triangle is the geometric mean of the two pieces of the hypotenuse. Here the right angle is at C, the altitude CD has length h, and it cuts the hypotenuse into pieces of lengths $a = 4$ and $b = 1$; the similar triangles ADC and CDB give $h = \sqrt{ab} = 2$.

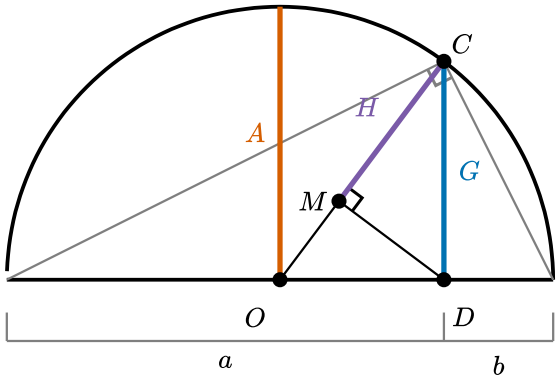


Figure 4.8.18 The three means of $a = 4$ and $b = 1$ in one picture. The semicircle has the segment of length $a + b$ as its diameter, so its radius is the arithmetic mean $A = 2.5$, drawn in red at the center O. The vertical segment DC erected where a meets b is the altitude of the inscribed right triangle, so its length is the geometric mean $G = 2$, drawn in blue. The perpendicular from D to the radius OC cuts off the piece CM, whose length is the harmonic mean $H = 1.6$, drawn in purple. Since a leg of a right triangle is never longer than its hypotenuse, $H \leq G \leq A$.

Chapter 5

Assignments and Review Problems

The ten assignments, the four review problem sets, two sample past exams, and Exam 1, in one place. Worked solutions appear beneath each problem once the assignment's due date has passed; solutions for the review sets are posted the same way.

Using the AI Tutor.

Stuck? A Socratic AI tutor is available on every assignment page. It knows the problems on the page and will coach you with questions and gentle hints — but it will never give away an answer. Here is how to use it.

Opening the tutor. On an assignment page, click the round "Tutor" button in the lower-right corner. A chat panel opens and stays in place as you scroll, so you can read a problem and talk to the tutor at the same time. The dash button in the panel's title bar minimizes it back to the button.

The tutor is available on each of the ten assignment pages. The review problem sets do not have it: those sets are practice for the exams, where you will be entirely on your own.

One-time setup. The first time you open the tutor it asks for a free Google AI Studio key (this takes about a minute):

1. Go to aistudio.google.com/apikey and sign in with a *personal* Google account — a regular Gmail. Do *not* use your `@scu.edu` account: the university has this service turned off, and you will see "Your account is managed by an organization."
2. Click *Create API key* and copy it.
3. Paste it into the tutor.

The key is stored only in your own browser and is sent only to Google; this site never sees it. Treat the key as yours alone: anyone who has it can spend your free quota, so do not share it or paste it anywhere else.

Privacy: what to keep out of the chat. Everything you type into the tutor is sent to Google and handled under Google's terms, not the university's. On the free tier your conversations *may be read by*

people at Google and used to improve their products, and they are not covered by the protections your school records have. Nothing you type is private, so treat the panel as a public place and keep it to mathematics.

- Do not type anything that identifies you: your name, student ID, SCU email, phone number, address, or date of birth. The tutor does not need to know who you are in order to help with a gradient.
- Do not type anything about anyone else — a classmate's name, their work, or their grade. That is their information to share, not yours.
- Keep out anything sensitive or unrelated to the course: health, finances, immigration or visa status, family matters, passwords, or anything you would not want quoted back to you.
- Before you paste or upload, look at what you are sending. A screenshot or a photo of your work can carry your name, a page header, or a browser tab in the background.
- The tutor is a study aid, not a counselor and not the instructor. If something personal is affecting your work in this course, bring it to office hours or to the university's student services instead — those conversations belong with a person, and they stay confidential.

None of this is a reason to avoid the tutor. Ask it anything about the mathematics — what you tried, where you are stuck, why a step feels wrong. Just keep yourself, and everyone else, out of the transcript.

Working with the tutor. Choose the problem you are working on from the drop-down at the top of the panel, then tell the tutor what you have tried so far — the more of your own thinking you show, the more useful its questions become. The tutor can nudge you toward a starting point, and it can point you to sections and worked examples in these notes to study.

If your answer is wrong, the tutor will tell you so and help you understand *why* it is wrong: which step breaks, what misconception is behind it, and what picture or small example will let you see the failure for yourself. Note that this requires showing your reasoning — a bare answer with no work, right or wrong, will simply get asked for the thinking behind it. And what the tutor will not do is tell you how to correct an error — no fixed-up step, no "instead, you should…" Finding the right path once you understand why the old one fails is your part of the work.

Two models. Next to the problem picker is a "Model" drop-down offering *Careful* and *Quick*. Both are Gemini Flash models. Careful is the default: it thinks before it answers, so it takes a few seconds longer and is better at staying in the role of a tutor rather than a solutions manual. Quick is lighter and replies almost instantly. Either will coach you and neither will hand you an answer; switching between them costs nothing and does not disturb your conversation or your hint count. Try both — and tell me which you prefer.

Hints are limited on purpose. The point of the tutor is to get you unstuck, not to walk you to the answer, so hints are deliberately scarce. The exact rules:

- You get at most *three hints per problem, total* — the budget never resets. The panel always shows how many you have left on the problem you selected ("Hints left: 2/3").
- *Every* reply from the tutor costs a hint — there are no free ones. Lead with what you have tried, so each reply can meet your actual thinking instead of spending itself asking for it.
- Restarting the conversation, reloading the page, or switching problems does not refill hints. Spend them thoughtfully — full solutions are posted after the due date.
- When your hints for a problem run out, bring your work to office hours — working through it together is exactly what office hours are for.

Tell me how it is going. Under every hint is a single question — "Was this hint helpful?" — with a thumb up and a thumb down. One tap is the whole thing, and it is the most useful second you can spend on this course: it tells me which hints land and which ones miss, hint by hint. Nothing but your answer and which problem you were on is sent.

For anything longer there is the "feedback" button at the top of the panel, which sends me a short note about how the tutor is working for you: whether it helped, and — in a box you can write as much as you like in — what would make it better. Tapping the thumb down offers it too. All of it is anonymous, it has nothing to do with your grade, and the tutor itself never sees it. If you want me to see the exchange you just had, there is a box to attach it; it is off unless you tick it. The tutor is new, and what you write is how it gets better.

Finally, know what the tutor will and will not do. It will not state answers, complete a solution, or turn a wrong answer into a right one — no matter how you ask. There is exactly one way to get an answer confirmed: state your conclusion *together with your own justification.* If your reasoning is sound and complete, the tutor will tell you that you have it; if there is a gap, it will point at the gap and let you close it. A bare answer, right or wrong, earns nothing.

5.1 Assignment 1

These problems exercise the hyperbolic identities, the derivatives of the hyperbolic functions and the inverse hyperbolic functions. If you would like to review the material first, see Hyperbolic Functions.

1. Show that $\tanh^2(x) + \operatorname{sech}^2(x) = 1$, for all real numbers x.

Solution. We write both terms over the common denominator $\cosh^2(x)$ and use the identity $\cosh^2(x) - \sinh^2(x) = 1$, i.e. $\sinh^2(x) + 1 = \cosh^2(x)$:

$$\begin{aligned}\tanh^2(x) + \operatorname{sech}^2(x) &= \frac{\sinh^2(x)}{\cosh^2(x)} + \frac{1}{\cosh^2(x)} \\ &= \frac{\sinh^2(x)+1}{\cosh^2(x)} = \frac{\cosh^2(x)}{\cosh^2(x)} = 1.\end{aligned}$$

Note that this is valid for every real x, since $\cosh(x) \geq 1 > 0$ and so the denominator is never zero.

2. Compute $\dfrac{d}{dx}\, 3\sinh\left(\dfrac{3}{x}\right)$.

Solution. We use $\dfrac{d}{dx}\sinh(u) = \cosh(u)\,\dfrac{du}{dx}$ with $u = \dfrac{3}{x} = 3x^{-1}$, so $\dfrac{du}{dx} = -3x^{-2} = -\dfrac{3}{x^2}$. Hence

$$\frac{d}{dx}\, 3\sinh\left(\frac{3}{x}\right) = 3\cosh\left(\frac{3}{x}\right)\left(-\frac{3}{x^2}\right) = -\frac{9}{x^2}\cosh\left(\frac{3}{x}\right).$$

3. Simplify $\cosh(\ln 2)$.

Solution. Using the definition $\cosh x = \dfrac{e^x + e^{-x}}{2}$ together with $e^{\ln 2} = 2$ and $e^{-\ln 2} = \dfrac{1}{e^{\ln 2}} = \dfrac{1}{2}$,

$$\cosh(\ln 2) = \frac{e^{\ln 2} + e^{-\ln 2}}{2} = \frac{2 + \frac{1}{2}}{2} = \frac{\frac{5}{2}}{2} = \frac{5}{4}.$$

4. Solve the equation $5\sinh(x) - 3\cosh(x) = 3$ for x.

Solution. We first replace the hyperbolic functions by their definitions:

$$\begin{aligned} & 5\sinh(x) - 3\cosh(x) = 3 \\ \Rightarrow \quad & \frac{5\,(e^x - e^{-x})}{2} - \frac{3\,(e^x + e^{-x})}{2} = 3 \\ \Rightarrow \quad & e^x - 4e^{-x} = 3.\end{aligned}$$

Multiplying through by e^x and writing $u = e^x$ turns this into a quadratic equation:

$$\begin{aligned} & u^2 - 3u - 4 = 0 \\ \Rightarrow \quad & (u-4)(u+1) = 0.\end{aligned}$$

Since $u = e^x$ is positive, the root $u = -1$ is impossible, and only $u = 4$ survives. Therefore

$$e^x = 4 \quad \Longrightarrow \quad x = \ln 4 = 2\ln 2.$$

5. Following the method of Example 1.4.2, show that the inverse hyperbolic tangent is given by

$$\tanh^{-1}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right), \qquad -1 < x < 1.$$

Solution. Unlike cosh, the function tanh is increasing on all of $\mathbb{R}$, so no restriction of its domain is needed. Set $y = \tanh(x)$ and solve for x. Multiplying the numerator and the denominator by e^x gives

$$y = \frac{e^x - e^{-x}}{e^x + e^{-x}} \cdot \frac{e^x}{e^x} = \frac{e^{2x} - 1}{e^{2x} + 1}.$$

Writing $u = e^{2x}$ and clearing the denominator,

$$y(u+1) = u - 1$$

$$\Rightarrow \quad y+1 = u - yu = u(1-y)$$
$$\Rightarrow \quad u = \frac{1+y}{1-y}.$$

Therefore $e^{2x} = \dfrac{1+y}{1-y}$, and taking the natural logarithm gives $2x = \ln\left(\frac{1+y}{1-y}\right)$. Interchanging the names of the two variables, so that $y = \tanh^{-1}(x)$ is the inverse function,

$$y = \tanh^{-1}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right).$$

The range of tanh is the interval $(-1, 1)$, which is exactly the set of x for which $\frac{1+x}{1-x}$ is positive, so this is the domain of $\tanh^{-1}$.

6. Use the substitution $x = \tanh(t)$ to calculate

$$\int \frac{dx}{1-x^2}, \qquad -1 < x < 1,$$

and then use Worksheet Exercise 5.1.5 to write your answer with a logarithm. *Hint:* $\frac{d}{dt}\tanh(t) = \operatorname{sech}^2(t)$, and $1 - \tanh^2(t) = \operatorname{sech}^2(t)$.

Solution. With $x = \tanh(t)$ we have $dx = \operatorname{sech}^2(t)\,dt$, and the denominator collapses by the identity in the hint:

$$\begin{aligned}\int \frac{dx}{1-x^2} &= \int \frac{\operatorname{sech}^2(t)}{1-\tanh^2(t)}\,dt \\ &= \int \frac{\operatorname{sech}^2(t)}{\operatorname{sech}^2(t)}\,dt \\ &= \int dt = t + C.\end{aligned}$$

Since $x = \tanh(t)$ means $t = \tanh^{-1}(x)$, and using Worksheet Exercise 5.1.5 to rewrite the inverse function,

$$\begin{aligned}\int \frac{dx}{1-x^2} &= \tanh^{-1}(x) + C \\ &= \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) + C.\end{aligned}$$

This agrees with the answer obtained by partial fractions, since $\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) = \frac{1}{2}\ln(1+x) - \frac{1}{2}\ln(1-x)$.

5.2 Assignment 2

These problems work with sequences, finite sums and geometric series, ask whether a series converges and what it sums to, and then use the Maclaurin series of the standard functions to recognise a sum. If you would like to review the material first, see Infinite Series and Taylor and Maclaurin Series.

1. The terms of a geometric series $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ satisfy

$$a_5 + a_7 = 1500 \quad \text{and} \quad a_{11} + a_{13} = 187500,$$

where $a_n = ar^{n-1}$ denotes the nth term. Find $a_1 + a_2 + a_3$.

Solution. Using $a_n = ar^{n-1}$,

$$\begin{aligned} a_5 + a_7 &= ar^4 + ar^6 = ar^4(1 + r^2) = 1500, \\ a_{11} + a_{13} &= ar^{10} + ar^{12} = ar^{10}(1 + r^2) = 187500. \end{aligned}$$

Dividing the two equations,

$$\frac{a_{11} + a_{13}}{a_5 + a_7} = \frac{ar^{10}(1 + r^2)}{ar^4(1 + r^2)} = \frac{187500}{1500} \;\Rightarrow\; r^6 = 125 = 5^3 \;\Rightarrow\; r^2 = 5.$$

Substituting back,

$$ar^4(1 + r^2) = 1500 \;\Rightarrow\; a(25)(1 + 5) = 1500 \;\Rightarrow\; a = \frac{1500}{150} = 10.$$

Since $r^2 = 5$, we have $r = \pm\sqrt{5}$, and therefore

$$\begin{aligned} a_1 + a_2 + a_3 &= a + ar + ar^2 = a(1 + r + r^2) \\ &= 10\left(1 \pm \sqrt{5} + 5\right) = 10\left(6 \pm \sqrt{5}\right). \end{aligned}$$

2. What is the sum of all multiples of 7 or 11 less than 1000?

Solution. Note that we need to subtract the multiples of 77, since they have been added twice:

$$\begin{aligned} &\underbrace{(7 + 14 + 21 + \cdots + 77 + \cdots + 994)}_{\text{multiples of 7}} \\ &\quad + \underbrace{(11 + 22 + \cdots + 77 + \cdots + 990)}_{\text{multiples of 11}} \\ &\quad - \underbrace{(77 + 154 + \cdots + 924)}_{\text{multiples of 77}} \\ &= 7(1 + 2 + 3 + \cdots + 142) + 11(1 + 2 + 3 + \cdots + 90) \\ &\quad - 77(1 + 2 + \cdots + 12) \end{aligned}$$

Using $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$:

$$\begin{aligned} &= 7\,\frac{(142)(143)}{2} + 11\,\frac{(90)(91)}{2} - 77\,\frac{(12)(13)}{2} \\ &= 7(71)(143) + 11(45)(91) - 77(6)(13) \\ &= (7)(71)(13)(11) + (11)(13)(7)(45) - (7)(11)(6)(13) \\ &= (7)(11)(13)(71 + 45 - 6) = (7)(11)(13)(110) = 110110, \end{aligned}$$

where we used $143 = (13)(11)$ and $91 = (13)(7)$.

3. What can we conclude by applying the nth term test in the following series?

A. $\displaystyle\sum_{n=1}^{\infty} \frac{(5n+3)^{2n}}{(7n^2+2)^n}$

B. $\displaystyle\sum_{n=1}^{\infty} \frac{2\ln(n)}{3n}$

Solution.

A.

$$\begin{aligned}\lim_{n\to\infty} \frac{(5n+3)^{2n}}{(7n^2+2)^n} &= \lim_{n\to\infty} \left(\frac{(5n+3)^2}{7n^2+2}\right)^n \\ &= \lim_{n\to\infty} \left(\frac{25n^2+30n+9}{7n^2+2}\right)^n \\ &= \lim_{n\to\infty} \left(\frac{25}{7}\right)^n = \infty \;\Rightarrow\; \text{the series diverges.}\end{aligned}$$

B.

$$\begin{aligned}\lim_{n\to\infty} \frac{2\ln(n)}{3n} &\overset{\text{H}}{=} \frac{2}{3}\lim_{n\to\infty} \frac{1/n}{1} = 0 \\ &\Rightarrow \text{the } n\text{th Term Test is inconclusive.}\end{aligned}$$

4. Evaluate the following sums or show that they diverge.

A. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$

B. $\displaystyle\sum_{n=0}^{\infty} e^{-2n}$

C. $\displaystyle\sum_{n=0}^{\infty} e^{2n}$

D. $\displaystyle\sum_{n=1}^{\infty} \left(1+\frac{4}{n}\right)^n$

E. $\displaystyle\sum_{n=2}^{\infty} \frac{2}{3^n}$

Solution.

A. We use partial fractions:

$$\begin{aligned}\frac{1}{n(n+1)(n+2)} &= \frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+2} \\ &= \frac{A(n+1)(n+2) + Bn(n+2) + Cn(n+1)}{n(n+1)(n+2)}.\end{aligned}$$

Setting $n=0$ gives $2A=1$, so $A=\frac{1}{2}$; setting $n=-1$ gives $-B=1$, so $B=-1$; and setting $n=-2$ gives $2C=1$, so $C=\frac{1}{2}$. Hence

$$\begin{aligned}&\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} \\ &= \sum_{n=1}^{\infty} \frac{1}{2}\left(\frac{1}{n} - \frac{2}{n+1} + \frac{1}{n+2}\right)\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right) - \frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{1}{n+1}-\frac{1}{n+2}\right)\\
&= \frac{1}{2}\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right) - \frac{1}{2}\sum_{n=2}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right)\\
&= \frac{1}{2}\left(1-\frac{1}{2}\right)\\
&\quad + \underbrace{\frac{1}{2}\sum_{n=2}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right) - \frac{1}{2}\sum_{n=2}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right)}_{\text{zero}}\\
&= \frac{1}{4}.
\end{aligned}
$$

B. This is a geometric series with $a = 1$ and $r = \frac{1}{e^2}$:

$$
\begin{aligned}
\sum_{n=0}^{\infty} e^{-2n} &= \sum_{n=0}^{\infty}\left(\frac{1}{e^2}\right)^n = \sum_{n=1}^{\infty}\left(\frac{1}{e^2}\right)^{n-1}\\
&= \frac{1}{1-\frac{1}{e^2}} = \frac{e^2}{e^2-1}.
\end{aligned}
$$

C. $\lim_{n\to\infty} a_n = \lim_{n\to\infty} e^{2n} = \infty$, so by the nth Term Test the series diverges.

D. Let $L = \lim_{n\to\infty}\left(1+\frac{4}{n}\right)^n$. Then

$$
\begin{aligned}
\ln L &= \lim_{n\to\infty} n\ln\left(1+\frac{4}{n}\right) = \lim_{n\to\infty}\frac{\ln\left(1+\frac{4}{n}\right)}{\frac{1}{n}}\\
&\overset{\text{H}}{=} \lim_{n\to\infty}\frac{\frac{-4/n^2}{1+4/n}}{-\frac{1}{n^2}} = \lim_{n\to\infty}\frac{4}{1+\frac{4}{n}} = 4.
\end{aligned}
$$

Hence $L = e^4 \neq 0$, so by the nth Term Test the series diverges.

E.

$$
\begin{aligned}
\sum_{n=2}^{\infty}\frac{2}{3^n} &= \sum_{n=2}^{\infty} 2\left(\frac{1}{3}\right)^n\\
&= \frac{2}{1-\frac{1}{3}} - \left(2+\frac{2}{3}\right) = 3 - \frac{8}{3} = \frac{1}{3},
\end{aligned}
$$

where the terms for $n = 0$ and $n = 1$ need to be subtracted.

5. Answer questions A and B below for the following infinite series:

$$\sum_{k=1}^{\infty}\left(\frac{6}{k^2}-\frac{6}{(k+1)^2}\right).$$

A. Does the nth-term test apply? Remember to fully justify your answer.

B. Evaluate the series or show that it diverges.

Solution.

A. We compute the limit of the terms:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}\frac{6}{n^2} - \frac{6}{(n+1)^2} = \lim_{n\to\infty}\frac{6(n+1)^2 - 6n^2}{n^2(n+1)^2}$$

$$= \lim_{n\to\infty} \frac{12n+6}{n^2(n+1)^2} = \lim_{n\to\infty} \frac{12n}{n^4} = 0.$$

Hence the nth-term test does *not* apply: since the limit of the terms is 0, the test is inconclusive.

B. The series telescopes. The partial sum is

$$\begin{aligned} s_n &= \left(\frac{6}{1^2} - \frac{6}{2^2}\right) + \left(\frac{6}{2^2} - \frac{6}{3^2}\right) + \cdots \\ &\quad + \left(\frac{6}{(n-1)^2} - \frac{6}{n^2}\right) + \left(\frac{6}{n^2} - \frac{6}{(n+1)^2}\right) \\ &= 6 - \frac{6}{(n+1)^2}, \end{aligned}$$

and therefore

$$\sum_{k=1}^{\infty} \left(\frac{6}{k^2} - \frac{6}{(k+1)^2}\right) = \lim_{n\to\infty} s_n = \lim_{n\to\infty} 6 - \frac{6}{(n+1)^2} = 6 - 0 = 6.$$

6. Express $8.81818181818\ldots$ as a rational number, i.e. in the form $\frac{p}{q}$, where p and q are positive integers with no common factors.

Solution. We write the repeating decimal as a geometric series:

$$8.81818181818\ldots = 8 + \frac{81}{100} + \frac{81}{10000} + \cdots = 8 + 81\sum_{n=1}^{\infty} \left(\frac{1}{100}\right)^n.$$

Using the geometric series with $a = \frac{1}{100}$ and $r = \frac{1}{100}$,

$$8 + 81\left(\frac{\frac{1}{100}}{1 - \frac{1}{100}}\right) = 8 + \frac{81}{99} = 8 + \frac{9}{11} = \frac{97}{11}.$$

Hence $p = 97$ and $q = 11$.

7. Find the sum of the convergent series $\displaystyle\sum_{n=0}^{\infty} \frac{(-25)^n}{(2n)!}$.

Solution. Recall $\cos x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$. With $x = 5$,

$$\begin{aligned} \cos(5) &= \sum_{n=0}^{\infty} \frac{(-1)^n 5^{2n}}{(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n 25^n}{(2n)!} \\ &= \sum_{n=0}^{\infty} \frac{(-25)^n}{(2n)!}, \end{aligned}$$

so the sum is $\cos(5)$.

8. Find the sum of the series $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n!}$.

Solution. The Maclaurin series of the exponential function is

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \text{valid for all } x \in \mathbb{R}.$$

Setting $x = 1$ gives

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e^1 = e.$$

9. Find the sum of the series

$$5 + 5^2 + \frac{5^3}{2!} + \frac{5^4}{3!} + \cdots + \frac{5^{n+1}}{n!} + \cdots$$

Solution. The general term is $\dfrac{5^{n+1}}{n!}$ starting at $n = 0$. Let us check the first few: at $n = 0$ we get $5^1/0! = 5$; at $n = 1$ we get $5^2/1! = 5^2$; at $n = 2$ we get $5^3/2!$. These match, so the series is

$$\sum_{n=0}^{\infty} \frac{5^{n+1}}{n!}.$$

Now pull out one factor of 5, which does not depend on n:

$$\sum_{n=0}^{\infty} \frac{5^{n+1}}{n!} = \sum_{n=0}^{\infty} \frac{5 \cdot 5^n}{n!} = 5 \sum_{n=0}^{\infty} \frac{5^n}{n!}.$$

The remaining series is the Maclaurin series $e^x = \displaystyle\sum_{n=0}^{\infty} \frac{x^n}{n!}$ evaluated at $x = 5$, which converges for every x. Therefore

$$5 + 5^2 + \frac{5^3}{2!} + \cdots = 5e^5.$$

5.3 Assignment 3

These problems bound the error in a Taylor approximation with the Remainder Theorem and use Taylor series to compute limits and integrals. Problems marked with a star (∗) are for the interested reader: they do not need to be handed in and will not be examined. If you would like to review the material first, see Taylor Series and its convergence.

The starred problems at the end use the binomial series to approximate a physical law; for those, see The Binomial Series and Applications of Taylor Series.

1. For approximately what values of x can we replace $\sin(x)$ by $x - \dfrac{x^3}{6}$ with an error of magnitude no greater than 4×10^{-3}?

Solution. Here $a = 0$, and the cubic $x - \dfrac{x^3}{3!}$ is $T_3(x)$. But the Maclaurin series of sin has no x^4 term, so the same polynomial is also the fourth order Taylor polynomial:

$$\begin{aligned} T_4(x) &= x - \frac{x^3}{3!} + \frac{f^{(4)}(0)}{4!}x^4 \\ &= x - \frac{x^3}{3!} = T_3(x), \qquad \text{since } f^{(4)}(0) = \sin(0) = 0. \end{aligned}$$

We should therefore take $n = 4$, not $n = 3$: both choices describe the very same approximation, but the remainder R_4 is one order smaller than R_3 and so gives the sharper estimate.

The derivatives of $f(x) = \sin(x)$ are

$$\begin{aligned} f'(x) &= \cos(x), & f''(x) &= -\sin(x), \\ f^{(3)}(x) &= -\cos(x), & f^{(4)}(x) &= \sin(x), \\ f^{(5)}(x) &= \cos(x), \end{aligned}$$

and therefore $\left|f^{(5)}(c)\right| = |\cos(c)| \le 1$, so we can take $M = 1$.

By the Remainder Theorem with $n = 4$,

$$\text{Error} = |R_4(x)| \le M\,\frac{|x-a|^5}{5!} = \frac{|x|^5}{5!} \le 4 \times 10^{-3},$$

which gives

$$\begin{aligned} |x| &\le \sqrt[5]{(5!)(0.004)} = \sqrt[5]{0.48} \approx 0.863, \\ \Rightarrow \quad & x \in [-0.863,\ 0.863]. \end{aligned}$$

See Figure 5.3.1.

For comparison, stopping at $n = 3$ would have given $|x| \le \sqrt[4]{(4!)(0.004)} \approx 0.556$, a needlessly small interval for exactly the same polynomial.

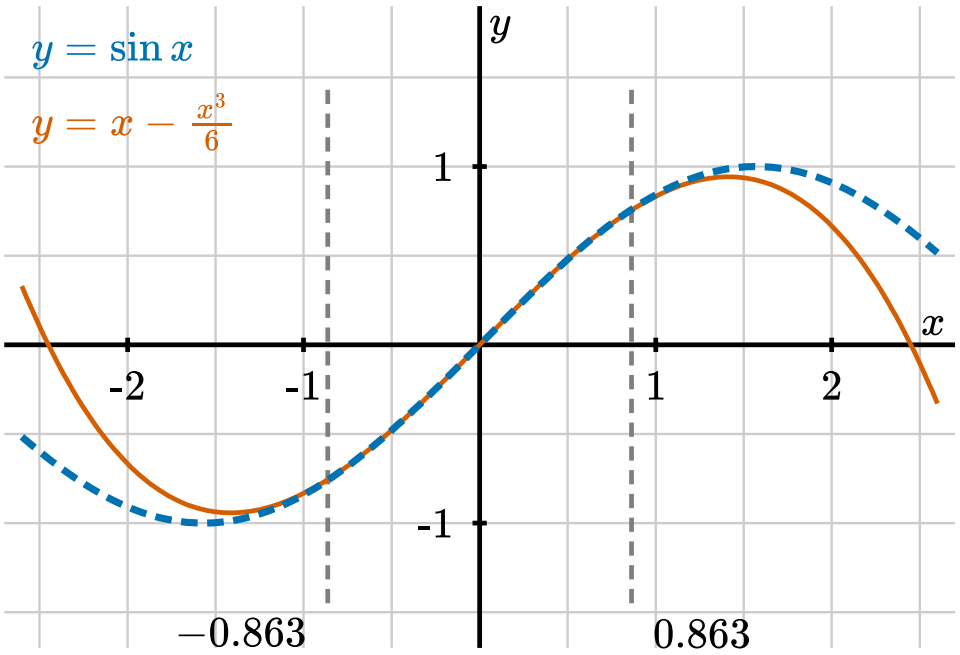


Figure 5.3.1 The function $y = \sin x$ and the cubic $y = x - \frac{x^3}{6}$. Between the dashed lines $x = \pm 0.863$ the two graphs differ by at most 4×10^{-3}.

2. Use the remainder estimation theorem to estimate the maximum error when approximating $f(x) = e^x$ by $T_2(x) = 1 + x + \dfrac{x^2}{2}$ on the interval $\left[-\dfrac{5}{6}, \dfrac{5}{6}\right]$.

Solution. From the remainder theorem, we know that

$$\text{Error} = |R_2(x)| = \left| \frac{f^{(3)}(c)}{3!}\, x^3 \right| \leq \frac{e^c}{3!}\, |x|^3.$$

Also, since $c \in \left[-\frac{5}{6}, \frac{5}{6}\right]$, we have $|c| \leq \frac{5}{6}$ and hence $e^c \leq e^{5/6}$:

$$\text{Error} = |R_2(x)| \leq \frac{e^{5/6}}{3!} \left(\frac{5}{6}\right)^3.$$

We know that $e^{5/6} < e^1 < 3$, hence

$$\text{Error} = |R_2(x)| \leq \frac{e^{5/6}}{3!} \left(\frac{5}{6}\right)^3 < \frac{3}{3!} \left(\frac{5}{6}\right)^3 \approx 0.289.$$

Hence, an estimate for the upper bound of the error is $\dfrac{3}{3!} \left(\dfrac{5}{6}\right)^3 \approx 0.289$. See Figure 5.3.2.

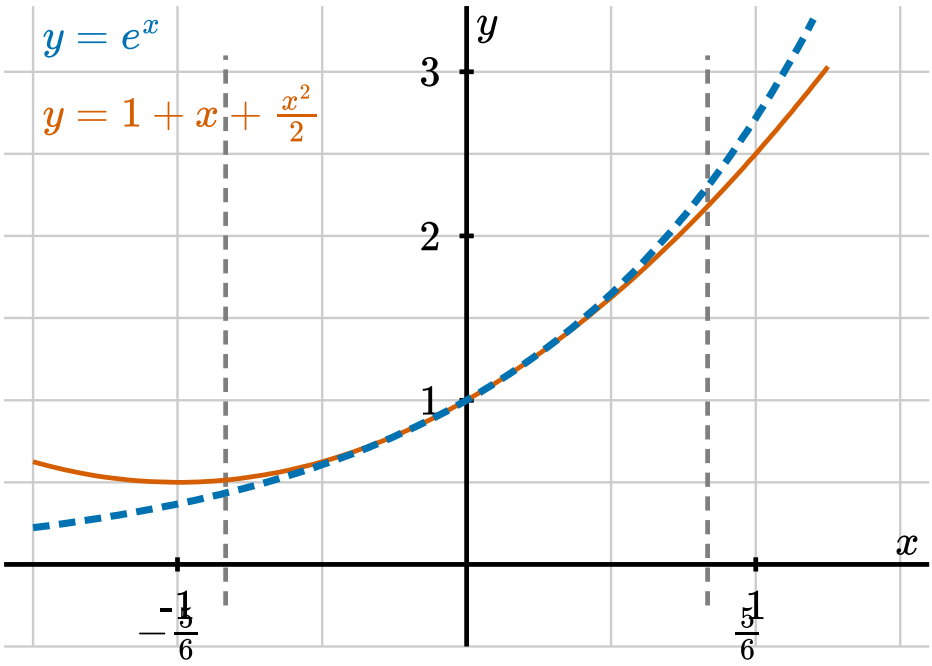


Figure 5.3.2 The function $y = e^x$ and $y = T_2(x) = 1 + x + \frac{x^2}{2}$ on the interval $\left[-\frac{5}{6}, \frac{5}{6}\right]$, marked by the dashed lines.

3. Consider the function $f(x) = \ln(1 + 2x)$.

A. Find the second order Taylor polynomial $T_2(x)$ centered at 0.

B. Estimate the maximum error when approximating $f(x) = \ln(1 + 2x)$ with the second order Taylor polynomial $T_2(x)$ centered at 0, on the interval $[-0.25, 0.25]$.

Solution.

A. We compute the derivatives at 0:

$$f'(x) = \frac{2}{1+2x} = 2(1+2x)^{-1}, \qquad f'(0) = 2,$$
$$f''(x) = -4(1+2x)^{-2}, \qquad f''(0) = -4.$$

Hence

$$T_2(x) = f(0) + f'(0)(x-0) + \frac{f''(0)}{2!}(x-0)^2 = 0 + 2x - 2x^2,$$

i.e. $T_2(x) = 2x - 2x^2$.

B. We have $f^{(3)}(x) = 16(1+2x)^{-3}$ and $-\frac{1}{4} \le x \le \frac{1}{4}$. Note that $f^{(4)}(x) = -96(1+2x)^{-4}$ is negative, which means $f^{(3)}$ is a decreasing function, hence at $x = -\frac{1}{4}$ the value $f^{(3)}(c)$ is maximized, i.e.,

$$f^{(3)}(c) \le f^{(3)}(-0.25) = 16\left(1 - \frac{1}{2}\right)^{-3} = 128.$$

Also

$$-\frac{1}{4} \le x \le \frac{1}{4} \;\Rightarrow\; |x^3| \le \left(\frac{1}{4}\right)^3 = \frac{1}{4^3} = \frac{1}{64},$$

and therefore

$$\text{Error} = |R_2(x)| \le \left|\frac{f^{(3)}(c)}{3!}\,x^3\right| \le \frac{1}{3!}\cdot\frac{1}{64}\cdot 128 = \frac{1}{3}.$$

This means that the error in estimating $f(x) = \ln(1+2x)$ for $-\frac{1}{4} \le x \le \frac{1}{4}$ by $T_2(x) = 2x - 2x^2$ is definitely smaller than $\frac{1}{3}$. See Figure 5.3.3.

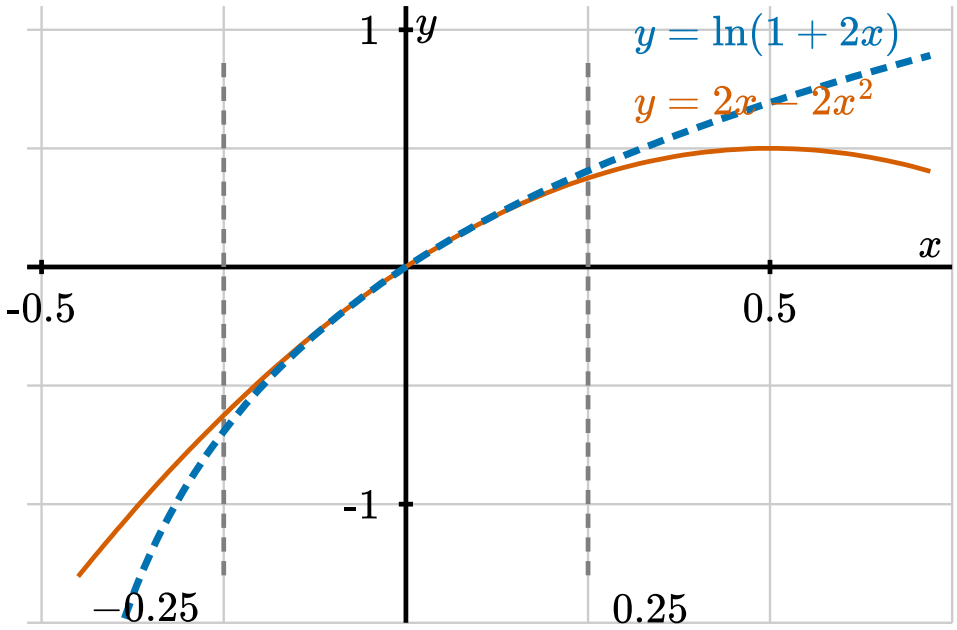


Figure 5.3.3 The function $y = \ln(1+2x)$ and $y = T_2(x) = 2x - 2x^2$. On the interval $[-0.25, 0.25]$, marked by the dashed lines, the two graphs are nearly identical.

4. Let

$$f(x) = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots$$

Find the exact value of $\displaystyle\int_0^{1/13} f(x)\,dx$.

Solution. First we identify f. Its general term is $(n+1)x^n$, so

$$f(x) = \sum_{n=0}^{\infty} (n+1)x^n.$$

This is exactly what you get by differentiating the geometric series term by term:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \qquad |x| < 1,$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1} = 1 + 2x + 3x^2 + \cdots$$

So $f(x) = \dfrac{1}{(1-x)^2}$ for $|x| < 1$. Since $\frac{1}{13} < 1$, the whole interval of integration lies inside the interval of convergence.

Now we integrate. An antiderivative of $(1-x)^{-2}$ is $(1-x)^{-1}$, so

$$\begin{aligned}\int_0^{1/13} \frac{dx}{(1-x)^2} &= \left[\frac{1}{1-x}\right]_0^{1/13} \\ &= \frac{1}{1-\frac{1}{13}} - \frac{1}{1-0} \\ &= \frac{1}{\frac{12}{13}} - 1 = \frac{13}{12} - 1 = \frac{1}{12}.\end{aligned}$$

5. Evaluate $\displaystyle\int \frac{\cos x - 1}{x}\,dx$ as an infinite series. Give the first three nonzero terms.

Solution. The integrand has no elementary antiderivative, so expand it as a series first. Starting from $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \dfrac{x^6}{6!} + \cdots$, subtracting 1 kills the constant term, so every surviving power of x is even and at least 2. Dividing by x therefore leaves an honest power series:

$$\begin{aligned}\frac{\cos x - 1}{x} &= \frac{1}{x}\left(-\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots\right)\\ &= -\frac{x}{2!} + \frac{x^3}{4!} - \frac{x^5}{6!} + \cdots\end{aligned}$$

Integrating term by term,

$$\begin{aligned}\int \frac{\cos x - 1}{x}\,dx &= C - \frac{x^2}{2\cdot 2!} + \frac{x^4}{4\cdot 4!} - \frac{x^6}{6\cdot 6!} + \cdots\\ &= C - \frac{x^2}{4} + \frac{x^4}{96} - \frac{x^6}{4320} + \cdots\end{aligned}$$

6. Use Taylor series to evaluate

$$\lim_{x\to 0} \frac{\cos(2x) - 1}{x^2 e^x}.$$

Solution. The quotient has the indeterminate form $\frac{0}{0}$. Replace the numerator and the denominator by their Maclaurin series. Putting $2x$ in place of x in $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$,

$$\begin{aligned}&\cos(2x) = 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \cdots = 1 - 2x^2 + \frac{2}{3}x^4 - \cdots\\ \Rightarrow\quad &\cos(2x) - 1 = -2x^2 + \frac{2}{3}x^4 - \cdots\end{aligned}$$

and from $e^x = 1 + x + \frac{x^2}{2!} + \cdots$,

$$x^2 e^x = x^2\left(1 + x + \frac{x^2}{2!} + \cdots\right) = x^2 + x^3 + \frac{x^4}{2} + \cdots.$$

Both vanish to exactly second order, so cancel the common factor of x^2 — legitimate, since the limit never uses the value at $x = 0$ itself:

$$\begin{aligned}\lim_{x\to 0}\frac{\cos(2x) - 1}{x^2 e^x} &= \lim_{x\to 0}\frac{x^2\left(-2 + \frac{2}{3}x^2 - \cdots\right)}{x^2\left(1 + x + \frac{x^2}{2!} + \cdots\right)}\\ &= \lim_{x\to 0}\frac{-2 + \frac{2}{3}x^2 - \cdots}{1 + x + \frac{x^2}{2!} + \cdots} = \frac{-2}{1} = -2.\end{aligned}$$

Each remaining series is continuous at 0 with a nonzero value there, so only the two leading coefficients matter.

7. Show that

$$\int_0^1 \frac{\sin(\ln t)}{\ln t}\,dt = \frac{\pi}{4}.$$

(The integrand is continuous on $(0,1)$. As $t \to 1^-$ we have $\ln t \to 0$, so the integrand tends to 1; as $t \to 0^+$ we have $\left|\dfrac{\sin(\ln t)}{\ln t}\right| \le \dfrac{1}{|\ln t|} \to 0$. So the integrand is bounded and the integral exists. Its antiderivative is not elementary.)

Hint. Two facts from Calculus II, both proved by integration by parts[1]: for every integer $k \ge 0$,

$$I_k = \int_0^\infty x^k e^{-x}\,dx = k!,$$

$$J_k = \int_0^1 (\ln t)^k\,dt = (-1)^k\,k!.$$

Either substitute $t = e^{-x}$ and use Example 2.3.13, or expand $\dfrac{\sin(\ln t)}{\ln t}$ as a series in $\ln t$ and integrate term by term with J_k.

Solution. *Solution 1 (substitution).* Let $t = e^{-x}$, so $x = -\ln t$ and $dt = -e^{-x}\,dx$. As t runs from 0 to 1, x runs from ∞ to 0. Since $\ln t = -x$,

$$\frac{\sin(\ln t)}{\ln t} = \frac{\sin(-x)}{-x} = \frac{\sin x}{x}.$$

Therefore

$$\begin{aligned}\int_0^1 \frac{\sin(\ln t)}{\ln t}\,dt &= \int_\infty^0 \frac{\sin x}{x}\,\left(-e^{-x}\right)dx \\ &= \int_0^\infty e^{-x}\,\frac{\sin x}{x}\,dx = \frac{\pi}{4}\end{aligned}$$

by Example 2.3.13.

Solution 2 (directly with series). Substituting $u = \ln t$ into the series $\dfrac{\sin u}{u} = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n u^{2n}}{(2n+1)!}$,

$$\frac{\sin(\ln t)}{\ln t} = \sum_{n=0}^{\infty} \frac{(-1)^n (\ln t)^{2n}}{(2n+1)!}.$$

Integrating term by term and using J_k with $k = 2n$ (so $(-1)^{2n} = 1$),

$$\begin{aligned}\int_0^1 \frac{\sin(\ln t)}{\ln t}\,dt &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}\,(2n)! \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \arctan 1 = \frac{\pi}{4}.\end{aligned}$$

Justification. With $S_N(x) = \displaystyle\sum_{n=0}^{N} \frac{(-1)^n x^{2n}}{(2n+1)!}$, the Remainder Estimation Theorem gives $\left|\dfrac{\sin x}{x} - S_N(x)\right| \le \dfrac{x^{2N+2}}{(2N+3)!}$, exactly as in Example 2.3.13. Replacing x by $u = \ln t$,

$$\left|\frac{\sin(\ln t)}{\ln t} - S_N(\ln t)\right| \le \frac{(\ln t)^{2N+2}}{(2N+3)!}.$$

Integrating over $[0,1]$ with J_k, $k = 2N+2$, bounds the error after $N+1$ terms by

$$\frac{(2N+2)!}{(2N+3)!} = \frac{1}{2N+3} \longrightarrow 0,$$

so the partial sums converge to the integral, and the integral equals $\pi/4$.

8. ∗Approximating The Electric Field due to an Electric Dipole. *Starred problem.* This one is for the interested reader. It will not be examined.

Consider an electric dipole consisting of two charges, $+q$ and $-q$, separated by a distance d. The electric field at a point P located at a distance r from the positive charge along the axis of the dipole is given by:

$$E = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r^2} - \frac{q}{(r+d)^2}\right)$$

[1]For I_k: $I_0 = \int_0^\infty e^{-x}\,dx = 1$, and for $k \ge 1$ integration by parts with $u = x^k$ and $dv = e^{-x}\,dx$ gives $I_k = \Big[-x^k e^{-x}\Big]_0^\infty + k\,I_{k-1} = k\,I_{k-1}$, since $x^k e^{-x} \to 0$ as $x \to \infty$; by induction $I_k = k!$. For J_k: $J_0 = 1$, and for $k \ge 1$ integration by parts with $u = (\ln t)^k$ and $dv = dt$, so $du = k(\ln t)^{k-1}\,\frac{dt}{t}$ and $v = t$, gives $J_k = \Big[t(\ln t)^k\Big]_0^1 - k\,J_{k-1} = -k\,J_{k-1}$. The bracket vanishes at $t = 1$ because $\ln 1 = 0$, and at $t = 0^+$ because $t(\ln t)^k \to 0$ (by L'Hôpital's rule). By induction, $J_k = (-k)(-(k-1))\cdots(-1)\,J_0 = (-1)^k k!$.

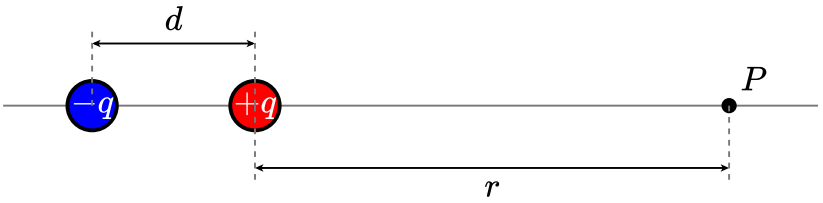


Figure 5.3.4 An electric dipole: charges $+q$ and $-q$ separated by a distance d, with the field point P a distance r from the positive charge along the axis.

Use the Taylor series to approximate the electric field E at point P for $d \ll r$. Show that the leading term in the approximation is proportional to d/r^3.

Solution. To approximate the electric field E at point P for $d \ll r$, we can use the Taylor series expansion for the function $f(x) = \frac{1}{(r+x)^2}$ around $x = 0$.

We have:

$$f(x) = \frac{1}{(r+x)^2} = \frac{1}{r^2}\left(1 + \frac{x}{r}\right)^{-2}$$

Using the binomial series expansion, we can write:

$$\left(1 + \frac{x}{r}\right)^{-2} = 1 - 2\frac{x}{r} + 3\frac{x^2}{r^2} - 4\frac{x^3}{r^3} + \cdots$$

Substituting $x = d$ and $x = 0$, we get:

$$f(d) = \frac{1}{(r+d)^2} = \frac{1}{r^2}\left(1 - 2\frac{d}{r} + 3\frac{d^2}{r^2} - 4\frac{d^3}{r^3} + \cdots\right)$$

$$f(0) = \frac{1}{r^2}$$

Now, we can write the electric field E as:

$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r^2} - \frac{q}{(r+d)^2}\right) \\ &= \frac{q}{4\pi\epsilon_0}\left(\frac{1}{r^2} - \frac{1}{(r+d)^2}\right) \end{aligned}$$

Substituting the Taylor series expansion for $f(d)$ and $f(0)$, we get:

$$E = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{r^2} - \frac{1}{r^2}\left(1 - 2\frac{d}{r} + 3\frac{d^2}{r^2} - 4\frac{d^3}{r^3} + \cdots\right)\right)$$

Simplifying, we have:

$$E = \frac{q}{4\pi\epsilon_0}\left(2\frac{d}{r^3} - 3\frac{d^2}{r^4} + 4\frac{d^3}{r^5} - \cdots\right)$$

Thus, the leading term in the approximation is:

$$E \approx \frac{q}{4\pi\epsilon_0} \cdot 2\frac{d}{r^3}$$

which shows that the leading term is proportional to d/r^3.

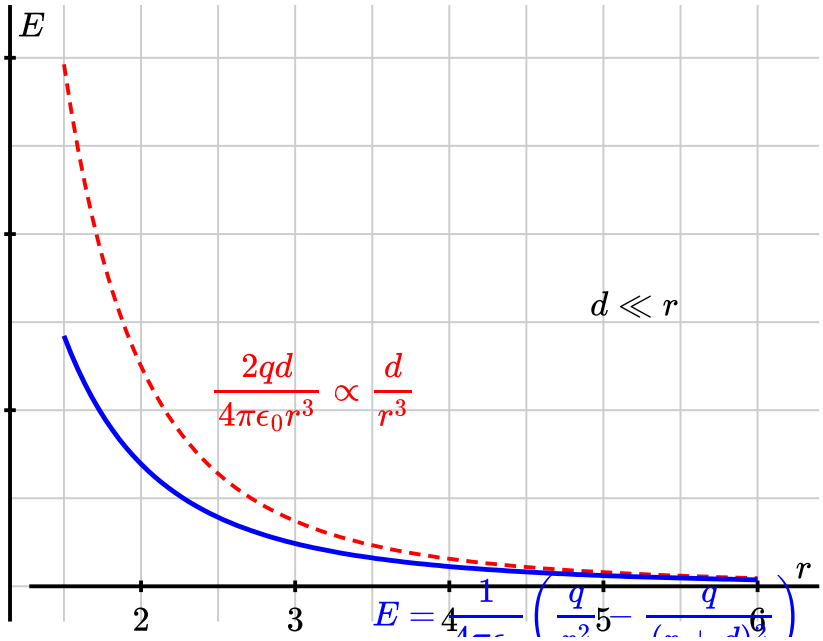


Figure 5.3.5 The axial field of a dipole, $E = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r^2} - \frac{q}{(r+d)^2}\right)$, compared with its leading Taylor term for $d \ll r$. The leading term $\frac{2qd}{4\pi\epsilon_0 r^3} \propto \frac{d}{r^3}$ overshoots slightly at small r but converges to the exact field as r grows.

9. **∗A binomial series for a rational function.** *Starred problem.* This one is for the interested reader. It will not be examined.

Use a **binomial series** to represent the function $f(x) = \dfrac{2x}{1-x}$ as a series. Write your final answer in sigma notation.

Solution. We start from the binomial series with $m = -1$:

$$\frac{1}{1-x} = (1-x)^{-1} = \sum_{k=0}^{\infty} \binom{-1}{k} (-x)^k.$$

The binomial coefficients are

$$\begin{aligned}
\binom{-1}{0} &= 1 = (-1)^0, \\
\binom{-1}{1} &= -1 = (-1)^1, \\
\binom{-1}{2} &= \frac{(-1)(-2)}{2!} = (-1)^2, \\
\binom{-1}{3} &= \frac{(-1)(-2)(-3)}{3!} = (-1)^3,
\end{aligned}$$

and in general $\binom{-1}{k} = (-1)^k$. Hence

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} (-1)^k (-x)^k = \sum_{k=0}^{\infty} x^k.$$

Therefore,

$$f(x) = 2x\,\frac{1}{1-x} = 2x \sum_{k=0}^{\infty} x^k = \sum_{k=0}^{\infty} 2x^{k+1}.$$

10. ∗**Maclaurin series from the binomial series.** *Starred problem.* This one is for the interested reader. It will not be examined.

Use the binomial series (2.4.6) to obtain the Maclaurin series for

A. $\dfrac{1}{(1+x)^3}$. Write your answer in sigma notation, $\displaystyle\sum_{k=0}^{\infty}(\ \cdots\ )$.

B. $\sqrt[7]{1+x}$. Give the first four terms only.

Solution.

A. Here $m=-3$, so

$$\begin{aligned}\frac{1}{(1+x)^3} &= (1+x)^{-3}\\ &= 1-3x+\frac{(-3)(-4)}{2!}x^2+\frac{(-3)(-4)(-5)}{3!}x^3+\cdots\\ &= 1-3x+6x^2-10x^3+\cdots\end{aligned}$$

The coefficient of x^k is

$$\begin{aligned}\frac{(-3)(-4)\cdots(-3-k+1)}{k!} &= (-1)^k\,\frac{3\cdot 4\cdots(k+2)}{k!}\\ &= (-1)^k\,\frac{(k+2)!}{2\,k!},\end{aligned}$$

so

$$\begin{aligned}\frac{1}{(1+x)^3} &= \sum_{k=0}^{\infty}(-1)^k\frac{(k+2)!}{2\,k!}\,x^k\\ &= \sum_{k=0}^{\infty}(-1)^k\frac{(k+1)(k+2)}{2}\,x^k.\end{aligned}$$

As a check, $k=0,1,2,3$ give the coefficients $1,-3,6,-10$ found above.

B. Here $m=\frac{1}{7}$, so

$$\begin{aligned}\sqrt[7]{1+x} &= (1+x)^{1/7}\\ &= 1+\frac{1}{7}x+\frac{\left(\frac{1}{7}\right)\left(-\frac{6}{7}\right)}{2!}x^2\\ &\quad+\frac{\left(\frac{1}{7}\right)\left(-\frac{6}{7}\right)\left(-\frac{13}{7}\right)}{3!}x^3+\cdots\\ &= 1+\frac{x}{7}-\frac{6}{7^2\,2!}x^2+\frac{(6)(13)}{7^3\,3!}x^3+\cdots\\ &= 1+\frac{x}{7}-\frac{3}{49}x^2+\frac{13}{343}x^3+\cdots\end{aligned}$$

11. ∗**Expanding a cube root.** *Starred problem.* This one is for the interested reader. It will not be examined.

Expand the quantity $\sqrt[3]{P+t}$ about 0 in powers of $\dfrac{t}{P}$. Give four nonzero terms.

Solution. The binomial series expands $(1+x)^m$, so first factor P out of the radicand to produce a $1+$ something:

$$\begin{aligned}\sqrt[3]{P+t} &= \left(P+P\left(\frac{t}{P}\right)\right)^{1/3}\\ &= \left(P\left(1+\frac{t}{P}\right)\right)^{1/3}\end{aligned}$$

$$= \sqrt[3]{P}\left(1+\frac{t}{P}\right)^{1/3}.$$

Now apply the binomial series with $m=\frac{1}{3}$ and $x=\frac{t}{P}$:

$$\begin{aligned}\sqrt[3]{P+t} &= \sqrt[3]{P}\left(1+\frac{1}{3}\cdot\frac{t}{P}+\frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\left(\frac{t}{P}\right)^2\right.\\ &\qquad\left.+\frac{\frac{1}{3}\left(\frac{1}{3}-1\right)\left(\frac{1}{3}-2\right)}{3!}\left(\frac{t}{P}\right)^3+\cdots\right)\\ &= \sqrt[3]{P}\left(1+\frac{1}{3}\,\frac{t}{P}-\frac{1}{9}\left(\frac{t}{P}\right)^2+\frac{5}{81}\left(\frac{t}{P}\right)^3+\cdots\right).\end{aligned}$$

12. ∗Gravity above the surface of the Earth. *Starred problem.* This one is for the interested reader. It will not be examined.

The gravitational force exerted by the Earth on an object of mass m at height h above the surface is

$$F=\frac{mgR^2}{(R+h)^2},$$

where R is the radius of the Earth and g is the acceleration due to gravity at the surface.

(a) Express F as a Maclaurin series in the variable h/R, writing out the first four non-zero terms.

(b) What approximation to F do you get by keeping only the first term? When is it reasonable?

Solution. **(a)** The point of the problem is to get the expression into the form $(1+u)^p$ with $u=h/R$. Factor R out of the denominator:

$$F=\frac{mgR^2}{(R+h)^2}=\frac{mgR^2}{R^2\left(1+\frac{h}{R}\right)^2}=mg\left(1+\frac{h}{R}\right)^{-2}.$$

Now use the binomial series $(1+u)^p=\displaystyle\sum_{n=0}^{\infty}\binom{p}{n}u^n$ with $p=-2$. The coefficients are

$$\begin{aligned}\binom{-2}{0}&=1,\\ \binom{-2}{1}&=-2,\\ \binom{-2}{2}&=\frac{(-2)(-3)}{2!}=3,\\ \binom{-2}{3}&=\frac{(-2)(-3)(-4)}{3!}=-4,\end{aligned}$$

and in general $\binom{-2}{n}=(-1)^n(n+1)$. Hence

$$(1+u)^{-2}=1-2u+3u^2-4u^3+\cdots,\qquad |u|<1,$$

and so

$$F=mg\left[1-2\left(\frac{h}{R}\right)+3\left(\frac{h}{R}\right)^2-4\left(\frac{h}{R}\right)^3+\cdots\right].$$

This is valid for $0\le h<R$.

(b) Keeping only the first term gives

$$F\approx mg,$$

the familiar constant weight of an object near the surface of the Earth. It is reasonable when h/R is very small, that is when the object stays close to the surface: the first term we discard is $-2mgh/R$, so the relative error is about $2h/R$. With $R\approx 6371$ km, even at $h=10$ km the relative error is only about 0.3%.

5.4 Review Problems #1

These are the leftover problems from the first four topics of the course, collected here for review. They are not assigned, but they are fair game on an exam. Problems marked with a star (∗) are the exception: those are for the interested reader and will not be examined. The topics they cover, with links to the notes:

- Hyperbolic Functions
- Infinite Series
- Taylor and Maclaurin Series
- Taylor Series and its convergence
- The Binomial Series and Applications of Taylor Series

1. **True or False.** For any real number x, we have $\cosh^2(x) + \sinh^2(x) = \cosh(2x)$.
Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. True.

Solution. True.
First,

$$\cosh(2x) = \frac{1}{2}\left(e^{2x} + e^{-2x}\right).$$

Second,

$$\begin{aligned}
&\cosh^2(x) + \sinh^2(x) \\
&= \frac{1}{4}\left(e^{x} + e^{-x}\right)^2 \\
&\quad + \frac{1}{4}\left(e^{x} - e^{-x}\right)^2 \\
&= \frac{1}{4}\left(2e^{2x} + 2e^{-2x}\right) \\
&= \frac{1}{2}\left(e^{2x} + e^{-2x}\right).
\end{aligned}$$

Comparing the two results gives $\cosh^2(x) + \sinh^2(x) = \cosh(2x)$.

2. If $a, b, a+b$ and ab are positive numbers that form 4 consecutive terms in a geometric sequence, find a.

Solution. Since consecutive ratios in a geometric sequence are equal,

$$\frac{a}{b} = \frac{b}{a+b} = \frac{a+b}{ab}.$$

From the first equality,

$$\begin{aligned}
\frac{a}{b} = \frac{b}{a+b} \;&\Rightarrow\; a^2 + ab = b^2 \;\Rightarrow\; b^2 - ab - a^2 = 0 \\
&\overset{\div a^2}{\Rightarrow}\; \left(\frac{b}{a}\right)^2 - \left(\frac{b}{a}\right) - 1 = 0 \\
\frac{b}{a} = \frac{1 \pm \sqrt{1+4}}{2} \;&\overset{\text{since } b/a>0}{\Rightarrow}\; \frac{b}{a} = \frac{1+\sqrt{5}}{2}. \quad (*)
\end{aligned}$$

From the equality of the first and third fractions,

$$\begin{aligned}
\frac{a}{b} = \frac{a+b}{ab} \;\Rightarrow\; a^2 = a + b \;&\overset{\div a}{\Rightarrow}\; a = 1 + \frac{b}{a} \\
&\overset{(*)}{=}\; 1 + \frac{1+\sqrt{5}}{2} = \frac{3+\sqrt{5}}{2}.
\end{aligned}$$

3. Consider the sequence defined by $a_1 = 1$ and $a_{n+1} = a_n + 3n^2 + 3n + 1$. Find a formula for a_n and use it to compute a_{200}.

Solution. We compute the first few terms:

$$\begin{aligned} n = 1: &\quad a_2 = a_1 + (3 + 3 + 1) = 1 + 7 = 8 \;\Rightarrow\; a_2 = 2^3 \\ n = 2: &\quad a_3 = a_2 + (12 + 6 + 1) = 8 + 19 = 27 \;\Rightarrow\; a_3 = 3^3 \end{aligned}$$

We notice that $a_1 = 1^3$, $a_2 = 2^3$, and $a_3 = 3^3$, hence it seems that $a_n = n^3$.

Let us verify by computing $a_{n+1} - a_n$ using $a_n = n^3$:

$$\begin{aligned} a_{n+1} - a_n &= (n+1)^3 - n^3 \\ &= n^3 + 3n^2 + 3n + 1 - n^3 = 3n^2 + 3n + 1, \end{aligned}$$

which confirms that when $a_n = n^3$, then $a_{n+1} = a_n + 3n^2 + 3n + 1$. Hence,

$$a_{200} = (200)^3 = 8 \times 10^6.$$

4. What can we conclude by applying the nth term test to the series $\displaystyle\sum_{n=1}^{\infty} \frac{3n}{\sqrt{6n^2 + 4n + 5}}$?

Solution.

$$\begin{aligned} \lim_{n\to\infty} \frac{3n}{\sqrt{6n^2 + 4n + 5}} &= \lim_{n\to\infty} \frac{3n}{\sqrt{6}\, n} = \sqrt{\frac{3}{2}} \neq 0 \\ &\Rightarrow \text{ the series diverges.} \end{aligned}$$

5. Evaluate the following sums or show that they diverge.

A. $\displaystyle\sum_{n=0}^{10} e^{-2n}$

B. $\displaystyle\sum_{n=1}^{\infty} \frac{6}{(2n-1)(2n+1)}$

C. $\displaystyle\sum_{n=1}^{\infty} \frac{2n+1}{n^2(n+1)^2}$

D. $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{n}{2n+1}\right)$

E. $\displaystyle\sum_{n=3}^{\infty} \frac{2^n}{3^n}$

Solution.

A. This is a finite geometric sum. Using $\displaystyle\sum_{n=1}^{N} ar^{n-1} = a\,\frac{1-r^N}{1-r}$, which was proved when we first studied geometric sums,

$$\begin{aligned} \sum_{n=0}^{10} e^{-2n} &= \sum_{n=1}^{11} \left(\frac{1}{e^2}\right)^{n-1} \\ &= \frac{1 - \left(e^{-2}\right)^{11}}{1 - e^{-2}} = \frac{1 - e^{-22}}{1 - e^{-2}}. \end{aligned}$$

B. Partial fractions give

$$\frac{6}{(2n-1)(2n+1)} = \frac{A}{2n-1} + \frac{B}{2n+1} = \frac{A(2n+1) + B(2n-1)}{(2n-1)(2n+1)},$$

and setting $n = \frac{1}{2}$ gives $2A = 6$, $A = 3$, while $n = -\frac{1}{2}$ gives $-2B = 6$, $B = -3$. The sum telescopes:

$$\begin{aligned} s_k &= \left(\frac{1}{1} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{7}\right) + \cdots \\ &\quad + \left(\frac{1}{2k-1} - \frac{1}{2k+1}\right) = 1 - \frac{1}{2k+1}, \\ \text{so} \quad & 3\sum_{n=1}^{\infty}\left(\frac{1}{2n-1} - \frac{1}{2n+1}\right) = 3\lim_{k\to\infty} s_k \\ &= 3\lim_{k\to\infty}\left(1 - \frac{1}{2k+1}\right) = 3. \end{aligned}$$

C. Partial fractions (or direct observation) give

$$\frac{2n+1}{n^2(n+1)^2} = \frac{1}{n^2} - \frac{1}{(n+1)^2},$$

so the sum telescopes:

$$\begin{aligned} s_k &= \sum_{n=1}^{k}\left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) \\ &= \left(1 - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{9}\right) + \left(\frac{1}{9} - \frac{1}{16}\right) + \cdots \\ &\quad + \left(\frac{1}{k^2} - \frac{1}{(k+1)^2}\right) = 1 - \frac{1}{(k+1)^2} \\ \Rightarrow \quad & \sum_{n=1}^{\infty}\frac{2n+1}{n^2(n+1)^2} = \lim_{k\to\infty}\left(1 - \frac{1}{(k+1)^2}\right) = 1. \end{aligned}$$

D. $\lim_{n\to\infty} \ln\left(\frac{n}{2n+1}\right) = \ln\left(\frac{1}{2}\right) \neq 0$, so by the nth Term Test the series diverges.

E.

$$\begin{aligned} \sum_{n=3}^{\infty}\frac{2^n}{3^n} &= \sum_{n=4}^{\infty}\left(\frac{2}{3}\right)^{n-1} = \frac{1}{1-\frac{2}{3}} - \left(\frac{4}{9} + \frac{2}{3} + 1\right) \\ &= 3 - \frac{4+6+9}{9} = 3 - \frac{19}{9} = \frac{27-19}{9} = \frac{8}{9}, \end{aligned}$$

where the subtracted terms correspond to $n = 3$, $n = 2$, and $n = 1$ of the shifted series.

6. The government has decided to give a \$1,000 tax rebate to each household in order to stimulate the economy. The government statistics say that each household will spend 80% of the rebate in goods and services. The businesses and individuals who benefited from that 80% will then spend 80% of what they received and so on. The result is called the multiplier effect. What is the total effect of the rebate on the economy?

Solution. The rounds of spending form a geometric series with $a = 1000$ and $r = 0.8$:

$$\begin{aligned} &1000 + 0.8(1000) + (0.8)^2(1000) + \cdots = \sum_{n=1}^{\infty} 1000\,(0.8)^{n-1} \\ &= \frac{1000}{1-0.8} = \frac{1000}{0.2} = 5000 \text{ dollars.} \end{aligned}$$

The successive rounds of spending are shown in Figure 5.4.1.

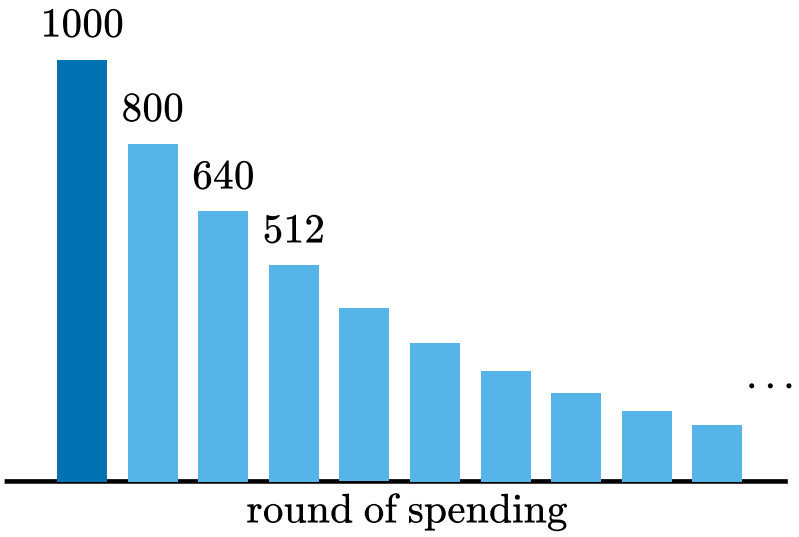


Figure 5.4.1 The multiplier effect. Each round of spending is 80% of the previous one, and the total of all the bars is $\frac{1000}{1-0.8} = 5000$ dollars.

7. Determine whether the series is convergent or divergent and if it is convergent calculate its sum.

A. $\displaystyle\sum_{n=0}^{\infty} \sin\left(\frac{1}{n+1}\right) - \sin\left(\frac{1}{n+2}\right)$

B. $\displaystyle\sum_{n=0}^{\infty} \frac{1+5^n}{3^n}$

Solution.

A. The series telescopes. The partial sum is

$$\begin{aligned} s_k &= \sum_{n=0}^{k} \sin\left(\frac{1}{n+1}\right) - \sin\left(\frac{1}{n+2}\right) \\ &= \sin(1) - \sin\left(\frac{1}{2}\right) + \sin\left(\frac{1}{2}\right) - \sin\left(\frac{1}{3}\right) \\ &\quad + \sin\left(\frac{1}{3}\right) - \sin\left(\frac{1}{4}\right) + \cdots \\ &\quad + \sin\left(\frac{1}{k+1}\right) - \sin\left(\frac{1}{k+2}\right) \\ &= \sin(1) - \sin\left(\frac{1}{k+2}\right). \end{aligned}$$

Hence the series converges and

$$\begin{aligned} \sum_{n=0}^{\infty} \sin\left(\frac{1}{n+1}\right) - \sin\left(\frac{1}{n+2}\right) &= \lim_{k\to\infty} s_k \\ &= \sin(1) - \sin(0) = \sin(1). \end{aligned}$$

B. We split the series into two geometric series:

$$\sum_{n=0}^{\infty} \frac{1+5^n}{3^n} = \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n + \sum_{n=0}^{\infty} \left(\frac{5}{3}\right)^n.$$

Since $\lim_{n\to\infty} \left(\frac{5}{3}\right)^n = \infty$, by the nth term test we can conclude that the series diverges.

8. Determine whether the series converges, and if so find its sum:

$$\sum_{n=1}^{\infty}\left(\frac{n}{n+5}\right)^n.$$

Solution. Note that

$$\begin{aligned}\lim_{n\to\infty} a_n &= \lim_{n\to\infty}\left(1+\frac{5}{n}\right)^{-n}\\ &= \lim_{n\to\infty}\left[\left(1+\frac{5}{n}\right)^{\frac{n}{5}}\right]^{-5} = e^{-5}\neq 0,\end{aligned}$$

since $\left(1+\frac{5}{n}\right)^{n/5}\to e$. Indeed, setting $x=\frac{n}{5}$ and $L=\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$,

$$\begin{aligned}\ln L &= \lim_{x\to\infty} x\ln\left(1+\frac{1}{x}\right) = \lim_{x\to\infty}\frac{\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}}\\ &\overset{\text{H}}{=} \lim_{x\to\infty}\frac{\frac{-1/x^2}{1+1/x}}{-\frac{1}{x^2}} = \lim_{x\to\infty}\frac{1}{1+\frac{1}{x}} = 1\\ &\Rightarrow\ L = e.\end{aligned}$$

Therefore, the series diverges by the nth Term Test.

9. Consider the following series. Answer the following questions.

$$\sum_{n=0}^{\infty}\frac{(x+4)^n}{2^n}$$

A. Find the values of x for which the series converges.

B. Find the sum of the series for those values of x.

Solution.

A. We can write the series as a geometric series:

$$\sum_{n=0}^{\infty}\frac{(x+4)^n}{2^n} = \sum_{n=0}^{\infty}\left(\frac{x+4}{2}\right)^n = \sum_{n=0}^{\infty}\left(\frac{x}{2}+2\right)^n,$$

which converges when $\left|\frac{x}{2}+2\right|<1$:

$$-1<\frac{x}{2}+2<1\ \Rightarrow\ -3<\frac{x}{2}<-1\ \Rightarrow\ -6<x<-2,$$

i.e. the series converges on the interval $(-6,-2)$.

B. When $x\in(-6,-2)$, the geometric series with $a=1$ and $r=\frac{x+4}{2}$ gives

$$\sum_{n=0}^{\infty}\left(\frac{x+4}{2}\right)^n = \frac{1}{1-\frac{x+4}{2}} = \frac{2}{2-x-4} = -\frac{2}{x+2}.$$

10. True or False. A geometric series $\sum_{n=1}^{\infty} ar^{n-1}$ converges if and only if $|r|\le 1$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

As a counterexample consider $\sum_{n=1}^{\infty} 1^{n-1}$, which clearly diverges even though $|r| = 1$. The correct condition is $|r| < 1$.

11. Consider the series $\sum_{k=0}^{\infty} \frac{3^{k-1}}{5^k}$. Does the series converge? If so, what is the result?

Solution. We first factor out $\frac{1}{3}$ so that the powers match:

$$\sum_{k=0}^{\infty} \frac{3^{k-1}}{5^k} = \frac{1}{3}\sum_{k=0}^{\infty} \frac{3^k}{5^k} = \frac{1}{3}\sum_{k=0}^{\infty} \left(\frac{3}{5}\right)^k.$$

Splitting off the $k = 0$ term so that the sum starts at $k = 1$,

$$\frac{1}{3}\sum_{k=0}^{\infty} \left(\frac{3}{5}\right)^k = \frac{1}{3}\left[1 + \sum_{k=1}^{\infty} \left(\frac{3}{5}\right)\left(\frac{3}{5}\right)^{k-1}\right].$$

The remaining sum is a geometric series with $a = \frac{3}{5}$ and $r = \frac{3}{5}$. Since $|r| = \frac{3}{5} < 1$, it converges and $\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}$, so

$$\frac{1}{3}\left[1 + \frac{\frac{3}{5}}{1-\frac{3}{5}}\right] = \frac{1}{3}\left[1 + \frac{3}{2}\right] = \frac{1}{3}\cdot\frac{5}{2} = \frac{5}{6}.$$

12. Find the sum of all of the convergent series.

A. $\sum_{n=0}^{\infty} \frac{\sin\left(\frac{\pi}{2}n\right)}{3^{n+1}}$

B. $\sum_{n=1}^{\infty} \frac{-2}{n(n+1)}$

Solution.

A. The factor $\sin\left(\frac{\pi}{2}n\right)$ cycles through $0, 1, 0, -1$ as $n = 0, 1, 2, 3, \ldots$, so only the odd n contribute, with alternating signs. Writing $n = 2k + 1$,

$$\begin{aligned}\sum_{n=0}^{\infty} \frac{\sin\left(\frac{\pi}{2}n\right)}{3^{n+1}} &= \frac{1}{3^2} - \frac{1}{3^4} + \frac{1}{3^6} - \cdots \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{3^{2k+2}} = \frac{1}{9}\sum_{k=0}^{\infty} \left(-\frac{1}{9}\right)^k.\end{aligned}$$

This is a geometric series with $a = 1$ and $r = -\frac{1}{9}$, and $|r| < 1$, so

$$\frac{1}{9}\cdot\frac{1}{1-\left(-\frac{1}{9}\right)} = \frac{1}{9}\cdot\frac{1}{\frac{10}{9}} = \frac{1}{10}.$$

B. By the constant multiple rule, $\sum_{n=1}^{\infty} \frac{-2}{n(n+1)} = -2\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$. The remaining series telescopes: since $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$,

$$\begin{aligned}s_n &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1} \longrightarrow 1.\end{aligned}$$

Hence the sum is $-2(1) = -2$.

13. True or False. If we manage to find the maximum value of $|f^{(n+1)}(c)|$ for c in the interval between a and x, then we can find the exact error in the Taylor polynomial approximation.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

The statement is incorrect. While finding the maximum value of $|f^{(n+1)}(c)|$ allows us to find an upper bound for the error, it does not give the exact error.

14. For the following functions, give the Taylor *series* generated by the function at $x = 0$ AND give the values of x for which the Taylor series converges to the given function.

A. $f(x) = \cos(x)$

B. $g(x) = \dfrac{1}{1-x}$

C. $h(x) = x^2 e^{x^2}$

D. Using the Taylor series you found for $h(x)$ in part C, find $h^{(12)}(0)$, i.e. the 12th order derivative of $h(x)$ at 0.

Solution.

A. The derivatives of $\cos(x)$ at 0 cycle through $1, 0, -1, 0$, so only even powers survive and

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots,$$

which converges to $\cos(x)$ for all $x \in \mathbb{R}$.

B. This is the geometric series with $a = 1$ and $r = x$:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots,$$

which converges to $\dfrac{1}{1-x}$ exactly when $|x| < 1$, i.e. for $-1 < x < 1$.

C. We start from $e^u = \displaystyle\sum_{n=0}^{\infty} \frac{u^n}{n!}$, valid for all u, and substitute $u = x^2$:

$$e^{x^2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}.$$

Multiplying by x^2,

$$h(x) = x^2 e^{x^2} = \sum_{n=0}^{\infty} \frac{x^{2n+2}}{n!} = x^2 + x^4 + \frac{x^6}{2!} + \frac{x^8}{3!} + \cdots,$$

which converges to $h(x)$ for all $x \in \mathbb{R}$.

D. In the Taylor series $h(x) = \displaystyle\sum_{k=0}^{\infty} \frac{h^{(k)}(0)}{k!} x^k$, the coefficient of x^{12} is $\dfrac{h^{(12)}(0)}{12!}$. From part C, the term with x^{12} occurs when $2n + 2 = 12$, i.e. $n = 5$, and its coefficient is $\dfrac{1}{5!}$. Equating the two,

$$\frac{h^{(12)}(0)}{12!} = \frac{1}{5!} \;\Rightarrow\; h^{(12)}(0) = \frac{12!}{5!} = \frac{479001600}{120} = 3991680.$$

15. Given $f(x) = \ln(1 - 2x)$.

A. Find the 2nd order Maclaurin polynomial for $f(x)$.

B. Use Taylor's Remainder Theorem to find an upper bound on the magnitude of the error if the 2nd order Maclaurin polynomial is used to approximate $f(x)$ for $|x| \le \frac{1}{4}$.

Solution.

A. We compute the derivatives at 0:

$$\begin{aligned} f(x) &= \ln(1-2x), & f(0) &= 0, \\ f'(x) &= \frac{-2}{1-2x} = -2(1-2x)^{-1}, & f'(0) &= -2, \\ f''(x) &= -4(1-2x)^{-2}, & f''(0) &= -4. \end{aligned}$$

Hence

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 = -2x - 2x^2.$$

B. By Taylor's Remainder Theorem,

$$|R_2(x)| \le M\,\frac{|x|^3}{3!},$$
$$\text{where } M \ge \left|f^{(3)}(c)\right| \text{ for all } c \text{ between 0 and } x.$$

Here $f^{(3)}(x) = -16(1-2x)^{-3}$, so $\left|f^{(3)}(c)\right| = \dfrac{16}{|1-2c|^3}$. For $|c| \le \frac{1}{4}$ we have $\frac{1}{2} \le 1-2c \le \frac{3}{2}$, and the fraction is largest when the denominator is smallest, i.e. at $c = \frac{1}{4}$:

$$\left|f^{(3)}(c)\right| \le \frac{16}{\left(\frac{1}{2}\right)^3} = 16 \cdot 8 = 128 = M.$$

Therefore, for $|x| \le \frac{1}{4}$,

$$|R_2(x)| \le 128 \cdot \frac{\left(\frac{1}{4}\right)^3}{3!} = \frac{128}{64 \cdot 6} = \frac{1}{3}.$$

16. What function has Maclaurin series $x - \dfrac{x^3}{2!} + \dfrac{x^5}{4!} - \dfrac{x^7}{6!} + \dfrac{x^9}{8!} - \cdots$?

Solution. Factoring x out of every term,

$$x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^7}{6!} + \cdots = x\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots\right),$$

and the bracket is exactly the Maclaurin series of $\cos(x)$. Hence the function is

$$x\cos(x).$$

17.

A. Use a third Taylor polynomial at $x = 100$ to approximate $\sqrt{102}$.

B. Give an upper bound for the error in using this approximation.

Solution.

A. Let $f(x) = \sqrt{x} = x^{1/2}$ and $a = 100$. Then

$$\begin{aligned} f(x) &= x^{1/2}, & f(100) &= 10, \\ f'(x) &= \frac{1}{2}x^{-1/2}, & f'(100) &= \frac{1}{20}, \\ f''(x) &= -\frac{1}{4}x^{-3/2}, & f''(100) &= -\frac{1}{4000}, \end{aligned}$$

$$f'''(x) = \frac{3}{8}x^{-5/2}, \qquad f'''(100) = \frac{3}{800000}.$$

Hence

$$T_3(x) = 10 + \frac{x-100}{20} - \frac{(x-100)^2}{8000} + \frac{(x-100)^3}{1600000},$$

and evaluating at $x = 102$ (so $x - 100 = 2$),

$$\begin{aligned}\sqrt{102} &\approx 10 + \frac{2}{20} - \frac{4}{8000} + \frac{8}{1600000} \\ &= 10 + \frac{1}{10} - \frac{1}{2000} + \frac{1}{200000}.\end{aligned}$$

B. By Taylor's Remainder Theorem, $|R_3(x)| \le M\dfrac{|x-100|^4}{4!}$, where M bounds $\left|f^{(4)}\right|$ on $[100, 102]$. Since $f^{(4)}(x) = -\frac{15}{16}x^{-7/2}$ and $x^{-7/2}$ is decreasing, the largest value occurs at $x = 100$:

$$M = \frac{15}{16}\cdot 100^{-7/2} = \frac{15}{16\cdot 10^7}.$$

Therefore

$$|R_3(102)| \le \frac{15}{16\cdot 10^7}\cdot\frac{2^4}{24} = \frac{15}{24\cdot 10^7} = \frac{1}{16000000}.$$

18. Use Taylor series to evaluate

$$\lim_{x\to 0}\frac{\arctan x - x}{\sin x - x}.$$

Solution. Numerator and denominator both tend to 0, and each vanishes to third order, so l'Hôpital's rule would have to be applied three times. Replacing each by its Maclaurin series is quicker. From $\arctan x = x - \dfrac{x^3}{3} + \dfrac{x^5}{5} - \cdots$ and $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \cdots$, subtracting x removes the leading term of each:

$$\begin{aligned}\arctan x - x &= -\frac{x^3}{3} + \frac{x^5}{5} - \cdots \\ \sin x - x &= -\frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\end{aligned}$$

Both series begin with x^3. The limit never sees $x = 0$ itself, so we may cancel a factor of x^3 from top and bottom:

$$\begin{aligned}\lim_{x\to 0}\frac{\arctan x - x}{\sin x - x} &= \lim_{x\to 0}\frac{-\frac{1}{3} + \frac{x^2}{5} - \cdots}{-\frac{1}{3!} + \frac{x^2}{5!} - \cdots} \\ &= \frac{\lim\limits_{x\to 0}\left(-\frac{1}{3} + \frac{x^2}{5} - \cdots\right)}{\lim\limits_{x\to 0}\left(-\frac{1}{3!} + \frac{x^2}{5!} - \cdots\right)} \\ &= \frac{-\frac{1}{3}}{-\frac{1}{3!}} = \frac{-\frac{1}{3}}{-\frac{1}{6}} = 2.\end{aligned}$$

Each of the two remaining series is a power series with a nonzero constant term, so it is continuous at 0 and the limit is just that constant term. Only the leading coefficients matter: the answer is the ratio $\left(-\frac{1}{3}\right) / \left(-\frac{1}{6}\right)$.

19. ***Three Maclaurin series from the binomial series.** *Starred problem.* This one is for the interested reader. It will not be examined.

Use the binomial series (2.4.6) to obtain the Maclaurin series for each of the following.

A. $(1+x)^5$. Give every nonzero term, and say what happens to the terms beyond them.

B. $(1+x)^{2/3}$. Give the first four terms only.

C. $\dfrac{1}{\sqrt[4]{1+x}}$. Give the first four terms only.

Solution.

A. Here $m = 5$, a positive integer:

$$\begin{aligned}(1+x)^5 &= 1 + 5x + \frac{(5)(4)}{2!}x^2 + \frac{(5)(4)(3)}{3!}x^3 \\ &\quad + \frac{(5)(4)(3)(2)}{4!}x^4 + \frac{(5)(4)(3)(2)(1)}{5!}x^5 + \cdots \\ &= 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5.\end{aligned}$$

The $k = 6$ coefficient contains the factor $m - k + 1 = 5 - 6 + 1 = 0$, so it vanishes, and so does every coefficient after it. When m is a positive integer the binomial series terminates and reproduces the ordinary binomial theorem: it is a polynomial of degree m, valid for every x, not just for $|x| < 1$.

B. Here $m = \frac{2}{3}$:

$$\begin{aligned}(1+x)^{2/3} &= 1 + \frac{2}{3}x + \frac{\left(\frac{2}{3}\right)\left(-\frac{1}{3}\right)}{2!}x^2 \\ &\quad + \frac{\left(\frac{2}{3}\right)\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)}{3!}x^3 + \cdots \\ &= 1 + \frac{2}{3}x - \frac{1}{9}x^2 + \frac{4}{81}x^3 + \cdots\end{aligned}$$

The factors $m - k + 1$ are never zero now, so the series does not terminate.

C. Write the radical in the denominator as a negative power, so that the binomial series applies with $m = -\frac{1}{4}$:

$$\begin{aligned}\frac{1}{\sqrt[4]{1+x}} &= (1+x)^{-1/4} \\ &= 1 - \frac{1}{4}x + \frac{\left(-\frac{1}{4}\right)\left(-\frac{5}{4}\right)}{2!}x^2 \\ &\quad + \frac{\left(-\frac{1}{4}\right)\left(-\frac{5}{4}\right)\left(-\frac{9}{4}\right)}{3!}x^3 + \cdots \\ &= 1 - \frac{1}{4}x + \frac{5}{32}x^2 - \frac{15}{128}x^3 + \cdots\end{aligned}$$

With m negative the signs alternate, as they did for $\dfrac{1}{(1+x)^3}$.

20. ∗A product of two series. *Starred problem.* This one is for the interested reader. It will not be examined.

Find the first four nonzero terms of the Taylor series about 0 for the function $f(x) = \sqrt{1+x}\,\cos(5x)$. Note that you may want to find these in a manner other than by direct differentiation of the function.

Solution. Differentiating the product four times is painful; multiplying two series you already know is not. The binomial series with $m = \frac{1}{2}$ gives

$$\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \cdots,$$

and replacing x by $5x$ in $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$ gives

$$\cos(5x) = 1 - \frac{(5x)^2}{2!} + \frac{(5x)^4}{4!} - \cdots = 1 - \frac{5^2}{2}x^2 + \cdots.$$

Only the constant and x^2 terms of $\cos(5x)$ can contribute below degree 4, so multiplying and collecting powers of x up to x^3,

$$\sqrt{1+x}\,\cos(5x) = \left(1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \cdots\right)$$

$$\times \left(1 - \frac{5^2}{2}x^2 + \cdots\right)$$
$$= 1 + \frac{x}{2} - \left(\frac{1}{8} + \frac{5^2}{2}\right)x^2 + \left(\frac{1}{16} - \frac{5^2}{4}\right)x^3 + \cdots$$
$$= 1 + \frac{x}{2} - \frac{101}{8}x^2 - \frac{99}{16}x^3 + \cdots .$$

21. ∗How small must the separation be? *Starred problem.* This one is for the interested reader. It will not be examined.

A dipole is often treated as a **point dipole** by replacing the exact axial field

$$E = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r^2} - \frac{q}{(r+d)^2}\right)$$

with its leading-order Taylor term for $d \ll r$,

$$E \approx \frac{1}{4\pi\epsilon_0} \cdot \frac{2qd}{r^3}.$$

Suppose this approximation must agree with the exact field to within a relative error of 1%. If the field point P is $r = 3.0$ cm from the positive charge, how small must the charge separation d be for the point-dipole approximation to be valid?

Hint. The dominant error comes from the *next* term in the expansion. Using $E = \dfrac{q}{4\pi\epsilon_0}\left(\dfrac{2d}{r^3} - \dfrac{3d^2}{r^4} + \cdots\right)$, form the relative error of the leading term and keep only the largest contribution.

Answer. $d \lesssim \dfrac{2}{3}(0.01)\, r \approx 0.020 \text{ cm} = 0.20 \text{ mm}$.

Solution. From the Taylor expansion,

$$E = \frac{q}{4\pi\epsilon_0}\left(\frac{2d}{r^3} - \frac{3d^2}{r^4} + \cdots\right),$$

the leading term is $E_{\text{lead}} = \dfrac{q}{4\pi\epsilon_0} \cdot \dfrac{2d}{r^3}$, and the first neglected term is $\dfrac{q}{4\pi\epsilon_0} \cdot \dfrac{3d^2}{r^4}$. The relative error of the approximation is therefore

$$\frac{|E_{\text{lead}} - E|}{E} \approx \frac{\dfrac{3d^2}{r^4}}{\dfrac{2d}{r^3}} = \frac{3}{2} \cdot \frac{d}{r}.$$

Notice that q and ϵ_0 cancel, so the relative error depends only on the ratio d/r. Requiring this to be at most $1\% = 0.01$ gives

$$\frac{3}{2} \cdot \frac{d}{r} \le 0.01 \qquad \Longrightarrow \qquad d \le \frac{2}{3}(0.01)\, r.$$

With $r = 3.0$ cm,

$$d \le \frac{2}{3}(0.01)(3.0 \text{ cm}) = 0.020 \text{ cm} = 0.20 \text{ mm}.$$

So the point-dipole approximation is accurate to 1% only when the charges are separated by less than about 0.2 mm at this distance—consistent with the plot of the exact and approximate fields in the dipole problem of Assignment 3, where the two curves visibly merge as r grows relative to d.

5.5 Sample Past Exam 1

A sample past exam, with worked solutions. It was a 65-minute, closed-book exam worth 20 points: no aids of their own (formula sheets, notes, phones, tablets, smart watches or similar) were permitted, and students were asked to write their work in a neat and organized format and to fully justify every step. The formula sheet at the end of this page will be the last page of the exam booklet on the day of the exam. Try the exam under the same conditions before you look at the solutions. The topics it covers, with links to the notes:

- Hyperbolic Functions
- Infinite Series
- Taylor and Maclaurin Series
- Taylor Series and its convergence

1. 4 points. Answer parts (a) and (b) for the following infinite series:

$$\sum_{k=1}^{\infty} \left(e^{1-k} - e^{-k}\right).$$

(a) Does the nth-term test apply? Remember to fully justify your answer. (2 points)

Solution.

$$\lim_{k\to\infty} a_k = \lim_{k\to\infty} \left(e^{1-k} - e^{-k}\right) = 0,$$

so the nth-term test does *not* apply: a limit of 0 is inconclusive.

(b) Evaluate the series or show that it diverges. (2 points)

Solution. The partial sums telescope:

$$\begin{aligned} S_n &= \underbrace{\left(e^0 - e^{-1}\right)}_{k=1} + \underbrace{\left(e^{-1} - e^{-2}\right)}_{k=2} + \underbrace{\left(e^{-2} - e^{-3}\right)}_{k=3} \\ &\quad + \cdots + \underbrace{\left(e^{2-n} - e^{1-n}\right)}_{k=n-1} + \underbrace{\left(e^{1-n} - e^{-n}\right)}_{k=n} \\ &= e^0 - e^{-n} = 1 - \frac{1}{e^n}. \end{aligned}$$

Hence

$$\sum_{k=1}^{\infty} \left(e^{1-k} - e^{-k}\right) = \lim_{n\to\infty} \left(1 - \frac{1}{e^n}\right) = 1.$$

2. 5 points. Consider the Taylor series for the function $f(x) = \sinh(x)$ about $x = 0$.

(a) Find $p_5(x)$, i.e. the Taylor polynomial of order 5. (2 points)

Solution.

$$\begin{aligned} f(x) &= \sinh(x), & f(0) &= 0, \\ f'(x) &= \cosh(x), & f'(0) &= 1, \\ f''(x) &= \sinh(x), & f''(0) &= 0, \\ f^{(3)}(x) &= \cosh(x), & f^{(3)}(0) &= 1, \\ f^{(4)}(x) &= \sinh(x), & f^{(4)}(0) &= 0, \\ f^{(5)}(x) &= \cosh(x), & f^{(5)}(0) &= 1. \end{aligned}$$

Therefore

$$p_5(x) = f(0) + f'(0)(x-0) + \frac{f''(0)}{2!}(x-0)^2$$

$$+\cdots+\frac{f^{(5)}(0)}{5!}(x-0)^5$$
$$=x+\frac{x^3}{3!}+\frac{x^5}{5!}.$$

(b) Express the Taylor series for the function $f(x)=\sinh(x)$ about $x=0$ in sigma notation. (1 point)

Solution.
$$\sum_{k=0}^{\infty}\frac{x^{2k+1}}{(2k+1)!}$$

(c) Assuming that $-2\le x\le 2$, find an upper bound on the error in estimating $f(x)$ using $p_5(x)$. (2 points)

Note: an answer in the form $\left(e^5+e^{-3}\right)\dfrac{3^{10}}{9!}$ suffices.

Solution. Collect the ingredients of the Remainder Estimation Theorem:

(I) $-2\le x\le 2$, so $|x|\le 2$.

(II) $n=6$, since $p_5(x)=p_6(x)$ for $f(x)=\sinh(x)$ (the coefficient of x^6 is $f^{(6)}(0)/6!=0$).

(III) $a=0$.

(IV) $f^{(n+1)}(t)=f^{(7)}(t)=\cosh(t)=\dfrac{e^t+e^{-t}}{2}$, so for $-2\le t\le 2$,

$$\left|f^{(7)}(t)\right|\le\frac{e^2+e^{-2}}{2},\qquad\text{and we take}\qquad M=\frac{e^2+e^{-2}}{2}.$$

Therefore
$$\text{Error}=|R_5(x)|=|R_6(x)|\le\frac{M\,|x-0|^7}{7!}\le\left(\frac{e^2+e^{-2}}{2}\right)\frac{2^7}{7!},$$

which is the upper bound on the error.

3. 4 points. Consider the series
$$\sum_{k=1}^{\infty}\left(\frac{2}{x}\right)^k.$$

(a) For which values of x does the series converge? (2 points)

Solution.
$$\sum_{k=1}^{\infty}\left(\frac{2}{x}\right)^k=\sum_{k=1}^{\infty}\left(\frac{2}{x}\right)\left(\frac{2}{x}\right)^{k-1}$$

is a geometric series with $r=\dfrac{2}{x}$ and $a=\dfrac{2}{x}$. It converges when

$$|r|<1\ \Rightarrow\ \left|\frac{2}{x}\right|<1\ \Rightarrow\ |x|>2\ \Rightarrow\ x\in(-\infty,-2)\cup(2,\infty).$$

(b) Evaluate the series for the x values that you found in part (a). (2 points)

Solution. Since
$$\sum_{n=1}^{\infty}ar^{n-1}=\frac{a}{1-r}\quad\text{if }|r|<1,$$

we get
$$\sum_{k=1}^{\infty}\left(\frac{2}{x}\right)^k=\frac{\frac{2}{x}}{1-\frac{2}{x}}=\frac{2}{x-2}.$$

4. **3 points.** Compute the result of

$$\sum_{n=1}^{\infty} \frac{2^{(n+1)}}{3^n}.$$

Solution.

$$\sum_{n=1}^{\infty} \frac{2^{n+1}}{3^n} = 2\sum_{n=1}^{\infty} \frac{2^n}{3^n} = 2\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)^{n-1}$$
$$= 2\,\frac{\frac{2}{3}}{1-\frac{2}{3}} = 2\,\frac{\frac{2}{3}}{\frac{1}{3}} = 4.$$

The last sum is a geometric series with $a = \frac{2}{3}$ and $r = \frac{2}{3}$, which converges to $a/(1-r)$.

5. **4 points.** Find the result of the following series or show that the series diverges.

(a) $-\ln(3) + \frac{(\ln(3))^2}{2} - \frac{(\ln(3))^3}{6} + \frac{(\ln(3))^4}{24} - \cdots$ (2 points)

Solution. Add and subtract 1:

$$-\ln(3) + \frac{(\ln(3))^2}{2} - \frac{(\ln(3))^3}{6} + \frac{(\ln(3))^4}{24} - \cdots$$
$$= -1 + \left[1 - \ln(3) + \frac{(\ln(3))^2}{2} - \frac{(\ln(3))^3}{6} + \frac{(\ln(3))^4}{24} - \cdots\right].$$

The bracket is the Maclaurin series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

with $x = -\ln(3)$. Hence the series equals

$$-1 + e^{-\ln(3)} = -1 + \frac{1}{e^{\ln(3)}} = -1 + \frac{1}{3} = -\frac{2}{3}.$$

(b) $e + e^2 + e^4 + e^6 + \cdots$ (2 points)

Solution.

$$e + e^2 + e^4 + e^6 + \cdots = e + \sum_{n=1}^{\infty} e^{2n},$$

and $\sum_{n=1}^{\infty} e^{2n}$ is a geometric series with $r = e^2$. Hence it diverges, since $|r| > 1$.

Formula Sheet.

Taylor's Formula. If f has derivatives of all orders in an open interval I containing a, then for each positive integer n and for each x in I,

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2$$
$$+ \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

where $R_n(x) = \dfrac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$ and c is between a and x.

The Remainder Estimation Theorem. If there is a positive constant M such that $\left|f^{(n+1)}(c)\right| \le M$ for

all c between x and a, inclusive, then the remainder term $R_n(x)$ in Taylor's Theorem satisfies the inequality

$$\text{Error} = |R_n(x)| \leq M \frac{|x-a|^{n+1}}{(n+1)!}.$$

If this condition holds for every n and the other conditions of Taylor's Theorem are satisfied by f, then the series converges to $f(x)$.

Maclaurin series.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad x \in (-1,1)$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

5.6 Exam 1

Exam 1 of this quarter (Fall 2026), with solutions. These were the instructions on the exam:

- You have ***65 minutes***. The exam has 5 problems and is worth ***20 points***.
- This is a closed-book exam. No aids of your own are permitted: no formula sheets, notes, phones, tablets, smart watches, or similar. ***Calculators of any kind are not permitted.***
- Write your work in a neat and organized format and ***fully justify every step***. An answer without supporting work may receive little or no credit.
- Give exact answers. Decimal approximations are not required.
- If you need more room, use the blank page at the end of the booklet and say so on the page of the problem.
- The last page of the booklet is a formula sheet. It is reproduced at the end of this page.
- Any violation of academic integrity, such as the use of AI, calculators, etc., may result in receiving an F and being reported to the Office of Student Life.

The topics it covers, with links to the notes:

- Hyperbolic Functions
- Infinite Series
- Taylor and Maclaurin Series
- Taylor Series and its convergence

1. **4 points.** The centerline of an arch is modeled by the curve

$$y = 50 - 30\cosh\left(\frac{x}{30}\right),$$

where x and y are measured in meters. The ground is the line $y = 0$, and the arch is the part of the curve with $y \geq 0$.

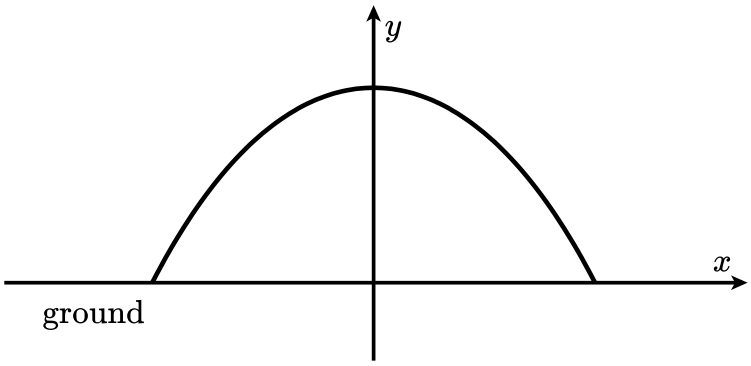


Figure 5.6.1 The arch $y = 50 - 30\cosh\left(\frac{x}{30}\right)$ above the ground $y = 0$.

(a) Find the height of the arch, that is, its maximum height above the ground. Justify why this is the maximum. (1 point)

Solution. Since $\cosh u \geq 1$ for every real u, with equality only at $u = 0$ (the range of cosh is $[1, \infty)$ and its minimum is at 0),

$$y = 50 - 30\cosh\left(\tfrac{x}{30}\right) \leq 50 - 30 = 20,$$

with equality exactly when $x = 0$. The arch is ***20 m*** tall.

(b) Find the width of the arch at ground level. (2 points)

Solution. The arch meets the ground where $y = 0$, that is, where $\cosh\left(\frac{x}{30}\right) = \frac{5}{3}$. Put $u = e^{x/30} > 0$, so that $e^{-x/30} = \frac{1}{u}$:

$$\frac{e^{x/30} + e^{-x/30}}{2} = \frac{5}{3} \quad \Longleftrightarrow \quad u + \frac{1}{u} = \frac{10}{3}.$$

Multiplying by $3u$:

$$\begin{aligned} 3u^2 - 10u + 3 = 0 &\implies (3u-1)(u-3) = 0 \\ &\implies u = 3 \text{ or } u = \tfrac{1}{3}. \end{aligned}$$

So $\frac{x}{30} = \ln 3$ or $\frac{x}{30} = \ln \frac{1}{3} = -\ln 3$, i.e. the arch meets the ground at $x = \pm 30 \ln 3$. The width is

$$30 \ln 3 - (-30 \ln 3) = 60 \ln 3 \text{ m} \quad (\approx 66 \text{ m}).$$

(c) Find the slope of the arch where it meets the ground on the right. (1 point)

Solution.

$$\frac{dy}{dx} = -30 \sinh\left(\frac{x}{30}\right) \cdot \frac{1}{30} = -\sinh\left(\frac{x}{30}\right).$$

At the right base $x = 30 \ln 3$,

$$\begin{aligned} \left.\frac{dy}{dx}\right|_{x=30\ln 3} &= -\sinh(\ln 3) = -\frac{e^{\ln 3} - e^{-\ln 3}}{2} \\ &= -\frac{3 - \frac{1}{3}}{2} = -\frac{4}{3}. \end{aligned}$$

2. 4 points. For each series below, determine whether it converges or diverges. If it converges, find its sum. Remember to fully justify your answers.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{\sinh(n)}{e^n}$ (2 points)

Solution. Using the definition of sinh,

$$\begin{aligned} a_n = \frac{\sinh(n)}{e^n} = \frac{e^n - e^{-n}}{2e^n} &= \frac{1 - e^{-2n}}{2} \\ &\longrightarrow \frac{1}{2} \neq 0 \quad (n \to \infty). \end{aligned}$$

By the nth Term Test, the series ***diverges***.

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{\sinh(n)}{e^{3n}}$ (2 points)

Solution. Here

$$\frac{\sinh(n)}{e^{3n}} = \frac{e^n - e^{-n}}{2e^{3n}} = \frac{1}{2}\left(e^{-2n} - e^{-4n}\right).$$

Both $\sum_{n\geq 1} e^{-2n}$ and $\sum_{n\geq 1} e^{-4n}$ are geometric series with $a = r$ and $|r| < 1$, so they converge:

$$\begin{aligned} \sum_{n=1}^{\infty} e^{-2n} &= \frac{e^{-2}}{1 - e^{-2}} = \frac{1}{e^2 - 1}, \\ \sum_{n=1}^{\infty} e^{-4n} &= \frac{e^{-4}}{1 - e^{-4}} = \frac{1}{e^4 - 1}. \end{aligned}$$

Since both converge, the difference and constant multiple rules give that the series ***converges***, and

$$\sum_{n=1}^{\infty} \frac{\sinh(n)}{e^{3n}} = \frac{1}{2}\left(\frac{1}{e^2 - 1} - \frac{1}{e^4 - 1}\right)$$

$$= \frac{1}{2} \cdot \frac{(e^2+1)-1}{e^4-1}$$
$$= \frac{e^2}{2\,(e^4-1)} \quad \left(= \frac{1}{4\sinh 2} \approx 0.069\right).$$

(The nth Term Test is inconclusive here, since the terms tend to 0.)

3. **3 points.** A patient takes a 60 mg dose of a medication at the same time every day. By the time the next dose is taken, the body has eliminated three quarters of the medication, so exactly $\frac{1}{4}$ of the amount that was in the body just after a dose is still there when the next dose is taken. Let Q_n be the amount of medication (in mg) in the body just after the nth dose, so that $Q_1 = 60$.

(a) Find a formula for Q_n that does not contain a sum (no "$\cdots$" and no $\sum$). (2 points)

Solution. Just after each dose, the amount is 60 plus a quarter of the amount just after the previous dose: $Q_n = 60 + \frac{1}{4}Q_{n-1}$. Hence

$$\begin{aligned} Q_1 &= 60, \\ Q_2 &= 60 + 60\left(\tfrac{1}{4}\right), \\ Q_3 &= 60 + 60\left(\tfrac{1}{4}\right) + 60\left(\tfrac{1}{4}\right)^2, \\ Q_n &= 60 + 60\left(\tfrac{1}{4}\right) + \cdots + 60\left(\tfrac{1}{4}\right)^{n-1} \\ &= \sum_{k=1}^{n} 60\left(\tfrac{1}{4}\right)^{k-1}. \end{aligned}$$

This is the nth partial sum of the geometric series with $a = 60$ and $r = \frac{1}{4}$, so

$$\begin{aligned} Q_n &= a\,\frac{1-r^n}{1-r} = 60 \cdot \frac{1-\left(\frac{1}{4}\right)^n}{\frac{3}{4}} \\ &= 80\left(1-\left(\tfrac{1}{4}\right)^n\right) \text{ mg.} \end{aligned}$$

(Check: $Q_1 = 80 \cdot \frac{3}{4} = 60$ and $Q_2 = 80 \cdot \frac{15}{16} = 75$.)

(b) What happens to Q_n in the long run? (1 point)

Solution. Since $\left(\frac{1}{4}\right)^n \to 0$, $\lim_{n\to\infty} Q_n = 80$ ***mg***. In the long run the amount in the body just after a dose levels off at (approaches, but never exceeds) 80 mg. This is the sum of the series $\sum_{k=1}^{\infty} 60\left(\frac{1}{4}\right)^{k-1} = \frac{60}{1-\frac{1}{4}}$.

4. **5 points.** Let $f(x) = \ln x$.

(a) Find the Taylor polynomial $p_3(x)$ of order 3 generated by f at $x = 1$. (2 points)

Solution.

$$\begin{aligned} f(x) &= \ln x, & f(1) &= 0, \\ f'(x) &= x^{-1}, & f'(1) &= 1, \\ f''(x) &= -x^{-2}, & f''(1) &= -1, \\ f'''(x) &= 2x^{-3}, & f'''(1) &= 2. \end{aligned}$$

Therefore

$$\begin{aligned} p_3(x) &= f(1) + f'(1)(x-1) \\ &\quad + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 \\ &= (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}. \end{aligned}$$

(b) For each n, let $p_n(x)$ be the Taylor polynomial of order n generated by f at $x = 1$. Use the Remainder Estimation Theorem to find the smallest n for which the theorem guarantees that the absolute value of the error in approximating $\ln x$ by $p_n(x)$ is no greater than 0.01 for every x in $\left[1, \frac{3}{2}\right]$. (3 points)

Solution. The derivatives follow a pattern: $f^{(k)}(x) = (-1)^{k-1}\,\frac{(k-1)!}{x^k}$ for $k \geq 1$, so

$$f^{(n+1)}(c) = (-1)^n\,\frac{n!}{c^{\,n+1}}.$$

For x in $\left[1, \frac{3}{2}\right]$, the number c lies between 1 and x, so $1 \leq c \leq \frac{3}{2}$ and

$$\left|f^{(n+1)}(c)\right| = \frac{n!}{c^{\,n+1}} \leq n!,$$

the largest value occurring at $c = 1$. Take $M = n!$. Since $|x-1| \leq \frac{1}{2}$, the Remainder Estimation Theorem gives

$$\begin{aligned} |\ln x - p_n(x)| = |R_n(x)| &\leq \frac{M\,|x-1|^{n+1}}{(n+1)!} \\ &\leq \frac{n!}{(n+1)!}\left(\frac{1}{2}\right)^{n+1} = \frac{1}{(n+1)\,2^{\,n+1}}. \end{aligned}$$

We need $\frac{1}{(n+1)\,2^{\,n+1}} \leq 0.01 = \frac{1}{100}$, that is, $(n+1)\,2^{\,n+1} \geq 100$:

$$\begin{aligned} n = 3: &\quad 4 \cdot 16 = 64 \text{ (too small)}, \\ n = 4: &\quad 5 \cdot 32 = 160 \text{ (large enough)}. \end{aligned}$$

The smallest such order is $n = 4$. (In fact p_3 really is not good enough: $\left|\ln \frac{3}{2} - p_3\left(\frac{3}{2}\right)\right| = \left|\ln \frac{3}{2} - \frac{5}{12}\right| \approx 0.0112$.)

5. 4 points.

(a) Use Taylor series to evaluate

$$\lim_{x \to 0} \frac{x - \sin x}{x\,(1 - \cos x)}.$$

(2 points)

Solution. From the Maclaurin series of $\sin x$ and $\cos x$,

$$\begin{aligned} x - \sin x &= x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right) \\ &= \frac{x^3}{6} - \frac{x^5}{120} + \cdots, \\ x\,(1 - \cos x) &= x\left(\frac{x^2}{2!} - \frac{x^4}{4!} + \cdots\right) \\ &= \frac{x^3}{2} - \frac{x^5}{24} + \cdots. \end{aligned}$$

Dividing numerator and denominator by x^3,

$$\begin{aligned} \lim_{x \to 0} \frac{x - \sin x}{x\,(1 - \cos x)} &= \lim_{x \to 0} \frac{\frac{1}{6} - \frac{x^2}{120} + \cdots}{\frac{1}{2} - \frac{x^2}{24} + \cdots} \\ &= \frac{1/6}{1/2} = \frac{1}{3}. \end{aligned}$$

(b) ***True or False?*** If $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both diverge, then $\sum_{n=1}^{\infty}(a_n + b_n)$ diverges. Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false. (2 points)

Solution. ***False.*** Take $a_n = 1$ and $b_n = -1$ for every n. Both $\sum a_n$ and $\sum b_n$ diverge by the nth Term Test, since $a_n \to 1 \neq 0$ and $b_n \to -1 \neq 0$ (their partial sums are n and $-n$). But $a_n + b_n = 0$ for every n, so

$$\sum_{n=1}^{\infty}(a_n + b_n) = 0 + 0 + 0 + \cdots = 0$$

converges. The Sum Rule $\sum(a_n + b_n) = \sum a_n + \sum b_n$ holds only when both series converge.

Formula Sheet.

Geometric series. The sum of the first n terms is

$$s_n = a + ar + ar^2 + \cdots + ar^{n-1} = a\,\frac{1-r^n}{1-r}, \qquad r \neq 1.$$

The geometric series converges to $\dfrac{a}{1-r}$ if $|r| < 1$, i.e.

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \cdots = \frac{a}{1-r}, \qquad |r| < 1,$$

and diverges if $|r| \geq 1$.

Taylor's Formula. If f has derivatives of all orders in an open interval I containing a, then for each positive integer n and for each x in I,

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

where $R_n(x) = \dfrac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$ and c is between a and x.

The Remainder Estimation Theorem. If there is a positive constant M such that $\left|f^{(n+1)}(c)\right| \leq M$ for all c between x and a, inclusive, then the remainder term $R_n(x)$ in Taylor's Theorem satisfies the inequality

$$\text{Error} = |R_n(x)| \leq M\,\frac{|x-a|^{n+1}}{(n+1)!}.$$

If this condition holds for every n and the other conditions of Taylor's Theorem are satisfied by f, then the series converges to $f(x)$.

Maclaurin series.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad x \in (-1,1)$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

Answers at a glance.

(a) (a) 20 m; (b) $60\ln 3$ m; (c) $-\frac{4}{3}$.

(b) (a) diverges (the terms tend to $\frac{1}{2}$); (b) converges to $\dfrac{e^2}{2\,(e^4-1)} = \dfrac{1}{4\sinh 2}$.

(c) (a) $Q_n = 80\left(1 - \left(\frac{1}{4}\right)^n\right)$ mg; (b) 80 mg.

(d) (a) $p_3(x) = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}$; (b) $n = 4$.

(e) (a) $\frac{1}{3}$; (b) False (for example, $a_n = 1$ and $b_n = -1$).

5.7 Assignment 4

These problems set up coordinates in space, describe regions and solids by equations and inequalities, do algebra with vectors, and use the dot product to measure angles and to split a vector into a piece along a given direction and a piece orthogonal to it. If you would like to review the material first, see Three-Dimensional Coordinate Systems, Vectors and The Dot Product.

1.

A. Find an equation of the set of all points equidistant from the points $A(-1,5,3)$ and $B(6,2,-2)$.

B. Describe the geometrical shape corresponding to the set.

Solution.

A. We want to find all points $P(x,y,z)$ that have the same distance from A and B. Squaring both distances (which is equivalent, since distances are non-negative) gives

$$(x+1)^2+(y-5)^2+(z-3)^2=(x-6)^2+(y-2)^2+(z+2)^2.$$

Expanding both sides,

$$\begin{aligned} &x^2+2x+1+y^2-10y+25+z^2-6z+9\\ &=x^2-12x+36+y^2-4y+4+z^2+4z+4,\\ &2x-10y-6z+35=-12x-4y+4z+44, \end{aligned}$$

and collecting terms,

$$14x-6y-10z=9.$$

B. This is the equation of a plane. Its normal vector is $\langle 14,-6,-10\rangle = 2\langle 7,-3,-5\rangle$, which is parallel to $\vec{AB}=\langle 6-(-1),\,2-5,\,-2-3\rangle=\langle 7,-3,-5\rangle$. Moreover, the midpoint of $\overline{AB}$,

$$C\left(\frac{-1+6}{2},\frac{5+2}{2},\frac{3+(-2)}{2}\right)=C\left(\frac{5}{2},\frac{7}{2},\frac{1}{2}\right),$$

satisfies the equation, since $14\left(\frac{5}{2}\right)-6\left(\frac{7}{2}\right)-10\left(\frac{1}{2}\right)=35-21-5=9$. Hence the set is the plane that passes through the midpoint of $\overline{AB}$ and is orthogonal to $\vec{AB}$, i.e. the **perpendicular bisector plane** of the segment $\overline{AB}$. See Figure 5.7.1.

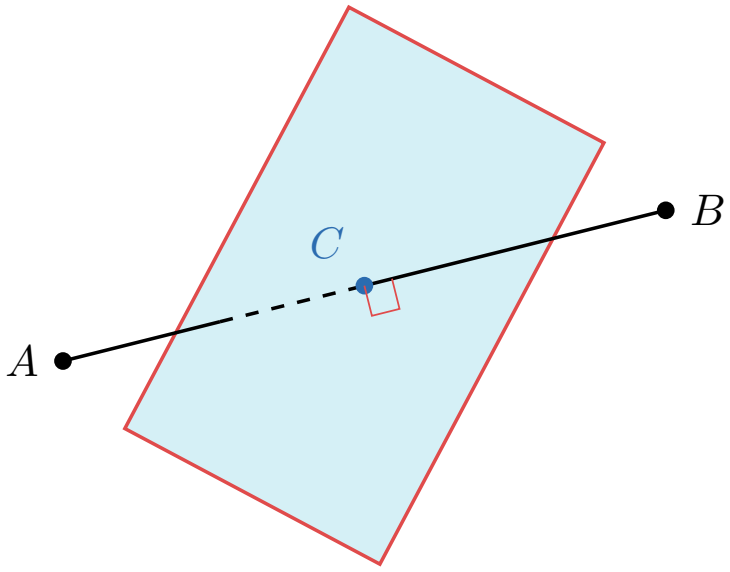


Figure 5.7.1 The set of points equidistant from A and B is the plane through the midpoint C of $\overline{AB}$ perpendicular to $\vec{AB}$.

2. Describe the following regions and graphs.

A. $x^2 + y^2 + z^2 \leq 4$, $z \geq 0$

B. $x^2 + y^2 + z^2 \leq 4$, $z = 0$

C. $x^2 + y^2 + z^2 = 4$, $z \geq 0$

D. $x^2 + y^2 + z^2 = 4$, $z = 0$

Solution. The key is to notice whether the first condition is an *inequality* (a solid region) or an *equation* (a surface), and then what the second condition does to it.

A. The solid ball of radius 2 centered at the origin, restricted to $z \geq 0$: the *upper half of the solid ball*, including the flat disk in the xy-plane.

B. Setting $z = 0$ in the solid ball gives $x^2 + y^2 \leq 4$: the *solid disk* of radius 2 centered at the origin, lying in the xy-plane.

C. The sphere of radius 2 restricted to $z \geq 0$: the *upper hemisphere* (a surface only), together with the circle where it meets the xy-plane.

D. Setting $z = 0$ on the sphere gives $x^2 + y^2 = 4$: the *circle* of radius 2 centered at the origin in the xy-plane.

3. Consider the point $(3, 7, -5)$.

(a) What is its distance to the yz-plane?

(b) What is its distance to the y-axis?

Solution. **(a)** The yz-plane is the plane $x = 0$. The closest point of that plane to $(3, 7, -5)$ is obtained by simply setting the x-coordinate to zero, giving $(0, 7, -5)$. Hence the distance is

$$\sqrt{(3-0)^2 + (7-7)^2 + (-5+5)^2} = |3| = 3.$$

In general the distance from (x_0, y_0, z_0) to the yz-plane is $|x_0|$.

(b) The y-axis consists of the points $(0, y, 0)$. The closest such point to $(3, 7, -5)$ is the one with the same y-coordinate, namely $(0, 7, 0)$, so the distance is

$$\sqrt{(3-0)^2 + (7-7)^2 + (-5-0)^2} = \sqrt{9+25} = \sqrt{34}.$$

In general the distance from (x_0, y_0, z_0) to the y-axis is $\sqrt{x_0^2 + z_0^2}$: we drop the coordinate along the axis and keep the other two. See Figure 5.7.2.

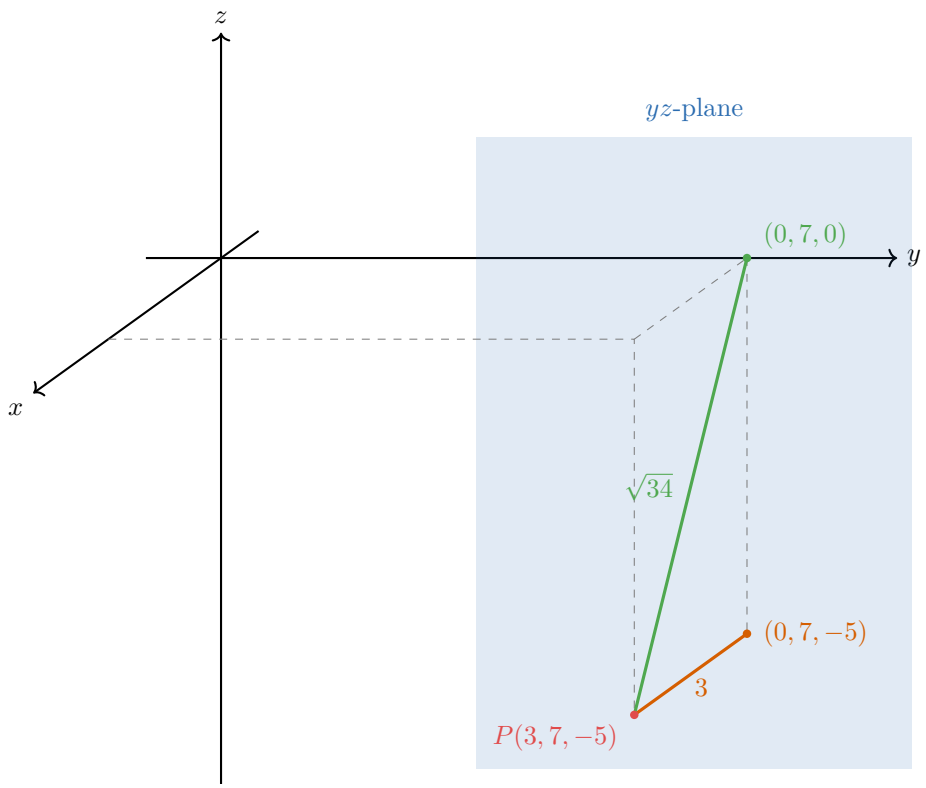


Figure 5.7.2 The point $P(3, 7, -5)$, its distance 3 to the yz-plane (measured horizontally, parallel to the x-axis), and its distance $\sqrt{34}$ to the y-axis (measured perpendicular to that axis).

4. A Solid Bounded by a Hemisphere, a Plane, and a Cylinder. Consider the solid region bounded below by the hemisphere $z = -\sqrt{1 - x^2 - y^2}$, above by the plane $z = 1$, and on the outside by the cylinder $x^2 + y^2 = 1$. Describe this solid.

Solution. Squaring $z = -\sqrt{1 - x^2 - y^2}$ gives $x^2 + y^2 + z^2 = 1$, and the minus sign keeps only the points with $z \le 0$: the surface is the lower half of the unit sphere centered at the origin, with its lowest point at $(0, 0, -1)$. It reaches $z = 0$ exactly on the circle $x^2 + y^2 = 1$, the same circle where the cylinder meets the xy-plane.

The solid consists of the points inside the cylinder that lie *above* the hemisphere and *below* the plane:

$$x^2 + y^2 \le 1, \qquad -\sqrt{1 - x^2 - y^2} \le z \le 1.$$

At the center $(x, y) = (0, 0)$, this range for z is the full interval $-1 \le z \le 1$. At the rim $x^2 + y^2 = 1$, it shrinks to $0 \le z \le 1$. The part with $z \ge 0$ is the solid cylinder $x^2 + y^2 \le 1$, $0 \le z \le 1$, and the part with $z \le 0$ is the solid lower half of the unit ball, $x^2 + y^2 + z^2 \le 1$, $z \le 0$. The solid is therefore a can of radius 1 and height 1 with a hemispherical bowl attached beneath it along the unit disk in the xy-plane, like the rounded end of a test tube, shown in Figure 5.7.3.

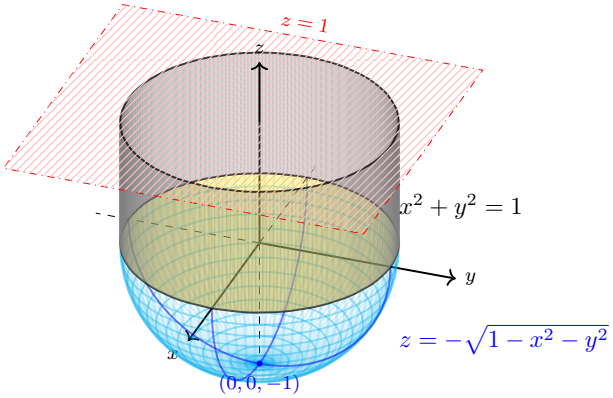


Figure 5.7.3 The solid bounded below by the hemisphere $z = -\sqrt{1 - x^2 - y^2}$, above by the plane $z = 1$, and on the outside by the cylinder $x^2 + y^2 = 1$: a solid cylinder of height 1 sitting on a solid half-ball, the two joined along the unit disk in the xy-plane.

5. Write each combination of vectors as a single vector. See Figure 5.7.4.

A. $\vec{PQ} + \vec{QR}$

B. $\vec{QS} - \vec{PS}$

C. $\vec{RS} + \vec{SP} + \vec{PQ}$

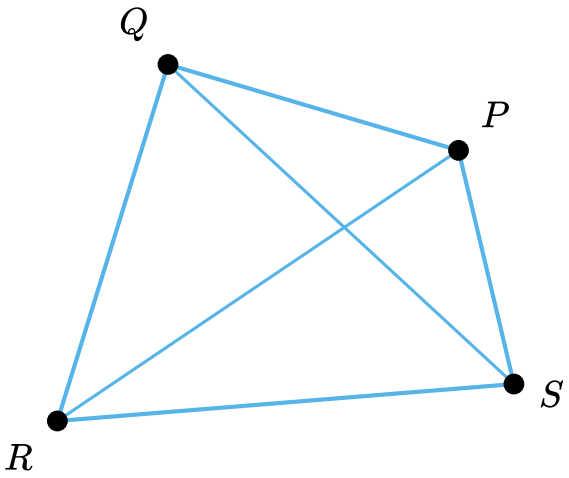


Figure 5.7.4 The quadrilateral with vertices P, Q, R, and S, with its diagonals.

Solution.

A. By the triangle law, placing $\vec{QR}$ at the tip of $\vec{PQ}$,

$$\vec{PQ} + \vec{QR} = \vec{PR}.$$

B. Since $-\vec{PS} = \vec{SP}$,

$$\vec{QS} - \vec{PS} = \vec{QS} + \vec{SP} = \vec{QP}.$$

C. Chaining the three vectors tip to tail,

$$\vec{RS} + \vec{SP} + \vec{PQ} = \vec{RQ}.$$

6. Find a vector of length 3 pointing in the opposite direction from the vector $2\,\mathbf{i} - 4\,\mathbf{j} + 4\,\mathbf{k}$.

Solution. The given vector has length $\sqrt{4 + 16 + 16} = \sqrt{36} = 6$, so the unit vector in its direction is

$\frac{1}{6}\langle 2,-4,4\rangle = \left\langle \frac{1}{3},-\frac{2}{3},\frac{2}{3}\right\rangle$. Multiplying by -3 reverses the direction and gives length 3:

$$-3\left\langle \frac{1}{3},-\frac{2}{3},\frac{2}{3}\right\rangle = \langle -1,2,-2\rangle = -\mathbf{i}+2\,\mathbf{j}-2\,\mathbf{k}.$$

7. Let $\vec{v}$ be the vector that goes from the point $\left(5,-4,\sqrt{3}\right)$ to the point $(3,-1,0)$. Find a vector of length $\frac{1}{3}$ that points in the same direction as $\vec{v}$.

Solution. The vector from $P(5,-4,\sqrt{3})$ to $Q(3,-1,0)$ is obtained by subtracting the tail from the head:

$$\vec{v} = \vec{PQ} = \left\langle 3-5,\ -1-(-4),\ 0-\sqrt{3}\right\rangle = \left\langle -2,\ 3,\ -\sqrt{3}\right\rangle.$$

Its length is

$$\|\vec{v}\| = \sqrt{(-2)^2+3^2+\left(-\sqrt{3}\right)^2} = \sqrt{4+9+3} = \sqrt{16} = 4.$$

The unit vector in the direction of $\vec{v}$ is therefore

$$\hat{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{4}\left\langle -2,\ 3,\ -\sqrt{3}\right\rangle.$$

Scaling by $\frac{1}{3}$ gives a vector of length $\frac{1}{3}$ pointing the same way:

$$\frac{1}{3}\,\hat{u} = \frac{1}{12}\left\langle -2,\ 3,\ -\sqrt{3}\right\rangle = \left\langle -\frac{1}{6},\ \frac{1}{4},\ -\frac{\sqrt{3}}{12}\right\rangle.$$

8. **A parallelogram in space.** Consider the four points

$$A(1,2,5), \qquad B(4,3,3), \qquad C(6,7,4), \qquad D(3,6,6).$$

A. Show that $ABCD$ is a parallelogram.

B. Find the point M at which the two diagonals of $ABCD$ meet.

C. Find the perimeter of $ABCD$.

Solution.

A. A quadrilateral is a parallelogram exactly when one pair of opposite sides is given by the *same* vector, since equal vectors have both the same direction and the same length. Along the sides AB and DC,

$$\vec{AB} = \langle 4-1,\ 3-2,\ 3-5\rangle = \langle 3,\ 1,\ -2\rangle,$$
$$\vec{DC} = \langle 6-3,\ 7-6,\ 4-6\rangle = \langle 3,\ 1,\ -2\rangle.$$

Since $\vec{AB} = \vec{DC}$, these two sides are parallel and have equal length, so $ABCD$ is a parallelogram. The other pair of sides matches as well:

$$\vec{AD} = \langle 2,\ 4,\ 1\rangle = \vec{BC}.$$

See Figure 5.7.5.

B. The diagonals of a parallelogram bisect each other, so M is the midpoint of AC. Using position vectors and $\vec{AC} = \langle 5,\ 5,\ -1\rangle$,

$$\vec{OM} = \vec{OA} + \tfrac{1}{2}\vec{AC} = \langle 1,\ 2,\ 5\rangle + \tfrac{1}{2}\langle 5,\ 5,\ -1\rangle$$
$$= \left\langle \tfrac{7}{2},\ \tfrac{9}{2},\ \tfrac{9}{2}\right\rangle.$$

Travelling along the other diagonal instead, with $\vec{BD} = \langle -1,\ 3,\ 3\rangle$,

$$\vec{OB} + \tfrac{1}{2}\vec{BD} = \langle 4,\ 3,\ 3\rangle + \tfrac{1}{2}\langle -1,\ 3,\ 3\rangle = \left\langle \tfrac{7}{2},\ \tfrac{9}{2},\ \tfrac{9}{2}\right\rangle,$$

the same point, which confirms both the answer and part A. Hence $M\left(\frac{7}{2},\ \frac{9}{2},\ \frac{9}{2}\right)$.

C. Opposite sides have equal length, so the perimeter is $2\left(\left\|\vec{AB}\right\| + \left\|\vec{AD}\right\|\right)$, where

$$\left\|\vec{AB}\right\| = \sqrt{3^2 + 1^2 + (-2)^2} = \sqrt{14},$$
$$\left\|\vec{AD}\right\| = \sqrt{2^2 + 4^2 + 1^2} = \sqrt{21}.$$

The perimeter is therefore $2\left(\sqrt{14} + \sqrt{21}\right) \approx 16.65$. Since $\sqrt{14} \neq \sqrt{21}$, this parallelogram is not a rhombus.

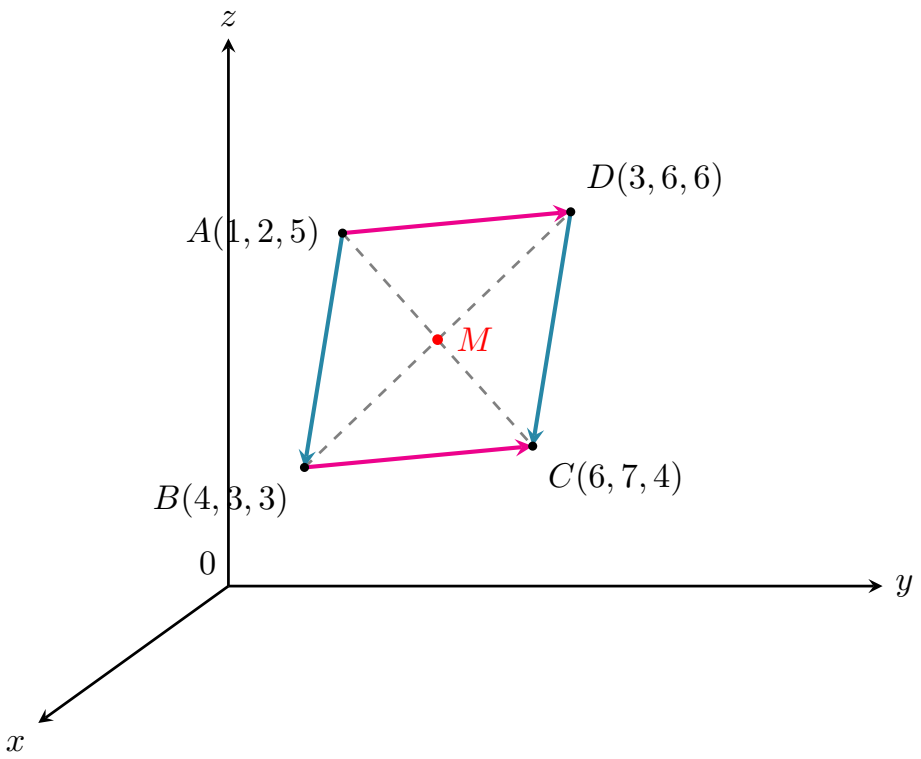


Figure 5.7.5 The parallelogram $ABCD$, with $\vec{AB} = \vec{DC} = \langle 3, 1, -2\rangle$ drawn in blue and $\vec{AD} = \vec{BC} = \langle 2, 4, 1\rangle$ in magenta. The diagonals AC and BD cross at their common midpoint $M\left(\frac{7}{2}, \frac{9}{2}, \frac{9}{2}\right)$.

9. Consider the two vectors $\mathbf{u} = 3\,\mathbf{j} + 4\,\mathbf{k}$ and $\mathbf{v} = \mathbf{i} + \mathbf{j}$. Write $\mathbf{u}$ as the sum of a vector parallel to $\mathbf{v}$ and a vector orthogonal to $\mathbf{v}$.

Solution. We have $\vec{u} \cdot \vec{v} = 3$ and $\|\vec{v}\| = \sqrt{1+1} = \sqrt{2}$. The component of $\vec{u}$ parallel to $\vec{v}$ is

$$\vec{u}_{\|} = \text{proj}_{\vec{v}}\,\vec{u} = \left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2}\right)\vec{v} = \frac{3}{2}\,(\mathbf{i} + \mathbf{j}) = \frac{3}{2}\,\mathbf{i} + \frac{3}{2}\,\mathbf{j},$$

and the component orthogonal to $\vec{v}$ is

$$\vec{u}_{\perp} = \vec{u} - \vec{u}_{\|} = (3\,\mathbf{j} + 4\,\mathbf{k}) - \left(\frac{3}{2}\,\mathbf{i} + \frac{3}{2}\,\mathbf{j}\right) = -\frac{3}{2}\,\mathbf{i} + \frac{3}{2}\,\mathbf{j} + 4\,\mathbf{k}.$$

Here $\vec{u}_{\|}$ is parallel to $\vec{v}$ and $\vec{u}_{\perp}$ is orthogonal to $\vec{v}$. Note that $\vec{u}_{\|} + \vec{u}_{\perp} = \vec{u}$.

10. Determine whether the following pairs of vectors are orthogonal, parallel, or neither.

A. $\vec{u}_1 = -3\,\mathbf{i} + 9\,\mathbf{j} + 6\,\mathbf{k}$ and $\vec{u}_2 = 4\,\mathbf{i} - 12\,\mathbf{j} - 8\,\mathbf{k}$

B. $\vec{v}_1 = \mathbf{i} + \mathbf{j} - 2\,\mathbf{k}$ and $\vec{v}_2 = \mathbf{i} + \mathbf{j} + \mathbf{k}$

C. $\vec{w}_1 = a\,\mathbf{i} + b\,\mathbf{j}$ and $\vec{w}_2 = c\,\mathbf{k}$, where a, b, and c are non-zero real numbers.

Solution.

A. **Parallel**, since $\vec{u}_2 = -\frac{4}{3}\,\vec{u}_1$: indeed $-\frac{4}{3}\langle -3, 9, 6\rangle = \langle 4, -12, -8\rangle$. (Two vectors are parallel exactly when one is a scalar multiple of the other.)

B. **Orthogonal**, since $\vec{v}_1 \cdot \vec{v}_2 = 1 + 1 - 2 = 0$.

C. **Orthogonal**, since $\vec{w}_1 \cdot \vec{w}_2 = \langle a, b, 0\rangle \cdot \langle 0, 0, c\rangle = 0$. They are not parallel, because $\vec{w}_2$ has a non-zero $\mathbf{k}$ component while $\vec{w}_1$ does not.

11. A 100-meter dash is run on a track in the direction of the vector $2\,\mathbf{i} + 6\,\mathbf{j}$. The wind velocity is $\mathbf{v} = 5\,\mathbf{i} + 2\,\mathbf{j}$ km/hr. The rules say that a legal wind speed measured in the direction of the dash must not exceed 5 km/hr.

A. Will the race results be disqualified due to an illegal wind? Justify your answer.

B. Find the velocity vector of the wind in the direction of the dash.

Solution. Let $\vec{d} = \langle 2, 6\rangle$ be the direction of the dash and $\vec{v} = \langle 5, 2\rangle$ the wind velocity. Then

$$\vec{v} \cdot \vec{d} = 10 + 12 = 22, \qquad \|\vec{d}\| = \sqrt{4 + 36} = \sqrt{40} = 2\sqrt{10}.$$

A. The wind speed measured in the direction of the dash is the length of the projection of $\vec{v}$ onto $\vec{d}$:

$$\left\|\text{proj}_{\vec{d}}\,\vec{v}\right\| = \frac{\vec{v} \cdot \vec{d}}{\|\vec{d}\|} = \frac{22}{2\sqrt{10}} = \frac{11}{\sqrt{10}} \approx 3.48 \text{ km/hr}.$$

Since $3.48 < 5$, the wind is legal and the race results will *not* be disqualified.

B. The velocity vector of the wind in the direction of the dash is the vector projection

$$\begin{aligned}\text{proj}_{\vec{d}}\,\vec{v} &= \left(\frac{\vec{v} \cdot \vec{d}}{\|\vec{d}\|^2}\right)\vec{d} = \frac{22}{40}\langle 2, 6\rangle \\ &= \left\langle \frac{11}{10}, \frac{33}{10}\right\rangle = 1.1\,\mathbf{i} + 3.3\,\mathbf{j} \text{ km/hr}.\end{aligned}$$

As a check, its length is $\sqrt{1.1^2 + 3.3^2} = 1.1\sqrt{10} = \frac{11}{\sqrt{10}}$, matching part A.

12. A constant force is

$$\mathbf{F} = \langle 2, -1, 3\rangle\,.$$

A particle is constrained to move in directions parallel to the plane

$$2x - y + 2z = 0.$$

For a displacement of one unit, the work done by the force is $W = \mathbf{F} \cdot \mathbf{d}$, where $\mathbf{d}$ is an allowed unit direction lying in the plane.

(a) Find the unit direction in the plane that maximizes the work.

(b) Find the maximum possible work per unit distance.

(c) Find one unit direction in the plane for which the work is zero.

Solution. Reading the coefficients of the plane, a normal vector is

$$\mathbf{n} = \langle 2, -1, 2\rangle\,, \qquad \|\mathbf{n}\| = \sqrt{4 + 1 + 4} = 3.$$

A vector is parallel to the plane exactly when it is orthogonal to $\mathbf{n}$.

(a) Decompose the force into a normal piece and a piece lying in the plane,

$$\mathbf{F} = \text{proj}_{\mathbf{n}}\,\mathbf{F} + \mathbf{F}_{\parallel}.$$

The first piece comes from the projection formula. Since

$$\mathbf{F} \cdot \mathbf{n} = \langle 2, -1, 3\rangle \cdot \langle 2, -1, 2\rangle = 2(2) + (-1)(-1) + 3(2)$$

$$= 4 + 1 + 6 = 11,$$

and $\|\mathbf{n}\|^2 = 9$, we get

$$\text{proj}_{\mathbf{n}}\,\mathbf{F} = \frac{11}{9}\langle 2, -1, 2\rangle = \left\langle \frac{22}{9},\ -\frac{11}{9},\ \frac{22}{9} \right\rangle.$$

Subtracting,

$$\mathbf{F}_{\parallel} = \mathbf{F} - \text{proj}_{\mathbf{n}}\,\mathbf{F} = \langle 2, -1, 3\rangle - \left\langle \frac{22}{9},\ -\frac{11}{9},\ \frac{22}{9} \right\rangle$$
$$= \left\langle -\frac{4}{9},\ \frac{2}{9},\ \frac{5}{9} \right\rangle = \frac{1}{9}\langle -4,\ 2,\ 5\rangle.$$

As a check, this really is parallel to the plane: $\langle -4, 2, 5\rangle \cdot \langle 2, -1, 2\rangle = -8 - 2 + 10 = 0$.

Now let $\mathbf{d}$ be any allowed unit direction. Because $\text{proj}_{\mathbf{n}}\,\mathbf{F}$ is orthogonal to every vector lying in the plane,

$$W = \mathbf{F} \cdot \mathbf{d} = \left(\text{proj}_{\mathbf{n}}\,\mathbf{F} + \mathbf{F}_{\parallel}\right) \cdot \mathbf{d} = \mathbf{F}_{\parallel} \cdot \mathbf{d}.$$

The normal part of the force is simply wasted: it presses the particle against the constraint and does no work. Writing $\mathbf{F}_{\parallel} \cdot \mathbf{d} = \|\mathbf{F}_{\parallel}\|\,\|\mathbf{d}\| \cos\varphi = \|\mathbf{F}_{\parallel}\| \cos\varphi$, the largest value occurs when $\cos\varphi = 1$, that is when $\mathbf{d}$ points in the same direction as $\mathbf{F}_{\parallel}$. Since

$$\|\langle -4, 2, 5\rangle\| = \sqrt{16 + 4 + 25} = \sqrt{45} = 3\sqrt{5},$$

the maximizing unit vector is

$$\mathbf{d}_{\max} = \frac{\langle -4,\ 2,\ 5\rangle}{3\sqrt{5}}.$$

See Figure 5.7.6.

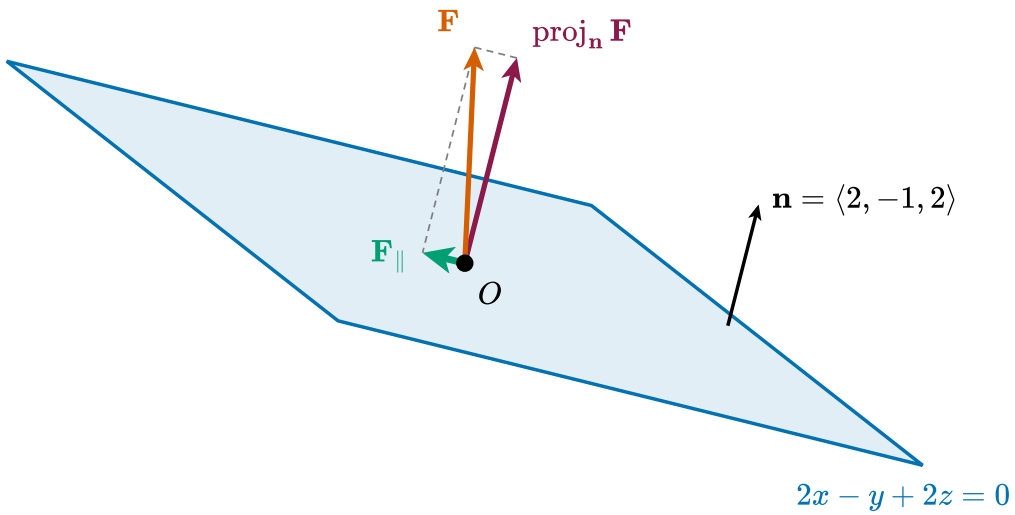


Figure 5.7.6 The force $\mathbf{F}$ splits into a piece $\text{proj}_{\mathbf{n}}\,\mathbf{F}$ along the normal, which does no work, and a much shorter piece $\mathbf{F}_{\parallel}$ lying in the plane, which is the only part that can do work. The picture is drawn to scale: $\|\text{proj}_{\mathbf{n}}\,\mathbf{F}\| = \frac{11}{3}$ while $\|\mathbf{F}_{\parallel}\| = \frac{\sqrt{5}}{3}$.

(b) The maximum work per unit displacement is the length of the in-plane component,

$$W_{\max} = \|\mathbf{F}_{\parallel}\| = \left|\frac{1}{9}\langle -4, 2, 5\rangle\right|$$
$$= \frac{1}{9}\left(3\sqrt{5}\right) = \frac{\sqrt{5}}{3} \approx 0.745.$$

We can confirm this directly from the original force:

$$\mathbf{F}\cdot\mathbf{d}_{\max} = \langle 2,-1,3\rangle\cdot\frac{\langle -4,2,5\rangle}{3\sqrt{5}} = \frac{-8-2+15}{3\sqrt{5}}$$
$$= \frac{5}{3\sqrt{5}} = \frac{\sqrt{5}}{3}.$$

Compare this with $\|\mathbf{F}\| = \sqrt{14} \approx 3.74$: the constraint costs us about 80% of the force.

(c) We want a unit vector $\mathbf{d} = \langle x, y, z\rangle$ satisfying two conditions at once: it must lie in the plane, and it must do no work. That is,

$$\mathbf{d}\cdot\mathbf{n} = 0 \qquad \text{and} \qquad \mathbf{d}\cdot\mathbf{F} = 0,$$

or explicitly

$$2x - y + 2z = 0,$$
$$2x - y + 3z = 0.$$

Subtracting the first equation from the second gives $z = 0$, and then $2x - y = 0$, so $y = 2x$. Taking $x = 1$ gives the non-zero vector $\langle 1, 2, 0\rangle$, whose length is $\sqrt{1+4} = \sqrt{5}$. Hence one suitable unit vector is

$$\mathbf{d}_0 = \frac{1}{\sqrt{5}}\langle 1,\ 2,\ 0\rangle.$$

Indeed $\mathbf{F}\cdot\mathbf{d}_0 = \frac{1}{\sqrt{5}}(2-2+0) = 0$, and the opposite direction $-\mathbf{d}_0$ works equally well. Geometrically, $\mathbf{d}_0$ is the direction in the plane perpendicular to $\mathbf{F}_{\|}$: moving along it, the particle travels neither with nor against the usable part of the force.

5.8 Assignment 5

These problems use the cross product to build normal vectors, areas and orthogonal directions, parametrize plane curves and curves of intersection in space, and intersect lines with planes and measure distances between them. If you would like to review the material first, see The Cross Product, Parametrization of Plane Curves and Lines and Planes in Space.

1. **True or False.** If $\vec{u}$ and $\vec{v}$ are two non-zero vectors in $\mathbb{R}^3$, then $\vec{u} \cdot (\vec{u} \times \vec{v}) = \|\vec{u}\|^2 + \vec{u} \cdot \vec{v}$.
Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.
Counterexample: take $\vec{u} = \mathbf{i}$ and $\vec{v} = \mathbf{i}$. Then

$$\mathbf{i} \cdot \underbrace{(\mathbf{i} \times \mathbf{i})}_{\vec{0}} = 0, \quad \text{whereas} \quad \|\mathbf{i}\|^2 + \mathbf{i} \cdot \mathbf{i} = 1 + 1 = 2,$$

and $0 \neq 2$.

2. Let $\mathbf{v}$ be the vector $3\,\mathbf{i} - 4\,\mathbf{j} + 5\,\mathbf{k}$.

A. Find a non-zero vector $\mathbf{v}'$ orthogonal to $\mathbf{v}$.

B. Find a vector orthogonal to both $\mathbf{v}$ and $\mathbf{v}'$.

Solution.

A. We need $\langle a, b, c\rangle$ with $3a - 4b + 5c = 0$. Any solution will do; taking $c = 0$ leaves $3a = 4b$, so $a = 4$, $b = 3$ works:

$$\mathbf{v}' = 4\,\mathbf{i} + 3\,\mathbf{j}, \qquad \mathbf{v} \cdot \mathbf{v}' = 12 - 12 + 0 = 0.$$

(There are infinitely many correct answers.)

B. The cross product is orthogonal to both:

$$\begin{aligned} \mathbf{v} \times \mathbf{v}' &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -4 & 5 \\ 4 & 3 & 0 \end{vmatrix} \\ &= (0 - 15)\mathbf{i} - (0 - 20)\mathbf{j} + (9 + 16)\mathbf{k} \\ &= -15\,\mathbf{i} + 20\,\mathbf{j} + 25\,\mathbf{k}. \end{aligned}$$

Dividing by 5 gives the simpler answer $-3\,\mathbf{i} + 4\,\mathbf{j} + 5\,\mathbf{k}$.

3. Suppose $\|\vec{u}\| = 3$, $\|\vec{v}\| = 4$ and $\|\vec{u} \times \vec{v}\| = 6$. Find

(a) $|\vec{u} \cdot \vec{v}|$;

(b) $\|\text{proj}_{\vec{u}}\, \vec{v}\|$.

Solution. **(a)** Let θ be the angle between the two vectors, $0 \leq \theta \leq \pi$. From $\|\vec{u} \times \vec{v}\| = \|\vec{u}\|\,\|\vec{v}\| \sin\theta$,

$$6 = (3)(4)\sin\theta \quad \Longrightarrow \quad \sin\theta = \frac{6}{12} = \frac{1}{2}.$$

Since $\sin^2\theta + \cos^2\theta = 1$,

$$|\cos\theta| = \sqrt{1 - \tfrac{1}{4}} = \frac{\sqrt{3}}{2}.$$

(We cannot determine the *sign* of $\cos\theta$, since θ could be $30°$ or $150°$; this is exactly why the problem asks for the absolute value.) Therefore

$$|\vec{u} \cdot \vec{v}| = \|\vec{u}\|\,\|\vec{v}\|\,|\cos\theta| = (3)(4)\frac{\sqrt{3}}{2} = 6\sqrt{3}.$$

(b) The projection is $\text{proj}_{\vec{u}}\,\vec{v} = \left(\dfrac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|^2}\right)\vec{u}$, so its length is

$$\|\text{proj}_{\vec{u}}\,\vec{v}\| = \frac{|\vec{u}\cdot\vec{v}|}{\|\vec{u}\|^2}\,\|\vec{u}\| = \frac{|\vec{u}\cdot\vec{v}|}{\|\vec{u}\|} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}.$$

As a sanity check, $2\sqrt{3} \approx 3.46$, which is less than $\|\vec{v}\| = 4$, as a shadow must be.

4. Find the Cartesian equation $f(x, y) = 0$ for the curve represented by the following parametric equations, then graph it:

$$\begin{aligned} x(t) &= e^{2t}, \\ y(t) &= e^t - 1, \\ 0 &\le t \le \ln(2). \end{aligned}$$

Solution. From the second equation, $y + 1 = e^t$, and hence

$$x = e^{2t} = \left(e^t\right)^2 = (y+1)^2 \;\Rightarrow\; y + 1 = \sqrt{x} \;\Rightarrow\; y = \sqrt{x} - 1.$$

For the range, note that $0 \le t \le \ln(2)$ gives

$$e^0 \le e^t \le e^{\ln(2)} \;\Rightarrow\; 1 \le e^t \le 2 \;\Rightarrow\; 0 \le e^t - 1 \le 1 \;\Rightarrow\; 0 \le y \le 1,$$

and correspondingly $1 \le x \le 4$. The curve is the portion of $y = \sqrt{x} - 1$ from $(1, 0)$ to $(4, 1)$, shown in Figure 5.8.1.

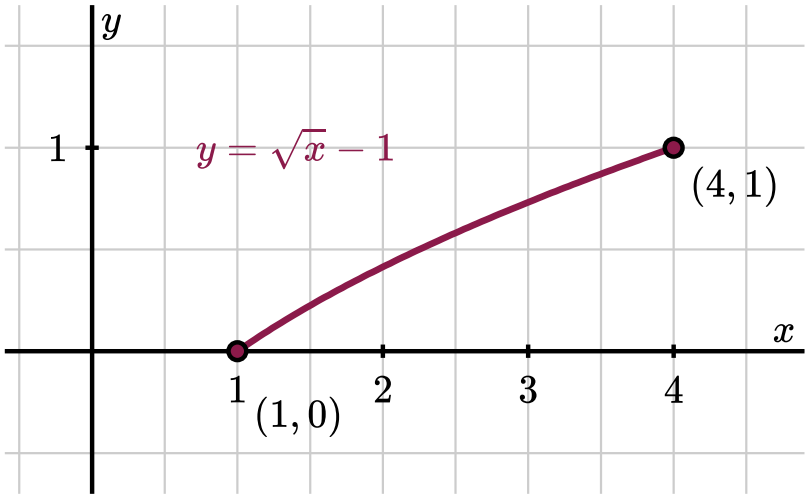


Figure 5.8.1 The curve $y = \sqrt{x} - 1$ traced by the parametric equations for $0 \le t \le \ln(2)$, from $(1, 0)$ to $(4, 1)$.

5. Provide a parametrized form of the curve given by

$$x^2 + 4y^2 + 8y - 2\sqrt{21}\,x = 0.$$

Assume that the curve is traced **counter-clockwise** as the parameter increases.

Solution. We complete the squares:

$$\begin{aligned} \left(x - \sqrt{21}\right)^2 - 21 + 4(y+1)^2 - 4 &= 0 \\ \left(x - \sqrt{21}\right)^2 + 4(y+1)^2 &= 25. \end{aligned}$$

This is an ellipse centered at $\left(\sqrt{21}, -1\right)$. Setting

$$x - \sqrt{21} = 5\cos t \quad \text{and} \quad 2(y+1) = 5\sin t$$

gives the counter-clockwise parametrization

$$\begin{aligned} x(t) &= 5\cos t + \sqrt{21} \\ y(t) &= \frac{5}{2}\sin t - 1 \\ 0 &\le t \le 2\pi. \end{aligned}$$

See Figure 5.8.2.

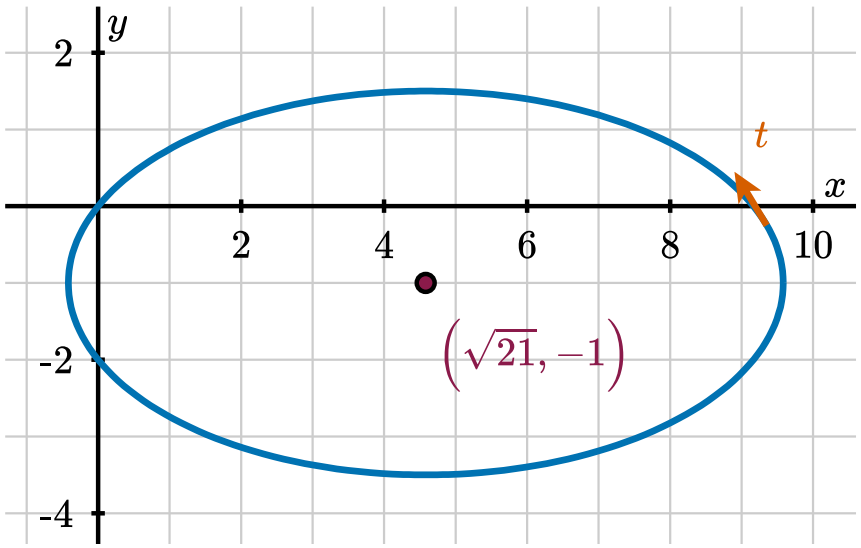


Figure 5.8.2 The ellipse $\left(x-\sqrt{21}\right)^2 + 4(y+1)^2 = 25$, centered at $\left(\sqrt{21}, -1\right)$ with semi-axes 5 and $\frac{5}{2}$, traced counter-clockwise.

6.

A. Parametrize the intersection of $x^2 + y^2 + z^2 = 25$ and the plane $x = 4$.

B. Which axis is perpendicular to the plane $x = 4$?

Solution.

A. Substituting $x = 4$ into the equation of the sphere,

$$16 + y^2 + z^2 = 25 \;\Rightarrow\; y^2 + z^2 = 9,$$

a circle of radius 3 lying in the plane $x = 4$. Hence

$$x = 4, \qquad y = 3\cos(t), \qquad z = 3\sin(t), \qquad 0 \le t \le 2\pi.$$

B. The plane $x = 4$ has normal vector $\langle 1, 0, 0\rangle$, so the x-axis is perpendicular to it.

7. Consider the curve $25(y+4)^2 + 400(x-1)^2 = 100$.

(a) Sketch the curve and label its important points.

(b) Parametrize the entire curve using one parametrization.

Solution. **(a)** Divide both sides by 100 to put the equation in standard form:

$$\begin{aligned} \frac{25(y+4)^2}{100} + \frac{400(x-1)^2}{100} &= 1 \\ \frac{(y+4)^2}{4} + 4(x-1)^2 &= 1 \end{aligned}$$

$$\frac{(x-1)^2}{\left(\frac{1}{2}\right)^2} + \frac{(y+4)^2}{2^2} = 1.$$

This is an ellipse with centre $(1,-4)$, semi-axis $a = \frac{1}{2}$ in the x-direction and semi-axis $b = 2$ in the y-direction. Since $b > a$, the major axis is vertical. The four vertices are

$$\left(\tfrac{1}{2},-4\right), \quad \left(\tfrac{3}{2},-4\right), \quad (1,-2), \quad (1,-6),$$

and the foci lie on the vertical axis at distance $c = \sqrt{b^2-a^2} = \sqrt{4-\frac{1}{4}} = \frac{\sqrt{15}}{2} \approx 1.94$ from the centre. See Figure 5.8.3.

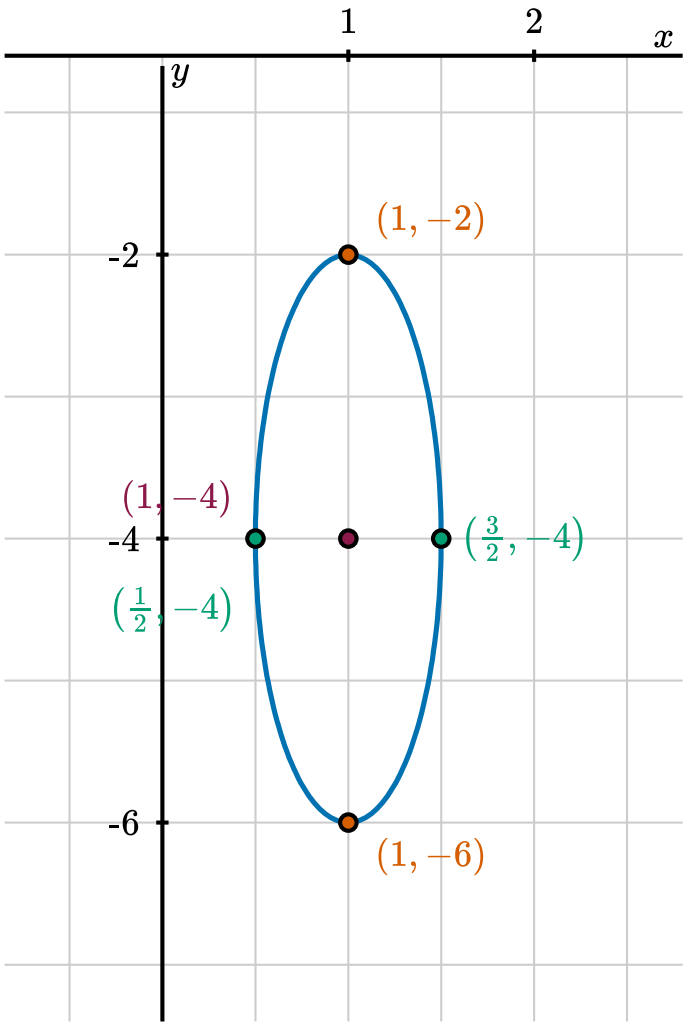


Figure 5.8.3 The ellipse $25(y+4)^2+400(x-1)^2 = 100$, with centre $(1,-4)$, horizontal semi-axis $\frac{1}{2}$ and vertical semi-axis 2.

(b) For a circle $x^2+y^2=r^2$ we would use $x = r\cos t$, $y = r\sin t$. Here the standard form tells us that

$$\frac{x-1}{\frac{1}{2}} \quad \text{and} \quad \frac{y+4}{2}$$

play the roles of $\cos t$ and $\sin t$, since their squares add to 1. So we set

$$\frac{x-1}{\frac{1}{2}} = \cos t, \qquad \frac{y+4}{2} = \sin t,$$

which gives

$$x = 1 + \frac{1}{2}\cos t, \qquad y = -4 + 2\sin t, \qquad 0 \le t \le 2\pi.$$

Letting t run over a full period of length 2π traces the entire ellipse exactly once, counterclockwise, starting at the right vertex $\left(\frac{3}{2}, -4\right)$.

8. Find the vector equation of the line of intersection of the following planes:

$$2x + y - z = 6,$$
$$x - y + 3z = 4.$$

Solution. The normal vectors are $\vec{n}_1 = \langle 2, 1, -1\rangle$ and $\vec{n}_2 = \langle 1, -1, 3\rangle$, so the direction vector of the line of intersection is

$$\mathbf{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -1 \\ 1 & -1 & 3 \end{vmatrix} = \langle 2, -7, -3\rangle.$$

Next we need a point on the line. Setting $x = 0$ in the equations of the planes gives

$$y - z = 6$$
$$-y + 3z = 4$$

Adding the two equations gives $2z = 10$, so $z = 5$ and $y = 11$. Hence the point $(0, 11, 5)$ is on the line, and the vector equation of the line of intersection is

$$\mathbf{r}(t) = \langle 0, 11, 5\rangle + t\,\langle 2, -7, -3\rangle.$$

9. Find an equation of a plane containing the line $\mathbf{r} = \langle -4, 5, -1\rangle + t\langle -4, 6, -3\rangle$ and parallel to the plane $-3x - y + 2z = 0$, in which the coefficient of x is -3.

Solution. Since the desired plane is parallel to $-3x - y + 2z = 0$, it has the same normal vector

$$\vec{n} = \langle -3, -1, 2\rangle = \langle A, B, C\rangle.$$

(Note that $\vec{n} \cdot \langle -4, 6, -3\rangle = 12 - 6 - 6 = 0$, so the line is indeed parallel to such a plane.) When $t = 0$, the point $(-4, 5, -1)$ is on the plane, so from the point-normal form,

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$
$$\Rightarrow \quad -3(x + 4) - (y - 5) + 2(z + 1) = 0.$$

10. Calculate the distance between the following two lines:

$$L_1: \quad x = 4 - t, \quad y = 3 + 2t, \quad z = -5 + 3t$$
$$L_2: \quad x = 5 - t, \quad y = 3 + 2t, \quad z = -5 + 3t$$

Solution. We notice that the two lines are parallel (both have direction vector $\mathbf{v} = \langle -1, 2, 3\rangle$) and therefore we can find the distance between a point on L_2 and the line L_1. Taking $S(5, 3, -5)$ on L_2 and $P(4, 3, -5)$ on L_1, we get $\vec{PS} = \langle 1, 0, 0\rangle$, and $\|\mathbf{v}\| = \sqrt{1 + 4 + 9} = \sqrt{14}$.

Using the cross product,

$$\vec{PS} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ -1 & 2 & 3 \end{vmatrix} = -3\,\mathbf{j} + 2\,\mathbf{k}, \qquad \left\|\vec{PS} \times \mathbf{v}\right\| = \sqrt{9 + 4} = \sqrt{13},$$

and the distance is

$$d = \frac{\left\|\vec{PS} \times \mathbf{v}\right\|}{\|\mathbf{v}\|} = \frac{\sqrt{13}}{\sqrt{14}}.$$

11. Find parametric equations for the line through $(2, 1, 1)$ which is normal to the plane containing $(0, 1, 1)$, $(-1, 2, 4)$ and $(3, 0, 1)$.

Solution. From Worksheet Exercise 5.10.10, the plane through those three points has normal vector $\vec{n} = \langle 3, 9, -2\rangle$. A line normal to the plane has $\vec{n}$ as its direction vector, so through $(2, 1, 1)$,

$$x = 2 + 3t, \qquad y = 1 + 9t, \qquad z = 1 - 2t, \qquad t \in \mathbb{R}.$$

12. Let L be the line given by

$$x = 4 + t, \qquad y = -1 - 2t, \qquad z = 5 + t.$$

Find the point on L that is closest to $(-2, 2, -1)$.

Solution. Write $Q = (-2, 2, -1)$ and let $P(t) = (4+t,\ -1-2t,\ 5+t)$ be the general point of L. The direction vector of the line is $\vec{d} = \langle 1,\ -2,\ 1\rangle$.

The closest point is the foot of the perpendicular from Q, so we want the value of t for which $\vec{QP}$ is orthogonal to $\vec{d}$. First,

$$\begin{aligned}\vec{QP} &= \langle (4+t) - (-2),\ (-1-2t) - 2,\ (5+t) - (-1)\rangle \\ &= \langle 6 + t,\ -3 - 2t,\ 6 + t\rangle\,.\end{aligned}$$

Now impose $\vec{QP} \cdot \vec{d} = 0$:

$$\begin{aligned}(6+t)(1) + (-3-2t)(-2) + (6+t)(1) &= 0 \\ 6 + t + 6 + 4t + 6 + t &= 0 \\ 18 + 6t &= 0 \\ t &= -3.\end{aligned}$$

Substituting $t = -3$ into the parametric equations gives the closest point

$$P(-3) = (4 - 3,\ -1 + 6,\ 5 - 3) = (1,\ 5,\ 2).$$

As a check, at $t = -3$ we get $\vec{QP} = \langle 3,\ 3,\ 3\rangle$, and

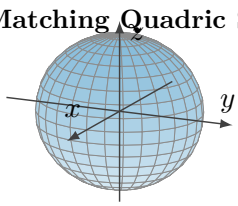
$$\vec{QP} \cdot \vec{d} = \langle 3, 3, 3\rangle \cdot \langle 1, -2, 1\rangle = 3 - 6 + 3 = 0,$$

as required. The distance from Q to the line is $\|\vec{QP}\| = \sqrt{9 + 9 + 9} = 3\sqrt{3}$.

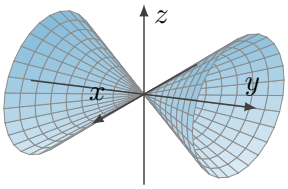
5.9 Assignment 6

These problems match equations to pictures, identify a quadric surface from its equation by slicing it with the coordinate planes and sketch it, and identify conic sections from their equations. If you would like to review the material first, see Quadric Surfaces and Conic Sections.

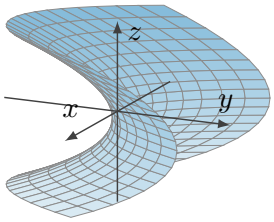
1. Matching Quadric Surfaces.



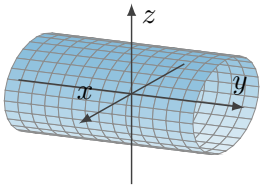
A



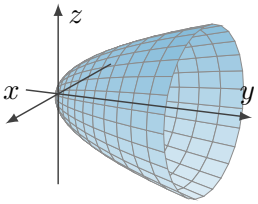
B



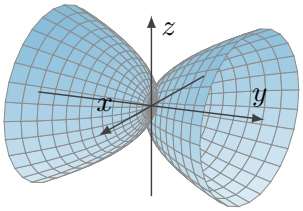
C



D



E



F

Match the equations of the surfaces with the graphs A–F shown above by entering a letter from A to F in each blank.

$y = 2x^2 + z^2$ matches graph ____.
$y^2 = x^2 + 2z^2$ matches graph ____.
$x^2 + 2z^2 = 1$ matches graph ____.
$y = x^2 - z^2$ matches graph ____.

Answer 1. E

Answer 2. B

Answer 3. D

Answer 4. C

Solution. We identify each surface by slicing, just as in this section. Here the special axis is the y-axis, so we slice with the planes $y = c$, $x = c$, and $z = c$.

For $y = 2x^2 + z^2$, the plane $y = c$ gives $2x^2 + z^2 = c$: no trace for $c < 0$, a single point when $c = 0$, and ellipses that grow as c increases. The planes $x = c$ and $z = c$ give the parabolas $y = z^2 + 2c^2$ and $y = 2x^2 + c^2$, both opening in the positive y direction. Ellipses on one side only together with parabolas: an elliptical paraboloid opening along the positive y-axis, graph E.

For $y^2 = x^2 + 2z^2$, the plane $y = c$ gives $x^2 + 2z^2 = c^2$: an ellipse for every $c \neq 0$, on both sides of the origin, shrinking to a single point when $c = 0$. The plane $z = 0$ gives $y^2 = x^2$, the two crossing lines $y = \pm x$. Ellipses collapsing to a point, together with crossing lines: an elliptic cone along the y-axis, graph B.

For $x^2 + 2z^2 = 1$, the variable y is missing, so the plane $y = c$ gives the *same* ellipse $x^2 + 2z^2 = 1$ for every value of c. Identical elliptical slices at every station: a cylinder parallel to the y-axis, graph D.

For $y = x^2 - z^2$, the plane $z = c$ gives the parabola $y = x^2 - c^2$ opening in the positive y direction, while the plane $x = c$ gives the parabola $y = c^2 - z^2$ opening in the negative y direction, and the plane $y = 0$ gives the crossing lines $z = \pm x$. Parabolas opening in opposite directions: a hyperbolic paraboloid, a saddle, graph C.

Graphs A and F match none of the equations. Slicing the ellipsoid in A gives bounded ellipses in every direction, which none of the equations produce. Slicing the two bowls in F with planes $y = c$ gives ellipses whose size grows like $\sqrt{|c|}$, unlike the straight-sided linear growth of the cone in B.

2. Identify and sketch the surface corresponding to the following equation:

$$z^2 - y^2 - x^2 = 0.$$

Solution. We can rewrite the equation as $z^2 = x^2 + y^2$. Let us look at its traces:

- $z = c$: the horizontal traces $x^2 + y^2 = c^2$ are circles of radius $|c|$ (for $c \neq 0$).
- $x = 0$: $z = \pm y$, a pair of lines intersecting at the origin.
- $y = 0$: $z = \pm x$, a pair of lines intersecting at the origin.

The surface is a **double cone** with its vertex at the origin, opening along the z-axis. See Figure 5.9.1.

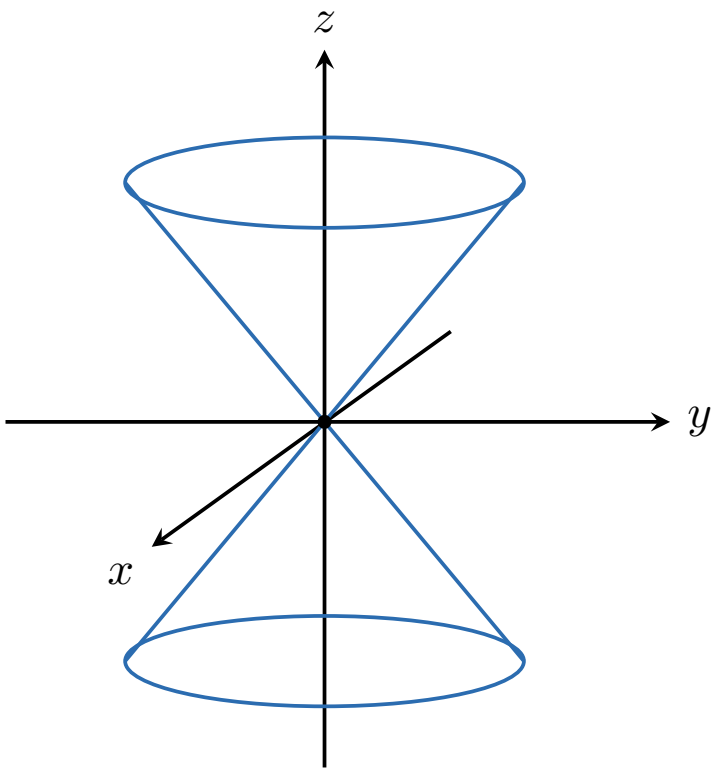


Figure 5.9.1 The surface $z^2 = x^2 + y^2$: a double cone with vertex at the origin. Horizontal cross sections are circles of radius $|z|$.

3. Identify and sketch the surface $x^2 - y^2 + z^2 - 4 = 0$.

Solution. We can rewrite the equation as $x^2 + z^2 = y^2 + 4$. Let us look at its traces:

- $y = c$: $x^2 + z^2 = c^2 + 4$, circles with radius $\sqrt{c^2 + 4} \geq 2$.
- $x = c$: $z^2 - y^2 = 4 - c^2$, hyperbolas.
- $z = c$: $x^2 - y^2 = 4 - c^2$, hyperbolas (for $c = 0$: $x^2 - y^2 = 4$).

The surface is a **hyperboloid of one sheet** whose axis is the y-axis. See Figure 5.9.2.

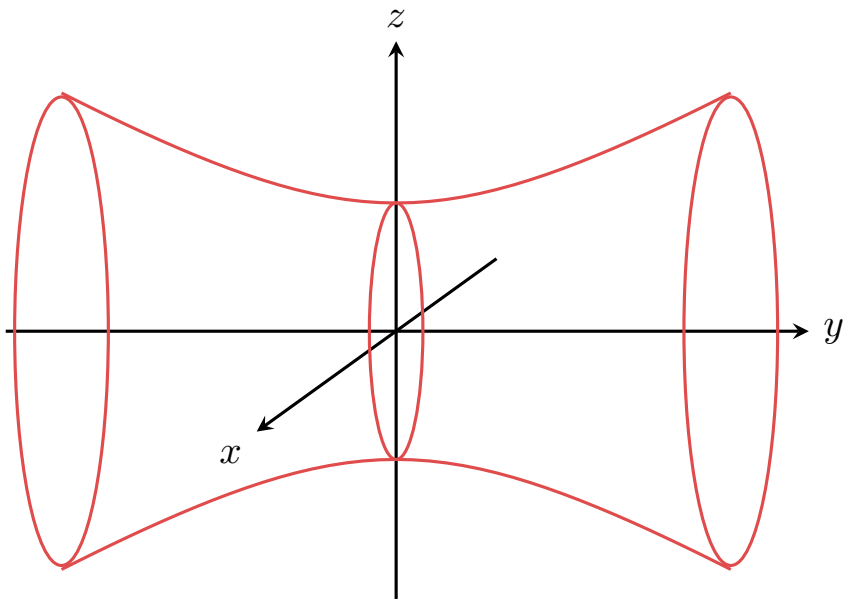


Figure 5.9.2 The hyperboloid of one sheet $x^2 + z^2 = y^2 + 4$, opening along the y-axis. The narrowest circle, of radius 2, occurs at $y = 0$.

4. Identify and sketch the surface corresponding to the equation $2x^2 - y + z^2 = 0$. Give reasons for your choice.

Solution. We can rewrite the equation as $y = 2x^2 + z^2$. Let us look at its traces:

- $y = 0$: $2x^2 + z^2 = 0$ forces $x = z = 0$, the single point $(0, 0, 0)$.
- $x = 0$: $y = z^2$, a parabola in the yz-plane.
- $z = 0$: $y = 2x^2$, a parabola in the xy-plane.
- $y = c > 0$: $2x^2 + z^2 = c$, ellipses.

The surface is an **elliptic paraboloid** with vertex at the origin, opening along the positive y-axis. See Figure 5.9.3.

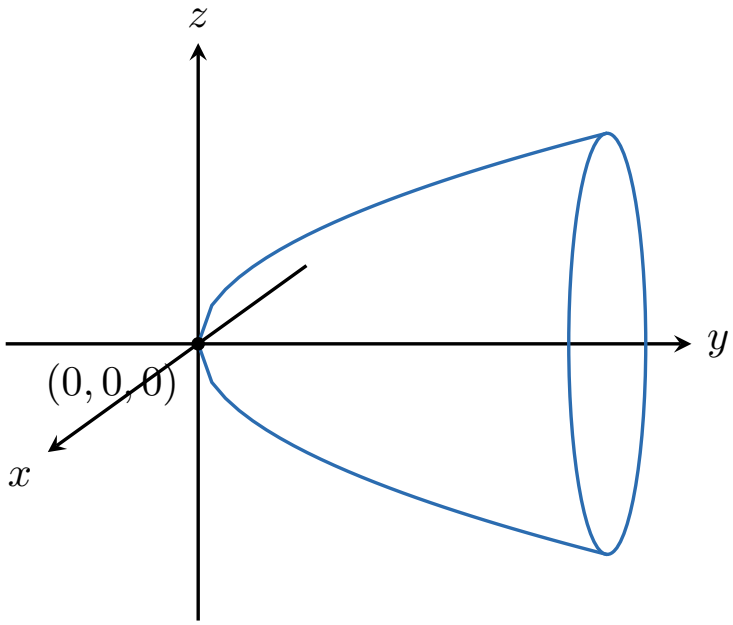


Figure 5.9.3 The elliptic paraboloid $y = 2x^2 + z^2$: a bowl with its vertex at the origin, opening along the positive y-axis. Cross sections $y = c > 0$ are ellipses.

5. Sketch the surface $4x^2 - y^2 = -4 - z^2$ in three dimensions. Label any points where the surface crosses the coordinate axes.

Solution. Collect everything on the left and divide by -4:

$$\begin{aligned} 4x^2 - y^2 + z^2 &= -4 \\ -x^2 + \frac{y^2}{4} - \frac{z^2}{4} &= 1 \\ \frac{y^2}{4} - x^2 - \frac{z^2}{4} &= 1. \end{aligned}$$

Two minus signs and one plus sign, with the constant 1 on the right, identify this as a **hyperboloid of two sheets**. The axis is the one belonging to the positive term, so the axis is the y-axis.

The traces confirm this.

- Setting $y = k$ gives $4x^2 + z^2 = k^2 - 4$, which is an ellipse when $|k| > 2$, a single point when $|k| = 2$, and empty when $|k| < 2$. So there is nothing at all in the slab $-2 < y < 2$: this is why there are two separate sheets.
- Setting $z = 0$ gives $y^2/4 - x^2 = 1$, a hyperbola opening along the y-axis, and setting $x = 0$ gives $y^2/4 - z^2/4 = 1$, likewise.

Axis crossings. Setting $y = z = 0$ gives $4x^2 = -4$, impossible; setting $x = y = 0$ gives $z^2 = -4$, impossible. Setting $x = z = 0$ gives $y^2 = 4$, so the surface meets the axes only at the two vertices

$$(0,\ 2,\ 0) \qquad \text{and} \qquad (0,\ -2,\ 0).$$

See Figure 5.9.4.

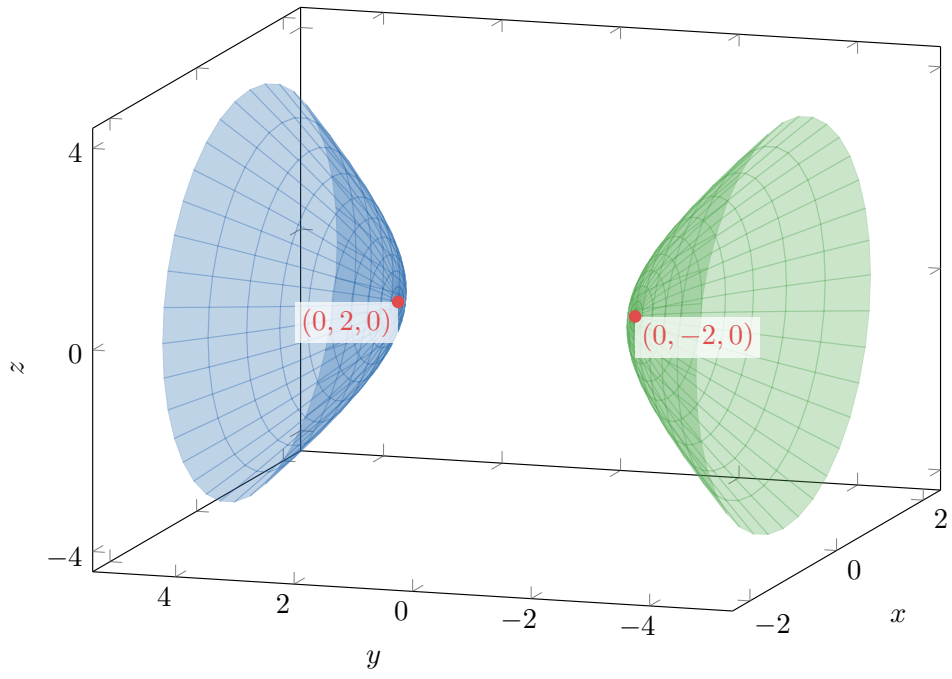


Figure 5.9.4 The hyperboloid of two sheets $\frac{y^2}{4} - x^2 - \frac{z^2}{4} = 1$, with vertices at $(0, \pm 2, 0)$. Nothing at all lies in the slab $-2 < y < 2$, which is why the two sheets are separate.

6. Find the equation of the hyperbola with asymptotes $y = 3x$ and $y = -3x$ passing through the point $(0, 2)$.

Solution. Since the hyperbola meets the y-axis (at $(0, 2)$) it opens up and down, so it has the form

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1,$$

whose asymptotes are $y = \pm\frac{a}{b}x$. Setting $x = 0$, $y = 2$ gives $a = 2$, and matching the asymptote slopes,

$$\frac{a}{b} = 3 \;\Rightarrow\; b = \frac{a}{3} = \frac{2}{3}.$$

Therefore

$$\frac{y^2}{4} - \frac{x^2}{\frac{4}{9}} = 1 \;\Rightarrow\; \frac{y^2}{4} - \frac{9x^2}{4} = 1 \;\Rightarrow\; y^2 - 9x^2 = 4.$$

See Figure 5.9.5.

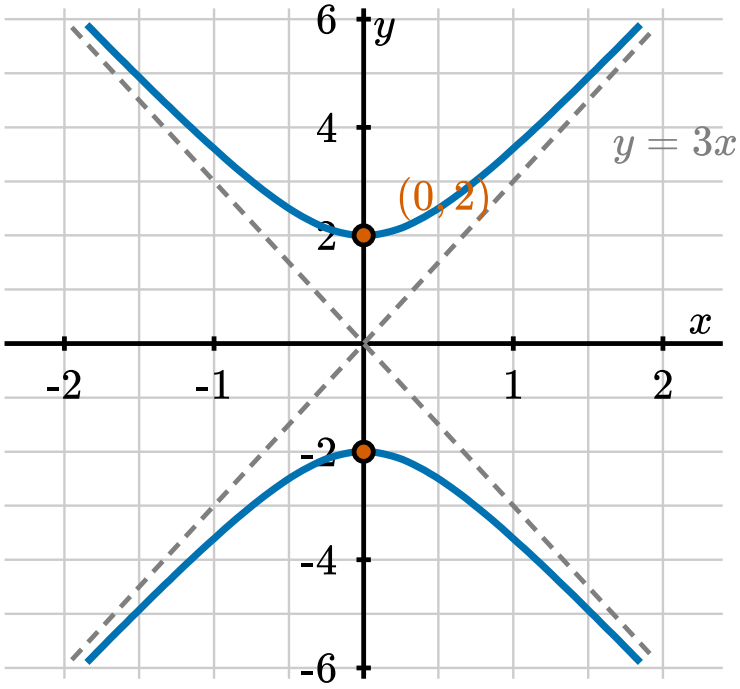


Figure 5.9.5 The hyperbola $y^2 - 9x^2 = 4$ with its asymptotes $y = \pm 3x$, passing through $(0, 2)$ and $(0, -2)$.

7. Find the equation of the parabola with vertex at the origin and axis of symmetry the y-axis, passing through the point $(4, 2)$. Then find its focus and its directrix.

Solution. The vertex is at the origin and the axis of symmetry is the y-axis, so the parabola has the form

$$x^2 = 4py,$$

whose focus is $(0, p)$ and whose directrix is the line $y = -p$. Only the one number p is unknown, and the point $(4, 2)$ pins it down:

$$4^2 = 4p(2) \;\Rightarrow\; 16 = 8p \;\Rightarrow\; p = 2.$$

Therefore

$$x^2 = 8y, \qquad \text{that is} \qquad y = \frac{x^2}{8}.$$

Since $p = 2 > 0$ the parabola opens upward, its focus is $(0, 2)$ and its directrix is the horizontal line $y = -2$.

As a check on the defining property of a parabola, the point $(4, 2)$ is at distance $\sqrt{(4-0)^2 + (2-2)^2} = 4$ from the focus and at distance $2 - (-2) = 4$ from the directrix, and the two agree. See Figure 5.9.6.

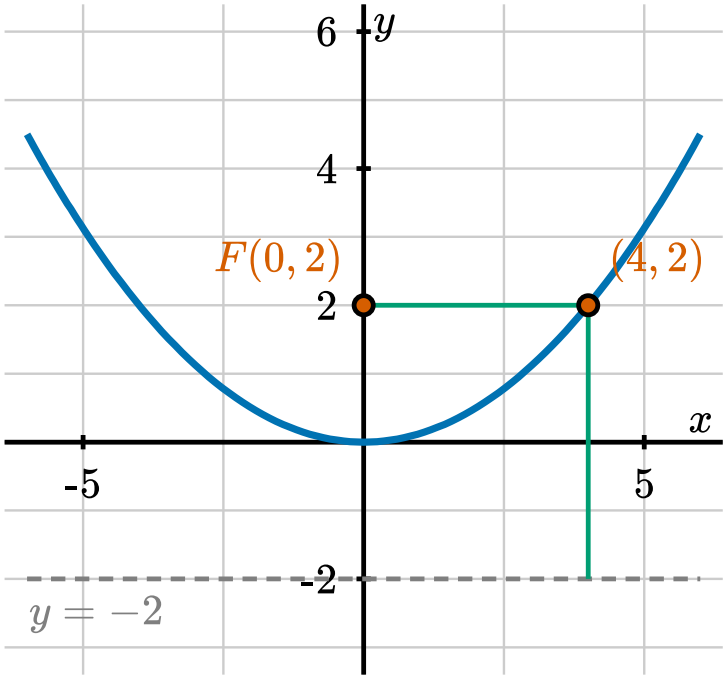


Figure 5.9.6 The parabola $x^2 = 8y$ through $(4, 2)$, with focus $(0, 2)$ and directrix $y = -2$. The point of the curve is the same distance from the focus as from the directrix.

8. Two trees grow 100 feet apart. A 120 foot rope is attached to the first tree at a height of 100 feet and to the second tree at a height of 50 feet. A child folds an empty plastic water bottle over the rope and rappels from the high end of the rope to the low end. At each moment, the rope is taut. The point where the water bottle is folded over the rope traces a path. Ignoring the numbers, state what kind of curve the path makes: line, ellipse, parabola or hyperbola.

Solution. Let A and B be the two attachment points and P the point where the bottle folds over the taut rope. Because the rope is taut and has fixed total length, the two straight segments $\overline{AP}$ and $\overline{PB}$ together always use up the whole rope:

$$|AP| + |PB| = 120 \text{ feet} \qquad \text{(constant)}.$$

This is precisely the defining property of an **ellipse** with foci A and B. (The curve exists because $120 > |AB|$: the distance between the attachment points is $\sqrt{100^2 + 50^2} \approx 111.8 < 120$ feet.) So the path is an arc of an **ellipse**, lying in the vertical plane containing the two trees. Since the bottle hangs below the rope, P stays on the lower half of the ellipse. See Figure 5.9.7 and Figure 5.9.8.

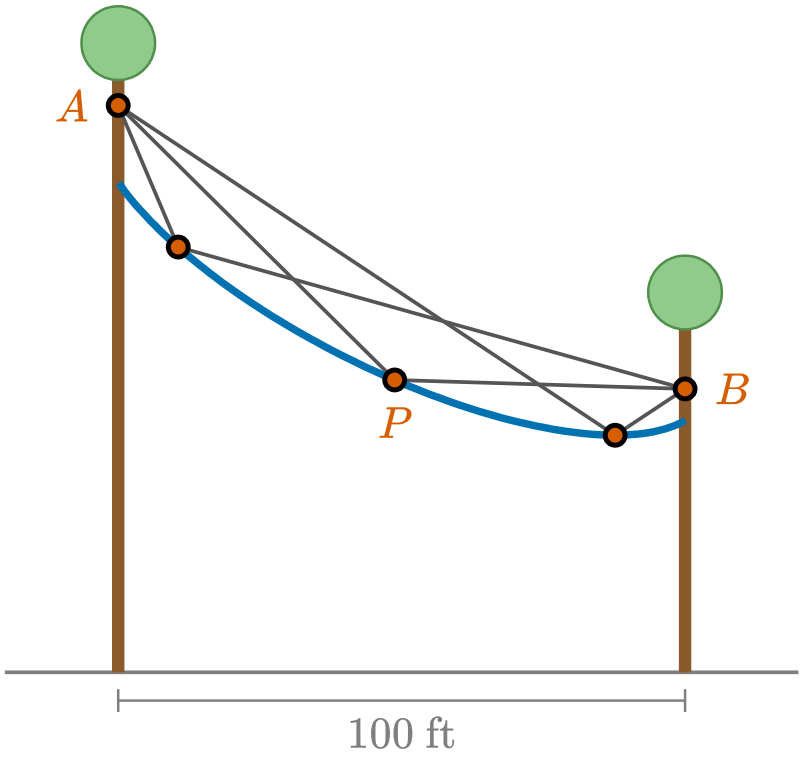


Figure 5.9.7 The two trees, with the rope tied at A, 100 feet up the first tree, and at B, 50 feet up the second, and the taut rope at three moments. The point P where the bottle folds over the rope traces the blue curve.

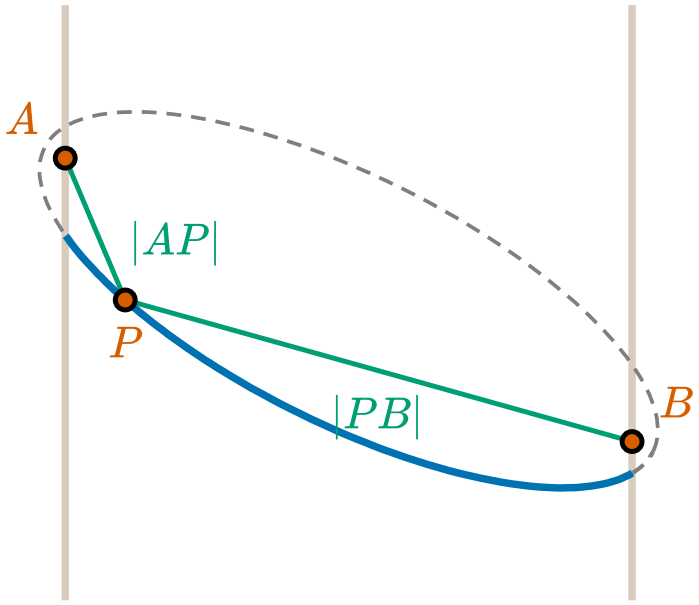


Figure 5.9.8 The curve of Figure 5.9.7 is part of the ellipse with foci A and B: wherever P is, the two straight pieces of rope add up to $|AP| + |PB| = 120$ feet. The bottle only travels the solid part, between the two trees; the dashed rest of the ellipse is out of its reach.

5.10 Review Problems #2

These are the problems from the Chapter 3 topics that were not assigned in Assignments 4-6, collected here for review. They are not assigned, but they are fair game on an exam. Problems marked with a star ($*$) are the exception: those are for the interested reader and will not be examined.

The topics they cover, with links to the notes:

- Three-Dimensional Coordinate Systems
- Vectors
- The Dot Product
- The Cross Product
- Parametrization of Plane Curves
- Lines and Planes in Space
- Conic Sections
- Quadric Surfaces

1. Describing More Equations and Inequalities Geometrically. Describe the geometrical meaning of the following equalities and inequalities.

A. $y = -2$

B. $y = -2,\ z = 1$

C. $(x-2)^2 + (y+1)^2 + z^2 = 9$

D. $(x-2)^2 + (y+1)^2 + z^2 \leq 9,\ y \geq -1$

E. $y^2 + z^2 = 4$

F. $y^2 + z^2 = 4,\ 0 \leq x \leq 5$

G. $x = y$

Solution. ***A.*** This is the set of all points whose y-coordinate equals -2, that is $\{(x, y, z) \in \mathbb{R}^3 : y = -2\}$. Both x and z are unrestricted, so it is a plane parallel to the xz-plane, sitting two units to the negative side of it, shown in Figure 5.10.1.

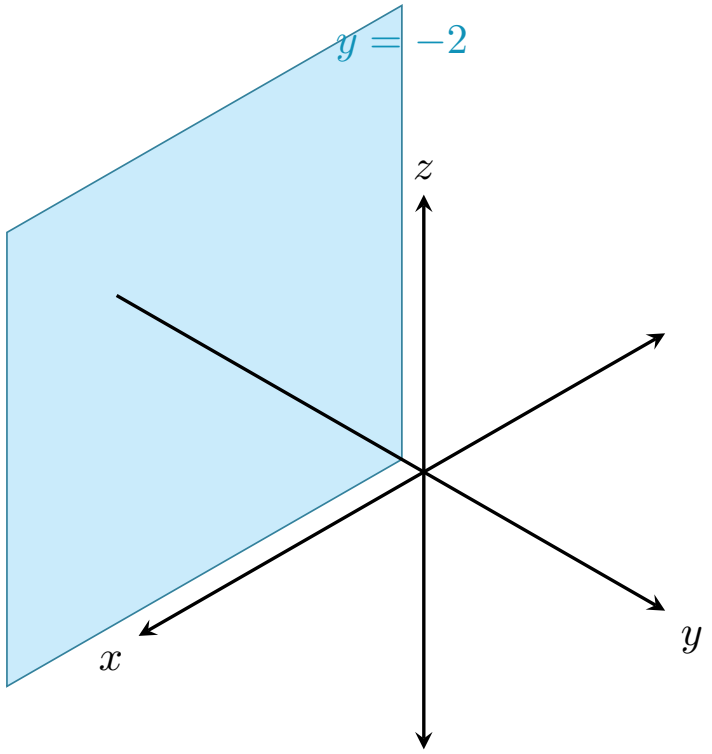


Figure 5.10.1 Part ***A***: the plane $y = -2$, parallel to the xz-plane.

B. This is the intersection of the two planes $y = -2$ and $z = 1$, which is a line. Only x is unrestricted, so the line is parallel to the x-axis and passes through the point $(0, -2, 1)$, as shown in Figure 5.10.2.

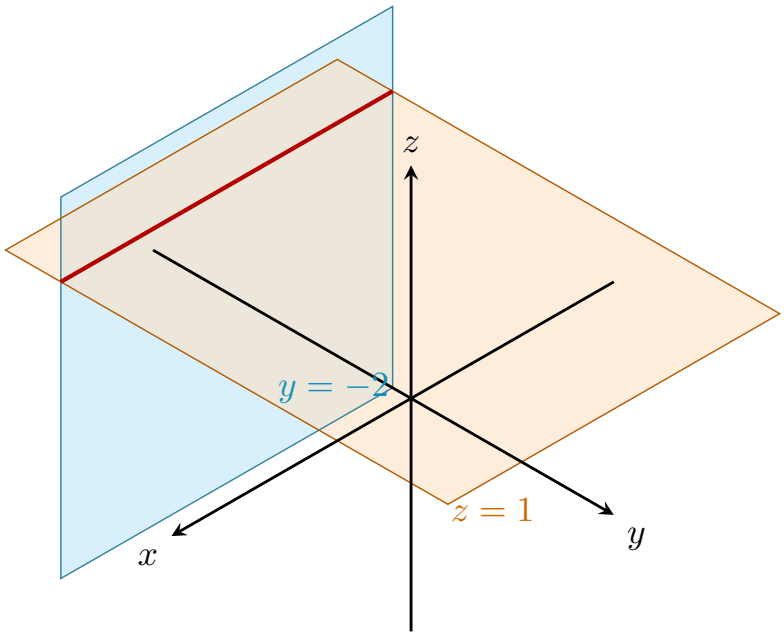
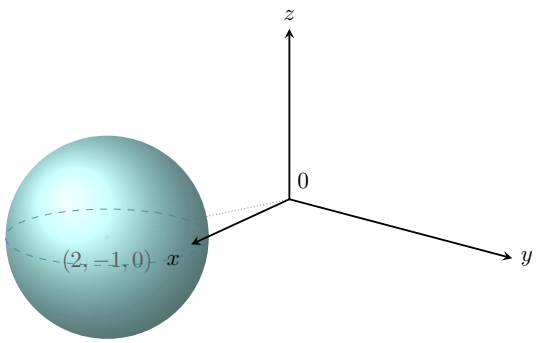


Figure 5.10.2 Part ***B***: the planes $y = -2$ and $z = 1$ meet in a line parallel to the x-axis, highlighted in red.

C. Comparing with the standard equation of a sphere, this is the sphere of radius 3 centered at $(2, -1, 0)$. Note the center lies in the xy-plane, and since the radius is 3 the sphere reaches from $z = -3$ up to $z = 3$, as shown in Figure 5.10.3.



$(x-2)^2+(y+1)^2+z^2=9$

Figure 5.10.3 Part ***C***: the sphere $(x-2)^2+(y+1)^2+z^2=9$, of radius 3 centered at $(2,-1,0)$.

D. Replacing the equality by $\leq$ gives the solid ball: every point whose distance from $(2,-1,0)$ is at most 3. The extra condition $y \geq -1$ keeps only the points on one side of the plane $y=-1$. That plane passes through the *center* of the ball, so exactly half of the ball survives: the region is a solid half-ball of radius 3, whose flat face is the disk of radius 3 cut out of the plane $y=-1$, shown in Figure 5.10.4.

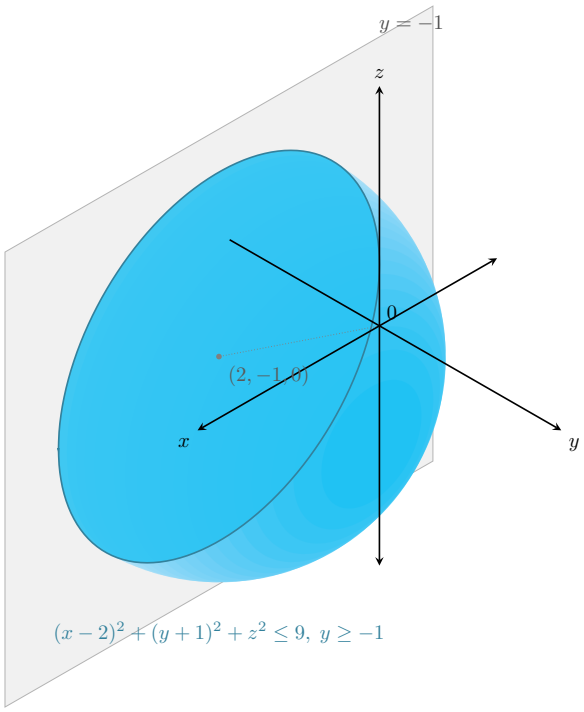
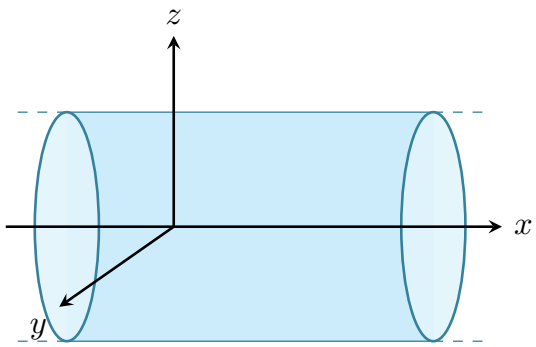


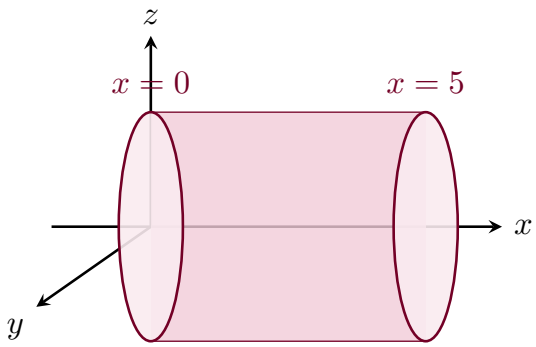
Figure 5.10.4 Part ***D***: the half-ball of radius 3 centered at $(2,-1,0)$, cut by the condition $y \geq -1$, a plane through its center.

E. In the yz-plane, $y^2+z^2=4$ is a circle of radius 2 centered at the origin. In space x is unrestricted, so the equation represents an infinitely long cylinder of radius 2 whose axis is the x-axis. Had the missing variable been z instead, the axis would have been the z-axis: the axis of the cylinder is always the axis of the variable that does not appear. The left half of Figure 5.10.5 shows this cylinder.

F. The extra condition $0 \leq x \leq 5$ cuts the cylinder of part ***E*** down to a piece of length 5, running from the circle in the plane $x=0$ to the circle in the plane $x=5$, as on the right of Figure 5.10.5. It is a tube, not a solid: the two end disks are not included, since the equation forces $y^2+z^2=4$ exactly.



$y^2 + z^2 = 4$



$y^2 + z^2 = 4,\ 0 \le x \le 5$

Figure 5.10.5 Parts ***E*** and ***F***. On the left, the infinite cylinder $y^2 + z^2 = 4$, whose axis is the x-axis; the dashed lines indicate that it continues without bound in both directions. On the right, the same cylinder cut down to $0 \le x \le 5$.

G. The condition $x = y$ says nothing about z, so whenever a point satisfies it, so does the whole vertical line through that point. In the xy-plane the equation $x = y$ describes a line through the origin, and sweeping that line vertically gives a plane: the plane containing the z-axis that cuts the xy-plane along the line $y = x$, shown in Figure 5.10.6. It is *not* parallel to any coordinate plane, unlike part ***A***.

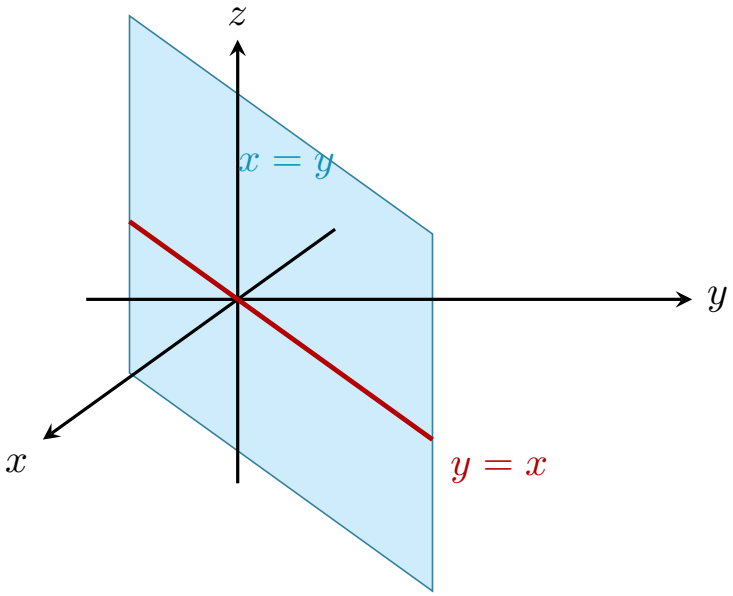


Figure 5.10.6 Part ***G***: the plane $x = y$. It contains the z-axis and meets the xy-plane in the line $y = x$, drawn in red.

2. A Solid Bounded by a Cylinder and a Plane. Consider the solid region bounded by the three surfaces

$$z = 0, \qquad x^2 + y^2 = 4, \qquad x + y + z = 4,$$

restricted to $z \ge 0$. Describe each of the three bounding surfaces, and describe the solid region they enclose.

Solution. The equation $x^2 + y^2 = 4$ is missing z, so it is the infinite cylinder of radius 2 whose axis is the z-axis. The equation $z = 0$ is the xy-plane, and $x+y+z = 4$ is a plane tilted with respect to all three coordinate axes.

Solving the third equation for z gives $z = 4 - x - y$. Over the disk $x^2 + y^2 \le 4$ enclosed by the cylinder, this height stays positive: the largest value of $x + y$ on that disk is $2\sqrt{2} \approx 2.83$, attained in the direction $(1, 1)$, so $z = 4 - x - y \ge 4 - 2\sqrt{2} > 0$ throughout the disk. The tilted plane therefore never dips below the xy-plane over the region enclosed by the cylinder.

The solid is the piece of the solid cylinder $x^2 + y^2 \le 4$ lying between the flat base $z = 0$ and the slanted lid $z = 4 - x - y$: an oblique, truncated cylinder whose base is the disk of radius 2 in the xy-plane and whose

slanted top is an ellipse, shown in Figure 5.10.7.

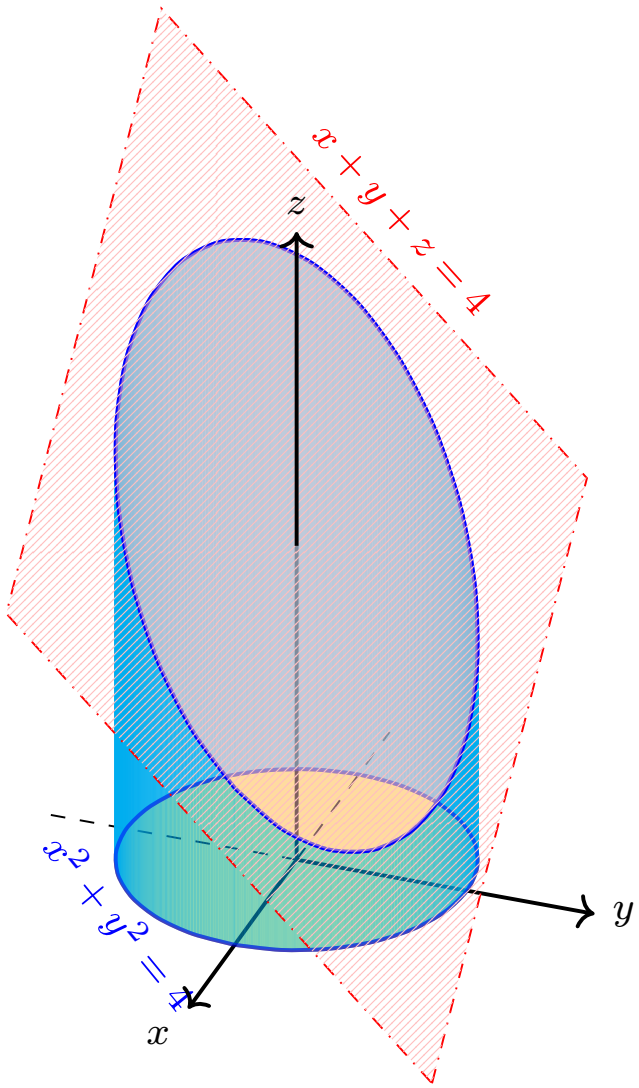


Figure 5.10.7 The solid bounded by $z = 0$, the cylinder $x^2 + y^2 = 4$, and the tilted plane $x + y + z = 4$. The disk in the xy-plane is the flat base, and the plane, hatched in pink, cuts the cylinder in an ellipse that forms the slanted top.

3. The vectors $2\,\mathbf{i} + 3\,\mathbf{j}$, $4\,\mathbf{i} + \mathbf{j}$ and $5\,\mathbf{i} + y\,\mathbf{j}$ have their tails at the origin. Find the value of y that will make their heads collinear.

Three points are **collinear** when they all lie on one straight line. Two points always do, so the condition is a condition on the third: it has to lie on the line already determined by the other two.

Solution. The heads are the points $(2, 3)$, $(4, 1)$, and $(5, y)$, and we want all three on a single line. Two of them are already given, so the line is fixed: the line through the first two has slope

$$m = \frac{1 - 3}{4 - 2} = -1.$$

For $(5, y)$ to lie on that same line, the slope from $(4, 1)$ to $(5, y)$ must also be -1:

$$\frac{y - 1}{5 - 4} = -1 \;\Rightarrow\; y - 1 = -1 \;\Rightarrow\; y = 0.$$

Equivalently, in vector language: the three heads are collinear exactly when the displacement vectors between them are parallel. Here $\langle 4, 1\rangle - \langle 2, 3\rangle = \langle 2, -2\rangle$ and $\langle 5, y\rangle - \langle 4, 1\rangle = \langle 1, y - 1\rangle$, and these are parallel exactly when $\langle 1, y - 1\rangle = \frac{1}{2}\langle 2, -2\rangle$, that is $y - 1 = -1$, giving $y = 0$ again.

4. Two vectors $\vec{u}$ and $\vec{v}$ are drawn with a common tail; $\vec{u}$ points up and to the right, $\vec{v}$ points up and to the left, and the angle between them is slightly larger than a right angle. Sketch and clearly label a diagram showing $\vec{u}$, $\vec{v}$, and $\vec{v}-\vec{u}$.

Solution. To make the picture concrete let us take $\vec{u} = \langle 4, 2\rangle$ and $\vec{v} = \langle -3, 5\rangle$, which have roughly the directions described. Everything below is a statement about the *geometry*, so the same picture is correct for any pair of vectors in that configuration.

The difference $\vec{v}-\vec{u}$ is the vector that, added to $\vec{u}$, gives $\vec{v}$. So when $\vec{u}$ and $\vec{v}$ are drawn from a common tail, $\vec{v}-\vec{u}$ is the arrow that runs *from the tip of* $\vec{u}$ *to the tip of* $\vec{v}$. (Here it is $\langle -7, 3\rangle$.) It is easy to get this backwards; the check is that $\vec{u} + (\vec{v}-\vec{u}) = \vec{v}$. See Figure 5.10.8.

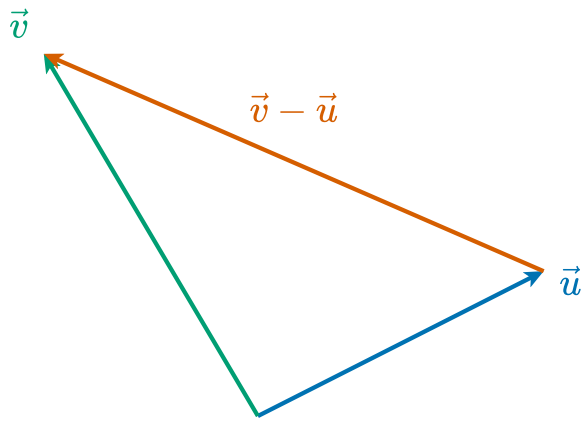


Figure 5.10.8 $\vec{v}-\vec{u}$ runs from the tip of $\vec{u}$ to the tip of $\vec{v}$.

5. Find the projection of $-\mathbf{i}+5\,\mathbf{j}-2\,\mathbf{k}$ on $3\,\mathbf{i}-\mathbf{j}$.

Solution. With $\vec{u} = \langle -1, 5, -2\rangle$ and $\vec{v} = \langle 3, -1, 0\rangle$,

$$\vec{u}\cdot\vec{v} = -3-5+0 = -8, \qquad \|\vec{v}\|^2 = 9+1+0 = 10,$$

so

$$\begin{aligned}\text{proj}_{\vec{v}}\,\vec{u} &= \left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2}\right)\vec{v} = -\frac{8}{10}\langle 3,-1,0\rangle \\ &= \left\langle -\frac{12}{5}, \frac{4}{5}, 0\right\rangle = -\frac{12}{5}\,\mathbf{i}+\frac{4}{5}\,\mathbf{j}.\end{aligned}$$

The negative scalar tells us the projection points opposite to $\vec{v}$, i.e. the angle between $\vec{u}$ and $\vec{v}$ is obtuse.

6. Let $\vec{v}$ be the vector from $(-4, 3, -1)$ to $(-2, -1, 3)$.

(a) Find a non-zero unit vector that is orthogonal to $\vec{v}$.

(b) Find the angle between $\vec{v}$ and $\vec{w} = 3\mathbf{i}-\mathbf{k}$.

Solution. First,

$$\begin{aligned}\vec{v} &= \langle -2-(-4),\ -1-3,\ 3-(-1)\rangle = \langle 2,\ -4,\ 4\rangle, \\ \|\vec{v}\| &= \sqrt{4+16+16} = \sqrt{36} = 6.\end{aligned}$$

(a) We need any non-zero $\vec{u}$ with $\vec{u}\cdot\vec{v} = 0$. Writing $\vec{u} = \langle a, b, c\rangle$, the condition is

$$2a-4b+4c = 0.$$

This is one equation in three unknowns, so there are infinitely many answers; we just pick a convenient one. Taking $c = 0$ gives $2a = 4b$, so $a = 2$, $b = 1$ works:

$$\vec{u} = \langle 2,\ 1,\ 0\rangle, \qquad \vec{u}\cdot\vec{v} = 4-4+0 = 0.$$

Finally we normalize:

$$\hat{u} = \frac{\vec{u}}{\|\vec{u}\|} = \frac{1}{\sqrt{5}}\langle 2,\ 1,\ 0\rangle = \left\langle \frac{2}{\sqrt{5}},\ \frac{1}{\sqrt{5}},\ 0 \right\rangle.$$

(b) With $\vec{w} = \langle 3, 0, -1\rangle$ we have

$$\vec{v}\cdot\vec{w} = (2)(3) + (-4)(0) + (4)(-1) = 6 - 4 = 2,$$
$$\|\vec{w}\| = \sqrt{9 + 0 + 1} = \sqrt{10}.$$

Therefore

$$\cos\theta = \frac{\vec{v}\cdot\vec{w}}{\|\vec{v}\|\,\|\vec{w}\|} = \frac{2}{6\sqrt{10}} = \frac{1}{3\sqrt{10}} \approx 0.1054,$$

so

$$\theta = \cos^{-1}\left(\frac{1}{3\sqrt{10}}\right) \approx 1.465 \text{ radians} \approx 83.9^\circ.$$

The angle is slightly less than a right angle, which matches the fact that the dot product is positive but small.

7. Let $A = (0, 0, 1)$ and $B = (0, 2, 3)$. Find an equation for the set of all points $P = (x, y, z)$ such that $\vec{AP}$ is orthogonal to $\vec{BP}$. Then sketch the resulting surface in three dimensions.

Solution. The two vectors are

$$\vec{AP} = \langle x - 0,\ y - 0,\ z - 1\rangle = \langle x,\ y,\ z - 1\rangle,$$
$$\vec{BP} = \langle x,\ y - 2,\ z - 3\rangle.$$

Orthogonality means the dot product vanishes:

$$\vec{AP}\cdot\vec{BP} = 0$$
$$x^2 + y(y-2) + (z-1)(z-3) = 0$$
$$x^2 + y^2 - 2y + z^2 - 4z + 3 = 0.$$

Now complete the square in y and in z:

$$x^2 + \left(y^2 - 2y + 1\right) + \left(z^2 - 4z + 4\right) = -3 + 1 + 4$$
$$x^2 + (y-1)^2 + (z-2)^2 = 2.$$

This is the sphere of radius $\sqrt{2}$ centred at $(0, 1, 2)$. That centre is exactly the midpoint of A and B, and the radius is exactly half of $\|\vec{AB}\| = \|\langle 0, 2, 2\rangle\| = 2\sqrt{2}$. This is the three-dimensional version of the familiar fact from plane geometry: the points from which a given segment subtends a right angle form the circle having that segment as a diameter. See Figure 5.10.9.

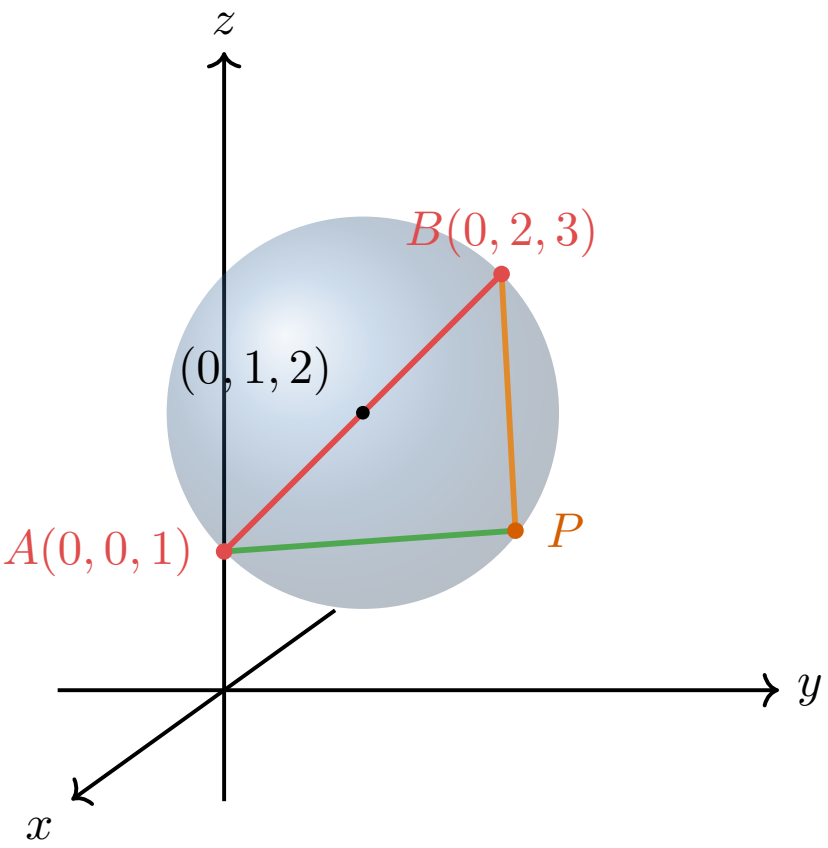


Figure 5.10.9 The set of points P with $\vec{AP} \perp \vec{BP}$ is the sphere having $\overline{AB}$ as a diameter, namely $x^2 + (y-1)^2 + (z-2)^2 = 2$.

8. Two vectors $\vec{u}$ and $\vec{v}$ are drawn with a common tail; $\vec{u}$ points up and to the right, $\vec{v}$ points up and to the left, and the angle between them is slightly larger than a right angle. Sketch and clearly label a diagram showing $\vec{u}$, $\vec{v}$, and $\text{proj}_{\vec{u}}\,\vec{v}$.

Solution. To make the picture concrete let us take $\vec{u} = \langle 4, 2\rangle$ and $\vec{v} = \langle -3, 5\rangle$, which have roughly the directions described. Everything below is a statement about the *geometry*, so the same picture is correct for any pair of vectors in that configuration.

The projection of $\vec{v}$ onto $\vec{u}$ is the shadow that $\vec{v}$ casts on the line through $\vec{u}$, obtained by dropping a perpendicular from the tip of $\vec{v}$ to that line. Algebraically,

$$\begin{aligned}\text{proj}_{\vec{u}}\,\vec{v} &= \left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|^2}\right)\vec{u} = \left(\frac{-12+10}{20}\right)\langle 4, 2\rangle \\ &= -\frac{1}{10}\,\langle 4, 2\rangle = \langle -0.4,\ -0.2\rangle\,.\end{aligned}$$

Because the angle between $\vec{u}$ and $\vec{v}$ is *obtuse*, the dot product is negative and the projection points in the direction *opposite* to $\vec{u}$. This is the feature of the picture the problem is testing. See Figure 5.10.10.

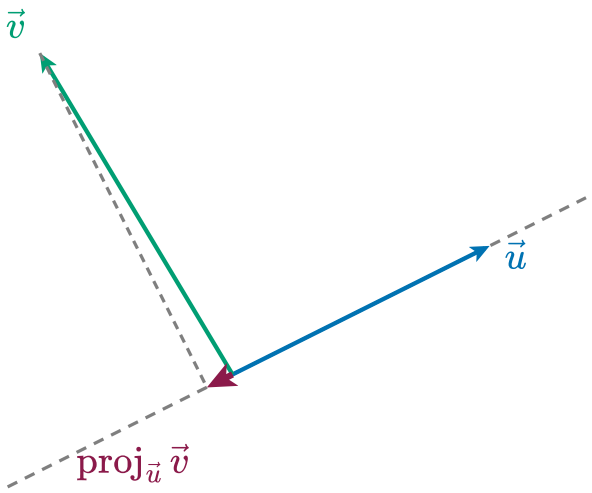


Figure 5.10.10 Since the angle between $\vec{u}$ and $\vec{v}$ is obtuse, $\text{proj}_{\vec{u}}\,\vec{v}$ points opposite to $\vec{u}$.

9. Suppose $\mathbf{u}$ and $\mathbf{v}$ are non-zero vectors satisfying

$$\|\mathbf{u}+\mathbf{v}\| = 5, \qquad \|\mathbf{u}-\mathbf{v}\| = 13,$$

and

$$\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = 97, \qquad \|\mathbf{u}\| = 7.$$

Without finding the coordinates of either vector:

(a) Compute $\mathbf{u}\cdot\mathbf{v}$.

(b) Find $\|\mathbf{v}\|$.

(c) Find the angle θ between $\mathbf{u}$ and $\mathbf{v}$.

(d) Determine whether the angle is acute, right, or obtuse.

Solution. The whole problem rests on one idea: the square of a length is a dot product, $\|\mathbf{w}\|^2 = \mathbf{w}\cdot\mathbf{w}$. Expanding with the distributive property,

$$\|\mathbf{u}+\mathbf{v}\|^2 = (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) = \|\mathbf{u}\|^2 + 2\,\mathbf{u}\cdot\mathbf{v} + \|\mathbf{v}\|^2,$$
$$\|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) = \|\mathbf{u}\|^2 - 2\,\mathbf{u}\cdot\mathbf{v} + \|\mathbf{v}\|^2.$$

These two identities are the algebraic form of the parallelogram in Figure 5.10.11: the diagonals of the parallelogram built on $\mathbf{u}$ and $\mathbf{v}$ are $\mathbf{u}+\mathbf{v}$ and $\mathbf{u}-\mathbf{v}$.

(a) Subtracting the second identity from the first makes the $\|\mathbf{u}\|^2$ and $\|\mathbf{v}\|^2$ terms cancel, leaving only the dot product:

$$\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4\,\mathbf{u}\cdot\mathbf{v}.$$

Substituting the two given lengths,

$$\begin{aligned} 5^2 - 13^2 &= 4\,\mathbf{u}\cdot\mathbf{v}, \\ 25 - 169 &= 4\,\mathbf{u}\cdot\mathbf{v}, \\ -144 &= 4\,\mathbf{u}\cdot\mathbf{v}, \end{aligned}$$

and therefore $\mathbf{u}\cdot\mathbf{v} = -36$.

It is worth pausing to check that the data are consistent. *Adding* the two identities instead of subtracting gives the parallelogram law,

$$\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\left(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\right),$$

and indeed $5^2 + 13^2 = 25 + 169 = 194 = 2(97)$, exactly as it should be.

(b) We are told $\|\mathbf{u}\| = 7$ and $\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = 97$, so

$$\begin{aligned} 7^2 + \|\mathbf{v}\|^2 &= 97, \\ 49 + \|\mathbf{v}\|^2 &= 97, \\ \|\mathbf{v}\|^2 &= 48. \end{aligned}$$

A length is never negative, so $\|\mathbf{v}\| = \sqrt{48} = 4\sqrt{3} \approx 6.93$.

(c) For non-zero vectors, $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\| \cos\theta$, so

$$\begin{aligned} \cos\theta &= \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|} = \frac{-36}{(7)\,(4\sqrt{3})} \\ &= -\frac{9}{7\sqrt{3}} = -\frac{3\sqrt{3}}{7} \approx -0.7423, \end{aligned}$$

and therefore

$$\theta = \cos^{-1}\left(-\frac{3\sqrt{3}}{7}\right) \approx 2.407 \text{ radians} \approx 137.9^\circ.$$

(d) Since $\mathbf{u}\cdot\mathbf{v} = -36 < 0$ and the lengths are positive, $\cos\theta < 0$. Hence $\frac{\pi}{2} < \theta < \pi$ and the angle is **obtuse**. Notice that we could have answered this part immediately after part (a), without knowing either length: the *sign* of the dot product already decides the type of angle.

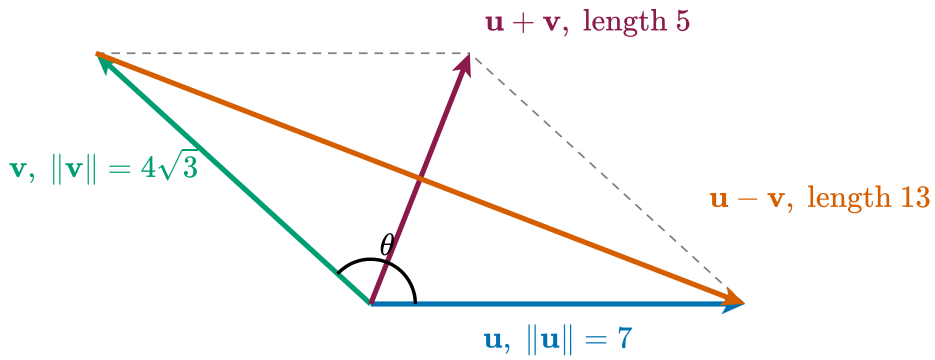


Figure 5.10.11 The parallelogram built on $\mathbf{u}$ and $\mathbf{v}$. Its diagonals are $\mathbf{u}+\mathbf{v}$, of length 5, and $\mathbf{u}-\mathbf{v}$, of length 13. One short diagonal together with one long one forces the angle θ between the sides to be obtuse.

A remark on the data. The two diagonal lengths by themselves already determine $\mathbf{u}\cdot\mathbf{v}$, and the condition $\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = 97$ then follows automatically from the parallelogram law. So the individual lengths are *not* pinned down by that data alone: the pair $\|\mathbf{u}\| = 4\sqrt{3}$, $\|\mathbf{v}\| = 7$ fits equally well and gives the same angle, while other splits of 97 give different angles. This is why the extra condition $\|\mathbf{u}\| = 7$ is needed for parts (b) and (c).

10.

A. Find the area of the triangle with vertices $(0,1,1)$, $(-1,2,4)$, $(3,0,1)$.

B. Find the equation of the plane containing the triangle.

Solution. Take $A(0,1,1)$ as the base vertex and form two edge vectors:

$$\vec{a} = \vec{AB} = \langle -1, 1, 3\rangle, \qquad \vec{b} = \vec{AC} = \langle 3, -1, 0\rangle.$$

Their cross product is

$$\vec{a}\times\vec{b}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-1&1&3\\3&-1&0\end{vmatrix}=(0+3)\mathbf{i}-(0-9)\mathbf{j}+(1-3)\mathbf{k}=\langle 3,9,-2\rangle.$$

A. The magnitude of the cross product is the area of the parallelogram spanned by $\vec{a}$ and $\vec{b}$, so the triangle has half that area:

$$\text{Area}=\frac{1}{2}\left\|\vec{a}\times\vec{b}\right\|=\frac{1}{2}\sqrt{9+81+4}=\frac{\sqrt{94}}{2}.$$

B. The cross product is normal to the plane, so the plane has the form $3x+9y-2z=D$. Substituting the point $(0,1,1)$ gives $D=0+9-2=7$, hence

$$3x+9y-2z=7.$$

11. Two vectors $\vec{u}$ and $\vec{v}$ are drawn with a common tail; $\vec{u}$ points up and to the right, $\vec{v}$ points up and to the left, and the angle between them is slightly larger than a right angle. Sketch and clearly label a diagram showing $\vec{u}$, $\vec{v}$, and $\vec{u}\times\vec{v}$.

Solution. To make the picture concrete let us take $\vec{u}=\langle 4,2\rangle$ and $\vec{v}=\langle -3,5\rangle$, which have roughly the directions described. Everything below is a statement about the *geometry*, so the same picture is correct for any pair of vectors in that configuration.

The cross product $\vec{u}\times\vec{v}$ is perpendicular to both, hence perpendicular to the page. The right-hand rule (curl the fingers of the right hand from $\vec{u}$ towards $\vec{v}$ through the smaller angle) shows that it points *out of the page*, which we draw as a circle with a dot in it. Indeed, treating the vectors as lying in the xy-plane,

$$\vec{u}\times\vec{v}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\4&2&0\\-3&5&0\end{vmatrix}=\langle 0,\ 0,\ (4)(5)-(2)(-3)\rangle=26\,\mathbf{k},$$

and the **k**-component is positive. Its length, 26, is the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$. See Figure 5.10.12.

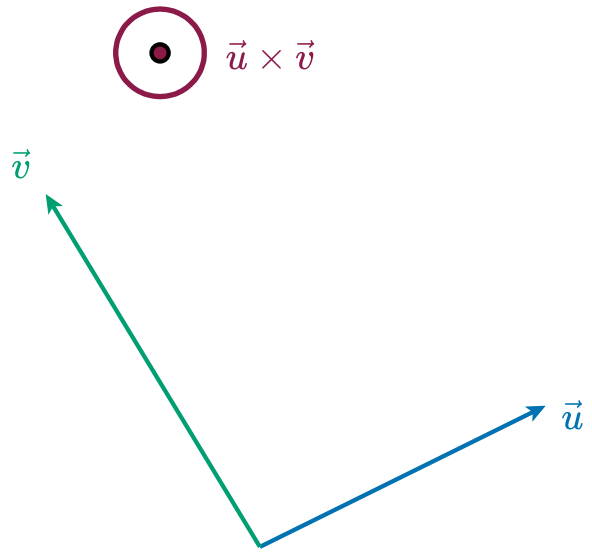


Figure 5.10.12 $\vec{u}\times\vec{v}$ points out of the page, drawn as a circle with a dot at its centre.

12. ∗**Circular Motion from a Perpendicular Force.** *Starred problem.* This one is for the interested reader. It will not be examined.

Through this guided exercise, we prove that *a force of constant magnitude that always points perpendicular to the velocity drives the particle around a circle.* We then apply it to a charged particle in a magnetic field.

Both of the products you have met so far appear, and they do genuinely different jobs. Watch for them:

- the **dot product** is used, through $\vec{F}\cdot\vec{v}$, to determine whether the speed of a particle acted upon by a perpendicular force can change,
- the **cross product** is used to determine the direction and magnitude of the force that keeps a charged particle moving in a uniform magnetic field.

(a) A particle of mass m moves under a force $\vec{F}$ of constant magnitude that is always perpendicular to its velocity $\vec{v}$. Show that the particle's *speed* is constant.

Solution. Since $\vec{F}$ is perpendicular to $\vec{v}$, we have

$$\vec{F}\cdot\vec{v}=0.$$

Now consider the square of the speed, $v^2=\vec{v}\cdot\vec{v}$. Using the product rule for the dot product[1] and then Newton's second law $m\dfrac{d\vec{v}}{dt}=\vec{F}$,

$$\begin{aligned}\frac{d}{dt}\left(v^2\right)&=\frac{d}{dt}\left(\vec{v}\cdot\vec{v}\right)=2\,\vec{v}\cdot\frac{d\vec{v}}{dt}\\&=\frac{2}{m}\,\vec{v}\cdot\vec{F}=0.\end{aligned}$$

So v^2 never changes, and hence neither does the speed v.

Why the dot product is the right tool here. The quantity $\vec{F}\cdot\vec{v}$ is the rate at which the force does work on the particle. A force with a forward component speeds the particle up; a force with a backward component slows it down. Only the part of $\vec{F}$ lying along $\vec{v}$ can change the speed, and the dot product is exactly the mathematical tool that isolates that part. When it vanishes, can only change the direction of the velocity but not its magnitude, and so can only *turn* the particle, while the speed remains constant. See Figure 5.10.13.

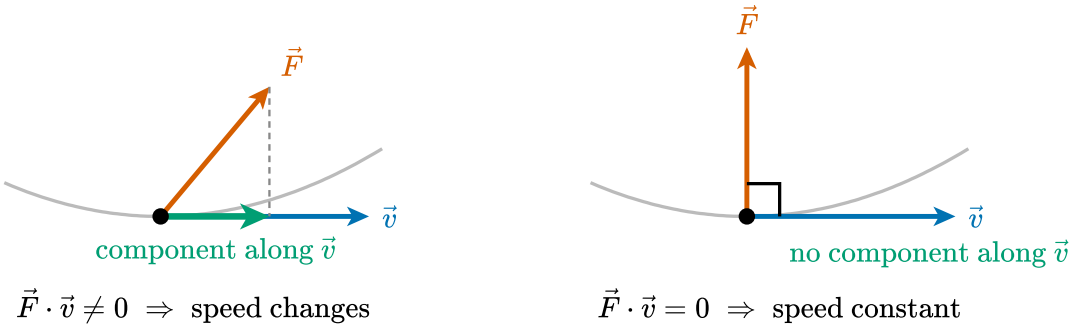


Figure 5.10.13 Only the component of $\vec{F}$ along $\vec{v}$ can change the speed. On the left that component is non-zero, so the particle speeds up; on the right $\vec{F}\perp\vec{v}$, the component is zero, and the speed cannot change.

(b) During a short time interval Δt, show that the *direction* of the velocity turns through the angle

$$\Delta\theta=\frac{F}{mv}\,\Delta t.$$

(Hint: draw the two velocity vectors, which have equal length, from a common tail.)

Solution. By the previous task the velocity never changes length; it only swings around. So $\vec{v}(t)$ and $\vec{v}(t+\Delta t)$ are two vectors of the *same* length v, separated by the small angle $\Delta\theta$ we are after. Drawn from a common tail they form an isosceles triangle whose third side is $\Delta\vec{v} = \vec{v}(t+\Delta t) - \vec{v}(t)$, as in Figure 5.10.14.

The tips of the two vectors both lie on a circle of radius v. The straight side $\Delta\vec{v}$ is the chord cutting off the arc of angle $\Delta\theta$ on that circle, and for a small angle a chord is very nearly the arc it subtends:

$$\|\Delta\vec{v}\| = 2v\sin\frac{\Delta\theta}{2} \approx v\,\Delta\theta.$$

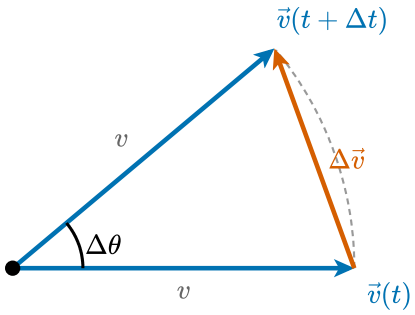
On the other hand Newton's second law over the short interval gives

$$\|\Delta\vec{v}\| \approx \frac{F}{m}\,\Delta t.$$

Equating the two expressions for $\|\Delta\vec{v}\|$,

$$v\,\Delta\theta = \frac{F}{m}\,\Delta t \qquad \Longrightarrow \qquad \Delta\theta = \frac{F}{mv}\,\Delta t,$$

which is what we wanted. Notice the rate of turning $\Delta\theta/\Delta t = F/(mv)$ is a *constant*, since F, m and v all are: the particle swings around at a steady angular rate.



equal lengths $\Rightarrow$ $\|\Delta\vec{v}\| \approx v\,\Delta\theta$

Newton: $\|\Delta\vec{v}\| = \frac{F}{m}\Delta t \;\Rightarrow\; \Delta\theta = \frac{F}{mv}\Delta t$

Figure 5.10.14 The two velocities have equal length v (dashed arc), so the triangle is isosceles and the chord $\Delta\vec{v}$ has length $\approx v\,\Delta\theta$.

(c) Combine $\Delta s = v\,\Delta t$ with the arc-length formula $\Delta s = R\,\Delta\theta$ to prove that

$$R = \frac{mv^2}{F}.$$

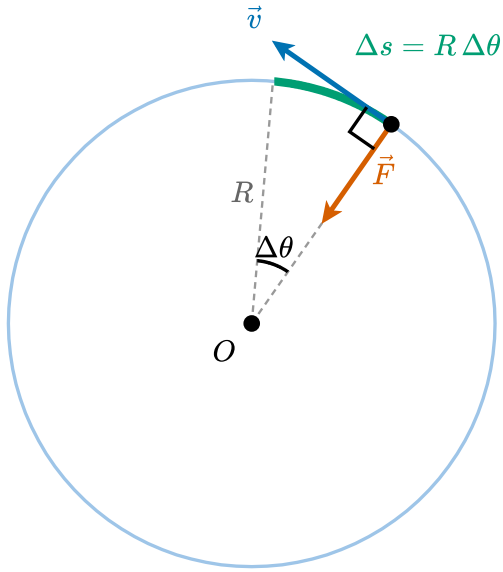
Solution. In time Δt the particle covers the arc length $\Delta s = v\,\Delta t$. If that arc bends through the angle $\Delta\theta$, then the radius R of the arc is defined by $\Delta s = R\,\Delta\theta$. Solving for R and substituting both expressions,

$$R = \frac{\Delta s}{\Delta\theta} = \frac{v\,\Delta t}{\dfrac{F}{mv}\,\Delta t} = \frac{mv^2}{F}.$$

(d) Conclude that the trajectory is a circle.

Solution. The quantity $R = mv^2/F$ is built from m, v and F, every one of which is constant — v because of the first task, and F by hypothesis. So the path bends through equal angles over equal arc lengths, everywhere along the trajectory: its radius of curvature is the same constant R at every point.

A plane curve with constant radius of curvature is a circle. Concretely, the centre of the turn always lies a fixed distance R from the particle in the direction of $\vec{F}$.



$$\Delta s = v\,\Delta t \quad \text{and} \quad \Delta\theta = \tfrac{F}{mv}\Delta t$$

$$\Rightarrow \quad R = \frac{\Delta s}{\Delta\theta} = \frac{mv^2}{F} \quad \text{— a constant}$$

Figure 5.10.15 The velocity is tangent to the circle and the force points straight at the centre. Equal arcs Δs subtend equal angles $\Delta\theta$, and the ratio of the two is the constant radius $R = mv^2/F$.

(e) A particle of charge q enters a uniform magnetic field $\vec{B}$ with its velocity *perpendicular* to the field. The magnetic force is

$$\vec{F}_B = q\,\vec{v} \times \vec{B}.$$

Show that this force has magnitude $|q|\,vB$.

Solution. The length of a cross product is $\left\|\vec{v} \times \vec{B}\right\| = vB\sin\alpha$, where α is the angle between $\vec{v}$ and $\vec{B}$. Here the velocity is perpendicular to the field, so $\alpha = 90°$ and $\sin\alpha = 1$. Therefore

$$\left\|\vec{F}_B\right\| = |q|\left\|\vec{v} \times \vec{B}\right\| = |q|\,vB\sin 90° = |q|\,vB.$$

Since v is constant (we are about to confirm this) and B is uniform, this magnitude is constant — the first hypothesis of part a.

(f) Explain why this force is *always* perpendicular to the velocity — not just at the initial instant, but at every moment of the motion.

Solution. This is the defining property of the cross product: $\vec{v} \times \vec{B}$ is perpendicular to *both* factors, whatever they happen to be. So $\vec{F}_B = q\,\vec{v} \times \vec{B}$ is perpendicular to $\vec{v}$ automatically, at every instant, with no computation and no special assumption about where the particle is or which way it is heading.

If you would like it verified rather than quoted, the dot product does it in one line.[2] Taking $\vec{c} = \vec{v}$,

$$\vec{F}_B \cdot \vec{v} = q\left(\vec{v} \times \vec{B}\right) \cdot \vec{v} = q\begin{vmatrix} v_1 & v_2 & v_3 \\ B_1 & B_2 & B_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = 0.$$

So such a magnetic field can *never* change the speed of a charged particle, no matter how strong it is. Here the magnetic force does no work and it solely changes the direction of the velocity.

(g) Deduce that the particle moves in a circle, and find the radius of that circle.

Solution. The previous two tasks establish exactly the two hypotheses of part a: the magnetic force has constant magnitude $|q|vB$, and it is always perpendicular to the velocity. The argument of parts a–d therefore applies verbatim, and the particle travels in a circle at constant speed.

For the radius, put $F = |q|vB$ into $R = mv^2/F$:

$$r = \frac{mv^2}{|q|vB} = \frac{mv}{|q|B}.$$

Equivalently, the magnetic force supplies precisely the inward force mv^2/r that circular motion demands:

$$|q|vB = \frac{mv^2}{r} \qquad \Longrightarrow \qquad r = \frac{mv}{|q|B}.$$

See Figure 5.10.16. Faster particles and heavier particles travel on wider circles; stronger fields and larger charges wind them tighter.

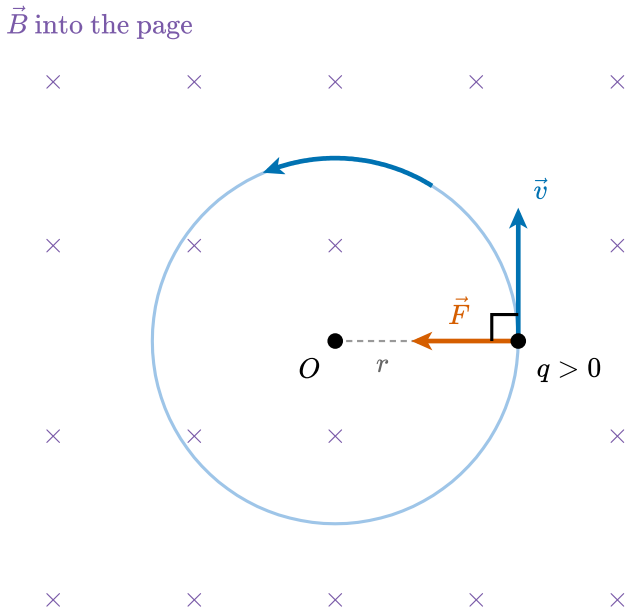


Figure 5.10.16 A positive charge in a uniform field $\vec{B}$ directed into the page. The force $q\,\vec{v} \times \vec{B}$ is perpendicular to $\vec{v}$ automatically, so it never changes the speed — it only bends the path into a circle of radius $r = mv/(|q|B)$.

13. ∗Speed and direction of a projectile. *Starred problem.* This one is for the interested reader. It will not be examined.

A particle is projected from O with speed v_0 at an elevation θ to the horizontal, as in Figure 5.10.17. At time t it is at the point P, moving with speed v in a direction making an angle ϕ with the horizontal. Show

[2]The product rule holds for the dot product just as it does for ordinary multiplication: $\frac{d}{dt}(\vec{u}\cdot\vec{v}) = \frac{d\vec{u}}{dt}\cdot\vec{v} + \vec{u}\cdot\frac{d\vec{v}}{dt}$. Taking $\vec{u} = \vec{v}$ makes the two terms equal, which gives the step used here, $\frac{d}{dt}(\vec{v}\cdot\vec{v}) = 2\,\vec{v}\cdot\frac{d\vec{v}}{dt}$.

[3]Two facts are doing the work. First, the scalar triple product $\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}$ equals the determinant whose rows are $\vec{a}$, $\vec{b}$ and $\vec{c}$. Second, a determinant with two equal rows is zero, since interchanging those two rows both leaves the determinant unchanged and reverses its sign.

that

$$v = \sqrt{v_0^2 - 2\,v_0\,g\,t\sin\theta + g^2t^2}$$

and

$$\tan\phi = \frac{v_0\sin\theta - g\,t}{v_0\cos\theta}.$$

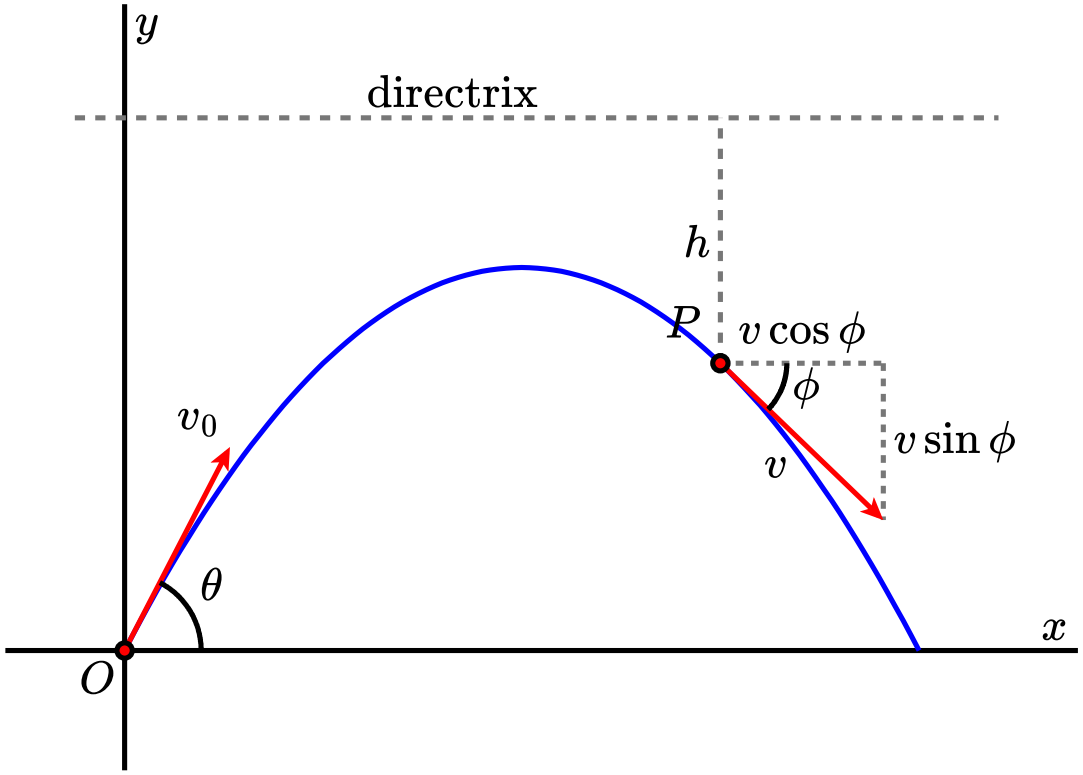


Figure 5.10.17 A particle projected from O with speed v_0 at elevation θ. At the point P reached after time t, the velocity v is tangent to the path and makes an angle ϕ with the horizontal; h is the depth of P below the directrix.

Solution. Take the origin at O with horizontal and vertical axes. The coordinates of P at time t are

$$x = v_0\cos\theta\,t, \qquad y = v_0\sin\theta\,t - \tfrac{1}{2}g\,t^2.$$

The horizontal and vertical components of the velocity are the time derivatives of these, so

$$v\cos\phi = \frac{dx}{dt} = v_0\cos\theta, \qquad v\sin\phi = \frac{dy}{dt} = v_0\sin\theta - g\,t.$$

Squaring and adding eliminates ϕ:

$$v^2 = v_0^2\cos^2\theta + \big(v_0\sin\theta - g\,t\big)^2 = v_0^2 - 2\,v_0\,g\,t\sin\theta + g^2t^2,$$

which gives the speed at time t. Dividing the vertical component by the horizontal component gives the direction,

$$\tan\phi = \frac{v_0\sin\theta - g\,t}{v_0\cos\theta},$$

that is, $\phi = \arctan\left(\dfrac{v_0\sin\theta - g\,t}{v_0\cos\theta}\right)$.

14. ∗Free fall from the directrix. *Starred problem.* This one is for the interested reader. It will not be examined.

The **directrix** of the parabolic path is the horizontal line at height $\dfrac{v_0^2}{2g}$ above O. Using the previous exercise, show that the speed of the particle at P is equal to the speed it would acquire by falling freely from the level

of the directrix down to P. That is, if h denotes the depth of P below the directrix (see Figure 5.10.17), then $v^2 = 2\,g\,h$.

Solution. From the previous exercise,

$$v^2 = v_0^2 - 2\,v_0\,g\,t\sin\theta + g^2t^2 = v_0^2 - 2g\big(v_0\,t\sin\theta - \tfrac{1}{2}g\,t^2\big) = v_0^2 - 2gy,$$

since $y = v_0\,t\sin\theta - \frac{1}{2}g\,t^2$ is the height of P. Writing this as

$$v^2 = 2g\left(\frac{v_0^2}{2g} - y\right)$$

and recognizing $\dfrac{v_0^2}{2g}$ as the height of the directrix above O, the quantity $h = \dfrac{v_0^2}{2g} - y$ is exactly the depth of P below the directrix. Hence $v^2 = 2\,g\,h$, the square of the speed gained by falling freely through the height h.

15. ∗A collision on the directrix. *Starred problem.* This one is for the interested reader. It will not be examined.

Two objects move under gravity. The first is the projectile of the two previous exercises: launched from O with speed v_0 at an elevation θ to the horizontal. The second is a ball released *from rest* at a point B on the directrix, the horizontal line at height $\dfrac{v_0^2}{2g}$ above O. We want the two to collide at the moment the projectile is at half of its maximum height, on the way *down*. See Figure 5.10.18.

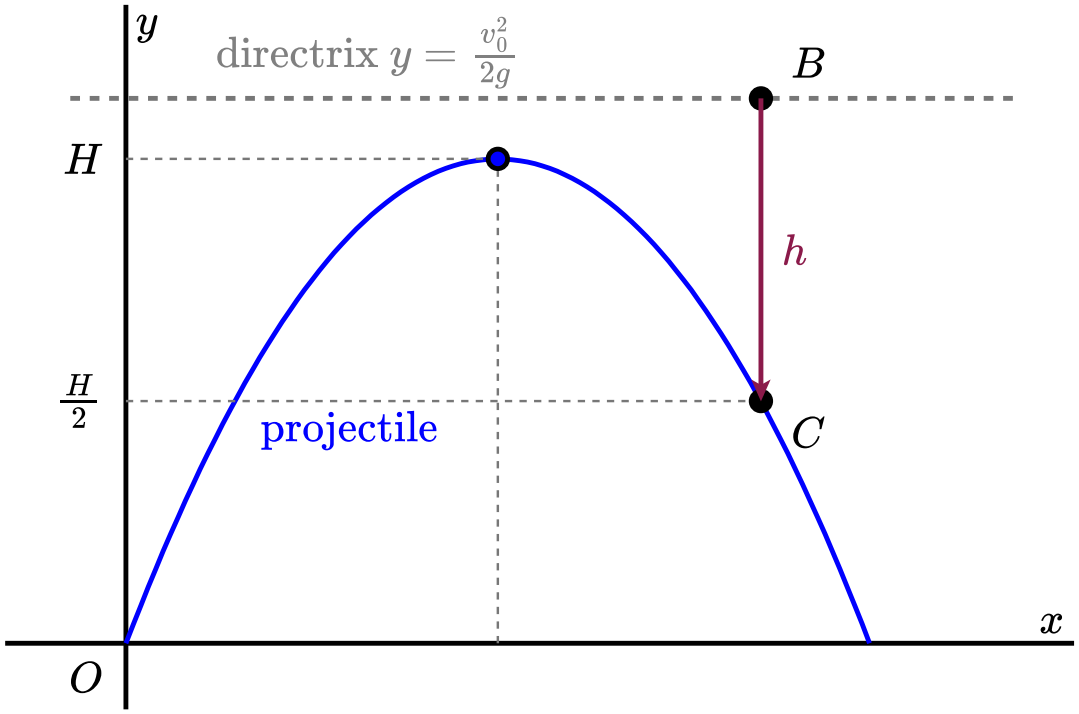


Figure 5.10.18 The projectile reaches half of its maximum height H at the point C on the way down. The ball is released from rest at the point B vertically above C on the directrix, and falls the depth h to meet it.

(a) Show that the projectile is at half of its maximum height at the two times

$$t_\pm = \frac{v_0\sin\theta}{g}\left(1 \pm \frac{\sqrt{2}}{2}\right)$$

and say which of these is on the descending branch.

Solution. The vertical motion is $y = v_0\sin\theta\,t - \frac{1}{2}gt^2$. It is greatest when $\dot{y} = v_0\sin\theta - gt = 0$, that is at $t = v_0\sin\theta/g$, which gives the maximum height

$$H = \frac{v_0^2\sin^2\theta}{2g}.$$

Setting $y = H/2$ gives the quadratic equation

$$t^2 - \frac{2v_0 \sin\theta}{g}\, t + \frac{v_0^2 \sin^2\theta}{2g^2} = 0,$$

whose discriminant is $2v_0^2 \sin^2\theta / g^2$. Hence

$$t_\pm = \frac{v_0 \sin\theta}{g} \pm \frac{\sqrt{2}\, v_0 \sin\theta}{2g} = \frac{v_0 \sin\theta}{g}\left(1 \pm \frac{\sqrt{2}}{2}\right).$$

These straddle the apex time $v_0 \sin\theta / g$ symmetrically, as they must. The descending crossing is the later one,

$$T = t_+ = \frac{\left(2+\sqrt{2}\right) v_0 \sin\theta}{2g}.$$

(b) Find the point B on the directrix from which the ball must be released.

Solution. Released from rest, the ball has no horizontal velocity, so it falls along a vertical line: B must be *directly above* the collision point C. The projectile's horizontal coordinate at time T is

$$x_C = v_0 \cos\theta \cdot T = \frac{\left(2+\sqrt{2}\right) v_0^2 \sin 2\theta}{4g},$$

so the release point is

$$B = \left(\frac{\left(2+\sqrt{2}\right) v_0^2 \sin 2\theta}{4g},\ \frac{v_0^2}{2g}\right).$$

In terms of the range $R = \dfrac{v_0^2 \sin 2\theta}{g}$ this is $x_C = \frac{2+\sqrt{2}}{4} R \approx 0.854\, R$—comfortably past the apex at $R/2$ and short of the landing point, as Figure 5.10.18 shows.

(c) Find how long after the projectile is launched the ball must be released.

Solution. The depth of C below the directrix is

$$h = \frac{v_0^2}{2g} - \frac{H}{2} = \frac{v_0^2\left(2 - \sin^2\theta\right)}{4g}.$$

Falling from rest through h takes a time τ with $h = \frac{1}{2} g\tau^2$, so

$$\tau = \sqrt{\frac{2h}{g}} = \frac{v_0}{g}\sqrt{\frac{2 - \sin^2\theta}{2}}.$$

The ball must therefore be released

$$t_{\text{rel}} = T - \tau = \frac{v_0}{g}\left[\frac{\left(2+\sqrt{2}\right)\sin\theta}{2} - \sqrt{\frac{2 - \sin^2\theta}{2}}\right]$$

seconds after the projectile is launched.

(d) Show that at the instant of the collision the two objects have exactly the same speed.

Solution. By Worksheet Exercise 5.10.14, a projectile at depth h below its directrix has speed $v_{\text{proj}} = \sqrt{2gh}$. The ball starts from rest *on* the directrix and has fallen the same depth h, so its speed is

$$v_{\text{ball}} = g\tau = g\sqrt{\frac{2h}{g}} = \sqrt{2gh}.$$

The two are equal. This is not a coincidence of our particular choice of C: the projectile always carries exactly the speed it *would* have gained by falling freely from the directrix, and the ball is doing precisely that fall. Wherever on the path we arrange the collision, the two objects meet with the same speed, here

$$v = \sqrt{2gh} = v_0\sqrt{\frac{2-\sin^2\theta}{2}}.$$

Their *directions* of course differ: the ball moves straight down, while the projectile moves down and forward at the angle ϕ of Worksheet Exercise 5.10.13.

(e) Is t_{rel} always positive? Find the elevations θ for which the ball is released after the projectile is launched, and explain what a negative t_{rel} would mean.

Solution. Requiring $t_{\text{rel}} \geq 0$ means

$$\frac{\left(2+\sqrt{2}\right)\sin\theta}{2} \geq \sqrt{\frac{2-\sin^2\theta}{2}},$$

and squaring gives $\sin^2\theta\left(4+2\sqrt{2}\right) \geq 2$, that is

$$\sin^2\theta \geq \frac{1}{2+\sqrt{2}} = \frac{2-\sqrt{2}}{2} \quad\Longleftrightarrow\quad \theta \geq \arcsin\sqrt{\frac{2-\sqrt{2}}{2}} \approx 32.8^\circ.$$

For flatter launches the required release time is negative: the ball would have to be dropped *before* the projectile is fired, because a shallow trajectory reaches its (low) half-height point sooner than the ball can fall the long way down from the directrix.

An animation of the whole scenario — the projectile rising to H, the ball released from B at the time t_{rel}, and the two meeting at C with velocity arrows of equal length — accompanies the online version of these notes.

16. Consider the curve of intersection of the surfaces $4x^2 + z^2 = 16$ and $y = z$.

(a) Parametrize the curve.

(b) Find the projection of the curve onto the xy-plane, and give a Cartesian equation for it.

Solution. **(a)** The first surface does not involve y; dividing by 16,

$$\frac{x^2}{4} + \frac{z^2}{16} = 1,$$

which is an ellipse in the xz-plane with semi-axes 2 (in x) and 4 (in z). It is parametrized by $x = 2\cos(t)$, $z = 4\sin(t)$. The second surface simply forces $y = z$, so

$$x = 2\cos(t), \qquad y = 4\sin(t), \qquad z = 4\sin(t), \qquad 0 \leq t \leq 2\pi.$$

(b) Projecting onto the xy-plane just means forgetting the z-coordinate, so the projected curve is parametrized by the first two components alone,

$$x = 2\cos(t), \qquad y = 4\sin(t), \qquad 0 \leq t \leq 2\pi.$$

To turn that into a Cartesian equation, eliminate t with $\cos^2 t + \sin^2 t = 1$:

$$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{4}\right)^2 = \cos^2 t + \sin^2 t = 1, \qquad \text{that is} \qquad \frac{x^2}{4} + \frac{y^2}{16} = 1.$$

The shadow is an ellipse in the xy-plane with semi-axes 2 in the x-direction and 4 in the y-direction. This is the same ellipse we started from in the xz-plane, which is no accident: the plane $y = z$ carries the z-coordinate over to the y-coordinate unchanged.

17. Find the intersection of the line $\mathbf{r}(t) = \langle 1, 0, 1\rangle + t\langle -1, 1, 1\rangle$ and the plane $2x - y + z = 1$.

Solution. The parametric equations of the line are

$$x(t) = 1 - t, \qquad y(t) = t, \qquad z(t) = 1 + t.$$

Substituting into the equation of the plane,

$$2x - y + z = 2(1 - t) - t + (1 + t) = 3 - 2t = 1 \;\Rightarrow\; t = 1,$$

and $x(1) = 0$, $y(1) = 1$, $z(1) = 2$. The intersection is the point $(0, 1, 2)$.

18. Find the distance between the two planes

$$x + 2y + z = 1 \qquad \text{and} \qquad 2x + 4y + 2z = 7.$$

Solution. Dividing the second equation by 2 gives $x+2y+z = \frac{7}{2}$, so both planes have the same normal vector $\vec{n} = \langle 1, 2, 1\rangle$ and are therefore *parallel*. (They are distinct, since $1 \neq \frac{7}{2}$.)

Pick any point on the first plane, say $P(1, 0, 0)$, and any point on the second, say $S\left(\frac{7}{2}, 0, 0\right)$. Then $\vec{PS} = \left\langle \frac{5}{2}, 0, 0\right\rangle$, and the distance between the planes is the length of the projection of $\vec{PS}$ onto $\vec{n}$:

$$d = \left|\vec{PS} \cdot \frac{\vec{n}}{\|\vec{n}\|}\right| = \frac{\left|\frac{5}{2} \cdot 1 + 0 + 0\right|}{\sqrt{1 + 4 + 1}} = \frac{5}{2\sqrt{6}}.$$

19. Find the angle between the planes $2x + 2y = 7$ and $y + z = -1$.

Solution. The angle between two planes equals the angle between their normal vectors, $\vec{n}_1 = \langle 2, 2, 0\rangle$ and $\vec{n}_2 = \langle 0, 1, 1\rangle$:

$$\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{\|\vec{n}_1\|\,\|\vec{n}_2\|} = \frac{0 + 2 + 0}{\sqrt{8} \cdot \sqrt{2}} = \frac{2}{4} = \frac{1}{2},$$

so $\theta = \dfrac{\pi}{3}$.

20. There is a line L parametrized by $x = 1 + 2t$, $y = 3 + 4t$, $z = 5 + 6t$. The line passes through some point P whose z-coordinate is -7. Find an equation for the plane through P normal to L.

Solution. First locate P. Setting $z = -7$,

$$5 + 6t = -7 \;\Rightarrow\; t = -2,$$

so $x = 1 + 2(-2) = -3$ and $y = 3 + 4(-2) = -5$, i.e. $P(-3, -5, -7)$.

A plane normal to L has the direction vector of L as its normal vector, $\vec{n} = \langle 2, 4, 6\rangle$. Therefore the plane is

$$2(x + 3) + 4(y + 5) + 6(z + 7) = 0.$$

21. A surveyor determines that the three points

$$A = (-1, 2, 16), \qquad B = (2, -2, 17), \qquad C = (2, 0, 16)$$

all lie on a flat surface W.

(a) Find the point of W that is closest to the point $(4, 4, -3)$.

(b) Find parametric equations for a line that passes through $(4, 4, -3)$, never crosses W, and satisfies $x(t) = y(t)$.

Solution. **Finding the plane.** Two vectors lying in W are

$$\vec{AB} = \langle 3,\ -4,\ 1\rangle, \qquad \vec{AC} = \langle 3,\ -2,\ 0\rangle.$$

A normal vector is their cross product:

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -4 & 1 \\ 3 & -2 & 0 \end{vmatrix}$$

$$= \langle (-4)(0) - (1)(-2),\ (1)(3) - (3)(0),\ (3)(-2) - (-4)(3) \rangle$$
$$= \langle 2,\ 3,\ 6 \rangle .$$

Note $\|\vec{n}\| = \sqrt{4+9+36} = 7$, which will be convenient. Using the point A, the plane is

$$2(x+1) + 3(y-2) + 6(z-16) = 0$$
$$2x + 3y + 6z = 100.$$

(Check: B gives $4 - 6 + 102 = 100$ and C gives $4 + 0 + 96 = 100$.)

(a) The closest point of a plane to a given point $Q = (4, 4, -3)$ is the foot of the perpendicular from Q, which we reach by moving from Q in the direction of the normal. So consider

$$\vec{r}(t) = Q + t\,\vec{n} = (4 + 2t,\ 4 + 3t,\ -3 + 6t)$$

and find the value of t for which this point lies on W:

$$2(4 + 2t) + 3(4 + 3t) + 6(-3 + 6t) = 100$$
$$8 + 4t + 12 + 9t - 18 + 36t = 100$$
$$2 + 49t = 100$$
$$t = 2.$$

Therefore the closest point is

$$\vec{r}(2) = (4 + 4,\ 4 + 6,\ -3 + 12) = (8,\ 10,\ 9).$$

(Check: $16 + 30 + 54 = 100$.) Incidentally, the distance from Q to W is $|t|\,\|\vec{n}\| = 2 \cdot 7 = 14$. See Figure 5.10.19.

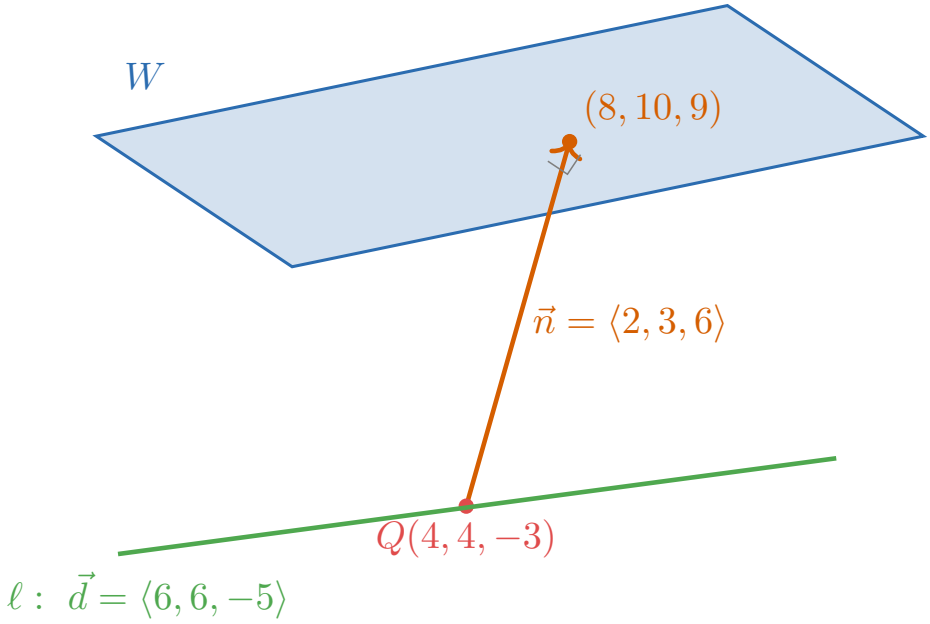


Figure 5.10.19 Schematic picture of Worksheet Exercise 5.10.21. From $Q(4, 4, -3)$ we travel along the normal direction $\vec{n} = \langle 2, 3, 6 \rangle$ until we hit W, arriving at $(8, 10, 9)$. Any line through Q whose direction is perpendicular to $\vec{n}$ stays parallel to W and never meets it.

(b) A line never crosses the plane W exactly when it is parallel to W and does not lie in it. Since $Q = (4, 4, -3)$ is not on W (we saw $2(4) + 3(4) + 6(-3) = 2 \neq 100$), we only need the direction vector $\vec{d} = \langle a, b, c \rangle$ to be perpendicular to the normal:

$$\vec{d} \cdot \vec{n} = 2a + 3b + 6c = 0.$$

The extra requirement $x(t) = y(t)$ forces the two lines $x = 4 + at$ and $y = 4 + bt$ to agree for all t, which happens exactly when $a = b$. Substituting $b = a$,

$$2a + 3a + 6c = 0 \quad \Longrightarrow \quad 5a = -6c.$$

Choosing $a = 6$ and $c = -5$ gives $\vec{d} = \langle 6,\ 6,\ -5\rangle$, and the line is

$$x = 4 + 6t, \qquad y = 4 + 6t, \qquad z = -3 - 5t.$$

Indeed $x(t) = y(t)$ for every t, and $\vec{d}\cdot\vec{n} = 12 + 18 - 30 = 0$, so the line is parallel to W and, starting from a point off W, never reaches it.

22. ∗A Parabolic Antenna. *Starred problem.* This one is for the interested reader. It will not be examined.

A parabolic antenna has the shape of a paraboloid of revolution: it is created by rotating part of a parabola around its axis. By the reflection property, all signals arriving parallel to the axis are concentrated at the focus after reflecting off the dish, so that is where the receiver is mounted. The axial cross-section of the dish is described by two measurements: the *diameter* d of the dish and its *depth* h. Placing the vertex at the origin with the dish opening upward, the cross-section is the parabola $x^2 = 4py$, and the rim passes through the points $\left(\pm\frac{d}{2},\, h\right)$.

Consider a dish antenna with diameter $d = 120$ cm and depth $h = 20.3$ cm, suitable for the amateur radio band at 5.76 GHz.

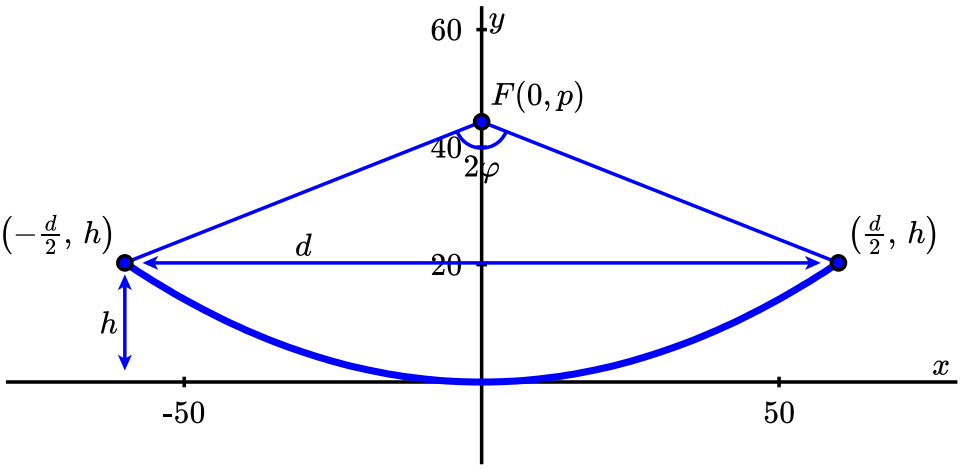


Figure 5.10.20 Cross-section of the parabolic antenna: diameter d, depth h, focus $F(0,p)$, and opening angle 2φ.

(a) Determine the optimal location for the receiver, that is, find the distance from the vertex of the dish to the focus.

Solution. The receiver must sit at the focus $(0,p)$. The rim point $\left(\frac{d}{2}, h\right) = (60,\ 20.3)$ lies on the parabola $x^2 = 4py$, so

$$60^2 = 4p(20.3) \quad\Longrightarrow\quad p = \frac{3600}{81.2} \approx 44.3.$$

The receiver should be mounted on the axis about 44.3 cm above the vertex of the dish. Note that $p > h$ here, so the focus sits *above* the rim of this shallow dish.

(b) Find the quadratic function $y = f(x)$ (explicit form) that describes the curvature of the dish, and graph it (for example, in GeoGebra or Desmos).

Solution. Solving $x^2 = 4py$ for y with $4p = \frac{3600}{20.3} \approx 177.3$ gives

$$y = \frac{1}{4p}\,x^2 = \frac{20.3}{3600}\,x^2 \approx \frac{x^2}{177.3}, \qquad -60 \le x \le 60.$$

The graph captures the true curvature of the dish provided both axes use the same scale.

(c) The *opening angle* 2φ of the dish is the angle at which the two edges of the rim are seen from the focus. Compute it.

Solution. The focus $F(0,p)$, the point $(0,h)$, and the rim point $\left(\frac{d}{2},h\right)$ form a right triangle with legs $p-h$ (vertical) and $\frac{d}{2}$ (horizontal), and φ is the angle at the focus. Hence

$$\tan\varphi = \frac{d/2}{p-h} = \frac{60}{44.3-20.3} = \frac{60}{24} \quad\Longrightarrow\quad \varphi \approx 68.2^\circ.$$

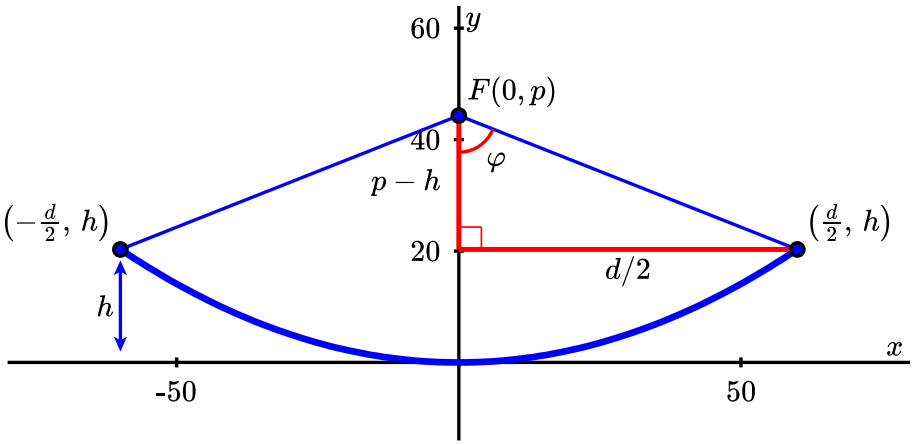


Figure 5.10.21 The right triangle used to compute the half opening angle φ: legs $p-h$ and $d/2$. The opening angle of the dish is $2\varphi \approx 136.4^\circ$.

23. Identify and sketch the surface $4x^2 - y^2 + 2z^2 + 4 = 0$.

Solution. Dividing by -4, we can rewrite the equation as

$$x^2 - \frac{y^2}{4} + \frac{z^2}{2} = -1, \qquad \text{i.e.} \qquad \frac{y^2}{4} = 1 + x^2 + \frac{z^2}{2},$$

which forces $y^2 \geq 4$, i.e. $y \geq 2$ or $y \leq -2$. Let us look at the traces:

- Intersection with the plane $x = c_1$: $\dfrac{y^2}{4} - \dfrac{z^2}{2} = 1 + c_1^2$, a hyperbola.
- Intersection with the plane $y = c_2$: $x^2 + \dfrac{z^2}{2} = \dfrac{c_2^2}{4} - 1$; when $|c_2| > 2$, we have ellipses as intersections.
- Intersection with the plane $z = c_3$: $\dfrac{y^2}{4} - x^2 = 1 + \dfrac{c_3^2}{2}$, a hyperbola.

The surface is a **hyperboloid of two sheets** along the y-axis, with vertices $(0,-2,0)$ and $(0,2,0)$. See Figure 5.10.22.

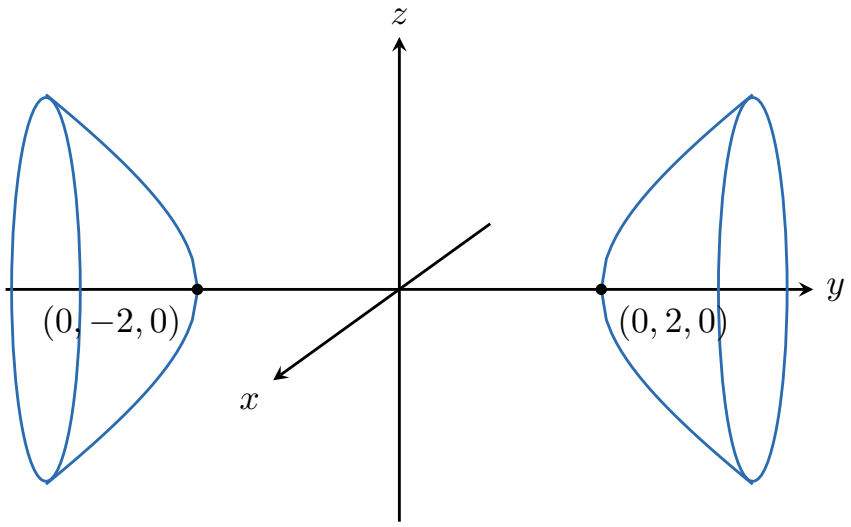


Figure 5.10.22 The hyperboloid of two sheets $\frac{y^2}{4} - x^2 - \frac{z^2}{2} = 1$: two bowl-shaped sheets opening along the y-axis, with vertices at $(0, \pm 2, 0)$.

24. Which of These Are Hyperboloids? None of the equations below is in standard position. Complete the square where necessary, then select *every* equation whose surface is a hyperboloid, of one sheet or of two.

A. $x^2 + y^2 - z^2 - 2x = 0$

B. $x^2 + y^2 + 4z^2 - 4y = 0$

C. $x^2 - y^2 - z^2 = 1$

D. $x^2 + y^2 - z^2 = 0$

E. $z + 1 = x^2 - y^2$

Answer. A, C.

Solution 1. Completing the square in the first and second,

$$x^2 + y^2 - z^2 - 2x = 0 \implies (x-1)^2 + y^2 - z^2 = 1,$$
$$x^2 + y^2 + 4z^2 - 4y = 0 \implies x^2 + (y-2)^2 + 4z^2 = 4.$$

Reading off the signs: one negative square against $+1$ is a hyperboloid of one sheet; two negative squares against $+1$ is a hyperboloid of two sheets. So the first and third are hyperboloids. The second is an ellipsoid, the fourth equals 0 and so is a cone, and the fifth is linear in z and so is a hyperbolic paraboloid.

Solution 2.

A. *Correct.*

Correct. Completing the square gives $(x-1)^2 + y^2 - z^2 = 1$: one negative square, equal to a positive constant, so a hyperboloid of *one* sheet.

B. *Incorrect.*

No. This is $x^2 + (y-2)^2 + 4z^2 = 4$ — every square is positive, so it is an ellipsoid.

C. *Correct.*

Correct. *Two* negative squares against a positive constant gives a hyperboloid of two sheets.

D. *Incorrect.*

No — and this is the trap. It has the same signs as the first option, but it equals 0 rather than a positive constant. That degenerates the hyperboloid into an *elliptic cone*, the surface the two families of hyperboloids approach asymptotically.

E. *Incorrect.*

No. The word "hyperbolic" is a red herring: this is linear in z, so it is a *hyperbolic paraboloid* (a saddle) with vertex at $(0, 0, -1)$, not a hyperboloid.

25. Graphing a Parabolic Cylinder. Sketch the graph of $z = 4 - x^2$ over the rectangle $-2 \le x \le 2$, $0 \le y \le 4$ in the xy-plane. Why is this surface called a **parabolic cylinder**?

Solution. Since the equation $z = 4 - x^2$ does not involve y, every cross-section of the surface at a fixed value of y is the *same* parabola $z = 4 - x^2$ in the xz-plane, opening downward from its vertex at $(0, 4)$. Sliding that parabola along the y-axis from $y = 0$ to $y = 4$ sweeps out the surface.

Over $-2 \le x \le 2$, the parabola satisfies $4 - x^2 \ge 0$, with equality exactly at $x = \pm 2$, so the surface touches the xy-plane along the two edges $x = \pm 2$ of the rectangular region and rises to a maximum height of 4 along the ridge $x = 0$. Because it is generated by sliding a fixed curve along a straight line, rather than curving in the y direction as well, it is called a **parabolic cylinder**, shown in Figure 5.10.23.

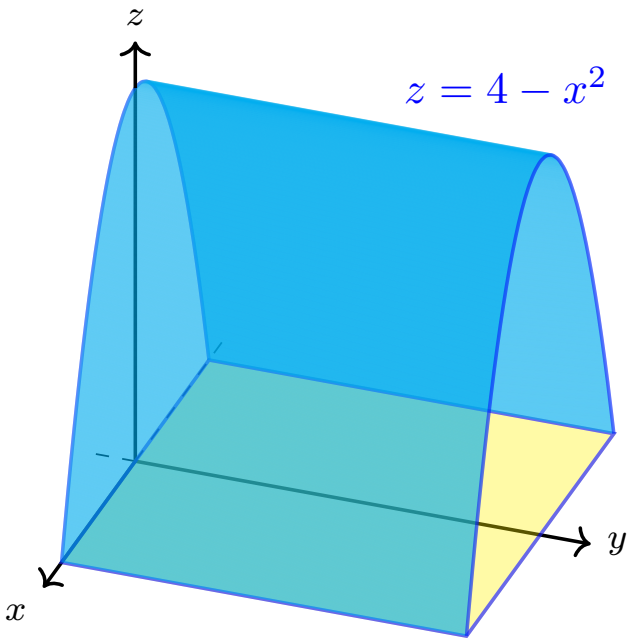


Figure 5.10.23 The parabolic cylinder $z = 4 - x^2$ over the rectangle $-2 \le x \le 2$, $0 \le y \le 4$. Every slice at a fixed y is the same downward-opening parabola.

26. A Solid Bounded by a Cylinder and a Paraboloid. Consider the solid bounded by the three surfaces

$$z = 0, \qquad z = x^2 + y^2, \qquad x^2 + y^2 = 4.$$

(a) Describe each of the three bounding surfaces.

(b) Describe the solid region they enclose.

(c) Describe the cross-section of the solid in the plane $y = 0$.

Solution. **(a)** The equation $z = 0$ is the xy-plane. The equation $z = x^2 + y^2$ is a paraboloid opening upward from its vertex at the origin. The equation $x^2 + y^2 = 4$ is missing z, so it is the infinite cylinder of radius 2 whose axis is the z-axis. The paraboloid meets that cylinder exactly where $x^2 + y^2 = 4$, at height $z = 4$.

(b) The solid consists of the points inside the cylinder lying between the flat base and the paraboloid:

$$x^2 + y^2 \le 4, \qquad 0 \le z \le x^2 + y^2.$$

Near the z-axis this range for z is almost nothing, since $x^2 + y^2 \approx 0$ there, while at the rim $x^2 + y^2 = 4$ it is the full interval $0 \le z \le 4$. The solid is therefore shaped like a shallow bowl carved out from underneath the paraboloid: thin at the center and rising to its full height 4 at the cylinder wall, shown in Figure 5.10.25.

(c) Setting $y = 0$ restricts the solid's defining inequalities to $-2 \le x \le 2$ and $0 \le z \le x^2$: the region under the parabola $z = x^2$, above the x-axis, between $x = -2$ and $x = 2$, shown in Figure 5.10.24.

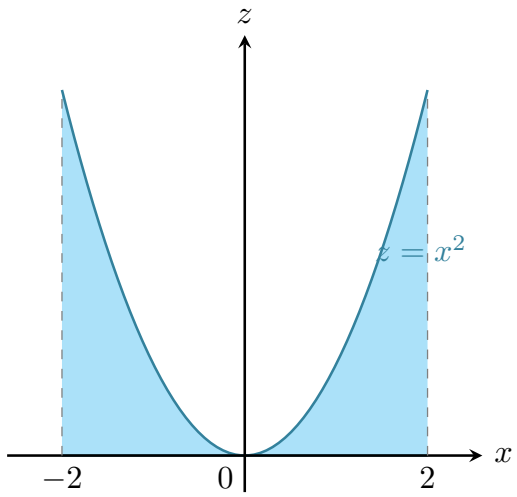


Figure 5.10.24 The cross-section of the solid in the plane $y = 0$: the region under the parabola $z = x^2$ for $-2 \le x \le 2$.

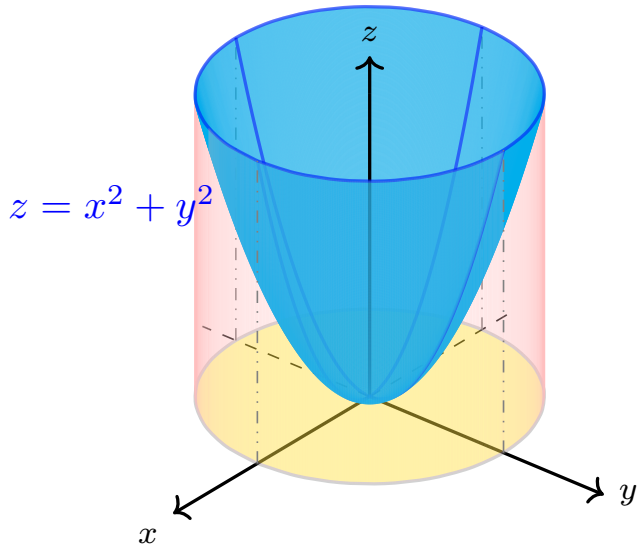


Figure 5.10.25 The solid bounded by $z = 0$, the paraboloid $z = x^2 + y^2$, and the cylinder $x^2 + y^2 = 4$. The solid fills the region under the paraboloid, thin near the z-axis and reaching full height where the paraboloid meets the cylinder wall.

27. Which Identity Fails? Exactly one of the following is *not* valid for all vectors $\vec{a}, \vec{b}, \vec{c}$ in $\mathbb{R}^3$. Which one?

A. $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$

B. $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$

C. $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$

D. $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\,\vec{b} - (\vec{a} \cdot \vec{b})\,\vec{c}$

Answer. B.

Solution.

A. *Incorrect.*

This one is valid. Both sides equal the determinant with rows $\vec{a}, \vec{b}, \vec{c}$, so the dot and the cross may be interchanged.

B. *Correct.*

Correct — the cross product is *not associative*. Take $\vec{a} = \vec{b} = \mathbf{i}$ and $\vec{c} = \mathbf{j}$. Then

$$\mathbf{i} \times (\mathbf{i} \times \mathbf{j}) = \mathbf{i} \times \mathbf{k} = -\mathbf{j}, \qquad (\mathbf{i} \times \mathbf{i}) \times \mathbf{j} = \vec{0} \times \mathbf{j} = \vec{0}.$$

The left side lies in the plane of $\vec{b}$ and $\vec{c}$; the right side lies in the plane of $\vec{a}$ and $\vec{b}$. There is no reason for those to agree.

C. *Incorrect.*

This one is valid — the cross product *is* distributive over addition. (It is only associativity that fails.)

D. *Incorrect.*

This one is valid: it is the "BAC-CAB" triple product expansion. It looks the most exotic of the four, which is exactly why it is worth remembering that it is true.

28. Distance Between Skew Lines. Consider the two lines

$$L_1: \quad \vec{r} = \langle 1, 0, 2\rangle + t\,\langle 2, -1, 1\rangle,$$
$$L_2: \quad \vec{r} = \langle 0, 3, -1\rangle + s\,\langle 1, 2, -1\rangle.$$

They are skew. Let $\vec{u}, \vec{v}$ be the direction vectors and let $\vec{w}$ join a point of L_1 to a point of L_2.
The absolute value of the scalar triple product $|\vec{w} \cdot (\vec{u} \times \vec{v})|$ equals _____.
The distance between the two lines, to three decimal places, is _____.

Hint. The vector $\vec{u} \times \vec{v}$ is perpendicular to both lines, so the distance is the length of the projection of $\vec{w}$ onto it: $d = |\vec{w} \cdot (\vec{u} \times \vec{v})| \,/\, |\vec{u} \times \vec{v}|$.

Solution 1. With $\vec{u} = \langle 2, -1, 1\rangle$ and $\vec{v} = \langle 1, 2, -1\rangle$,

$$\vec{u} \times \vec{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix} = \langle -1,\, 3,\, 5\rangle, \qquad |\vec{u} \times \vec{v}| = \sqrt{35}.$$

Taking $\vec{w} = \langle 0, 3, -1\rangle - \langle 1, 0, 2\rangle = \langle -1, 3, -3\rangle$,

$$\vec{w} \cdot (\vec{u} \times \vec{v}) = 1 + 9 - 15 = -5, \qquad d = \frac{5}{\sqrt{35}} = \frac{\sqrt{35}}{7} \approx 0.845.$$

Solution 2. Consider the two lines

$$L_1: \quad \vec{r} = \langle 1, 0, 2\rangle + t\,\langle 2, -1, 1\rangle,$$
$$L_2: \quad \vec{r} = \langle 0, 3, -1\rangle + s\,\langle 1, 2, -1\rangle.$$

They are skew. Let $\vec{u}, \vec{v}$ be the direction vectors and let $\vec{w}$ join a point of L_1 to a point of L_2.
The absolute value of the scalar triple product $|\vec{w} \cdot (\vec{u} \times \vec{v})|$ equals _5_.
The distance between the two lines, to three decimal places, is _5_.
Correct.
Correct.

29. Why a Magnetic Field Cannot Change a Particle's Speed. Arrange the steps below into a correct proof that the magnetic force does no work, and therefore that a charged particle in a magnetic field moves at constant speed.

- Let a particle of mass m and charge q move with velocity $\vec{v}$ through a magnetic field $\vec{B}$.
- The force on the particle is $\vec{F} = q\,\vec{v} \times \vec{B}$.
- By definition, a cross product is perpendicular to both of its factors, so $\vec{v} \times \vec{B}$ is perpendicular to $\vec{v}$.
- Hence $\vec{F} \cdot \vec{v} = q\left(\vec{v} \times \vec{B}\right) \cdot \vec{v} = 0$.
- The kinetic energy satisfies $\dfrac{d}{dt}\left(\frac{1}{2} m\,\vec{v} \cdot \vec{v}\right) = m\,\vec{v} \cdot \dfrac{d\vec{v}}{dt} = \vec{F} \cdot \vec{v}$.

- Therefore $\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = 0$, so v is constant: the field changes the particle's direction but never its speed.

Solution.

- Let a particle of mass m and charge q move with velocity $\vec{v}$ through a magnetic field $\vec{B}$.
- The force on the particle is $\vec{F} = q\,\vec{v} \times \vec{B}$.
- By definition, a cross product is perpendicular to both of its factors, so $\vec{v} \times \vec{B}$ is perpendicular to $\vec{v}$.
- Hence $\vec{F} \cdot \vec{v} = q\left(\vec{v} \times \vec{B}\right) \cdot \vec{v} = 0$.
- The kinetic energy satisfies $\frac{d}{dt}\left(\frac{1}{2}m\,\vec{v} \cdot \vec{v}\right) = m\,\vec{v} \cdot \frac{d\vec{v}}{dt} = \vec{F} \cdot \vec{v}$.
- Therefore $\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = 0$, so v is constant: the field changes the particle's direction but never its speed.

5.11 Sample Past Exam 2

A sample past exam, with worked solutions. It was a 65-minute, closed-book exam: no aids of their own (formula sheets, notes, phones, tablets, smart watches or similar) were permitted, and students were asked to write their work in a neat and organized format and to fully justify every step. Try the exam under the same conditions before you look at the solutions. The topics it covers, with links to the notes:

- Vectors
- The Dot Product
- The Cross Product
- Parametrization of Plane Curves
- Lines and Planes in Space
- Conic Sections

1.

(a) Compute the vector $\vec{v}$ whose tail and tip are at $(2, 0, -1)$ and $\left(3, 2, -1+\sqrt{11}\right)$, respectively.

Solution.
$$\vec{v} = \left\langle 3-2,\ 2-0,\ -1+\sqrt{11}-(-1) \right\rangle = \left\langle 1,\ 2,\ \sqrt{11} \right\rangle,$$
and
$$\|\vec{v}\| = \sqrt{1+4+11} = \sqrt{16} = 4.$$

(b) Find a vector $\vec{w}$ that points in the opposite direction to $\vec{v}$ (see part (a)) and whose length is triple the length of $\vec{v}$.

Solution. Opposite direction and three times as long:
$$\vec{w} = -3\vec{v} = \left\langle -3,\ -6,\ -3\sqrt{11} \right\rangle$$
(check: $\|\vec{w}\| = 3 \cdot 4 = 12$).

(c) Find the unit vector in the direction of $\vec{v} + \vec{w}$ (see part (b)).

Solution.
$$\vec{v} + \vec{w} = \vec{v} - 3\vec{v} = -2\vec{v} = \left\langle -2,\ -4,\ -2\sqrt{11} \right\rangle.$$
Therefore
$$\begin{aligned} \frac{\vec{v}+\vec{w}}{\|\vec{v}+\vec{w}\|} &= \frac{-2\vec{v}}{2\|\vec{v}\|} = -\frac{\vec{v}}{\|\vec{v}\|} = -\frac{1}{4}\left\langle 1,\ 2,\ \sqrt{11} \right\rangle \\ &= \left\langle -\frac{1}{4},\ -\frac{1}{2},\ -\frac{\sqrt{11}}{4} \right\rangle. \end{aligned}$$
Note that $\vec{v} + \vec{w}$ points opposite to $\vec{v}$.

2. Consider the two vectors $\vec{u} = \langle 2, -1, 3 \rangle$ and $\vec{v} = \langle 1, 2, 2 \rangle$. Write the vector $\vec{u}$ as the sum of two vectors, one of which is parallel to $\vec{v}$ and the other one is perpendicular to it.

Solution.
$$\vec{u} \cdot \vec{v} = 2 - 2 + 6 = 6 \qquad \text{and} \qquad \|\vec{v}\|^2 = 1 + 4 + 4 = 9.$$
Then
$$\begin{aligned} \vec{u}_{\parallel} &= \operatorname{proj}_{\vec{v}} \vec{u} = \left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \right) \vec{v} = \frac{6}{9}\langle 1, 2, 2 \rangle = \left\langle \frac{2}{3},\ \frac{4}{3},\ \frac{4}{3} \right\rangle, \\ \vec{u}_{\perp} &= \vec{u} - \vec{u}_{\parallel} = \langle 2, -1, 3 \rangle - \left\langle \frac{2}{3},\ \frac{4}{3},\ \frac{4}{3} \right\rangle = \left\langle \frac{4}{3},\ -\frac{7}{3},\ \frac{5}{3} \right\rangle. \end{aligned}$$

Check: $\vec{u}_\perp \cdot \vec{v} = \frac{4}{3} - \frac{14}{3} + \frac{10}{3} = 0$. Hence

$$\underbrace{\langle 2,-1,3\rangle}_{\vec{u}} = \underbrace{\left\langle \frac{2}{3}, \frac{4}{3}, \frac{4}{3}\right\rangle}_{\vec{u}_\parallel} + \underbrace{\left\langle \frac{4}{3}, -\frac{7}{3}, \frac{5}{3}\right\rangle}_{\vec{u}_\perp}.$$

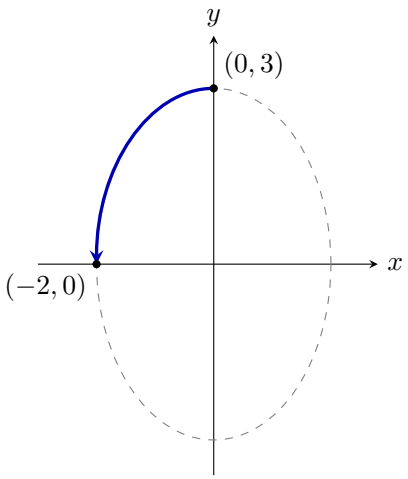
3. Parametrize the following curves.

(a) The quarter-ellipse from $(0,3)$ to $(-2,0)$ centered at the origin.

Solution. The ellipse is $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$, so

$$x(t) = 2\cos t, \qquad y(t) = 3\sin t, \qquad \frac{\pi}{2} \le t \le \pi.$$

Check: $t = \frac{\pi}{2}$ gives $(0,3)$ and $t = \pi$ gives $(-2,0)$.

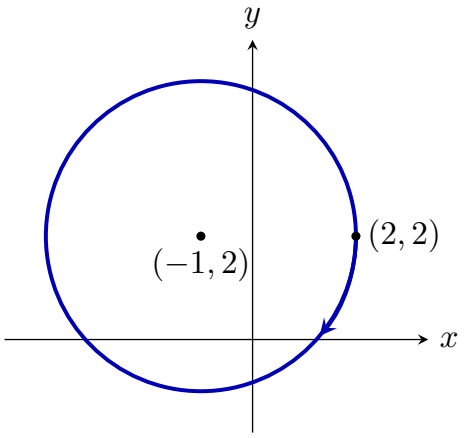


(b) The circle of radius 3 centered at $(-1,2)$, traced once *clockwise* starting at $(2,2)$.

Solution. $(x+1)^2 + (y-2)^2 = 3^2$. A minus sign on the sine term makes the motion clockwise:

$$x(t) = -1 + 3\cos t, \qquad y(t) = 2 - 3\sin t, \qquad 0 \le t \le 2\pi.$$

Check: $t = 0$ gives $(2,2)$; $t = \frac{\pi}{2}$ gives $(-1,-1)$, so the point moves down first (clockwise).



(c) The line segment from $(2,-1)$ to $(-1,3)$.

Solution.

$$\begin{aligned}\vec{r}(t) &= \langle 2,-1\rangle + t\,\langle -1-2,\ 3-(-1)\rangle \\ &= \langle 2,-1\rangle + t\,\langle -3,4\rangle,\end{aligned}$$

i.e.

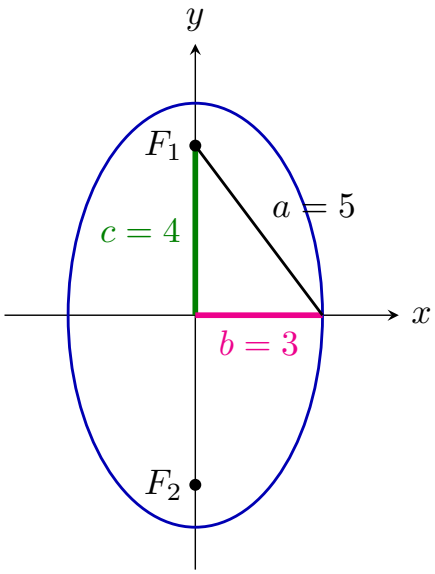
$$x(t) = 2 - 3t, \qquad y(t) = -1 + 4t, \qquad 0 \le t \le 1.$$

Check: $t = 0$ gives $(2, -1)$ and $t = 1$ gives $(-1, 3)$.

4. Consider an ellipse whose ***center*** is at the origin. Assume that it has ***foci*** $(0, -4)$ and $(0, 4)$, and suppose that the length of its ***minor axis*** is 6. Assume that the major axis is in the y-direction. What is the equation of this ellipse?

Solution. Foci on the y-axis: $c = 4$. Minor axis $2b = 6$, so $b = 3$. For an ellipse, $a^2 = b^2 + c^2 = 9 + 16 = 25$, so $a = 5$. With the major axis vertical,

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \quad \Longrightarrow \quad \boxed{\frac{x^2}{3^2} + \frac{y^2}{5^2} = 1}$$



5. Find a unit vector that is perpendicular to the xz-plane and points in the negative y-direction. If the tail of this vector is placed at $(2, -1, 3)$, where is its tip?

Solution. Perpendicular to the xz-plane means parallel to the y-axis: $\langle 0, c, 0\rangle$. Length 1 and pointing in the negative y-direction gives $c = -1$:

$$-\hat{\jmath} = \langle 0, -1, 0\rangle.$$

Tip = tail + vector $= (2,\ -1 - 1,\ 3) = (2, -2, 3)$.

6. Mark the following statements as True or False. If a statement is true, prove it and if a statement is false, provide a counter-example.

(a) Let $\vec{u}$ and $\vec{v}$ be any two vectors in $\mathbb{R}^3$. Then $\|\vec{u} + \vec{v}\|^2 + \|\vec{u} - \vec{v}\|^2 = 2\|\vec{u}\|^2 + 2\|\vec{v}\|^2$.

Solution. ***True.*** Use $\|\vec{a}\|^2 = \vec{a} \cdot \vec{a}$ and the distributive and commutative properties of the dot product:

$$\|\vec{u} + \vec{v}\|^2 = (\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v}) = \|\vec{u}\|^2 + 2\,\vec{u} \cdot \vec{v} + \|\vec{v}\|^2,$$
$$\|\vec{u} - \vec{v}\|^2 = (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) = \|\vec{u}\|^2 - 2\,\vec{u} \cdot \vec{v} + \|\vec{v}\|^2.$$

Adding, the $\pm 2\,\vec{u} \cdot \vec{v}$ terms cancel, leaving $2\|\vec{u}\|^2 + 2\|\vec{v}\|^2$. (Geometrically: the sum of the squares of the diagonals of a parallelogram equals the sum of the squares of its four sides.)

(b) Let $\vec{u}$, $\vec{v}$, and $\vec{w}$ be non-zero vectors such that $\vec{u} \cdot \vec{v} = \vec{u} \cdot \vec{w}$. Then $\vec{v} = \vec{w}$.

Solution. ***False.*** Counter-example: $\vec{u} = \langle 1, 0, 0\rangle$, $\vec{v} = \langle 1, 2, 0\rangle$, $\vec{w} = \langle 1, 0, 5\rangle$. Then $\vec{u} \cdot \vec{v} = 1 = \vec{u} \cdot \vec{w}$, but $\vec{v} \neq \vec{w}$. (In general $\vec{u} \cdot (\vec{v} - \vec{w}) = 0$ only says $\vec{v} - \vec{w}$ is perpendicular to $\vec{u}$, not that it is $\vec{0}$.)

7. Consider the points $P(1, 0, 1)$, $Q(2, 1, 0)$, and $R(0, 2, 3)$.

(a) Find an equation of the plane that contains P, Q, and R.

Solution. $\overrightarrow{PQ} = \langle 1, 1, -1\rangle$ and $\overrightarrow{PR} = \langle -1, 2, 2\rangle$. A normal vector is

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 1 & -1 \\ -1 & 2 & 2 \end{vmatrix}$$
$$= (2+2)\,\hat{\imath} - (2-1)\,\hat{\jmath} + (2+1)\,\hat{k} = \langle 4, -1, 3\rangle.$$

Using the point P: $4(x-1) - (y-0) + 3(z-1) = 0$, i.e.

$$\boxed{4x - y + 3z = 7}.$$

Check: Q: $8 - 1 + 0 = 7$ ✓; R: $0 - 2 + 9 = 7$ ✓.

(b) Find parametric equations of the line through $S(5, 2, 5)$ that is perpendicular to the plane in part (a).

Solution. The line is parallel to the normal $\vec{n} = \langle 4, -1, 3\rangle$:

$$x(t) = 5 + 4t, \qquad y(t) = 2 - t, \qquad z(t) = 5 + 3t, \qquad t \in \mathbb{R}.$$

(c) Find the point where the line in part (b) meets the plane, and use it to find the distance from S to the plane.

Solution. Substitute into $4x - y + 3z = 7$:

$$4(5 + 4t) - (2 - t) + 3(5 + 3t) = 7 \quad \Longrightarrow \quad 33 + 26t = 7 \quad \Longrightarrow \quad t = -1.$$

The line meets the plane at $F = (5-4,\ 2+1,\ 5-3) = (1, 3, 2)$ (check: $4 - 3 + 6 = 7$ ✓). The distance from S to the plane is

$$\left\|\overrightarrow{FS}\right\| = \|\langle 4, -1, 3\rangle\| = \sqrt{16 + 1 + 9} = \sqrt{26} \approx 5.10.$$

(Same answer from the projection formula: $\dfrac{|\vec{n} \cdot \overrightarrow{PS}|}{\|\vec{n}\|} = \dfrac{|\langle 4, -1, 3\rangle \cdot \langle 4, 2, 4\rangle|}{\sqrt{26}} = \dfrac{26}{\sqrt{26}} = \sqrt{26}$.)

5.12 Assignment 7

These problems sketch domains of functions of two variables, reason about level curves, identify and sketch level surfaces, and ask when a limit of a function of two variables exists and how to show that one does not, and compute and estimate higher order partial derivatives. If you would like to review the material first, see Functions of Several Variables, Limits and Continuity in Higher Dimensions and Partial Derivatives.

1. Sketch the domain of $f(x,y) = \sqrt{y - 3x}$.

Solution. The square root is defined exactly when its argument is non-negative, so the domain is

$$\{(x,y) : y - 3x \geq 0\} = \{(x,y) : y \geq 3x\},$$

the closed half-plane on and above the line $y = 3x$. See Figure 5.12.1.

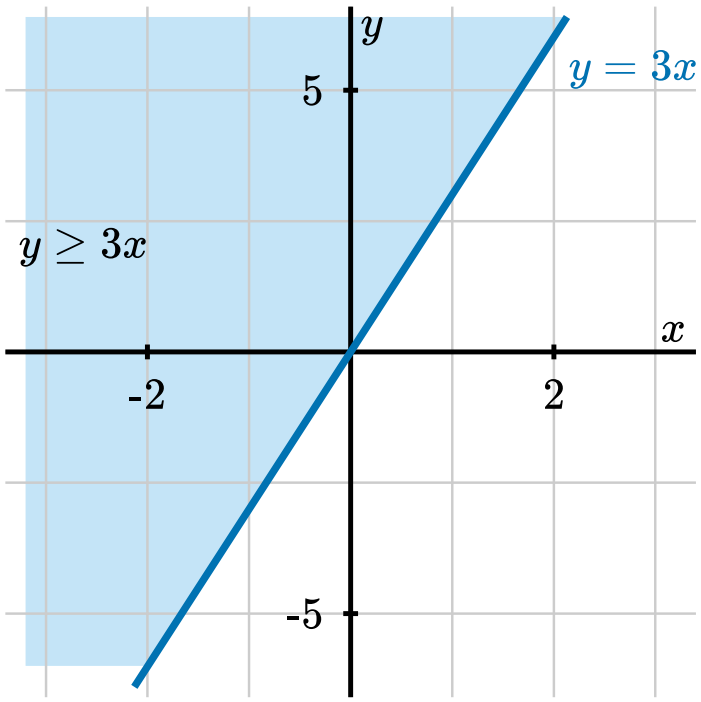


Figure 5.12.1 The domain of $f(x,y) = \sqrt{y - 3x}$ is the shaded closed half-plane $y \geq 3x$, including the boundary line itself.

2. In Figure 5.12.2 are level curves of $g(x,y) = 2$ and $g(x,y) = 3$ for some function $g(x,y)$. What's wrong?

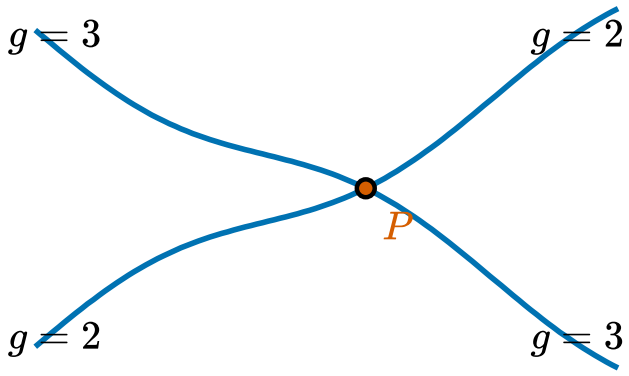


Figure 5.12.2 Two curves labelled $g = 2$ and $g = 3$ that cross each other.

Solution. Two level curves belonging to *different* values can never intersect. At the point P where the drawn curves cross, the first curve says $g(P) = 2$ while the second says $g(P) = 3$. Since g is a function, it assigns exactly one output to the point P, so $2 = 3$ — which is impossible. Hence no function g can have the level curves shown.

3. Let g be a function of 3 variables defined by

$$g(x, y, z) = \sqrt{4x^2 + y^2 + 9z^2}.$$

Find the equation for the level surface of g through the point $\left(2, \sqrt{11}, 1\right)$ and sketch this level surface. Label any points where the surface crosses a coordinate axis AND label the value of g on the surface.

Solution. Evaluating g at the given point,

$$g\left(2, \sqrt{11}, 1\right) = \sqrt{4(4) + 11 + 9(1)} = \sqrt{36} = 6,$$

so the level surface is $g(x, y, z) = 6$, i.e.

$$\begin{aligned} \sqrt{4x^2 + y^2 + 9z^2} = 6 \;&\Rightarrow\; 4x^2 + y^2 + 9z^2 = 36 \\ &\Rightarrow\; \frac{x^2}{9} + \frac{y^2}{36} + \frac{z^2}{4} = 1. \end{aligned}$$

This is an **ellipsoid** with semi-axes 3, 6, and 2. It crosses the axes at $(\pm 3, 0, 0)$, $(0, \pm 6, 0)$, and $(0, 0, \pm 2)$. See Figure 5.12.3.

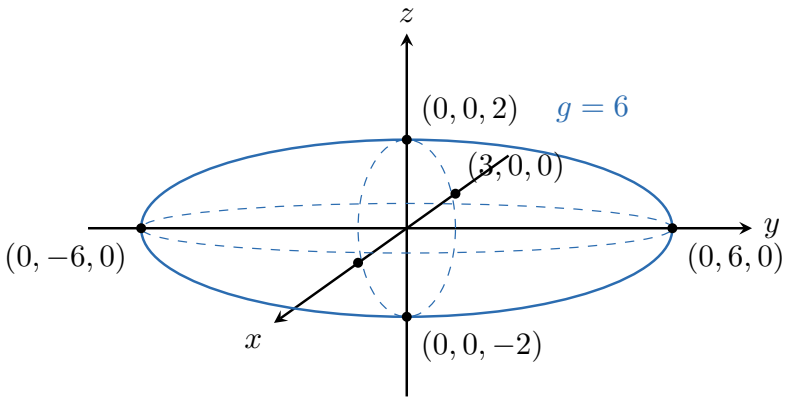


Figure 5.12.3 The level surface $g = 6$, i.e. the ellipsoid $\frac{x^2}{9} + \frac{y^2}{36} + \frac{z^2}{4} = 1$, with its axis intercepts.

4. Sketch the level surface $f(x, y, z) = z^2 - x^2 - y^2$ for $f(x, y, z) = 4$.

Solution. The level surface is $z^2 - x^2 - y^2 = 4$, i.e.

$$\frac{z^2}{4} - \frac{x^2}{4} - \frac{y^2}{4} = 1.$$

Rewriting as $z^2 = 4 + x^2 + y^2$ shows that $z^2 \geq 4$, so $z \geq 2$ or $z \leq -2$: no part of the surface lies between the two horizontal planes $z = \pm 2$. The traces are

- $z = c$ with $|c| > 2$: $x^2 + y^2 = c^2 - 4$, circles;
- $x = c$: $z^2 - y^2 = 4 + c^2$, hyperbolas;
- $y = c$: $z^2 - x^2 = 4 + c^2$, hyperbolas.

Hence the level surface is a **hyperboloid of two sheets**, symmetric about the z-axis, with vertices $(0, 0, \pm 2)$. See Figure 5.12.4.

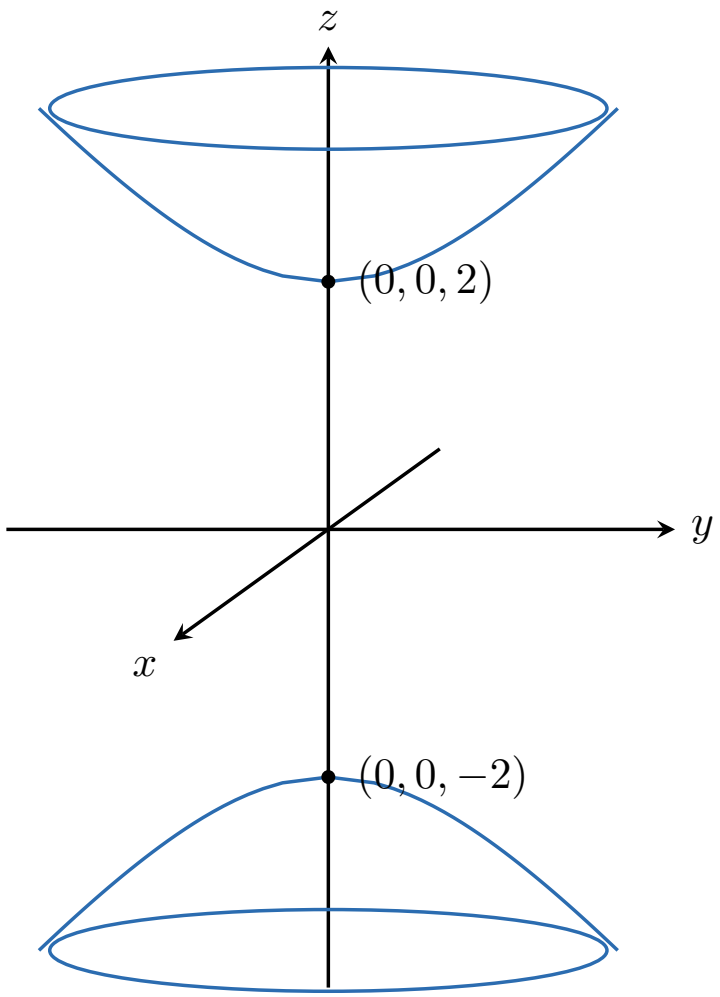


Figure 5.12.4 The level surface $z^2 - x^2 - y^2 = 4$: a hyperboloid of two sheets about the z-axis with vertices $(0, 0, \pm 2)$.

5. True or False. If $\lim_{(x,y)\to(a,b)} f(x, y)$ exists, the function $f(x, y)$ must be defined or continuous at (a, b).

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

We can for example define $f(x, y)$ to be zero when $(x, y) \neq (0, 0)$ and to be 1 when $(x, y) = (0, 0)$. If we then pick $(a, b) = (0, 0)$, the limit of $f(x, y)$ for $(x, y) \to (a, b)$ exists but it is not equal to $f(a, b)$. We could also leave $f(x, y)$ undefined at the origin to give another counterexample.

6. **True or False.** If $f(x,y) \to L$ as $(x,y) \to (a,b)$ along any line through (a,b), then $\lim_{(x,y)\to(a,b)} f(x,y) = L$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

One counterexample is

$$f(x,y) = \begin{cases} 0 & \text{if } y \le 0 \text{ or } y \ge 2x^2 \\ 1 & \text{if } 0 < y < 2x^2 \end{cases}$$

Along every straight line passing through the origin this will be constant 0 near the origin, but the limit approaching via the parabola $y = x^2$ is 1. See Figure 5.12.5.

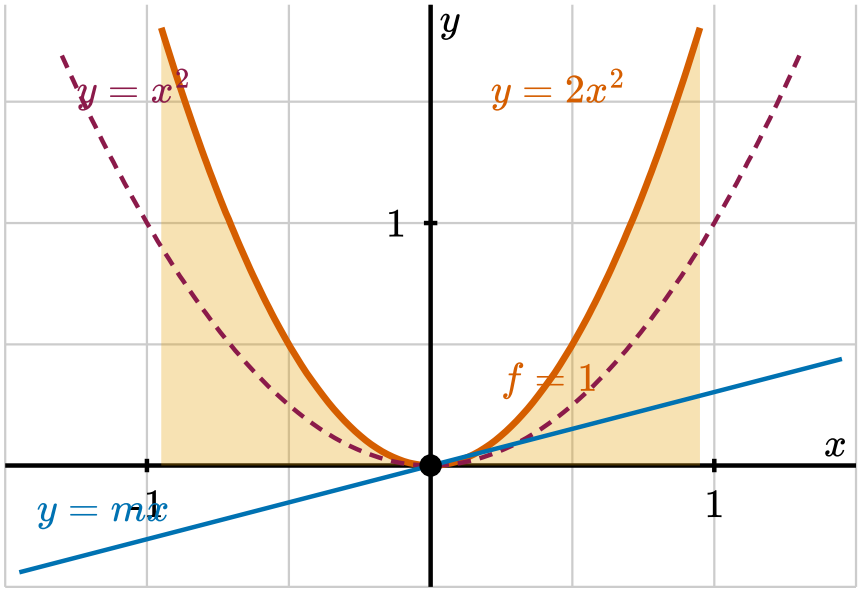


Figure 5.12.5 The counterexample. The function is 1 on the shaded region $0 < y < 2x^2$ and 0 everywhere else. Every line through the origin leaves the shaded region near the origin, but the parabola $y = x^2$ stays inside it.

7. Discuss continuity of the following functions:

A. $\dfrac{2x^4 y}{x^8 + y^2}$

B. $\dfrac{x^2 \cos^2 x}{x^2 + (y-1)^2}$

Solution.

A. This is obviously continuous whenever the denominator is nonzero, i.e. everywhere except at the origin. But at the origin it is not continuous (or rather we can't extend the function to the origin in a way which makes it continuous): if we approach the origin via any of the coordinate axes we get a limit of 0, but if we approach it via the path $y = x^4$ we get a limit

$$\lim_{x\to 0} \frac{2x^4 x^4}{x^8 + x^8} = 1.$$

Hence the limit $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

B. This is continuous when the denominator is nonzero, so everywhere except at $(0,1)$. At $(0,1)$ it can't be defined in a way which makes the function continuous: approaching $(0,1)$ via the y-axis, we get the limit

$$\lim_{y\to 1} \frac{0 \cdot \cos^2(0)}{(y-1)^2} = 0.$$

On the other hand, if we approach $(0,1)$ on the line $y=1$ we get the limit

$$\lim_{x\to 0}\frac{x^2\cos^2 x}{x^2}=\cos^2(0)=1.$$

8. Evaluate each of the following limits or show that it does not exist.

A. $\displaystyle\lim_{(x,y)\to(0,0)}\frac{xy^5}{x^8+y^{10}}$

B. $\displaystyle\lim_{(x,y)\to(1,1)}\frac{x^8-y^8}{x^4-y^4}$

Solution.

A. Along the path $x=y^5$:

$$\lim_{(x,y)\to(0,0)}\frac{xy^5}{x^8+y^{10}}=\lim_{y\to 0}\frac{y^{10}}{y^{40}+y^{10}}=\lim_{y\to 0}\frac{1}{y^{30}+1}=1.$$

Along the path $x=0$:

$$\lim_{(x,y)\to(0,0)}\frac{xy^5}{x^8+y^{10}}=\lim_{y\to 0}\frac{0}{0+y^{10}}=0.$$

Since $1\neq 0$, the limit does not exist.

B. We factor the numerator as a difference of squares:

$$\begin{aligned}\lim_{(x,y)\to(1,1)}\frac{x^8-y^8}{x^4-y^4}&=\lim_{(x,y)\to(1,1)}\frac{\left(x^4-y^4\right)\left(x^4+y^4\right)}{x^4-y^4}\\&=\lim_{(x,y)\to(1,1)}x^4+y^4=2.\end{aligned}$$

9. Find the limit, if it exists, or show that it does not exist:

$$\lim_{(x,y)\to(1,1)}\frac{x^2+y^2-2}{|x-1|+|y-1|}.$$

Solution. Along the path $x=1$, $y\to 1^+$ (so that $|y-1|=y-1$):

$$\lim_{y\to 1^+}\frac{y^2-1}{y-1}=\lim_{y\to 1^+}\frac{(y-1)(y+1)}{y-1}=\lim_{y\to 1^+}(y+1)=2.$$

Along the path $x=1$, $y\to 1^-$ (so that $|y-1|=-(y-1)$):

$$\lim_{y\to 1^-}\frac{y^2-1}{-(y-1)}=\lim_{y\to 1^-}-(y+1)=-2.$$

Since the limits computed along the two paths are not equal, the limit does not exist.

10. Find the following limit, if it exists, or show that it does not exist:

$$\lim_{(x,y)\to(0,0)}\frac{y^4}{x^4+3y^4}.$$

Solution. Along the path $x=0$:

$$\lim_{(x,y)\to(0,0)}\frac{y^4}{x^4+3y^4}=\lim_{y\to 0}\frac{y^4}{3y^4}=\frac{1}{3}.$$

Along the path $y=x$:

$$\lim_{(x,y)\to(0,0)}\frac{y^4}{x^4+3y^4}=\lim_{x\to 0}\frac{x^4}{x^4+3x^4}=\frac{1}{4}.$$

Since $\frac{1}{3}\neq\frac{1}{4}$, the limit does not exist.

11. Higher order computation. Prove that $c(x,t) = \dfrac{1}{\sqrt{Dt}}\, e^{-\frac{x^2}{4Dt}}$ is a solution of the diffusion equation

$$\frac{\partial c}{\partial t} = D\,\frac{\partial^2 c}{\partial x^2}.$$

Solution. We compute the needed derivatives:

$$\begin{aligned}
\frac{\partial c}{\partial t} &= -\frac{1}{2\sqrt{D}\,t^{3/2}}\, e^{-x^2/4Dt} + \frac{1}{\sqrt{Dt}}\,\frac{x^2}{4Dt^2}\, e^{-x^2/4Dt} \\
\frac{\partial c}{\partial x} &= -\frac{x}{2Dt}\,\frac{1}{\sqrt{Dt}}\, e^{-x^2/4Dt} \\
\frac{\partial^2 c}{\partial x^2} &= -\frac{1}{2Dt\sqrt{Dt}}\, e^{-x^2/4Dt} + \frac{x^2}{4D^2t^2}\,\frac{1}{\sqrt{Dt}}\, e^{-x^2/4Dt}
\end{aligned}$$

We see that the right hand side of the third equation is $\dfrac{1}{D}$ times the right hand side of the first, i.e. multiplying $\dfrac{\partial^2 c}{\partial x^2}$ by D gives exactly $\dfrac{\partial c}{\partial t}$, showing that c satisfies the diffusion equation.

12. Let $f(x,y,z) = xy + 3y^2z^3 + \ln(xz)$. Compute $\dfrac{\partial^2 f}{\partial x^2}$ and $\dfrac{\partial^2 f}{\partial y \partial z}$.

Solution. Note first that $\ln(xz) = \ln x + \ln z$, which makes the differentiation easier. Treating the other variables as constants,

$$\frac{\partial f}{\partial x} = y + \frac{1}{x} \;\Rightarrow\; \frac{\partial^2 f}{\partial x^2} = -\frac{1}{x^2}.$$

For the mixed partial, differentiate first with respect to z and then with respect to y:

$$\frac{\partial f}{\partial z} = 9y^2z^2 + \frac{1}{z} \;\Rightarrow\; \frac{\partial^2 f}{\partial y \partial z} = \frac{\partial}{\partial y}\left(9y^2z^2 + \frac{1}{z}\right) = 18yz^2.$$

13. A metal plate occupies the square $0 \le x \le 5$, $0 \le y \le 5$, and is heated so that the temperature at the point (x,y) is $T(x,y)$, measured in degrees Fahrenheit. Distances are measured in feet. The table below records some measured values of T.

Table 5.12.6 Measured temperatures $T(x,y)$ in °F

	$y=1$	$y=1.5$	$y=2$
$x=1$	70	65	58
$x=1.5$	74	68	59
$x=2$	79	75	65

(a) Approximate $T_x(1,1)$ and interpret your answer, with units, in the context of the problem.

(b) Approximate $T_{xy}(1,1)$, and give its units.

(c) An ant walks across the plate along the path

$$x = 1 - \frac{\ln(2t-3)}{8}, \qquad y = \frac{2^t}{4}, \qquad 1.5 < t < 4.$$

Approximate the rate at which the temperature the ant feels is changing at the moment it is at the point $(x,y) = (1,1)$, and give the units.

Solution. **(a)** A partial derivative is a limit of difference quotients, so with the data we have we approximate it by a difference quotient. Holding $y = 1$ fixed and stepping forward in x from $x = 1$ to $x = 1.5$,

$$T_x(1,1) \approx \frac{T(1.5,1) - T(1,1)}{1.5 - 1} = \frac{74 - 70}{0.5} = 8.$$

The units are degrees Fahrenheit per foot. The interpretation: at the point $(1,1)$, if the ant moves in the positive x-direction while its y-coordinate stays fixed, the temperature increases at a rate of about 8 °F per foot.

(b) By definition $T_{xy} = (T_x)_y$, so we compute T_x at two different values of y and then take a difference quotient in y. We already have $T_x(1,1) \approx 8$, and in the same way

$$T_x(1,1.5) \approx \frac{T(1.5,1.5) - T(1,1.5)}{0.5} = \frac{68-65}{0.5} = 6.$$

Therefore

$$T_{xy}(1,1) \approx \frac{T_x(1,1.5) - T_x(1,1)}{1.5-1} = \frac{6-8}{0.5} = -4.$$

The units are degrees Fahrenheit per foot per foot, i.e. $^\circ\text{F/ft}^2$. This says that as we move in the y-direction, the rate at which the temperature rises in the x-direction is itself decreasing.

(c) First we find the value of t at which the ant is at $(1,1)$. We need

$$\begin{aligned} 1 - \frac{\ln(2t-3)}{8} = 1 \quad &\Longrightarrow \quad \ln(2t-3) = 0 \\ &\Longrightarrow \quad 2t-3 = 1 \quad \Longrightarrow \quad t = 2, \end{aligned}$$

and indeed $y = 2^2/4 = 1$ at $t = 2$, as required. Note that $t = 2$ lies in the given interval $1.5 < t < 4$.

Next we differentiate the path:

$$\frac{dx}{dt} = -\frac{1}{8} \cdot \frac{2}{2t-3} = -\frac{1}{4(2t-3)}, \qquad \left.\frac{dx}{dt}\right|_{t=2} = -\frac{1}{4},$$

$$\frac{dy}{dt} = \frac{2^t \ln 2}{4}, \qquad \left.\frac{dy}{dt}\right|_{t=2} = \frac{4\ln 2}{4} = \ln 2.$$

We also need $T_y(1,1)$, which we approximate from the table just as before, stepping forward in y:

$$T_y(1,1) \approx \frac{T(1,1.5) - T(1,1)}{0.5} = \frac{65-70}{0.5} = -10.$$

Now the chain rule gives

$$\begin{aligned} \frac{dT}{dt} &= T_x \frac{dx}{dt} + T_y \frac{dy}{dt} \\ &\approx (8)\left(-\frac{1}{4}\right) + (-10)(\ln 2) \\ &= -2 - 10\ln 2 \approx -8.93. \end{aligned}$$

The units are degrees Fahrenheit per unit of t (per minute, if t is measured in minutes). So the ant is cooling off at about $8.9\,^\circ\text{F}$ per unit of time. This makes sense: the ant is moving slightly in the $-x$ direction, which cools it a little, and it is moving in the $+y$ direction, which cools it a lot. Figure 5.12.7 shows the situation.

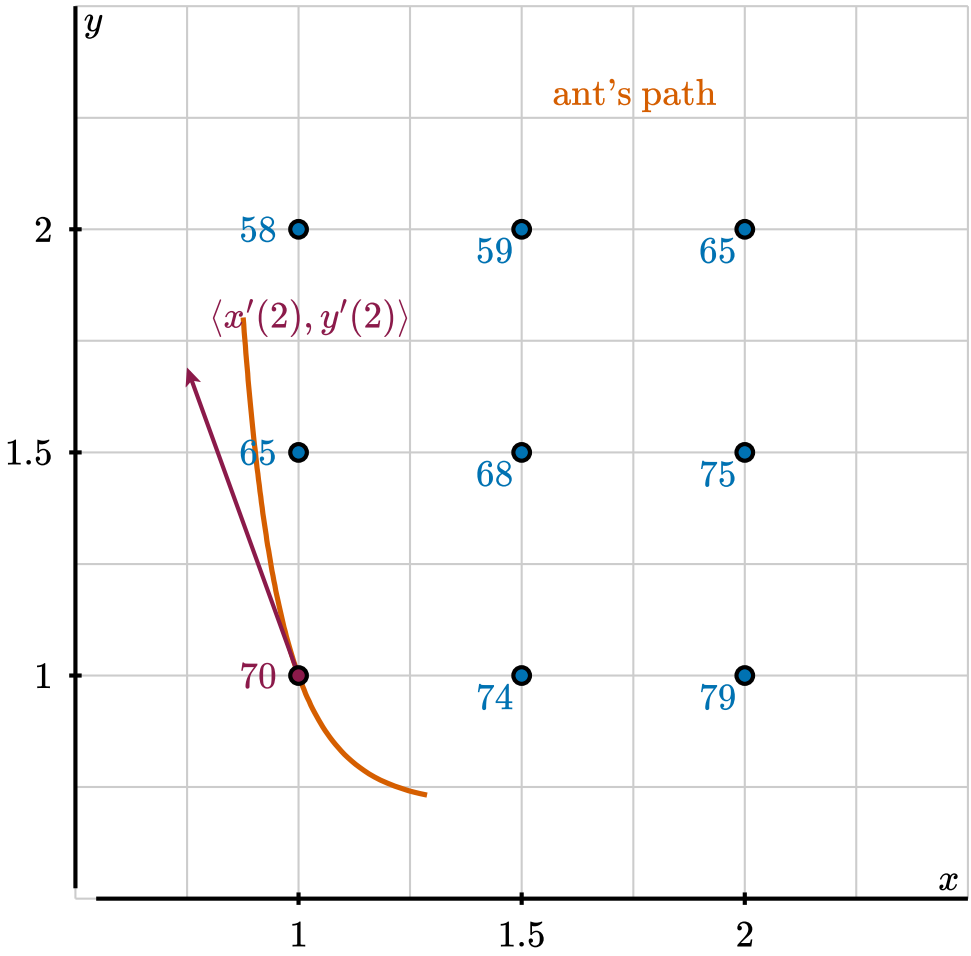


Figure 5.12.7 The nine data points of Table 5.12.6 and the ant's path. At $t = 2$ the ant is at $(1, 1)$ and its velocity is $\langle -\frac{1}{4}, \ln 2 \rangle$. Temperatures fall as we move up and to the left, so the ant is cooling.

5.13 Assignment 8

These problems apply the chain rule to functions of several variables: differentiating a composition both by the chain rule and by substitution, deciding a True/False statement about a rate along a curve, proving Euler's relation for homogeneous functions, and working through a dependency tree with two intermediate levels. They then use the gradient to compute directional derivatives, find directions of fastest change, and read a contour map. If you would like to review the material first, see The Chain Rule and Directional Derivatives and Gradient Vectors.

1. Compute the partial derivatives of f with respect to x, y for

$$f(s,t) = s^2 - t^2, \qquad s = \frac{x+y}{2}, \qquad t = \frac{x-y}{2},$$

using both the chain rule and by plugging in s and t.

Solution. Using the chain rule:

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial f}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial f}{\partial t}\frac{\partial t}{\partial x} \\ &= 2s \cdot \frac{1}{2} - 2t \cdot \frac{1}{2} = s - t = y \\ \frac{\partial f}{\partial y} &= \frac{\partial f}{\partial s}\frac{\partial s}{\partial y} + \frac{\partial f}{\partial t}\frac{\partial t}{\partial y} \\ &= 2s \cdot \frac{1}{2} + 2t \cdot \frac{1}{2} = s + t = x\end{aligned}$$

Plugging in, we see

$$f(s(x,y), t(x,y)) = \frac{1}{4}\left((x+y)^2 - (x-y)^2\right) = xy,$$

and easily obtain the same partial derivatives.

2. **True or False.** Consider $f(x,y) = (\cos x + (1+x)\tan y)\, e^{x^2-1+y}$. Then $\dfrac{d}{dt} f(t^2, t^3) = 0$ at $t = 0$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. True.

Solution. True.

Writing $x(t) = t^2$ and $y(t) = t^3$, we see that

$$\frac{dx}{dt} = 2t = 0 \quad \text{and} \quad \frac{dy}{dt} = 3t^2 = 0 \quad \text{at } t = 0.$$

Now we see from the chain rule that

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} = 0 \quad \text{at } t = 0.$$

3. **Homogeneous functions.** A function f is called **homogeneous of degree** n if it satisfies the equation

$$f(tx, ty) = t^n f(x,y) \tag{5.13.1}$$

for all t, where n is a positive integer and f has continuous second order derivatives.

A. Verify that $f(x,y) = x^2y + 2xy^2 + 5y^3$ is homogeneous of degree 3.

B. Show that if f is homogeneous of degree n then

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = nf(x,y).$$

(Hint: use the Chain Rule to differentiate $f(tx, ty)$ with respect to t.)

C. If f is homogeneous of degree n, show that $f_x(tx, ty) = t^{n-1} f_x(x,y)$ for $t > 0$.

Solution.

A.

$$\begin{aligned} f(tx,ty) &= t^2x^2\,ty + 2tx\,t^2y^2 + 5t^3y^3 \\ &= t^3\left(x^2y + 2xy^2 + 5y^3\right) = t^3 f(x,y). \end{aligned}$$

B. Using the chain rule we obtain

$$\begin{aligned} \frac{d}{dt}f(tx,ty) &= \frac{\partial f}{\partial x}(tx,ty)\,\frac{d}{dt}tx + \frac{\partial f}{\partial y}(tx,ty)\,\frac{d}{dt}ty \\ &= x\,\frac{\partial f}{\partial x}(tx,ty) + y\,\frac{\partial f}{\partial y}(tx,ty). \end{aligned} \tag{5.13.2}$$

On the other hand, the homogeneity condition (5.13.1) shows that

$$\frac{d}{dt}f(tx,ty) = \frac{d}{dt}\,t^n f(x,y) = nt^{n-1}f(x,y). \tag{5.13.3}$$

Now we can equate the right hand sides of equations (5.13.2) and (5.13.3) and set $t = 1$ to obtain the desired equation.

C. We take the partial u-derivative of both sides of the equation $f(tu,tv) = t^n f(u,v)$ (this is just (5.13.1) plugging in u and v instead of x and y):

$$\begin{aligned} \frac{\partial}{\partial u}\,t^n f(u,v) &= t^n f_x(u,v) \\ \frac{\partial}{\partial u}\,f(tu,tv) &= t\,f_x(tu,tv) \end{aligned}$$

Hence $t^{n-1}f_x(u,v) = f_x(tu,tv)$ after canceling one t, and plugging in $u = x$, $v = y$ shows the desired equation.

4. Let

$$w = 5xy + 2yz - 3xz, \qquad x = st, \qquad y = e^{st}, \qquad z = t^2.$$

Compute $\dfrac{\partial w}{\partial s}(-4,-4)$ and $\dfrac{\partial w}{\partial t}(-4,-4)$.

Solution. The dependency tree is shown in Figure 5.13.1. We first compute

$$\frac{\partial w}{\partial x} = 5y - 3z, \qquad \frac{\partial w}{\partial y} = 5x + 2z, \qquad \frac{\partial w}{\partial z} = 2y - 3x,$$

and at $(s,t) = (-4,-4)$ we have

$$x(-4,-4) = 16, \qquad y(-4,-4) = e^{16}, \qquad z(-4,-4) = 16.$$

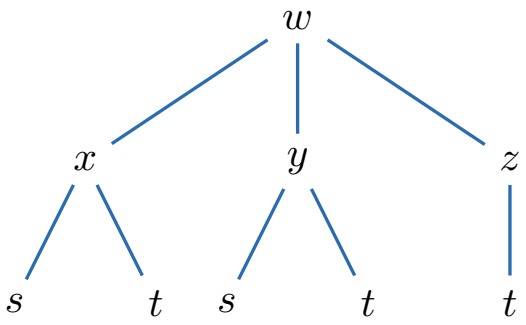


Figure 5.13.1 The tree diagram for $w = w(x,y,z)$ with $x = x(s,t)$, $y = y(s,t)$, and $z = z(t)$.

By the chain rule, since $\dfrac{\partial z}{\partial s} = 0$,

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s}$$

$$\begin{aligned}&= (5y-3z)(t) + (5x+2z)\left(te^{st}\right) + (2y-3x)(0)\\ &= \left(5e^{16}-48\right)(-4) + (80+32)\left(-4e^{16}\right)\\ &= -20e^{16} + 192 - 448e^{16} = -468e^{16} + 192.\end{aligned}$$

Similarly,

$$\begin{aligned}\frac{\partial w}{\partial t} &= \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}\\ &= (5y-3z)(s) + (5x+2z)\left(se^{st}\right) + (2y-3x)(2t)\\ &= \left(5e^{16}-48\right)(-4) + (112)\left(-4e^{16}\right) + \left(2e^{16}-48\right)(-8)\\ &= -20e^{16} + 192 - 448e^{16} - 16e^{16} + 384 = -484e^{16} + 576.\end{aligned}$$

5. Consider $f(x,y) = e^{-r^4}$ where $r = \sqrt{x^2+y^2}$. Compute its directional derivative at $(0,0)$ with respect to the unit vectors in (polar) directions $\theta = 0, \frac{\pi}{4}, \frac{\pi}{2}$. What about any other angle?

Solution. The function is symmetric with respect to rotations around the origin, so all the directional derivatives will be equal. Therefore it is sufficient to compute the derivative for $\theta = 0$, i.e. $\dfrac{\partial f}{\partial x}$:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} e^{-(x^2+y^2)^2} = -2(x^2+y^2)\cdot 2x\, e^{-(x^2+y^2)^2}.$$

Plugging in $x = y = 0$ we see that this is zero. Hence the directional derivative at $(0,0)$ is 0 in the directions $\theta = 0, \frac{\pi}{4}, \frac{\pi}{2}$, and by the same symmetry it is 0 for every other angle as well.

6. **True or False.** If $\hat{u} = \langle a, b\rangle$ is a unit vector and f has continuous second partials, then

$$D^2_{\hat{u}} f(x,y) = f_{xx}a^2 + 2f_{xy}ab + f_{yy}b^2,$$

where $D^2_{\hat{u}} f = D_{\hat{u}}\left[D_{\hat{u}} f(x,y)\right]$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. True.

Solution. True.

We have that

$$D^2_{\hat{u}} f = D_{\hat{u}}\left[D_{\hat{u}} f(x,y)\right] = D_{\hat{u}}\left[af_x + bf_y\right].$$

This simplifies to

$$\langle af_{xx} + bf_{xy},\ af_{xy} + bf_{yy}\rangle \cdot \langle a, b\rangle = f_{xx}a^2 + 2f_{xy}ab + f_{yy}b^2.$$

Note we needed f to have continuous second partials to invoke Clairaut's Theorem.

7. **A mountain lion on a mountain.** A mountain lion runs on a mountain whose height above the point (x,y) is $z = x^2 + \sin^2(xy)$.

A. In which direction(s) should the mountain lion run from point $(1,0,1)$ so that the height is increasing at the fastest possible rate? What is this fastest rate?

B. In which direction(s) should the mountain lion run from point $(1,0,1)$ so that the height is increasing at *half* of the fastest possible rate?

Solution.

A. The fastest possible rate of increase is attained in the direction of the gradient:

$$\nabla z = \left\langle \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right\rangle = \langle 2x + 2\sin(xy)\cos(xy)\, y,\ 2\sin(xy)\cos(xy)\, x\rangle.$$

At $(1,0,1)$, this gradient is simply $\nabla z(1,0) = \langle 2, 0\rangle$, and the fastest possible rate of change is the length of the gradient, i.e. $\|\nabla z(1,0)\| = \|\langle 2,0\rangle\| = 2$.

B. Half of the fastest possible rate of height increase at point $(1,0,1)$ is $\frac{1}{2}\cdot 2 = 1$. Let $\hat{u} = \langle a, b\rangle$ be a unit vector in the desired direction. Then the directional derivative $D_{\hat{u}}(1,0) = 1$ is the desired rate of change

of height at $(1,0,1)$ in the direction of $\hat{u}$; it can be calculated using the dot product with the gradient:

$$1 = D_{\hat{u}}(1,0) = \hat{u} \cdot \nabla z(1,0) = \langle a, b\rangle \cdot \langle 2, 0\rangle = 2a + 0b = 2a.$$

Thus, $a = \frac{1}{2}$ and b must be such that $\hat{u} = \left\langle \frac{1}{2}, b\right\rangle$ is unit, i.e. $\left(\frac{1}{2}\right)^2 + b^2 = 1$, from which $b^2 = \frac{3}{4}$ and $b = \pm\frac{\sqrt{3}}{2}$. Therefore, the desired directions in which the height is increasing at a rate equal to half of the fastest possible rate are $\hat{u}_1 = \left\langle \frac{1}{2}, \frac{\sqrt{3}}{2}\right\rangle$ and $\hat{u}_2 = \left\langle \frac{1}{2}, -\frac{\sqrt{3}}{2}\right\rangle$.

8. Suppose that $f(x,y)$ is a function of two variables and let

$$\hat{u} = \frac{1}{\sqrt{2}}\langle 1, -1\rangle, \qquad \hat{v} = \langle 0, 1\rangle.$$

Suppose that at a point $P_0(a,b)$,

$$\begin{aligned}(D_{\hat{u}} f)_{P_0} &= 0,\\ (D_{\hat{v}} f)_{P_0} &= 1.\end{aligned}$$

A. Find *the direction* of the maximum rate of change at $P_0(a,b)$.

B. Find the maximum rate of change at $P_0(a,b)$.

Solution.

A. Using $(D_{\hat{u}} f)_{P_0} = \nabla f\big|_{P_0} \cdot \hat{u}$, the two conditions become

$$\begin{aligned}(D_{\hat{u}} f)_{P_0} = 0 \;\Rightarrow\; \langle f_x, f_y\rangle \cdot \frac{1}{\sqrt{2}}\langle 1,-1\rangle = \frac{1}{\sqrt{2}}(f_x - f_y) = 0\\ \Rightarrow\; f_x = f_y,\\ (D_{\hat{v}} f)_{P_0} = 1 \;\Rightarrow\; \langle f_x, f_y\rangle \cdot \langle 0,1\rangle = f_y = 1.\end{aligned}$$

Hence $\nabla f\big|_{P_0} = \langle 1, 1\rangle$, and the direction of the maximum rate of change at P_0 is given by

$$\hat{u} = \frac{\nabla f\big|_{P_0}}{\left\|\nabla f\big|_{P_0}\right\|} = \frac{1}{\sqrt{2}}\langle 1, 1\rangle.$$

B. $\left\|\nabla f\big|_{P_0}\right\| = \sqrt{1+1} = \sqrt{2}$ is the maximum rate of change.

9. Consider the function $f(x,y) = x^2 e^{3y}$ and the point $P_0 = (-1, 0)$.

A. Find the direction in which $f(x,y)$ decreases the most rapidly at P_0.

B. Find the directional derivative of $f(x,y)$ at P_0 in the direction given by $\overrightarrow{v} = \langle -1, 1\rangle$.

Solution. The gradient is

$$\begin{aligned}\nabla f &= \left(2xe^{3y}\right)\hat{\imath} + \left(3x^2 e^{3y}\right)\hat{\jmath},\\ \nabla f\big|_{(-1,0)} &= -2\,\hat{\imath} + 3\,\hat{\jmath}, \qquad \left\|\nabla f\big|_{(-1,0)}\right\| = \sqrt{13}.\end{aligned}$$

A. The direction of most rapid decrease is opposite to the gradient:

$$-\frac{\nabla f}{\|\nabla f\|} = \left(\frac{2}{\sqrt{13}}\right)\hat{\imath} - \left(\frac{3}{\sqrt{13}}\right)\hat{\jmath} = \left\langle \frac{2}{\sqrt{13}}, -\frac{3}{\sqrt{13}}\right\rangle.$$

B. The unit vector in the direction of $\overrightarrow{v}$ is $\hat{u} = \dfrac{\overrightarrow{v}}{\|\overrightarrow{v}\|} = \left\langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right\rangle$, so

$$D_{\hat{u}} f = \nabla f \cdot \hat{u} = \langle -2, 3\rangle \cdot \left\langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right\rangle = \frac{5}{\sqrt{2}}.$$

10. Suppose the temperature (in degrees Celsius) at a point (x, y) is given by $T(x, y)$, where T is a continuous function that has continuous partial derivatives, and distances in the xy-plane are measured in meters. We are given the following information about T at the point $(x, y) = (1, 1)$:

- $T(1, 1) = 20$ °C,
- the derivative of T at $(1, 1)$ obtains its smallest (i.e. most negative) value in the direction toward the *point* $(-1, 2)$,
- the derivative of T at $(1, 1)$ in the direction of the *vector* $4\,\hat{\imath} + 3\,\hat{\jmath}$ is 3 °C/m.

A. Find the derivative of T at $(1, 1)$ in the direction toward the *point* $(4, -3)$.

B. If you start at $(1, 1)$ and move along a straight path toward the point $(4, -3)$, *approximately* how far will you have to go in this direction until the temperature reaches 24 °C?

Solution. First we determine ∇T at $(1, 1)$. The derivative is most negative in the direction of $-\nabla T$, and the direction from $(1, 1)$ toward $(-1, 2)$ is

$$\langle -1 - 1,\ 2 - 1\rangle = \langle -2, 1\rangle.$$

Hence $-\nabla T$ is a positive multiple of $\langle -2, 1\rangle$, i.e. $\nabla T = c\,\langle 2, -1\rangle$ for some $c > 0$.

The third piece of information fixes c. The unit vector in the direction of $4\,\hat{\imath} + 3\,\hat{\jmath}$ is $\frac{1}{5}\langle 4, 3\rangle$, since $\|\langle 4, 3\rangle\| = 5$, so

$$3 = \nabla T \cdot \frac{1}{5}\langle 4, 3\rangle = \frac{c}{5}\Big(8 - 3\Big) = c \ \Rightarrow\ c = 3.$$

Therefore

$$\nabla T\big|_{(1,1)} = 3\langle 2, -1\rangle = \langle 6, -3\rangle.$$

A. The direction from $(1, 1)$ toward $(4, -3)$ is $\langle 3, -4\rangle$, whose length is 5, so the unit vector is $\hat{u} = \left\langle \frac{3}{5}, -\frac{4}{5}\right\rangle$ and

$$(D_{\hat{u}}T)_{(1,1)} = \langle 6, -3\rangle \cdot \left\langle \frac{3}{5}, -\frac{4}{5}\right\rangle = \frac{18}{5} + \frac{12}{5} = 6\ ^\circ\text{C/m}.$$

B. The temperature must rise from 20 °C to 24 °C, i.e. $\Delta T = 4$ °C, and along this direction it rises at approximately 6 °C per meter. Using the linear approximation $\Delta T \approx (D_{\hat{u}}T) \cdot s$, where s is the distance travelled,

$$s \approx \frac{\Delta T}{D_{\hat{u}}T} = \frac{4}{6} = \frac{2}{3} \approx 0.67 \text{ meters}.$$

11. f is a "nice" function of x and y (i.e. f is continuous and its partials exist). In Figure 5.13.2 are shown the level curves $f(x, y) = 0, 2, 4, 6, 8, 10, 12$.

A. Let $u_1 = \frac{1}{\sqrt{2}}\hat{\imath} + \frac{1}{\sqrt{2}}\hat{\jmath}$. Estimate $D_{u_1} f(4, 3)$.

B. Estimate $D_{u_1} f(1, 2)$.

C. Draw on the diagram a unit vector pointing in the same direction as $\nabla f(1, 2)$.

D. At P, draw a unit vector and label it u_2, such that $D_{u_2} f(P) < 0$.

E. Estimate $\dfrac{\partial f}{\partial x}(P)$.

F. Which appears bigger, $\dfrac{\partial f}{\partial y}(Q)$ or $\dfrac{\partial f}{\partial y}(R)$?

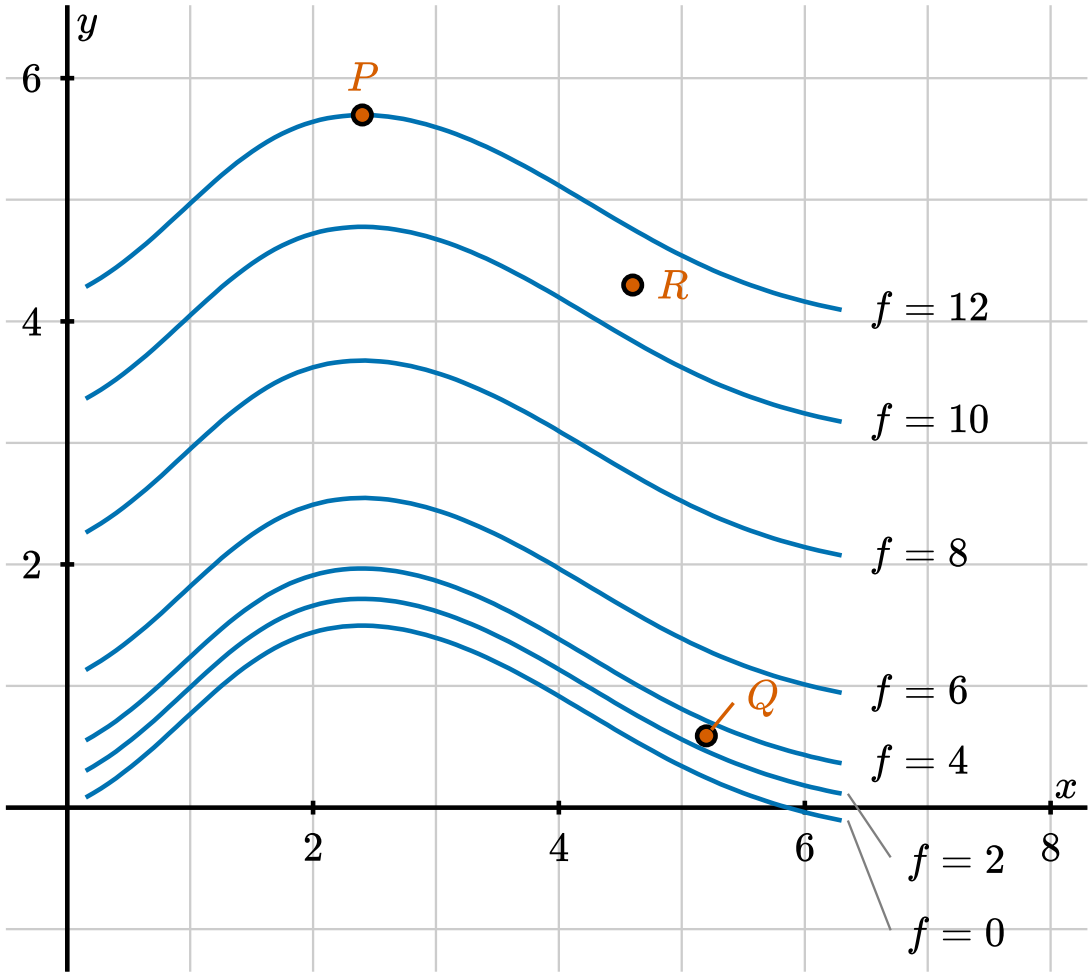


Figure 5.13.2 Level curves of f at the values $0, 2, 4, 6, 8, 10, 12$. The marked point P is the highest point of the $f = 12$ curve; Q lies midway between the $f = 2$ and $f = 4$ curves; R lies midway between the $f = 10$ and $f = 12$ curves.

Solution.

A. The directional derivative $D_{u_1} f(4, 3)$ tells us roughly how much f increases if we move *one unit* from $(4, 3)$ in the direction u_1, i.e. diagonally up and to the right at 45°. Starting at $(4, 3)$ and moving one unit that way lands us near $(4.7, 3.7)$, which is about two contours further along, and each contour step is $\Delta f = 2$ spread over roughly one unit of travel. Reading the map,

$$D_{u_1} f(4, 3) \approx 2.$$

B. At $(1, 2)$ the level curve through that point is climbing at roughly 45°, i.e. the level curve is *parallel* to u_1. Moving along a level curve does not change the value of f, so

$$D_{u_1} f(1, 2) \approx 0.$$

C. The gradient is perpendicular to the level curve and points in the direction of *increase*. At $(1, 2)$ the level curve runs up and to the right at about 45°, and f increases as we move upward, so $\nabla f(1, 2)$ points up and to the *left*, i.e. in the direction $\frac{1}{\sqrt{2}}\langle -1, 1\rangle$. See Figure 5.13.3.

D. We need any direction of *decrease* at P, i.e. any direction pointing toward the lower-valued level curves (toward $f = 10$, not away from it). Since the level curves below P have smaller values, $u_2 = \langle 0, -1\rangle$ (straight down) works. See Figure 5.13.3.

E. P sits at the very top of its level curve, where the curve is *horizontal*. Walking in the $+x$ direction from P therefore keeps us on the same level curve for a while, so f does not change to first order:

$$\frac{\partial f}{\partial x}(P) \approx 0.$$

F. $\dfrac{\partial f}{\partial y}$ measures how fast f changes as we move straight up, which is large where the contours are close together and small where they are far apart. At Q we only have to go up about 0.25 units to move from one contour to the next, so $\dfrac{\partial f}{\partial y}(Q) \approx \dfrac{2}{0.25} = 8$. At R we must go up about 1 unit for the same change, so $\dfrac{\partial f}{\partial y}(R) \approx \dfrac{2}{1} = 2$. Hence $\dfrac{\partial f}{\partial y}(Q)$ is bigger.

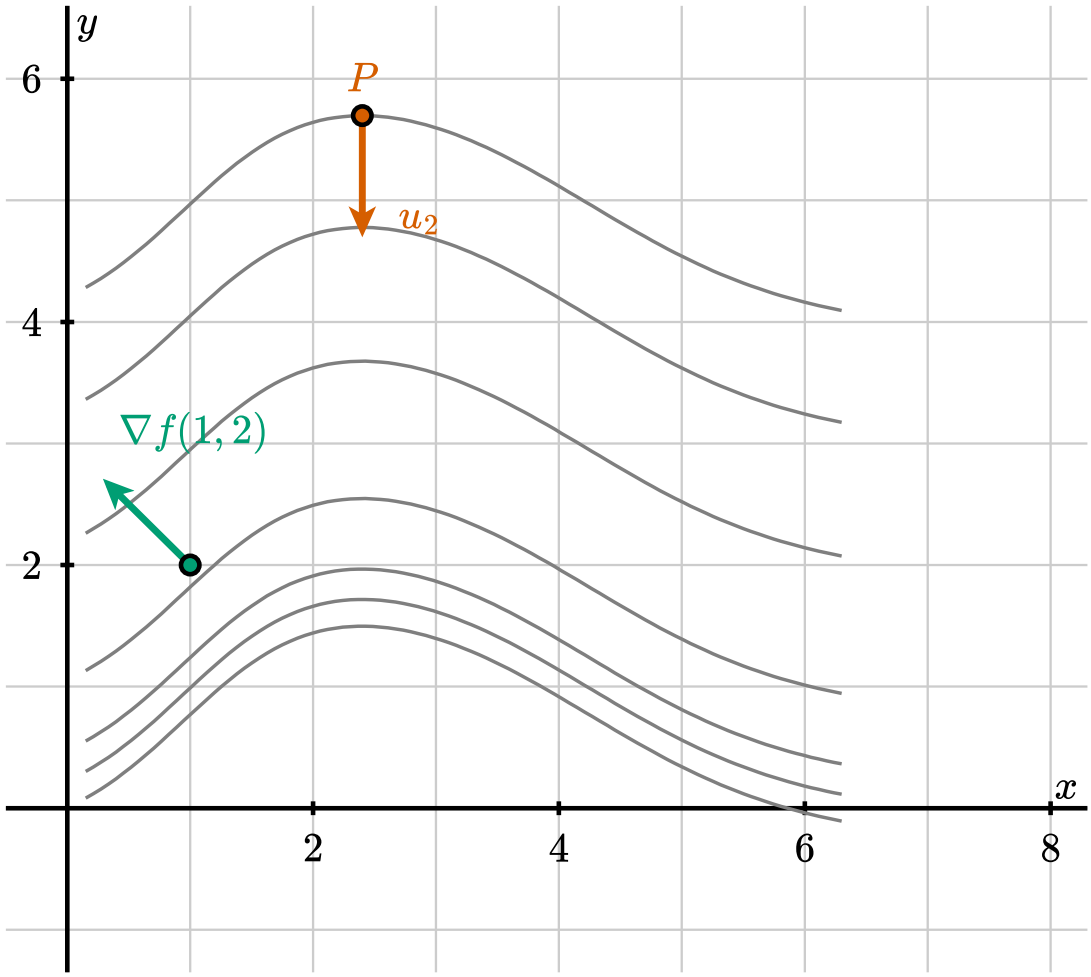


Figure 5.13.3 The vectors asked for in parts C and D: $\nabla f(1,2)$ points perpendicular to the level curve toward increasing f, while u_2 at P points toward decreasing f.

The two problems that close this assignment repeat the three questions of Example 4.5.14 in two more physical settings: find the direction of most rapid decrease, the rate along a prescribed direction, and the directions of no change.

12. Electric Potential on a Coated Plate. The electric potential on a thin coated plate, in volts, is

$$V(x,y) = 100 - 2x^2 - 6y^2,$$

with x and y in centimeters. Consider the point $P_0(2,1)$.

(a) Find ∇V at P_0 and the electric field $\mathbf{E} = -\nabla V$ there. Along which unit vector does the potential drop most rapidly, and at what rate?

(b) A test charge is nudged along $\hat{u} = \frac{4}{5}\hat{\imath} - \frac{3}{5}\hat{\jmath}$. Compute $D_{\hat{u}}V$ at P_0.

(c) Find the two unit vectors along which the potential does not change, and name the curve they are tangent to.

Answer.

(a) $\nabla V = -4x\,\hat{\imath} - 12y\,\hat{\jmath}$, so $(\nabla V)_{(2,1)} = -8\,\hat{\imath} - 12\,\hat{\jmath}$ and $\mathbf{E} = 8\,\hat{\imath} + 12\,\hat{\jmath}$ V/cm. The potential drops fastest along $\hat{u} = \frac{1}{\sqrt{13}}(2\,\hat{\imath} + 3\,\hat{\jmath})$, at the rate $-\|\nabla V\| = -4\sqrt{13} \approx -14.4$ V/cm.

(b)
$$D_{\hat{u}}V = (-8)\left(\tfrac{4}{5}\right) + (-12)\left(-\tfrac{3}{5}\right) = \tfrac{4}{5} = 0.8$$
in V/cm. The potential rises slightly along this heading.

(c) $\hat{u} = \pm\frac{1}{\sqrt{13}}(3\,\hat{\imath} - 2\,\hat{\jmath})$. These are tangent to the equipotential ellipse $2x^2 + 6y^2 = 14$ through P_0, and they are orthogonal to $\mathbf{E}$, as they must be.

An interactive figure of the plate — the equipotential route at $V = 86$ V and the field-line route side by side, with the potential read out as the point moves — accompanies the online version of these notes.

13. Sailing the Isobars of a Low. Near the center of a low-pressure system the sea-level pressure, in millibars, is modeled by
$$P(x, y) = 988 + 0.004\left(x^2 + y^2\right),$$
with x and y in miles from the center of the low. A ship sits at $P_0(20, 10)$.

(a) Find the pressure at the ship and ∇P there. The pressure-gradient force per unit volume of air is a positive multiple of $-\nabla P$; in which unit direction does it push, and what is $\|\nabla P\|$ in millibars per mile?

(b) A second ship 40 miles from the center reports a stronger wind. Using this model, explain why $\|\nabla P\|$ supports that report.

(c) The captain decides to hold a constant barometer reading. Give the two possible unit headings at P_0, and describe the shape of the full route.

Answer.

(a) $P(20, 10) = 988 + 0.004(500) = 990$ mb, and $\nabla P = 0.008x\,\hat{\imath} + 0.008y\,\hat{\jmath}$ gives $(\nabla P)_{(20,10)} = 0.16\,\hat{\imath} + 0.08\,\hat{\jmath}$. The force points along $-\mathbf{u} = -\frac{1}{\sqrt{5}}(2\,\hat{\imath} + \hat{\jmath})$, that is, inward toward the low, and $\|\nabla P\| = 0.08\sqrt{5} \approx 0.18$ mb/mi.

(b) Here $\|\nabla P\| = 0.008\sqrt{x^2 + y^2}$ grows linearly with distance from the center, so at 40 miles it is 0.32 mb/mi, nearly double the value at the first ship. (Real storms reverse this trend close to the eye; the model is only reasonable over a limited annulus.)

(c) $\mathbf{u} = \pm\frac{1}{\sqrt{5}}(\hat{\imath} - 2\,\hat{\jmath})$. Since the level curves of P are the circles $x^2 + y^2 =$ constant, holding the barometer steady means circling the storm at a fixed radius of $\sqrt{500} \approx 22.4$ miles.

An interactive figure of the pressure field — the two headings switchable, with the barometer responding as the ship moves — accompanies the online version of these notes.

5.14 Assignment 9

These problems find the tangent plane to a graph and lines lying in it, show that the radius of a sphere is orthogonal to its tangent plane, and use differentials to estimate a change in a function and to approximate a value. If you would like to review the material first, see Tangent Planes and Differentials.

1. Find an equation for the tangent plane to the graph of $f(x, y) = \cos(xy)$ passing through the point $\left(\frac{\pi}{2}, 1, 0\right)$.

Solution. We have $f_x = -y\sin(xy)$ and $f_y = -x\sin(xy)$, so the tangent plane at a general point (x_0, y_0) is given by

$$z - \cos(x_0y_0) = -y_0\sin(x_0y_0)(x - x_0) - x_0\sin(x_0y_0)(y - y_0).$$

Plugging in our point $\left(\frac{\pi}{2}, 1, 0\right)$, this becomes

$$z = -1 \cdot \left(x - \frac{\pi}{2}\right) - \frac{\pi}{2} \cdot (y - 1),$$

or equivalently

$$x + \frac{\pi}{2}\, y + z = \pi.$$

2. Find a parametric equation for a line contained in the tangent plane you found in the previous problem. (Any line will suffice.)

Solution. Note that the point $\left(\frac{\pi}{2}, 1, 0\right)$ lies on this plane, and the plane has normal vector $\vec{n} = \left\langle 1, \frac{\pi}{2}, 1\right\rangle$. Any vector orthogonal to the normal vector will point along this plane. By inspection, we see that

$$\langle 1, 0, -1\rangle \cdot \left\langle 1, \frac{\pi}{2}, 1\right\rangle = 0,$$

so the vector $\langle 1, 0, -1\rangle$ points along the plane. Thus $\langle 1, 0, -1\rangle$ is the direction vector of a line pointing along this plane. Combining this with the point on the plane that we have already found, we see that the line

$$\mathbf{r}(t) = \left\langle t + \frac{\pi}{2},\ 1,\ -t\right\rangle$$

is contained in the tangent plane. We could find other lines contained in the plane by making different choices of direction vector. See Figure 5.14.1.

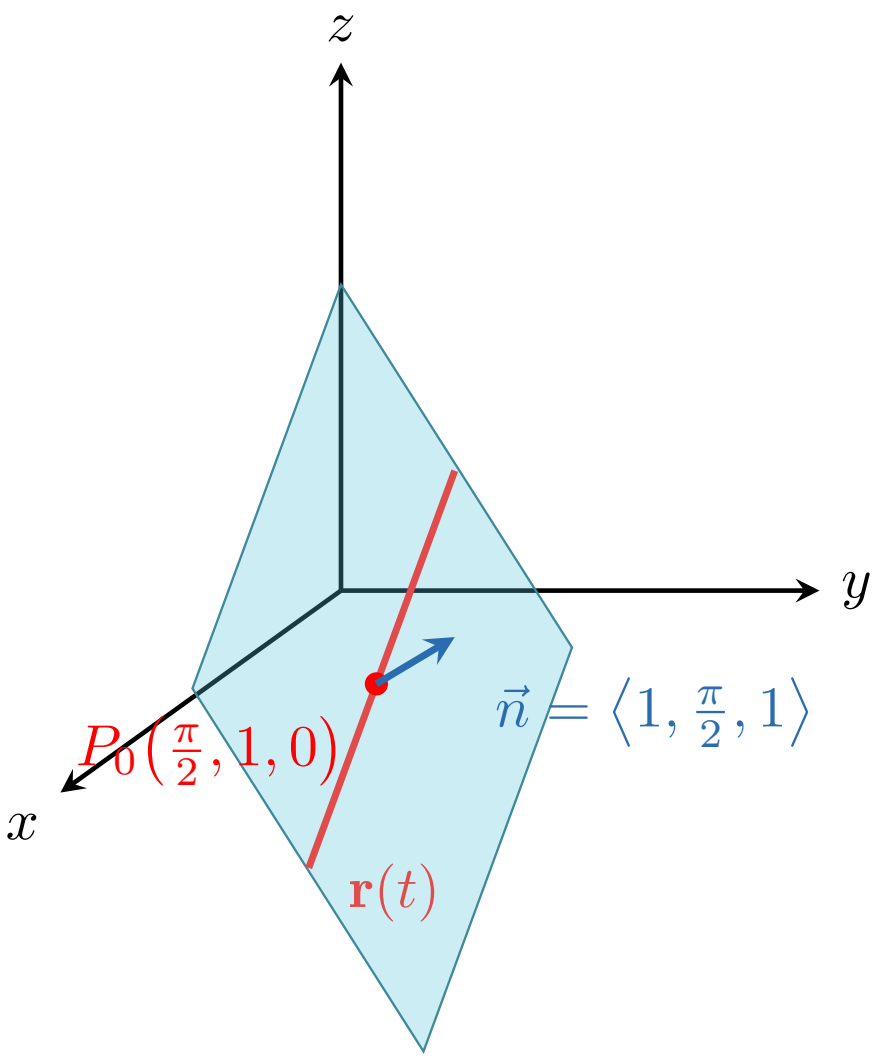


Figure 5.14.1 The tangent plane $x + \frac{\pi}{2}y + z = \pi$ to the graph of $f(x, y) = \cos(xy)$ at $P_0\left(\frac{\pi}{2}, 1, 0\right)$, its normal vector $\vec{n} = \langle 1, \frac{\pi}{2}, 1\rangle$, and the line $\mathbf{r}(t) = \langle t + \frac{\pi}{2}, 1, -t\rangle$ contained in the plane.

3. Find the equation of the tangent plane at the point $P_0(2, 0, 2)$ on the surface $2z - x^2 = 0$.

Solution. Let $w(x, y, z) = 2z - x^2 = 0$. Then

$$\nabla w = (-2x)\,\mathbf{i} + 2\,\mathbf{k},$$

and the normal to the tangent plane is

$$\vec{n} = \nabla w\big|_{P_0} = -4\,\mathbf{i} + 2\,\mathbf{k}.$$

The constant is

$$D = \vec{n} \cdot \vec{OP_0} = \langle -4, 0, 2\rangle \cdot \langle 2, 0, 2\rangle = -8 + 4 = -4,$$

so from $Ax + By + Cz = D$, the tangent plane at $P_0(2, 0, 2)$ is

$$-4x + 2z = -4 \qquad \text{or} \qquad 2x - z = 2.$$

4. **True or False.** Any tangent plane to a graph must meet that graph at exactly one point.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

Consider the plane $z = c$. The tangent plane to this graph at any point is the same as $z = c$, so it meets the graph at infinitely many points.

5. Let S be a sphere centered at the origin in $\mathbb{R}^3$, and consider any point P on S. Show that the vector $\vec{OP}$ is orthogonal to the tangent plane to S at P.

Solution. Let R be the radius of the sphere, and write $P = (x_0, y_0, z_0)$. Assume for the moment that P lies on the top half of the sphere (i.e. $z_0 > 0$). Note that the top half of the sphere is the same as the graph of the function $f(x, y) = \sqrt{R^2 - x^2 - y^2}$. We can compute

$$\frac{\partial f}{\partial x}(x_0, y_0) = -\frac{x_0}{\sqrt{R^2 - x_0^2 - y_0^2}},$$
$$\frac{\partial f}{\partial y}(x_0, y_0) = -\frac{y_0}{\sqrt{R^2 - x_0^2 - y_0^2}},$$

so that the tangent plane to the graph at P (which is the same as the tangent plane to S at P) is given by

$$z - z_0 = -\frac{x_0}{\sqrt{R^2 - x_0^2 - y_0^2}}(x - x_0) - \frac{y_0}{\sqrt{R^2 - x_0^2 - y_0^2}}(y - y_0).$$

We can rewrite this as

$$x_0(x - x_0) + y_0(y - y_0) + z_0(z - z_0) = 0,$$

where we use the fact that $z_0 = \sqrt{R^2 - x_0^2 - y_0^2}$ to simplify things. From this we see that $\langle x_0, y_0, z_0\rangle$ is a normal vector to the tangent plane. But $\langle x_0, y_0, z_0\rangle = \vec{OP}$, so this is exactly what we needed to show. See Figure 5.14.2.

A similar argument (with $-\sqrt{R^2 - x^2 - y^2}$ in place of $\sqrt{R^2 - x^2 - y^2}$) works when P is on the bottom half of the sphere (i.e. $z_0 < 0$). When $z_0 = 0$, we have to use a "sideways graph" of some function like $f(y, z) = \sqrt{R^2 - y^2 - z^2}$, but other than that, pretty much everything is the same.

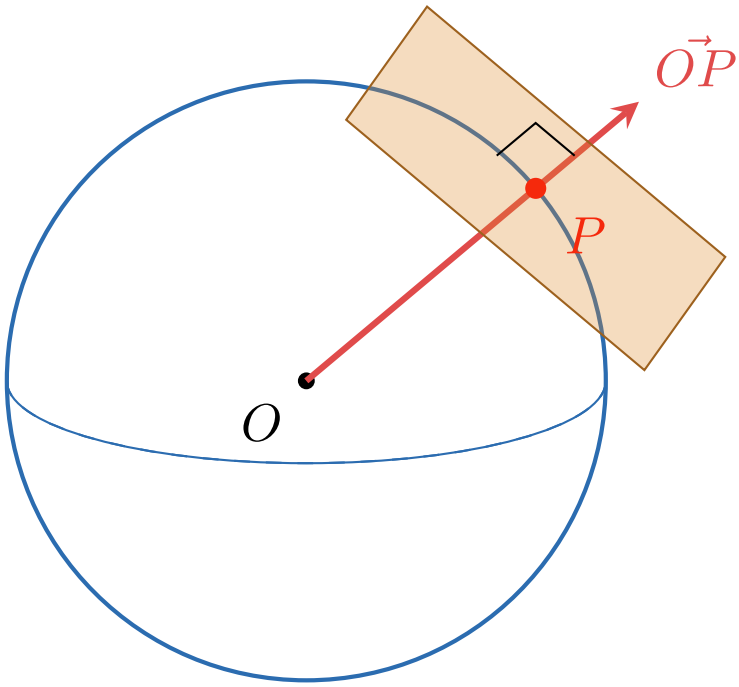


Figure 5.14.2 The vector $\vec{OP} = \langle x_0, y_0, z_0\rangle$ from the center of the sphere to the point P is a normal vector to the tangent plane to S at P.

6. Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$.

Solution. Write the sphere as the level surface $F(x, y, z) = x^2 + y^2 + z^2 = 14$. The gradient is normal to the level surface:

$$\nabla F = \langle 2x, 2y, 2z\rangle, \qquad \vec{n} = \nabla F\big|_{(1,2,3)} = \langle 2, 4, 6\rangle.$$

Hence the tangent plane is

$$2(x - 1) + 4(y - 2) + 6(z - 3) = 0,$$

which simplifies to $x + 2y + 3z = 14$. (Notice that the normal vector is parallel to the position vector $\langle 1, 2, 3\rangle$,

exactly as we proved for spheres centered at the origin.)

7. The tangent plane to $z = f(x, y)$ at the point $(1, 2, 9)$ is $z = 3x + 5y - 4$. Find $\nabla f(1, 2)$.

Solution. The tangent plane to the graph of f at (a, b) is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b),$$

so once it is expanded the coefficient of x is $f_x(a, b)$ and the coefficient of y is $f_y(a, b)$. Comparing with $z = 3x + 5y - 4$ gives

$$f_x(1, 2) = 3, \qquad f_y(1, 2) = 5,$$

and therefore

$$\nabla f(1, 2) = 3\,\mathbf{i} + 5\,\mathbf{j}.$$

(As a check, the plane passes through $(1, 2, 9)$: $3(1) + 5(2) - 4 = 9$ ✓.)

8. Let $f(x, y) = 2 + \ln(1 + 3x - 4y)$. Use a linearization to approximate $f(0.1, 0.2)$.

Solution. The point $(0.1, 0.2)$ is close to the origin, where the function is easy to work with, so we linearize at $(0, 0)$:

$$f(0, 0) = 2 + \ln(1) = 2, \qquad f_x = \frac{3}{1 + 3x - 4y}, \qquad f_y = \frac{-4}{1 + 3x - 4y},$$

so $f_x(0, 0) = 3$ and $f_y(0, 0) = -4$. The linearization is

$$L(x, y) = f(0, 0) + f_x(0, 0)x + f_y(0, 0)y = 2 + 3x - 4y,$$

and therefore

$$f(0.1, 0.2) \approx L(0.1, 0.2) = 2 + 0.3 - 0.8 = 1.5.$$

9. Let

$$g(x, y) = \frac{\sqrt{x}}{\sqrt[3]{y}},$$

and note that $g(36, 8) = 3$. Use an appropriate partial derivative to approximate $g(36, 6)$.

Solution. We move from $(36, 8)$ to $(36, 6)$. Only y changes, and it changes by

$$\Delta y = 6 - 8 = -2,$$

so the partial derivative we need is g_y. Writing $g(x, y) = x^{1/2}y^{-1/3}$,

$$g_y(x, y) = -\frac{1}{3}\,x^{1/2}y^{-4/3} = -\frac{\sqrt{x}}{3\,y^{4/3}}.$$

At $(36, 8)$ we have $\sqrt{36} = 6$ and $8^{4/3} = \left(8^{1/3}\right)^4 = 2^4 = 16$, so

$$g_y(36, 8) = -\frac{6}{3 \cdot 16} = -\frac{6}{48} = -\frac{1}{8}.$$

The linear approximation is

$$\begin{aligned} g(36, 6) &\approx g(36, 8) + g_y(36, 8)\,\Delta y \\ &= 3 + \left(-\frac{1}{8}\right)(-2) \\ &= 3 + \frac{1}{4} = \frac{13}{4} = 3.25. \end{aligned}$$

As a check, the exact value is $g(36, 6) = 6/\sqrt[3]{6} \approx 3.302$, so the approximation is good to about 1.6%.

5.15 Review Problems #3

These are problems from the Chapter 4 topics that were not assigned in Assignments 7, 8 and 9, collected here for review. They are not assigned, but they are fair game on an exam. If you would like to review the material first, see Functions of Several Variables, Limits and Continuity in Higher Dimensions, Partial Derivatives, The Chain Rule, Directional Derivatives and Gradient Vectors and Tangent Planes and Differentials.

1. Consider the function

$$h(x, y, z) = z + \sqrt{x^2 + y^2} + 1.$$

Sketch the level surface of h that passes through the point $(3, 4, -5)$. Clearly label the value of h on that surface, and label the coordinate axes.

Solution. The level surfaces of h are the surfaces $h(x, y, z) = c$. To find which one passes through $(3, 4, -5)$ we simply evaluate:

$$\begin{aligned} h(3, 4, -5) &= -5 + \sqrt{3^2 + 4^2} + 1 \\ &= -5 + \sqrt{25} + 1 \\ &= -5 + 5 + 1 = 1. \end{aligned}$$

So the level surface we want is $h(x, y, z) = 1$.

Now we simplify the equation of that level surface:

$$\begin{aligned} z + \sqrt{x^2 + y^2} + 1 &= 1 \\ z + \sqrt{x^2 + y^2} &= 0 \\ z &= -\sqrt{x^2 + y^2}. \end{aligned}$$

This is a cone. Indeed, squaring gives $z^2 = x^2 + y^2$, which is the full double cone; but the equation $z = -\sqrt{x^2 + y^2}$ forces $z \le 0$, so we keep only the *lower* nappe. The vertex is at the origin, and the horizontal trace at height $z = -k$ (with $k > 0$) is the circle $x^2 + y^2 = k^2$ of radius k. In other words the cone opens downwards, and the slope of every straight line drawn on it from the vertex is -1. The point $(3, 4, -5)$ lies on it, since $-\sqrt{3^2 + 4^2} = -5$. See Figure 5.15.1.

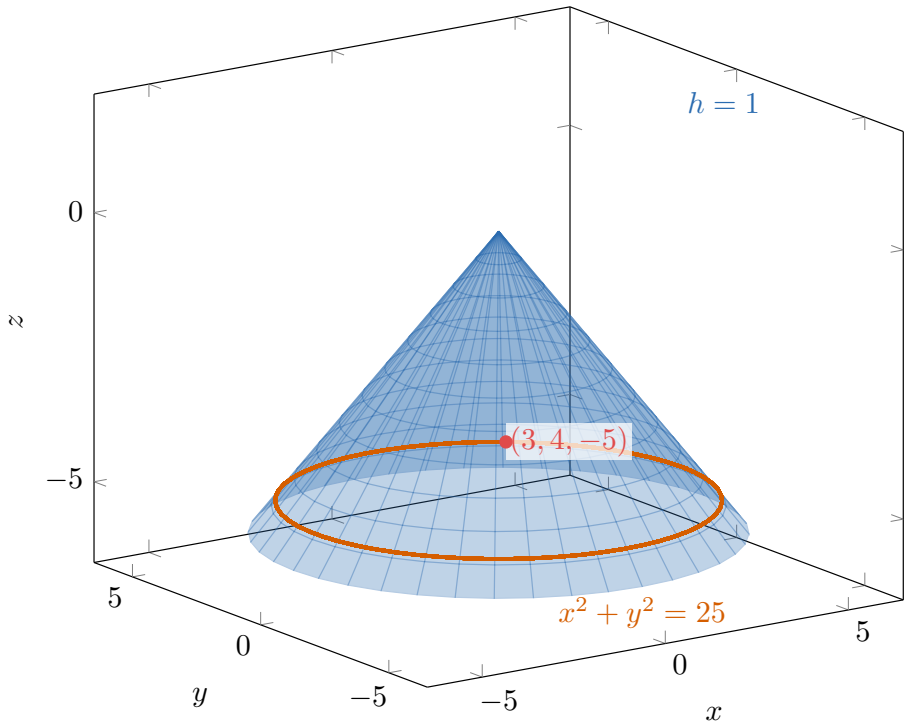


Figure 5.15.1 The level surface $h(x, y, z) = 1$, that is the downward cone $z = -\sqrt{x^2 + y^2}$. It passes through $(3, 4, -5)$, which lies on the circle $x^2 + y^2 = 25$ in the plane $z = -5$.

2. Sketch three level curves of

$$h(x,y) = \frac{y^2}{x},$$

including the one that passes through the point $(-1,2)$. Clearly label the value of h on each curve.

Solution. First we find the level through $(-1,2)$:

$$h(-1,2) = \frac{2^2}{-1} = -4.$$

So one of the three curves must be $h(x,y) = -4$.

In general the level curve $h(x,y) = c$ is

$$\frac{y^2}{x} = c \quad \Longleftrightarrow \quad y^2 = cx \quad \Longleftrightarrow \quad x = \frac{y^2}{c} \qquad (c \neq 0).$$

Each of these is a *parabola opening along the x-axis*, with vertex at the origin. If $c > 0$ it opens to the right, and if $c < 0$ it opens to the left. Note that the domain of h excludes $x = 0$, so the vertex itself, the origin, must be removed from every level curve.

Let us take the three levels $c = -4$, $c = 1$ and $c = 4$:

$h = -4$:	$x = -\dfrac{y^2}{4}$	(opens left),
$h = 1$:	$x = y^2$	(opens right, narrow),
$h = 4$:	$x = \dfrac{y^2}{4}$	(opens right, wide).

The larger $|c|$ is, the wider the parabola. These are drawn in Figure 5.15.2.

One more remark: the level $c = 0$ is special. There $y^2 = 0$, so the level curve $h = 0$ is not a parabola at all but the x-axis with the origin removed.

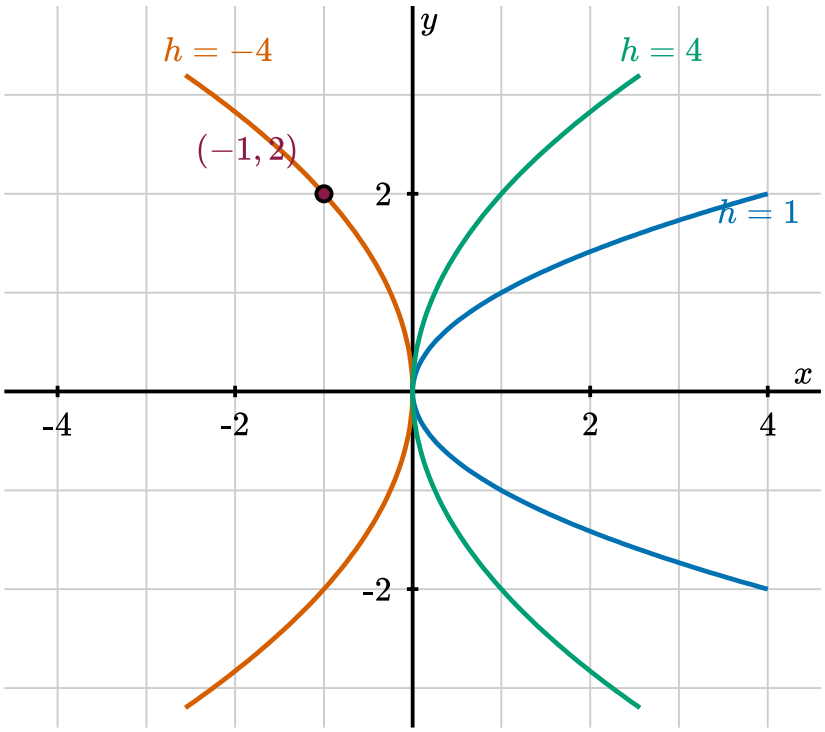


Figure 5.15.2 Three level curves of $h(x,y) = y^2/x$. The curve through $(-1,2)$ is the level $h = -4$, which opens to the left. The origin is excluded from every level curve, since h is undefined when $x = 0$.

3. **True or False.** To show that the limit at a point (a, b) exists, it suffices to find two paths to the point (a, b) where the limits of $f(x, y)$ agree.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

We would have to show the limit agrees on *every* path, but this is not feasible.

4. Show that the limit does not exist:

$$\lim_{(x,y)\to(1,0)} \frac{x+y^2}{(x-1)^3+y^4}.$$

Solution. Approaching the limit along the x-axis, i.e. $y = 0$, we need to consider

$$\lim_{x\to 1} \frac{x}{(x-1)^3},$$

which does not exist since the numerator has limit 1 but the denominator has limit 0.

5. Evaluate the following limit or show that the limit does not exist:

$$\lim_{(x,y)\to(0,0)} \frac{x^2+y^4}{x^2+y^2}.$$

Solution. Along the line $y = x$:

$$\lim_{(x,y)\to(0,0)} \frac{x^2+y^4}{x^2+y^2} = \lim_{x\to 0} \frac{x^2+x^4}{2x^2} = \lim_{x\to 0} \frac{1}{2}\left(1+x^2\right) = \frac{1}{2}.$$

Along the line $y = 0$:

$$\lim_{(x,y)\to(0,0)} \frac{x^2+y^4}{x^2+y^2} = \lim_{x\to 0} \frac{x^2}{x^2} = 1.$$

Since $1 \neq \frac{1}{2}$, the limit does not exist.

6. Show that the following limit does not exist:

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^4+y^2}.$$

Solution. Along $x = 0$, we have

$$f(0, y) = \frac{0}{0+y^2} = 0,$$

hence $f(x, y)$ approaches zero as $(x, y) \to (0, 0)$ along the line $x = 0$. Along $y = x^2$, we have

$$f\left(x, x^2\right) = \frac{2x^4}{x^4+x^4} = \frac{x^4}{x^4} = 1,$$

and hence $f(x, y) \to 1$ as $(x, y) \to (0, 0)$ along $y = x^2$. Since the limits computed along two different paths are not equal, we can conclude that $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist.

7. **True or False.** If two functions $f(x, y)$ and $g(x, y)$ are both continuous at the point P_0, then the quotient $\dfrac{f}{g}$ is also continuous at P_0.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

Counterexample:

$$f(x, y) = x + y \quad \text{and} \quad g(x, y) = x^2 - y^2$$

are both continuous at $(0, 0)$. However,

$$\frac{f}{g} = \frac{x+y}{x^2-y^2} = \frac{1}{x-y}$$

is *not* continuous at $(0,0)$, since for example along $y=0$,

$$\lim_{(x,y)\to(0,0)} \frac{1}{x-y} = \lim_{x\to 0^{\pm}} \frac{1}{x} = \pm\infty.$$

8. A metal plate occupies the square $0 \le x \le 5$, $0 \le y \le 5$, and is heated so that the temperature at the point (x,y) is $T(x,y)$, measured in degrees Fahrenheit. Distances are measured in feet. The table below records some measured values of T.

Table 5.15.3 Measured temperatures $T(x,y)$ in °F

	$y=1$	$y=1.5$	$y=2$
$x=1$	70	73	79
$x=1.5$	65	69	75
$x=2$	58	60	65

(a) Approximate $T_x(1,1)$ and interpret your answer, with units, in the context of the problem.

(b) Approximate $T_{xy}(1,1)$, and give its units.

(c) An ant walks across the plate along the path

$$x = 1 - \frac{\ln(2t-3)}{4}, \qquad y = \frac{2^t}{4}, \qquad 1.5 < t < 4.$$

Approximate the rate at which the temperature the ant feels is changing at the moment it is at the point $(x,y)=(1,1)$, and give the units.

Solution. **(a)** Holding $y=1$ and stepping forward in x,

$$T_x(1,1) \approx \frac{T(1.5,1)-T(1,1)}{0.5} = \frac{65-70}{0.5} = -10 \quad \text{°F/ft}.$$

So at $(1,1)$, moving in the positive x-direction with y held fixed, the temperature *drops* at a rate of about 10 °F per foot.

(b) We need T_x at a second value of y:

$$T_x(1,1.5) \approx \frac{T(1.5,1.5)-T(1,1.5)}{0.5} = \frac{69-73}{0.5} = -8.$$

Then

$$T_{xy}(1,1) \approx \frac{T_x(1,1.5)-T_x(1,1)}{0.5} = \frac{-8-(-10)}{0.5} = 4 \quad \text{°F/ft}^2.$$

(c) As in Worksheet Exercise 5.12.13, the ant is at $(1,1)$ when $\ln(2t-3)=0$, that is at $t=2$, and then $y = 2^2/4 = 1$ as required. Differentiating the path,

$$\frac{dx}{dt} = -\frac{1}{4}\cdot\frac{2}{2t-3} = -\frac{1}{2(2t-3)}, \qquad \left.\frac{dx}{dt}\right|_{t=2} = -\frac{1}{2},$$

$$\frac{dy}{dt} = \frac{2^t \ln 2}{4}, \qquad \left.\frac{dy}{dt}\right|_{t=2} = \ln 2.$$

From the table,

$$T_y(1,1) \approx \frac{T(1,1.5)-T(1,1)}{0.5} = \frac{73-70}{0.5} = 6.$$

By the chain rule,

$$\begin{aligned}\frac{dT}{dt} &= T_x\frac{dx}{dt} + T_y\frac{dy}{dt} \\ &\approx (-10)\left(-\frac{1}{2}\right) + (6)(\ln 2) \\ &= 5 + 6\ln 2 \approx 9.16,\end{aligned}$$

in degrees Fahrenheit per unit of t. Here the ant is *warming up*: it moves in the $-x$ direction, where the plate is hotter, and in the $+y$ direction, where the plate is also hotter, so both contributions are positive. Compare this with Worksheet Exercise 5.12.13, where the same computation gave a negative rate.

9. Let $f(x,z) = x^2 z + e^{xz+z^2}$ and $x = r\cos(\theta)$, $z = r\sin(\theta)$. Use the chain rule to compute $\dfrac{\partial f}{\partial \theta}$ at $(r,\theta) = \left(3, \frac{\pi}{2}\right)$.

Solution. The chain rule gives

$$\begin{aligned}\frac{\partial f}{\partial \theta} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial \theta} \\ &= \left[2xz + z\,e^{xz+z^2}\right](-r\sin\theta) \\ &\quad + \left[x^2 + (x+2z)e^{xz+z^2}\right](r\cos\theta).\end{aligned}$$

At $(r,\theta) = \left(3, \frac{\pi}{2}\right)$ we have $x = 3\cos\frac{\pi}{2} = 0$ and $z = 3\sin\frac{\pi}{2} = 3$, so $(x,z) = (0,3)$ and $xz + z^2 = 0 + 9 = 9$. Also $-r\sin\theta = -3$ and $r\cos\theta = 0$. Hence

$$\frac{\partial f}{\partial \theta} = \left[0 + 3e^9\right](-3) + \left[0 + 6e^9\right](0) = -9e^9.$$

10. Find all points at which the direction of fastest change of the function $f(x,y) = x^2 + y^2 - 2x - 4y$ is $\langle 1,1\rangle$.

Solution. The direction of fastest change is $\nabla f(x,y) = \langle 2x-2,\, 2y-4\rangle$, so we need to find all points (x,y) such that $\nabla f(x,y) = k\langle 1,1\rangle$. Then

$$2x - 2 = k \quad \text{and} \quad 2y - 4 = k,$$

so

$$2x - 2 = 2y - 4 \;\Rightarrow\; y = x + 1.$$

Hence, the direction of fastest change is $\langle 1,1\rangle$ at all points on the line $y = x + 1$. See Figure 5.15.4.

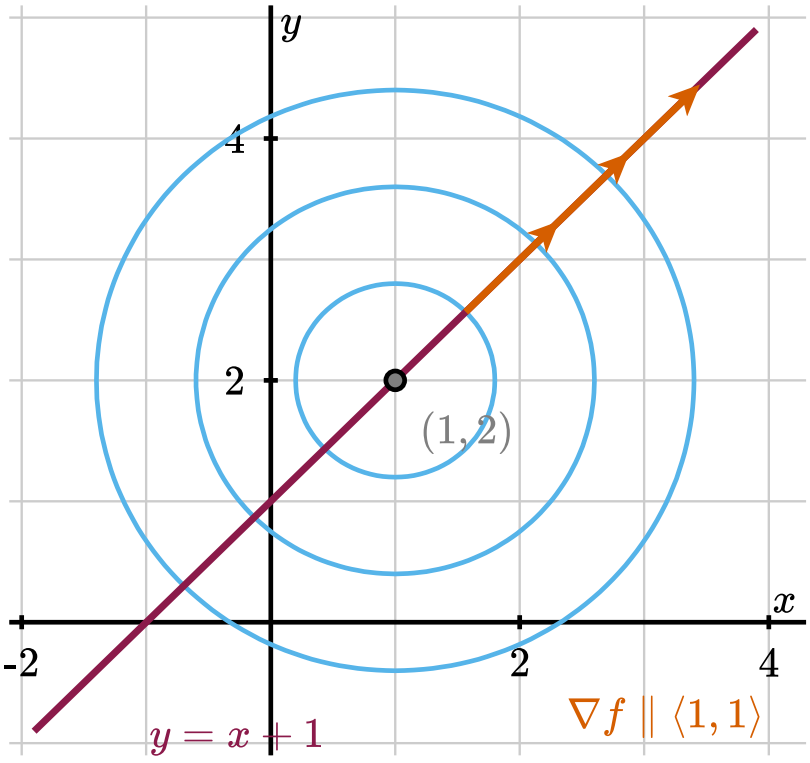


Figure 5.15.4 Level curves of $f(x,y) = x^2 + y^2 - 2x - 4y$, which are circles centered at $(1,2)$. Along the line $y = x + 1$ the gradient ∇f points in the direction $\langle 1,1\rangle$.

11. **True or False.** $D_{\hat{\imath}+\hat{\jmath}} f(x,y) = f_x + f_y$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. True.

Solution. True.

Since $\hat{\imath}+\hat{\jmath}=\langle 1,1\rangle$,

$$D_{\hat{\imath}+\hat{\jmath}}\,f(x,y)=\nabla f(x,y)\cdot\langle 1,1\rangle=f_x+f_y.$$

12. A metal plate is placed on the xy-plane in such a way that the temperature T at any point (x,y) is given by $T=e^x(\sin x+\sin y)$ °C.

A. What is the rate of change in temperature at $(0,0)$ in the direction of $3\,\hat{\imath}-4\,\hat{\jmath}$?

B. At $(0,0)$, in what direction is the temperature increasing most rapidly?

C. At $(0,0)$, in what direction is the temperature decreasing most rapidly?

Solution.

A. We first compute the gradient:

$$\begin{aligned} T_x &= e^x(\sin x+\sin y)+e^x\cos x \\ &= e^x(\sin x+\cos x+\sin y), \qquad T_x(0,0)=1, \\ T_y &= e^x\cos y, \qquad T_y(0,0)=1, \end{aligned}$$

so $\nabla T\big|_{(0,0)}=\langle 1,1\rangle$. The direction vector is $\overrightarrow{u}=\langle 3,-4\rangle$ with $\|\overrightarrow{u}\|=\sqrt{9+16}=5$, so the unit vector is $\hat{u}=\dfrac{\overrightarrow{u}}{\|\overrightarrow{u}\|}=\left\langle\dfrac{3}{5},-\dfrac{4}{5}\right\rangle$. Hence

$$(D_{\hat{u}}T)_{(0,0)}=\nabla T\big|_{(0,0)}\cdot\hat{u}=\langle 1,1\rangle\cdot\left\langle\frac{3}{5},-\frac{4}{5}\right\rangle=-\frac{1}{5}.$$

B. The temperature increases most rapidly in the direction of the gradient:

$$\hat{u}=\frac{\nabla T\big|_{(0,0)}}{\left\|\nabla T\big|_{(0,0)}\right\|}=\left\langle\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right\rangle.$$

C. The temperature decreases most rapidly in the direction opposite to the gradient:

$$\hat{u}=-\frac{\nabla T\big|_{(0,0)}}{\left\|\nabla T\big|_{(0,0)}\right\|}=\left\langle-\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right\rangle.$$

13. Calculate the maximum rate of change of $f(x,y)=\ln(x^2+y^2)$ at the point $(-4,3)$ and determine the direction in which it occurs.

Solution. The gradient is

$$\nabla f=\left(\frac{2x}{x^2+y^2}\right)\hat{\imath}+\left(\frac{2y}{x^2+y^2}\right)\hat{\jmath},$$

so at $(-4,3)$,

$$\begin{aligned} \nabla f(-4,3) &= \left(-\frac{8}{25}\right)\hat{\imath}+\left(\frac{6}{25}\right)\hat{\jmath}, \\ \|\nabla f(-4,3)\| &= \sqrt{\frac{100}{(25)^2}}=\frac{10}{25}=\frac{2}{5}. \end{aligned}$$

The direction of maximum increase is

$$\hat{u}=\frac{\nabla f(-4,3)}{\|\nabla f(-4,3)\|}=\left(-\frac{4}{5}\right)\hat{\imath}+\left(\frac{3}{5}\right)\hat{\jmath},$$

and the maximum rate of increase is

$$(D_{\hat{u}}f)_{P_0}=\|\nabla f(-4,3)\|=\frac{2}{5}.$$

14. Consider the function $f(x,y) = x^2 + xy + y^2$.

A. Find the directions in which $f(x,y)$ increases and decreases most rapidly at $P_0(-1,1)$.

B. Find the derivatives of $f(x,y)$ in these directions at $P_0(-1,1)$.

Solution.

A. The gradient is

$$\nabla f = (2x+y)\,\hat{\imath} + (x+2y)\,\hat{\jmath},$$
$$(\nabla f)_{(-1,1)} = \langle -1, 1\rangle, \qquad \left\|(\nabla f)_{(-1,1)}\right\| = \sqrt{2}.$$

The function increases most rapidly in the direction of the gradient,

$$\hat{u}_1 = \frac{(\nabla f)_{(-1,1)}}{\left\|(\nabla f)_{(-1,1)}\right\|} = \left\langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle,$$

and decreases most rapidly in the opposite direction,

$$\hat{u}_2 = -\frac{(\nabla f)_{(-1,1)}}{\left\|(\nabla f)_{(-1,1)}\right\|} = \left\langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right\rangle.$$

B. The directional derivatives are

$$(D_{\hat{u}_1} f)_{P_0} = (\nabla f)_{(-1,1)} \cdot \hat{u}_1 = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2} = \left\|(\nabla f)_{(-1,1)}\right\|,$$
$$(D_{\hat{u}_2} f)_{P_0} = (\nabla f)_{(-1,1)} \cdot \hat{u}_2 = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\sqrt{2} = -\left\|(\nabla f)_{(-1,1)}\right\|.$$

As expected, the largest possible rate of change is $\|\nabla f\|$ and the smallest is $-\|\nabla f\|$.

15. In the xy-plane, the temperature at the point (x,y) is given by $T(x,y) = 3x^2 + 8xy + y^2$. You're at $(1,0)$ where the temperature is 3°. Brrrr!

A. In what direction should you go, from $(1,0)$, in order to warm up the fastest?

B. What is the directional derivative in that direction at the point $(1,0)$?

C. If you walk 0.2 units in that direction, by about how many degrees will you warm up?

Solution.

A. A function increases fastest in the direction of its gradient:

$$\nabla T = \langle 6x + 8y,\ 8x + 2y\rangle, \qquad \nabla T\big|_{(1,0)} = \langle 6, 8\rangle = 6\,\hat{\imath} + 8\,\hat{\jmath}.$$

So head in the direction $6\,\hat{\imath} + 8\,\hat{\jmath}$ (equivalently, the unit vector $\left\langle \frac{3}{5}, \frac{4}{5}\right\rangle$).

B. In the direction of the gradient, the directional derivative equals the length of the gradient:

$$(D_{\hat{u}} T)_{(1,0)} = \left\|\nabla T\big|_{(1,0)}\right\| = \sqrt{36+64} = 10.$$

C. The directional derivative is the rate of change of temperature per unit of distance travelled, so by the linear approximation

$$\Delta T \approx (D_{\hat{u}} T) \cdot s = 10 \times 0.2 = 2°.$$

16. A sheet of metal of varying density occupies the xy-plane. At the point (x,y), the density is $100 + 0.1xy + 0.1y^2$. We move away from the point $(10,10)$ in the direction $\frac{3}{5}\hat{\imath} + \frac{4}{5}\hat{\jmath}$. Use the directional derivative to estimate about how far we'll have to go for the density to increase by 6.

Solution. Let $\rho(x,y) = 100 + 0.1xy + 0.1y^2$. Then

$$\nabla\rho = \langle 0.1y,\ 0.1x + 0.2y\rangle, \qquad \nabla\rho\big|_{(10,10)} = \langle 1,\ 1+2\rangle = \langle 1, 3\rangle.$$

The given direction is already a unit vector, since $\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = 1$, so

$$D_{\hat{u}}\rho(10,10) = \langle 1,3\rangle \cdot \left\langle \frac{3}{5}, \frac{4}{5}\right\rangle = \frac{3}{5} + \frac{12}{5} = 3.$$

So the density increases at about 3 units of density per unit of distance. To gain 6 units of density we therefore need to travel approximately

$$s \approx \frac{6}{3} = 2 \text{ units}.$$

The problem that closes this set repeats the three questions of Example 4.5.14 in a different physical setting: find the direction of most rapid decrease, the rate along a prescribed direction, and the directions of no change.

17. The Fall Line on a Ski Slope. A mountainside has elevation, in feet,

$$h(x,y) = 3000 - 0.0004x^2 - 0.001y^2,$$

with x and y in feet. A skier stands at $P_0(200, 100)$. Her gravitational potential energy is $U = mgh$, so the downhill force she feels is a positive multiple of $-\nabla h$.

(a) Find the **fall line**, the unit vector of steepest descent, and the grade of the slope there, expressed as a percent (feet dropped per hundred feet travelled).

(b) A traverse is a route on which she neither climbs nor descends. Find the two unit vectors that accomplish this.

(c) She wants a gentler line of exactly 12% grade. Using $D_{\hat{u}}h = \|\nabla h\|\cos\theta$, find the angle θ between her heading and the fall line.

Answer.

(a) $\nabla h = -0.0008x\,\hat{\imath} - 0.002y\,\hat{\jmath}$, so $(\nabla h)_{(200,100)} = -0.16\,\hat{\imath} - 0.20\,\hat{\jmath}$. Steepest descent is along $\hat{u} = \frac{1}{\sqrt{41}}(4\,\hat{\imath} + 5\,\hat{\jmath})$, and $\|\nabla h\| = \sqrt{0.0656} \approx 0.256$, a grade of about 25.6%.

(b) $\hat{u} = \pm\frac{1}{\sqrt{41}}(5\,\hat{\imath} - 4\,\hat{\jmath})$, tangent to the contour line of constant elevation through P_0.

(c) $\cos\theta = 0.12/0.256 \approx 0.469$, so $\theta \approx 62^\circ$ off the fall line, on either side.

Two interactive figures accompany the online version of these notes: one shows the grade as the angle α of the tangent line cut by a swinging vertical plane through P_0, and the other compares the fall line, the level traverse, and the 12% line on the mountainside itself.

18. You measure the magnitude of a force F to be 6, but there could be an error of at most ± 0.3. You measure a mass M to be 3 with an error of at most ± 0.1. You are trying to use these two to approximate the magnitude of the acceleration A. We have $A = \dfrac{F}{M}$. Use the principle of the differential to find an upper bound on the absolute value of the error when assuming $A = 2$.

Solution. Treat A as a function of the two measured quantities, $A(F,M) = \dfrac{F}{M}$. Its differential is

$$dA = \frac{\partial A}{\partial F}\,dF + \frac{\partial A}{\partial M}\,dM = \frac{1}{M}\,dF - \frac{F}{M^2}\,dM.$$

At the measured values $F = 6$ and $M = 3$ (where indeed $A = 2$), the coefficients are $\frac{1}{3}$ and $-\frac{6}{9} = -\frac{2}{3}$.

Taking absolute values and using the triangle inequality with the worst-case errors $|dF| \le 0.3$ and $|dM| \le 0.1$,

$$|dA| \le \frac{1}{3}|dF| + \frac{2}{3}|dM| \le \frac{1}{3}(0.3) + \frac{2}{3}(0.1) = 0.1 + \frac{1}{15} = \frac{1}{6}.$$

So the error in the value $A = 2$ is at most about $\frac{1}{6} \approx 0.167$.

19. Let $g(x,y) = \sin(xy)$, and note that $g\left(\pi, \frac{1}{3}\right) = \frac{\sqrt{3}}{2}$. Use an appropriate partial derivative to approximate $g\left(\pi, \frac{1}{3} + \frac{1}{20}\right)$.

Solution. Only y changes, by $\Delta y = \frac{1}{20}$, so we need g_y. Treating x as a constant and using the chain rule,

$$g_y(x,y) = x\cos(xy).$$

At the point $\left(\pi, \frac{1}{3}\right)$ we have $xy = \frac{\pi}{3}$ and $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$, so

$$g_y\left(\pi, \tfrac{1}{3}\right) = \pi \cdot \frac{1}{2} = \frac{\pi}{2}.$$

Therefore

$$\begin{aligned} g\left(\pi, \tfrac{1}{3} + \tfrac{1}{20}\right) &\approx g\left(\pi, \tfrac{1}{3}\right) + g_y\left(\pi, \tfrac{1}{3}\right) \cdot \frac{1}{20} \\ &= \frac{\sqrt{3}}{2} + \frac{\pi}{2} \cdot \frac{1}{20} \\ &= \frac{\sqrt{3}}{2} + \frac{\pi}{40} \approx 0.944. \end{aligned}$$

The exact value is $\sin\left(\frac{23\pi}{60}\right) \approx 0.933$, so again the approximation is quite good.

20. **True or False.** If every directional derivative $D_{\mathbf{u}}f(P_0)$ exists at a point P_0, then f must be continuous at P_0. Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.

False, and the failure is dramatic. Let

$$f(x,y) = \frac{x^2 y}{x^4 + y^2} \quad \text{for } (x,y) \neq (0,0), \qquad f(0,0) = 0.$$

For a unit vector $\mathbf{u} = \langle a, b\rangle$ with $b \neq 0$,

$$D_{\mathbf{u}}f(0,0) = \lim_{t\to 0} \frac{f(ta,tb)}{t} = \lim_{t\to 0} \frac{a^2 b}{t^2 a^4 + b^2} = \frac{a^2}{b},$$

and $D_{\mathbf{u}}f(0,0) = 0$ when $b = 0$. So *every* directional derivative exists. Yet along the parabola $y = mx^2$,

$$f(x, mx^2) = \frac{m}{1+m^2},$$

which depends on m — different parabolas approach different values, so $\lim_{(x,y)\to(0,0)} f$ does not exist and f is not continuous at the origin. Existence of directional derivatives is a statement about behavior along *straight lines* only; continuity requires control along *every* approach.

21. **Find the Error.** Below is a student's computation of the derivative of $f(x,y) = x^2 y$ at the point $(1,2)$ in the direction of $\vec{v} = \langle 3, 4\rangle$. Exactly one line contains the error. Click on that line.

$\nabla f = \langle 2xy,\ x^2\rangle$
$\nabla f(1,2) = \langle 4,\ 1\rangle$
$D_{\vec{v}}f(1,2) = \nabla f(1,2) \cdot \vec{v} = (4)(3) + (1)(4) = 16$
So the directional derivative equals 16.

Answer. Correct: $D_{\vec{v}}f(1,2) = \nabla f(1,2) \cdot \vec{v} = (4)(3) + (1)(4) = 16$. Incorrect: ~~$\nabla f = \langle 2xy,\ x^2\rangle$~~; ~~$\nabla f(1,2) = \langle 4,\ 1\rangle$~~; ~~So the directional derivative equals 16.~~.

The error is in the third line. The formula $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ requires a *unit* vector. Since $|\vec{v}| = 5$, the correct direction is $\mathbf{u} = \left\langle \frac{3}{5}, \frac{4}{5}\right\rangle$, giving

$$D_{\mathbf{u}}f(1,2) = 4 \cdot \tfrac{3}{5} + 1 \cdot \tfrac{4}{5} = \frac{16}{5} = 3.2,$$

not 16. Lines 1, 2 and 4 are each correct given the line before them — line 4 faithfully reports the (wrong) number produced by line 3.

Solution. ~~$\nabla f = \langle 2xy,\ x^2\rangle$~~

~~$\nabla f(1,2) = \langle 4, 1 \rangle$~~
$D_{\vec{v}} f(1,2) = \nabla f(1,2) \cdot \vec{v} = (4)(3) + (1)(4) = 16$
~~So the directional derivative equals 16.~~

5.16 Assignment 10

These problems find absolute extrema on a closed region, classify critical points with the second derivative test, and use Lagrange multipliers to optimize a function subject to a constraint. If you would like to review the material first, see Extreme Values and Saddle Points and Lagrange Multipliers.

1. Find the absolute maximum and minimum values of

$$f(x,y) = x^2 - xy + y^2 - 3y$$

on the region bounded by the x- and y-axes and the line $x + y = 4$.

Solution. The gradient is $\nabla f = \langle 2x - y,\ -x + 2y - 3\rangle$, which vanishes only at $(1,2)$, a point in the interior of the region. There $f(1,2) = -3$. Since this is the only critical point on all of $\mathbb{R}^2$ and it is a minimum (by the second derivative test), it is a global minimum. We now examine the three boundary segments to look for possible maxima.

On the x-axis with $0 \le x \le 4$, we have $f(x,0) = x^2$, which is maximized at $f(4,0) = 16$.

On the y-axis with $0 \le y \le 4$, we have $f(0,y) = y^2 - 3y$, which is maximized at $f(0,4) = 4$.

Finally, on the line $x = 4 - y$ with $0 \le y \le 4$, we have $f(4-y,\ y) = 3y^2 - 15y + 16$, which is maximized at $f(4,0) = 16$.

Altogether, there is an absolute minimum $f(1,2) = -3$ and an absolute maximum $f(4,0) = 16$.

2. Find and classify all the critical points of $f(x,y) = xy\, e^{x^2+y^2}$.

Solution. The gradient is

$$\nabla f(x,y) = \left\langle \left(1 + 2x^2\right) y\, e^{x^2+y^2},\ x\left(1 + 2y^2\right) e^{x^2+y^2} \right\rangle.$$

Since the exponential factors, as well as $1 + 2x^2$ and $1 + 2y^2$, are always positive, we might as well ignore them when finding the zeros. So we are left with $\langle x, y\rangle = \langle 0, 0\rangle$, i.e. the only critical point is $(0,0)$. The second derivatives of f are

$$\begin{aligned} f_{xx} &= \left(6x + 4x^3\right) y\, e^{x^2+y^2} \\ f_{xy} &= \left(1 + 2x^2\right)\left(1 + 2y^2\right) e^{x^2+y^2} \\ f_{yy} &= x\left(6y + 4y^3\right) e^{x^2+y^2} \end{aligned}$$

Hence $D(0,0) = f_{xx} f_{yy} - f_{xy}^2 = -1$, so $(0,0)$ is a saddle point.

3. Find the maximum and minimum values of the function $f(x,y) = 2x^2 + 3y^2 - 4x - 5$ on the domain $x^2 + y^2 \le 121$.

Solution. *Critical points in the interior:*

$$f_x = 4x - 4 = 0 \ \Rightarrow\ x = 1, \qquad f_y = 6y = 0 \ \Rightarrow\ y = 0,$$

so $(1,0)$ is the only critical point, with $f(1,0) = -7$.

The boundary: on the circle $x^2 + y^2 = 121$ we have $y^2 = 121 - x^2$ with $-11 \le x \le 11$, so

$$f = 2x^2 + 3\left(121 - x^2\right) - 4x - 5 = 358 - 4x - x^2,$$

and

$$\frac{d}{dx}\left(358 - 4x - x^2\right) = -4 - 2x = 0 \ \Rightarrow\ x = -2,$$

giving $y^2 = 117$, i.e. $y = \pm\sqrt{117}$, with

$$f\left(-2, \pm\sqrt{117}\right) = 358 + 8 - 4 = 362.$$

At the endpoints $x = \pm 11$ (where $y = 0$), $f(11,0) = 193$ and $f(-11,0) = 281$.

Comparing all the values, the maximum value of f is 362, attained at $\left(-2, \sqrt{117}\right)$ and $\left(-2, -\sqrt{117}\right)$, and the minimum value is -7, attained at $(1,0)$. See Figure 5.16.1.

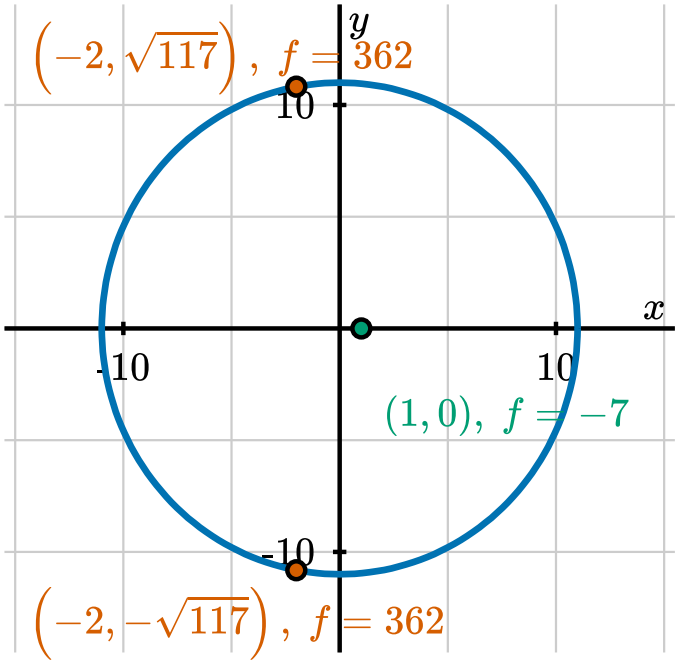


Figure 5.16.1 The disk $x^2 + y^2 \leq 121$, the interior minimum at $(1, 0)$, and the boundary maxima at $\left(-2, \pm\sqrt{117}\right)$.

4. The business manager for Acme Corporation plots a grid on a map of the region Acme serves and determines that the company's three most important customers are located at points $A(1, 5)$, $B(0, 0)$, and $C(8, 0)$, where units are in miles. At what point $W(x, y)$ should a warehouse be located in order to minimize *the sum of the squares of the distances* (NOT the square of the sum) from W to A, B, and C?

Solution. The sum of the squares of the distances is

$$\begin{aligned} f(x, y) &= (x-1)^2 + (y-5)^2 + x^2 + y^2 + (x-8)^2 + y^2 \\ &= 3x^2 - 18x + 3y^2 - 10y + 90. \end{aligned}$$

Setting the partial derivatives equal to zero,

$$\begin{aligned} f_x = 6x - 18 = 0 \ &\Rightarrow\ x = 3, \\ f_y = 6y - 10 = 0 \ &\Rightarrow\ y = \frac{5}{3}, \end{aligned}$$

so $\left(3, \frac{5}{3}\right)$ is the only critical point. The second derivatives are $f_{xx} = 6 > 0$, $f_{yy} = 6$, $f_{xy} = 0$, so $D = f_{xx} f_{yy} - f_{xy}^2 = 36 > 0$. Since $D > 0$ and $f_{xx} > 0$, the point $\left(3, \frac{5}{3}\right)$ is a local minimum.

Also, if we complete the squares we can see that this point is where the absolute minimum happens:

$$f(x, y) = 3\left(y - \frac{5}{3}\right)^2 + 3(x-3)^2 + \underbrace{90 - \frac{25}{3} - 27}_{>0},$$

where the first two terms are non-negative and vanish exactly at $\left(3, \frac{5}{3}\right)$. See Figure 5.16.2.

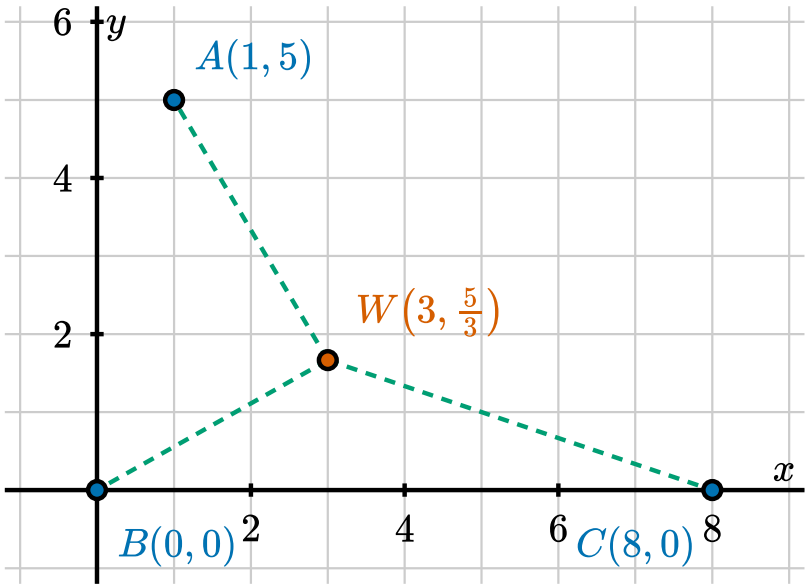


Figure 5.16.2 The three customers $A(1,5)$, $B(0,0)$, $C(8,0)$, and the warehouse location $W\left(3, \frac{5}{3}\right)$ that minimizes the sum of the squares of the distances.

5. Find all local maxima, minima and saddle points for $f(x,y) = \frac{1}{3}x^3 - x + xy^2$.

Solution. The first partials are

$$f_x = x^2 - 1 + y^2, \qquad f_y = 2xy.$$

Setting $f_y = 0$ forces $x = 0$ or $y = 0$.

- If $x = 0$: $f_x = -1 + y^2 = 0$ gives $y = \pm 1$, so the critical points are $(0,1)$ and $(0,-1)$.
- If $y = 0$: $f_x = x^2 - 1 = 0$ gives $x = \pm 1$, so the critical points are $(1,0)$ and $(-1,0)$.

The second partials are $f_{xx} = 2x$, $f_{yy} = 2x$, and $f_{xy} = 2y$, so the discriminant is

$$D = f_{xx}f_{yy} - f_{xy}^2 = 4x^2 - 4y^2.$$

Applying the second derivative test:

- $(0, \pm 1)$: $D = -4 < 0$, so both are **saddle points**.
- $(1,0)$: $D = 4 > 0$ and $f_{xx} = 2 > 0$, so this is a **local minimum**.
- $(-1,0)$: $D = 4 > 0$ and $f_{xx} = -2 < 0$, so this is a **local maximum**.

6. Tangent Planes Parallel to a Given Plane. Find the points on the surface

$$x^2 + y^2 + 3z^2 = 1$$

at which the tangent plane is parallel to the plane $-x - 5y - 2z = -3$.

Solution. Multiplying by -1, the given plane is $x + 5y + 2z = 3$, with normal vector $\mathbf{n} = \langle 1, 5, 2 \rangle$. The tangent plane to the ellipsoid is parallel to this plane exactly at the points of the ellipsoid that are closest to and farthest from it, so we minimize and maximize the distance to the plane subject to the constraint of lying on the ellipsoid.

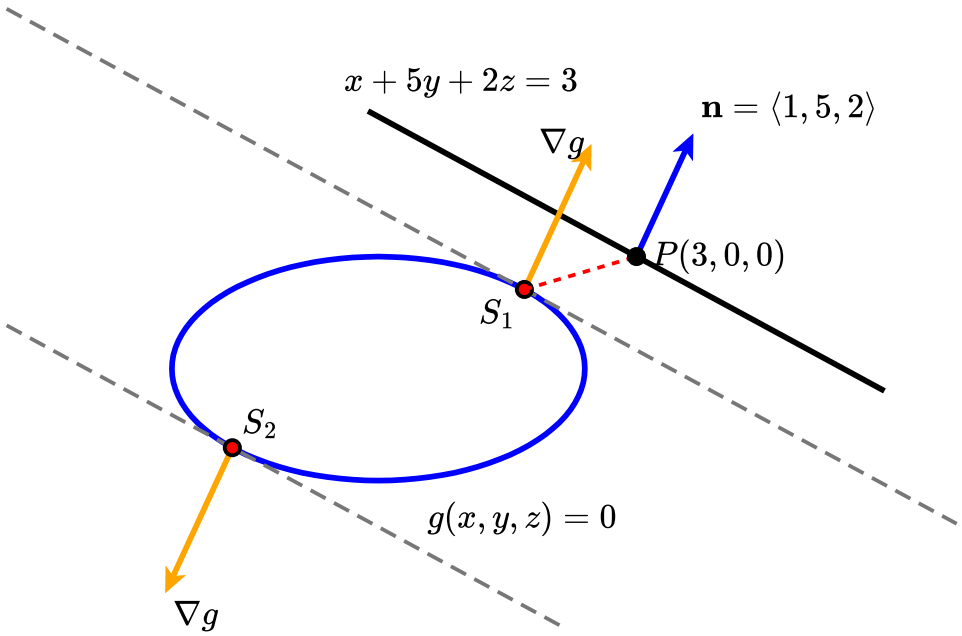


Figure 5.16.3 A schematic cross-section. The tangent plane to the ellipsoid is parallel to the given plane exactly at the closest point S_1 and the farthest point S_2, where ∇g is parallel to the plane's normal $\mathbf{n}$.

Choose a point on the plane, say $P(3, 0, 0)$. For a point $S(x, y, z)$ on the ellipsoid, the distance from S to the plane is

$$\frac{\overrightarrow{PS} \cdot \mathbf{n}}{|\mathbf{n}|} = \frac{\langle x - 3,\ y,\ z \rangle \cdot \langle 1, 5, 2 \rangle}{|\mathbf{n}|}.$$

Since $|\mathbf{n}|$ is constant, it suffices to optimize the numerator

$$f(x, y, z) = x - 3 + 5y + 2z$$

subject to the constraint

$$g(x, y, z) = x^2 + y^2 + 3z^2 - 1 = 0.$$

The Lagrange condition $\nabla f = \lambda \nabla g$ gives

$$\langle 1, 5, 2 \rangle = \lambda \langle 2x,\ 2y,\ 6z \rangle,$$

so

$$x = \frac{1}{2\lambda}, \qquad y = \frac{5}{2\lambda}, \qquad z = \frac{1}{3\lambda}.$$

Substituting into the constraint $x^2 + y^2 + 3z^2 = 1$,

$$\begin{aligned} \frac{1}{4\lambda^2} + \frac{25}{4\lambda^2} + \frac{3}{9\lambda^2} &= 1 \\ \frac{3 + 75 + 4}{12\lambda^2} &= 1 \\ \frac{1}{\lambda^2} &= \frac{12}{82}, \end{aligned}$$

so that

$$\frac{1}{\lambda} = \pm 2\sqrt{\frac{3}{82}}.$$

Therefore

$$x = \pm\sqrt{\frac{3}{82}}, \qquad y = \pm 5\sqrt{\frac{3}{82}}, \qquad z = \pm\sqrt{\frac{2}{123}},$$

and the two points are

$$S_1\left(\sqrt{\tfrac{3}{82}},\ 5\sqrt{\tfrac{3}{82}},\ \sqrt{\tfrac{2}{123}}\right) \qquad \text{and}$$

$$S_2\left(-\sqrt{\tfrac{3}{82}},\ -5\sqrt{\tfrac{3}{82}},\ -\sqrt{\tfrac{2}{123}}\right).$$

The positive signs give S_1, the point closest to the plane, and the negative signs give S_2, the farthest. At both points the tangent plane is parallel to the plane $x + 5y + 2z = 3$.

7. **Maximizing a Cobb-Douglas utility.** A consumer has \$600 to spend on two commodities, the first of which costs \$20 per unit and the second \$30 per unit. Suppose that the utility derived by the consumer from x units of the first commodity and y units of the second commodity is given by the Cobb-Douglas utility function

$$U(x, y) = 10\, x^{0.6} y^{0.4}.$$

How many units of each commodity should the consumer buy to maximize utility?

Solution. The budget constraint is $20x + 30y = 600$, so we take

$$g(x, y) = 20x + 30y - 600 = 0,$$

with $g_x = 20$ and $g_y = 30$, and maximize $U(x, y) = 10\, x^{0.6} y^{0.4}$, whose partial derivatives are

$$U_x = 6\, x^{-0.4} y^{0.4}, \qquad U_y = 4\, x^{0.6} y^{-0.6}.$$

The Lagrange condition $\nabla U = \lambda \nabla g$ gives the system

$$6\, x^{-0.4} y^{0.4} = 20\lambda, \qquad 4\, x^{0.6} y^{-0.6} = 30\lambda.$$

Dividing the first equation by the second eliminates λ:

$$\frac{3}{2} \cdot \frac{y}{x} = \frac{2}{3} \qquad \Longrightarrow \qquad y = \frac{4}{9}\, x.$$

Substituting into the budget constraint,

$$\begin{aligned} 20x + 30\left(\frac{4}{9}x\right) &= 600 \\ 20x + \frac{40}{3}x &= 600 \\ \frac{100}{3}x &= 600, \end{aligned}$$

so $x = 18$ and $y = 18 \cdot \frac{4}{9} = 8$. The consumer should buy 18 units of the first commodity and 8 units of the second.

8. You have 24 square inches of cardboard and want to build a box in the shape of a rectangular prism. Find the dimensions of the box, if you would like it to have the largest possible volume. See Figure 5.16.4.

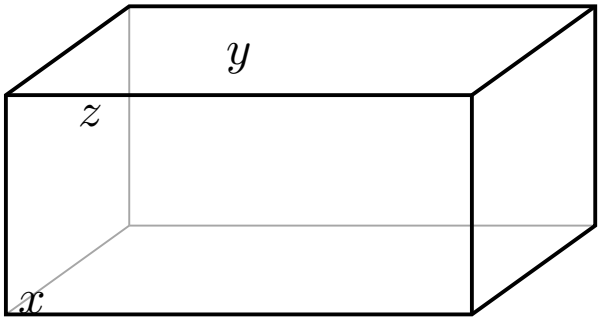


Figure 5.16.4 A rectangular prism with dimensions x, y, and z.

Solution. The surface area of the box is

$$2xy + 2xz + 2yz = 24,$$

so the constraint is $g(x, y, z) = xy + xz + yz - 12 = 0$, and the function to be maximized is $f(x, y, z) = xyz$. We

compute

$$\nabla f = \langle f_x, f_y, f_z\rangle = \langle yz, xz, xy\rangle,$$
$$\nabla g = \langle y+z, x+z, x+y\rangle.$$

The Lagrange system $\nabla f = \lambda \nabla g$ is

$$yz = \lambda(y+z)$$
$$xz = \lambda(x+z)$$
$$xy = \lambda(x+y)$$

Multiplying the equations by x, y, and z respectively gives

$$xyz = \lambda x(y+z), \qquad xyz = \lambda y(x+z), \qquad xyz = \lambda z(x+y).$$

Equating the first two, $x(y+z) = y(x+z)$, which simplifies to $xz = yz$, i.e. $x = y$. Equating the last two, $y(x+z) = z(x+y)$, which simplifies to $xy = xz$, i.e. $y = z$.

So $x = y = z$, and substituting into the constraint $xy + xz + yz - 12 = 0$ gives $3x^2 = 12$, or $x^2 = 4$. Hence

$$x = y = z = 2 \text{ inches}$$

and these dimensions (a cube) will maximize the volume.

9. Use the method of Lagrange Multipliers to find the maximum and minimum values of $f(x,y) = \sqrt{2}\,x - \sqrt{2}\,y$ subject to the constraint $x^2 + y^2 = 4$.

Solution. With $g(x,y) = x^2 + y^2 - 4 = 0$, we have

$$\nabla f = \sqrt{2}\,\mathbf{i} - \sqrt{2}\,\mathbf{j}, \qquad \nabla g = (2x)\,\mathbf{i} + (2y)\,\mathbf{j}.$$

The system $\nabla f = \lambda \nabla g$ is

$$\sqrt{2} = 2\lambda x \;\Rightarrow\; x = \frac{1}{\sqrt{2}\,\lambda},$$
$$-\sqrt{2} = 2\lambda y \;\Rightarrow\; y = -\frac{1}{\sqrt{2}\,\lambda}.$$

Substituting into the constraint,

$$\left(\frac{1}{\sqrt{2}\,\lambda}\right)^2 + \left(-\frac{1}{\sqrt{2}\,\lambda}\right)^2 = 4 \;\Rightarrow\; \frac{1}{\lambda^2} = 4 \;\Rightarrow\; \lambda = \pm\frac{1}{2}.$$

For $\lambda_1 = \frac{1}{2}$: $x = \sqrt{2}$, $y = -\sqrt{2}$. For $\lambda_2 = -\frac{1}{2}$: $x = -\sqrt{2}$, $y = \sqrt{2}$. Evaluating f,

$$f\left(\sqrt{2}, -\sqrt{2}\right) = 2 + 2 = 4 \qquad \text{absolute maximum,}$$
$$f\left(-\sqrt{2}, \sqrt{2}\right) = -2 - 2 = -4 \qquad \text{absolute minimum.}$$

See Figure 5.16.5.

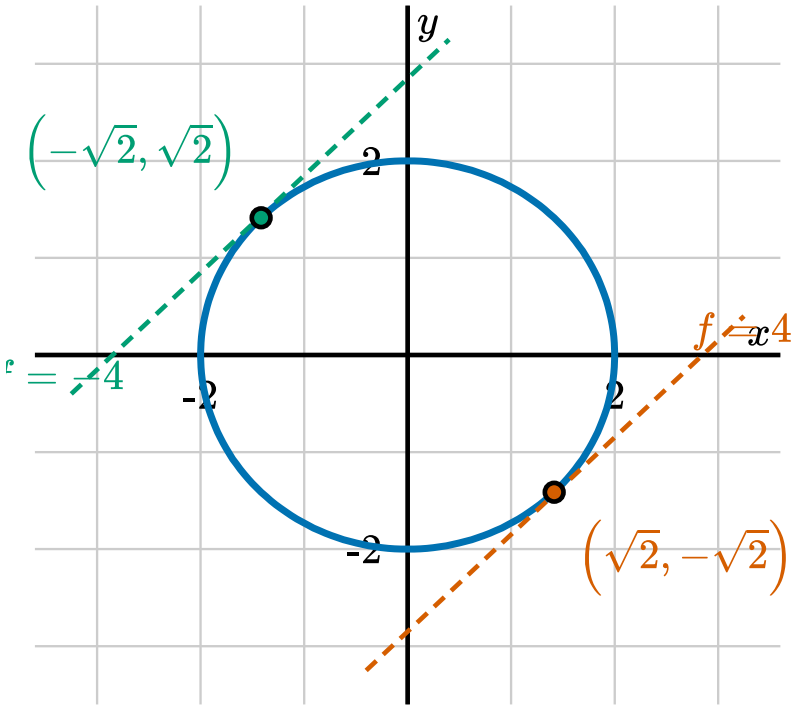


Figure 5.16.5 The constraint circle $x^2 + y^2 = 4$ and the level lines $\sqrt{2}\,x - \sqrt{2}\,y = \pm 4$ of f, which are tangent to the circle at the extrema $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$.

10. Given $f(x, y) = -x^2 + 4x - y^2 - 4y$.

A. Find the maximum and minimum values of $f(x, y)$ on the circle $x^2+y^2 = 9$, first using Lagrange multipliers, then again by parametrizing the circle.

B. Find the maximum and minimum values of $f(x, y)$ over the disk $x^2 + y^2 \le 9$.

Solution.

A. *Lagrange multipliers.* With $g(x, y) = x^2 + y^2 - 9 = 0$, the system $\nabla f = \lambda \nabla g$ reads

$$\begin{aligned} -2x + 4 = 2\lambda x \qquad &\Rightarrow\ x = \frac{2}{1+\lambda}, \\ -2y - 4 = 2\lambda y \qquad &\Rightarrow\ y = \frac{-2}{1+\lambda}. \end{aligned}$$

Hence $y = -x$, and substituting into the constraint, $2x^2 = 9$, so $x = \pm\frac{3}{\sqrt{2}}$ and $y = \mp\frac{3}{\sqrt{2}}$. Evaluating f at these two points, and noting that $-x^2 - y^2 = -9$ on the circle,

$$f = -9 + 4x - 4y = -9 + 4(x - y) = -9 \pm \frac{24}{\sqrt{2}}.$$

So the maximum on the circle is $-9 + \frac{24}{\sqrt{2}}$ at $\left(\frac{3}{\sqrt{2}}, -\frac{3}{\sqrt{2}}\right)$ and the minimum is $-9 - \frac{24}{\sqrt{2}}$ at $\left(-\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$.

By parametrizing. On the circle, $x = 3\cos t$, $y = 3\sin t$, so

$$\begin{aligned} f &= -9\cos^2 t + 12\cos t - 9\sin^2 t - 12\sin t \\ &= -9 + 12(\cos t - \sin t) = -9 + 12\sqrt{2}\cos\left(t + \frac{\pi}{4}\right). \end{aligned}$$

Since cos ranges over $[-1, 1]$, this ranges over $-9 \pm 12\sqrt{2} = -9 \pm \frac{24}{\sqrt{2}}$, agreeing with the Lagrange computation.

B. Over the disk we must also check the interior. Setting

$$f_x = -2x + 4 = 0 \;\Rightarrow\; x = 2, \qquad f_y = -2y - 4 = 0 \;\Rightarrow\; y = -2,$$

gives the critical point $(2, -2)$, which does lie inside the disk since $4 + 4 = 8 < 9$. There $f(2, -2) = -4 + 8 - 4 + 8 = 8$. (It is a local maximum: $f_{xx} = -2 < 0$, $f_{yy} = -2$, $f_{xy} = 0$, so $D = 4 > 0$.)

Comparing with the boundary values from part A, the maximum over the disk is 8 at $(2, -2)$, and the minimum is $-9 - \frac{24}{\sqrt{2}} \approx -25.97$ at $\left(-\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$. See Figure 5.16.6.

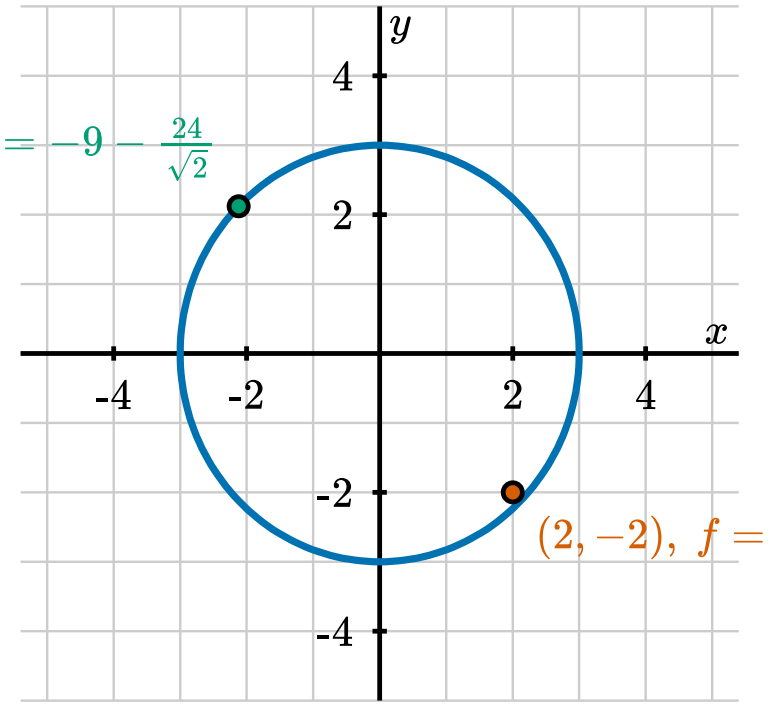


Figure 5.16.6 The disk $x^2 + y^2 \le 9$. The maximum of f occurs at the interior critical point $(2, -2)$, and the minimum at the boundary point $\left(-\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$.

11. You are in charge of erecting a radio telescope on a newly discovered planet. To minimize interference, you want to place it where the magnetic field of the planet is weakest. The planet is spherical, with a radius of 6 units. Based on a coordinate system whose origin is at the center of the planet, the strength of the magnetic field is given by

$$M(x, y, z) = 6x - y^2 + xz + 60.$$

Where should you locate the radio telescope?

Solution. We must minimize M subject to the constraint that we stay on the surface of the planet, $g(x, y, z) = x^2 + y^2 + z^2 = 36$. The Lagrange system $\nabla M = \lambda \nabla g$ is

$$\begin{aligned} 6 + z &= 2\lambda x, \\ -2y &= 2\lambda y, \\ x &= 2\lambda z. \end{aligned}$$

The second equation factors as $2y(\lambda + 1) = 0$, so either $\lambda = -1$ or $y = 0$.

Case $\lambda = -1$. The first and third equations become $6 + z = -2x$ and $x = -2z$. Substituting, $6 + z = 4z$, so $z = 2$ and $x = -4$. The constraint then gives $16 + y^2 + 4 = 36$, i.e. $y = \pm 4$. At these points

$$M(-4, \pm 4, 2) = -24 - 16 - 8 + 60 = 12.$$

Case $y = 0$. From the third equation $\lambda = \dfrac{x}{2z}$ (with $z \neq 0$; if $z = 0$ then $x = 0$ and the first equation gives $6 = 0$, which is impossible). Substituting into the first, $6 + z = \dfrac{x^2}{z}$, i.e. $x^2 = 6z + z^2$. Combined with

$x^2 + z^2 = 36$ this gives

$$6z + z^2 = 36 - z^2 \;\Rightarrow\; 2z^2 + 6z - 36 = 0 \;\Rightarrow\; (z+6)(z-3) = 0,$$

so $z = -6$ (giving $x = 0$, with $M = 60$) or $z = 3$ (giving $x = \pm 3\sqrt{3}$, with $M = 60 \pm 27\sqrt{3} \approx 13.2$ or 106.8).

Comparing all the candidate values $\{12,\ 60,\ 13.2,\ 106.8\}$, the magnetic field is weakest where $M = 12$. So the radio telescope should be located at

$$(-4, 4, 2) \qquad \text{or} \qquad (-4, -4, 2).$$

5.17 Review Problems #4

These are the problems on extreme values and Lagrange multipliers that were not assigned in Assignment 10, collected here for review. They are not assigned, but they are fair game on an exam. If you would like to review the material first, see Extreme Values and Saddle Points and Lagrange Multipliers.

1. **True or False.** If $f(x,y)$ has two local maxima, then it must have a local minimum too.
Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Answer. False.

Solution. False.
This is true for single-variable functions but no longer holds in two dimensions. For example, the function $f(x,y) = -x^4 + x^2 - \frac{1}{2}y^2$, shown in Figure 5.17.1, has two local maxima at $\left(\pm\frac{1}{\sqrt{2}}, 0\right)$ with a saddle point at the origin lying between them, and no local minimum.

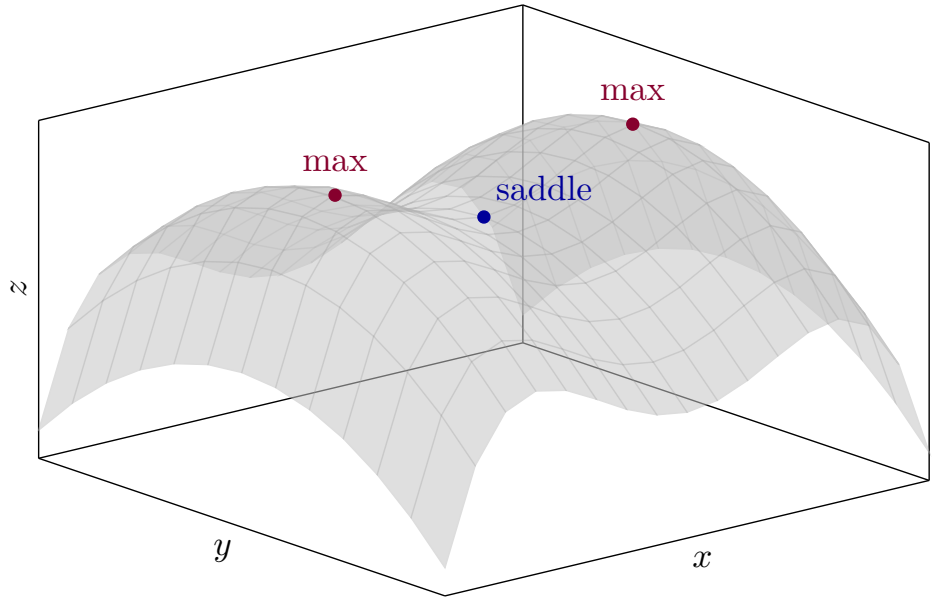


Figure 5.17.1 The graph of $f(x,y) = -x^4 + x^2 - \frac{1}{2}y^2$: two local maxima with a saddle point between them, and no local minimum.

2. Let $f(x,y) = x^2 - y^2 - 2x + 4y + 6$. Find all the critical points and classify them as local maxima, local minima, and saddle points.

Solution. The first and second partial derivatives are

$$f_x = 2x - 2, \qquad f_{xx} = 2, \qquad f_{xy} = f_{yx} = 0,$$
$$f_y = -2y + 4, \qquad f_{yy} = -2.$$

Setting $f_x = 0$ gives $x = 1$ and $f_y = 0$ gives $y = 2$, so the only critical point is $(1,2)$. The Hessian is

$$H(x,y) = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ 0 & -2 \end{vmatrix} = -4 < 0,$$

so $(1,2)$ is a saddle point since the Hessian is negative.

3. Find the absolute maximum and absolute minimum of $f(x,y) = 2 + xy - 2x - \frac{1}{4}y^2$ in the closed triangular region with vertices given by $(0,0)$, $(1,0)$, and $(0,2)$.

Solution. *Critical points:*

$$f_x = y - 2 = 0 \;\Rightarrow\; y = 2, \qquad f_y = x - \frac{y}{2} = 0 \;\Rightarrow\; x = 1,$$

so the only critical point is $(1,2)$. Note that this point is NOT in the closed region R, so we only need to check the boundary.

The boundary: we check the three sides of the triangle.

- L_1: $x = 0$, $0 \le y \le 2$. Here $f(0, y) = 2 - \frac{y^2}{4}$, with maximum 2 at $(0, 0)$ and minimum 1 at $(0, 2)$.
- L_2: $y = 0$, $0 \le x \le 1$. Here $f(x, 0) = 2 - 2x$, with maximum 2 at $(0, 0)$ and minimum 0 at $(1, 0)$.
- L_3: $y = -2x + 2$, $0 \le x \le 1$. Here

$$g(x) = f(x, -2x + 2) = 2 - 2x^2 - (x - 1)^2 = 1 - 3x^2 + 2x,$$

and $g'(x) = -6x + 2 = 0$ gives $x = \frac{1}{3}$, $y = \frac{4}{3}$, with $f\left(\frac{1}{3}, \frac{4}{3}\right) = 2 + \frac{4}{9} - \frac{2}{3} - \frac{4}{9} = \frac{4}{3}$. The endpoints of L_3 give the previously found values 1 and 0.

Comparing all the values, the absolute maximum of f is 2, attained at $(0, 0)$, and the absolute minimum is 0, attained at $(1, 0)$. See Figure 5.17.2.

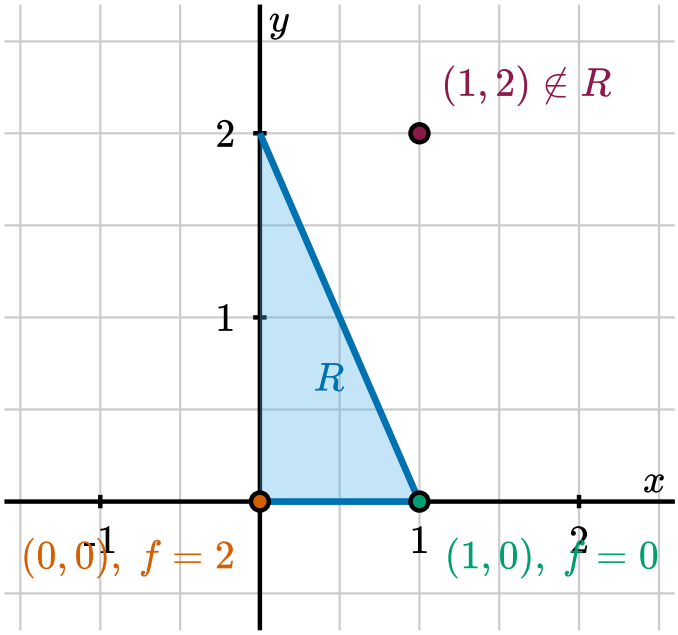


Figure 5.17.2 The triangular region R with vertices $(0, 0)$, $(1, 0)$, and $(0, 2)$. The critical point $(1, 2)$ lies outside R; the absolute maximum 2 occurs at $(0, 0)$ and the absolute minimum 0 at $(1, 0)$.

4. The porosity of the soil beneath an elliptical section of a field, $x^2 + 4y^2 \le 25$, is given by

$$f(x, y) = 2x^2 - 9x + 2y^2 + 20.$$

Find the (x, y) location(s) in this section of the field where water will drain the fastest and slowest, i.e. the locations where porosity is an absolute maximum or absolute minimum.

Solution. The region is closed and bounded and f is a polynomial, hence continuous, so by the Extreme Value Theorem absolute extrema exist. We check the interior and the boundary.

Interior critical points:

$$f_x = 4x - 9 = 0 \;\Rightarrow\; x = \frac{9}{4}, \qquad f_y = 4y = 0 \;\Rightarrow\; y = 0.$$

The point $\left(\frac{9}{4}, 0\right)$ satisfies $\left(\frac{9}{4}\right)^2 = \frac{81}{16} \approx 5.06 < 25$, so it lies inside the region, and

$$\begin{aligned} f\left(\frac{9}{4}, 0\right) &= 2\left(\frac{81}{16}\right) - 9\left(\frac{9}{4}\right) + 20 \\ &= \frac{81}{8} - \frac{81}{4} + 20 = -\frac{81}{8} + 20 = \frac{79}{8} = 9.875. \end{aligned}$$

The boundary: on $x^2 + 4y^2 = 25$ we have $y^2 = \dfrac{25 - x^2}{4}$ with $-5 \le x \le 5$, so

$$f = 2x^2 - 9x + 2 \cdot \frac{25 - x^2}{4} + 20 = \frac{3}{2}x^2 - 9x + \frac{65}{2},$$

a function of the single variable x. Setting its derivative to zero,

$$3x - 9 = 0 \;\Rightarrow\; x = 3 \;\Rightarrow\; y^2 = \frac{25 - 9}{4} = 4 \;\Rightarrow\; y = \pm 2,$$

with $f(3, \pm 2) = \frac{27}{2} - 27 + \frac{65}{2} = 19$. At the endpoints of the interval,

$$f(5, 0) = \frac{75}{2} - 45 + \frac{65}{2} = 25, \qquad f(-5, 0) = \frac{75}{2} + 45 + \frac{65}{2} = 115.$$

Comparing all the values $\left\{\frac{79}{8}, 19, 25, 115\right\}$, the porosity is an absolute *maximum* of 115 at $(-5, 0)$ — where water drains the fastest — and an absolute *minimum* of $\frac{79}{8}$ at $\left(\frac{9}{4}, 0\right)$ — where water drains the slowest. See Figure 5.17.3.

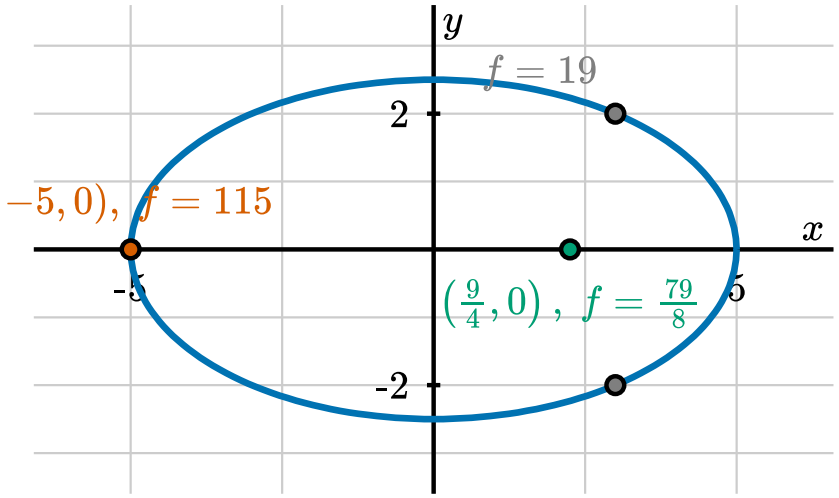


Figure 5.17.3 The elliptical section $x^2 + 4y^2 \le 25$. The porosity is largest at the boundary point $(-5, 0)$ and smallest at the interior critical point $\left(\frac{9}{4}, 0\right)$.

5. A product on a circle. Find the maximum and minimum values of the function $f(x, y) = 4xy$ subject to the constraint $x^2 + y^2 = 8$.

Solution. Write the constraint as

$$g(x, y) = x^2 + y^2 - 8 = 0,$$

so that $g_x = 2x$ and $g_y = 2y$, while $f_x = 4y$ and $f_y = 4x$. The Lagrange condition $\nabla f = \lambda \nabla g$ gives

$$4y = 2\lambda x, \qquad 4x = 2\lambda y.$$

Solving each equation for λ,

$$\lambda = \frac{2y}{x} = \frac{2x}{y} \qquad \Longrightarrow \qquad x^2 = y^2.$$

Substituting into the constraint $x^2 + y^2 - 8 = 0$ gives $2x^2 = 8$, so

$$x = \pm 2, \qquad y = \pm 2.$$

Hence there are four extrema: $(2, 2)$, $(2, -2)$, $(-2, 2)$, $(-2, -2)$.

Evaluating f at the four candidates:

(x, y)	$f(x, y)$
$(2, 2)$	16
$(-2, -2)$	16
$(2, -2)$	-16
$(-2, 2)$	-16

The maximum value is 16, attained at $(2, 2)$ and $(-2, -2)$; the minimum value is -16, attained at $(2, -2)$ and $(-2, 2)$.

Geometric interpretation. A rectangle with corners $(\pm x, \pm y)$ inscribed in the circle $x^2 + y^2 = 8$ has area exactly $4xy = f(x, y)$. As the corner (x, y) moves along the circle, we obtain rectangles of different areas.

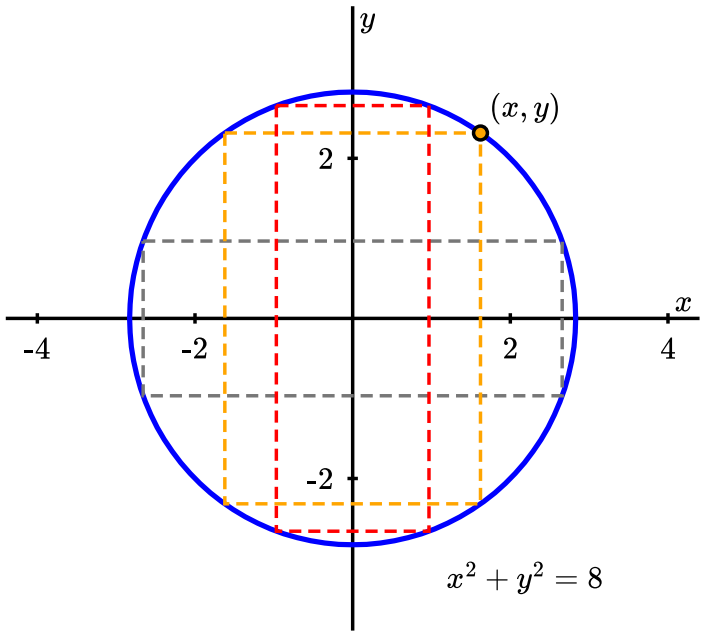


Figure 5.17.4 Some of the possible rectangles inscribed in the circle $x^2 + y^2 = 8$. Each has corners $(\pm x, \pm y)$ for a point (x, y) on the circle, and therefore area $4xy = f(x, y)$.

The values of (x, y) that maximize f are exactly the values that maximize the area of the inscribed rectangle.

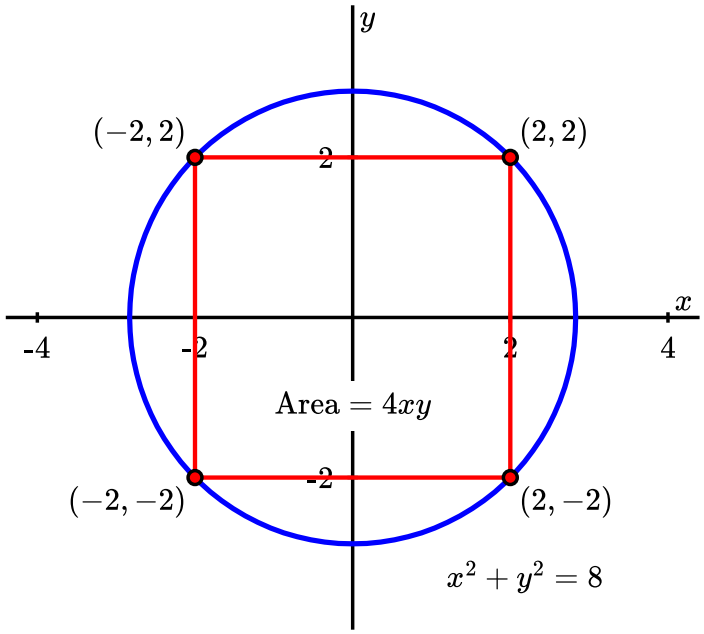


Figure 5.17.5 The optimal solution: the inscribed rectangle with corners $(\pm 2, \pm 2)$, whose area $4xy = f(2,2) = 16$ is the largest possible.

6. Consider $f(x,y) = xy$ and $x^2 - y = 12$. We assume $y \leq 0$.

A. Why do we need $y \leq 0$ here?

B. Find the extreme values of f subject to the above constraints.

Solution.

A. Else a minimum and maximum would not exist, since we could just pick y very large and $x = \pm\sqrt{y+12}$ would also have a very large magnitude, resulting in a very large or very negative value of $f(x,y)$.

B. The constraint describes the portion of the parabola $y = x^2 - 12$ in the third and fourth quadrant. This shape is closed and bounded so we are guaranteed the existence of global maxima and minima. The two boundary points, which occur at $y = 0$, need to be handled separately; this part is not very hard, since we immediately see that f has value 0 on them.

Now we proceed to find the critical points of f on the parabola $g(x,y) = x^2 - y - 12 = 0$. The system of equations we need to solve is obtained from $\nabla f(x,y) = \langle y, x\rangle$ and $\nabla g(x,y) = \langle 2x, -1\rangle$. It is

$$\begin{aligned} y &= 2\lambda x \\ x &= -\lambda \\ x^2 - y &= 12 \end{aligned}$$

Combining the first two equations we see that $y = -2x^2$, and using the third, $3x^2 = 12$, so $x = \pm 2$ and $y = -8$. Computing the value of f at these points, we see that the global maximum of f is at $(-2,-8)$ with a value of 16, while the global minimum with value -16 is at $(2,-8)$. See Figure 5.17.6.

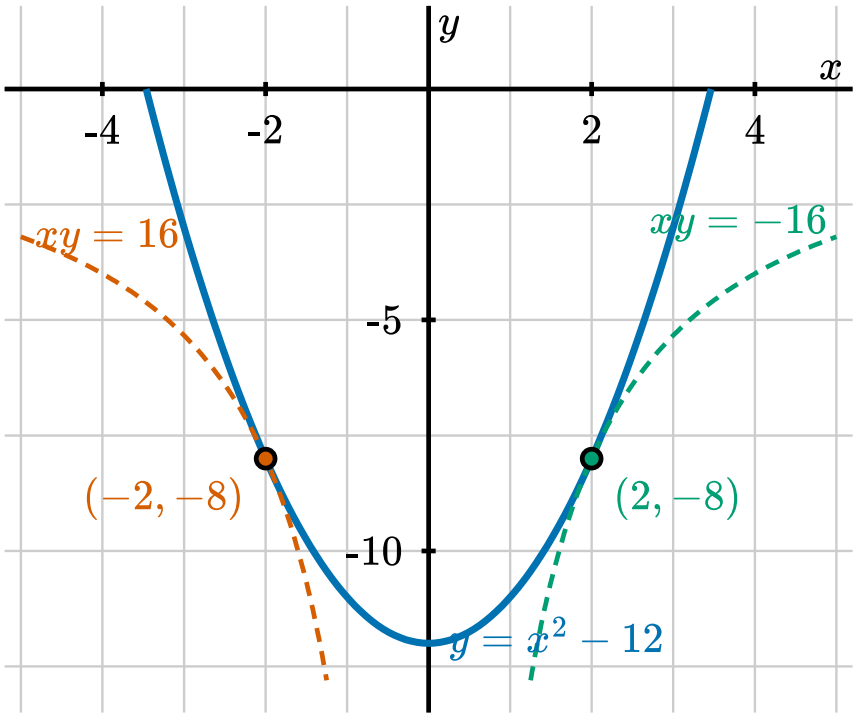


Figure 5.17.6 The constraint curve $y = x^2 - 12$ with $y \le 0$, and the level curves $xy = 16$ and $xy = -16$ of f. The extrema occur at $(-2, -8)$ and $(2, -8)$, where the level curves are tangent to the parabola.

7. Find the point(s) on the surface $xyz = 1$ closest to the origin.

Solution. The distance of a point from the origin is $\sqrt{x^2 + y^2 + z^2}$, so it suffices to minimize $f(x, y, z) = x^2 + y^2 + z^2$ with the constraint $g(x, y, z) = xyz - 1 = 0$. The Lagrange system $\nabla f = \lambda \nabla g$ is

$$\langle 2x, 2y, 2z \rangle = \langle \lambda yz, \lambda xz, \lambda xy \rangle.$$

Multiplying the first equation by x and using $xyz = 1$,

$$2x = \lambda yz \;\Rightarrow\; 2x^2 = \lambda xyz \overset{xyz=1}{\Rightarrow} 2x^2 = \lambda \;\Rightarrow\; x^2 = \frac{\lambda}{2},$$

and similarly $y^2 = \dfrac{\lambda}{2}$ and $z^2 = \dfrac{\lambda}{2}$, hence $x^2 = y^2 = z^2$.

Substituting into the constraint,

$$x^2 y^2 z^2 = 1 \;\Rightarrow\; x^6 = 1 \;\Rightarrow\; x^2 = 1.$$

Since $xyz = 1$, we know that x, y, and z are either all positive or two are negative and the third one is positive. Hence, the closest points are

$$(1, 1, 1), \quad (1, -1, -1), \quad (-1, 1, -1), \quad (-1, -1, 1).$$

8. Find the point $P(x, y, z)$ closest to the origin on the plane $2x + y - z - 5 = 0$.

Solution. The distance from the origin is $\sqrt{x^2 + y^2 + z^2}$, so it suffices to minimize the function $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraint $g(x, y, z) = 2x + y - z - 5 = 0$. We compute

$$\nabla f = \langle 2x, 2y, 2z \rangle, \qquad \nabla g = \langle 2, 1, -1 \rangle.$$

The Lagrange system $\nabla f = \lambda \nabla g$ is

$$\begin{aligned} 2x &= 2\lambda && \Rightarrow\; x = \lambda, \\ 2y &= \lambda && \Rightarrow\; y = \frac{\lambda}{2}, \end{aligned}$$

$$2z = -\lambda \qquad \Rightarrow\ z = -\frac{\lambda}{2}.$$

Substituting into the equation of the plane,

$$2\lambda + \frac{\lambda}{2} + \frac{\lambda}{2} - 5 = 0 \ \Rightarrow\ 3\lambda = 5 \ \Rightarrow\ \lambda = \frac{5}{3}.$$

Hence the closest point is

$$P\left(\frac{5}{3},\ \frac{5}{6},\ -\frac{5}{6}\right).$$

(Only one critical point arises, and the distance certainly has a minimum on the plane but no maximum, so this point must be the minimizer.)

9. Find the minimum value of $x + y^2 + z^3$ along the part of the plane $x + y + z = 2$ in the first octant $(x, y, z \geq 0)$. The solution found by Lagrange multipliers gives the minimum. Use common sense for the maximum value.

Solution. Let $f(x, y, z) = x + y^2 + z^3$ and $g(x, y, z) = x + y + z - 2 = 0$. Then

$$\nabla f = \left\langle 1, 2y, 3z^2 \right\rangle, \qquad \nabla g = \langle 1, 1, 1 \rangle,$$

and $\nabla f = \lambda \nabla g$ gives

$$1 = \lambda, \qquad 2y = \lambda, \qquad 3z^2 = \lambda.$$

Hence $\lambda = 1$, $y = \frac{1}{2}$, and $z = \frac{1}{\sqrt{3}}$ (taking the positive root, since $z \geq 0$). From the constraint, $x = 2 - \frac{1}{2} - \frac{1}{\sqrt{3}} = \frac{3}{2} - \frac{1}{\sqrt{3}}$, which is positive, so the point is in the first octant.

The minimum value is therefore

$$\begin{aligned} f &= \left(\frac{3}{2} - \frac{1}{\sqrt{3}}\right) + \frac{1}{4} + \frac{1}{3\sqrt{3}} \\ &= \frac{7}{4} - \frac{1}{\sqrt{3}} + \frac{1}{3\sqrt{3}} = \frac{7}{4} - \frac{3}{3\sqrt{3}} + \frac{1}{3\sqrt{3}} \\ &= \frac{7}{4} - \frac{2}{3\sqrt{3}} = \frac{7}{4} - \frac{2}{3}\sqrt{\frac{1}{3}}. \end{aligned}$$

For the maximum, note that the region is a triangle with vertices $(2, 0, 0)$, $(0, 2, 0)$, and $(0, 0, 2)$. Since the largest power dominates, the largest value should occur at the vertex where all the "weight" is on z. Checking the three vertices,

$$f(2, 0, 0) = 2, \qquad f(0, 2, 0) = 4, \qquad f(0, 0, 2) = 8,$$

so the maximum value is 8, attained at $(0, 0, 2)$.

10. Classify the Critical Points. The function $f(x, y) = x^3 - 3x + y^3 - 3y$ has four critical points. Drag each one into the correct category.

$(1, 1)$	Local minimum
$(-1, -1)$	Local maximum
$(1, -1)$	Saddle point
$(-1, 1)$	

Solution 1. From $f_x = 3x^2 - 3 = 0$ and $f_y = 3y^2 - 3 = 0$ we get $x = \pm 1$ and $y = \pm 1$, giving four critical points. Since $f_{xx} = 6x$, $f_{yy} = 6y$ and $f_{xy} = 0$,

$$D = f_{xx} f_{yy} - f_{xy}^2 = 36xy.$$

So $D = 36 > 0$ at $(1, 1)$ and $(-1, -1)$, while $D = -36 < 0$ at $(1, -1)$ and $(-1, 1)$ — both saddles. Among the two with $D > 0$, the sign of f_{xx} decides: $f_{xx}(1, 1) = 6 > 0$ gives a local minimum, and $f_{xx}(-1, -1) = -6 < 0$ gives a local maximum.

Solution 2.

- Local minimum

- $(1, 1)$

- Local maximum
  - $(-1, -1)$
- Saddle point
  - $(1, -1)$
  - $(-1, 1)$